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Engineering and Design
Geotechnical Design of Shallow Foundations on Soils

FOR THE COMMANDER:

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Purpose. This engineer manual provides guidance on geotechnical design calculations for shallow foundations supporting various types of structures. Performance criteria are also provided. Discussion of construction methods and equipment are generally limited to those aspects that are most important to the design engineers. The manual is not intended to replace the judgment of the design engineer.

Applicability. This manual applies to Headquarters, U.S. Army Corps of Engineers, Office of the Corps of Engineers elements, major subordinate commands, districts, laboratories, and field operating activities having responsibilities for the design and construction of military and civil works projects.

Distribution statement. Approved for public release; distribution is unlimited.

Proponent and exception authority. The proponent of this manual is the Headquarters, U.S. Army Corps of Engineers, Directorate of Engineering and Construction. The proponent has the authority to approve exceptions or waivers to this manual that are consistent with controlling law and regulations. Only the proponent of a publication or form may modify it by officially revising or rescinding it.

*This manual supersedes EM 1110-1-1904, dated 30 September 1990 and EM 1110-1-1905, dated 30 October 1992.

Summary of Change

EM 1110-1-1905
Geotechnical Design of Shallow Foundations on Soils

Methods presented in this manual have been made more consistent with EM 1110-2-2100, Stability Analysis of Concrete Structures; EM 1110-2-2502, Retaining and Floodwalls; and utilized in current U.S. Army Corps of Engineers software.

This manual is an update and compilation of the information pertaining to shallow foundations presented in EM 1110-1-1904, Settlement Analysis, dated 30 September 1990, and EM 1110-1-1905, Bearing Capacity of Soils, dated 30 October 1992. These documents are now combined and incorporate the following changes:

- All chapters of the new manual were updated to incorporate and discuss current design philosophies and standards for geotechnical design of shallow foundations.
- The following chapters are new and address subject matter that was not previously or fully covered by either manual:
    - Chapter 2 – Design and Performance Requirements.
    - Chapter 4 – Site Exploration.
    - Chapter 5 – Ultimate Limit States for Static Analysis.
    - Chapter 8 – Mat Foundations.
    - Chapter 9 – Seismic Design.
    - Chapter 11 – Ground Improvement.
    - Chapter 12 – Earthwork Observation, Testing, Observation, and Documentation.
- The following content from EM 1110-1-1904 (1990) and EM 1110-1-1905 (1992) was removed and the content was not relocated to another chapter in the current version:
    - Deep foundations.
    - Design of embankments.
    - Design of shallow foundations on expansive and collapsible soil.
    - Laboratory consolidometer tests.
    - Computer programs.
- References were updated.

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Glossary of Terms

Chapter 1
Overview

1–1. Purpose

This engineer manual provides guidance on geotechnical design calculations for shallow foundations supporting various types of structures. Performance criteria are also provided. Discussion of construction methods and equipment are generally limited to those aspects that are most important to the design engineers. The manual is not intended to replace the judgment of the design engineer.

1–2. Distribution statement

Approved for public release; distribution is unlimited.

1–3. References

See Appendix A.

1–4. Records management (recordkeeping) requirements

The records management requirement for all record numbers, associated forms, and reports required by this publication are addressed in the Army Records Retention Schedule. Detailed information for all related record numbers is located on the U.S. Army Corps of Engineers (USACE) Records Management Site https://usace.dps.mil/sites/INTRA-CIOG6/SitePages/Records-Management.aspx. If any record numbers, forms, and reports are not current, addressed, and/or published correctly, see DA Pam 25-403 for guidance.

1–5. Associated publications

Guidance for the analysis and design of floodwalls and other hydraulic structures is provided in EM 1110-2-2100, EM 1110-2-2502, and the referenced applicable portions of this manual.

1–6. Mandatory requirements and deviations

Due to USACE's unique structures and their associated risks, policy standards in this manual may exceed industry requirements. The minimum standards are intended to ensure reasonable uniformity throughout USACE; however, the designer may exceed the minimum standards to enhance resiliency as risk and economics dictate.

a. *Mandatory requirements*. In certain cases, design guidance is mandatory as discussed in ER 1110-2-1150. These mandatory requirements are intended to ensure project safety and function. These cases will be identified as "mandatory," or the word "must" will be used. The term "should" indicates a strong preference for a given criterion and a USACE recommendation.

b. Exceptions. No standard can be considered universally applicable. When a deviation from mandatory requirements is deemed necessary, the designer must follow the policy requirements outlined in ER 1110-2-1150 and ER 1110-345-100.

1–7. Applicability

This manual is applicable to military and civil works structures.

a. Soil foundation material. Analyses within this manual are applicable to foundations supported on soil. Soil is a mixture of irregularly shaped mineral particles of various sizes containing voids between particles. These voids might contain water if the soil is saturated, water and air if unsaturated, and air if dry. Materials include sands, silts, clays, gravels, and mixtures of these components.

b. Rock foundation material. This manual is not applicable to shallow foundations on intact, jointed, layered, and fractured rock materials. Design on these materials is discussed in EM 1110-1-2908. Highly fractured rock and rockfill can behave like dense cohesionless sands and gravels, and methods in this manual are applicable. The distinction is generally made when the median particle (or rock block) size exceeds 1/50th of the footing width.

c. Load application. Foundation design methods in this manual assume static loading conditions unless due to earthquake. For seismic design, failure zones are mostly based on an assumed static failure surface (pseudo-static) and various parameters are used to characterize these loads. Guidance for shallow foundations on soils subjected to seismic loads is provided in this manual.

d. Problematic ground conditions and ground improvement. Guidance for shallow foundation design for problematic ground conditions and ground improvement is not fully covered in this manual. Available references are provided.

e. Settlement analysis. The methods presented in this manual cover the general theory of settlement, immediate (or distortion) settlement, primary consolidation settlement, and secondary settlement (or creep). Stress distributions are included in this manual for embankments as well as shallow foundations. Settlement calculation procedures in this manual are applicable to the settlement of embankments when using appropriate stress distributions.

f. Static and seismic slope stability. No guidance is provided in this manual for static and seismic slope stability. The designer is referred to EM 1110-2-1902 for static slope stability.

g. Design coordination. This manual pertains primarily to the geotechnical design components of shallow foundations. However, due to soil-structure interactions which are present to some degree where any structural element is in contact with the ground, foundation design requires collaboration with the structural engineer and other design disciplines.

Chapter 2
Design and Performance Requirements

2–1. Introduction

Shallow foundations are commonly used for support of light and moderately loaded structures. The typical design process for conventional shallow foundations and common foundation types are described in this chapter. Criteria is provided for evaluating the design loads with consideration to the design life, geotechnical limit states, factor of safety, reliability, intended structure function, and minimum code requirements.

2–2. Design process

Like many design processes, shallow foundation design requires coordination with other design disciplines so that the geotechnical aspects can be integrated into the overall design. Where design requirements for shallow foundations cannot be achieved, deep foundations and ground improvement should be considered. Critical design aspects include:

a. Uncertainty and risk. Considerable uncertainty is inherent in geotechnical design. This uncertainty is primarily associated with load uncertainties, soil property uncertainty, testing method uncertainties, and the fact that the geotechnical engineer's understanding of the subsurface are developed based on a very small sampling of the site. This uncertainty results in a range of possible design solutions, each with a different associated risk. Risk is further discussed in paragraph 2–3.

b. Geotechnical investigations. For foundation design, geotechnical investigations are performed so that subsurface conditions can be identified in the early stages of design. Insufficient geotechnical information could result in either an overly or non-conservative design, shifting responsibility to the contractor that could lead to differing site claims, and an increase in project scope. Some guidance is provided in Chapter 4. For more extensive site exploration details, reference EM 1110-1-1804.

c. Geotechnical site assessment. Key objectives of a geotechnical site assessment for shallow foundation support are to establish baseline information, expected subsurface and hydrologic conditions, and evaluate engineering characteristics. A subsurface design model is an effective means of site assessment that is accomplished by summarizing and illustrating pertinent geological conditions and soil properties. The extent, complexity, and accuracy of the model is project specific and should consider quantity and quality of information available as well as the failure mode or limit state under consideration.

d. Engineering evaluations. Foundations must be designed using the appropriate design loads and must satisfy requirements for ultimate and serviceability limit states presented in this manual. In addition, foundations must meet certain constructability and economic requirements. Once the shallow foundation support system is selected, it

must be designed to provide the necessary support against expected and probable external loadings and should align with the site's geologic and subsurface conditions.

e. Final design. The structural engineer often prepares the foundation design based on structural requirements and the site-specific recommendations provided by the geotechnical engineer. Prior to final design, the geotechnical engineer should review the project information and contract documents to confirm their original design assumptions and recommendations were implemented as intended. If significant changes have occurred (such as changes in site grade, loading, structure location/layout), modifications to the foundation design or additional evaluation should be made so that the foundation can perform as intended.

f. Construction observations and testing. Earthwork observations and testing are considered an extension of the geotechnical design and should be performed so that the field conditions can be reconciled with the design assumptions. The geotechnical engineer should coordinate with the foundation and earthwork field inspector and assist in resolving discrepancies encountered during construction that could conflict with the design intent. Additional guidance for earthwork observation and testing is provided in Chapter 12.

2–3. Risk

The term "risk," as defined by USACE, is a measure of the probability and consequences of uncertain future events. Risk provides a framework for comparing project choices and informing decisions by a combined evaluation of hazard, performance, and consequences, considering their inherent uncertainties.

a. Risk assessment. Risk assessment can be performed during design or for existing structures. The USACE Risk Management Center (RMC) supports the development of methodology to support risk assessment for a variety of USACE missions as described in ER 10-1-55. The RMC has developed risk assessment procedures and has compiled practice guidelines for the purposes of dam and levee safety that can serve as a framework to support risk assessment of other project structure types. Contact the RMC for guidance if a risk analysis is planned.

(1) *Risk assessment for new structures*.

(a) Prescribed factors of safety in Allowable Stress Design (ASD) and Load and Resistance Factor Design (LRFD) should be used in designing foundations. These factors were established by various methods (such as statistical analysis, observations, experience) to better manage risk. The factors consider typical uncertainty of the loads, typical uncertainty of the resistance, and the typical consequences of a limit state being exceeded. For most structure designs, the prescribed procedures will result in acceptable risk.

(b) For special cases, such as where a foundation failure would result in unusually severe consequences such as loss of life or irreparable environmental damage, a risk analysis may help evaluate if a design using conventional factors is adequate to

manage risk. Performing risk analysis does not allow reduction in standard design criteria. For scenarios in which the conventional design criteria are proposed to be reduced based on risk, guidance is provided in paragraph 1–6b.

(2) *Risk assessment of existing structures*. Risk assessments may be performed to estimate the risk associated with an existing structure and to inform study decisions. Example project decisions that risk assessment can help inform include those associated with reduction in economic or safety risks.

b. Geotechnical uncertainty. Geotechnical uncertainty is the uncertainty of the estimation of the ground conditions due to lack of knowledge of ground conditions between and beyond exploration locations and due to limitations in the ability to measure and predict properties and behavior. Geotechnical uncertainty is adequately represented in most cases using an expected value and standard deviation terms. For example, the uncertainty of the effective friction angle of a sand could described as having expected value of 34 degrees and a standard deviation of 1 degree and an appropriate probability distribution model. Geotechnical uncertainty can be reduced by additional study such as exploration, laboratory testing, and performance monitoring (paragraph 2–6d). Geotechnical uncertainty is not the same as variability.

c. Geotechnical variability. Geotechnical variability is the actual range of geotechnical conditions that exist for a project and is sometimes called explained variation or natural variability. Variability is observed in material type, material spatial distribution, and material properties. The variability of the site cannot be reduced by additional study. The variability of the site is considered in foundation design by designing a foundation that has acceptable performance for the range of expected geotechnical conditions of a project, or by applying different foundation designs to different areas of the project.

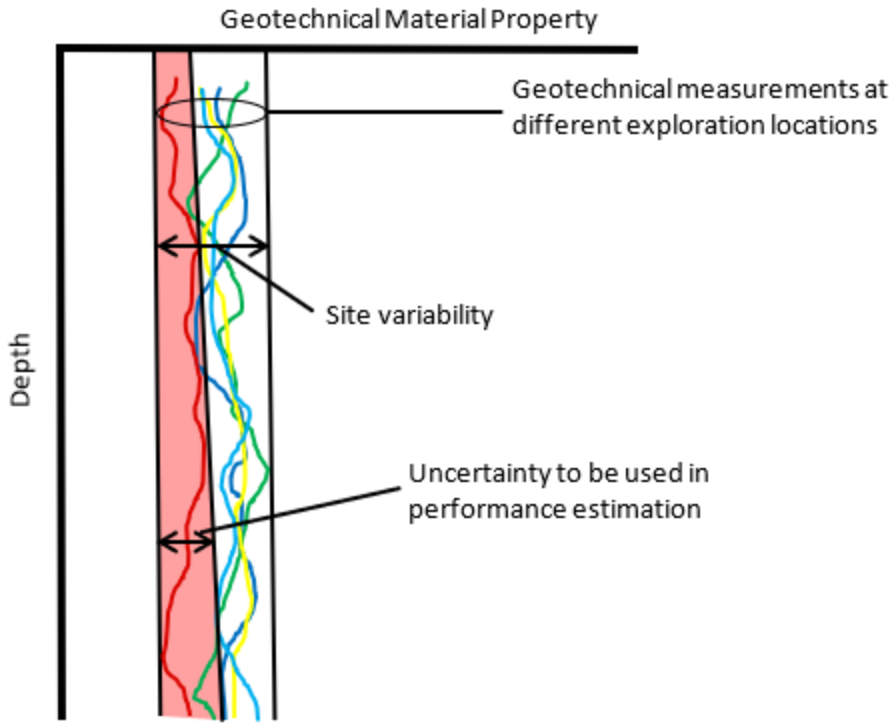


Figure 2–1. Example of site variability vs uncertainty; red line denotes geotechnical material property for critical condition

d. Reliability analysis. Reliability analysis consists of calculating a reliability index and probability of failure considering the loading and uncertainty of the geotechnical behavior. The reliability index is the probability that a limit state is not exceeded for a given load, and the probability of failure is the opposite. Duncan (2000) provides several geotechnical examples of reliability and probability of failure calculations. When performing reliability analysis, a common mistake is to confuse variability with uncertainty. If a weak or problematic ground condition is known to exist, the uncertainty of this critical condition should be used for evaluating reliability, not the variability of the conditions encountered at the site. Figure 2–1 illustrates this concept.

2–4. Constructability and economic requirements

Shallow foundations should be constructable using state of the practice means and methods. In general, the foundation design should be more conservative than the superstructure since poor performance impacts both the foundation and superstructure. In most instances, foundation repair costs are significant due to access constraints and impacts to building function which can require technically challenging solutions. In general, the appropriate type of foundation should represent a balance between performance, design, construction cost, and time. Of these, design cost is often the least significant. Where common construction methods or foundation design is excluded, a cost-benefit comparison with ground improvement and deep foundation support should be performed.

2–5. Shallow foundations

Shallow foundations transfer and spread loads from the superstructure to the earth at shallow depth. They consist of isolated or combined spread footings, continuous (or strip) footings, ring foundations, and mats. For shallow foundations, the bearing area can, theoretically, be increased up to any practical size; however, there are some cases in which a shallow foundation should be analyzed as a mat, particularly if the footing is excessively long and wide, Refer to paragraph 8–2.

a. Footing depth. The minimum vertical distance from the ground surface to the base of the footing is known as the depth of foundation (D_f). This depth must be sufficient to meet strength and serviceability criteria, and the requirements of paragraph 2–8. Problematic ground conditions (such as expansive, frozen soils) and other functional requirements (such as adjacent structures) may require additional embedment. Guidance for problematic ground is provided in Chapter 10. Terzaghi 1943, defined a shallow foundation as having a width, B, equal to or greater than its depth, D_f; however, subsequent research suggests D_f/B could be as large as 2 to 4. For constructability purposes, maximum depths should generally not exceed about 6 to 8 ft.

b. Isolated spread footing. An isolated footing provides support for an individual column, where the footing width (B) and length (L) have dimensions of $B \leq L \leq 10B$ (Figure 2–2a). Isolated spread footings may be circular; however, they are generally approximately circular (octagonal) for easier constructability.

c. Continuous spread footing. A continuous spread footing (also known as a wall or strip footing) has a length to width ratio of 10 or greater ($L > 10B$), (Figure 2–2b) and approximately meets the theoretical assumptions for plane strain conditions (see paragraph 6–6b) parallel to its long axis. Continuous footings typically support a single row of columns or a bearing wall. Continuous footings may be stepped to accommodate elevation changes in the structure or foundation bearing conditions if structural and code requirements are met.

d. Combined footing.

(1) *General*. A combined footing consists of two or more columns that are supported by a rectangular (Figure 2–2c), trapezoidal, or T-shaped footing in lieu of two isolated spread footings. Combined footings are often used when the column spacing is very close and their individual footings or influence zones would overlap and can be used to resist large overturning moments in the longitudinal direction of the column row. For uniform distribution of pressure below a combined footing, the centroid of the footing should coincide with the resultant of the column loads.

(2) *Strap footing*. A strap footing is a special type of combined footing that consists of two spread footings (proportioned with approximately equal bearing pressures) with a beam or strap connecting the foundations (Figure 2–2d). When two columns are far apart, a strap foundation can be used to transfer eccentric moment between two

columns. For design, the strap beam should be rigid and unable to transfer load by not being in contact with the soil at its bottom contact surface. To reduce differential settlement, large differences in footing size should be avoided.

e. Ring foundation. A ring foundation is essentially a continuous footing that is formed into a circle (Figure 2–3); and may incorporate an enlarged base. Ring foundations can be used to resist lateral loads and increase stability against overturning. The stress distribution under ring foundations is more complex than for a circular or rectangular foundation. Ring foundations are often used to support above ground storage tanks and silos and have no interior loading.



Figure 2-2a. Isolated footing, $B \leq L \leq 10B$

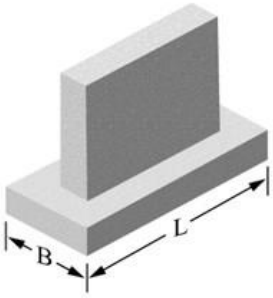


Figure 2-2b. Continuous footing, $L > 10B$

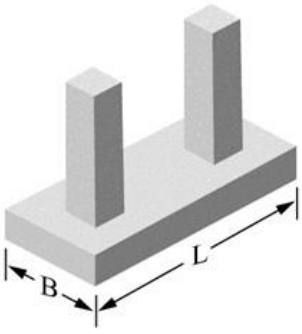


Figure 2-2c. Combined footing

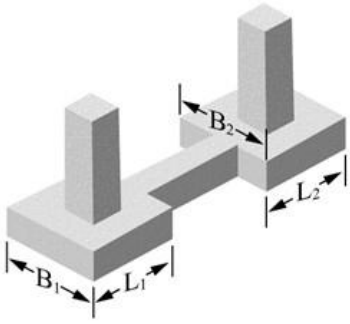


Figure 2-2d. Strap footing

Figure 2–2. Types of shallow foundations

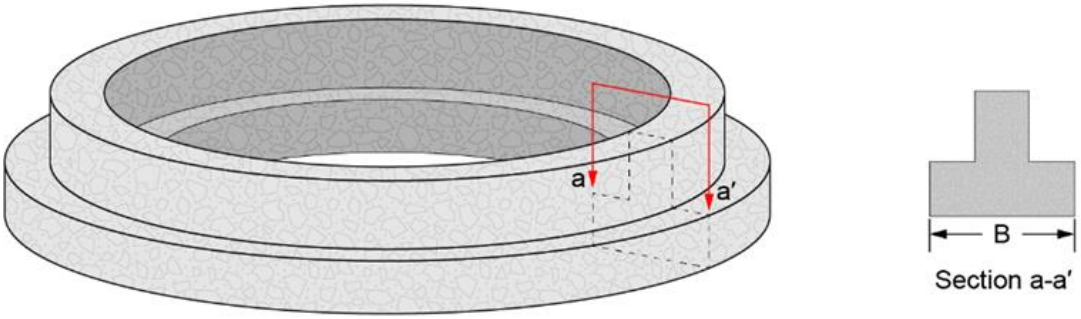


Figure 2–3. Example of ring foundation

f. Mat foundations. Mats are large concrete slabs that supports column or line (such as wall or line of columns) loads in two directions and generally require negative moment reinforcing steel. Mats can be (1) thin (less than 1 ft thickness) supporting light structures, (2) ribbed or reinforced with cross beams, or (3) thick (greater than 1 ft thickness). Mats are generally considered when individual foundations occupy over 30 percent of the structure footprint area, where columns or walls would be placed so closely that overlap would occur, and where it is desired to reduce stresses in the soils. Mat foundations can be placed at the surface or extend to great depths (such as a fully compensated mat). Common mat types are shown in Figure 2–4.

2–6. Ultimate limit state requirements

a. Ultimate limit state. The ultimate limit state (ULS), also referred to as strength limit state, corresponds to a condition where the structure is judged to be unsafe. Geotechnical failure modes associated with this condition include: (1) bearing capacity, (2) sliding, (3) overturning moment and eccentricity, and (4) uplift.

b. Allowable Stress Design. This manual incorporates ASD for working with design loads. For ASD, also known as Working Stress Design (WSD), a single, global factor of safety (FS) is used in foundation design to combine the uncertainties of loads acting on the foundation and the uncertainty of the resistance of the soil (or surcharge) to the load effects. Safety against failure is expressed, in the general form, as equation 2–1.

$$Load\ Effects, Q \leq \frac{Resistance\ of\ Soil, R}{Global\ Factor\ of\ Safety, FS} \tag{2–1}$$

(1) *Minimum factors of safety*. Minimum factors of safety for shallow foundations are shown in Table 2–1 with exception to structures that are governed by EM 1110-2-2100 and EM 1110-2-2502.

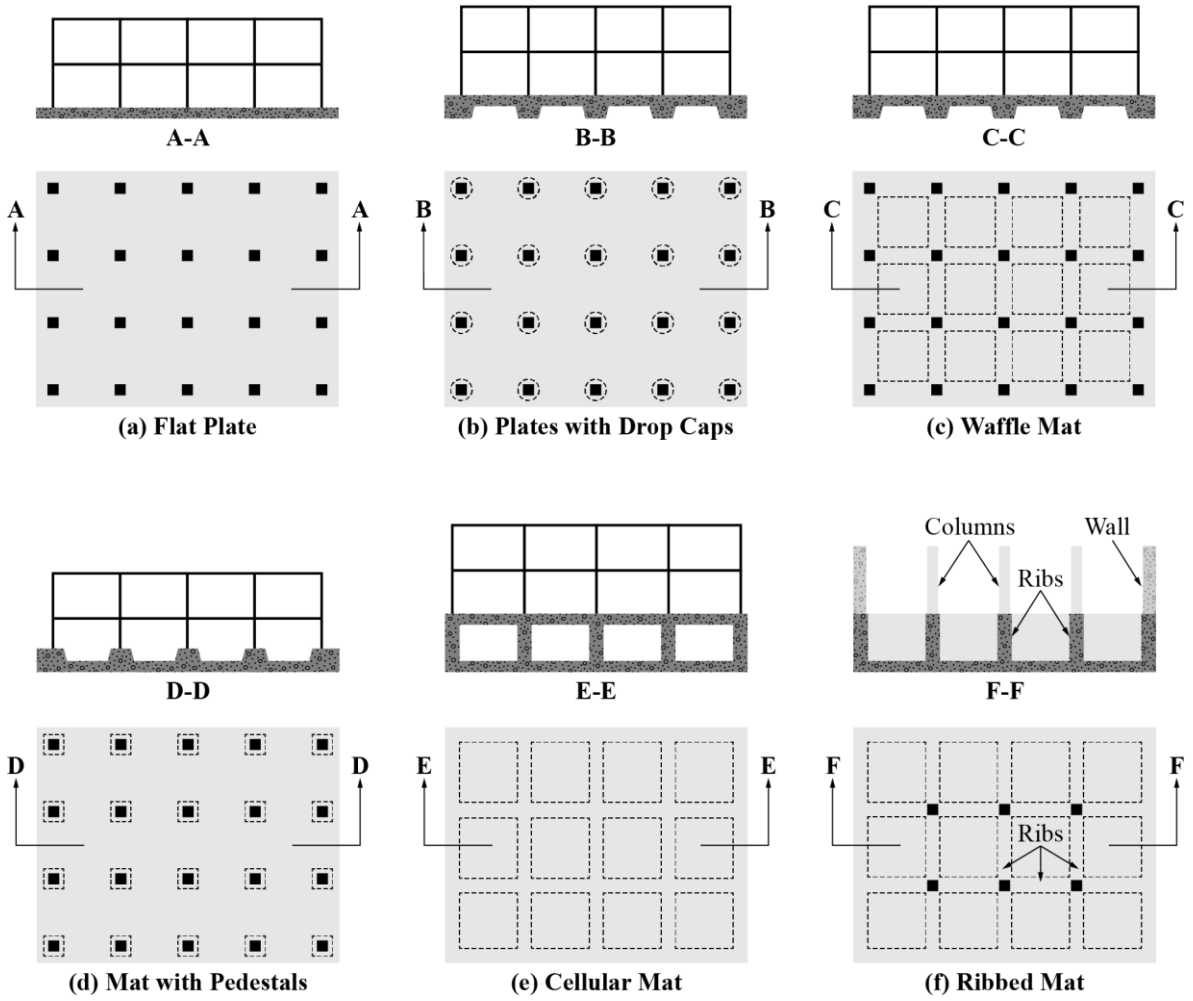


Figure 2–4. Common type of mat foundations (a) flat plate, (b) plates with drop caps, (c) waffle mat, (d) mat with pedestals, (e) cellular mat, (f) ribbed mat

Table 2–1
Minimum factors of safety for ultimate limit state

ULS Failure Mode	Minimum Factor of Safety, FS
Bearing capacity	3.0[1]
Sliding	2.0[2]
Overturning moment and eccentricity	Note 3
Uplift	1.5

Notes:

[1] Increases in allowable stress (for wind or seismic) must comply with the stipulated building code and UFC requirements (paragraph 2–8). Increases are not permitted where bearing soils are susceptible to shear strength loss (SSL), reference paragraph 9–10a. If a FS < 3.0 is used for design, additional settlement could occur due to partial yielding or plastic flow.

[2] Sliding (and passive resistance) are evaluated per paragraph 5–6. Factor of safety may be reduced to 1.1 for load combinations that include earthquake loads.

[3] For overturning stability, the factor of safety concept does not strictly apply when the resultant moves beyond the third point; eccentricity governs and is required to be within the inside middle-third of the footing.

(2) *Limitations of Allowable Stress Design*. For ASD, design parameters (such as material properties, strengths, loads) are assigned a single, deterministic value when the actual value is, in fact, uncertain. This means that the probability of failure (exceeding an ultimate limit state) cannot be directly estimated with this type of design analysis. To better support risk-informed decisions, risk and reliability analyses may be incorporated, reference paragraph 2–3.

(3) *Performance monitoring*. Performance monitoring (such as settlement monitoring) may be used to better quantify the design assumptions. Performance data (visual, measured) should be assessed at appropriate stages to provide confirmation or denial of the characteristics assumed in design. If unacceptable performance is identified, the design should be modified until acceptable conditions exist.

2–7. Serviceability limit state requirements

The serviceability limit state (SLS) refers to a condition beyond which a structure becomes unfit for service and is judged to be no longer useful for its intended function. Serviceability requirements typically govern the design of shallow foundations and include total and differential foundation movements and durability. The selection of serviceability criteria should be coordinated with the structural engineer, architect, and owner, as appropriate, depending on the structural form and intended use. Guidance for allowable movements is provided in the following sections.

a. Allowable movements. Total and differential movements must be calculated for shallow foundations. Movements that negatively affect strength or serviceability of a structure are unacceptable; however, architectural considerations (such as appearance, cosmetic damage) may also influence acceptable tolerances. Allowable movements presented in this manual should be interpreted as “routine limits” and not as “rigid criteria.” Routine limits should not be applied to structures which are out of the ordinary or where loading conditions are markedly non-uniform. Allowable movements due to static loading conditions are presented in this section. Allowable movements due to earthquake loads are discussed in Chapter 9.

(1) *Total settlement*. For most structures supported by spread foundations, total settlement should not exceed 2 inches; however, a typical specification for commercial buildings is 1 inch (Lambe and Whitman 1969). Mat-supported structures are commonly designed with total settlements not exceeding 2 to 4 inches; however, these limits may vary considerably depending on structure stiffness and serviceability requirements. For some structures (such as large flexible storage tanks on flexible bases), total settlements of 12 inches and greater can be tolerated; however, stricter tolerances are often stipulated due to serviceability requirements (such as piping connections, drainage). Total settlement should be calculated for a number of points across the footprint of the structure to establish the expected differential settlement pattern. Where differential settlements do not meet project requirements, total settlement should be reduced.

(2) *Differential settlement*.

(a) The physical basis for structure damage is maximum angular distortion because maximum differential settlements may not necessarily be adjacent (may not result in maximum angular distortion). Both deflection ratio and angular distortion can give similar results as indicators for damage where hogging, sagging, and tilt are treated in a similar manner; however, angular distortion is generally considered more readily evaluated. During the design stage, angular distortion can be difficult to predict due to complex soil-structure interactions. Consequently, correlations with maximum total settlement and maximum differential settlement have been established for some common conditions (see paragraph 2–7a(2)(g)).

(b) Control of differential settlement is generally accomplished by reducing total settlement to tolerable levels. For example, a maximum total settlement of 1 inch would generally limit maximum differential settlements to less than 1/2 inch for most buildings. Where cut/fill activities are planned to achieve site grades, total and differential settlements should account for potential rebound due to excavation and settlement due to fills.

(c) For estimating movements associated with expansive and collapsible soils, reference American Society for Testing and Materials (ASTM) D4546 and ASTM D5333. Potential conditions that should be evaluated include:

1. *Soil heave and non-uniform wetting*. Sum of the maximum settlement without soil wetting and maximum potential heave of the wetted soil.

2. *Soil heave and uniform wetting*. Difference between the maximum and minimum soil swell potential and depth of expansive soil subject to moisture changes and surcharge loads.

3. *Soil collapse or shrinkage*. Difference between the maximum and minimum total settlement with foundation and surcharge loading, assuming collapse (or shrinkage) does and does not occur.

(d) Deflection ratio. The deflection ratio is a measure of differential movement (see Figure 2–5). The deflection ratio is defined by equation 2–2 as the maximum vertical deflection (Δ) divided by the horizontal length (L) over which the deflection occurs. The deflection ratio is considered a critical factor for unreinforced load-bearing walls which generally fail by sagging or hogging. In this respect, the length-to-height (L/H) ratio of the wall is significant; the higher the L/H ratio, the higher the limiting value of deflection ratio. Deflection ratio limits for unreinforced masonry walls are provided in Table 2–2. Larger deflection ratios than those shown in Table 2–2 can be accommodated if some of the settlement has occurred before installation of architectural finishes.

$$Deflection\ Ratio = \frac{\Delta}{L} \qquad (2–2)$$

where:

Δ = maximum vertical deflection between the tilt line of the building and two points on the imposed settlement curve (ft)

L = horizontal length between two points (L_{sag} or L_{hog}) on imposed settlement curve (ft)

Note. Actual deformations are perpendicular to the chord line L_{sag} or L_{hog}; however, the difference between the actual deflection and the calculated deflection (Δ) is inconsequential, and therefore, not considered.

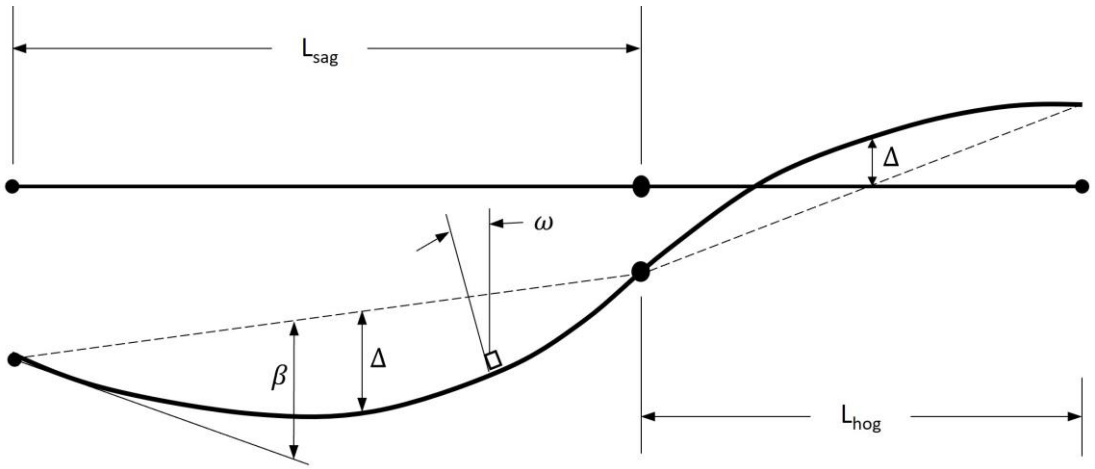


Figure 2–5. Deflection ratio

Table 2–2
Differential settlement limits (deflection ratio) to avoid potential damages[1]

Type of Structure/Condition	Deflection Ratio, Δ/L	Source
Buildings with plain brick walls[2]		(Polshin and Tokar 1957)
Sagging, $L/H \leq 3$	1/2,500 to 1/3,333	
Sagging, $L/H \geq 5$	1/2,000 to 1/1,430	
One-story mills; between columns for brick-clad column frames	1/1,000	
Unreinforced load-bearing walls[2]		(Burland and Wroth 1975)
Sagging, $L/H = 1$	1/2,500	
Sagging, $L/H = 5$	1/1,250	
Hogging, $L/H = 1$	1/5,000	
Hogging, $L/H = 5$	1/2,500	
Unreinforced load-bearing walls		(Meyerhof 1982)
Sagging (safe limit)	1/1,000	
Hogging (safe limit)	1/2,000	

Notes:
[1] Damage to a structure is very subjective and depends on the function of the building and reaction of users.
[2] Length of deflected part of wall (in feet) and height of wall from footing (in feet) denoted as L and H, respectively.

(e) Angular distortion. Angular distortion (β), also known as relative rotation, is a measure of differential movement and is defined by equation 2–3. Angular distortion does not distinguish between hogging and sagging, and it implies that damage is due to shear distortion within the structure, which may not necessarily be the case. Table 2–3 provides angular distortion limits for various structures. Larger angular distortions than those shown in Table 2–3 can be accommodated if some of the settlement has occurred before installation of architectural finishes. Empirical data suggests that typical buildings have architectural damage at $\beta \approx 1/300$ and structural damage at $\beta \approx 1/150$ (Skempton and MacDonald 1956).

$$\beta = {}^{\delta}/_{l} \tag{2–3}$$

where:

β = angular distortion; also referred to as relative rotation (ft/ft)
δ = differential movement between two points (ft)
l = distance between two points, over which settlement has occurred (ft)

Note. For evaluation purposes, uniform settlement and tilt should be excluded (Grant et al. 1974).

Table 2–3
Differential settlement limits (angular distortion) to avoid potential damages[1]

Type of Structure/Condition	Angular Distortion Limits, β (Ratio)	Source
Steel or reinforced concrete frame structures with sensitive interior or exterior finish (such as plaster, ornamental stone, or tile facing)	1/500 to 1/1,000	UFC 3-220-10
Difficulties with machinery sensitive to settlement to be feared[2,3]	1/750	Bjerrum 1963
Multi-story, heavy concrete rigid frame on structural mat foundation, ± 4 ft thick	1/666	NAVFAC 7.1 1971
Steel and reinforced concrete frame buildings		Skempton and McDonald 1956
Recommended limit (includes FS = 1.5)	1/500	
Cracking in wall panels (aesthetic damage)	1/300	
Structural damage to columns and beams	1/150	
Tall and slender rigid structures (such as stacks, silos, water tanks on rigid, circular mat or ring footing). [cross-slope or rigid foundation]	1/500	UFC 3-220-10
Circular steel tanks with flexible base and floating top[3]	1/333 to 1/500	UFC 3-220-10
Slab-on-grade structure (cracking in range of 1/8 inch)		Marsh and Thoeny 1999
Non-reinforced slabs	1/240	
Lightly reinforced slab	1/200	

Type of Structure/Condition	Angular Distortion Limits, β (Ratio)	Source
Post-tensioned slab	1/130	
Overhead crane rails[3] [occurring longitudinally along track]	1/300	Bjerrum 1963
Considerable cracking of panel and brick walls. Danger of structural damage to general buildings.	1/150	Bjerrum 1963
Circular steel tanks on flexible base w/ fixed top[3]	1/125	UFC 3-220-10

Notes:

[1] Damage to a structure is very subjective and depends on the function of the building and reaction of users. Allowable angular distortion in the superstructure should exceed the maximum angular distortion expected in the foundation to avoid structural distress.

[2] Large machines (such as turbines) often have their own foundation separate from the building. A major problem with machinery is vibration due to reciprocating loads; a problem not addressed in this manual.

[3] Consult with manufacturer, stricter tolerances may be required.

(f) Location of the maximum angular distortion. For uniformly loaded, non-rigid foundations on cohesive soils, the maximum angular distortion typically occurs at the corner (Figure 2–6). For uniform loading on cohesionless soils and assuming a non-rigid foundation, the maximum angular distortion can occur anywhere on the foundation because the settlement profile is usually erratic (Figure 2–7). The location of maximum angular distortion does not necessarily correspond to the location of maximum settlement.

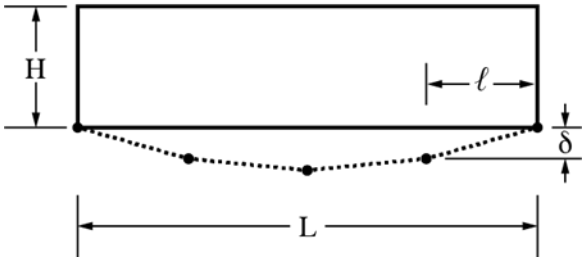


Figure 2–6. Regular settlement profile

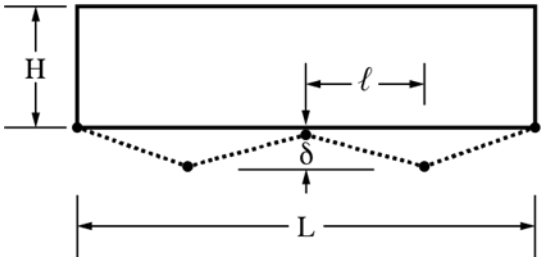


Figure 2–7. Irregular settlement profile

(g) Correlations with maximum angular distortion. For steel or reinforced concrete frame buildings and load-bearing wall structures, Skempton and MacDonald (1956) and Grant et al. (1974) reported correlations with maximum angular distortion (equations 2–4 and 2–7). A crude correlation was reported for the footings bearing on clay and a fair correlation was reported for the footings bearing on sand. In general, the maximum angular distortion for a given maximum settlement is smaller for clays than for sands or fills. For mat foundations, correlations with maximum angular distortion are more complex due to the influence of mat rigidity and should be considered directly by soil-structure interaction, refer to Chapter 8.

Isolated foundations on clay*

$$\rho_{max} = 1{,}200 \times \beta_{max} \tag{2–4}$$

$$\Delta_{max} = 550 \times \beta_{max} \tag{2–5}$$

Isolated foundations on sand or fill*

$$\rho_{max} = 650 \times \beta_{max} \tag{2–6}$$

$$\Delta_{max} = 350 \times \beta_{max} \tag{2–7}$$

where:

ρ_{max} = maximum total settlement (in)
*not applicable where deep-seated clay contributes to settlement
Δ_{max} = maximum differential settlement (ft/ft)
β_{max} = maximum angular distortion; also referred to as relative rotation (ft/ft)

(h) Tilt. The rotation of the entire superstructure or a well-defined portion of the structure resulting from differential movement is referred to as tilt (ω) and is defined by equation 2–6. Rigid body tilt can be difficult to evaluate since it depends on the deformation capacity of the structure, which is influenced by the type of foundation. Limits for tolerable tilt of a building are usually related to serviceability rather than collapse. For low-rise buildings, the maximum acceptable tilt should not exceed 1/400. Variations in tilt limits for some other structures are listed in Table 2–4.

$$\omega = \frac{S_V}{L} \tag{2–8}$$

where:

ω = tilt (ratio)
S_V = differential movement (edge to edge) of entire structure or defined portion (in)
L = span of entire structure or well-defined portion (in)

Table 2–4
Serviceability limits of tilt ω for some structures (Charles and Skinner 2004)

Structure or Component	Tilt (ratio)
Radar system	1/50,000
Satellite antenna tower	1/6,000
Machine operation: turbine	1/5,000
Warehouse high racking	1/2,000
Concrete tanks	1/500
Crane rails	1/333
Tilting of high-rise buildings (such as chimneys, towers) visible	1/250
Stacking of goods	1/100
Floor drainage	1/100 to 1/50

(3) *Lateral movement and uplift*. A serviceability limit state check is not required for shallow foundations. It is intended that shallow foundations have very little lateral and uplift movement. Guidance provided in paragraphs 5–6 and 5–7 is intended to conform to this condition. For retaining and floodwall structures, tolerable movements are described in EM 1110-2-2502.

(4) *Heave*. Soil heave may occur from expansive soil activity “swell” or due to excavation-induced stress relief or “rebound.” For expansive soils, maximum allowable swell potential due to expansive soils (even without foundation surcharge), should not exceed 1 inch for foundations placed above where volume change occurs or above expansive soils (O’Neill and Ghazzaly 1977). Differential movements should not exceed those based on foundation settlement. Typically, ground improvement or other mitigation efforts (such as drainage improvements) are necessary for suitable shallow foundation support. Where soil rebound may occur, foundation total and differential settlements should meet the criteria presented in this chapter.

(5) *Collapse and shrinkage*. Maximum allowable settlement potential (under full surcharge and foundation loading) due to wetting-induced soil collapse or shrinkage of expansive soils should meet requirements for total and differential settlement. Typically, ground improvement or other mitigation efforts (such as drainage improvements) are necessary for suitable shallow foundation support.

b. Durability-related serviceability limit. The integrity of foundations embedded in soil can be adversely affected by chemical and biological processes that can cause deterioration. A durability-related limit state is intended to ensure foundations can resist these processes and provide adequate performance over the design life. The durability limit state is often associated with corrosion of steel reinforcement or degradation of concrete due to environmental (such as freezing/thawing, water) or chemical (such as sulfates, chlorides) exposure.

(1) *Steel reinforcement*. Corrosion resistance of reinforcing steel is primarily provided by concrete cover over the reinforcing steel that reduces contact with the surrounding soil and water. Minimum clear cover requirements for reinforcing steel and alternative methods of protecting steel reinforcement are provided in American Concrete Institute (ACI) 318 and should be verified prior to concrete placement, see also EM 1110-2-2104 for hydraulic concrete structures.

(2) *Concrete exposure*.

(a) Shallow foundations are normally constructed of reinforced concrete, which can remain intact for many years in most soils. In certain environments, where concrete is exposed to freezing/thawing conditions or chemical exposure (such as sulfates, chlorides, salt) degradation can occur. Exposure categories and classes, and requirements for concrete by exposure class, are provided in ACI 318 for some common conditions. In some cases, the exposure categories can be influenced by design functions or construction activities and may not be solely a function of pre-construction site subsurface conditions.

(b) The exposure category and class should be designated by the structural engineer based on the site-specific subsurface conditions reported by the geotechnical engineer and the design conditions. The geotechnical engineer should be familiar with the requirements of ACI 318. For especially severe exposure (such as acid exposure), a corrosion engineer should be engaged.

2–8. Unified Facilities Criteria and Building Code requirements

a. General. United Facilities Criteria, UFC 1-200-1 is the overarching document for buildings and facilities used by the Department of Defense (DoD). UFC 1-200-1 directs use of the International Building Code® (IBC), the International Existing Building Code® (IEBC), Core UFC, other UFC as applicable to the building, facility, structure, or system, and Facility Criteria (FC) as they pertain to the applicable DoD component. USACE has a wide range of building types in the Civil Works Directorate Building Portfolio, and some building types (such as pumping stations, powerhouses) require an enhanced level of analysis or design due to associated risk. Reference ER 1110-2-8161 and ER 1110-2-1806.

b. Presumptive load-bearing values. Presumptive values should not be used in final design of shallow foundations; however, they may be used for preliminary evaluation of shallow foundation feasibility. Presumptive values do not account for footing configuration, eccentricity, or load/footing inclination and could vary greatly from actual field conditions.

2–9. Design life

The design life for USACE military projects is typically 25 to 50 years, and for civil works projects is typically 50 to 100 years. In some cases, major projects can remain in use indefinitely, see ER 1110-2-8159.

2–10. Foundation design loads

For ASD, foundation loads (often referred to as unfactored loads) are provided by the structural engineer for evaluation of the ultimate limit state (strength design). Building codes provide very little commentary on serviceability loads because code requirements are intended to provide a reasonable level of safety and property protection and are not intended for economic viability of structures.

a. Load application. Various types of loads (such as static, dynamic) can be applied to a shallow foundation. Foundation design methods in this manual assume static load conditions unless due to earthquake. For seismic design, failure zones are mostly based on an assumed static failure surface (pseudo-static) and various parameters are used to characterize these loads. Guidance for shallow foundations subjected to seismic loads is provided in Chapter 9.

b. Design loads for ultimate limit state. For military structures, criteria for design loads used for evaluating ultimate limit states are provided in UFC 3-301-01. For new civil works buildings, requirements are outlined in ER 1110-2-8161. For civil works hydraulic structures (such as dams, spillways, locks, pumping stations, floodwalls), guidance is provided in EM 1110-2-2100 and EM 1110-2-2502.

c. Loads causing settlement.

(1) For settlement evaluations, service loads are used. Typically, the dead load plus some portion of live load is recommended. There is a significant difference between live loads estimated from building codes for evaluating the strength limit state (ultimate limit state) and those necessary for evaluating the serviceability limit state. Live loads used in building codes for evaluating the strength limit state represent the maximum live loads that could occur during the life of the structure and are intentionally higher than have been measured during arbitrary live load surveys (these live loads are higher than would be expected to be sustained for long periods).

(2) For settlement evaluations, live loads should not be reduced by more than 50 percent. The amount of live load reduction should be based on knowledge of the intended use of the structure (such as duration of live load, time-settlement behavior).

d. Typical building foundation loads. Typical line and column loads for buildings are described in Table 2–5 and are based on a survey of some engineering firms. An understanding of typical loads is useful as a general reference during design or for preliminary studies; however, should not be used in final design unless verified by the structural engineer.

Table 2–5
Some typical loads on building foundations

Structure	Line Load (kips/ft)	Column Load (kips)
Apartments	1 to 2	60
Individual housing	1 to 2	< 1
Warehouses	2 to 4	100 to 200
Retail spaces and two-story buildings	2 to 4	100 to 150
Multi-story buildings (< 4 stories)	4 to 10	200 to 300
Schools and administrative buildings	2 to 6	100 to 150
Industrial facilities	varies	100 to 200

Notes:

1. Tall multi-story structures can have column loads exceeding 2,000 kips.
2. Column spacings are often 20 to 25 ft or more.
3. The average pressure applied per story of a building often varies from 0.2 to 0.4 tsf.

Chapter 3
Soil Properties

3–1. Introduction

Engineering properties of soils are those properties which can be used for evaluating the ultimate and serviceability limit states (such as shear strength, compressibility, permeability). This chapter considers engineering properties of soil that are useful in designing shallow foundations using static methods. Limited discussions on dynamic soil properties are included in this chapter; more details of dynamic properties are discussed in Chapter 9.

a. Influence depths. Supporting soils can be influenced to certain depths depending on foundation type, size, load application, and other soil-structure interactions. These depths are identified as influence zones and can vary based on the limit state evaluated. Soil properties should be evaluated within the load influence zones. In addition, the geotechnical engineer should also be familiar with major geologic processes that have and can affect the project site so that soil and rock conditions outside of the influence zones do not negatively impact foundation performance.

b. Methods of determining soil properties. Engineering soil properties of geologic strata or imported fill are typically estimated using (1) in situ test data from field explorations, (2) laboratory testing, and (3) empirical correlations. In situ and laboratory tests should be performed using ASTM procedures.

c. Purpose of guidance tables and charts. Tables and charts based on index properties (such as grain size, Atterberg limits, and water contents) are useful for initial estimates or checking the reasonableness of design parameters. However, these correlations can incorporate a higher level of uncertainties and should be used with caution. The final design parameters should be based on in situ test results (such as standard penetration test [SPT], cone penetration test [CPT], Becker Penetration Test [BPT], and vane shear tests [VST]), laboratory test results on relatively undisturbed or disturbed soils based on soil types, or appropriate empirical relationships. Use of empirical correlations should be based on in situ or laboratory test data and should be evaluated based on successful applications in local geologic conditions (whichever is more reliable).

d. Reference guidelines. Several national agencies including USACE have developed guidelines for soil parameters that should be reviewed by the practitioners for background, development, and application of soil parameters.

3–2. Main categories of soil parameters analyses and design

Soil parameters for static analyses of shallow foundations fall into four main categories: (1) water flow characteristics, hydraulic conductivity, and coefficient of consolidation, (2) strength, (3) volume change (compressibility), and (4) shear distortion (stiffness).

a. Water flow characteristics. The rate at which water moves through the soil as a function of a hydraulic gradient or a change in volume. These are generally characterized using vertical and horizontal hydraulic conductivity (k_v and k_h, respectively) and coefficient of consolidation (vertical C_v and horizontal C_h). Coefficient of consolidation (C_v) provides estimates of rate of consolidation due to loading and is generally measured from deformation versus time data from laboratory consolidation testing.

b. Shear strength. Soil shear strength is defined as the ultimate resistance to shear stresses imposed by structures in static or dynamic conditions. A key consideration for evaluation shear strength is whether the soil will be drained or undrained during loading.

(1) *Drained conditions (effective stress parameters)*. For coarse-grained soils with relatively high permeability, or where the rate of loading is relatively slow, drained conditions should be evaluated. Pore pressures near equilibrium can be estimated from the position of the water table, or by flownet solution if flow is present. Effective stress parameters are used for evaluation of drained conditions. For moderately consolidated to overconsolidated clays and plastic silts, both drained and undrained strength parameters should be evaluated, and design should reflect both end of construction and long-term conditions. For coarse-grained soils (such as sand, gravel), drained strength is often more critical. Drained conditions are also critical for settlement analyses.

(2) *Undrained conditions (total stress parameters)*. For fine-grained soil with relatively low permeability, or if the rate of loading is relatively fast, undrained conditions should be evaluated. For undrained conditions, pore pressures will be unknown and only total stresses can be estimated. Therefore, total stress parameters are used for evaluation of undrained conditions. For normally consolidated to slightly overconsolidated clays and plastic silt with overconsolidation ratio, *OCR* < 4, undrained soil strengths could be more critical. Overconsolidation ratio (*OCR*) is defined as the ratio of the preconsolidation stress (σ_p') to the existing vertical effective overburden stress (σ_{vo}').

(3) *Peak strength condition*. Soil strength for peak strength condition is generally characterized by using shear stress at failure (τ_f) through the undrained strength ($\tau_f = s_u$), drained friction angle ($\tau_f = \sigma_n' \, tan \, \phi'$), or combination of effective stress cohesion and drained friction angle for peak strength ($\tau_f = c' + \sigma_n' \, tan \, \phi'$). Drained soil parameters are also identified as effective strength parameters and thus, “drained strength parameter” and “effective strength parameter” are used interchangeably.

(4) *Design strength*. In general, shallow foundations are designed for peak soil strengths using drained soil parameters (c' and ϕ') or undrained strength (s_u). However, for seismic analysis, or evaluating performance of an existing foundation, other soil strengths associated with higher strains such as such as critical state strength, fully softened strength, and residual strength may be needed. EM 1110-2-1902 should be used for guidance for these strength parameters. For dynamic analyses of shallow foundations, guidance is provided in Chapter 9.

c. Volume change (compression).

(1) *General*. Volume changes can occur due to changes in effective stresses. This can be visualized as the change in size of a soil element. The effective stress level with shallow foundation loading relative to the maximum effective stress level the soil has experienced is important to estimate the volume change due to foundation loading. Chapter 7 provides detail procedures to estimate settlement using laboratory testing. This chapter provides typical values of compression indexes.

(2) *Consolidation test parameters*. Volume change in soil due to shallow foundation loadings in fine-grained soils are generally characterized by consolidation-related parameters (such as compression index, C_c, recompression index, C_r, and coefficient of secondary settlement, C_α). These parameters are generally estimated from consolidation testing. Definition of these parameters are as follows,

(a) Compression index (C_c), is the slope of the virgin compression part of the void ratio (e) versus effective vertical stress (σ_z') curve from the consolidation test.

(b) Recompression index (C_r), is the slope of the recompression part of the void ratio (e) versus effective vertical stress (σ_z') curve from the consolidation test.

(c) Secondary compression index (C_α), is the ratio of the change in void ratio (Δe) and logarithm of time curve between time of interest (t), expressed as $[\Delta log(t) = log(t_1) - log(t_2)]$ from a part of the void ratio versus logarithm of time curve.

d. Stiffness. Stiffness properties of soil are relationships that relate stress to strain. The four main stiffness relationships used in foundation analysis and design are the shear modulus (*G*), elastic modulus (*E*), constrained modulus (M) and bulk modulus (*K*). These different stiffness relationships are related, and if one is known, the others can be determined using Poison's ratio. Stiffness parameters and typical values are discussed in paragraph 3–9. Figure 3–13 illustrates the different stiffness definitions. The stiffness parameters can be determined by laboratory testing, in situ testing, and correlation with other properties.

(1) *Shear modulus*. The shear modulus (*G*) is the ratio of shear stress to shear strain. This is described as shear distortion and can be visualized as the change in shape of a soil element. The shear modulus is used primarily in numerical modeling of excavations, large embankments, retaining structures, and other situations where soil shear stresses and shear distortion are of concern.

(2) *Elastic modulus*. The elastic modulus, *E* (also Young's modulus), is the ratio of normal stress to normal strain in the same direction. *E* is used in many foundation analysis settlement solutions in which the ground is treated as an infinite elastic half-space. Soil behavior generally has a non-linear, stress-strain relationship, and therefore, care should be taken to apply *E* for the stress range under consideration.

(3) *Constrained modulus*. The constrained modulus (M) is the ratio of normal stress to strain in the loading direction when the ground is constrained from movement in the perpendicular directions. Some settlement analysis formulas use M to estimate vertical displacements under foundation loads, under the assumption that there is no horizontal ground strain.

(4) *Bulk modulus*. Bulk modulus (*K*) is the ratio of the mean stress to the volumetric strain. The bulk modulus is used primarily in certain numerical constitutive model formulations methods for estimating ground deformation patterns, however, most routine shallow foundation problems do not use the bulk modulus in settlement estimates.

3–3. Unit weight and phase relationships

Unit weight of soil is required to estimate the in situ stress conditions of soils. Some common phase relationships are provided in Table 3–1. Care should be exercised when using several typical values (such as unit weights, moisture, void ratio) as some are not independent of others.

Table 3–1
Unit weight and phase relationships

Moisture content (w)	$w = \frac{W_w}{W_s} \cdot 100\%$ $w = \frac{W_{sat} - W_s}{W_s} \cdot 100\%\ (saturated)$
Void ratio (e)	$e = \frac{V_v}{V_s}$ $e = w \cdot G_s\ (\text{saturated})$ $e = \frac{w \cdot G_s}{S_r}\ (\text{partially saturated})$
Porosity	$n = \frac{V_v}{V} = \frac{e}{1+e}$
Degree saturation (S_r) (S_r = 1.0 if saturated)	$S_r = \frac{V_w}{V_v} \cdot 100\%$
Specific gravity of solids (G_s)	$G_s = \frac{W_s}{V_s \cdot \gamma_w}$
Moist unit weight (γ_m) (also referred to as total, wet, and bulk)	$\gamma_m = \frac{G_s\,(1+w)}{1+e}\ \gamma_w = \gamma_d(1+w)$

Dry unit weight (γ_d)	$\gamma_d = \frac{\gamma_w \cdot G_s}{1+e} = \frac{\gamma_b}{1+w}$
Saturated unit weight (γ_{sat})	$\gamma_{sat} = \frac{(G_s+e)}{1+e}\gamma_w$
Submerged unit weight (γ') (also referred to as effective or buoyant)	$\gamma' = \gamma_w\left(\frac{G_s-1}{1+e}\right) = \gamma_{sat} - \gamma_w$
Relative density	$R.D. = \frac{e_{max}-e}{e_{max}-e_{min}}$

Notes:
W_w = weight of water, W_s = weight of soil solids, W_{sat} = weight of soil saturated, V_w = volume of water, V_v = volume of voids, V_s = volume of soil solids, Υ_w = unit weight of water

a. Specific gravity. General ranges of specific gravity are provided in Table 3–2. Washed silica sand (such as quartz) has a specific gravity of about 2.65. Organic soils have a lower specific gravity, and soils with heavy metallic minerals have a higher specific gravity. In practice, in situ unit weight is generally calculated from in situ water content (w) and dry unit weight (γ_d).

Table 3–2
General ranges of specific gravity

Soil Type	Specific Gravity
Sand	2.63 to 2.67
Silty sand	2.67 to 2.70
Silt	2.65 to 2.70
Clay and silty clay	2.67 to 2.90
Organic soils	< 2.0

b. Unit weight and void ratio. Typical values of unit weights of different soils are presented in Table 3–3. Table 3–4 presents typical values of range of maximum and minimum void ratio, dry and saturated unit weights. Care should be exercised when using several typical values (such as unit weights, moisture, void ratio) as some are not independent of others.

Table 3–3
Typical values of unit weights for different soils γ_{sat} and γ_{dry} (Carter and Bentley 1991)

Soil Type	Saturated Unit Weight (pcf)	Dry Unit Weight (pcf)
Sand and Gravels		
Very loose	106 to 112	81 to 87
Loose	112 to 119	87 to 94
Medium dense	119 to 131	94 to 112
Dense	125 to 137	106 to 125
Very dense	137 to 144	125 to 137
Poorly graded sands	106 to 119	81 to 94
Well-graded sands	112 to 144	87 to 137
Well-graded sand/gravel mix	119 to 144	94 to 137
Clays		
Unconsolidated muds	100 to 106	56 to 69
Soft, open-structured	106 to 119	69 to 87
Typical, normally consolidated	112 to 137	81 to 119
Boulder clays (overconsolidated)	125 to 150	106 to 137

Table 3–4
Typical values of range of void ratio, dry unit weight, and saturated unit weight for natural soils (Kulhawy and Mayne 1990)

Soil Description	Particle Size and Uniformity Coefficient (C_u) (See Note)		Range of Void Ratio (e)		Range of Dry Unit Weight (γ_d)		Range of Saturated Unit Weight (γ_{sat})	
–	$\mathbf{D_{10}}$ (mm)	C_u	min	max	min	max	min	max
Uniform Granular Soil								
a) Equal sphere (theoretical)	–	1.0	0.35	0.92	91.8	109.9	93.0	131.1
b) Standard Ottawa Sand	0.67	1.1	0.50	0.80	83.0	118.0	84.3	136.1
c) Clean, uniform Sand	–	1.2 to 2.0	0.40	1.00	79.9	118.0	81.2	136.1
d) Uniform, inorganic Silt	0.012	1.2 to 2.0	0.40	1.10	86.8	127.4	88.0	142.3
Well-Graded Granular Soil								
a) Silty Sand	0.02	5 to 10	0.30	0.90	84.9	138.0	86.2	148.0
b) Clean, fine to coarse Sand	0.09	4 to 6	0.20	0.95	76.2	119.9	76.8	138.0
c) Micaceous Sand	–	–	0.40	1.20	89.3	146.1	89.9	154.8
d) Silty Sand and Gravel	0.02	15 to 300	0.14	0.85	59.9	134.8	99.9	147.3
Fine-Grained Soil								
Silty or sandy Clay	0.003	10 to 30	0.25	1.80	84.3	139.8	114.9	151.1
Gap-graded silty Clay with Gravel or larger	–	–	0.20	1.00	99.9	148.0	124.9	156.1
Well-graded Gravel, Sand, Silt, and Clay	0.002	25 to 1000	0.13	0.70	49.9	111.7	94.3	133.0
Clay (30 to 50% < 2μm size)	0.001	–	0.50	2.40	13.1	106.1	71.2	128.0
Colloidal Clay (over 50% < 2μm size)	–	–	0.60	12.00	40.0	109.9	86.8	131.1
Organic Silt	–	–	0.55	3.00	30.0	99.9	81.2	124.9
Organic Clay (30 to 50% < 2μm size)	–	–	0.70	4.40	0.70	–	–	131.1

Note: Uniformity coefficient, $C_u = D_{60}/D_{10}$, where D_{60} and D_{10} = particle size corresponding to 10 percent and 60 percent finer on the cumulative particle size distribution curve, respectively (mm).

3–4. Relative density

Relative density (D_R) represents the relative denseness or looseness of a soil layer and applies to soils having less than 15 percent fines. Laboratory and in situ tests can be performed to estimate relative density of soils. Different in situ soil index parameters (such as normalized SPT blow counts $(N_1)_{60cs}$, CPT tip resistance, CPT-q_c) can provide quantitative measures for relative denseness of coarse-grained soils.

a. Relative density and void ratio of coarse-grained soils. Relative density of coarse-grained soils compares the void ratio of a given soil with the maximum void ratio (e_{max}) and minimum void ratio (e_{min}). The void ratio (e) is defined by equation 3–1. Typical values of void ratios for sand vary from about 0.4 to 1.0, and typical values of clays vary from about 0.3 to 1.5. Higher void ratios can be present for some organic soils (Holtz et al. 2011).

$$e = V_v/V_s \tag{3–1}$$

where:

e = void ratio (expressed as a decimal)
V_v = volume of voids (ft^3)
V_s = volume of solids (ft^3)

b. Coarse-grained sand and gravelly soils. For coarse-grained sand and gravelly soils, in situ density tests in combination with laboratory maximum and minimum density tests can generally provide more reliable results than laboratory or in situ methods alone. Table 3–5 provides a general range of denseness of cohesionless soils, expressed in terms of relative density.

Table 3–5
Relative density (D_R) of cohesionless soils

Descriptive Term	**Relative Density, D_R (%)**	
	(Lambe and Whitman 1969)	(Meyerhof 1956)
Very Loose	0 to 15	0 to 20
Loose	15 to 35	20 to 40
Medium Dense	35 to 65	40 to 60
Dense	65 to 85	60 to 80
Very Dense	85 to 100	80 to 100

c. Relative density of coarse-grained soils from in situ and laboratory testing. The relative density (D_R) is commonly expressed in terms of void ratio as indicated in equations 3–2 and 3–3.

$$D_R = \frac{e_{max} - e}{e_{max} - e_{min}} \times 100(\%) \tag{3–2}$$

$$D_R = \frac{1/\gamma_{d,min} - 1/\gamma_d}{1/\gamma_{d,min} - 1/\gamma_{d,max}} \times 100\ (\%) \tag{3–3}$$

where:

D_R = relative density (%)
e_{max} = void ratio of cohesionless soil in its loosest state
e_{min} = void ratio of cohesionless in its densest state

e = void ratio of cohesionless in its natural existing state
$\gamma_{d,max}$ = maximum dry density (pcf) of cohesionless soil in most compact state
$\gamma_{d,min}$ = minimum dry density (pcf) of cohesionless soil in most loose state
γ_d = dry density (pcf) of cohesionless soil in its natural state

(1) *Laboratory test methods*.

(a) The maximum and minimum dry density of a soil can be obtained from laboratory testing. The maximum dry density is obtained from laboratory testing ASTM D4253, Standard Test Methods for Maximum Index Density and Unit Weight of Soils Using a Vibratory Table. The minimum dry density is obtained from laboratory testing ASTM D4254, Standard Test Methods for Minimum Index Density and Unit Weight of Soils and Calculation of Relative Density.

(b) These test methods (ASTM D4253 and ASTM D4254) are applicable to soils that contain up to 15 percent, by dry mass, of soil particles passing a No. 200 sieve, provided they still have cohesionless, free-draining characteristics. Further, these test methods are applicable to soils in which 100 percent, by dry mass, of soil particles pass a 3-inch sieve. In case of soils that contain larger particles, scalping of particles larger than 3 inches is performed for this testing.

(2) *In situ test methods*. The in situ void ratio (e) or dry density (γ_d) of soil can be obtained from a sand cone test (ASTM D1556) or a large-diameter ring density test (ASTM D4914 or ASTM D5030). Sand cone tests (ASTM D1556) are not suitable for soils consisting of unbound granular materials that will not maintain stable sides in the test hole, soils containing appreciable amounts of coarse materials larger than 1.5 inches, and granular soils having high void ratios (very loose soils). When materials contain appreciable amounts of particles larger than 1.5 inch, or when test hole volumes larger than 0.1 ft^3 are required, test method ASTM D4914 or ASTM D5030 are used.

d. Relative density of coarse-grained soils from standard penetration test and laboratory.

(1) Relative density of coarse-grained soils provides quantitative measurements of relative denseness of soils. In situ methods such as SPT N-value or CPT-q_c provide method-specific measurements that can be correlated to relative density of soils. It is important for practitioners to have background in relative density approaches and expected values.

(2) Terzaghi and Peck (1948) first developed a relationship between relative density (D_R) and SPT N-values. Meyerhof (1957) developed a relationship between SPT N-values and relative density of clean sands based on chamber tests. Skempton (1986) used laboratory and field test data from the United States and Japan to provide guidance on the relationship between SPT N-values and relative density. SPT N-values corrected for overburden pressure $(N_1)_{60}$ and relative density (D_R) are related based on equation 3–4.

$$D_R = \sqrt{\frac{(N_1)_{60}}{C_d}} \times 100(\%) \quad (3\text{–}4)$$

where:

D_R = relative density (%)

$(N_1)_{60}$ = normalized SPT N-Value, $(N_1)_{60} = N_{60} \cdot C_N$
Skempton (1986) suggested that for fine sands $(N_1)_{60}$ values should be reduced in the ratio of 55/60 and for coarse sands, $(N_1)_{60}$ values should be increased in the ratio of 65/60.

N_{60} = SPT N-value (blows/ft) corrected to 60% of the theoretical free fall hammer energy, = $N_m \cdot (\text{Energy Ratio}/60)$, equation 4–4

N_m = measured SPT N-value in field per ASTM D1586 (blows/ft)

C_N = overburden stress correction factor, equation 4–6

C_d = parameter representing the grain-size effects and is a function of $e_{max} - e_{min}$, see Section 3–4d(3)

(3) Widely recognized empirical relations for relative density were developed by Skempton (1986) and Cubrinovski and Ishihara (1999).

(a) Skempton (1986) found that the relative density of sand depends on overburden pressure, SPT N_{60} value, particle, size, soil aging, and overconsolidation. Selection of appropriate C_d is an important part of relative density estimation from in situ index properties such as $(N_1)_{60}$. Skempton found good conformance with Terzaghi and Peck's (1948) classification using $C_d \approx 60$.

(b) Cubrinovski and Ishihara (1999) compiled results of 61 relative density measurements performed on sands and gravel samples, where SPT data of known $(N_1)_{78}$ blow counts (Japanese energy ratio of 78 percent) were available. The relationship of $e_{max} - e_{min}$ for sand and gravel (equation 3–5), suggested by Cubrinovski and Ishihara (1999), is based on D_{50} values (particle diameter at 50 percent in the cumulative distribution). However, as discussed in paragraph 3–4c, $e_{max} - e_{min}$ values can be more accurately developed using laboratory tests ASTM D4253 and ASTM D4254. Figure 3–1 shows a relationship between $e_{max} - e_{min}$ and D_{50} of sand and gravel.

(4) Equation 3–6 can be used for C_d estimation for relating to SPT $(N_1)_{60}$ values. Equation 3–6 has been converted from original Cubrinovski and Ishihara (1999) relationship for use in U.S. practice of SPT $(N_1)_{60}$. In general, C_d values of 45 reflect silty sand, 60 reflects clean sands to gravelly sand, and 60 to 70 reflects gravelly soil to gravel.

$$e_{max} - e_{min} = 0.23 + \frac{0.06}{D_{50}} \quad (3\text{–}5)$$

where:

e_{max} = void ratio of cohesionless soil in its loosest state

e_{min} = void ratio of cohesionless in its densest state

D_{50} = particle size corresponding to 50 percent finer on the cumulative particle size distribution curve (mm)

$$C_d = \frac{11.7}{(e_{max} - e_{min})^{1.7}} \; for \; hammer \; energy \; ratio = 60\% \qquad (3\text{–}6)$$

where:

C_d = parameter representing the grain-size effects, and therefore, function of $e_{max} - e_{min}$

e_{max} = void ratio of cohesionless soil in its loosest state

e_{min} = void ratio of cohesionless in its densest state

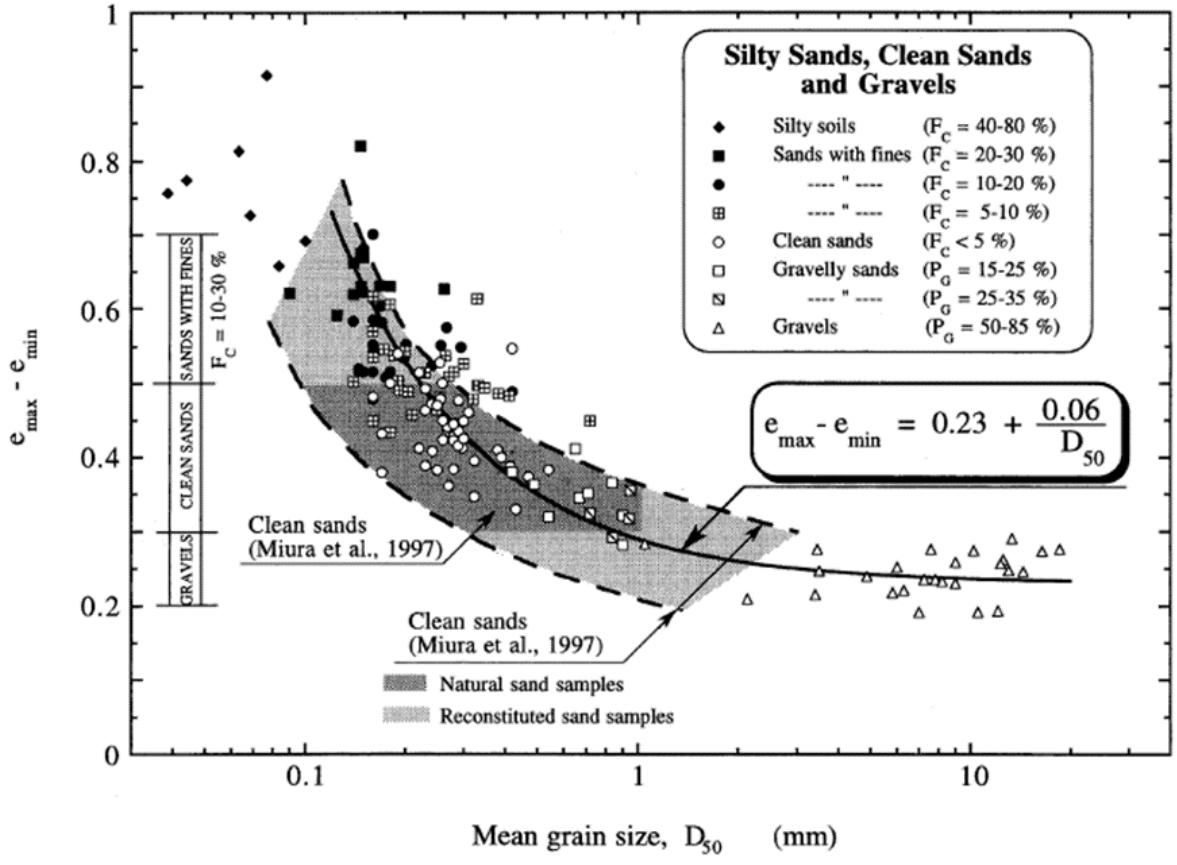


Figure 3–1. Relationship between void ratio range and mean grain diameter (Cubrinovski and Ishihara 1999)

e. Relative density of gravelly soils.

(1) In soils with gravel, the soil behavior will depend on quantity, distribution, and size of gravel particles (percent retained on #4 sieve) as well as the characteristics of the other soil particles. Gravels could be in particle-to-particle contact in soils with higher gravel contents or stay in a soil matrix without particle contacts. If gravel particles stay in a soil matrix, the soil behaviors could be predominantly influenced by the other soil types in the matrix, such as sand, silt, and clay. In this case, gravel particles could float in a soil matrix. However, if gravel particles are in particle-to-particle contact, the probability of particle breakage could increase due to stress concentration at smaller contact areas.

(2) In both cases, SPT or CPT measurements tend higher than for a sand at a similar relative density. Gravel or larger particle contents as low as 5 to 10 percent can result in higher SPT N_{60} or CPT-tip resistance, whereas the soil behavior tends be influenced by other soil types in the matrix.

(3) Relative density for gravel can be used to estimate the in situ state of gravelly materials. Rollins (1998) used $C_d = 70$ to estimate relative density (D_R) and shear modulus of gravelly soils, when using equation 3–4. Cubrinovski and Ishihara (1999) can be used for gravelly soils, as the database includes both sand and gravel soils (equation 3–5 and 3–6). Menq (2003) compiled ($e_{max} - e_{min}$) data from different studies in the United States and Japan (Figure 3–2) to develop a relationship for $e_{max} - e_{min}$, e_{max}, and e_{min}, based on C_u.

Note. It is not apparent from Menq (2003) whether the tests in the database were performed with scalping particles larger than 3 inches, as required in ASTM D4253 and ASTM D4254. A gradation of 3 inches minus should be used in equations 3–7 to 3–9 to appropriately capture the portion of the soil matrix that would most influence behaviors.

$$e_{max} - e_{min} = 0.35\left(1/C_u\right) + 0.21, if\ 200 \geq C_u \geq 2 \quad (3–7)$$

$$e_{max} = 0.95\left(1/C_u\right) + 0.43, if\ 200 \geq C_u \geq 1 \quad (3–8)$$

$$e_{min} = 0.60\left(1/C_u\right) + 0.22, if\ 200 \geq C_u \geq 2 \quad (3–9)$$

where:

e_{max} = void ratio of cohesionless soil in its loosest state
e_{min} = void ratio of cohesionless in its densest state
C_u = uniformity coefficient of soil from gradation tests (D_{60} / D_{10})
D_{60} = particle size corresponding to 60% finer on the cumulative particle size distribution curve (mm)
D_{10} = particle size corresponding to 10% finer on the cumulative particle size distribution curve (mm)

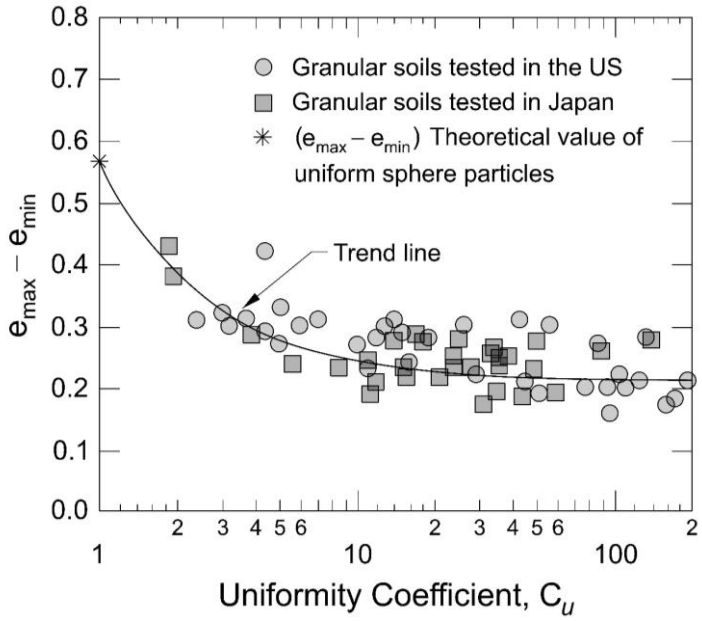


Figure 3–2. Variation of ($e_{max} - e_{min}$) with uniformity coefficient, C_u of gravelly materials tested in the United States and Japan (Winterkom and Fang 1975; Kokusho et al. 1995; Rix 1984) (Figure adopted from Menq 2003)

Note: As shown in Figure 3–2, scatter in data is significant. For example, for a uniformity coefficient, C_u = 50, $e_{max} - e_{min}$ results from the database vary between 0.2 and 0.32. To reduce unconservative bias in using this dataset, the gradation curves (after adjusting by removing materials greater than 3-inch particles) should be used to calculate C_u. It will be consistent with protocol for ASTM D4253 and ASTM D4254. The adjustment should provide a more realistic basis for the engineering properties for the soil portions that would control design and performance of a shallow foundation.

3–5. Drained shear strength parameters

Drained shear strength parameters are generally cohesion intercept (c') and drained friction angle (ϕ'). In situ and laboratory tests can be used for ϕ' estimation. Laboratory tests can be used to estimate cohesion intercept (c'). However, it is typical to neglect c' in shallow foundation design considering that the values are usually low. This section provides typical values and correlations for estimating drained friction angle.

a. Typical relative density and strength characteristics of silica sands at shallow depths.

(1) Evaluation of strength states or similar calculations for shallow foundation design require strength parameters at shallower depths up to the influence zones, with considerations for outside the influence zones. At these shallower depths, SPT blow counts or CPT tip resistance could be negatively impacted by the lack of confinement.

(2) With construction of the shallow foundation and structures, overburden stress increases. Stress-normalized strength relationships (based on overburden stress-corrected blow counts or tip resistance) have potential to overestimate drained friction angle of coarse-grained soils at shallower depths. Considering the limitations regarding

in situ testing and empirical relationships for shallow depths, Table 3–6 can be used to estimate drained friction angle of coarse-grained soils at shallow depths. The table is based on different geotechnical design textbooks and manuals.

Table 3–6
Typical relative density and strength characteristics of silica sands ($c' = 0$) at shallow depths

Density Class	Relative Density (%)	SPT N_{60} (blows/ft)	CPT q_t (tsf)	Peak Friction Angle (ϕ'_{peak})
Very Loose	< 20	< 4	10 to 25	28
Loose	20 to 40	4 to 10	25 to 60	30
Medium Dense	40 to 60	10 to 30	60 to 180	32
Dense	60 to 80	30 to 50	180 to 300	34
Very Dense	>80	>50	>300	36

Note: q_t = corrected tip resistance (tsf), reference Table 4–7.

b. Drained friction angle of sand from standard penetration test. The friction angle of sand can be expressed for different states such as peak friction angle (ϕ'_{peak}), critical state friction angle (ϕ'_{cv}), and residual state friction angle (ϕ'_{rs}). As shallow foundations are not designed for post-peak strength or large-strain conditions, peak friction angle is applicable for shallow foundation design. An SPT N-value-based relationship by Hatanaka and Uchida (1996) is widely used for estimating the drained friction angle of sand (Figure 3–3). The Hatanaka and Uchida (1996) database includes clean sand (~1 percent fines) to silty sand (up to 30 percent fines).

$$\phi' = 20^{\circ} + \sqrt{15.4\,(N_1)_{60}} \text{ (Hatanaka and Uchida 1996)} \quad (3\text{–}10)$$

where:

ϕ' = drained friction angle (degrees), see note
$(N_1)_{60}$ = normalized SPT N-Value (bpf), $(N_1)_{60} = N_{60} \cdot C_N$
N_{60} = SPT N-value corrected to 60 percent of the theoretical free-fall hammer energy (bpf), reference equation 4–4
C_N = overburden stress correction factor; equation 4–6

Note. This relationship has potential to overestimate drained friction angle for design at shallow depths. A comparison with Table 3–6 should be performed to estimate drained friction angle. Values greater than 36 degrees should be generally avoided for shallow foundation design.

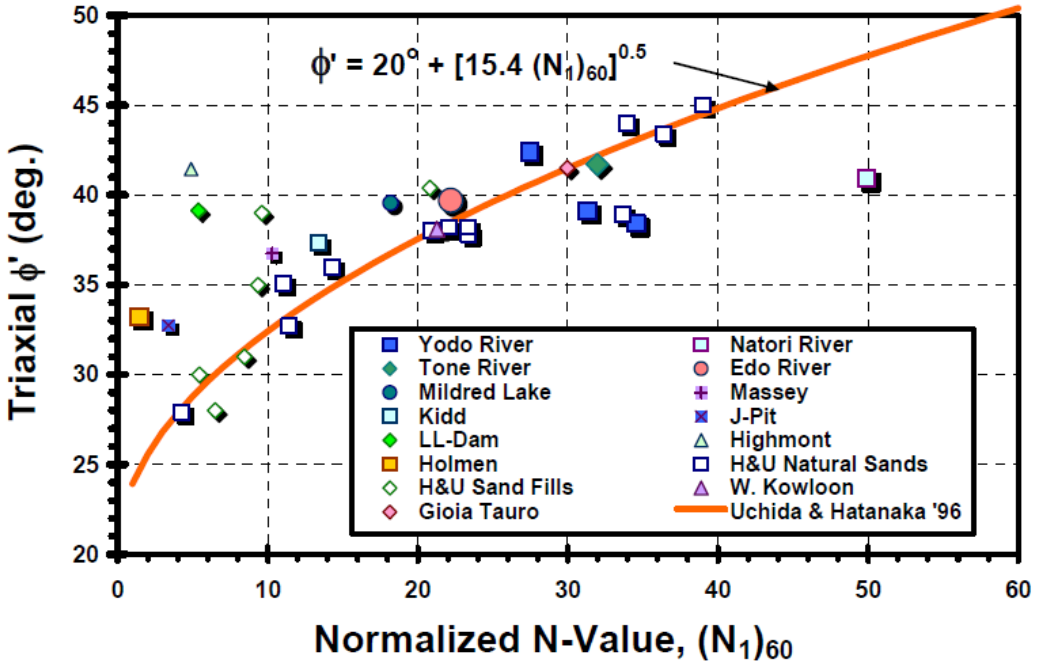


Figure 3–3. Effective friction angle of undisturbed sands vs. normalized standard penetration test-value based on Hatanaka and Uchida (1996) relationship (Mayne 2007b)

c. Drained friction angle of sand from CPTu. The drained friction angle (ϕ') of sand can be estimated from equation 3–11 (Uzielli et al. 2013) as illustrated in Figure 3–4. This equation is applicable for clean quartz to silica sands where the CPTu soundings are advanced at a rate of 0.787 inches/sec or less, pore water pressures are essentially hydrostatic (pore water ratio, $B_q = 0$, reference Table 4–7), and the soil behavior type index, $I_c < 2.6$.

$$\phi' = 25^{\circ}(q_{t1})^{0.10} \qquad (3\text{–}11)$$

(Uzielli et al. 2013)

where:

ϕ' = drained friction angle (degrees), see note

q_{t1} = stress-normalized CPT resistance for sands, $(q_t/P_{atm})/(\sigma'_{vo}/P_{atm})^{0.5}$ same units as P_{atm}

q_t = corrected total cone resistance (same unit as P_{atm})

P_{atm} = atmospheric pressure (at sea level P_a ≈ 2,116.2 psf or 101.3 kPa)

σ'_{vo} = in situ vertical effective stress (same unit as P_{atm})

Note. This relationship has potential to overestimate drained friction angle for design of shallow foundations at shallow depths. A comparison with Table 3–6 should be performed to estimate drained friction angle. Values greater than 36 degrees should generally be avoided for shallow foundation design.

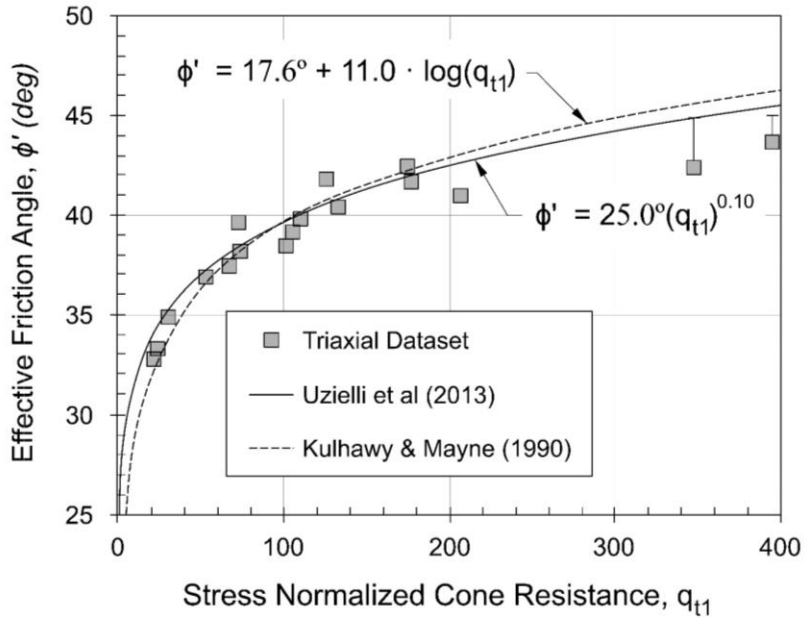


Figure 3–4. Effective triaxial friction angle of undisturbed sands vs. normalized cone tip resistance (Uzielli et al. 2013)

d. Drained friction angle of sand from dilatometer. A series of direct relationships between the effective stress friction angle of clean sands (ϕ') at peak strength to the dilatometer text (DMT) horizontal stress index (K_D) were developed by Marchetti (1997) as illustrated in Figure 3–5. The relationships were developed for several specific horizontal stress conditions. A lower bound solution from (Marchetti 1997) is shown in equation 3–12. For clean sands, equation 3–13 (Rix et al. 2018) can also be used.

$$\phi' = 28.0° + 14.6 \cdot \log(K_D) - 2.1 \cdot \log[(K_D)]^2 \quad \text{(3–12)}$$
(Marchetti 1997)

$$\phi' = 20.0° + [0.04 + 0.06 \cdot (K_D)^{-1}]^{-1} \quad \text{(3–13)}$$
(Rix et al.2018)

where:

ϕ' = drained friction angle (degrees), see note

K_D = horizontal stress index, see equation 4–14

Note. Equation 3–12 tends to underpredict the drained friction angle by approximately 2° to 4°. A comparison with Table 3–6 should be performed to estimate drained friction angle. Values greater than 36 degrees should be generally avoided for shallow foundation design.

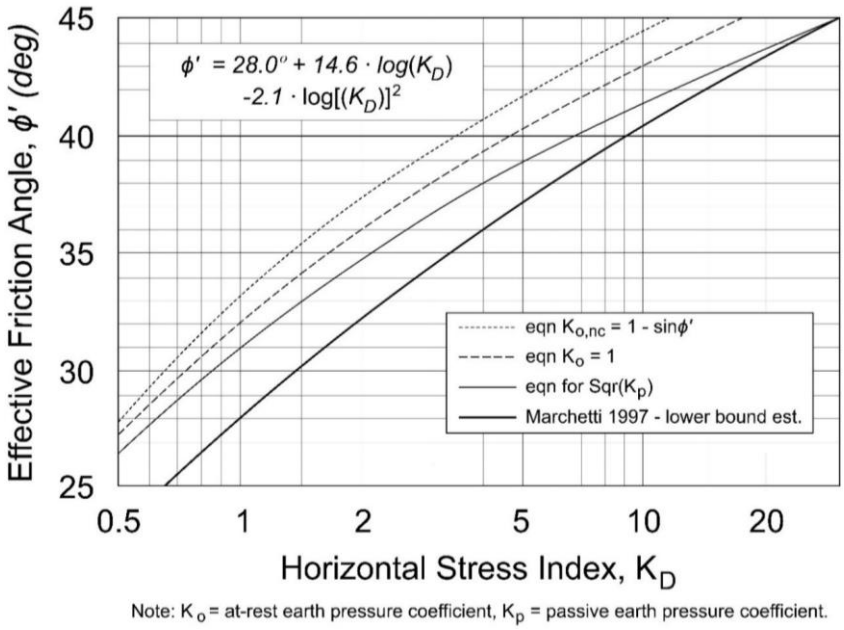


Figure 3–5. Sand friction angle ϕ' in terms of K_D for different lateral stress states (after Marchetti 1997)

e. Drained friction angle of sand and gravel from dry unit weight, void ratio, and relative density. The drained friction angle (ϕ') of sand and gravel can be estimated from Figure 3–6 if in situ or compacted dry unit weight, void ratio, and relative density are known.

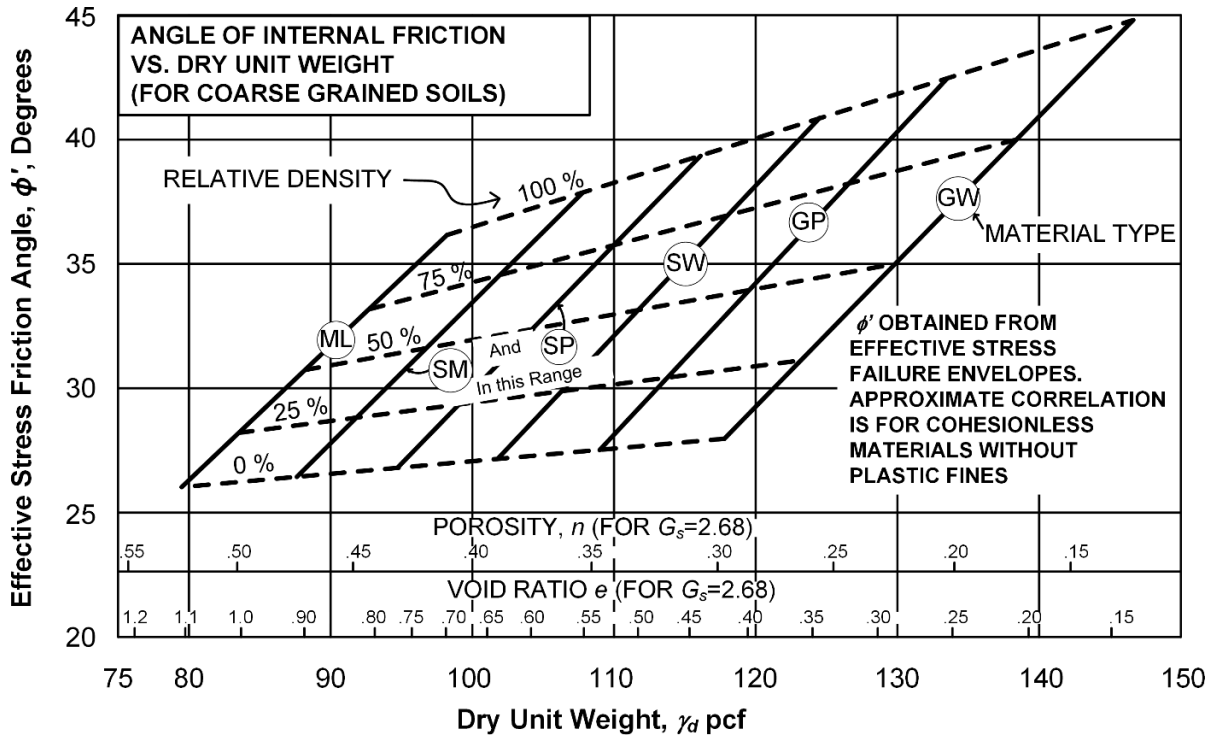


Figure 3–6. Angle of internal friction for coarse-grained soils based on dry unit weight, void ratio, and relative density (UFC 3-220-10)

f. Drained friction angle of clay from laboratory.

(1) Appendix D in EM 1110-2-1902 provides guidelines for using three different types of triaxial strength tests: Unconsolidated-Undrained (UU or Q), Consolidated-Undrained (CU or R), and Consolidated-Drained (CD or S). As relatively undisturbed soil samples can be collected in clayey soils, triaxial strength tests are regularly performed on clayey soils for important projects. Figure 3–7 shows drained or effective stress paths for normally consolidated and overconsolidated clays in p′-q space.

(2) As shown on Figure 3–7, an evaluation of stress paths can also be helpful to identify the stress history of undisturbed soil and establish whether the soil is normally consolidated or overconsolidated. Figure 3–8 shows determination of cohesion (c') and friction angle (ϕ') from a p′-q plot. The confining pressure for the triaxial tests is selected considering the in situ stress condition and design stress conditions.

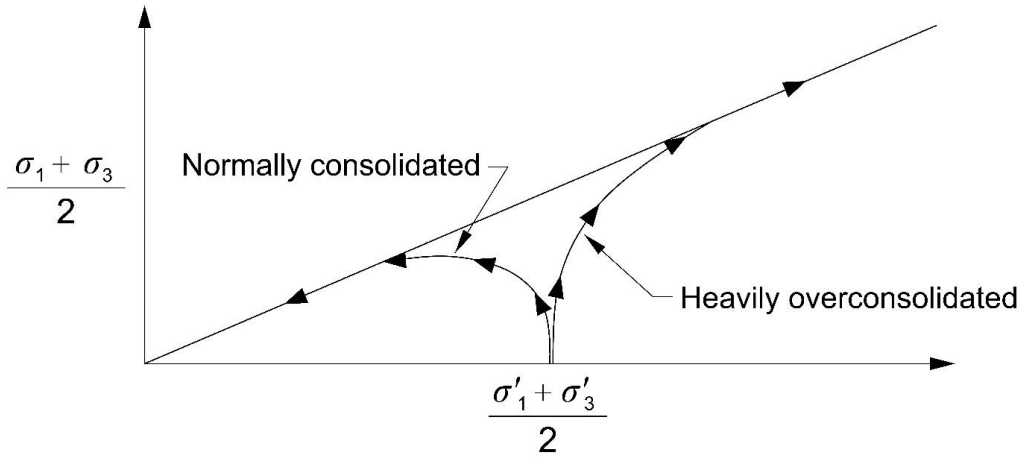


Figure 3–7. Effective stress paths for normally consolidated and heavily overconsolidated clay in p′-q space

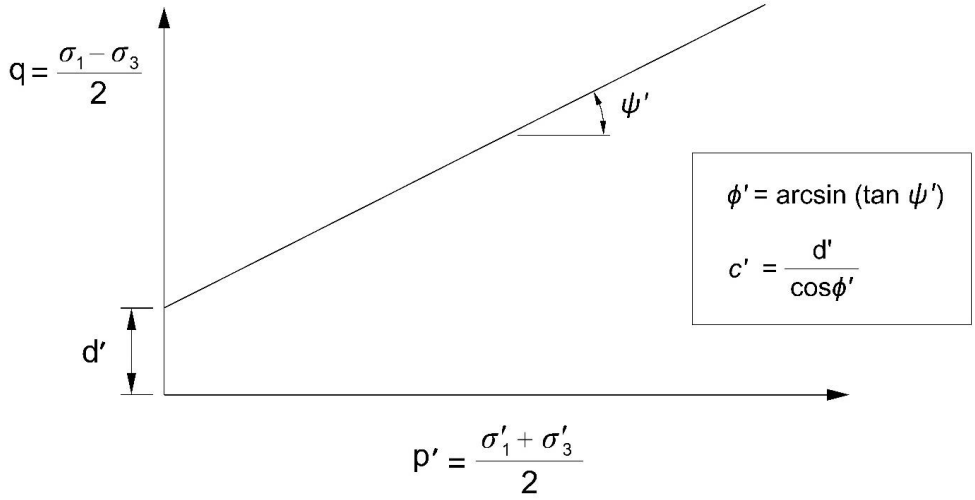


Figure 3–8. Effective stress parameters (effective cohesion intercept and drained friction angle) from triaxial strength tests

3–6. Undrained shear strength

Undrained shear strength (s_u) values vary in response to a particular loading condition, and is, therefore, stress path-dependent. Generally, s_u varies over depth as a function of in situ confining stress.

a. Undrained shear strength of clay from standard penetration test. For soft to firm intact clays and structured or sensitive clays, the dynamic energy of the SPT is inappropriate and correlations with undrained shear strength could be unreliable. In the case of stiff to hard, insensitive clays, the SPT may be used to estimate undrained shear strength. Even for these clays, site-specific relationships for a given clay in a

particular geologic setting should be developed, considering geologic setting, plasticity, origin, and other aspects of the clay formation (Rix et al. 2018).

b. Undrained shear strength of clay from cone penetration test. CPTs are commonly used for undrained shear strength (s_u) estimates using equation 3–14. The s_u estimates based on cone penetration data depend on the empirical cone factor.

$$s_u = \frac{q_t - \sigma_{vo}}{N_{kt}} \tag{3–14}$$

where:

s_u = undrained shear strength (tsf)
q_t = total cone resistance (tsf)
σ_{vo} = total in situ vertical stress (tsf)
N_{kt} = empirical cone factors, see Note

Note. Empirical cone factor (N_{kt}) values have been found to range between 6 and 30, with commonly used values ranging from 12 to 15 (Lunne et al. 1997). Robertson and Cabal (2022) recommends an average value of 14. Multiple studies suggest that N_{kt} increases with increasing plasticity and decreases with increasing soil sensitivity. For very sensitive, fine-grained soils, values as low as 6 are reported. In many projects, an N_{kt}value of 15 is assumed without consideration of the actual site-specific soil conditions or predominant shear modes. In critical projects, laboratory tests from adjacent borings should be performed to determine site-specific N_{kt} values.

c. Undrained shear strength of clay from dilatometer. Marchetti (1980) suggested estimating the undrained shear strength (s_u) as a function of the K_D-value as shown in equation 3–15 for cohesive soils ($I_D < 1.2$). This equation incorporates a value of $(s_u/\sigma'_{vc})_{nc} = 0.22$ which was obtained from field vane tests as suggested by Mesri (1975).

$$s_u = 0.22\sigma'_{vc}(0.5K_D)^{1.25} \tag{3–15}$$

where:

s_u = undrained shear strength (tsf)
σ'_{vc} = initial vertical effective stress, prior to blade penetration (tsf)
K_D = horizontal stress index, see equation 4–14

d. Typical undrained strength characteristics of clay. As discussed in paragraph 3–6a, s_u estimates using SPT N_{60} tend to be unreliable. Considering the limitations regarding in situ testing and empirical relationships for shallow depths, Table 3–7 can be used to estimate undrained shear strength of clayey soils at shallow depths for an initial estimate. Cross checking or confirmation of these values with laboratory test results is recommended.

Table 3–7
Typical undrained strength characteristics of clays[1]

Consistency	s_u, psf (kPa)	SPT N_{60} (blows/ft)	CPT q_n[2,3] tsf (MPa)	Field Test
Very Soft	0 to 250 (0 to 12)	0 to 2	0 to 2 (0 to 0.2)	Squeezes between fingers when fist is closed
Soft	250 to 500 (12 to 24)	2 to 4	2 to 4 (0.2 to 0.4)	Easily molded by fingers
Medium	500 to 1,000 (24 to 48)	4 to 8	4 to 8 (0.4 to 0.8)	Molded by strong pressure with fingers
Stiff	1,000 to 2,000 (48 to 96)	8 to 15	8 to 15 (0.8 to 1.5)	Dented slightly by finger pressure
Very Stiff	2,000 to 4,000 (96 to 192)	15 to 30	15 to 30 (1.5 to 3.0)	Dented only slightly by pencil point
Hard	> 4,000 (192)	> 30	> 30 (3.0)	–

Notes:
[1] Strength values in tables are not for use in design without additional verification, but to present the designer with general characteristics of the consistency of clays.
[2] Use q_n rather than q_1 or q_c in CPT undrained strength correlations to avoid depth bias.
[3] tsf = tons per square foot.

e. Undrained shear strength using SHANSEP (stress history and normalized soil engineering properties) method. Soil strength of plastic, fine-grained soils due to shallow foundation loading depends on stress history of soils. The overconsolidation ratio (*OCR*), established from an oedometer test, is a common measure of stress history of soil, reference paragraph 6–5d. A plastic, fine-grained soil that has experienced higher loading in the past due to geologic processes (such as glaciation) or artificial preloading or groundwater level change, or drainage, etc., has a higher *OCR* than a soil that has not been subjected to subsequent loading. If a soil is loaded and then unloaded, it retains part of its stress history, which can then influence soil behavior if it is later reloaded (Holtz et al. 2011).

f. Undrained shear strength ratio.

(1) *General*. The undrained shear strength ratio in clayey soils increases with increasing *OCR*. Ladd and Foote (1974) developed the SHANSEP method to estimate undrained strengths of clayey soils. This procedure also reduces impacts of disturbances that could occur in soil sample collection, transportation, sample preparation, and laboratory testing. Equation 3–16 relates undrained strength ratio with *OCR*. Figure 3–9 presents examples of s_u/σ'_{vc} vs. *OCR* for different soils (Ladd et al. 1977).

$$\frac{s_u}{\sigma'_{vo}} = S(OCR)^m \tag{3–16}$$

where:

s_u = undrained shear strength, (psf or kPa)
σ'_{vo} = in situ vertical effective stress (same unit as s_u)
S = undrained shear strength ratio $(s_u/\sigma'_{vo})_{NC}$, use $S = 0.22$ if no laboratory data
OCR = overconsolidation ratio
m = 0.8 ± 0.1 for most soils (if laboratory test data is not available, use 0.8)

(2) *Modified SHANSEP procedure.* Mayne (2005) developed a modified version of SHANSEP equation considering experimental data and the critical state soil mechanics (CSSM) framework (equation 3–17 and Figure 3–10). These strength parameters indicate direct simple shear (DSS) mode.

$$\left(\frac{s_u}{\sigma'_{vo}}\right)_{DSS} = \frac{sin\phi'}{2} OCR^{\Lambda} \quad (3\text{–}17)$$

where:

s_u = undrained shear strength (psf or kPa)
σ'_{vo} = in situ vertical effective stress (same unit as s_u)
OCR = overconsolidation ratio
ϕ' = effective or drained friction angle (degrees)
Λ = plastic volumetric strain ratio (1 – recompression index/compression index) clays of low to medium sensitivity exhibit $0.7 < \Lambda < 0.8$, whereas structured and sensitive soils show $0.9 < \Lambda < 1.0$

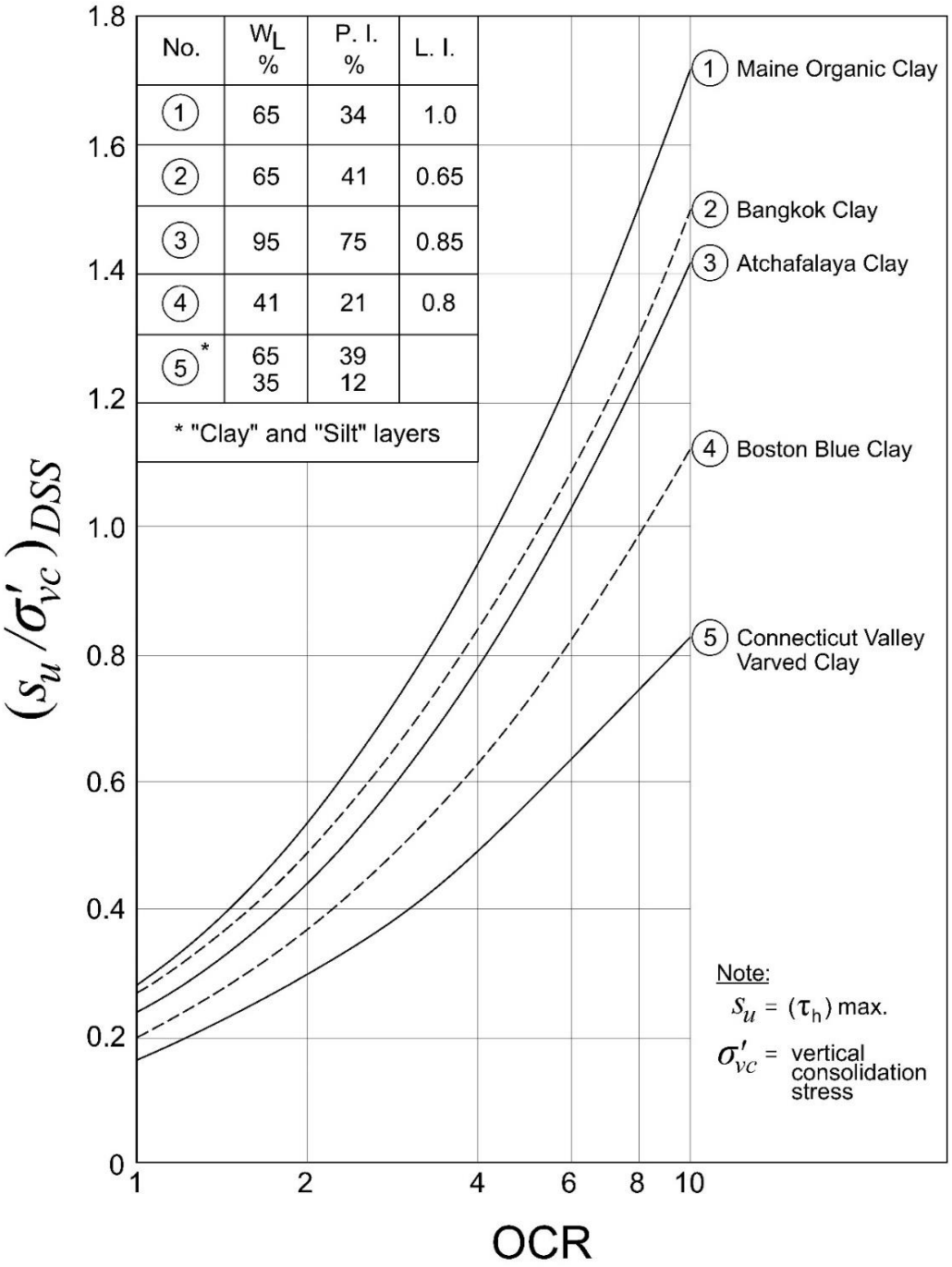


Figure 3–9. Undrained shear strength ratio, s_u/σ'_{vc} vs. *overconsolidation ratio* from consolidated-undrained (CK_oU) direct simple shear tests on five clays (after Ladd and Edgers 1972)

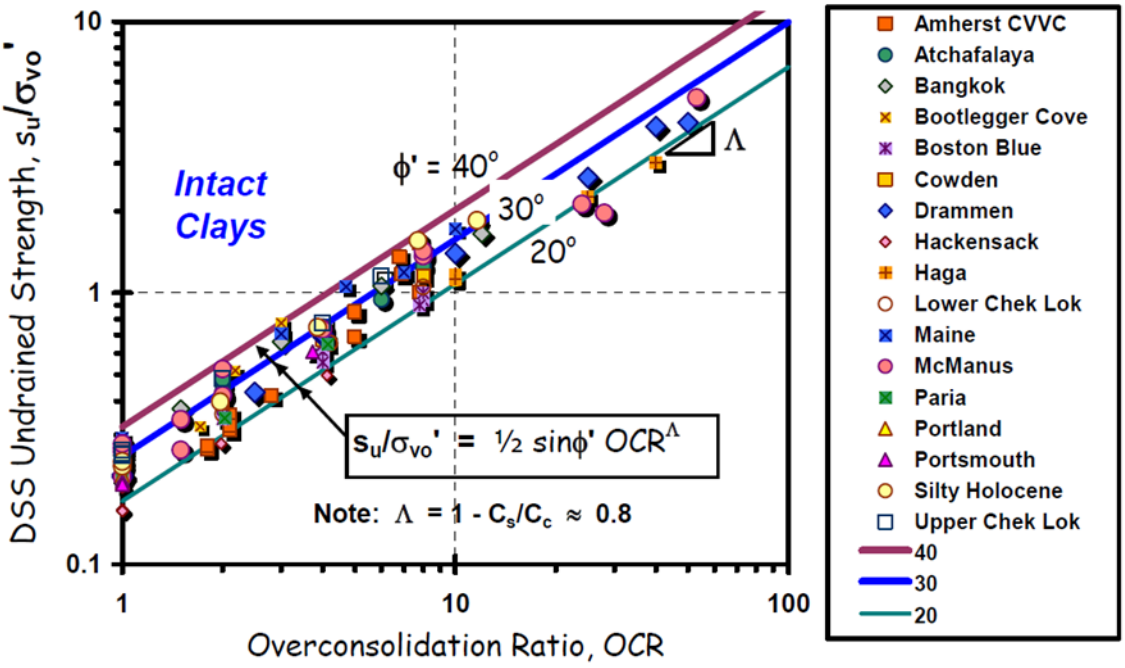


Figure 3–10. Experimental and critical state soil mechanics variation of undrained direct simple shear strengths with overconsolidation ratio (Mayne 2005)

g. Undrained shear strength from vane shear strength tests. Vane shear testing is a useful method for determining undrained shear strength (s_u), stress history, and sensitivity in soft clays, silts, and organic soils. VSTs are not applicable in sandy soils or non-plastic silts as these can allow drainage during testing. VSTs should be performed according to ASTM D2573. Field vane shear strength values require correction based on plasticity index (PI) values to calculate mobilized shear strength. Bjerrum's (1973) correction factor (μ) has been commonly used to calculate mobilized shear strength (equation 3–18 and Figure 3–11). ASTM D2573 also presents the vane shear correction factor (μ_R), as per Chandler (1988) and Aas et al. (1988) for determining s_u. The Chandler correction factor depends on time to failure, along with PI.

(1) *Determination of overconsolidation ratio from vane shear test results*. The undrained shear strength of softer soils, as measured by the VSTs can be used to estimate OCR using the SHANSEP relationship (Ladd and Foote 1974; Ladd et al. 1977; and Ladd 1991).

(2) *Sensitivity of soft soils*. Sensitivity of soft soils can be calculated as the ratio of peak to remolded shear strengths. Sensitivity of soft soils is a useful index to estimate residual strength at large deformations. It can be important for slope stability of shallow structures on slopes or conditions where construction of a structure on shallow foundation on soft soils may induce large deformations. Slope stability is not covered in this manual.

$$(s_u)_{fv}(mob) = \mu(s_u)_{fv} \quad (3\text{–}18)$$

where:

$(s_u)_{fv}(mob)$ = mobilized undrained shear strength from field vane shear test (tsf)

μ = correction factor (Bjerrum 1973; Chandler 1988; or Aas et al. 1988); see Figure 3–11 for Bjerrum 1972 plot after Ladd et al. 1977

$(s_u)_{fv}$ = peak undrained shear strength from field vane shear test (tsf)

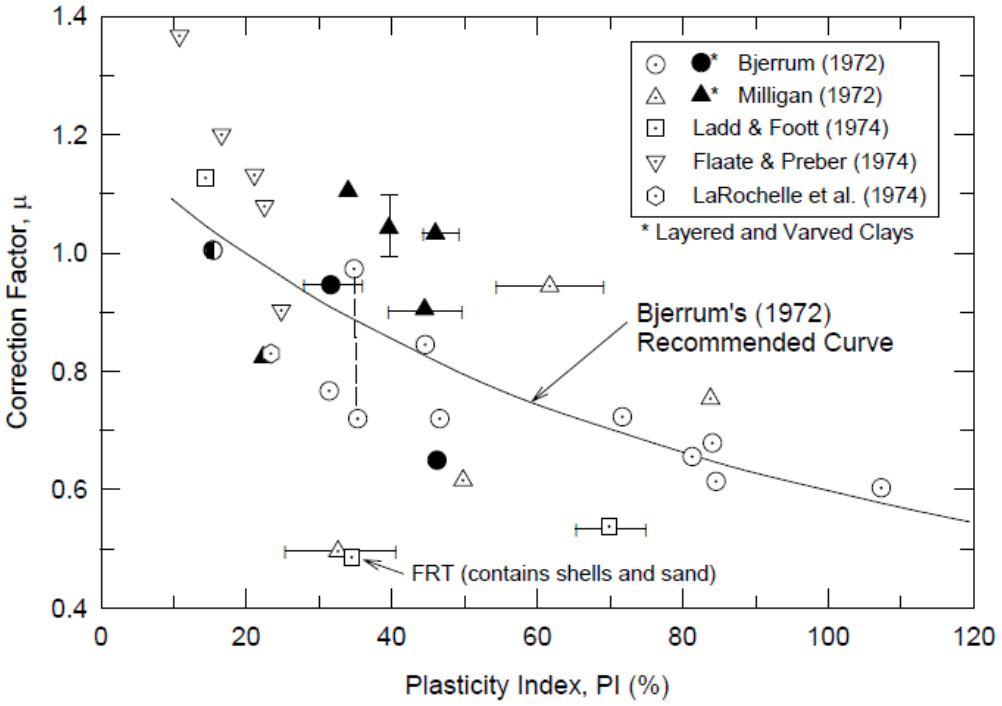


Figure 3–11. Field vane shear correction factor, μ vs plasticity index derived from embankment failures (after Ladd et al. 1977)

Note: Data in profile should include peak and remolded undrained shear strength values from field vane shear tests, $(s_u)_{fv}$ and $(s_{ur})_{fv}$, respectively. These values should be presented as the adjusted values after the correction factor is applied.

3–7. Consolidation and rate of consolidation parameters

Consolidation theory is described in geotechnical engineering textbooks. Section 7–9 provides step-by-step procedures for estimating consolidation parameters from laboratory consolidation test data. This section provides typical values and empirical relationships for consolidation and rate of consolidation parameters that can be used to evaluate the reasonableness of laboratory test results.

a. Compression (C_c) and recompression index (C_r). Typical values of compression index (C_c) of various natural soils are provided in Table 3–8. Empirical relationships are also provided in Table 3–9, based on soil index properties. The recompression index

(C_r) is typically 5 to 10 percent of the magnitude of C_c. Typical values of C_r range from 0.015 to 0.035, with nearly all results between 0.002 and 0.05 (Leonards 1976).

Table 3–8
Typical values of compression index (C_c)

Soil	Compression Index (C_c)	Source
Boston blue Clay, undisturbed (CL)	0.30 to 0.50	Modified after Lambe and Whitman 1969
Cincinnati Clay (CL)	0.17	
Louisiana Clay undisturbed	0.33	
New Orleans Clay, undisturbed (CH)	0.29	
Siburua Clay (CH)	0.21	
Na-Montmorillonite (CH)	0.21 to 0.26	
Normally consolidated medium sensitive Clays	0.2 to 0.5	Holtz et al. 2011
Chicago silty Clay (CL)	0.15 to 0.3	
Boston blue Clays (CL)	0.3 to 0.5	
Vicksburg buckshot Clay (CH)	0.5 to 0.6	
San Francisco Bay Mud (CL)	0.4 to 1.2	
San Francisco Old Bay Clays (CH)	0.7 to 0.9	
Bangkok Clay (CH)	0.4	
Swedish medium sensitive Clays (CL-CH)	1 to 3	
Canadian Leda Clays (CL-CH)	1 to 4	
Mexico City Clay (MH)	7 to 10	
Organic Clays (OH)	> 4.0	
Organic silt and clayey Silts (ML-MH)	1.5 to 4.0	
Peats (Pt)	10 to 15	
Uniform Sand, loose (SP)	0.05 to 0.06	EM 1110-1-1904 For $\sigma_v^{'} = 2{,}000\ to\ 8{,}000\ pcf$
Uniform Sand, dense (SP)	0.02 to 0.03	
Uniform Silts (ML)	0.20	

Table 3–9
Empirical relationships for compression index (C_c) based on soil index properties

Applicability	Equation	Source
For remolded Clay	C_c= 0.007 (LL – 7%)	Skempton 1944
Normally consolidated Clay with sensitivity less than 4	C_c= 0.009 (LL – 10%)	Terzaghi and Peck 1948
For organic soil, Peat	C_c= 0.0115 ω_n	Moran et al. 1959
All Clays	C_c= 1.15 (e_0 – 0.35)	Nishida 1956
Varved Clays	C_c= (1 + e_0) [0.1 + (ω_n – 25) 0.006]	NAVFAC DM 7.1

Note: ω_n = natural water content, LL = liquid limit, and e_0 = initial void ratio.

b. Compression and recompression ratios. Values of compression index (C_c) and recompression index (C_r) can be normalized with respect to the void space available in the soil by dividing by (1+e_0) as shown in equations 3–19 and 3–20. The resulting values are expressed in terms of vertical strain (such as $C_{\varepsilon c}$ and $C_{\varepsilon r}$) and can be used as an alternative to estimating the compression and recompression indices directly where the initial void ratio is known. For soils of similar geologic origin, site-specific correlations can be established for $C_{\varepsilon c}$ and $C_{\varepsilon r}$ versus initial void ratio. Ranges of compression ratio ($C_{\varepsilon c}$) and recompression ratio ($C_{\varepsilon r}$) for different classifications of soil compressibility are provided in Table 3–10 (Coduto et al. 2016).

$$\text{Compression Ratio}, C_{\varepsilon c} = \frac{C_c}{1 + e_0} \tag{3–19}$$

$$\text{Recompression Ratio}, C_{\varepsilon r} = \frac{C_r}{1 + e_0} \tag{3–20}$$

Table 3–10
Typical ranges of soil compressibility based on compression and recompression ratios ($C_{\varepsilon c}$ and $C_{\varepsilon r}$)

$\frac{C_c}{1+e_0}$ $(for\ NC\ clays)$ or $\frac{C_r}{1+e_0}$ $(for\ OC\ clays)$	Description of Soil Compressibility
0 to 0.05	Very slightly compressible
0.05 to 0.10	Slightly compressible
0.10 to 0.20	Moderately compressible
0.20 to 0.35	Highly compressible
> 0.35	Very highly compressible

c. Coefficient of consolidation. The coefficient of consolidation (C_v) is a key parameter used to describe the rate at which saturated clay or other soil undergoes consolidation when subjected to an increase in effective stress. Larger C_v values indicate faster consolidation. In general, C_v is greater for stiff clays than soft clays.

Often, consolidation is faster in the field than determined in a laboratory due to sand seams and fissures that can shorten the drainage path.

(1) *Laboratory determination of C_v and correlations with liquid limit*. For incremental stress consolidation tests (ASTM D2435), C_v can be calculated from the time curves using one of several different methods, reference paragraph 7–18c. For the constant rate of strain (CRS) consolidation test (ASTM D4186), C_v can be calculated at every point where the excess pore water pressure, average stress, and strain are known. The CRS consolidation test provides increased resolution of the compression curve in the vicinity of the preconsolidation stress. CRS testing also allows the determination of the coefficient of consolidation (C_v) over the entire range of loading as long as the excess pore water pressures are in an acceptable range. Table 3–11 presents typical values of coefficient of consolidation (C_v). Correlations between coefficient of consolidation and liquid limit are shown in Figure 3–12.

Table 3–11
Typical values of coefficient of consolidation (C_v)

Soil	**Coefficient of Consolidation (C_v) (cm^2/sec, x 10^{-4})**
Boston blue Clay (CL) (Source: Ladd and Luscher 1965)	40 ± 20
Chicago silty Clay (CL) (Source: Terzaghi and Peck 1967)	8 to 11
Organic Silts and Clays (OH) (Source: Sivakugan 1990)	1 to 10
Glacial lake Clay (CL) (Source: Wallace and Otto 1964)	6.5 to 8.7
Swedish medium sensitive Clay (CL-CH) (Source: Holtz and Broms 1972)	–
1. Laboratory	0.4 to 0.7
2. Field	0.7 to 3.0
San Francisco Bay Mud (CL) (Source: Holtz et al. 2011)	2 to 4
Mexico City Clay (MH) (Source: Leonards and Girault 1961)	0.9 to 1.5

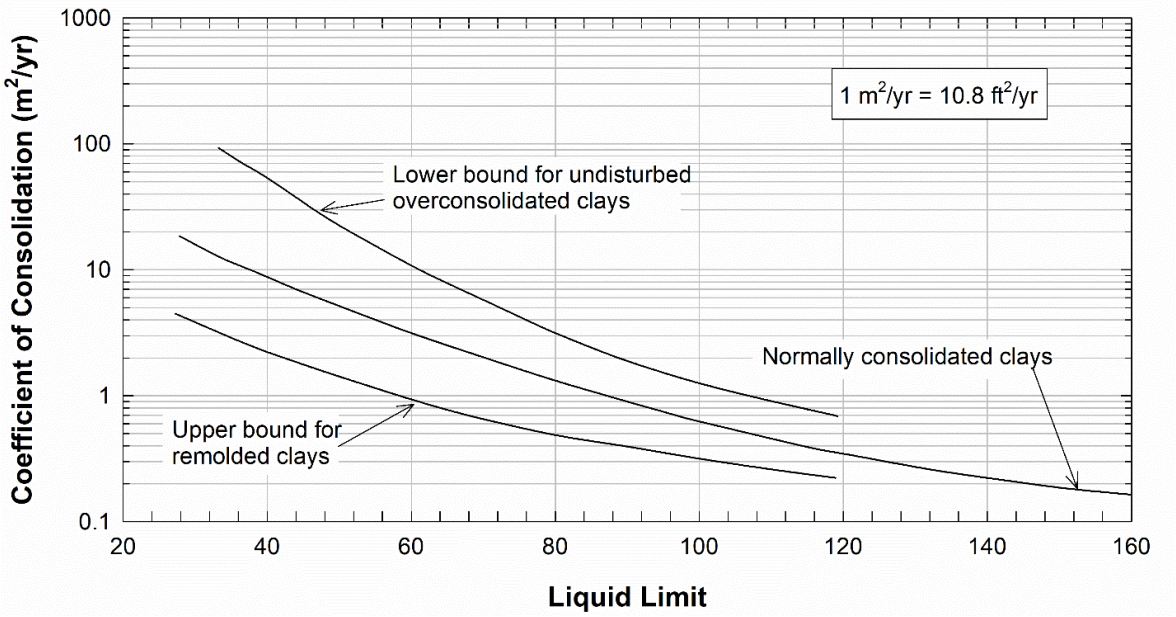


Figure 3–12. Approximate relationship between coefficient of consolidation (C_v) and liquid limit (UFC 3-220-10)

(2) *Hydraulic conductivity relationship with coefficient of consolidation*. Hydraulic conductivity (k) can be indirectly calculated at each loading step based on Terzaghi's consolidation theory per equation 3–21. Indirect methods tend to give much lower hydraulic conductivity values in the low-compressibility zone. Mesri et al. (1994) reported that the indirect method typically underestimates the value of permeability by a factor of 2.

$$\text{Hydraulic Conductivity}, k = C_v \cdot m_v \cdot \gamma_w \quad (3\text{–}21)$$

where:

k = hydraulic conductivity or permeability (ft/sec)
C_v = coefficient of consolidation (ft^2/sec), reference paragraph 7–18c
m_v = coefficient of volumetric consolidation (psf^{-1})
γ_w = unit weight of water (pcf)

d. Coefficient of secondary compression.

(1) *General*. The coefficient of secondary compression (C_α) is defined as the creep deformation over one log cycle of time, *t* following the end of primary consolidation. The magnitude of C_α is stress-dependent for a particular soil and should be determined at an effective stress similar to that expected in situ. Soil compressibility is characterized for C_α in Table 3–12. As with the compression ratios, the coefficient of secondary compression can be defined as function of strain ($C_{\varepsilon\alpha}$) or as a function of the void ratio C_α. Mesri (1973) summarized the data shown in Figure 3–13 to estimate the modified secondary compression index ($C_{\varepsilon\alpha}$) of NC clays using the natural water

content. Based on that data, $C_{\varepsilon\alpha}$ can be estimated using equation 3–22. For overconsolidated clays, $C_{\varepsilon\alpha}$ generally ranges from 0.0005 and 0.001 (Kulhawy and Mayne 1990). The procedure for estimating the coefficient of secondary compression from oedometer test data is addressed in paragraph 7–19.

$$C_{\varepsilon\alpha} = C_\alpha/(1+e) = 0.0001\omega_n \quad (3\text{–}22)$$

where:

$C_{\varepsilon\alpha}$ = modified secondary compression index
C_α = coefficient of secondary compression
ω_n = natural water content (%)

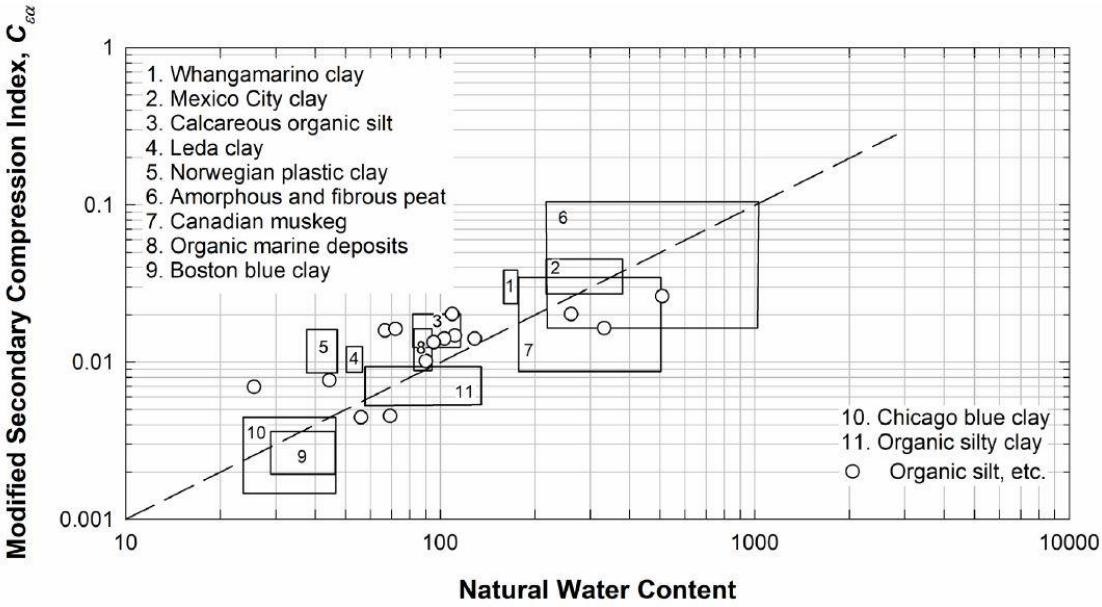


Figure 3–13. Modified secondary compression index vs. natural water content (after Mesri et al. 1973)

Table 3–12
Soil compressibility based on coefficient of secondary compression C_α (Mesri 1973)

C_α % (per log cycle)	Secondary Compression
< 0.2	Very low
0.4	Low
0.8	Medium
1.6	High
3.2	Very high
> 6.4	Extremely high

(2) *Compression index ratio, C_α/C_c*. Mesri and Godlewski (1977) observed that the value of C_α/C_c, referred to as the compression index ratio, is approximately a constant over the range of engineering stresses for a wide variety of natural soils. Empirical correlations are provided in Table 3–13 for coefficient of secondary compression and compression index ratio (C_α/C_c). The rate of secondary compression generally increases with increases in the liquid limit and natural water contents.

Table 3–13
Empirical correlation between C_α/C_c for various types of soil (Mesri and Godlewski 1977)

Grouping **Soil**	**Typical values of C_α/C_c**
Inorganic Clays and Silts	
Leda Clay (Canada)	0.03 to 0.06
Soft blue Clay (Victoria, B.C.)	0.026
Sensitive Clay, Portland, ME	0.025 to 0.055
Hudson River Silt	0.03 to 0.06
Mexico City Clay	0.03 to 0.035
San Francisco Bay Mud	0.04 to 0.06
New Liskeard (Canada) varved Clay	0.03 to 0.06
Organic Clays and Silts	
Organic Clays and Silts	0.04 to 0.06
New Haven organic clay Silt	0.040 to 0.075
Norfolk organic Silt	0.05
Post-glacial Swedish Clay	0.05 to 0.07
Peats	
Amorphous and fibrous Peat	0.035 to 0.083
Canadian Muskeg	0.09 to 0.10
Peat	0.05 to 0.085
Fibrous Peat	0.06 to 0.085

e. Preconsolidation stress and overconsolidation ratio from standard penetration test. Empirical correlations can be used to estimate the *OCR* of a sand. For *OCR* in sands, silty sands, and sandy silts, equation 3–23 (Mayne 2007a) can be used. Equation 3–24, suggested by Kulhawy and Chen 2007, is intended for gravels. For sandy soils, preconsolidation stress can be more difficult to estimate due to difficulties in separating the effects of relative density from *OCR*.

$$OCR = \sigma'_p/\sigma'_{zo} = \frac{0.468 P_a \cdot N_{60}^m}{\sigma'_{zo}} \quad \text{(3–23)}$$

sands, silty sand, sandy silts

$$OCR = \sigma_p'/\sigma_{zo}' = \frac{0.15 P_a \cdot N_{60}^m}{\sigma_{zo}'} \quad \text{(3–24) gravels}$$

where:

OCR = overconsolidation ratio
σ_p' = preconsolidation stress (tsf)
σ_{zo}' = initial vertical effective stress (tsf)
P_a = atmospheric pressure (tsf), $1\ P_a \approx 1$ tsf
N_{60} = corrected SPT N-value for field procedure and hammer efficiency (blows/ft)
m = exponent, 0.6 for clean quartzite sands or 0.8 for silty sands to sandy silts (such as Piedmont), 1.0 for intact clays and clayey silts

f. Preconsolidation stress from cone penetration test. Mayne 2014 suggested a correlation expressed as equation 3–25, which applies to clays and sands. The correlation was derived from cavity expansion—critical state solution for clays, large calibration chamber tests on sand, and field and laboratory data worldwide. The interpreted σ_p' results should be compared with other information, such as oedometer results on high-quality, undisturbed samples, as well as geologic stress history. Additional corroboration may be obtained by comparison with other in situ tests, such as DMT and/or VST.

$$\sigma_p' = 0.33\ (q_t - \sigma_{vo})^m \cdot (0.01\ p_a)^{1-m} \quad \text{(3–25) clays and sands}$$

where:

σ_p' = preconsolidation stress (psf)
q_c = measured end-bearing component of cone resistance, psf
q_t = see Table 4–7 (psf); for sandy soils $q_c = q_t$
σ_{vo} = total vertical stress (psf)
m = function of SBT I_c and typically
0.72 in clean quartz to silica sands,
0.80 in silty sands,
0.85 in silt mixtures,
0.9 in organic clays,
1.0 in intact clays of low sensitivity.
For fissured clays, m ≈1.1+ may be applicable.
$m \approx 1 - 0.28/[1 + (I_c/2.65)^{25}]$; where I_c= CPT material index, see Table 4–7
P_a = atmospheric pressure, $\approx 2{,}000$ psf

g. Overconsolidation ratio from dilatometer. *OCR* can be calculated for uncemented clays with $I_D < 1.2$ and sands/silts with $I_D > 1.2$ using equation 3–26 (Marchetti 1980). Kulhawy and Mayne (1990) showed test data suggesting that the coefficient 0.5 in equation 3–26 can vary in the range of 0.27 to 0.75 depending on the degree of fissuring, sensitivity, and geologic origin. For all soil types, equation 3–27 (Marchetti 1980) can be used.

$$OCR = (0.5K_D)^{1.56} \tag{3–26}$$

where:

OCR = overconsolidation ratio (dimensionless)
K_D = horizontal stress index, see equation 4–14

Note. For NC clays with *OCR* = 1, $K_D = 2$. This correlation does not apply to "cemented-structured-aged" clays. It is more difficult to correlate DMT data with *OCR* of sands, and only very rough estimates can be obtained using both M from DMT data and q_c from the CPT.

$$OCR \approx 0.51(p_o - u_o) \tag{3–27}$$

where:

OCR = overconsolidation ratio (dimensionless)
p_o = contact stress or corrected A-Reading (kPa), reference equation 4–10
u_o = hydrostatic pore water pressure (kPa), reference equation 4–12

h. Preconsolidation stress from pressuremeter. Briaud (2013) established relationships for determining the preconsolidation stress (σ_p') of clay and sand from PMT data, equations 3–28 and 3–29. The yield pressure (P_y) of the PMT (at the end of the straight line of the PMT curve) is a yield pressure comparable to the preconsolidation stress (σ_p').

$$\sigma_p' = 0.5p_l \text{ (clays)} \tag{3–28}$$

$$\sigma_p' = 0.33p_l \text{ (sands)} \tag{3–29}$$

where:

σ_p' = preconsolidation stress (tsf)
p_l = limit pressure (tsf), reference paragraph 4–10d(4)

Note. To limit creep deformations, designing beyond the yield pressure (P_y) is not recommended, reference paragraph 4–10d(4).

3–8. Hydraulic conductivity

a. General. The hydraulic conductivity (k) depends primarily on the size and shape of the soil grains, void ratio, shape and arrangement of voids, degree of saturation, and temperature. Hydraulic conductivity of soil in saturated conditions is typically used for steady-state seepage analyses. Time-dependent settlement of soil depends on its

hydraulic conductivity, among other factors. Hydraulic conductivity parameters are required to evaluate drainage conditions, uplift pressure, and time rate of settlement. EM 1110-2-1901 and EM 1110-2-1913 provide guidance for determining hydraulic conductivity of different types of soils. Hydraulic conductivity of fine-grained soils should be obtained from laboratory tests or guidance provided in EM 1110-2-1901 and EM 1110-2-1913.

b. Hydraulic conductivity of sand and gravel.

(1) Empirical equations such as Kozeny-Carman and Hazen equations have been modified recently to account for both relative density and grain size distributions of soils. These equations can be used to develop site-specific k-values.

(2) A modified version of the Kozeny-Carman equation, as proposed by Chapuis and Aubertin (2003) requires a complete gradation curve (both sieve and hydrometer), as it accounts for the surface area of the particles. In the absence of hydrometer tests, the estimated k-values could tend higher, as the tail portion of the grain size distribution impacts the estimate. Equation 3–30 (Chapuis 2004) provides a relationship for hydraulic conductivity values based on void ratio (e) and D_{10} (mm) values, which is considered a modification of the original Hazen (1911) equation. Equation 3–30 is applicable to sand and gravel specimens having an effective diameter D_{10} between 0.13 and 1.98 mm and a void ratio (e) between 0.4 and 1.5.

$$k = 2.4622[D_{10}^2 \cdot {e_0}^3/(1 + e_0)]^{0.7825} \quad (3\text{–}30)$$

where:

k = hydraulic conductivity (cm/s)
D_{10} = particle size corresponding to 10 percent finer on the cumulative particle size distribution curve (mm)
e_0 = in situ void ratio

3–9. Soil stiffness parameters

Soil stiffness parameters used in analysis depend on mode of loading and can be evaluated using laboratory testing, in situ testing, correlation from other soil parameters, or geophysical testing. Soil stiffness parameters are used for evaluating soil deformation due to static or dynamic loadings. Figure 3–14 illustrates the definitions of various forms of elastic moduli in terms of loading and applied boundary conditions. Based on theory of elasticity, equations 3–32 through 3–35 correlate, K, G, E, M with Poisson's ratio (υ). Units for G, E, M, and K are expressed in stress/strain format using same unit system, such as English or Metric system. Evaluation of bulk modulus and constrained modulus is complicated when Poisson's ratio (υ) approaches 0.5 due to the term $(1 - 2\upsilon)$ in the denominator of the equation.

$$Poisson's\ ratio, \upsilon = -\frac{\varepsilon_h}{\varepsilon_v} \quad (3\text{–}31)$$

$$Bulk\ modulus, K = \frac{E}{3(1-2\upsilon)} \quad (3\text{–}32)$$

$$Shear\ modulus, G = \frac{E}{2(1+\upsilon)} \quad (3\text{–}33)$$

$$Elastic\ modulus, E = \frac{9K \cdot G}{3K+G} \quad (3\text{–}34)$$

$$Constrained\ modulus, M = \frac{E(1-\upsilon)}{(1+\upsilon)(1-2\upsilon)} \quad (3\text{–}35)$$

where:

υ = Poisson's ratio
ε_h = horizontal strain, in/in
ε_v = vertical strain, in/in

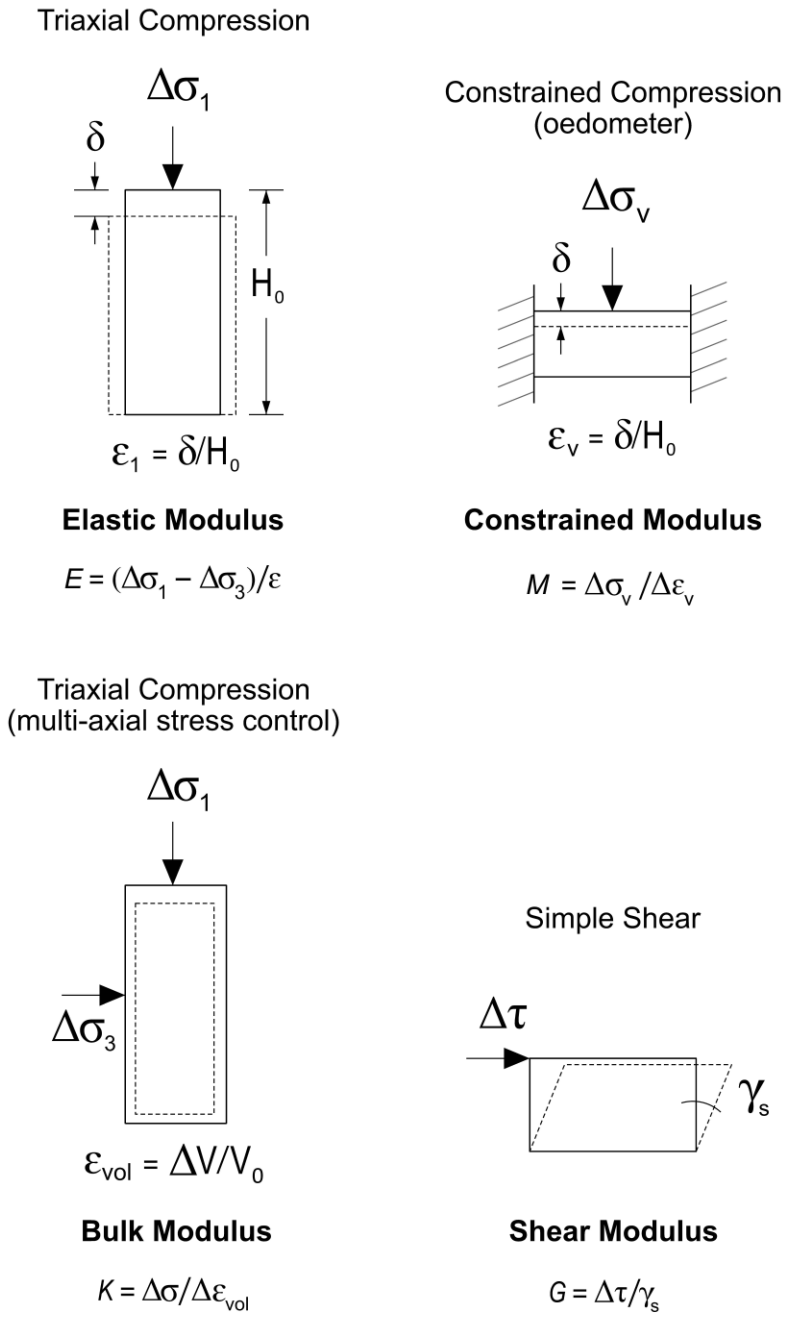


Figure 3–14. Definitions of elastic moduli in terms of loading and applied boundary conditions (adapted from Rix et al. 2018)

a. Poisson's ratio. Poisson's ratio is defined as the ratio of the change in width per unit width of a material, to the change in its length per unit length, as a result of strain. Typical values of Poisson's ratio (υ) are shown in Table 3–14. For isotropic elastic materials, the range of υ is from 0 to 0.5. A perfectly incompressible (no volume change under applied deformation) isotropic material deformed elastically at small strains would

have a Poisson's ratio of exactly 0.5. Materials that do not deform in the lateral direction (such as cork) under elastic compression have a Poisson's ratio that approaches zero.

Table 3–14
Range of Poisson's ratio (adapted from Coduto et al. 2016)

Type of Soil	Drainage Condition	Poisson's Ratio, υ
Clay (saturated)	undrained	0.49
Clay (partially saturated)	undrained	0.30 to 0.40
Dense Sand	drained	0.30 to 0.40
Loose Sand	drained	0.10 to 0.30
Sandstone	–	0.25 to 0.30
Granite	–	0.23 to 0.27
Hardened concrete	–	0.15 to 0.25

b. Elastic modulus. Modulus of elasticity is an important soil property for elastic settlement analysis of foundations and for estimation of dynamic soil behavior. In many foundation settlement formulas, the elastic modulus (E) is the stiffness parameter routinely used. The elastic modulus can be defined in different ways along the stress-strain curve as illustrated in Figure 3–16. The elastic modulus definition commonly referenced for shallow foundation design is the secant modulus determined at about 1/2 the deviator stress (Das 2008). When soils show nearly linear behavior in a stress range, the secant modulus definition and tangent definition are nearly the same.

(1) *Undrained clayey soils*.

(a) The elastic modulus of undrained clay (E_u) is commonly used to estimate the immediate settlement of a saturated clay due to foundation loading. The elastic modulus can be determined using in situ tests, triaxial shear strength testing on undisturbed samples, or estimated from correlations with other soil properties. E_u can be correlated with the undrained shear strength by the relationship shown in equation 3–36, with β being correlated to *OCR* and plasticity. D'Appolonia et al. (1971) published results of ten large-scale field studies that found correlation between E_u, undrained shear strength, soil sensitivity, overconsolidation ratio, and plasticity index.

(b) D'Appolonia et al. also concluded that for highly plastic clays, the laboratory-consolidated, undrained shear tests provided a close approximation of field behavior and unconfined compression. It was also observed that unconsolidated, undrained tests significantly underestimated E_u, and for most clays, consolidated undrained Triaxial tests (both CIU and CK_oU) resulted in similar E_u to the measured field conditions. Duncan and Buchagani (1976) advanced the work, compiling results of various studies and published a graph (Figure 3–15) and table relating the undrained elastic modulus of clay to undrained shear strength, soil plasticity, and overconsolidation, as shown in Table 3–15.

$$E_u = \beta \cdot s_u \tag{3–36}$$

where:

E_u = undrained elastic modulus of clay (psf or kPa)
β = dimensionless factor from Figure 3–5
s_u = undrained shear strength of clay (psf or kPa)

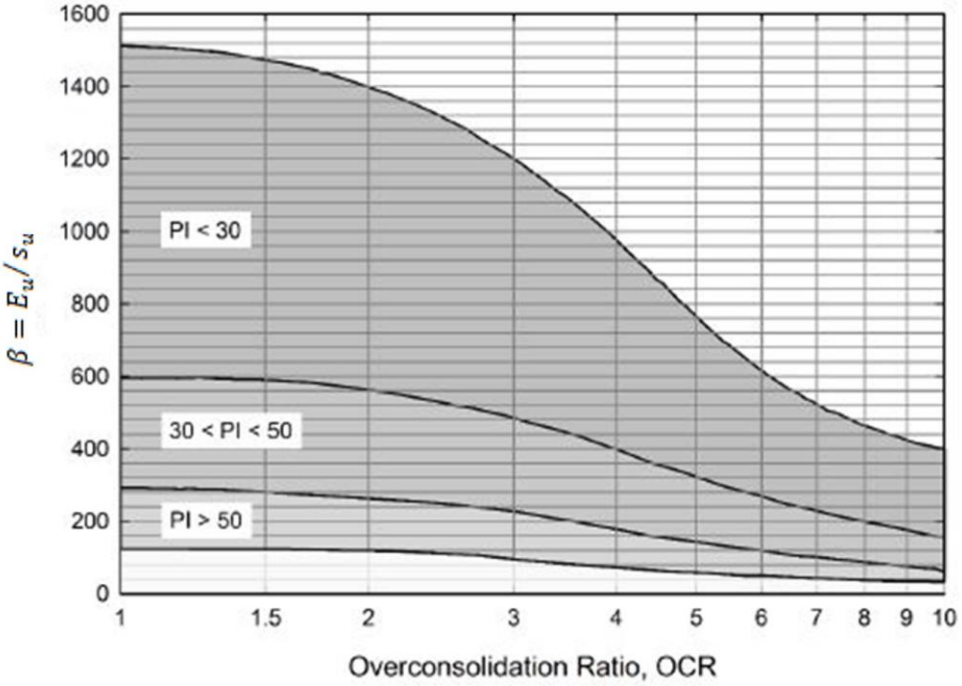


Figure 3–15. Elastic modulus for undrained clays (adapted from Duncan and Buchagani 1976)

Table 3–15
Elastic modulus of undrained clays (after Duncan and Buchagani 1976)

OCR	PI Range	β Range
1	$PI < 30$	600 to 1,500
1	$30 < PI < 50$	300 to 600
1	$PI > 50$	125 to 300
2	$PI < 30$	575 to 1,450
2	$30 < PI < 50$	275 to 575
2	$PI > 50$	115 to 75
4	$PI < 30$	400 to 975
4	$30 < PI < 50$	185 to 400
4	$PI > 50$	70 to 185
6	$PI < 30$	250 to 600
6	$30 < PI < 50$	115 to 250
6	$PI > 50$	60 to 115

(2) *Elastic modulus from in situ deformation tests*. Flat plate DMT and pressuremeter tests (PMT) are two in situ test “deformation” methods that are commonly used to test the stiffness of the ground in situ. The elastic modulus (E) has been related empirically to the pressuremeter modulus (E_o) by a rheological coefficient (see paragraph 4–10d(4)) and can be estimated by determining the shear modulus (G) using cavity expansion theory. If required, E can also be derived from the DMT modulus (M_{DMT}) using the theory of elasticity. For a Poisson’s ratio of ~ 0.2, theory of elasticity provides E = 0.9M . In view of this approximation, E and M are often used interchangeably.

(3) *Elastic modulus from laboratory triaxial tests*. Consolidated, undrained triaxial testing of clay soils reasonably approximates field conditions. As stiffness of soil depends on the confining pressure, strain level, stress level, and drainage conditions, different definitions of modulus of elasticity (E) can be used, considering the application. Figure 3–16 illustrates a representative stress-strain curve of a soil in triaxial compression mode with different types of modulus of elasticity values (Rix et al. 2018). The type of modulus of elasticity (E) is defined by the loading conditions and specific point on the stress-strain curve, such as (1) initial modulus (E_{max}), (2) secant modulus, $E_{sec} = q/\varepsilon_a$, (3) tangent modulus, $E_{tan} = \Delta q/\varepsilon_a$, and (4) unload-reload modulus, E_{ur}. For foundation design, it is typical to use the secant modulus at 1/2 of the deviator stress.

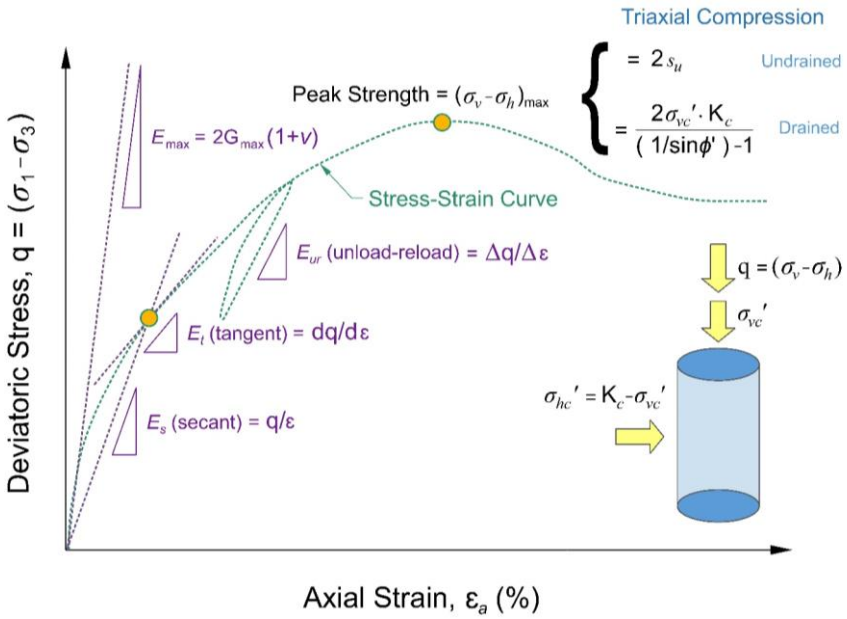


Figure 3–16. Representative modulus of elasticity in a typical stress-strain strength curve of soil in triaxial compression mode (Rix et al. 2018)

(4) *Drained sand and gravels*.

(a) The elastic modulus for sand and gravel is best determined by in situ testing, either directly or through correlation with strength measurements. Table 3–16 summarizes the general range of E that can be expected (Das 2017), and equations 3–37, 7–16, and 7–17 provide formulas correlating E with SPT and CPT measurements. The horizontal stress-strain can be directly measured using the flat plate DMT or E can be estimated from SPT and CPT measurements. DMT testing to estimate elastic modulus and constrained modulus is presented in paragraph 3–9c(2).

Table 3–16
General range of modulus of elasticity (E) for sand (after Das 2017)

Type	**E_s (psi)**
Coarse and Medium Coarse Sand	
Loose	3,626 to 5,076
Medium Dense	4,351 to 5,802
Dense	5,802 to 6,527
Fine Sand	
Loose	2,900 to 3,626
Medium Dense	3,626 to 5,076
Dense	5,076 to 5,802
Sandy Silt	
Loose	1,160 to 1,740
Medium Dense	1,450 to 1,740
Dense	1,740 to 2,476

(b) Correlations with standard penetration test data. Correlation of E with SPT blowcount data is inherently approximate and much scatter is observed in the literature. It is generally better to use CPT or DMT measurements to estimate E, due the higher degree of reliability and repeatability. Codutu (2016) identifies equation 3–37 to estimate E from SPT data, as it is noted to be generally conservative. In general, an *OCR* equal to 1 is assumed unless there is compelling evidence otherwise.

$$E = \beta_0\sqrt{OCR} + \beta_1 N_{60} \tag{3–37}$$

where:

E = elastic modulus (psf)
β_o and β_1 = factors related to soil classification, see Table 3–17
N_{60} = corrected SPT N-value for field procedures, per equation 4–4 (blows/ft)
OCR = overconsolidation ratio

Table 3–17
β_o and β_1 factors

Soil Classification	β_o (psf)	β_1 (psf)
Clean Sands (SW and SP)	100,000	24,000
Silty and Clayey Sands (SM and SC)	50,000	12,000

(5) *Unsaturated soils*. For soils, the elastic modulus varies in relation to degree of saturation. The higher the degree of saturation, the lower the elastic modulus. This is especially true for clay, and much less significant for sand. Lu and Kaya (2013) proposed a power-law relationship to relate the elastic modulus E to the degree of saturation. The elastic modulus is a maximum when the soil is dry and minimum when the specimen is wet. For some clays evaluated by Lu and Kaya, the ratio of E_{min} to E_{max} was as low as about 0.1 but was generally negligible for sand with intermediate levels of reduction for silts. For most static foundation designs on clay, it is conservative to use E from saturated undrained testing.

c. Constrained modulus. The constrained modulus (M) is used in the one-dimensional (1-D) analysis of consolidation settlement and is simply the ratio of axial stress to axial strain in confined compression. The constrained modulus (M) can be determined from laboratory oedometer test data (see paragraph 7–7) as the inverse of the coefficient of volume compressibility (m_v) or correlated with in situ test data. For an ideal elastic soil, M is related to the elastic modulus by equation 3–35. Codutu (2016) indicates that the constrained modulus is commonly used for settlement analysis (as opposed to the e-log-p method) when the soil is overconsolidated and behaves in a linear way, and when strain vs load is plotted on a linear scale.

(1) *Constrained modulus from cone penetration test*. Existing correlations between constrained modulus and cone resistance typically have the form shown in equation 3–38.

$$M = \alpha_M (q_t - \sigma_{v0}) \quad (3\text{–}38)$$

where:

M = constrained modulus (tsf)
α_M = constrained modulus cone factor (dimensionless), see Note
q_t = corrected tip resistance (tsf), see Table 4–7
σ_{v0} = total vertical stress (tsf)

Note. Mayne (2006) showed that α_M varies with soil type and net cone resistance with values of 1 to 10, where the low values apply to soft clays. Robertson and Cabal (2022) related the coefficient α_M to the normalized cone resistance (Q_t) and the soil behavior index (I_c) as shown in equations 3–39a, 3–39b, and 3–40.

$$\alpha_M = Q_t \text{ when } Q_t \leq 14 \quad (3\text{–}39a)$$
fine-grained soils, where $I_c > 2.2$

$$\alpha_M = 14 \text{ when } Q_t > 14 \quad (3\text{–}39b)$$
fine-grained soils, where $I_c > 2.2$

$$\alpha_M = 0.0188\left[10^{(0.55 I_c + 1.68)}\right] \quad (3\text{–}40)$$
coarse-grained soils, where $I_c < 2.2$

where:

α_M = constrained modulus cone factor (dimensionless), used in equation 3–38
Q_t = normalized cone resistance, see Table 4–7
I_c = soil behavior index, see Table 4–7

(2) *Constrained modulus from dilatometer*. The DMT constrained modulus (M *or* M_{DMT}) is obtained at the pre-insertion in situ vertical effective stress (σ'_{vo}). M_{DMT} is the vertical drained confined tangent modulus. M_{DMT} is calculated by applying a correction factor to the DMT modulus (E_D). M_{DMT} should be used in the same way as the constrained modulus determined from laboratory oedometer testing, reference paragraph 7–7. For sand or clay, M_{DMT} can be calculated per equation 3–41 using the correction factor R_M. The equations defining $R_M = f(I_D, K_D)$ were established by Marchetti (1980).

$$\text{Constrained Modulus, } M_{DMT} = R_M \cdot E_D \quad (3\text{–}41)$$

$$R_M = 0.14 + 2.36 \log K_D \quad \text{for } I_D \leq 0.6 \quad (3\text{–}41a)$$

$$R_M = R_{M,0} + \left(2.5 - R_{M,0}\right) \log K_D \quad \text{for } 0.6 < \mathrm{I_D} < 3 \quad (3\text{–}41b)$$
with $R_{M,0} = 0.14 + 0.15(I_D - 0.6)$

$$R_M = 0.5 + 2.0 \log K_D \quad \text{for } I_D > 3 \quad (3\text{–}41c)$$

$$R_M = 0.32 + 2.18 \log K_D \qquad \text{for } K_D > 10 \qquad \text{(3–41d)}$$

where:

M_{DMT} = constrained modulus from DMT (bar); note that 1 bar ≈ 1.044 tsf
R_M = correction factor for converting E_D to M_{DMT}; if R_M is calculated to be less than 0.85, set $R_M = 0.85$
E_D = DMT modulus, see equation 4–15
K_D = horizontal stress index, see equation 4–14
I_D = material index, see equation 4–13

Note. The correction factor (R_M) is a function of the intermediate DMT parameters (K_D and I_D). K_D typically varies in the range of 2 to 20, with corresponding values of R_M in the range of 1 to 3 (Marchetti et al. 2001). Typically, M_{DMT} varies in the range of 4 to 4,000 bar (~ 4.18 tsf to 4,177 tsf).

d. Bulk and shear modulus. Bulk modulus (K) and shear modulus (G) are not used in typical hand-calculation formulations for routine foundation design. For some numerical analyses, the constitutive models require the use of K and G as model inputs.

(1) *Bulk modulus*. The bulk modulus (K) is not used directly in most routine calculations for foundation design. The bulk modulus is more commonly used in non-linear elastic stress models such as Duncan and Chang (1970) to model stress-dependent stiffness changes, which is further described in Duncan (1980).

(2) *Shear strain modulus*. The shear strain modulus is a non-linear function, which is defined by the initial or small-strain modulus and relationships to describe the shear modulus reduction at larger strains. For static foundation problems, shear strain is typically about 1/2 percent. The initial, or small, strain shear modulus (G_{max}) is commonly determined using in situ shear wave velocity measurements. Geophysical tests such as crosshole seismic tests and suspension logging or surface geophysical methods (from free field) or downhole seismic tests (from free field) measure shear wave velocity (V_s), of soils. Shear wave velocity data can be used to calculate $G_{max}$of soil using equation 3–42. Typical values for small strain shear modulus are provided in Table 3–18.

$$G_{max} = \rho \cdot V_s^2 \qquad \text{(3–42)}$$

where:

G_{max} = initial or small strain shear modulus (psf)
ρ = density [total unit weight of soil, γ_t (pcf) / g (acceleration due to gravity, ft/sec^2)]
V_s = shear wave velocity (ft/sec)

Table 3–18
Typical values of small-strain shear modulus (AASHTO LRFD Bridge Design Specifications)

Soil Type	Small-Strain Shear Modulus G_{max} (kPa)
Soft clays	2,700 to 13,750
Firm clays	6,900 to 34,500
Silty sands	27,600 to 138,000
Dense sands and gravels	69,000 to 345,000

(a) Initial or small-strain shear modulus. Small-strain shear modulus (G) depends on in situ confining pressure and packing conditions of soil particles. A few common relationships to estimate the initial or maximum shear modulus are provided in equations 3–43 through 3–46. These relations use SPT N-values or relative density for coarse-grained soils and void ratio and stress history for fine-grained soils.

$$G_{max} = 21.7K_{2max} \cdot P_a(\sigma'_m/P_a)^{1/2} \quad (3\text{–}43)$$

$$K_{2max} = 20[(\mathrm{N1})_{60\mathrm{cs}}]^{0.33} \quad (3\text{–}44)$$
(Seed et al. 1984)

$$K_{2max} = 0.6D_R + 15 \quad (3\text{–}45)$$
(Byrne et al. 1987)

$$K_{2max} = 625\ \frac{OCR^k}{0.3 + 0.7e^2} \quad (3\text{–}46)$$
(Hardin 1978)

where:

K_{2max} = modulus coefficient for estimating G_{max}
P_a = atmospheric pressure (at sea level P_a ≈ 2,116 psf or 101.3 kPa)
σ'_m = mean normal effective stress (same unit as P_a)
$(N_1)_{60cs}$ = normalized SPT N-Value using Youd et al. (2001) or Boulanger and Idriss (2015) or Cetin et al. (2018) methods, as used in analysis
D_R = relative density (percent)
OCR = overconsolidation ratio (dimensionless)
k = function of PI and *OCR*
e = initial void ratio (dimensionless)

(b) G/G_{max} relationships. The shear modulus for the shallow foundation needs to be reduced from the G_{max}level, considering the loading and strain condition. A full shear modulus curve can be estimated using the G/G_{max} relationships by Darendeli (2001) for clay and sand and Menq (2003) relationship for gravel or other appropriate soil-type-specific relationships based on local experience or laboratory testing. A strength correction considering the soil strength should be performed according to Stewart et al. (2014). The full curves are commonly used in dynamic numerical analysis where the shear strain and modulus values change during the course of dynamic loading.

(c) Shear modulus for shallow foundations. Shear modulus (G) values vary with strain. The allowable strain depends on structure type and design objectives. Figure 3–17 conceptually shows shear modulus (G) versus shear strain relationship and the allowable stress range for different types of structures, including shallow foundations. For most static shallow foundation applications, G is about 0.1 to 0.05 of G_{max} (Das 1997), although sensitive buildings or machines may require smaller strain levels.

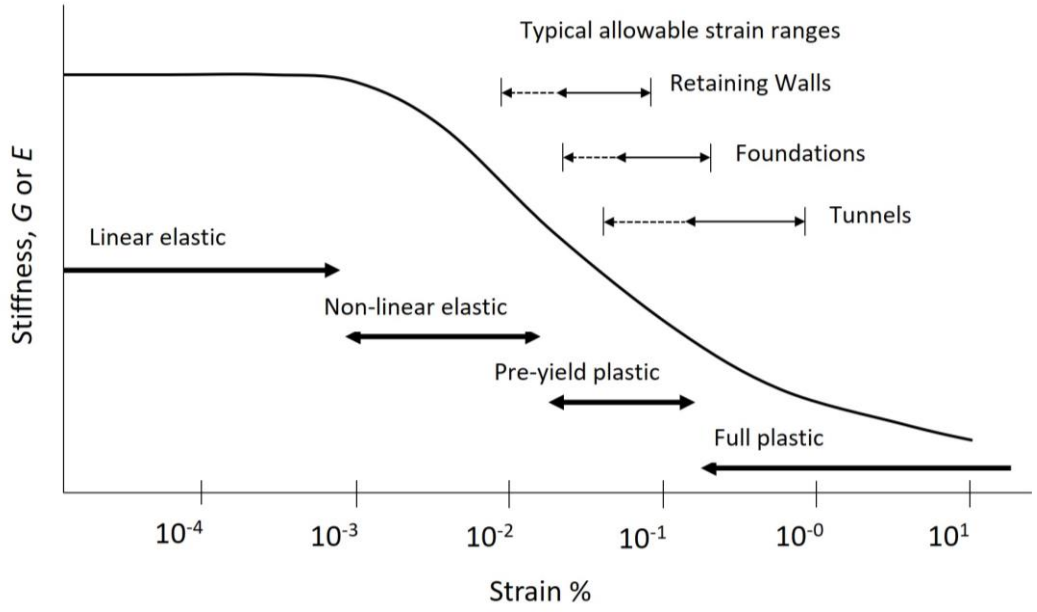


Figure 3–17. Shear modulus (G) vs. shear strain relationship, and typical allowable shear strain for different structures, including shallow foundations

e. Shear wave velocity from standard and cone penetration tests. Seismic CPT or other surface or borehole geophysical methods can measure shear wave velocity, which can then be used to calculate G_{max}. In the absence of field measurement of shear wave velocity (V_s) from seismic CPTs or other geophysical methods, empirical relationships can be used. Wair et al. (2012) provides empirical relationships for V_s estimation from SPT and CPT data. Some of these relationships were developed using SPT energy ratio of 67 percent or 78 percent, commonly used in Japan. A conversion factor of 1.12 or 1.3 between the U.S. SPT N-value and the Japanese N-value should be used [SPT N(U.S.) = 1.12 or 1.3 SPT N (Japan)] while converting relationships that have been developed using Japanese SPT N-values. Uncertainty in V_s to SPT or CPT is high and site-specific relationships are preferred.

Chapter 4
Site Exploration

4–1. Introduction

The design of shallow foundations requires an understanding of the surface and subsurface characteristics on which the supporting structure is founded. This chapter provides general guidance for site exploration and characterization for structures supported on shallow foundations. More detailed information on geotechnical investigations can be found in EM 1110-1-1804. For design of floodwalls and other hydraulic retaining walls, guidance for site exploration is provided in EM 1110-2-2502.

4–2. Geotechnical investigations

a. General. Geotechnical investigations should be implemented at the project site and scaled for the intended structure to obtain the engineering characteristics of the site and characterize the stratigraphy and nature of subsurface materials and their expected behavior under structure loadings. The goal of the investigation is to identify adverse subsurface conditions (if present) that could lead to construction difficulties, excessive maintenance, or possible failure of the foundation/structure. The scope of investigations depends on the nature and complexity of subsurface materials and the size, requirements for, and importance of the structure. An experienced geotechnical engineer who is familiar with the region is required to develop the scope of the investigation.

b. Problematic ground. Requirements for site exploration and characterization of problematic ground are not covered in this manual. Some considerations for problematic ground are listed in Chapter 10 of this manual.

4–3. Exploration phases

Site investigations can be divided into three major phases: review of available information (such as geologic maps), surface examination, and subsurface exploration. Site explorations should be performed with a basic understanding of the parameters that will be needed for design and the expected geologic variability. In some instances, site conditions will vary significantly from those assumed, based on local knowledge of the regional geology. When this occurs, the exploration plan should be modified to provide more detailed data for design relevant to the unanticipated conditions encountered.

a. Available information. To better understand site conditions, available information (such as geologic and topographic maps, boring data, site modification records) should be reviewed. These documents can better guide the site exploration and potentially reduce exploration costs. In most cases, geologic maps are readily available and should be reviewed so that the appropriate exploration techniques are implemented. Reliance on historical data for design (such as boring data, geologic maps) is not recommended unless supported by current exploration data.

b. Site reconnaissance. A site reconnaissance visit will help confirm findings from the data research phase and establish or guide exploration areas. Site observations should include topographical and (surface) geological characteristics of the site (such as surficial soils, rock outcrops), assessment of exploration access needs, surface and subsurface utility markings, site drainage, and apparent site modifications (such as partially exposed debris, cut, or fill slopes). Careful observation of damage to existing structures, such as cracks in buildings, could suggest foundation problems to be encountered or avoided.

c. Subsurface explorations. For shallow foundations, design parameters of primary concern include shear strength, compressibility, and those necessary to evaluate durability characteristics (such as pH, resistivity, chlorides), in addition to groundwater depths and presence of geologic hazards. Subsurface explorations are necessary to evaluate these conditions. A geotechnical engineer or geologist should be present during drilling activities to adjust sampling and testing (in situ or laboratory) requirements based on encountered conditions. Duties should include visual classification and description of geologic materials (such as hand penetrometer, torvane) and recording of field data.

(1) *Preliminary subsurface explorations*.

(a) The purpose of preliminary subsurface explorations is to obtain approximate soil profiles and representative samples from principal strata or to broadly characterize bedrock or stratigraphic profiles by indirect methods. Auger or split-spoon borings are commonly used for obtaining representatives samples. Geophysical methods (such as shear wave velocity, electrical resistivity, refraction) are often used in preliminary exploration of sites for large projects; however, borings should be performed to verify correlations with geophysical data.

(b) Preliminary reconnaissance explorations furnish data for planning detailed and special exploration of sites for large and important projects. The preliminary exploration could be sufficient for some construction purposes associated with foundation design, such as excavation or borrow materials. It could also be adequate for foundation design of small warehouses, residential buildings, and retaining walls located in areas where soil properties have been reasonably well established.

(2) *Detailed subsurface exploration*. For critical projects, complex subsurface conditions, and cases where preliminary subsurface explorations provide insufficient data for design, more detailed investigations should be performed. Detailed subsurface explorations are necessary to obtain more detailed geologic profiles, undisturbed samples, and cores for laboratory testing, or larger or more continuous representative samples of foundation materials. The amount of in situ or laboratory testing will vary depending on the availability of existing information, the subsurface characteristics, and the project requirements. Well-defined loading conditions should be established and referenced when determining the scope and extent of the investigation prior to implementing detailed subsurface exploration programs so that the appropriate foundation influence depths can be explored.

4–4. Exploration plan

A field exploration plan should be approved by the district geotechnical design element. The plan should be developed through close cooperation between the project geologist and geotechnical engineer. Plans should identify known safety hazards (such as utility locations/markings), right of way or permits of entry requirements, access routes, property lines, instructions for limiting site impacts, exploration locations, and procedures to include minimum sample and testing intervals. Plans must be approved by the district Dam Safety Officer/Levee Safety Officer (DSO/LSO) if drilling into, in close proximity to, or through embankment dam and levees and their foundations (see ER 1110-1-1807).

4–5. Exploration methods and extent

Numerous exploration methods are available for characterizing subsurface conditions and may be suitable. Exploration methods and their advantages/limitations are described in EM 1110-1-1804. For shallow foundation design, vertically drilled borings are generally necessary to efficiently sample, test, and characterize subsurface conditions. Minimum guidance for defining exploration extents, sample collection, testing intervals, and groundwater measurements is provided in paragraphs 4–5a through 4–8.

a. Exploration location and spacing. Reasonable locations for soil exploration (typically soil borings) should include, but not be limited to, near the center, edges, and corners of the proposed structure, and in potential borrow areas used to raise site grades above foundation areas. In certain cases, borings spaced at 40 ft or less could be necessary to adequately define erratic subsurface conditions (such as uncontrolled fill, lenses, boulders, bedrock irregularities).

b. Exploration depth. Minimum exploration depths should account for changes in effective stress that can influence foundation settlements. For induced stresses, the depth of significant influence is described in paragraph 6–6c. Since foundation dimensions are not usually known in advance, exploration depths should reflect conservative assignments of foundation size based on anticipated or presumptive ground conditions. For determining seismic site classification and adjusting mapped spectral accelerations, the site should be classified based on the upper 100 ft of the site profile as permitted by code. Often, adjustment of boring depth (or exploration scope) may be necessary due to:

(1) *Soft and loose soil deposits*. The thickness of soft and loose soils should be adequately determined (such as continuous or near-continuous sampling and testing) within the settlement zone of influence. Since the settlement zone of influence is not precisely known, borings are often extended some additional depth (to account for this uncertainty) or until more competent conditions are encountered.

(2) *Shallow refusal*. Depending on the drilling method, refusal at shallow depth can be an indicator of hard or dense zones (such as shallow rock, cobbles/boulders, buried

debris). Where these conditions are encountered (or suspected), offset borings and other exploratory methods (such as test pits, rock coring, geophysics) are recommended to better characterize the subsurface conditions.

(3) *Depth to primary formation*. Typically, depth of exploration need not exceed the depth of the primary formation where rock or soil of exceptional bearing capacity and compressibility characteristics is located.

4–6. In situ test intervals

In situ tests should be performed at significant changes in soil strata (such as stress state, soil type). Minimum in situ test intervals are predicated on standard test methods (ASTM) and are described in paragraph 4–10; however, larger intervals may be adequate. For SPT methods, test intervals should not exceed 2.5 ft in the upper 10 ft and 5 ft thereafter within foundation influence depths.

4–7. Soil sampling

Soil samples should be obtained at significant changes in soil strata (such as stress state, soil type). Commonly, thin-walled tube and disturbed samples are obtained during field exploration activities associated with shallow foundations; however, alternate sampling techniques (such as undisturbed block, ASTM D7015) may also be used. Table 4–1 and Table 4–2 may be used as a guide for determining the required sizes of thin-walled tube and disturbed samples, respectively. Sampling methods should adhere to standard practices (ASTM) to include preserving, storage, and transporting. In general, intact samples of clean sands and gravels cannot be obtained using conventional sampling methods. For this reason, in situ testing methods are commonly used in these soils.

a. Thin-walled tube samples. Thin-walled tube soil samplers are described in EM 1110-1-1804. Sampling of cohesive soils should be according to ASTM D1587. Strength index tests (such as lab vane) and visual observation should be performed above and below cohesive specimens obtained from thin-walled tubes to assess relative changes in sample quality. It may not be possible to obtain samples of very soft or hard soils using a thin-walled tube sampler. For these cases, in situ testing methods may be necessary for engineering evaluation.

(1) *Sample disturbance*. Thin-walled tube sampling results in a stress change, possible pore water change, and some structure alteration because of displacement effects of the sampler. Sample disturbance is influenced by sampling method, transportation, storage, and trimming. Practices for preserving and transporting soil samples to reduce disturbance should be adhered, reference ASTM D4220. The effects of sampling disturbance on laboratory-measured soil properties are detailed in Miscellaneous Paper (MP) GL-92-35 (U.S. Army Engineer Waterways Experiment Station). Sample disturbance can generally be evaluated using the following methods; however, no definitive method exists to determine the absolute sample quality.

(a) Area ratio and inside clearance ratio. Sample disturbance is related to the area ratio (A_r) as defined in equation 4–1 and to the inside clearance ratio (*ICR*), as defined in equation 4–2. For an acceptable degree of disturbance, the *ICR* should be approximately 1 percent and the area ratio should 10 percent or less. Thin-walled tubes are typically manufactured to meet these specifications, but a thicker-walled tube with an *ICR* of zero is commonly used in the Gulf states (Texas, Louisiana) to sample very stiff, overconsolidated clays. The thicker-walled tube is intended to minimize buckling of the sampler in the stiff deposits, and the *ICR* of zero criteria is intended to minimize sample expansion within the tube.

$$A_r = \frac{(D_o^2 - D_i^2)}{D_i^2} \text{ x } 100\% \quad (4\text{–}1)$$

$$ICR = \frac{(D_i - D_e)}{D_e} \quad (4\text{–}2)$$

where:

A_r = area ratio (dimensionless)
ICR = inside clearance ratio (percent)
D_o = outside diameter of sampler tube (in)
D_i = inside diameter of the cutting shoe (in)
D_e = diameter of the sampler cutting tip (in)

(b) Recovery ratio. Sample quality is indicated by recovery ratio (L_r) defined in equation 4–3. A value of L_r < 1 indicates that the sample was compressed or lost during recovery, and L_r >1 indicates the sample expanded during recovery or excess soil was forced into the sampler. Undisturbed samples should be handled and stored to preserve stratification or structure, water content, and in situ stresses.

$$\text{Recovery Ratio}, L_r = \frac{Length\ of\ recovered\ sample, in}{Length\ of\ sample\ tube\ pushed, in} \quad (4\text{–}3)$$

Table 4–1
Recommended thin-walled tube diameter

Test	Thin-Walled Tube Diameter, inch
Unit weight	3.0
Permeability	3.0
Consolidation	5.0
Triaxial compression	5.0
Unconfined compression	3.0
Direct shear	5.0
Direct simple shear	3.0

(2) *Sample quality*. Sample quality can be estimated from Table 4–2 by obtaining the vertical strain obtained from a standard oedometer test at a vertical pressure equal to the in situ vertical effective stress (σ'_{vo}) calculated for the appropriate field conditions. A sample quality designation (SQD) of 4 percent means that the oedometer sample strained 4 percent upon reaching a vertical stress that is equal to σ'_{vo} (the sample expanded about 4 percent upon extraction). Effects of sample disturbance on a consolidation test e-log-p curve are illustrated in for an OC clay. Practically, no soil sample can be obtained in a perfectly undisturbed state. The sample quality designation (SQD) should be rated as very good to excellent for use in strength evaluation or consolidation characteristics.

Table 4–2
Sample quality designation versus strain (%) on reloading to σ'_{vo} (Lunne et al. 2006)

$\Delta e/e_o$ **at** σ'_{vo}		**Rating**[1]
OCR = 1 to 2	*OCR* = 2 to 4	
< 0.04	< 0.03	Very good to excellent
0.04 to 0.07	0.03 to 0.05	Good to fair
0.07 to 0.14	0.05 to 0.10	Poor
> 0.14	> 0.10	Very poor

Note:
[1] The description refers to the use of the samples for measurement of mechanical properties.

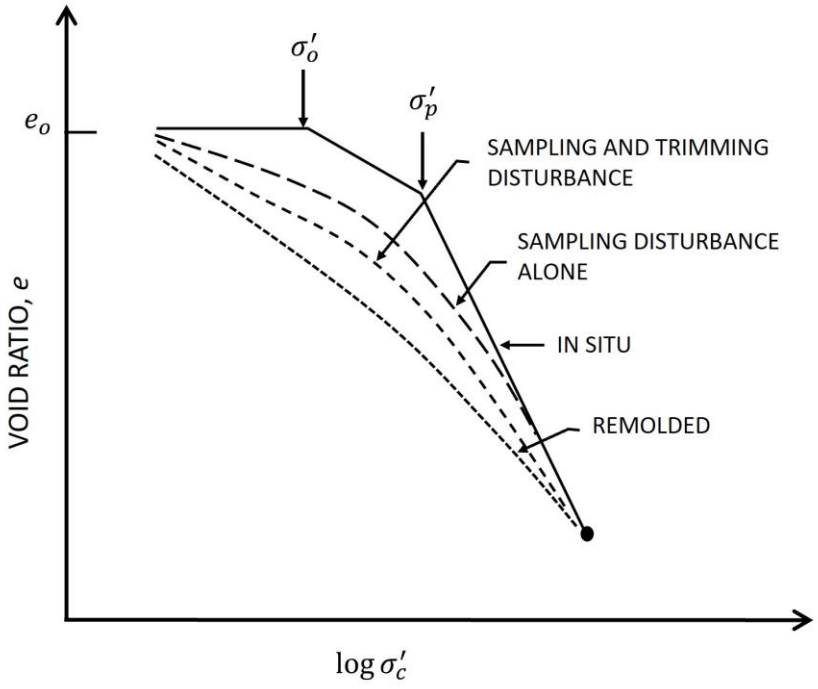


Figure 4–1. Effect of sample and specimen preparation on the laboratory e-log-p curve for overconsolidated clay (after Perloff and Baron 1976)

b. Disturbed samples. Disturbed soil samples, which alter the macro-structure of the soil, can be obtained by various methods (such as auger or mechanically pushed, or driven sampling). Thick-wall, solid, or split-barrel drive samplers can be used in most soils, excluding gravelly soils. Samples obtained with a drive sampler should be not less than 2 inches, and preferably 3 inches or more in diameter. Where loose sands or soft silts are encountered, a special sampler with a flap valve or a plunger should be used to hold the material in the barrel. The minimum recommended sample size for a disturbed sample is provided in Table 4–3.

Table 4–3
Minimum recommended sample size for disturbed soils

Test	Minimum Recommended Dry Weight (lb)[1]
Water content	0.5
Atterberg limits	0.2
Grain size analysis	0.5
Compaction	100.0
Direct shear	2.0
Permeability	2.0
4-inch diam. consolidation	2.0
2.8-inch diam. triaxial (4 pts)	8.0

Note:
[1] Fine-grained (minus No. 4 sieve). For material containing plus No. 4 sieve sizes, sampling requirements should be discussed with the laboratory.

4–8. Groundwater measurements

Groundwater levels should be obtained in boreholes during drilling to document groundwater levels for shallow foundation design and constructability purposes. Groundwater conditions can be complex due to artesian conditions, perched groundwater, and interbedded aquicludes and aquifers. Due to the low permeability of fine-grained soils, a long period of time may be required for water levels to stabilize. For low-permeability soils, water levels should be obtained after a period approximately 24 hours after drilling is completed. Instrumentation (typically monitoring wells or piezometers) should be installed to monitor groundwater levels if potential borehole collapse or “cave-in” could occur. Where groundwater levels are influenced by nearby water bodies (ocean, river), historical water level data should be reviewed and could influence design.

a. Observation wells. Observation (or monitoring) wells consist of vertical pipes with a slotted section at the bottom or a tube with a porous tip at the bottom (Figure 4–2a). Observation wells are appropriate in relatively uniform, pervious material. Often, the top of the borehole is sealed to prevent surface water from entering the borehole. Observation wells create a hydraulic connection between strata and could lead to misinterpreted groundwater conditions.

b. Piezometers. Piezometers consist of a vertical pipe with a screened (or measured) interval (Figure 4–2b). Seals are usually placed as bentonite clay or cement grout. Pipe joints should be watertight to prevent leakage. Various types of piezometers can be installed; however, open standpipes (observation wells with subsurface seals) are more common.

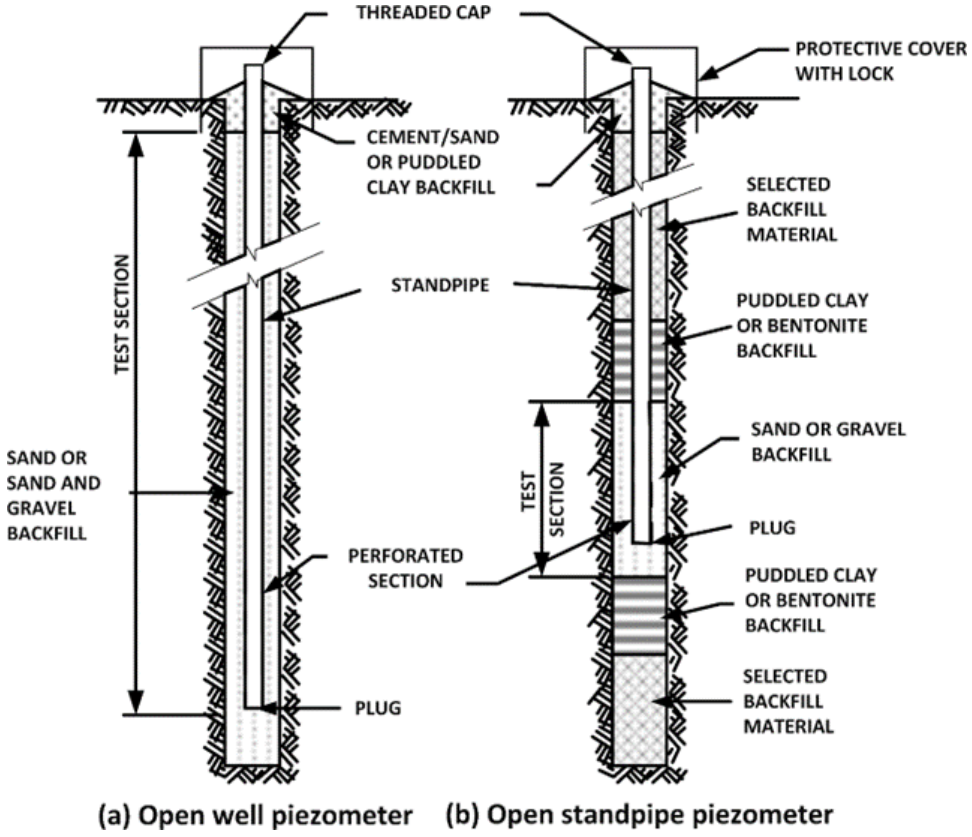


Figure 4–2. Schematic of open piezometers (UFC 3-220-10)

4–9. Borehole closure

Borings should be backfilled after completion of field activities. Backfilling should be performed by tremie grouting through drilling rods or other pipes. Backfilling should be performed according to state and environmental regulations.

4–10. In situ testing

Appropriate in situ testing methods should be assigned, see EM 1110-1-1804. Commonly, borehole penetration tests (such as SPT, CPT) and deformation tests (such as DMT, PMT, VST) are used to characterize subsurface conditions for shallow foundation design. See Table 4–4. In some cases, depending on equipment or test limitations, several methods may be necessary to adequately define subsurface conditions. However, direct comparison between different in situ or laboratory tests can be misleading due to different modes of deformation. For estimating foundation settlement, borehole deformation tests (such as DMT, PMT) are recommended due to

strain rate effects associated with penetration tests. For gravelly sites, reference EM 1110-1-1804 for guidance on in situ testing methods.

Table 4–4
Reference publications for in situ testing

In Situ Test	ASTM	Reference
SPT	ASTM D1586 ASTM D6066 ASTM D4633	*Performance and Use of the Standard Penetration Test in Geotechnical Engineering Practice*, 1998, McGregor, J.A., Duncan, J.M., Center for Geotechnical Practice and Research.
CPT, CPTu, SCPTu	ASTM D3341, D5778	*Guide to Cone Penetration Testing for Geotechnical Engineering*. P.K. Robertson and K.L. Cabal (Robertson), Gregg Drilling and Testing, Inc., 2022. *Cone-Penetration Testing in Geotechnical Practice*, Lunne, T., Robertson, P.K., and Powell, J.J.M. (1997), London: Blackie Academic and Professional.
DMT	ASTM D6635	*In Situ Test by Flat Dilatometer*, Marchetti, S., In situ Tests by Flat Dilatometer, ASCE Journal of Geotechnical Engineering, Vol. 106, 299–321, 1980. *The Flat Dilatometer Test in Soil Investigations*. A Report by the International Society for Soil Mechanics and Geotechnical Engineering (ISSMGE) Committee TC16. Marchetti et al. 2001. Schmertmann, J. H., (1988) Guidelines for Using the CPT, CPTu, and Marchetti DMT for Geotechnical Design: Volume III – DMT Test Methods and Data Reduction. FHWA-PA-87-024+84-24. FHWA (1992), *The Flat Dilatometer Test*. Report No. FHWA-SA-91-044, Riaund, J.L., and Miran, J., U.S. Department of Transportation.
PMT	ASTM D4719	The Pressuremeter by Louis Ménard, *Interpretation and Application of Pressuremeter Results*, 1975. Publisher: Techniques Louis Ménard, 1975. FHWA (1989), *The Pressuremeter Test for Highway Applications*. Publication No. FHWA IP-89-008, Briaud J.L., Federal Highway Administration, U.S. Department of Transportation.
VST	ASTM D2573	ASTM (1988) Special Technical Publication. *Vane Shear Strength Testing in Soils: Field and Laboratory Studies*, American Society for Testing Materials, Committee D-18 on Soils and Rock for Engineering Purposes, Philadelphia, PA.

a. Standard penetration test.

(1) *Application*. The SPT is an in situ penetration test described in ASTM D1586. Disturbed “split-spoon” samples (1.5-ft to 2-ft length) can be obtained during testing, continuously or intermittently. In general, SPT blow count data are most useful for fine-grained sands, with coarser sands and silty sands giving reasonably useful results. For gravelly and fine-grained soils, SPT can be unreliable for obtaining strength or stiffness properties. Engineering behavior of saturated or nearly saturated silty sands could be underestimated by the SPT. A typical production rate is 120 lineal feet per shift.

(2) *Standard penetration test N-value corrections*. Several factors contribute to variations of the standard penetration test (SPT) N-value at a given depth for similar soil profiles. These include SPT hammer efficiency, borehole diameter, sampling method, rod length, water table, and overburden pressure. Schmertmann et al. (1978) and Kovacs and Salomone (1982) identify the most significant factors affecting measured N-values as (1) the amount of energy delivered to the drill rods, and (2) the method used to drill and support the borehole. The engineer should be familiar with the N-value requirements of the correlations or direct design methods.

(3) Field procedures and hammer efficiency. SPT design correlations were popularized circa 1950 and are widely used today; however, testing methods have been modernized. N-value corrections allow these established correlations to remain valid and standardize drilling procedures. The measured SPT N-value is typically adjusted for hammer efficiency per equation 4–4a. Additional correction factors may also be needed to arrive at a more standardized value of N_{60} per equation 4–4b. The corrections are applicable for both cohesive and cohesionless soils. Hammer efficiency should be verified using the methods of ASTM D4633; typical energy correction factors are provided in Table 4–5.

$$N_{60} = N_m \cdot \frac{ETR}{60} \tag{4–4a}$$

$$N_{60} = N_m \cdot C_E \cdot C_B \cdot C_R \cdot C_S \tag{4–4b}$$

where:

N_{60} = standard penetration resistance adjusted to a 60% drill rod energy ratio (blows/ft); note: may also be corrected for field procedures

ETR = ratio of the measured energy transferred to the drill rods to the theoretical energy per ASTM D4633; the ratio $ETR/60$ is also referred to as an energy ratio, (C_E) in equation 4–4b and Table 4–5.

C_B, C_R, C_S = field procedure correction factors, see Table 4–6 (dimensionless)

N_m = measured SPT N-value in field (blows/foot) per ASTM D1586

Table 4–5
Typical energy correction (hammer efficiency) factors for standard penetration test N-values (NCEER 1998)

Correction Factor	Variable	Value
Energy ratio (C_E)	Automatic-trip, donut-type hammer	0.8 to 1.3
	Safety hammer	0.7 to 1.2
	Donut hammer	0.5 to 1.0

Table 4–6
Field correction factors for standard penetration test N-value (adapted from NCEER 1998)

Correction Factor	Variable	Value
Borehole diameter (C_B)[1]	65 to 115 mm (~2.6 to 4.5 inch)	1.00
	150 mm (~5.9 inch)	1.05
	200 mm (~7.9 inch)	1.15
Rod length (C_R)	< 3 m (~9.8 ft)	0.75
	3 to 4 m (~9.8 to 13.1 ft)	0.80
	4 to 6 m (~13.1 to 19.7 ft)	0.85
	6 to 10 m (~19.7 to 32.8 ft)	0.95
	10 to 30 m (~32.8 to 98.4 ft)	1.00
Sampling method (C_S)[2]	standard sampler	1.00
	U.S. sampler without liners	1.1 to 1.3

Notes:
[1] Large diameter boreholes allow for stress relaxation that can be more significant in sands but is probably negligible in cohesive soils (Sanglerat 1982). For clay-like soils, use $C_B = 1.0$. When drilling with hollow-stem augers, the inside diameter is used for the borehole diameter.
[2] If the sampler is not designed for liners or if the correct size liner is used, use $C_S = 1.0$.

(a) Overburden stress correction. In cohesionless soils, SPT N-values are affected by the effective overburden stress, and therefore, an overburden correction factor (C_N) is recommended per equation 4–5 (Liao and Whitman 1985). For cohesive soils, there is no need for overburden pressure correction (Peck et al. 1974).

$$(N_1)_{60} = C_N \cdot N_{60} \leq 2N_{60} \tag{4–5}$$

$$C_N = \left(\frac{P_a}{\sigma'_{vo}}\right)^{0.5} \tag{4–6}$$

where:

$(N_1)_{60}$ = penetration resistance, N_{60} normalized to a 1 ton/ft^2 stress level (blows/ft)
C_N = overburden stress correction factor (dimensionless)
N_{60} = corrected SPT N-value per equation 4–4 (blows/ft)
P_a = atmospheric pressure; at sea level P_a ≈ 2,116 psf (101.3 kPa)
σ'_{vo} = in situ vertical effective stress, same unit as P_a

(b) Correction of standard penetration test N-value for water table. For dense or very dense fine sand and silt below the water table, dilatancy effects can influence SPT N-values, resulting in abnormally high SPT N-values. For penetration resistance, $(N_1)_{60}$ greater than 15 blows per foot (bpf), equation 4–7 (Terzaghi and Peck 1967) may be used to correct for dilatancy effects. For coarse sand, this correction is not applicable. In applying this correction, overburden correction should be applied first and then corrected for dilatancy.

$$(N_1)_{60\,(corr)} = 15 + \frac{1}{2}[(N_1)_{60} - 15] \tag{4–7}$$

where:

$(N_1)_{60\,(corr)}$ = corrected $(N_1)_{60}$ value for dilatancy/water table effects (blows/ft)

$(N_1)_{60}$ = penetration resistance, N_{60} normalized to a 1 ton/ft^2 stress level (blows/ft)

(c) Gravel in the soil matrix. In 1987, Andrus and Youd (see MP GL-87-8) describe an alternate procedure for determining SPT N-Values in gravel. They suggest that the penetration per blow be measured and the cumulative penetration versus blow be plotted, reference Figure 4–3. With this procedure, changes in slope could be identified when gravel particles interfere with penetration.

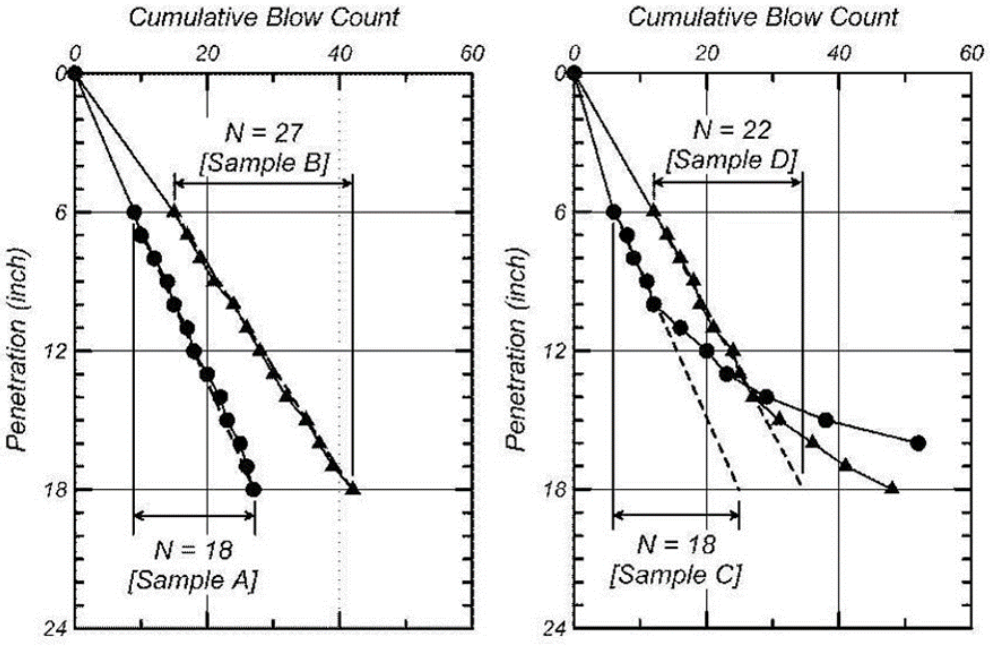


Figure 4–3. Examples of interpreting standard penetration test blow counts on a per-inch basis: (a) smooth driving patterns that do not require corrections to N-values and (b) strong increases in driving resistance that, along with sample recoveries, suggest that the sampler encountered large particles; this graph also shows the adjusted N-value, based on extrapolating the pre-obstruction driving rate

b. Cone penetration test. The CPT, ASTM D3441 and ASTM D5778, is a quasi-static penetration test. Test results are usually obtained at 1-inch to 2-inch depth intervals for estimating soil properties. The CPT is generally suitable in soft to medium dense sand, silt, and clay, and is not generally suitable in glacially overconsolidated soil, rock, and gravel or debris. For penetration of very dense and hard soils, pre-drilling or advancement through hollow stem augers may be necessary. Although not typical, discrete soil samples can be obtained using piston-type samplers that typically measure 12-inch length by 1-inch diameter. A typical production rate is 400 lineal feet per shift. Other applications include measurement of shear wave velocity (with a seismic module), and inclination or verticality measurements.

(1) *Piezocone penetration test*.

(a) Application. The Piezocone penetration test (CPTu) is a cone penetration test with pore water pressure measurements. Between one and three pore pressure sensors can be used. The pore pressure measurements help to better distinguish between sands and clays and can be used for determining deformation and permeability characteristics. The rate of penetration for the CPTu test is the same as the basic CPT test unless pore water dissipation testing is performed.

(b) Pore water dissipation tests. Dissipation tests measure the decay of pore water pressure over time in saturated soil. During a dissipation test, the CPTu is paused at specified intervals, and pore water pressure decay readings are recorded in situ. The test is usually performed until a time at which 50 percent of the pore water has dissipated (t_{50}) to assess hydraulic conductivity and coefficient of consolidation. If conditions are hydrostatic, the equilibrium water pressure (u_o) can be used to calculate the approximate depth of the ground water table. The water table (or phreatic surface) depth is defined as the distance below the soil surface at which pore pressure is equal to atmospheric pressure, per equation 4–8. The water head (h_w) is the height of the water above the cone, per equation 4–9.

$$d_{water} = d_{cone} - h_w \tag{4–8}$$

where:

d_{water} = depth of water table (ft)
d_{cone} = depth of piezocone (ft)
h_w = water head (ft)

$$h_w = u_o/\gamma_w + z \tag{4–9}$$

where:

h_w = water head (ft)
u_o = equilibrium pore pressure determined from dissipation test (psf)
γ_w = unit weight of water (pcf)
z = distance, if any, between pressure sensor location and depth reference point on the piezocone (ft)

(2) *Measured and interpreted parameters*. Numerous correlations have been developed to estimate soil parameters from CPT data. Common measured and interpreted parameters used in these correlations are shown in Table 4–7.

(3) *Soil behavior type*. The primary parameters for soil behavior type (SBT) are: (1) cone resistance (q_c, q_t, or Q_t), (2) friction ratio (R_f), and (3) pore water measurements (commonly measured at the u_2 position). SBT is categorized in Figure 4–4 and Table 4–8. Figure 4–4 depicts the shifts in SBTs based on adjustments when introducing pore pressure measurements and normalizing cone resistance. Other charts by Schmertmann (1978), Douglas and Olsen (1981), Robertson (1990), and Schneider et al. (2008) may also be used for evaluating soil type. Although SBT charts can be similar in appearance and results, each has its own limitations.

Table 4–7
Common cone penetration test-measured and -interpreted parameters

$\boldsymbol{q_c}$	=	cone tip resistance; measured end-bearing component of cone resistance, units of force/area
q_t	=	corrected cone tip resistance; corrected for water pressure acting behind the cone tip, units of force/area $q_t = q_c + u_2\,(1 - a)$
q_n	=	net cone resistance, $q_n = q_t - \sigma_{vo}$
u_o	=	static or equilibrium pore water pressure
u_1	=	pore water pressure measured on the midface or tip of the cone
u_2	=	pore pressure measured immediately behind the cone tip (standard location)
Δu	=	excess pore water pressure; water pressure in excess of the equilibrium pore water pressure caused by penetration of cone into ground
a_n	=	net area ratio from laboratory calibration, = A_n/A_c, typically ranges from 0.60 to 0.85
Q_t	=	normalized cone tip resistance (overburden correction applied); $Q_t = (q_t - \sigma_{vo})/\sigma'_{vo}$ where, σ'_{vo} = effective vertical stress, and σ_{vo} = total vertical stress
f_s	=	friction sleeve resistance; friction component of cone resistance developed on friction sleeve, units of force/area
R_f	=	friction ratio; the ratio of the friction sleeve resistance (f_s), to the cone tip resistance (q_c), with the latter measured at the depth for the middle of the friction sleeve
F_r	=	normalized friction ratio, $F_r = f_s/(q_t - \sigma_{vo})$
B_q	=	pore water pressure ratio, $B_q = \Delta u_2/(q_t - \sigma_{vo})$; where σ_{vo} = total vertical stress
I_c	=	soil behavior type index, $[(3.47 - \log_{10} Q_t)^2 + (\log_{10} F_r + 1.22)^2]^{0.5}$

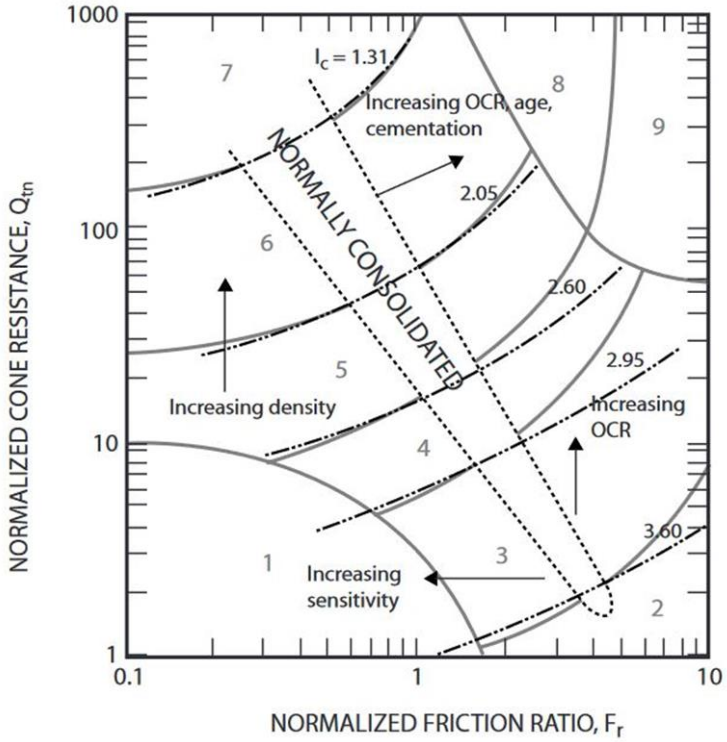


Figure 4–4. Normalized cone penetration test soil behavior type chart (Robertson and Cabal 2022)

Table 4–8
Soil behavior type based on cone penetration test soil behavior type index (I_c)

Zone	Soil Behavior Type	Soil Behavior Type Index, I_c
1	Sensitive, fine grained	Not Applicable
2	Organic soils – clay	> 3.6
3	Clays – silty clay to clay	2.95 to 3.6
4	Silt mixtures – clayey silt to silty clay	2.60 to 2.95
5	Sand mixtures – silty sand to sandy silt	2.05 to 2.6
6	Sands – clean sand to silty sand	1.31 to 2.05
7	Gravelly sand to dense sand	< 1.31
8	Very stiff sand to clayey sand[1]	Not Applicable
9	Very stiff, fine grained[1]	Not Applicable

Note:
[1] Heavily overconsolidated or cemented.

c. Flat plate dilatometer test.

(1) *Application*. The dilatometer test (ASTM D6635) is a calibrated, static deformation test that can be used to estimate soil parameters and provide information

on stress history. It is generally not suitable for testing gravel, hard rock, or soils with angular gravel size stone which can damage the membrane; however, it is commonly used in residual soils (SPT N < 30 bpf), which can be hard to penetrate.

(2) *Testing and sampling*. The DMT has a wedge-shaped tip for reduced soil disturbance (Figure 4–5). Testing is specified at approximate 8- to 12-inch increments (1 to 2 minutes per test) and the thrust pressure applied for advancement can be recorded to aid in soil characterization. A typical production rate (at ~8-inch intervals) is 400 feet per day per shift. Although not typical, it is possible to take small-diameter samples using the same pushing equipment used to insert the probe. Shear wave velocity measurements can also be obtained where a seismic module is incorporated (SDMT).

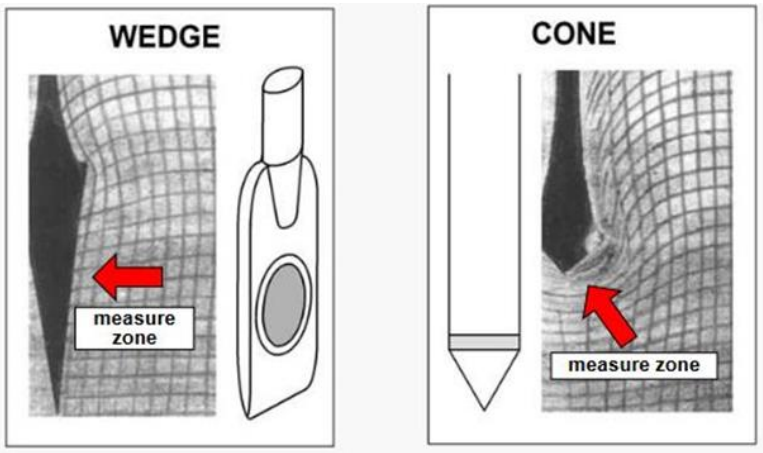


Figure 4–5. Deformed grids comparing dilatometer test wedge and cone penetration test cone (Baligh and Scott 1975)

(3) *Readings A and B corrections*. At each test interval, pressure readings (A, B, and C) are recorded as described in Table 4–9. The A, B, and C readings are corrected during calibration to account for membrane stiffness, pressure gage zero offset, and other factors using equipment calibration readings per equations 4–10 through 4–12 (Marchetti 1999).

Table 4–9
Flat plate dilatometer test (A, B, and C readings)

Reading No.	Description
Reading A	Gage gas pressure against the inside of the membrane when the center of the membrane has lifted above its support and moved laterally 0.05 mm into the soil surrounding the blade.
Reading B	Gage gas pressure against the inside of the membrane when the center of the membrane has lifted above its support and moved horizontally into the surrounding soil 1.1 mm.
Reading C (optional)	Gage gas pressure against the inside of the membrane obtained by slowly deflating the membrane until contact is reestablished.

$$Contact\ Stress,\ p_o = 1.05(\mathrm{A} - Z_M + \Delta\mathrm{A}) - 0.05(\mathrm{B} - Z_M - \Delta\mathrm{B}) \quad (4\text{–}10)$$

$$Expansion\ Stress,\ p_1 = \mathrm{B} - Z_M - \Delta\mathrm{B} \quad (4\text{–}11)$$

$$Equilibrium\ Pore\ Pressure, u_o \approx p_2 = \mathrm{C} - Z_M - \Delta\mathrm{A} \quad (4\text{–}12)$$

where:

p_o = contact stress or corrected first reading (bar)
A = reading A (bar)
$\Delta\mathrm{A}$ = vacuum pressure required to keep membrane in contact with its seating (bar)
Z_M = gauge pressure deviation from zero when vented to atmospheric pressure (bar), if $\Delta\mathrm{A}$ and $\Delta\mathrm{B}$ are measured with the same gage used for current readings, A and B, set $Z_M = 0$ (Z_M is compensated)
B = reading B (bar)
$\Delta\mathrm{B}$ = air pressure inside the membrane to expand the membrane 1.1 mm (bar)
p_1 = expansion stress or corrected B reading (bar)
u_o = pre-insertion equilibrium pore water pressure at center of membrane (bar). Often assumed hydrostatic below the water table surface. Estimated from ground water surface or Eq. 4–12 in free-draining soils
p_2 = corrected C reading, $p_2 \approx u_o$ in free-draining soils
C = reading C (bar), if the C reading exceeds the equilibrium pore water pressure, this is an indication of a non-free draining material.

(4) *Intermediate dilatometer parameters*. Four “intermediate” parameters are interpreted from the corrected DMT measurements, reference equations 4–13 through 4–16. The DMT modulus (E_D) is an uncorrected and intermediate parameter and should be converted to the constrained modulus (M_{DMT}) for use in settlement evaluations, reference paragraph 3–9c(2). The material index (I_D) is based on mechanical response without benefit of grain size distribution; however, it can be used to help estimate soil stratigraphy and soil type. Similarly, the pore water pressure index (U_D) can be used as an indicator of soil type as well as to correlate other soil parameters with DMT data (such as soil unit weight).

$$Material\ Index,\ I_D = \frac{(p_1 - p_o)}{(p_o - u_o)} \quad (4\text{–}13)$$

$$Horizontal\ Stress\ Index,\ K_D = \frac{(p_o - u_o)}{\sigma'_{vo}} \quad (4\text{–}14)$$

$$Dilatometer\ Modulus,\ E_D = 34.7\ (\mathrm{p}_1 - \mathrm{p}_o) \quad (4\text{–}15)$$

$$Pore\ Water\ Pressure\ Index,\ U_D = \frac{(p_2 - u_o)}{(p_o - u_o)} \quad (4\text{–}16)$$

where:

I_D = material index (dimensionless)
K_D = horizontal stress index (dimensionless)
E_D = DMT modulus (bar)
U_D = pore water pressure index (dimensionless); in free-draining soils, $p_2 \approx u_o$, $U_D \approx 1$. In non-free draining soils, p_2 will be higher than u_o and U_D.
σ'_{vo} = in situ effective vertical stress prior to blade penetration (bar)

Note. The DMT modulus (E_D) is obtained from p_o and p_1 by theory of elasticity for the 60 mm membrane and 1.1 mm displacement (Gravesen 1960). Deformation properties should be derived from M_{DMT} (paragraph 3–9c(2)) and not E_D (especially when E_D is an "uncorrected" modulus).

(5) *Dilatometer dissipation test*. Dissipation tests are not feasible in relatively permeable soils (such as silty sands). In clean sands, excess pore pressures are not generated during penetration of the DMT; however, in soft clays, the basic DMT data (p_o, p_1) are dominated by large penetration pore pressures (Campanella et al. 1985). The coefficient of consolidation in the horizontal direction (C_h) can be estimated from dissipation tests using various methods, reference ASTM D6635. The DMT-A method (Marchetti and Totani 1989) utilizes the A-reading and is generally preferred. Other methods include the DMT-C method and the DMT-A_2 method. An example plot is shown in Figure 4–6.

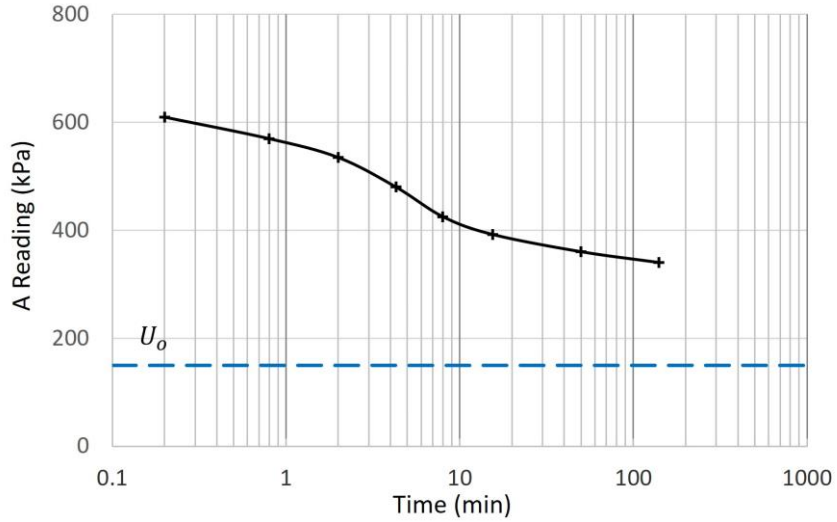


Figure 4–6. Example of dissipation test in cohesive soils, dilatometer test-A method (Totani and Monaco 1998)

(6) *Soil classification*. Soil behavioral response from DMT can be used to evaluate soil classification. A general relationship between material index (I_D), DMT modulus (E_D), and soil type is shown in Figure 4–7. Another general indicator of soil type is the

pore pressure index (U_D). Typically, coarse-grained (such as clean sands and gravels) will exhibit a U_D between 0.0 and 0.2. Fine-grained soils (such as lean and fat clays, elastic silts) will exhibit a U_D of 0.7 or greater. A wide range of soils can have an intermediate permeability that exhibit U_D between 0.2 and 0.7. U_D values are a general indication of soil type and should be used in conjunction with I_D to characterize soil type.

d. Pressuremeter test. Pressuremeter tests (PMTs) can be used to estimate the stress-strain response in a wide variety of soils and soft/weak rock. In cohesive soils, the PMT can be used to determine both the immediate and consolidation settlements together. Various types of pressuremeters are available (such as pre-bored, self-boring). Standard test methods for the pre-bored PMT are described in ASTM D4719 and consist of drilling a borehole, removing the drilling tool, and lowering the probe in the open borehole (Figure 4–8). Probe diameters are in the range of 2 to 3 inches, and the length of the inflatable part of the probe is in the range of 12 to 24 inches. Minimum spacing between consecutive tests (probe center to center) should not be less than 1.5 times the length of the inflatable part of the probe. For pre-bored methods, borehole disturbance in very soft clays and very loose sands can be significant. Production rate for PMT testing is typically 2 to 6 tests per shift.

(1) *Interpretation of pressuremeter test results*. During testing, a pressuremeter modulus and a limit pressure are obtained. Three characteristic segments of a pre-bored PMT pressure-volume change plot are illustrated in Figure 4–9 and include: (1) the re-loading phase; (2) the pseudo-elastic phase; and (3) the plastic phase. After the borehole is drilled and the augers are withdrawn, the borehole walls relax, and the cavity volume is reduced. As the pressuremeter probe is initially inflated, the walls of the borehole are pushed back to their original position. The point at which the volume of the borehole cavity has fully returned to its initial position corresponds to the in situ horizontal stress is given the coordinates, V_o, p_o. The pseudo-elastic phase is similar to the elastic behavior of steel or concrete. The plastic phase begins once the creep pressure has been reached, and which corresponds with coordinates, V_f, P_f. During the plastic phase, the data plots asymptotic to the limit pressure (P_l). The limit pressure is defined as the pressure reached for the infinite expansion of the cylinder.

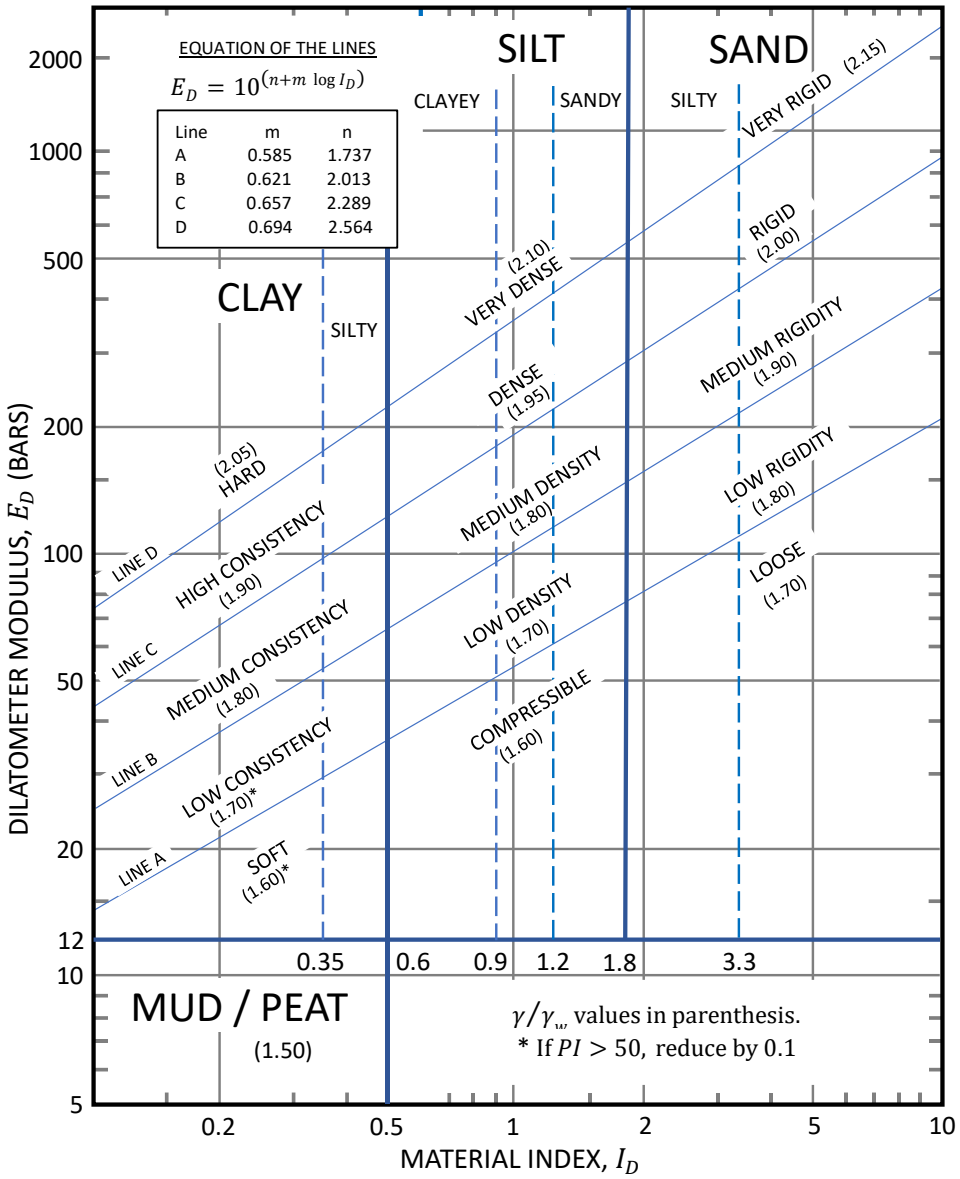


Figure 4–7. Dilatometer test chart for estimating soil description and unit weight (after Marchetti and Crapps 1981); dashed lines represent transition between soil types; note that 1 kg/cm² = 0.981 bar

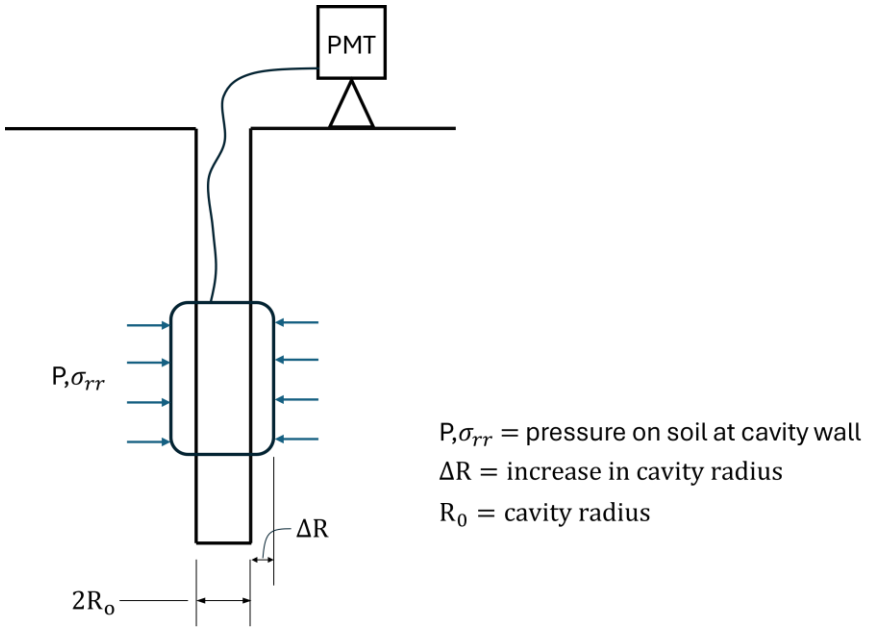


Figure 4–8. Pre-bored pressuremeter test setup (FHWA-IP-89-008)

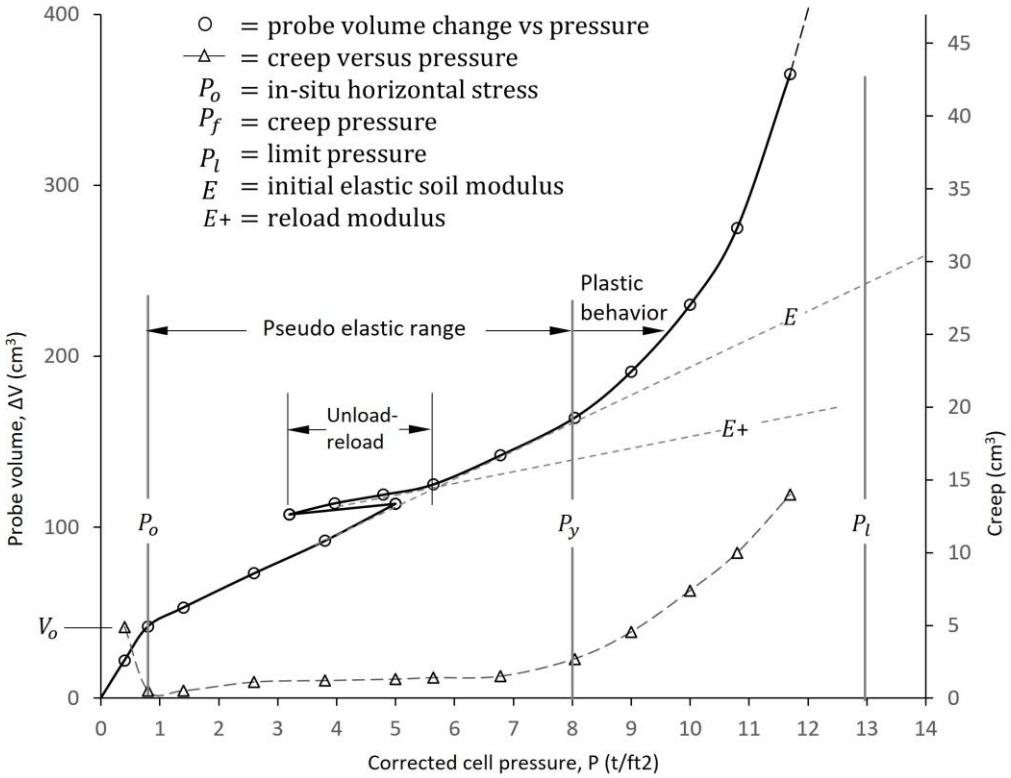


Figure 4–9. Typical pre-bored pressuremeter test plot

(a) Pressuremeter modulus. The pressuremeter modulus, E_o (also denoted E_p) is calculated over the pseudo-elastic range per ASTM D4719 and equation 4–17. Designing beyond the yield pressure (P_y) is not recommended. If it needs to be done, then the modulus used should be a secant modulus related to the line joining the beginning of the straight line in the PMT curve to the point corresponding to the pressure applied in the geotechnical problem. Harder and denser soils will have a higher PMT modulus, having a flatter pressure vs. volume curve. The pressuremeter modulus has been related empirically to the elastic modulus of the soil as $E_0/E = \alpha$, (Ménard 1965), in which α is termed by Ménard as the rheological coefficient and has a value between 0 and 1, reference Table 7–6.

$$\text{Pressuremeter Modulus, } E_o = 2(1+\upsilon)(V_o + V_m)\frac{\Delta P}{\Delta V} \qquad (4\text{–}17)$$

Note. The PMT provides a measure of the horizontal modulus in soils. In clays, the pressuremeter modulus is approximately equal to the undrained elastic modulus (Kulhawy and Mayne 1990). The "undrained" PMT modulus value obtained in fine-grained soils has been compared to case histories and load tests and typically gives the long-term settlement.[1]

where:

E_o = pressuremeter modulus, tsf; based on data reduction (see ASTM D4719)
υ = Poisson's ratio (0.33 recommended for pressuremeter applications)
V_o = volume of the measuring portion of the uninflated probe at 0 volume reading at ground surface, ft^3
V = corrected volume reading of the measuring portion of the probe, ft^3
$V_o + V$ = current volume of inflated probe
ΔP = corrected pressure increase in the center part of the straight-line portion of the pressure-volume curve (tsf), see Figure 4–8
ΔV = corrected volume increase in the center part of the straight-line portion of the pressure-volume curve corresponding to ΔP (ft^3), see Figure 4–8
V_m = corrected volume reading in the center portion of the ΔV volume increase (ft^3)

(b) Initial total horizontal pressure, P_o. For the pre-bored PMT, the pressure at the beginning of the pseudo-elastic phase (P_o) is often interpreted as the in situ horizontal stress (σ_{ho}). As shown in Figure 4–9, the initial expansion of the probe mobilizes a low resistance in seating the membrane to the soil and deforming a disturbed annular zone of the soil. Upon further expansion, the resistance increases, and the curve becomes somewhat flatter. The point where the curve becomes relatively linear can be interpreted as an approximate representation of σ_{ho} as the probe has reapplied the

[1] J.L. Briaud, personal email communication, May 8, 2023.

existing ground stress conditions. The more the borehole is disturbed, the less reliable the interpretation and overprediction of σ_{ho}.

(c) Limit pressure, P_L. The limit pressure is defined as the pressure reached for the infinite expansion of the cylinder (FHWA-IP-89-008). This is the point where no additional pressure is required to continue to apply strain to the soil (see Figure 4–9). The conventional limit pressure (P_L) is defined as the pressure where the probe volume reaches twice the original soil volume cavity volume, defined as $2V_o + V_i$, where V_i is the corrected volume reading at the pressure where the probe contacted the borehole. The limit pressure is usually not obtained by direct measurement due to limitation of probe expansion or excessively high pressure. If the volumetric increase at the end of the test is less than twice the cavity volume, extrapolation can be used to calculate P_L. Common values of the pressuremeter modulus and net limit pressure are summarized in Table 4–10.

(d) Net limit pressure, P_L^*. The net limit pressure is used for calculating bearing capacity and can be calculated using equation 4–18. Although the initial horizontal pressure (P_o) should equal the pressure corresponding to the volume required to inflate the pressuremeter and push soil to its original position (v_o), it is difficult to accurately establish P_o from the test data due to disturbance of the borehole walls and a lack of points at the beginning of the test.

$$Net\ Limit\ Pressure,\ P_L^* = P_L - P_o \tag{4–18}$$

where:

P_L^* = net limit pressure (tsf)
$\mathrm{P_L}$ = limit pressure (tsf)
P_o = initial total horizontal pressure in the ground (tsf) = $[(\gamma_m - u)z]K_o + u$
γ_m = moist unit weight of soil above test interval (tsf)
u = pore water pressure at test interval (tsf)
z = depth to middle of test probe below ground surface (ft)
K_o = at-rest lateral earth pressure coefficient

(e) Yield pressure, P_y. The yield pressure (P_y), also referred to as creep pressure (P_f), is found at the end of the straight line corresponding to the PMT modulus and is comparable to the preconsolidation pressure. Typically, P_y is $0.5P_L$ for clays and $0.33P_L$ for sands (Briaud 2013). At working loads, it is advisable to keep the pressure under these limits to avoid creep deformations. Designing beyond the yield pressure is not recommended.

(f) Soil type based on pressuremeter data. Soil type can be classified from soil samples obtained during drilling. Soil type can also be characterized by observing the soil cuttings and the general shape of the corrected pressuremeter curve as illustrated in Figure 4–10, and the ratio of the pressuremeter modulus (E_0), to the net limit pressure ($\mathrm{P_L^*}$), per equations 4–19 and 4–20.

$$\text{Clays: } (E_0/P_L^*) > 12 \quad (4\text{–}19)$$

$$\text{Sands: } 7 < (E_0/P_L^*) < 12 \quad (4\text{–}20)$$

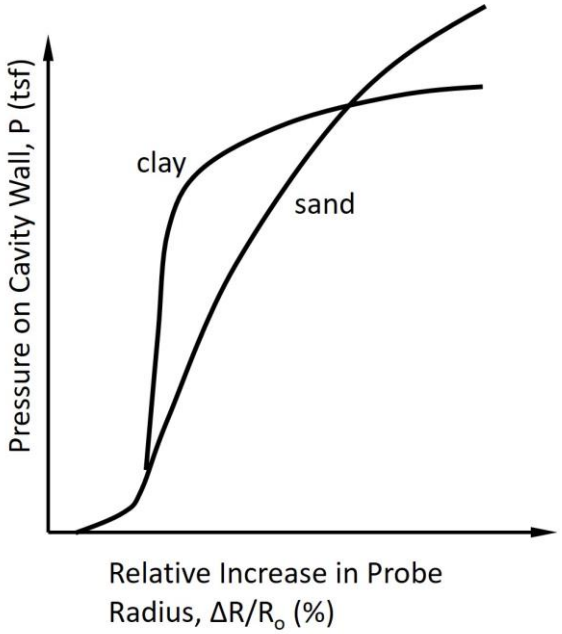


Figure 4–10. General pre-bored pressuremeter test curves for clay and sand (Briaud 1992)

Table 4–10
Common values of net limit pressure (P_L^*) and pressuremeter modulus (E_0^*) (Briaud 1992)

Clay	Soft	Medium	Stiff	Very Stiff	Hard
P_L^* (tsf)	0 to 2	2 to 4	4 to 8	8 to 16	> 16
E_0 (tsf)	0 to 25	25 to 50	50 to 120	120 to 150	> 250

Sand	Loose	Compact	Dense	Very Dense
P_L^* (tsf)	0 to 5	5 to 15	15 to 25	> 25
E_0 (tsf)	0 to 35	35 to 120	120 to 225	> 225

(2) *Pressuremeter holding test*. The pressuremeter holding test can be used to determine creep and consolidation characteristics of clays. During a PMT holding test, either the pressure or strain is held constant over a period during which pore water dissipation occurs. If the total pressure is held constant, it will expand (as the cavity diameter increases) and the excess pore pressures dissipate. Alternatively, the maximum strain (~10 percent) can be held constant, in what is known as the strain hold test (SHT) (Clarke et al. 1979). Holding the expansion of the probe constant is similar to a piezocone dissipation test and can be analyzed in the same manner. During the strain hold test, relaxation of the soil is observed by the decrease of measured excess pore pressure and total cavity pressure. An example holding test is shown in Figure 4–11.

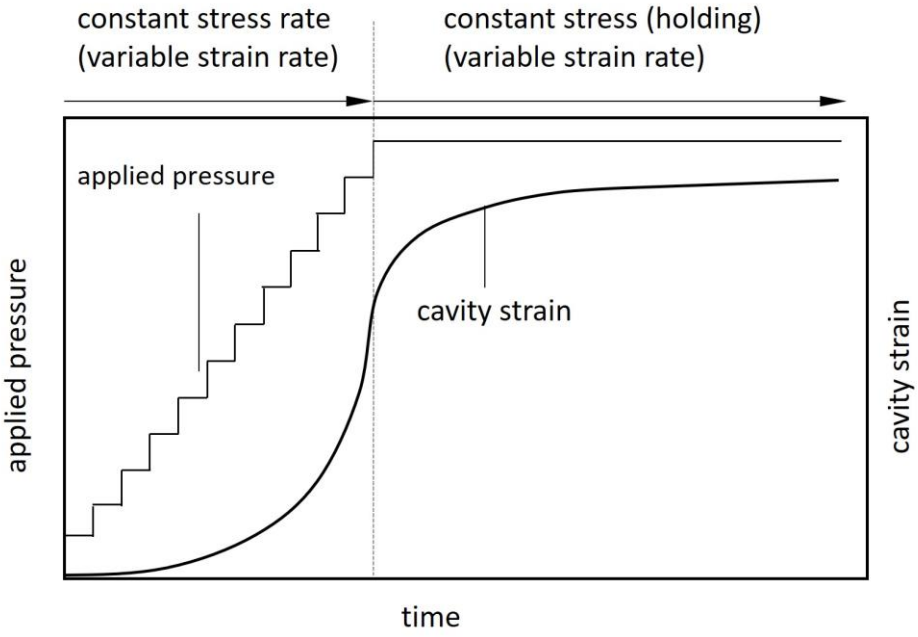


Figure 4–11. Constant stress holding test (Clarke 1994)

e. Field vane shear test.

(1) *Application*. Field vane shear testing (FVST) (ASTM D2573) is performed to evaluate the in situ undrained shear strength of fine-grained, cohesive soil. The apparatus consists of four blades on the end of a rod as shown in Figure 4–12 and Figure 4–13. VST is well suited for very soft to firm clays; however, is not applicable in sandy soils or non-plastic silts as these can allow drainage during testing. The presence of gravels can cause the results to be too high, and therefore, should be verified using other means (such as SHANSEP comparisons or other in situ or laboratory testing methods).

(2) *Companion tests*. The test is often performed in conjunction with SPT borings (or in companion borings to SPT borings), CPTs (such as to validate the cone factors), DMTs, and PMTs. Tests intervals should be assigned once the stratigraphy is characterized. The VST may be performed either at the bottom of a prebored hole or, in soft clays, by merely pushing the vane rods to the desired test depth. The latter requires a correction for rod friction.

f. Test pits and trenches. Test pits and trench excavations can allow visual inspection of subsurface conditions and can be advantageous in some cases (such as shallow testing/sampling, delineation of uncontrolled fills, colluvium, and shallow rock). Excavators can typically be used to excavate test pits or trenches to depths up to about 10 to 20 ft; however, dewatering could be required. In some cases, exploratory trench excavations can be used in fault evaluation studies. An extension of a rock fault into much younger overburden materials exposed by trenching is usually considered proof

of recent fault activity. Consult Occupational Health and Safety Administration (OSHA) guidelines on trenching/shoring.

g. Geophysical methods. Geophysical methods may be used to characterize subsurface conditions between borings. Common geophysical methods for use in shallow foundation design include measurement or estimation of shear wave (S-wave) velocity, seismic refraction (P-wave), electrical resistivity, and ground penetrating radar. Geophysical data should be used in conjunction with boring/penetration tests and interpreted by qualified experienced personnel. Geophysical methods are further discussed in EM 1110-1-1804.



Figure 4–12. Field vanes of various sizes (left) and manual penetration vane shear test rig (right)

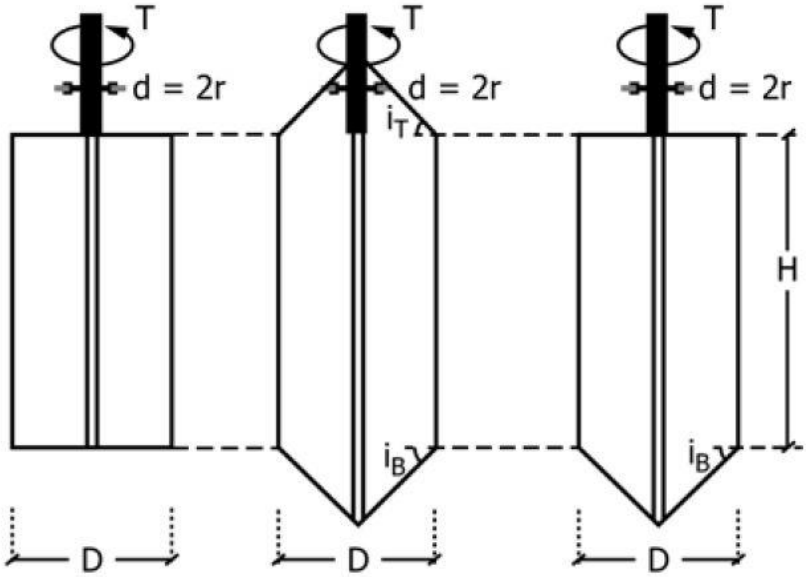


Figure 4–13. Geometry of field vanes (ASTM D2573)

4–11. Laboratory testing

Laboratory testing should be performed to establish engineering characteristics. Guidance for recommended testing procedures for making determinations of the soil properties to be used in design is provided in EM 1110-2-1906. Laboratory testing is generally necessary to verify visual classification of soil samples, for estimating or comparing strength and settlement parameters with in situ test data, and for determining corrosion characteristics (such as sulfate, sulfides). Depending on the subsurface conditions encountered, laboratory testing could range from simple laboratory index tests to performance tests (such as those used to characterize soil strength and compressibility). Laboratories performing geotechnical testing must be validated to comply with ER 11101-8100.

4–12. Report documentation

Exploration data should be provided in the geotechnical or site characterization report.

Chapter 5
Ultimate Limit State for Static Analysis

5–1. Introduction

This chapter presents calculation procedures for evaluating the ultimate limit or "strength state" for static analysis (especially when design loads are assumed to be static). Failure modes associated with the ultimate limit state include: (1) overturning moment and eccentricity, (2) bearing capacity, (3) sliding, (4) uplift, and (5) global stability (see EM 1110-2-1902).

5–2. Loads and pressure

a. Shallow foundations are influenced by a combination of vertical (V), horizontal (H), and moment (M) loads. The weight of the footing must be included in assessment of the loads. It is common to separate external vertical loads on a footing (Q) from the weight of the foundation (W_f). Many aspects of this chapter revolve around the total vertical load, expressed in equation 5–1.

$$V = Q + W_f \tag{5–1}$$

where:

V = downward vertical load induced by the foundation on the ground (units of kips and kips/ft for isolated spread and continuous footing loads, respectively)

Q = vertical column loading, Q is positive in compression and negative in uplift (units of kips and kips/ft for isolated spread and continuous footing loads, respectively)

W_f = weight of foundation, including weight of soil above foundation bearing elevation (units of kips and kips/ft for isolated spread and continuous footings, respectively)

b. The gross bearing pressure acting on the soil (q) is expressed differently for isolated and continuous footings. A continuous footing is based on loads per unit length out of plane, therefore, only the width of the footing is used to calculate pressure. For bearing capacity and settlement calculations, gross (q) and gross equivalent (q_{eq}) bearing pressure are calculated in a similar manner. The distinction is when loads are eccentric, the equivalent area (A') and equivalent width (B') are used to determine the gross "equivalent" bearing pressure (q_{eq}). Reference paragraph 5–4 for equivalent footing parameters for eccentrically loaded foundations. Equations 5–2 and 5–3 are presented as gross equivalent bearing pressure, however, $A = A'$, $B = B'$, and $q = q_{eq}$, when eccentricity is zero.

$$q_{eq} = \frac{Q + W_f}{A'} - u_d = \frac{V}{A'} - u_d = \frac{V}{B' \cdot L'} - u_d \tag{5–2}$$

Isolated Spread Footing

$$q_{eq} = \frac{Q + W_f}{B'} - u_d = \frac{V}{B'} - u_d \qquad \text{(5–3)}$$

Continuous Footing

where:

Q = vertical column loading (units of kips and kips/ft for isolated spread and continuous footing loads, respectively)

V = vertical load induced by the foundation on the ground (units of kips and kips/ft for isolated spread and continuous footing loads, respectively)

W_f = weight of foundation, including weight of soil above foundation bearing elevation (units of kips and kips/ft for isolated spread and continuous footings, respectively)

q_{eq} = gross equivalent bearing pressure (ksf); denoted as gross bearing pressure, q if loading is not eccentric

u_d = pore water pressure at base of foundation to account for uplift pressures; u_D = 0 if groundwater table is at or deeper than the base of foundation (ksf)

B' = effective base width of footing (ft); substitute B if loading is not eccentric

L' = effective base length of rectangular footing (ft); substitute L if loading is not eccentric

A' = effective area of circular type or rectangular footing (ft^2); substitute A if loading is not eccentric

5–3. Soil properties

Soil properties should be evaluated through assessment of design shear strength and unit weight profiles within the shear zone. The shear zone is the soil next to and below the footing within which shear strains are concentrated when an ultimate limit state is reached. The shear zone for each failure mode is discussed in the relevant sections of this chapter. Soil parameters used in analyses may vary and will depend on the limit state and soil conditions. Reference Table 5–1 for information on soil strength relevant to analyses.

Table 5–1
Shear strength for static design conditions

	Drainage Condition	**Drainage Condition**
Failure Mode	Free-draining soils (such as sand, silty sand, non-plastic silt)	Low permeability soils (such as clayey soil, plastic silt)
Bearing capacity	Use drained shear strengths related to effective stresses	Use minimum of (i) undrained strength ($\phi = 0$) related to total stresses; and (ii) drained soil shear strengths related to effective stresses
Sliding	Use drained-soil shear strengths related to effective stresses	Use minimum of (i) undrained strength ($\phi = 0$) related to total stresses; and (ii) drained soil shear strengths related to effective stresses
Overturning	NA – based on eccentricity	NA – based on eccentricity
Uplift	Use drained-soil shear strengths related to effective stresses	Use drained soil shear strengths related to effective stresses
Global stability	See EM 1110-2-1902	See EM 1110-2-1902

5–4. Overturning and eccentricity

Shallow foundation overturning is checked by calculating eccentricity. Overturning is illustrated in Figure 5–1. Limiting eccentricity leads to the full width of the footing in compression against the supporting soil. Furthermore, the eccentricity from an overturning assessment is used in subsequent bearing capacity analyses. The overturning and eccentricity ultimate limit state are, therefore, discussed first. A shallow foundation experiences moment loading when a vertical load is not concentric to a footing. Moment loading also occurs when a horizonal load is applied at some height above the point of rotation of a foundation.

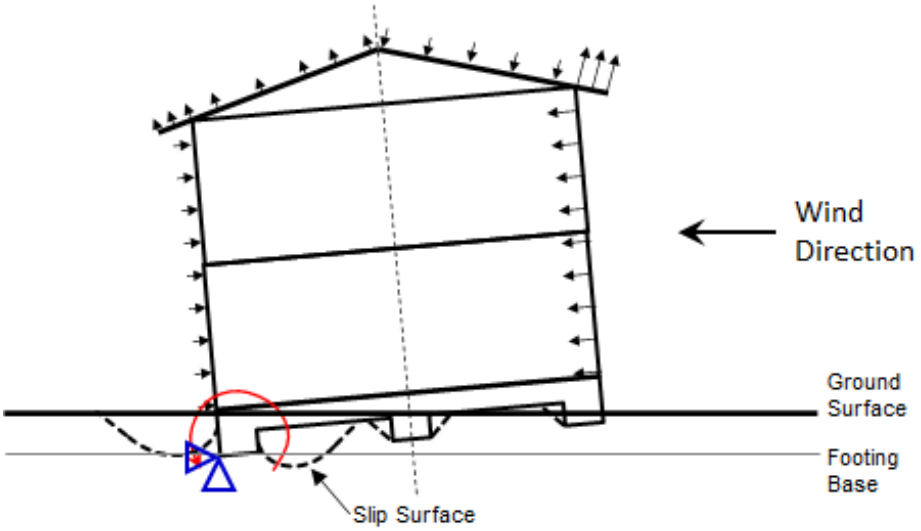


Figure 5–1. Overturning moment ultimate limit state for structure resting on three connected, continuous footings, assuming the corner is pinned; slip surfaces indicating soil separation and failure

a. Effective footing dimensions. A moment load is assessed by converting horizontal forces causing a moment about an axis of the footing and/or applied moments into an eccentricity. The effect of eccentricity must be analyzed by reducing the footing widths to an equivalent value for bearing capacity and settlement calculations. These values of effective dimensions are used in the calculations of bearing capacity and associated correction factors. Effective footing dimensions are illustrated in Figure 5–2, and are calculated as:

$$B' = B - 2 \cdot e_B \tag{5–4}$$

$$e_B = \frac{M_B}{V} \tag{5–5}$$

$$L' = L - 2 \cdot e_L \tag{5–6}$$

$$e_L = \frac{M_L}{V} \tag{5–7}$$

$$A' = B' \cdot L' \tag{5–8}$$

where:

B = base width of footing (ft)
B' = effective base width of footing (ft)
e_B = eccentricity away from the center line of the footing width (ft)
M_B = moment with respect to e_B (kip-ft)

V = vertical load induced by the foundation on the ground (units of kips and kips/ft for isolated spread and continuous footing loads, respectively)
L = base length of rectangular footing (ft)
L' = effective base length of rectangular footing (ft)
e_L = eccentricity away from the center line of the footing length (ft)
M_L = moment with respect to e_L (kip-ft)
A' = effective area of circular type or rectangular footing (ft^2)

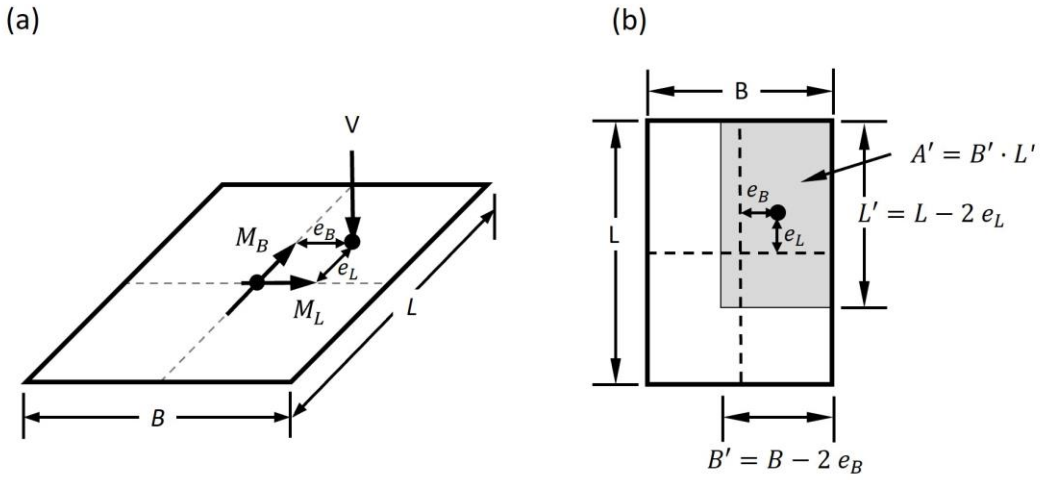


Figure 5–2. Eccentric load applied to a footing within (a) 3-D reference frame; (b) plan view to illustrate e_L and e_B

b. Effective dimensions of approximately circular footings. For circular and approximately circular footings (such as octagon), the effective area parameters are based on an ellipse, which is then converted to an equivalent rectangle. Effective footing dimensions for approximately circular footings are illustrated in Figure 5–3. The first step is to calculate the radius of an equivalent circle with the same area as the footing. From that, the effective area, length of the major axis, and length of the minor axis of the ellipse can be calculated. The ellipse is then converted to an equivalent rectangle for determination of correction factors for bearing capacity calculations.

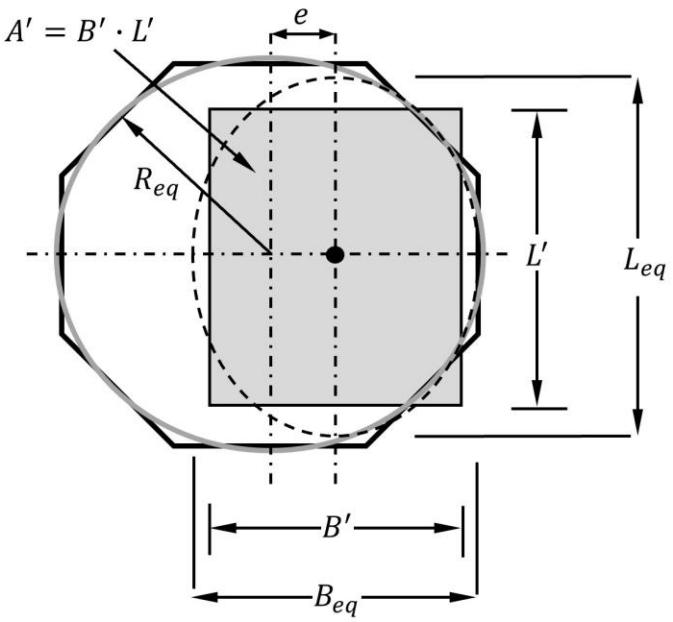


Figure 5–3. Effective area parameters for approximately circular footings

$$R_{eq} = \sqrt{\frac{A}{\pi}} \tag{5–9}$$

$$A' = 2\left[{R_{eq}}^2 cos^{-1}\left(\frac{e}{R_{eq}}\right) - e\sqrt{{R_{eq}}^2 - e^2}\right] \tag{5–10}$$

$$B_{eq} = 2\left(R_{eq} - e\right) \tag{5–11}$$

$$L_{eq} = 2R_{eq}\sqrt{1 - \left(1 - \frac{B_{eq}}{2R_{eq}}\right)^2} \tag{5–12}$$

$$L' = \sqrt{A'\left(\frac{L_{eq}}{B_{eq}}\right)} \tag{5–13}$$

$$B' = L'\left(\frac{B_{eq}}{L_{eq}}\right) \tag{5–14}$$

where:

π = ratio of circumference to diameter for a circle ≈ 3.14 (dimensionless)
A = base area of a footing (ft^2)
R_{eq} = equivalent radius of a circular footing (ft)
e = eccentricity (ft)

A' = effective area of circular type or rectangular footing (ft^2)
B_{eq} = equivalent width of an elliptical footing with A' (ft)
L_{eq} = equivalent length of an elliptical footing with A' (ft)
L' = effective base length of rectangular footing (ft)
B' = effective base width of footing (ft)

c. Overturning and eccentricity requirements. The overturning ULS is governed by eccentricity. When the eccentricity is in the middle 1/3 of the footing, the entire base of the footing will transmit compressive loads to the soil. Per Table 2–1, the eccentricity must be within the middle third of the footing, less than $B/6$ away from the centerline, to meet overturning requirements.

5–5. Ultimate bearing capacity

Ultimate bearing capacity is the normal load that soils near a foundation can resist. Bearing capacity failures are indicated by excessive settlement or rotation of the foundation. The ultimate pressure to result in a bearing capacity limit state is presented in equation 5–15 for an isolated spread footing, and equation 5–16 for continuous footings. This pressure is known as the ultimate bearing capacity and is the same as the ultimate gross bearing capacity. The ultimate bearing capacity must be used when calculating safety factors.

$$q_{ult} = q_{ult,gross} = q_u = q_l = q_{bL} = V_{ult}/A' \quad \text{(5–15)}$$
Isolated Spread Footing

$$q_{ult} = q_{ult,gross} = q_u = q_l = q_{bL} = V_{ult}/B' \quad \text{(5–16)}$$
Continuous Footing

where:

V_{ult} = ultimate vertical force, including foundation weight, for circular or rectangular footings (kips) or force/unit length for continuous footings (kips/ft)
q_{ult} = ultimate bearing capacity (ksf), sometimes denoted as $q_{ult,gross}$, q_u, q_l, q_{bL}
A' = effective area of circular type or rectangular footings (ft^2)
B' = effective base width of footing (ft)

a. Shear zone for bearing capacity. The shear zone for general shear bearing capacity is illustrated in Figure 5–4 and extends to a depth below the footing. The design strength should be based on materials within zones I, II, and III. The effective unit weight of the soil in Zones I, II, and III is applicable to the unit weight term in the bearing capacity equation. The unit weight of the soil in Zone IV is applicable to the surcharge term in the bearing capacity equation. The opportunity for encountering layers with different soil properties in the shear zone is more likely as foundation width or diameter increases. As eccentricity increases, the size of the shear zone decreases, as shown in Figure 5–4 and Figure 5–5. Effective width parameters are used to

estimate the size of the shear zone. If footings are spaced closely together, interference could occur and influence the shear zone, reference paragraph 6–6d.

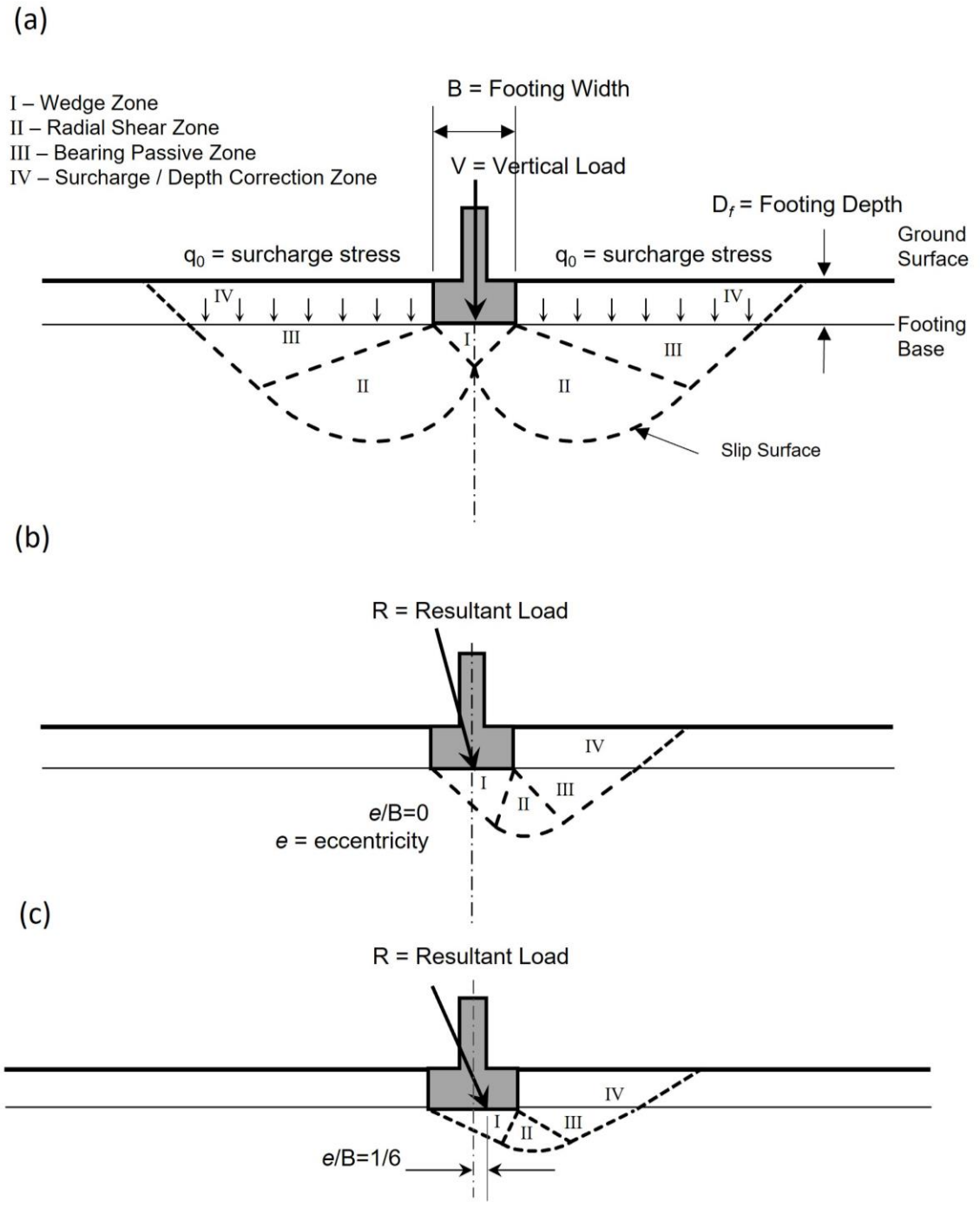


Figure 5–4. Slip surfaces for rough footings indicating size and shape of bearing capacity ultimate limit state shear zones for (a) concentric vertical loading; (b) combined vertical and horizontal loading; (c) combined vertical, horizontal, and moment loading; e = eccentricity; B = footing base width; arrow = loading direction

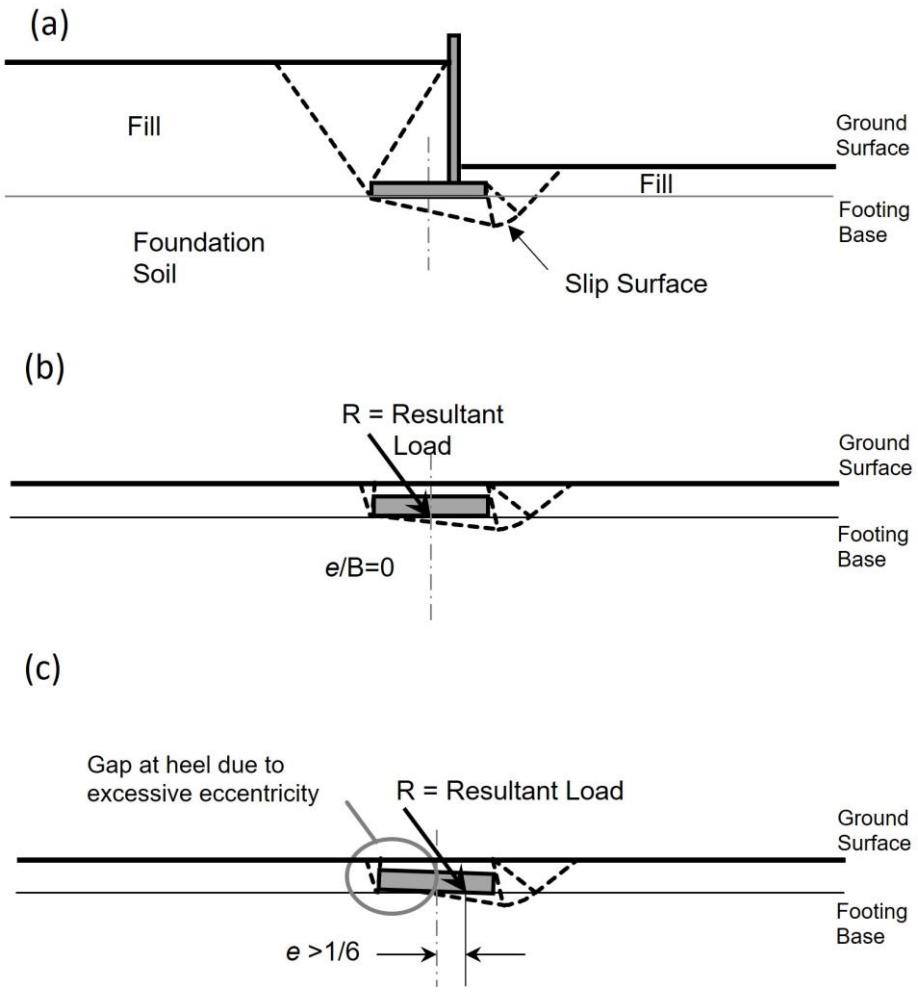


Figure 5–5. Influence of eccentricity on shear zone (a) shear zone for retaining wall failure; (b) shear zone for idealized footing modeled without eccentricity; (c) shear zone for idealized footing with excessive eccentricity (e)

(1) *Depth of shear zone*.

(a) For square and circular footings, the thickness of the shear zone tends to vary from 1/3 of the effective footing base width to twice the effective footing base width. Continuous footings tend to have a shear zone that is deeper, and which could extend up to twice as deep as a square or circular footing, or up to $4B'$ below the footing for continuous footings. For bearing capacity analyses, soil strength is typically assessed between the base of the footing to 1/2 of the effective footing width for circular and square footings, and one effective footing width below the base of continuous footings.

(b) Soil layering can also influence the size of the shear zone. If strong layers are below weak layers, the shear zone is typically smaller and mostly in the weak layer. If weak layers are below strong layers, the shear zone is typically larger. Analyzing bearing capacity based on the weakest layer in the shear zone is often sufficient for analysis. The lowest median, or geometric mean, of strength between the base of the

footing to a range between $(1/3)B'$ to $2B'$ for circular and square footings, or $(2/3)B'$ to $4B'$ for continuous footings can be assessed and applied in a more detailed analysis.

(2) *Punching failure for strong layer over weak clay*. Punching failures can occur where foundations are placed on strong soils overlying weaker soils. If there is a weaker layer below a strong layer between 0.5 B and $2B$ of the bottom of square or circular footings, or within $1B$ to $4B$ below the bottom of continuous footings, a punch-through analysis should be performed. The bearing capacity can be assessed by taking the minimum of the bearing capacity in the strong layer or the bearing capacity in a weak layer increased by a load spreading factor. The load spreading factor is the area of an equivalent footing at the top of the underlying layer when assuming a 2V:1H load spreading factor. Terms are illustrated in Figure 5–6.

$$q_{ult} = \min\left(q_{ult,1}, \frac{(B' + H_s) \cdot (L' + H_s)}{B \cdot L} q_{ult,2}\right) \quad \text{(5–17)}$$

Isolated Spread Footing

$$q_{ult} = \min\left(q_{ult,1}, \frac{B' + H_s}{B'} q_{ult,2}\right) \quad \text{(5–18)}$$

Continuous Footing

where:

q_{ult} = the ultimate bearing capacity (ksf), due to punching failure
B' = effective base width of footing (ft)
L' = effective base length of rectangular footing (ft)
H_s = height of a strong surface layer (Layer 1) (ft)
$q_{ult,1}$ = the ultimate bearing capacity using strength of the strong surface layer (ksf)
$q_{ult,2}$ = the ultimate bearing capacity using strength of the weak underlying layer (ksf)

(3) *Local and punching failures in homogeneous sand*. Punching failures can occur for strong soils overlying weaker soils, as discussed in paragraph 5–5a(2). For homogeneous sands, local and punching shear have also been historically discussed in relation to loose and compressible sandy soils. Punching failure calculations are not needed for homogeneous sands. Settlement analyses address the potential for a punching failure. Soil strength parameters for loose and compressible soils should be used without reduction.

(4) *Local and punching failures in homogeneous clays*. Local and punching failures are not typically discussed for homogeneous clays. As discussed by Vesic (1973), a local failure is the transition from general shear to punching shear. Punching shear occurs when the failure mechanism is localized close to the footing, similar to that of failure for deep foundation end bearing. Local and punching shear failures in homogeneous clays are accounted for by the depth correction factor in Table 5–4. The maximum depth correction factor of 1.5 relates to a deep failure mechanism. No additional calculations are necessary to account for local and punching failures in homogeneous clays.

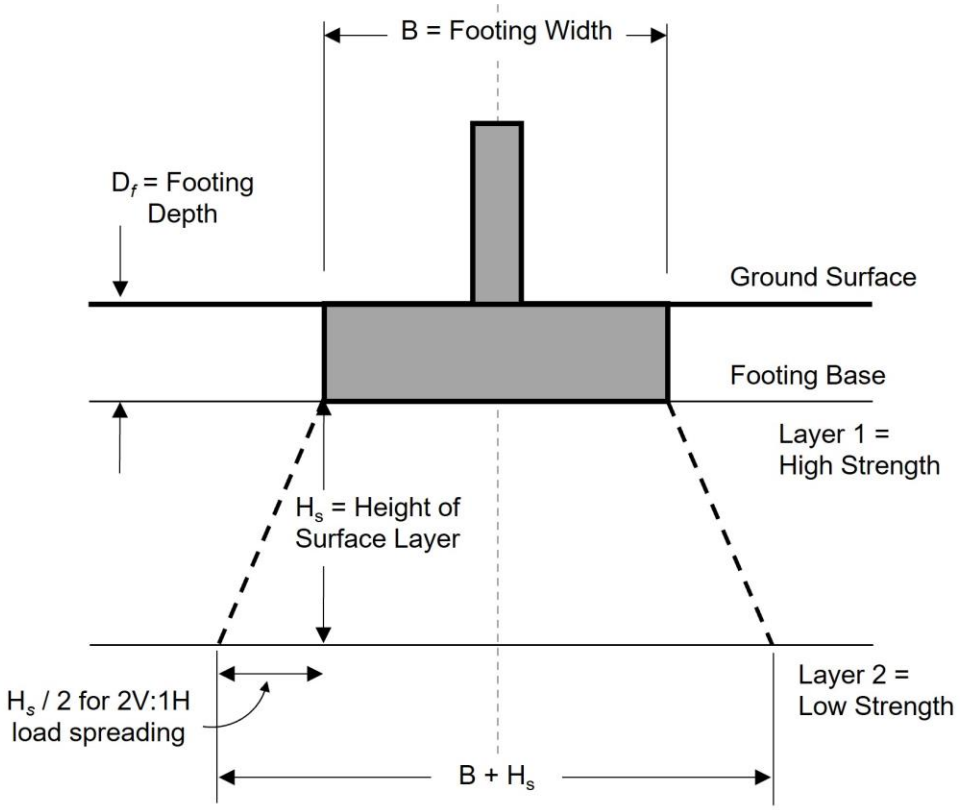


Figure 5–6. Illustration of load spreading for evaluating punching failure for strong over weak layers

(5) *Groundwater in the shear zone*. Soil strength is frictional, and therefore, controlled by effective stresses. The effect of soil self-weight on soil strength in the shear zone needs to account for water pressures. This can be accommodated by adjusting the total unit weight of the soil (γ_t) to an effective unit weight (γ'). If the water table is below the shear zone, then $\gamma' = \gamma_t$. If the water table is at the base of the footing, $\gamma' = \gamma_t - \gamma_w$, where γ_w is the unit weight of water. For intermediate water levels, the relationship in equation 5–19 can be used to assess γ'. To be consistent with method formulation, the Federal Highway Administration (FHWA) (2002) groundwater correction factor included in equation 5–20 should be used with the Association of State Highway and Transportation Officials (AASHTO) LRFD Bridge Design Specifications, 9th Ed.(reference herein as AASHTO 2020) method. For cases where seepage is involved, see EM 1110-2-2502.

$$\gamma' \approx \gamma_t \cdot min\left[0.45 + 0.55\left(\frac{\max\left(D_W - D_f, 0\right)}{B}\right), 1\right] \quad (5\text{–}19)$$

$$\gamma' \approx \gamma_t \cdot C_{W_\gamma} = \gamma_t \cdot min\left[0.5 + 0.5\left(\frac{D_W}{1.5B + D_f}\right), 1\right] \quad (5\text{–}20)$$

(FHWA 2002)

where:

D_W = depth of groundwater table below ground surface (ft)
D_f = depth of base of footing below ground surface (ft)
B = base width of footing (ft)
γ_t = total unit weight of the soil beneath the footing (kcf)
γ' = effective unit weight of the soil beneath the footing (kcf)
C_{W_γ} = dimensionless ground water correction factor

(6) *Groundwater above the shear zone*. Groundwater above the shear zone is accounted for through any uplift pressures on the base of the footing (equation 5–2 and 5–3), as well as through the surcharge term in the bearing capacity equation. Groundwater does not affect the surcharge term for undrained analyses ($q_o = \sigma_{vo}$). However, the pore pressures on the plane at the base of the footing reduce the total stress for the surcharge for a drained analysis ($q_0 = \sigma_{vo} - u_D = \sigma'_{vo}$).

b. Bearing capacity equation. Bearing capacity analysis assumes a general shear failure based on superposition of bearing capacity factors (N_c, N_q, and N_γ) (for example, Terzaghi 1943). Equation 5–21 is the "general equation" for shallow foundation bearing capacity. It is based on the following assumptions that the footing is (1) a rectangular rigid continuous footing (length > 10x width), (2) at ground surface, (3) resting on homogeneous soil, (4) under a centralized vertical load, and (5) level ground. Correction factors and application of the effective width for conditions outside of these assumptions are discussed in paragraph 5–5f.

$$q_{ult} = c \cdot N_c + q_0 \cdot N_q + \frac{1}{2} B \cdot \gamma' \cdot N_\gamma \qquad \text{(5–21)}$$

Continuous Footing

where:

q_{ult} = the ultimate bearing capacity (ksf)
c = cohesion parameter for the foundation soils in the shear zone, use undrained strength (s_u) for undrained analysis (ksf)
N_c = dimensionless bearing capacity factor for cohesion (c), or undrained strength
q_0 = surcharge pressure on the plane at the base of the footing (ksf)
N_q = dimensionless bearing capacity factor for surcharge
γ' = effective unit weight of the soil beneath the footing (kcf); see paragraph 5–5a(5)
N_γ = dimensionless bearing capacity factor for soil effective weight
B = base width of footing (ft)

c. Water pressures and groundwater conditions. Groundwater in the shear zone is accounted for by the effective unit weight term, discussed in paragraph 5–5a(5). Groundwater above the shear zone (N_q term) is accounted for by using the effective stress for q_0 for effective stress analyses with $\phi' > 0$. For total stress analyses based on undrained strength, $c = s_u$, q_0 should be the total stress at the base of the footing.

FHWA (2002) multiplies the total stress at the base of the footing by a correction factor (C_{W_q}), which essentially accounts for differences between total and effective stress. When the water table is above the base of the footing, buoyancy forces on the footing are used to reduce the applied load (equations 5–2 and 5–3).

d. Bearing capacity factors. Bearing capacity factors come from rigorous theoretical analysis of a simplified continuous footing geometry on homogeneous soil. There is general agreement for N_c and N_q. Published values of N_γ differ. For example, the Meyerhof (1963) N_γ equation can differ significantly from the Vesic (1973) N_γ equation. Correction factors are applied to account for differences between the solution assumptions for bearing capacity factors and typical design problems.

e. Application of design methods presented. Corrections factors are presented for the USACE method (Instructional Report K-82-7) and are based primarily on Meyerhof (1963) equations. Additionally, the AASHTO (2020)/FHWA (2002) method is presented, which is based primarily on the Vesic (1973) correction factors. It is recommended to perform analyses with both methods and take the average of results if bearing capacity controls design. When settlement controls design, calculations performed by either of the methods are sufficient. The difference in results indicate a level of uncertainty in factors used.

f. Correction factors. Many design applications do not meet the assumptions behind factors presented for equation 5–21, and correction factors need to be applied. Correction factors are applied for shape, embedment depth, load inclination, base inclination, and ground inclination. Equation 5–22 is the corrected equation for computing bearing capacity. Correction factors for drained analysis are presented in Table 5–2 and Table 5–3. Correction factors for undrained analyses are presented in Table 5–4. Details of correction factors are discussed individually in the following sections.

$$q_{ult} = s_c d_c i_c b_c g_c c N_c + s_q d_q i_q b_q g_q q_0 N_q + s_\gamma d_\gamma i_\gamma b_\gamma g_\gamma \frac{1}{2} B' \gamma' N_\gamma \qquad (5\text{–}22)$$

where:

q_{ult} = the ultimate bearing capacity (ksf)
c = cohesion parameter for the foundation soils in the shear zone, use undrained strength (s_u) for undrained analysis (ksf)
N_c = dimensionless bearing capacity factor for cohesion, c, or undrained strength
q_0 = surcharge pressure on the plane at the base of the footing (ksf), see paragraph 5–5a(6)
N_q = dimensionless bearing capacity factor for surcharge
γ' = effective unit weight of the soil beneath the footing (kcf), see paragraph 5–5a(4)
N_γ = dimensionless bearing capacity factor for soil effective weight
B' = effective base width of footing (ft)

s_c, s_q, s_γ = dimensionless correction factors for footing shape = 1 for continuous footing and increases with B/L for rectangular and circular-type footings

d_c, d_q, d_γ = dimensionless correction factors for footing depth = 1 for footing at the ground surface and increases with normalized embedment below the footing

i_c, i_q, i_γ = dimensionless correction factors for load inclination = 1 for vertical load and < 1 for inclined loads

b_c, b_q, b_γ = dimensionless correction factors for base inclination = 1 for a flat base and < 1 for an angled base

g_c, g_q, g_γ = dimensionless correction factors for ground inclination = 1 for flat ground and < 1 for ground sloping away from the footing

Additional parameters not mentioned in reference to equation 5–22, but in Table 5–2, Table 5–3, and Table 5–4 are:

N = dimensionless Flow Number = principal stress ratio, see Table 5–2

ϕ' = effective stress friction angle for the foundation soils in the shear zone (degrees)

e = base of the natural log, or Euler number ≈ 2.718 (dimensionless)

π = ratio of circumference to diameter for a circle ≈ 3.14 (dimensionless)

L' = effective base length of rectangular footing (ft)

D_f = depth to base of footing below ground surface (ft), see Figure 5–8

β = load inclination from vertical, (degrees), see Figure 5–9

V = vertical load induced by the foundation on the ground (units of kips and kips/ft for isolated spread and continuous footing loads, respectively)

H = horizontal component of the applied load on the foundation (kips)

θ = projected direction of load in the plane of the footing, measured from the side of length, L (degrees), see Figure 5–9

a_b = base inclination (degrees), see Figure 5–10

a_g = ground inclination (degrees), see Figure 5–11

d_g = distance from edge of footing to edge of sloping ground (ft)

Table 5–2
Drained bearing capacity and correction factors based on Flow Number N

Bearing and Correction Factors	USACE (Meyerhof Based)	AASHTO (2020)/FHWA (2002) (Vesic Based)
Continuous Footing Bearing		
N	$N = tan^2(45^o + \phi'/2) = \frac{1 + sin\phi'}{1 - sin\phi'}$	$N = tan^2(45^o + \phi'/2) = \frac{1 + sin\phi'}{1 - sin\phi'}$
N_q	$N_q = N \cdot e^{\pi \cdot tan\phi'}$	$N_q = N \cdot e^{\pi \cdot tan\phi'}$
N_c	$N_c = (N_q - 1) \cdot cot\phi'$	$N_c = (N_q - 1) \cdot cot\phi'$
N_γ	$N_\gamma = (N_q - 1) \cdot tan(1.4\phi')$	$N_\gamma = 2(N_q + 1) \cdot tan(\phi')$
Shape		
s_q	$s_q = 1 + 0.1N\frac{B'}{L'}$	$s_q = 1 + \left(\frac{B'}{L'} tan\phi\right)$
s_c	$s_c = 1 + 0.2N\frac{B'}{L'}$	$s_q = 1 + \left(\frac{B'}{L'}\right)\left(\frac{N_q}{N_c}\right)$
s_γ	$s_\gamma = 1 + 0.1N\frac{B'}{L'}$	$s_\gamma = 1 - 0.4\left(\frac{B'}{L'}\right)$
Depth[a]		
d_q	$d_q = 1 + 0.1\sqrt{N}\frac{D_f}{B'}$	$d_q = 1 + 2tan\,\phi'\,(1 - sin\phi')^2 atan\frac{D_f}{B'}$
d_c	$d_c = 1 + 0.2\sqrt{N}\frac{D_f}{B'}$	1
d_γ	$d_\gamma = 1 + 0.1\sqrt{N}\frac{D_f}{B'}$	1

Note:

[a] Depth factor, d_q for AASHTO (2020) ≤ 1.4, $atan(D/B')$ in radians, FHWA (2002) presents tabulated values.

Table 5–3
Drained bearing capacity inclination correction factors

Correction Factors	USACE (Meyerhof Based)	AASHTO (2020)/FHWA (2002) (Vesic Based)
Load Inclination		
β	$\beta = atan\left(\frac{H}{V}\right)$	$\beta = atan\left(\frac{H}{V + B'L' \cdot c \cdot cot\phi'}\right)$
n	Not Applicable	$n_B = \frac{2 + L'/B'}{1 + L'/B'}$ $n_L = \frac{2 + B'/L'}{1 + B'/L'}$ $n = n_L cos^2\theta + n_B sin^2\theta$
i_q	$i_q = \left[1 - \frac{\beta}{90^o}\right]^2$	$i_q = [1 - tan\beta]^n$
i_c	$i_c = i_q$	$i_c = i_q - \frac{1 - i_q}{N_q - 1}$
i_γ	$i_\gamma = \left[1 - \frac{\beta}{\phi'}\right]^2$	$i_q = [1 - tan\beta]^{n+1}$
Base Inclination[a]		
b_q	$b_q = \left[1 - \frac{\alpha_b}{57^o} \cdot tan\phi'\right]^2$	$b_q = \left[1 - \frac{\alpha_b}{57^o} \cdot tan\phi'\right]^2$
b_c	$b_c = b_q - \left[\left(1 - b_q\right)/(N_c \cdot tan\phi')\right]$	$b_c = b_q - \left[\left(1 - b_q\right)/(N_c \cdot tan\phi')\right]$
b_γ	$b_\gamma = \left[1 - \frac{\alpha_b}{57^o} \cdot tan\phi'\right]^2$	$b_\gamma = \left[1 - \frac{\alpha_b}{57^o} \cdot tan\phi'\right]^2$
Ground Inclination[b]		
$g_{q,0}$	$g_{q,0} = \left[1 - tan\alpha_g\right]^2$	Not Applicable
g_q	$g_q = min\left[1, g_{q,0} + \left(1 - g_{q,0}\right)\frac{d_g}{5B'}\right]$	See tables in AASHTO (2020)
$g_{c,0}$	$g_{c,0} = g_{q,0} - \left[\left(1 - g_{q,0}\right)/(N_c \cdot tan\phi')\right]$	Not Applicable
g_c	$g_c = min\left[1, g_{c,0} + \left(1 - g_{c,0}\right)\frac{d_g}{5B'}\right]$	See tables in AASHTO (2020)
$g_{\gamma,0}$	$g_{\gamma,0} = \left[1 - tan\alpha_g\right]^2$	Not Applicable
g_γ	$g_\gamma = min\left[1, g_{\gamma,0} + \left(1 - g_{\gamma,0}\right)\frac{d_g}{5B'}\right]$	See tables in AASHTO (2020)

Note:

[a] α_b is the base inclination from horizontal in degrees, base inclination is not presented in AASHTO (2020), but values are presented in FHWA (2002).

[b] α_g is the ground inclination from horizontal in degrees, subscript "0" indicates for footing directly adjacent to slope with FS > 1.3. Increases linearly from $g_x = g_{x,o}$ to a value of $g_x = 1$ at ratio of distance (d) from slope to effective footing (B') width of 5.

Table 5–4
Undrained ($\phi' = 0$) bearing capacity and correction factors

Bearing and Correction Factors	USACE (Meyerhof Based)	AASHTO (2020)/FHWA (2002) (Vesic Based)
Continuous Footing Bearing		
N_c	$N_c = (2 + \pi) = 5.14$	$N_c = (2 + \pi) = 5.14$
Shape[a]		
s_c	$s_c = 1 + 0.2\frac{B'}{L'}$	$s_c = 1 + 0.2\frac{B'}{L'}$
Depth[a,b]		
d_c	$d_c = 1 + 0.2\frac{D_f}{B'} \leq 1.5$	$d_c = 1$[b]
Load Inclination[a]		
n	N/A	$n_B = \frac{2 + L'/B'}{1 + L'/B'}$ $n_L = \frac{2 + B'/L'}{1 + B'/L'}$ $n = n_L cos^2\theta + n_B sin^2\theta$
i_c	$i_c = 1 - 1.3\frac{H}{V}$	$i_c = 1 - \frac{n \cdot H}{cB'L'N_c}$
Base Inclination[a,c]		
b_c	$b_c = 1 - \frac{\alpha_b}{147.3^o}$	$b_c = 1 - \frac{\alpha_b}{147.3^o}$
Ground Inclination[a,d]		
$g_{c,0}$	$g_{c,0} = 1 - \frac{\alpha_g}{147.3^o}$	Not Applicable
g_c	$g_c = min\left[1, g_{c,0} + \left(1 - g_{c,0}\right)\frac{d_g}{2B'}\right]$	See tables in AASHTO (2020)

Notes:

[a] Undrained corrections factors for *q* (specifically, s_q, d_q, i_q, b_q, g_q) are equal to 1 for both methods.

[b] FHWA recommends the Brinch Hansen (1970) depth factors for sand; however, does not present depth factors for ϕ' = 0 and states that "the depth correction factors can be conservatively omitted."

[c] α_b is the base inclination from horizontal in degrees, base inclination not presented in AASHTO (2020), but values presented in FHWA (2002).

[d] α_g is the ground inclination from horizontal in degrees, subscript "$_0$" indicates for footing directly adjacent to slope with FS > 1.3. Increases linearly from $g_c = g_{c,o}$ to a value of g_c = 1 at ratio of distance (d) from slope to effective footing (B') width of 2.

(1) *Shallow foundation shape*. Site and structure requirements could lead to footings with differing shapes. As the length of a footing becomes closer to its width, the additional resistance from shearing in the out-of-plane direction becomes more significant (Figure 5–7). Shallow foundation shape correction factors are presented in Table 5–2 and Table 5–4. Circular footings have a shape factor approximately 2 percent (undrained) to 6 percent (drained) higher than those for a square foundation. Square foundation shape factors should be used in design of circular foundations. The USACE method uses Meyerhof (1963) shape factors. The AASHTO (2020)/FHWA (2002) method uses Vesic (1973) shape factors.

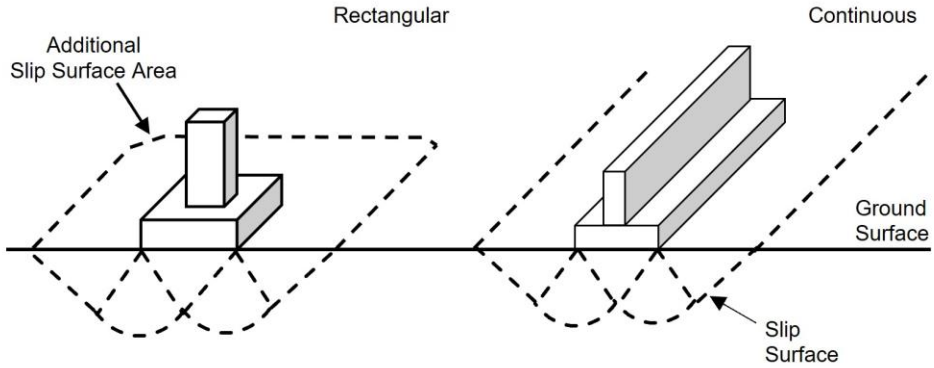


Figure 5–7. Difference in shape of shear zone that results in a shape factor (after Salgado 2008)

(2) *Shallow foundation depth*. When a shallow foundation is embedded below the surface, soil resistance, in addition to the surcharge pressure, increases the shallow foundation bearing capacity. Footings typically have an embedment of at least the thickness of the footing or the frost penetration depth, whichever is greater. Embedment also generally increases horizontal resistance. Shallow foundation depth correction factors are presented in Table 5–2 and Table 5–4. The USACE method uses Meyerhof (1963) depth factors. The AASHTO (2020)/FHWA (2002) method uses Vesic (1973) depth factors for sand. However, the AASHTO (2020)/FHWA (2002) method does not present depth factors for $\phi' = 0$ and states that "the depth correction factors can be conservatively omitted". An illustration of terms is shown in Figure 5–8.

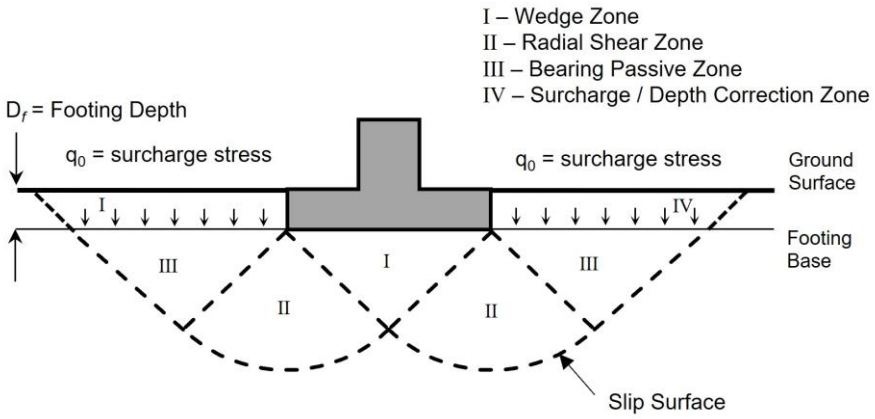


Figure 5–8. Definition parameters used to assess footing embedment

(3) *Load inclination*. When a load has a horizontal component, the ultimate bearing capacity reduces from the case with a purely vertical load. The influence of horizontal loads is analyzed using load inclination factors. Shallow foundation load inclination correction factors are presented in Table 5–3 and Table 5–4. An illustration of terms is shown in Figure 5–9. The USACE method uses Meyerhof (1963) shape factors. The AASHTO (2020)/FHWA (2002) method uses Vesic (1973) shape factors. FHWA (2002) recommends to not include both the inclination and shape factors for the AASHTO method. Including both factors could be overly conservative due to shape factor on N_γ for the AASHTO method, as well as because of depth correction factors. FHWA (2002) recommends to not include the inclination factor for the AASHTO method. Inclination and shape factors should be included for the USACE method.

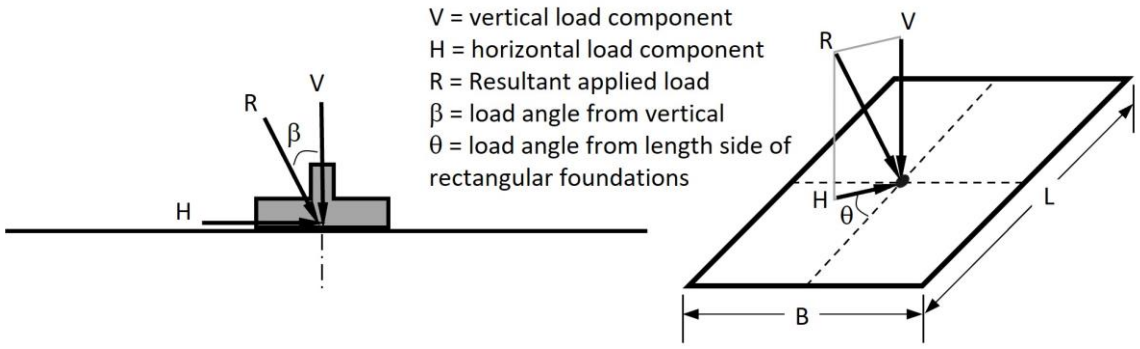


Figure 5–9. Footing with inclined load

(4) *Base inclination*. For constructability purposes, it is preferred to use foundations with flat bases. However, in some cases (such as T-walls) a key may be added for additional lateral stability. That increase in lateral stability can reduce vertical bearing capacity. In these cases, the base of the footing becomes inclined from a calculation perspective. Shallow foundation base inclination correction factors are presented in Table 5–3 and Table 5–4. An illustration of terms is shown in Figure 5–10. Both the USACE and AASHTO/FHWA methods use the Vesic (1973) base inclination factors.

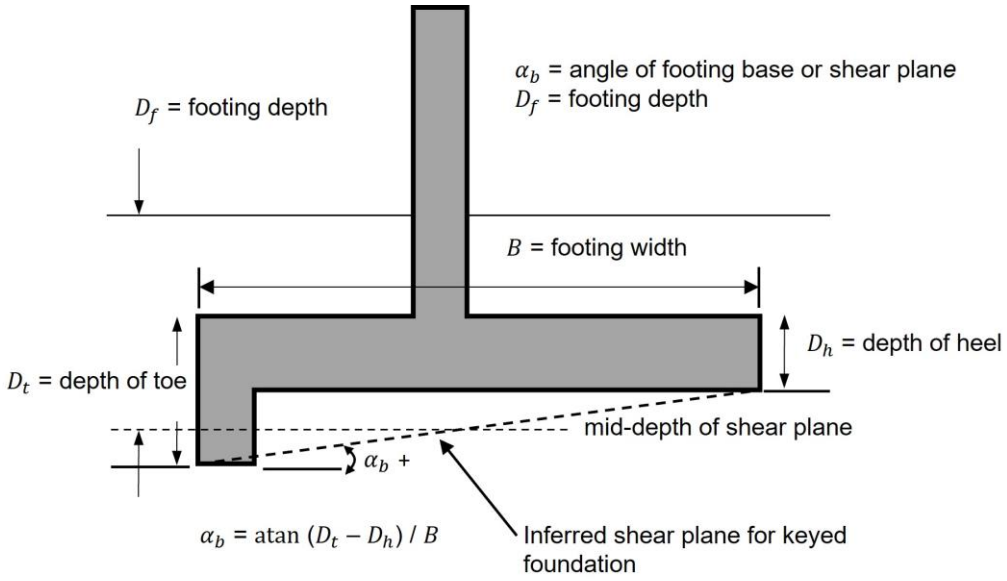


Figure 5–10. Footing with inclined base due to shear key

(5) *Ground inclination*. When a footing is adjacent to a slope, the bearing capacity differs from the level ground case. Sloping ground correction factors are presented in Table 5–3 and Table 5–4. As illustrated in Figure 5–11, when ground slopes down adjacent to a footing, the surcharge stress (q_0) decreases and bearing capacity decreases. The ground inclination factor increases the further the slope is from the footing. The USACE method uses Vesic (1973) ground inclination factors for footings adjacent to slopes. The AASHTO (2020)/FHWA (2002) method uses Leshchinsky (2015) and Leshchinsky and Xie (2017) ground inclination factors. Adjacent slopes must meet global stability criteria in EM 1110-2-1902.

(a) Drained analysis. If the top of the slope is five footing widths or more away from the nearest edge of the foundation, the ground inclination factor will equal 1.0.

(b) Undrained analysis. If the top of the slope is two footing widths or more away from the nearest edge of the foundation, the ground inclination factor will equal 1.0.

(6) *Ring footings*. There are no special required correction factors for ring footings. Ring footings are assumed to act as a circular footing with the pressure averaged over the full circular area based on the outer width of the ring. These assumptions are appropriate when the soil cannot fail upward into the middle of the ring due to concrete failure. Correction factors of circular footings are appropriate for ring footings. Isolated failure of the soil around the ring beam footing can create a non-uniform pressure distribution across a flexible foundation or foundation pad. This is known as dishing (D'Orazio and Duncan 1987). Dishing can progress to foundation failure due to edge shear of the slab or foundation pad within the ring foundation (Duncan and D'Orazio 1984). Techniques for assessing differential settlements and structural resistance of material within the ring are similar to those discussed for mat foundations in Chapter 8.

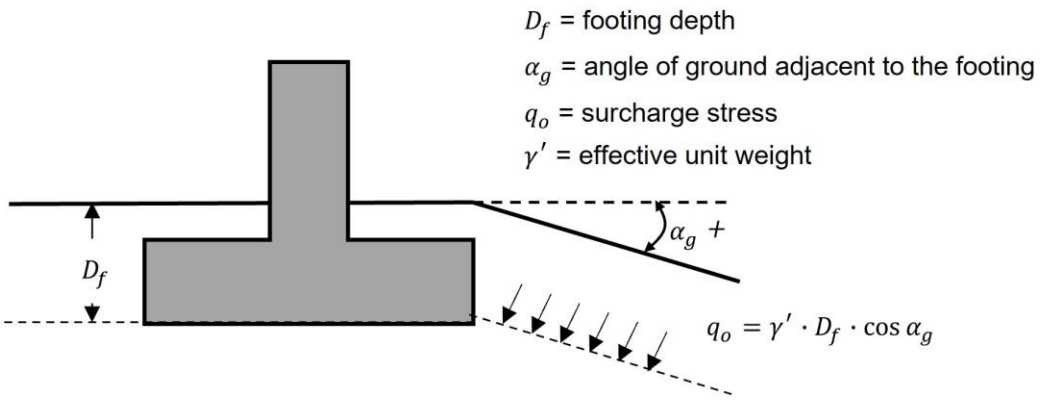


Figure 5–11. Footing adjacent to sloping ground

g. Groundwater and hydrostatic uplift. If the slip surface is located partially or fully below the water table, the effective stress of the soil is lower than if water is not present. Lower effective stress reduces strength and bearing capacity through surcharge stress (q_o) for embedded footings for the N_q term and through effective unit weight, γ' for the N_γ term for drained analysis. Groundwater correction factors and the effect of groundwater on surcharge stress (q_o) are discussed in paragraph 5–5a(5). Hydrostatic uplift acts as a reduction in footing stress through equations 5–2 and 5–3.

h. Safety factor for bearing capacity. The calculated factor of safety against a bearing capacity failure is defined in equation 5–23 and must meet the requirements in Table 2–1.

$$FS = \frac{q_{ult}}{q_{eq}} \tag{5–23}$$

where:

FS = calculated factor of safety (dimensionless)
q_{ult} = ultimate bearing pressure (ksf)
q_{eq} = gross equivalent bearing pressure (ksf)

i. Bearing capacity from in situ tests. Laboratory and in situ tests should be used to establish soil strength parameters that are applied in equation 5–22. Direct application of in situ test results to assess ultimate bearing capacity often provides similar results to equation 5–22. When the bearing pressure of the foundation is based on allowable settlement calculations, direct use of in situ test results can be used to assess the factor of safety for ultimate bearing capacity.

(1) *In situ test parameters*. Direct bearing capacity methods are available for the SPT, the CPT, and the PMT. Methods are discussed in detail in Briaud (1992), FHWA-IP-89-008, and Mayne (2007). The SPT method uses the equivalent $N_{60} \cdot p_{ref}$ value in the shear zone. The value of p_{ref} is a reference stress equal to 2 ksf. The CPT method uses the equivalent net corrected cone tip resistance ($q_{ne} = q_t - \sigma_{vo}$) in the

shear zone. The PMT method uses the equivalent net pressuremeter limit pressure ($p_{ne} = P_L - P_o$) in the shear zone.

(2) *Equivalent in situ test results*. In situ test results are likely not constant in the shear zone and should be based on the geometric mean of the values. Discussions of the size of the shear zone for bearing capacity are presented in paragraph 5–5a. The geometric mean for net pressuremeter limit pressure, for example, is defined in equation 5–24. Similar expressions would be used for $N_{60e} \cdot p_{ref}$ and q_n.

$$s = p_{ne} = \sqrt[y]{P_{L1}^* \times P_{L2}^* \times \ldots P_{Ly}^*} \tag{5–24}$$

where:

s = equivalent in situ testing parameter indicative of soil strength, in this case for the PMT
P_L^* = net limit pressure = $P_L - P_o$ (ksf); see also equation 4–17
p_{ne} = equivalent net limit pressure in the shear zone (ksf)
P_{L1}^* = 1st measured value of P_L^* within the bearing capacity shear zone (ksf)
P_{L2}^* = 2nd measured value of P_L^* within the bearing capacity shear zone (ksf)
P_{Ly}^* = y^{th} measured value of P_L^* within the bearing capacity shear zone (ksf)

(3) *In situ test bearing capacity equation*. The in situ test bearing capacity equation is often presented in a simplified form; however, this requires corrections for the same terms that are included in equation 5–22. The drainage conditions during critical foundation loading should be the same as those occurring during the in situ test. If this is not the case, equation 5–22 should be used with soil parameters that are consistent with drainage conditions during critical foundation loading. The in situ test bearing capacity equation is presented in equation 5–25 for drained conditions, and in equation 5–26 for undrained conditions. The value of k is presented in Table 5–5.

(4) *In situ test bearing capacity equation correction factors*. The USACE (Meyerhof-based) bearing capacity correction factors should be used for the in situ test bearing capacity equation (FHWA-IP-89-008). Correction factors for equation 5–25 and equation 5–26 are provided in Table 5–2 through Table 5–4. Briaud (1992) indicates depth correction factors for drained conditions that are greater than those in Table 5–2. Factors presented by Briaud (1992) include aspects of both shape and depth correction and should not be used. USACE (Meyerhof) depth and shape correction factors in Table 5–2 and Table 5–4 should be used in calculations.

$$q_{ult} = k \cdot s_q \cdot d_q \cdot i_{q\gamma} \cdot b_q \cdot g_q \cdot s + q_0 \tag{5–25}$$
drained

$$q_{ult} = k \cdot s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s + q_0 \tag{5–26}$$
undrained

$$i_{q\gamma} = \min\left(\frac{D_f}{B'}, 1\right) \cdot i_q + \max\left(1 - \frac{D_f}{B'}, 0\right) \cdot i_\gamma \tag{5–27}$$

where:

q_{ult} = ultimate bearing capacity (ksf)

k = dimensionless bearing capacity factor, see Table 5–5

s_q = dimensionless surcharge correction factor for footing shape, use 1 for continuous footing and increases with B/L for rectangular and approximately circular footings; in the absence of friction angle estimates, use Flow Number, N = 3 for in situ test methods, see Table 5–2

N = dimensionless Flow Number = principal stress ratio, use N = 3 for in situ test methods

d_q = dimensionless surcharge correction factor for footing depth, use 1 for footing at ground surface and increase w/normalized embedment below the footing, in the absence of friction angle estimates use N = 3 for in situ test methods, Table 5–2

$i_{q\gamma}$ = dimensionless surcharge and unit weight correction factors for load inclination, use 1 for vertical load and < 1 for inclined loads, see equation 5–27

b_q = dimensionless surcharge correction factor for base inclination, use 1 for a flat base and < 1 for an angled base, see Table 5–3

g_q = dimensionless surcharge correction factor for ground inclination, use 1 for flat ground and < 1 for ground sloping away from the footing, see Table 5–3

s = equivalent in situ testing parameters indicative of soil strength = $N_{60e} \cdot p_{ref}, q_{ne}, p_{ne}$

N_{60e} = equivalent SPT N_{60} value in the shear zone (blows/ft)

p_{ref} = reference stress equal to 2 (ksf)

q_{ne} = equivalent net cone tip resistance in the shear zone (ksf)

p_{ne} = equivalent net limit pressure in the shear zone (ksf)

q_o = surcharge pressure on the plane at the base of the footing (ksf)

s_c = dimensionless cohesion (undrained strength) correction factor for footing shape, use 1 for continuous footing and increase with B/L for rectangular and circular type footings, see Table 5–4

d_c = dimensionless (undrained strength) correction factor for footing depth, use 1 for footing at the ground surface and increase with normalized embedment below the footing, see Table 5–4

i_c = dimensionless (undrained strength) and unit weight correction factor for load inclination, use 1 for vertical load and < 1 for inclined loads, see Table 5–4

b_c = dimensionless (undrained strength) correction factor for base inclination, use 1 for a flat base and < 1 for an inclined base, see Table 5–4

g_c = dimensionless (undrained strength) correction factor for ground inclination, use 1 for flat ground and < 1 for ground sloping away from the footing, see Table 5–4

D_f = depth of base of footing below ground surface (ft)

B' = effective base width of footing (ft)

i_q = dimensionless surcharge correction factor for load inclination, use 1 for vertical load and < 1 for inclined loads, see Table 5–3

i_γ = dimensionless unit weight correction factor for load inclination, use 1 for vertical load and < 1 for inclined loads, see Table 5–3

Table 5–5
Uncorrected bearing capacity factor (k) using in situ tests (after FHWA-IP-89-008; Mayne 2007a; Briaud 2013)

In situ Test	s Parameter (ksf)	k, Drained (dimensionless)	k, Undrained (dimensionless)
SPT	$N_{60e} \cdot p_{ref}$	0.6	0.35
CPT	q_{ne}	0.1	0.35
PMT	p_{ne}	0.9	0.7

5–6. Sliding

Sliding resistance is the horizontal load that a foundation can resist. Sliding failures are indicated by excessive horizontal movement of the foundation.

a. Sliding shear zone. The shear zones for the sliding ultimate limit state are illustrated in Figure 5–12. There can be one or two components of shear resistance. The first is the soil-foundation interface at the base of the footing. The second is the passive resistance in the soil above the bottom of the footing. Each component has a distinct shear zone.

(1) *Soil-foundation interface shear zone*. Soil-foundation interface strength parameters should be representative of the strength in this zone. For sliding below walls with shear keys, the sliding shear zone extends through the soil from the tip of the shear key to the end of the footing (Figure 5–10). When the sliding shear zone passes through soil, the interface strength coefficients should equal the soil strength coefficients.

(2) *Passive soil shear zone*. Passive resistance is developed by the shear zone extending from the base of the foundation to the ground surface. In relatively uniform soils, a representative strength within the shear zone can be used. In layered soils, a multiple wedge analysis should be considered (see EM 1110-2-2100). Each layer is characterized separately by its representative strength for a multiple-wedge analysis.

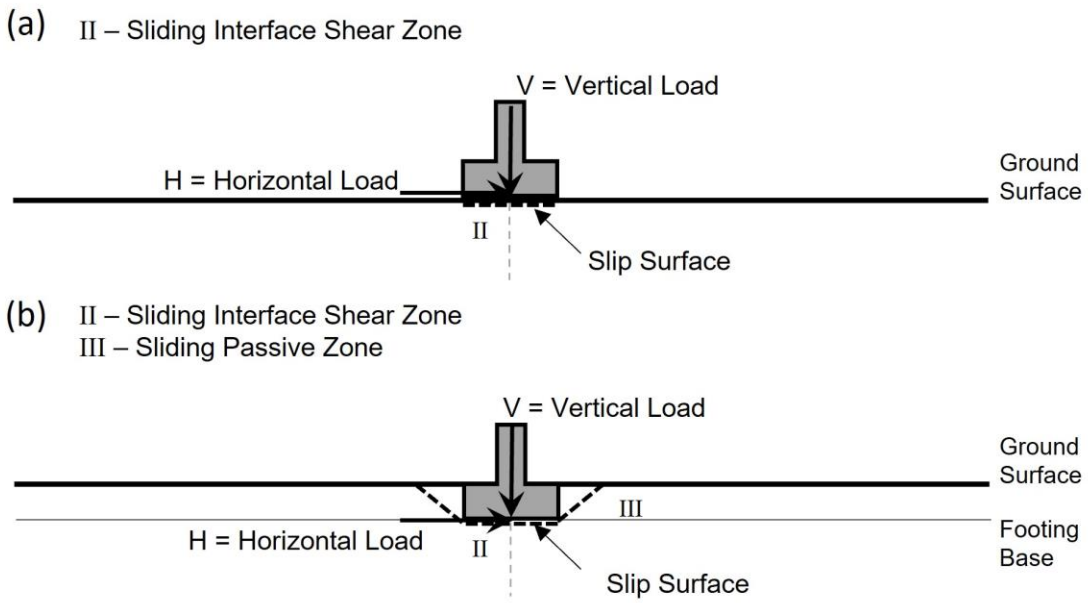


Figure 5–12. Slip surfaces for rough footings indicating size and shape of bearing capacity ultimate limit state sliding limit state shear zones for (a) surface footing; (b) embedded footing; vertical arrow = constant static load; horizontal arrow = loading direction

b. Soil-foundation interface strength coefficients. A sliding failure at the base of a foundation is controlled by the weaker of a failure within the soil mass or at the interface between the soil and the structure. The interface generally controls, and values lower than the friction angle (ϕ'), cohesion (c'), or undrained strength (s_u) should be used to assess the sliding resistance.

(1) *Fully-rough drained interface strength coefficients*. The interface friction angle (δ') is used in calculations based on effective stresses. Even for very dense sands against a rough interface, the peak interface friction angle is typically less than or equal to the large displacement (constant volume or critical state) soil-soil friction angle (Gómez et al. 2008). For rough concrete cast against typical sands, δ' = 28° is generally applicable regardless of initial density or grain size. If laboratory interface shear strength test data are available, direct measurements of δ' can be used. Alternatively, a δ' value that is 2° less than the large displacement (constant volume or fully softened) friction angle can be used for rough interfaces in sands, silty sands, clays, and other soil mixtures. Adhesion is typically assessed using the ratio of adhesion to effective stress cohesion (c_a/c'). For cast-in-place concrete footings c_a/c' is typically 0.8 for soils.

(2) *Undrained interface strength coefficients for smoother conditions*. For smoother interfaces, such as precast concrete footings, the interface friction coefficient will be lower than for rough footings (see EM 1110-2-2502). If laboratory interface shear strength test data are available, direct measurements of δ' can be used. For precast concrete and smoother footings, adhesion is zero.

(3) *Fully-rough undrained interface strength coefficients*. For undrained conditions, adhesion is assessed using the ratio of adhesion to undrained strength (c_a/s_u). For a fully rough cast in place concrete footing where the material has not been softened by construction activities, (c_a/s_u) is often taken as one.

(4) *Undrained interface strength coefficients for smoother conditions*. For smoother interfaces, such as precast concrete footings, (c_a/s_u) will be approximately equal to the ratio of the effective stress interface friction angle and compared to the drained soil effective stress friction angle, $\tan(\delta')/\tan(\phi')$. See paragraph 5–6b(2) for recommended values of δ'.

c. Applied shear force. The applied shear force (T) assessed using a single wedge is discussed in EM 1110-2-2100. For surface footings, the applied shear force is the component of force parallel to the base. For embedded footings, the applied shear force also includes the difference between the active and passive components of soil force using developed strengths. The applied shear force is shown in Figure 5–13a for the general case. Figure 5–13b shows the components of the applied shear force for the case of an embedded footing with no differential water pressure considered in this manual. See EM 1110-2-2502 for footings subject to horizontal hydraulic loads.

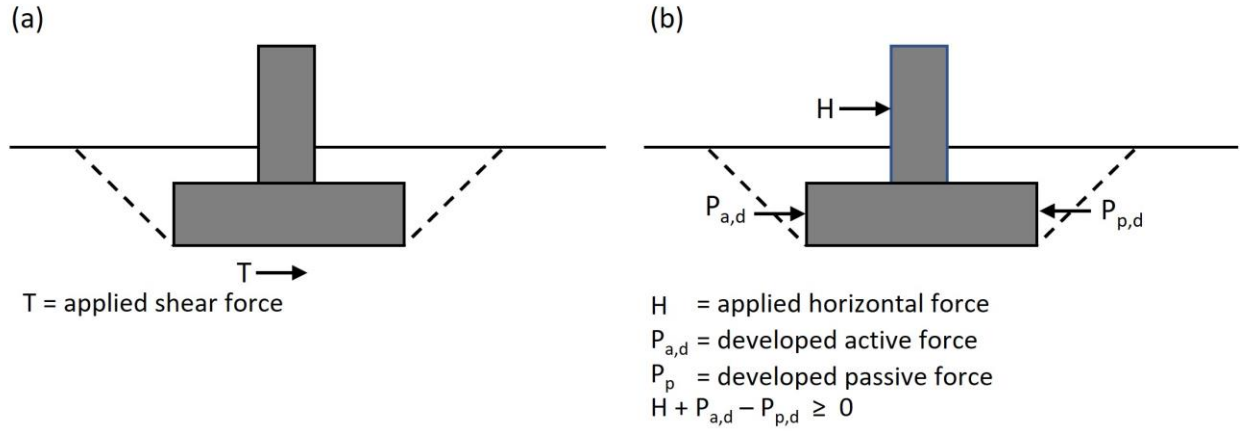


Figure 5–13. Components of single-wedge analysis used to assess the applied shear force (a) general case; (b) equivalent case for an embedded footing with no differential water pressure

d. Safety factor for sliding. The general expression for the sliding factor of safety is expressed in equation 5–28. Equation 5–29 is the drained factor of safety accounting for external horizontal loads and earth pressures. Equation 5–30 is the undrained factor of safety accounting for external horizontal loads and earth pressures. Earth pressures using developed soil parameters (paragraph 5–6e) must be used within equations 5–29 and 5–30.

$$FS = \frac{(N - U)\tan(\delta') + \left({}^{c_a}/_{c'}\right) \cdot c' \cdot A}{T} \quad (5\text{–}28)$$

General Expression for Drained Sliding Factor of Safety, after EM 1110-2-2100

$$FS = \frac{(N - U)\tan(\delta') + \left({}^{c_a}/_{c'}\right) \cdot c' \cdot A}{H + P_{a,d} - P_{p,d}} \quad (5\text{–}29)$$

Drained Sliding Factor of Safety

$$FS = \frac{\left({}^{c_a}/_{s_u}\right) \cdot A \cdot s_u}{H + P_{a,d} - P_{p,d}} \quad (5\text{–}30)$$

Undrained Sliding Factor of Safety $\phi = 0$

where:

N = component of the resultant force normal to the base (units of kips and kips/ft for isolated spread and continuous footing loads, respectively)

T = component of the resultant force parallel to the base (units of kips and kips/ft for isolated spread and continuous footing loads, respectively)

A = base area of an isolated spread footing (ft^2), use $B = 1\ ft$ for area of continuous footing (ft^2/ft), where B is the base width of the footing in contact with soil

U = water uplift force acting on the base of the structure (kips)

δ' = interface friction angle (degrees)

c' = effective stress cohesion intercept (ksf)

c_a/c' = ratio of adhesion component of interface strength to effective stress cohesion intercept

c_a = adhesion component of interface strength (ksf)

s_u = undrained strength for the foundation soils in the shear zone (ksf)

c_a/s_u = ratio of the adhesion component of interface strength to undrained strength

$P_{p,d}$ = developed passive soil force on the front of footing per unit width of wall (kips/ft)

$P_{a,d}$ = developed active soil force on the back of the footing (kips/ft)

H = externally applied horizontal force (units of kips and kips/ft for isolated spread and continuous footing loads, respectively), $H + P_{a,d} \geq P_{p,d}$

FS = calculated factor of safety (dimensionless)

e. Developed strength. To correctly compute factor of safety according to equation 5–28, soil strength is reduced by the factor of safety when calculating active and passive earth pressures for sliding calculations. The concept of developed strength (EM 1110-2-2100) is used for this purpose. The use of developed parameters results in a force larger than the active soil force on the driving side, and a force smaller than the passive soil force on the resisting side. Developed friction angle, developed cohesion, and developed undrained strength are defined in equations 5–31, 5–32, and 5–33, respectively. The minimum factor of safety of 2.0 for sliding presented in Table 2–1 or higher factors of safety can be used in analyses. The developed strengths should only be used for sliding calculations, not for bearing capacity calculations.

$$\phi'_d = \tan^{-1}\left(\frac{\tan\phi'}{FS_{slide}}\right) \quad (5\text{–}31)$$

$$c'_d = \frac{c'}{FS_{slide}} \quad (5\text{–}32)$$

$$s_{u,d} = \frac{s_u}{FS_{slide}} \quad (5\text{–}33)$$

where:

FS_{slide} = minimum factor of safety for sliding analysis from Table 2–1 (dimensionless)
ϕ' = effective stress friction angle for the foundation soils in the passive zone (degrees)
ϕ'_d = developed effective stress friction angle per equation 5–31 (degrees)
c' = effective cohesion parameter for the foundation soils in the passive zone (ksf)
c'_d = developed effective cohesion parameter per equation 5–32 (ksf)
s_u = undrained strength for the foundation soils in the passive zone (ksf)
$s_{u,d}$ = developed undrained strength per equation 5–33 (ksf)

f. Earth pressure resistance of embedded footings. Earth pressures act as a load and a resistance on footings. The developed passive pressure is a resistance and cannot exceed the sum of external horizontal loads and the developed active earth pressure.

(1) *Earth pressure coefficients*. Earth pressure coefficients may be based on the Rankine equation. To account for side friction, sloping ground, or sloping foundation sides, the log spiral method (Caquot and Kérisel 1948) upper bound or lower bound earth pressure coefficients as tabulated in EM 1110-2-2502 may be used in design. The Coulomb earth pressure coefficient equation may be used provided that the follow conditions are met.

(a) The interface friction coefficient (δ') used in calculations is less than $\phi'/2$ for passive earth pressures.

(b) The slope angle used in calculations must be equal to zero or be negative for passive earth pressures.

(2) *Soil force in unform soils*. When soil and water conditions between the ground surface and the base of the footing are generally uniform, the average soil pressure is calculated as half the earth pressure at the base of the footing. The passive force is calculated using equation 5–34. The active force is calculated using equation 5–35.

$$P_{p,d} = \left(\frac{1}{2}\gamma' \cdot D_f \cdot K_{p,d} + 2c'_d\right) D_f \cdot L \quad (5\text{–}34)$$

$$P_{a,d} = \left(\frac{1}{2}\gamma' \cdot D_f \cdot K_{a,d} - 2c'_d\right) D_f \cdot L \ \geq 0 \quad (5\text{–}35)$$

where:

$P_{p,d}$ = developed passive soil force on the front of the footing per unit width of wall (kips/ft)

γ' = effective unit weight of the between the base of the footing and the ground surface (kcf)

c'_d = developed effective cohesion parameter, use $s_{u,d}$ for undrained analysis (ksf)

D_f = depth to base of footing below ground surface (ft)

L = base length of rectangular footing, use *B* for circular footings or 1 for continuous footings (ft)

$K_{p,d}$ = passive earth pressure coefficient using developed soil parameters (dimensionless)

$P_{a,d}$ = developed active soil force on the back of the footing, must be ≥ 0 (kips/ft)

$K_{a,d}$ = active earth pressure coefficient using developed soil parameters (dimensionless)

5–7. Uplift

An uplift failure mode occurs when the upward vertical force exceeds the downward vertical force.

a. Shear zone for uplift. The shear zone for uplift is illustrated in Figure 5–14 and is confined to the interface strength between the foundation and structure and the surrounding soil. While the weight of soil above the foundation is included, soil strength is often not considered as part of uplift resistance. Interface soil strength resisting uplift could degrade with time due to the potential for gapping or cyclic degradation.

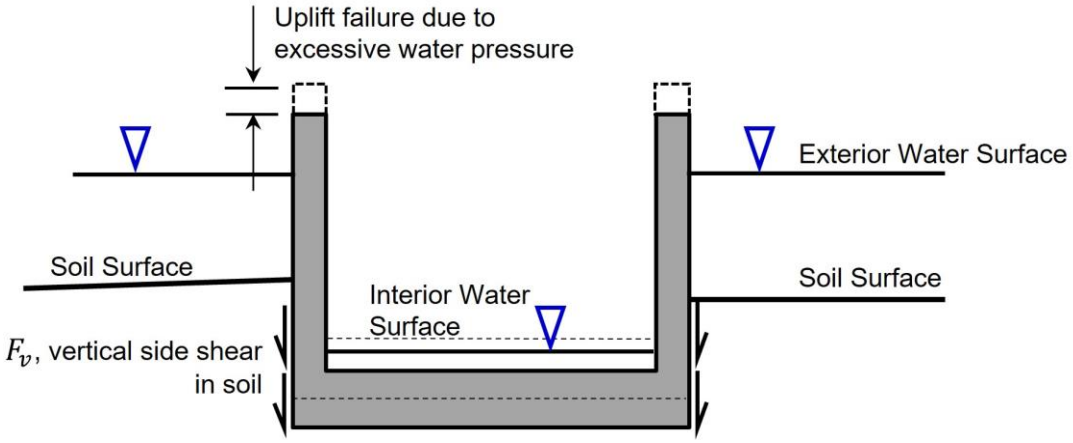


Figure 5–14. Separation of footing and soil for uplift failure due to water pressures

b. Factor of safety for uplift. Uplift is generally a result of wind or water loads. Permanent dead loads are considered only when assessing uplift resistance due to wind loads. Uplift calculations that account for vertical soil shear resistance are discussed in EM 1110-2-2100. The calculated factor of safety for uplift is defined by equation 5–36:

$$FS = \frac{W_f + W_s + W_c}{U} \tag{5–36}$$

where:

FS = calculated factor of safety (dimensionless)

W_f = weight of foundation including weight of soil above foundation bearing elevation (units of kips and kips/ft for isolated spread and continuous footings, respectively)

W_s = weight of the structure, including weights of the fixed equipment and soil above the top surface of the structure (units of kips and kips/ft for isolated spread and continuous footings, respectively)

W_c = weight of water contained within a hydraulic structure at the time of the design load case (units of kips and kips/ft for isolated spread and continuous footings, respectively)

U = uplift force acting on the base of the structure (kips)

Chapter 6
Stress Distribution

6–1. Introduction

When a load is applied by a foundation or similar external load, or when changes in the soils self-weight occurs, it influences the in situ stress within a soil mass. These changes in a soil's stress state can result in volume changes. This chapter describes the analysis of stress distribution within the ground. Static loading conditions are assumed.

6–2. Consolidation deformation of soils

a. Prior to loading, the laterally restrained soil is in equilibrium. When an increment of stress is applied to a saturated soil mass, it is assumed that all the external stress is first carried by the pore water pressure in the soil voids. As the pore water pressure dissipates, the stress is then gradually transferred to the soil skeleton, with an increase in effective stress equal to the same magnitude of stress as the pore water pressure decreases. The increase in effective stress results in a volume decrease. When the deformation ceases, the soil is assumed to be in equilibrium with the stress at this stable volume.

b. Since the solids are incompressible and only the volume of voids filled with water changes, the deformation is a measure of the void ratio change. Consequently, the problem concerning settlement becomes an estimation of the void ratio versus the vertical effective stress (σ_z') behavior for the soil. Therefore, the significant stress is effective stress, not total stress, because the soil is in volumetric equilibrium only when the pore water pressures are dissipated.

6–3. Sources of stress

Sources of stress are divided into two broad categories: geostatic and induced. To estimate foundation settlements, changes in both geostatic and induced stresses must be evaluated. Geostatic and induced stresses are discussed in paragraphs 6–4 and 6–6, respectively.

6–4. Geostatic stresses

Stresses caused by gravity acting on the soil or rock mass are known as vertical geostatic stresses. These are naturally present in the ground and can be changed or altered through human activity (such as mass excavation, areal fill, dewatering).

a. *Total vertical stress*. The total vertical stress (or total overburden stress) is the normal stress acting on a horizontal plane at some depth within the soil. The total vertical stress (σ_z) at a particular depth is calculated by multiplying the thickness of all overlying layers by the total unit weight of each layer. The total stress may be divided into two parts: the effective normal stress and the pore water pressure. Total stress is calculated per equation 6–1.

$$\sigma_z = \sum_{i=1}^{n} z_i \cdot \gamma_{t,i} \quad (6\text{–}1)$$

where:

σ_z = total vertical stress (tsf)
z_i = thickness of overlying layer, i (feet)
$\gamma_{t,i}$ = total unit weight of the overlying layer, i (tcf)
n = number of layers above point of interest

b. Pore water pressure. The pressure in a saturated soil mass that occurs between soil particles (pores) is referred to as pore water pressure. As opposed to soil stresses, which are directional, pore water pressures are omnidirectional. Under hydrostatic conditions, the pore water pressure above a point is equivalent to the head of water above the point of interest times the unit weight of water.

c. Effective vertical stress concept. The portion of total vertical stress carried by the soil grains themselves (particle-to-particle contact) is the vertical effective stress (σ_v' or σ_z'). It is this stress component that controls both volume change deformation and the shear strength of soil, since normal and shear stresses are transmitted across grain-to-grain contacts. The vertical effective stress of a saturated soil is calculated using equation 6–2 by subtracting the pore water pressure at any point from the total vertical stress at the same point. If unsaturated, the pore water pressure is generally assumed to be zero. Compression, permeability, and shear strength of soils are influenced by effective stress.

$$\sigma_v' = \sigma_z - u \quad (6\text{–}2)$$

where:

σ_v' = initial vertical effective stress (tsf), also denoted σ_{zo}'
σ_z = initial total vertical stress (tsf)
u = $\gamma_w \cdot z_w$ = pore pressure (tsf)
γ_w = unit weight of water = 0.031 tcf (62.4 pcf)
z_w = depth below water table (feet)

6–5. Initial state of stress and stress history

The initial state of stress and stress history are principal factors in determining the settlement magnitude of shallow foundations. For shallow foundation support, the geotechnical site characterization process should include an assessment of the initial stress state and stress history. These initial stresses include estimates of the initial vertical effective stress (σ_{zo}') and preconsolidation stress (σ_p') of the soil. For sandy soils, preconsolidation stress could be more difficult to estimate due to difficulties in separating the effects of relative density from OCR.

a. Initial vertical effective stress. The initial vertical effective stress can be obtained from equation 6–2. The initial vertical effective stress should be determined prior to changes in geostatic stress conditions associated with recent site work (such as fill surcharge, dewatering).

b. Preconsolidation stress.

(1) The preconsolidation stress (σ_p') is the maximum vertical effective stress to which the in situ soil has been consolidated in the past and is the boundary between recompression and virgin consolidation. Mechanisms that influence preconsolidation stress include changes in total stress, changes in pore water pressure, changes in soil structure due to aging, and chemical weathering. The preconsolidation stress has great importance because it separates elastic and reversible deformations from inelastic and only partially irreversible deformations and marks the starting point of high compressibility. Where soils are subjected to pressures less than the preconsolidation stress, magnitudes of settlement are relatively less severe.

(2) Preconsolidation stress (σ_p') cannot be measured directly; however, it can be estimated from oedometer test results. Commonly, the Casagrande and Schmertmann procedures are used (paragraph 7–9a(2)). Other methods for determining preconsolidation stress from oedometer data include Janbu (1969), Pacheco Silva (1970), Becker et al. (1987), Oikawa (1987), Burland (1990), and Boone (2010). These methods have in common the assumption of a change in soil stiffness from a stiffer to a softer response near the σ_p'. Preconsolidation stress (σ_p') can also be estimated from empirical methods; however, the results are generally less reliable. Sample disturbance tends to decrease the apparent preconsolidation stress, reference paragraph 4–7a. Previous testing or geologic information should be used to validate estimates of σ_p'.

c. Normally consolidated versus overconsolidated soils. Soils are either normally consolidated or overconsolidated. If the current overburden pressure is equal to the preconsolidation stress (σ_p'), the soil is normally consolidated (NC). If the preconsolidation stress (σ_p') is larger than the current vertical effective stress (σ_z'), the soil is overconsolidated (OC). Soils cannot be underconsolidated. The term “underconsolidated” is a misnomer for soils that are “undergoing consolidation,” where the current effective overburden stress that is consolidating is equal to the preconsolidation stress (Fellenius 2009). The natural moisture content of a normally consolidated clay is usually closer to its liquid limit, while for an overconsolidated clay, it is usually closer to its plastic limit.

d. Overconsolidation ratio and overconsolidation margin. The degree of overconsolidation is expressed as the ratio of the maximum overburden pressure it has experienced in the past to the current overburden pressure as defined in equation 6–3, or the overconsolidation margin (σ_m') as defined in equation 6–4. The overconsolidation margin should be relatively constant throughout a stratum of the same geologic origin, but the OCR can vary as a function of depth. Therefore, knowing the overconsolidation margin (σ_m') and the initial vertical effective stress (σ_{zo}'), one can estimate the

preconsolidation stress at other depths in the same stratum of the same geologic origin. The state of stress is often quantified by the OCR and overconsolidation margin as indicated in Table 6–1.

$$\mathrm{OCR} = \sigma_p'/\sigma_z' \qquad (6\text{–}3)$$

Overconsolidation Ratio

$$\sigma_m' = \sigma_p' - \sigma_{zo}' \qquad (6\text{–}4)$$

Overconsolidation Margin

where:

OCR = overconsolidation ratio (dimensionless)
σ_m' = overconsolidation margin (tsf)
σ_p' = preconsolidation stress (tsf)
σ_z' = current vertical effective stress (tsf)
σ_{zo}' = initial vertical effective stress (tsf)

Table 6–1
In situ state of stress based on overconsolidation ratio and overconsolidation margin (σ_m')

Description	State of Stress	OCR	σ_m' (ksf)
Normally consolidated	$\sigma_z' = \sigma_p'$	1	0
Lightly Overconsolidated	$\sigma_z' < \sigma_p'$	1 to 3	0 to 2.0
Moderately overconsolidated	$\sigma_z' < \sigma_p'$	3 to 6	2.0 to 8.0
Heavily overconsolidated	$\sigma_z' < \sigma_p'$	> 6	> 8.0

6–6. Induced stresses

Stress changes at depth caused by external loads applied on or near the surface (such as structural foundations, embankment loads) are known as induced stresses ($\Delta\sigma_z$). Induced stresses within the soil vary spatially, decreasing with distance from the center and depth of load application and must be determined to estimate settlement of shallow foundations. The distribution of vertical stress due to induced stresses can be approximated using elastic theory solutions, approximate methods, or numerical analyses.

a. Bearing pressure and contact stress. When an external load (such as bearing pressure) is applied to a soil mass, a restoring force develops within the soil to resist deformation. The ratio of the restoring force (or soil reaction) between the foundation and soil that is perpendicular to the surface area over which the load is applied is known as contact stress. The total contact stress must equal the applied force if equilibrium exists; however, the contact stress is not distributed evenly. For isolated and spread footings, bearing pressure and the resulting contact stress are generally assumed uniform. For mats and large footings, contact stress distribution can have a more significant influence, reference paragraph 6–6a(3).

(1) *Net bearing pressure*. An alternative way to define bearing pressure is “net bearing pressure.” The net bearing pressure (q') is the difference between the gross bearing pressure, q (or gross equivalent bearing pressure (q_{eq}) for eccentrically loaded foundations) and the initial vertical effective stress at the bearing elevation of the footing (σ'_{zD}). Net bearing pressure is calculated per equation 6–5. Gross and gross equivalent bearing pressure are defined in paragraph 5–2b. Therefore, the net bearing pressure can be defined as the increase in the vertical effective stress at the bearing elevation of the footing. Both the gross and net bearing pressures will produce the identical design if used consistently and correctly.

$$q' = q - \sigma'_{zD} \tag{6–5}$$

where:

q' = net bearing pressure (tsf)

q = gross bearing pressure (tsf); use gross equivalent bearing pressure (q_{eq}) for eccentrically loaded foundations

σ'_{zD} = initial vertical effective stress at the bearing elevation of the footing (tsf)

(2) *Contact stress distribution*. Contact stress distribution depends principally on foundation stiffness, soil stiffness, and loading condition. For a uniformly loaded and perfectly flexible foundation, the distribution of contact stress is also uniform regardless of soil type, see Figure 6–1a and Figure 6–1b. For a symmetrically loaded and perfectly rigid foundation, contact stress and stresses in the soil mass must be such to ensure strain compatibility, resulting in uniform settlement. This contact stress distribution will depend on the soil stiffness behavior illustrated in Figure 6–1c and Figure 6–1d. For soils that are linear elastic in behavior, contact stress and settlement behavior is expected to be similar to Figure 6–1b and Figure 6–1d. For soils with stiffness that is higher under higher stresses, behavior such as illustrated in Figure 6–1a and Figure 6–1c is expected.

(3) *Foundation rigidity*.

(a) Induced stresses are typically computed using the theory of elasticity and assuming a perfectly flexible foundation. This assumption produces larger variations of deflections than expected, since nearly all foundations have some degree of rigidity. Stress variations as great as 20 to 30 percent between theoretical and field conditions could occur due to rigidity effects. To account for footing rigidity, empirical methods (that account for footing rigidity) can be used for estimating settlement. Alternatively, rigidity corrections (paragraph 6–6a(4)) can be applied.

(b) For mat foundations and some large footings that generally have more flexibility than isolated footings, contact stress distribution can be of more significance due to bending moments and differential settlements that can develop. For these foundations, contact stress distribution should be determined using the design methods provided in Chapter 8 (especially when rigidity corrections should not be applied).

(4) *Foundation rigidity corrections*. The stress distribution variations due to rigidity effects are applicable only to stresses directly near the footing and are derived from and applicable to elastic half-space theory. At depth, the overall stress distribution bulb generally matches flexible loading assumptions as illustrated in Figure 6–2. In practice for conventional spread footing design, the nuance of the contact stress distribution is not required to be considered. In a special situation, if it is needed to more accurately predict soil stresses directly beneath a footing, additional details can be found in literature (for example, Das 2017; Coduto et al. 2016; Tsytovich 1976).

b. Plane strain and axisymmetric stress distribution. Footing shape will influence the induced stress distribution within the soil. Circular and approximately circular footings (such as square, octagonal) subjected to vertical loadings are assumed to have an axisymmetric stress distribution where the bearing pressures are dispersed laterally in all directions (see Figure 6–3a). Strip footings where the base length, L, is greater than 10 times the base width, B (L/B >10), are assumed to have a plane strain stress distribution where deformation occurs only in the planes perpendicular to the long axis (see Figure 6–3b). Rectangular footings with length-to-width ratios between 1 and 10 will have stress distributions varying linearly between the assumed axisymmetric and plane strain stress distributions.

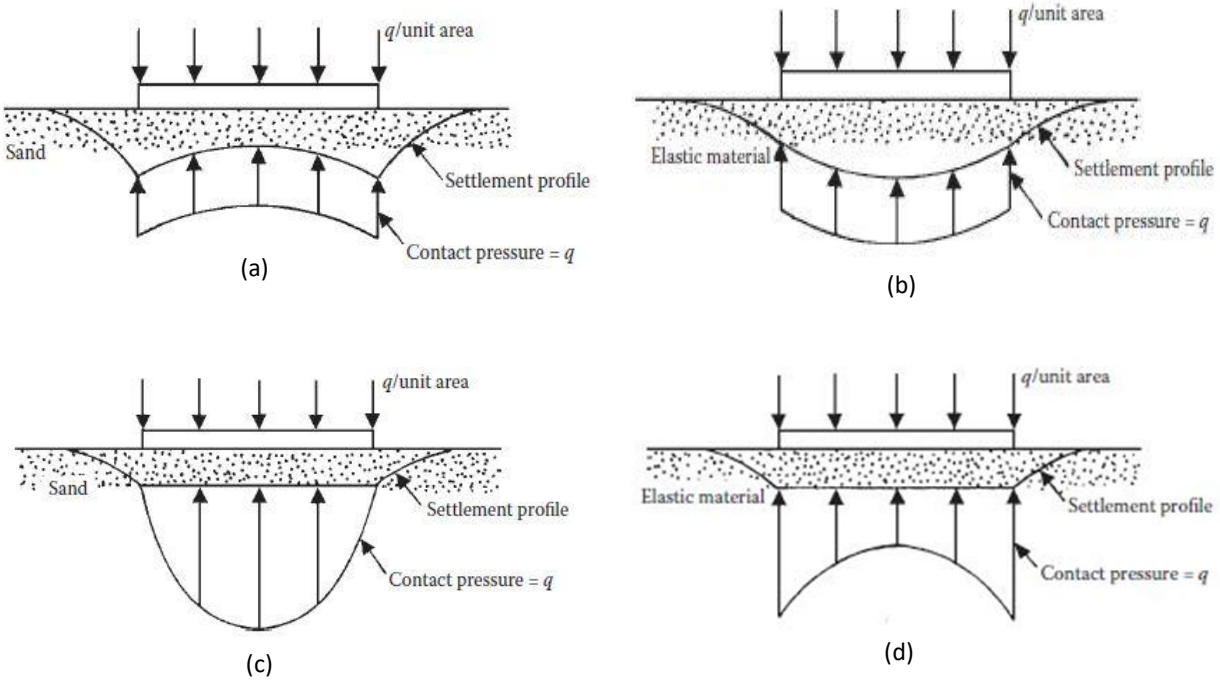


Figure 6–1. Theoretical contact stress and settlement distribution under uniform loading for (a) flexible foundation on sand, (b) flexible foundation on clay or elastic medium, (c) rigid foundation on sand, and (d) rigid foundation on clay or elastic medium (Das 2017)

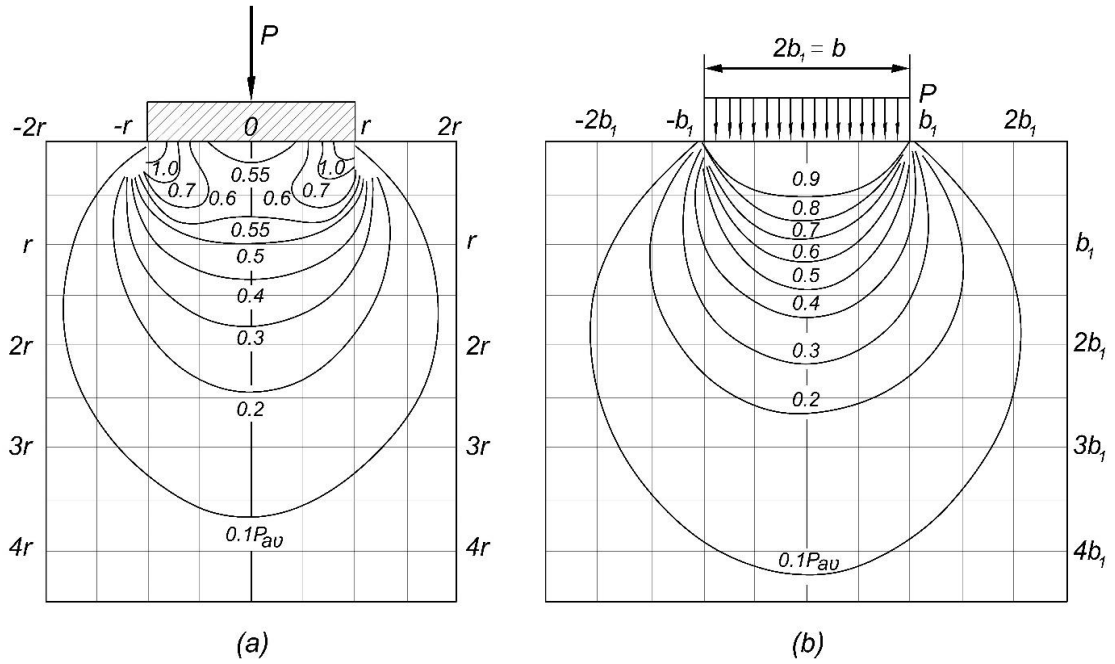


Figure 6–2. Isobars in soil under foundations (a) absolutely rigid foundation; (b) flexible foundation (Tsytovich 1976)

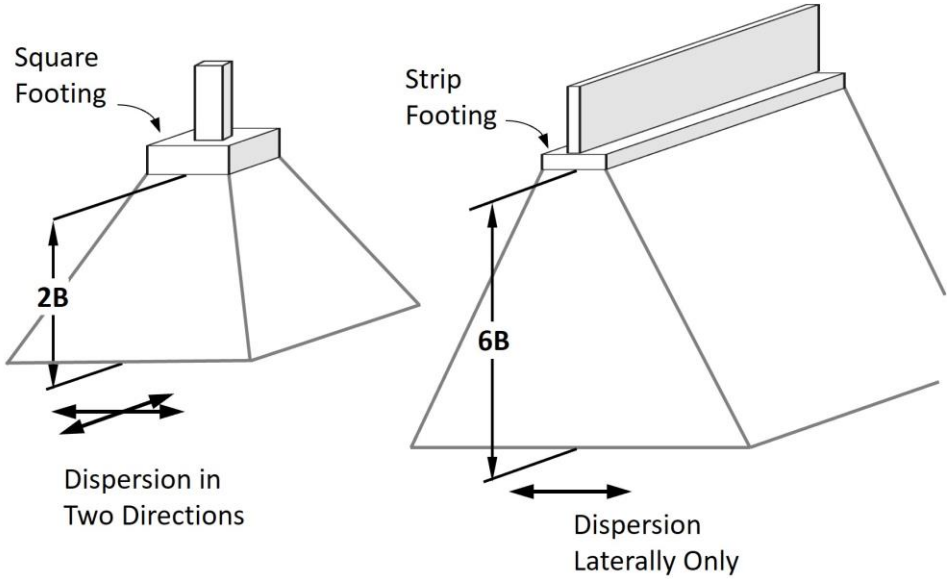


Figure 6–3. Stress distribution due to induced load to depth of significant influence for (a) approximately circular (such as square footing), and (b) strip footing

c. Depth of significant influence due to induced stresses. At a certain depth below a foundation, induced stresses have a negligible effect on foundation settlement. This generally occurs at a depth where the vertical stress increase from the induced load (Δq) is reduced to less than 10 percent of the stress intensity applied at the surface

(specifically, $< 0.1q$). For square footings (or fills with a similar length-to-width ratio), the depth of significant influence is approximately twice the footing (or fill) width ($2B$) as illustrated in Figure 6–4a. For strip footings (or embankments with a length-to-width ratio ≥ 10), the depth of significant influence is approximately $6B$.

d. Interference of adjacent foundations.

(1) When several foundations are spaced closely together, the stress isobars can overlap and lead to interference. If several footings are spaced closely enough, the individual isobars can combine and merge into one large isobar as illustrated in Figure 6–4b. The effect of footing interference can lead to an increase in settlement for the same magnitude of load applied on each footing and a decrease in bearing capacity. Induced vertical stresses below foundations due to interference can be approximated using the principle of superposition (paragraph 6–6e(2)). Where the clear spacing (S) is less than $0.5B$ (for footings of similar size), maximum magnitudes of settlement and tilt should be expected.

(2) Foundation spacing is based on ULS and SLS requirements. Clear spacing (horizontal) between new footings should exceed 1.5 times the width of the larger footing to avoid significant stress interactions that can influence bearing capacity and settlement. Where new footings will be placed adjacent existing footings, induced stresses could result in unacceptable settlements due to footing interference and should be evaluated using the methods presented in this manual.

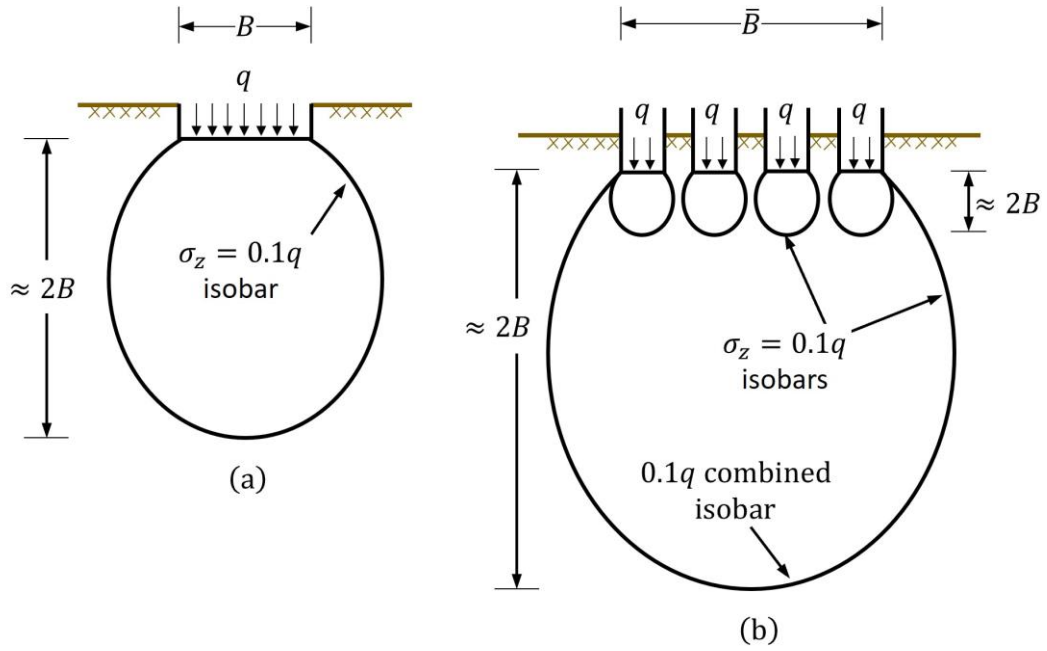


Figure 6–4. (a) depth of significant influence for isolated square foundation, (b) depth of significant influence for closely spaced foundations

e. Computation of induced vertical stresses.

(1) *Beneath foundation or uniformly loaded area*. The induced vertical stress ($\Delta\sigma_z$) beneath a footing or uniformly loaded area should be computed in relation to the net bearing pressure ($q - \sigma'_{zD}$) and influence factor (I_σ) per equation 6–6. The influence factor for vertical stress (I_σ) beneath a footing or uniformly loaded area can be determined from elastic solutions (for example, Boussinesq or Westergaard).

$$\Delta\sigma_z = \mathrm{I}_\sigma(q - \sigma'_{zD}) \tag{6–6}$$

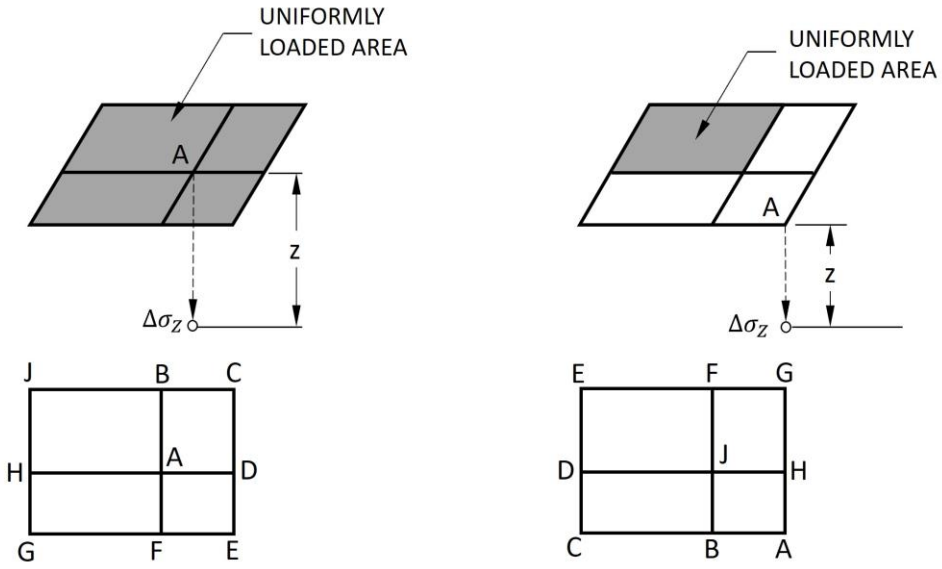
where:

$\Delta\sigma_z$ = induced vertical stress (tsf)
I_σ = influence factor for vertical stress (dimensionless)
q = gross bearing pressure (tsf)
σ'_{zD} = vertical effective stress (tsf), at the bearing elevation of the footing

(2) *Principle of superposition*.

(a) Boussinesq and Westergaard chart solutions for computing the influence factor (I_σ) beneath the corner of a uniformly loaded rectangle are illustrated in Figure 6–8 and Figure 6–10, respectively. These figures can be used to determine the influence factor for vertical stress (I_σ) at the point within and beyond the footprint of the footing or uniformly loaded area using the principle of superposition as illustrated in Figure 6–5.

(b) Suitable sub-areas are chosen such that their sum is the loaded area causing the stress, and each has the same corner over the point in question. For example, to determine the influence factor for vertical stress (I_σ) beneath the center of a footing, the footing is divided into four sub-areas, each a rectangle that is 1/4 of the larger one, and the individual influence factors, I_σ for these four smaller areas are summed to find I_σ for the larger rectangle at the location of the same corner in question. This principle is especially useful for determining the change in stress below and outside of loaded areas.



INFLUENCE FACTOR	RECTANGLE
I_1	ABCD
I_2	ABJH
I_3	AFGH
I_4	ADEF
$I_A = I_1 + I_2 + I_3 + I_4$	

(a) Point below middle of rectangular area

INFLUENCE FACTOR	RECTANGLE
I_5	ACEG
I_6	ABFG
I_7	ACDH
I_8	ABJH
$I_A = I_5 - I_6 - I_7 + I_8$	

(b) Point outside of rectangular area

Figure 6–5. Use of superposition to determine change in vertical stress

6–7. Elastic solutions for vertical stresses due to external loads

The induced vertical stress due to an external load can be analyzed using elastic theory. According to elastic theory, it is assumed that constant ratios exist between stresses and strain. This assumption is appropriate for the estimation of foundation deformations when an adequate factor of safety is applied to the ultimate limit state failure modes (especially when shear stresses throughout the soil mass will be significantly below the shear strength of the soil). Estimation of in situ stresses presented in this chapter assume that elastic conditions prevail. With respect to these assumptions and the limitations of each method, the Boussinesq and Westergaard solutions satisfactorily compute the stresses within the soil for estimating settlements.

a. Boussinesq solution.

(1) *General*. Boussinesq (1885) developed a solution for stress distribution in a homogeneous and isotropic soil subjected to a concentrated load at the surface. The theory assumes an elastic foundation on an elastic medium (modulus, *E* constant). Also, the soil mass is assumed to be semi-infinite (extends all directions below a level surface), the soil obeys Hooke's law, and the soil is weightless. For the Boussinesq solution, it is observed that the vertical stress decreases with the increase in depth from the ground surface and in distance from the axis of loading, expressed as an influence factor (I_σ). The Boussinesq stress distribution is illustrated in Figure 6–6 for an infinitely long and square area, and Figure 6–7 for a uniformly loaded circular area. The stress distribution below the corner of a uniformly loaded rectangular area is illustrated in Figure 6–8. Chart solutions for embankment loads based on Boussinesq are provided in paragraph 6–8.

(2) *Limitations of the Boussinesq theory*. Boundary conditions can differ substantially from idealized conditions to invalidate solutions by elasticity theory. Elastic solutions such as the Boussinesq solution assumes a weightless material not subject to initial stress. Initial stress always exists in situ because of overburden weight of overlying soil, past stress history, and environmental effects such as desiccation. These initial stresses through Poisson's ratio, nonlinear elastic modulus, and soil anisotropy significantly influence in situ stress and strain that occur through additional applied loads. Actual stresses beneath the center of shallow footings can exceed Boussinesq values by 15 to 30 percent in clays and 20 to 30 percent in sands.

b. Westergaard solution. Westergaard (1939) developed a stress/displacement theory for varved clays (clays with numerous horizontal seams of silt and fine sand). According to the theory, soils that are stratified with strong layers generally reinforce soft layers so that the resulting stress intensity at deeper depths are less than that formulated for an isotropic soil after the Boussinesq solution.

(1) *Comparison of Westergaard to Boussinesq*. The main effect is that Westergaard's theory produces a lower stress directly below the center of a load as compared to Boussinesq; however, at a certain distance away from the center, stresses begin to exceed those calculated using Boussinesq. The Westergaard stress distribution is illustrated in Figure 6–9 for both an infinitely long and square area, and in Figure 6–10 for the corner of a uniformly loaded rectangular area. For a uniformly loaded square footing (with Poisson's ratio, $\upsilon = 0$), comparative values (to Boussinesq) are provided in Table 6–2.

(2) *Limitations of Westergaard theory*. Soil is assumed to consist of a linear elastic material horizontally reinforced by inextensible membranes of negligible thickness and spaced very closely together. The Westergaard medium assumes isotropy with respect to the elastic modulus, *E,* but anisotropy with respect to shear modulus, *G*. Westergaard's theory does not satisfy one of the necessary stress boundary conditions, which is that the surface (z = 0) should be free of shear stresses.

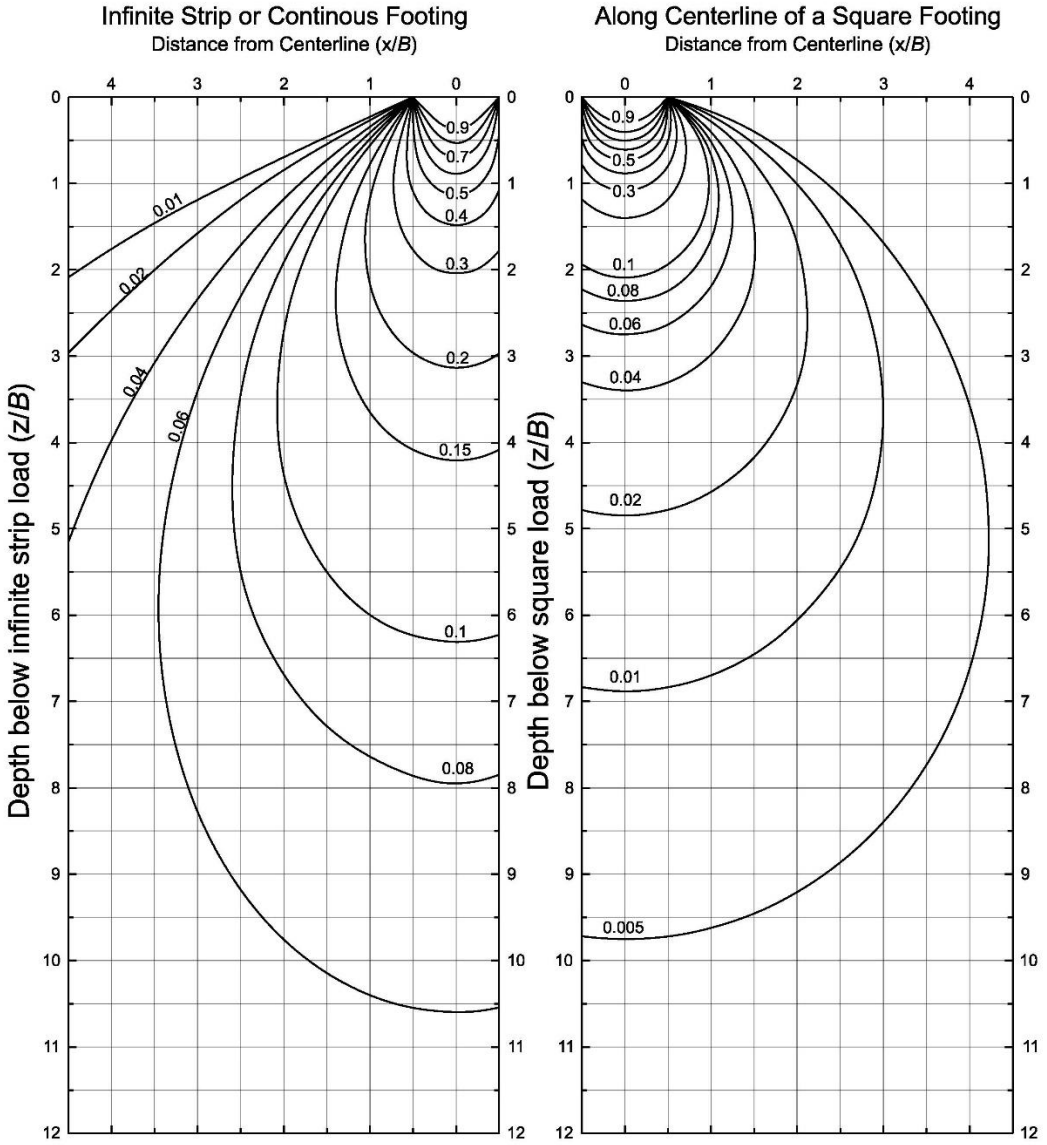


Figure 6–6. Contours of induced vertical stress beneath a foundation in a semi-infinite elastic solid by the Boussinesq solution

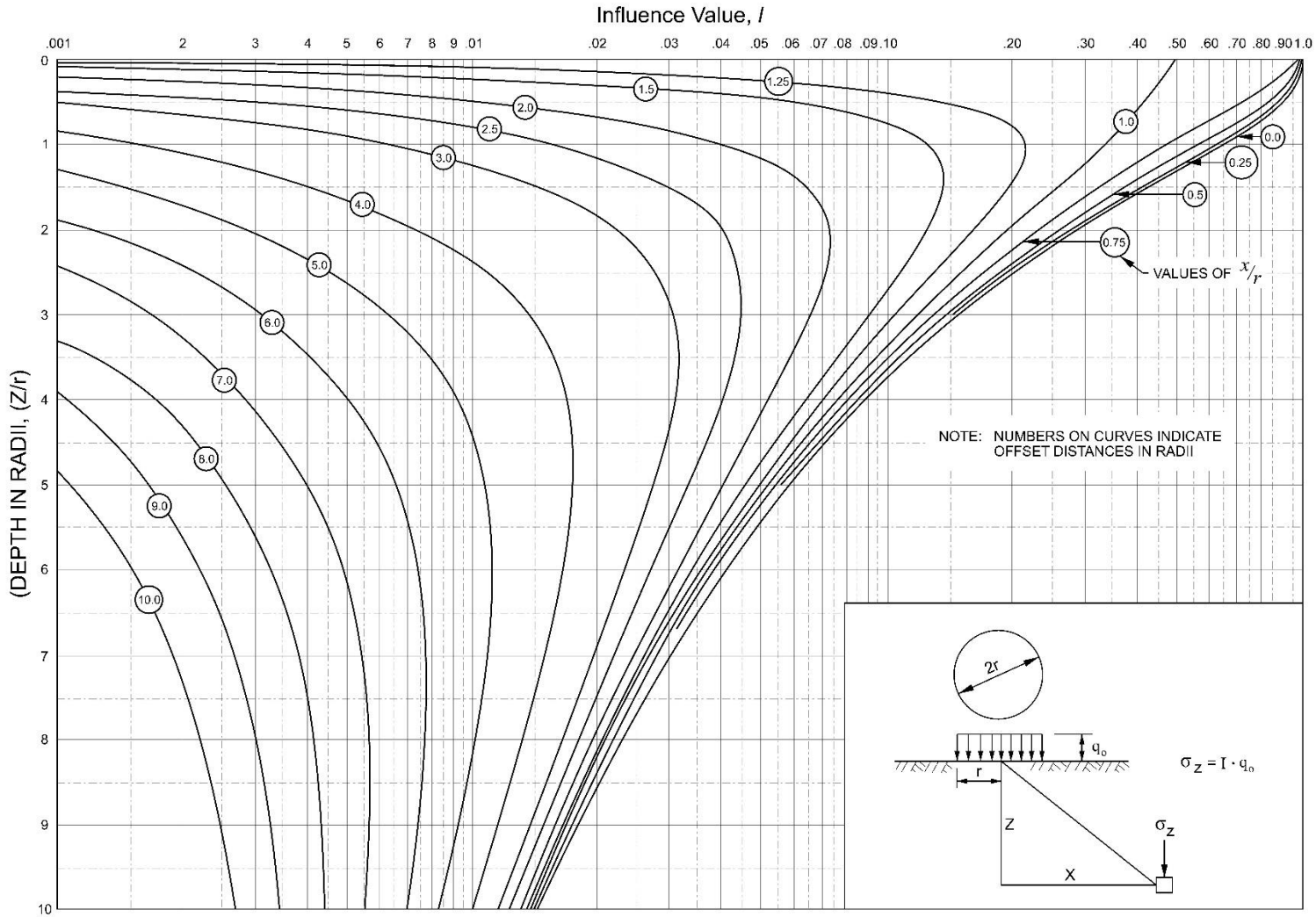


Figure 6–7. Influence factor (I_σ) for computing induced vertical stress beneath a uniformly loaded circular area for the Boussinesq solution (adapted from Foster and Ahlvin 1954)

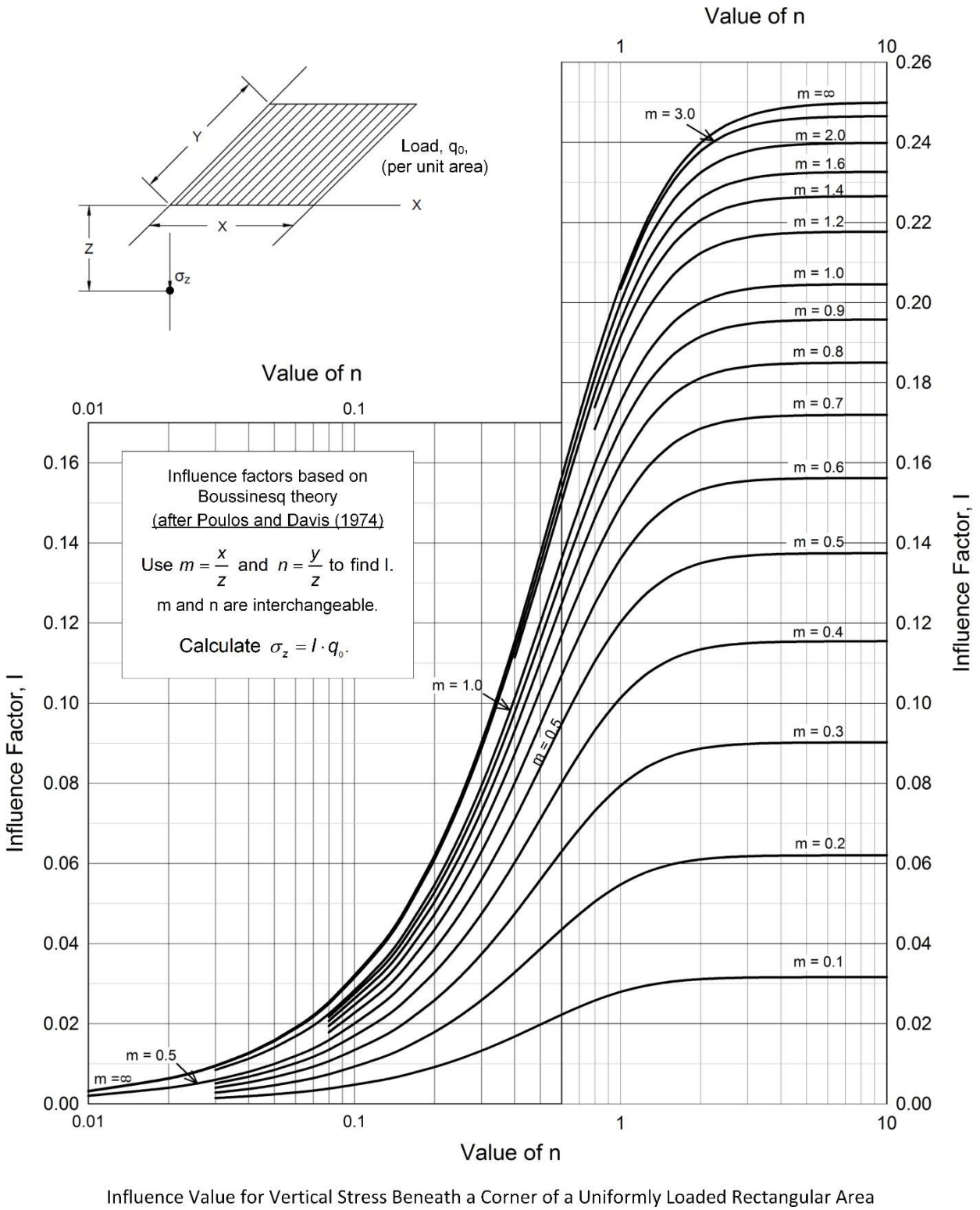


Figure 6–8. Influence factor (I_σ) for computing induced vertical stress beneath a corner of a uniformly loaded rectangular area for the Boussinesq solution (adapted from Fadum 1948)

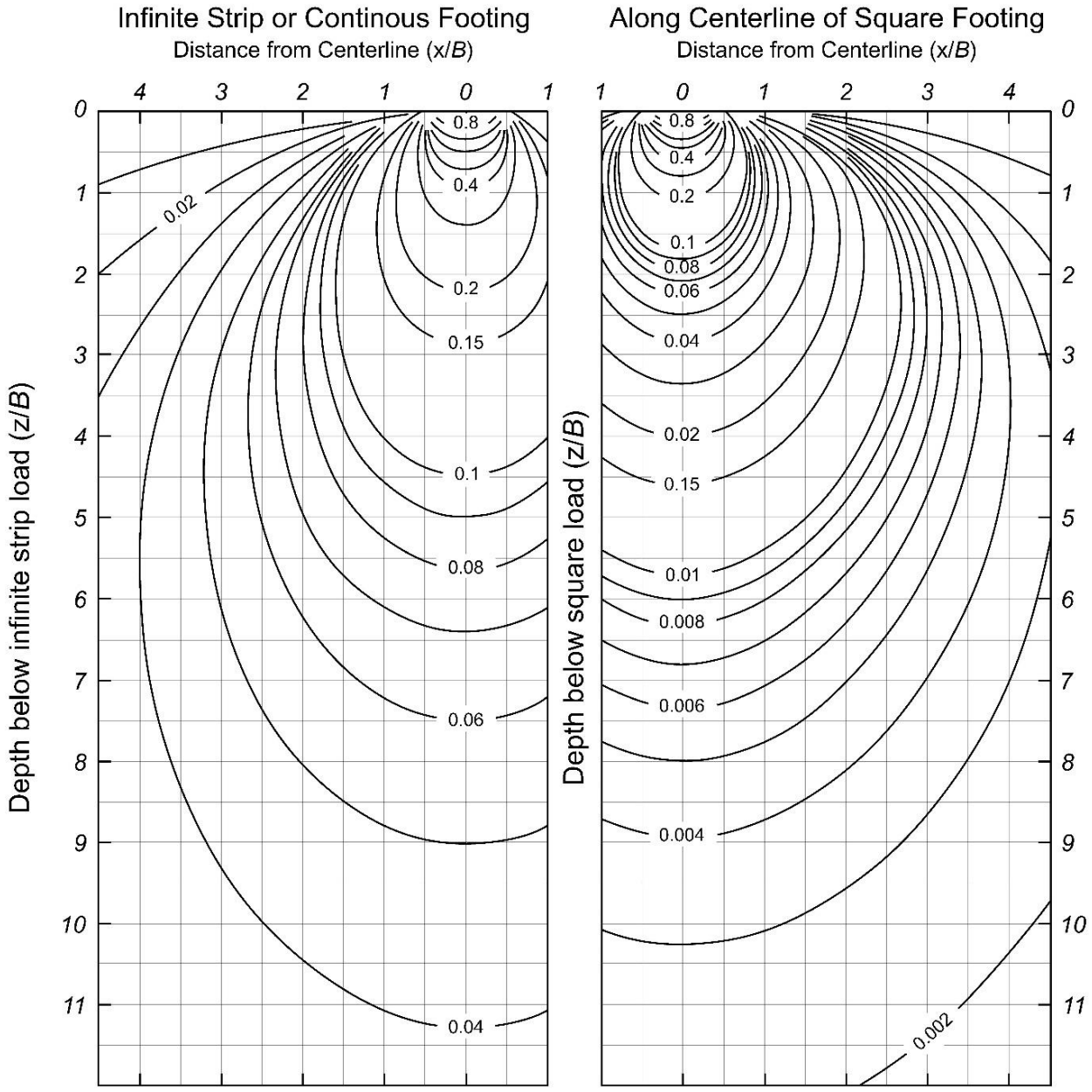


Figure 6–9. Contours of induced vertical stress beneath a foundation in a semi-infinite elastic solid by the Westergaard solution

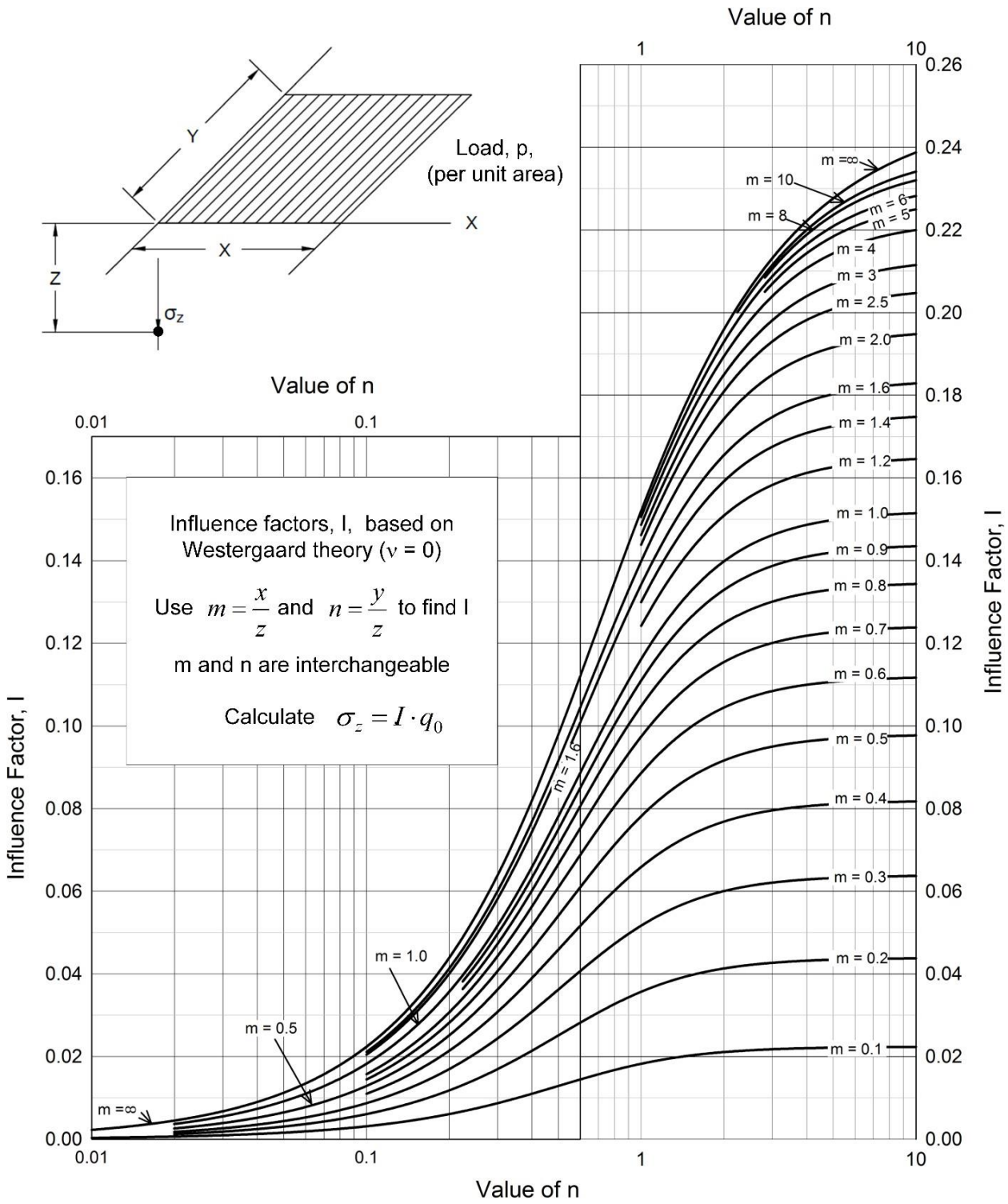


Figure 6–10. Influence factor (I_σ) for computing induced vertical stress beneath a corner of a uniformly loaded rectangular area for the Westergaard solution (adapted from Fadum 1948)

Table 6–2
Comparison of influence factors (I_σ) beneath the center of a square uniformly loaded area (after Duncan and Buchignani 1976; assumes a Poisson's ratio $\upsilon = 0$)

$\frac{\text{Ftg. Width } (B)}{\text{Ftg. Depth } (z)}$	I_σ	I_σ
	Boussinesq Theory	Westergaard Theory
∞	1.0000	1.0000
20	0.9992	0.9365
16	0.9984	0.9199
12	0.9968	0.8944
10	0.9944	0.8734
8	0.9892	0.8435
6	0.9756	0.7926
5	0.9604	0.7525
4	0.9300	0.6971
3.6	0.9096	0.6659
3.2	0.8812	0.6309
2.8	0.8408	0.5863
2.4	0.7832	0.5328
2.0	0.7008	0.4647
1.8	0.6476	0.4246
1.6	0.5844	0.3794
1.4	0.5108	0.3291
1.2	0.4276	0.2858
1.0	0.3360	0.2165
0.8	0.2410	0.1560
0.6	0.1494	0.0999
0.4	0.0716	0.0477
0.2	0.0188	0.0127
0	0.0000	0.0000

6–8. Stress increase due to embankment loads

The induced vertical stress distribution under a non-uniform embankment load can also be analyzed using elastic solutions. However, the change in stress is complicated by embankment geometry (side slopes). Embankments are generalized into two categories: (1) those having an infinite length, which is represented as trapezoidal, and (2) those having a finite length with end slopes.

a. Infinite embankment. Osterberg (1957) developed a chart for determining the influence factor for an infinitely long sloping embankment (Figure 6–12) by integration of the Boussinesq equations. The influence factor (I_σ) and corresponding induced vertical stresses ($\Delta\sigma_{zfill}$) beneath the centerline of an infinite embankment with side slopes can be computed per equation 6–7, see also Figure 6–11. The influence factor is a function of the embankment dimensions. To determine the influence factor (I_σ) and corresponding $\Delta\sigma_{zfill}$ at a location other than directly beneath the centerline of the embankment, the principle of superposition may be used.

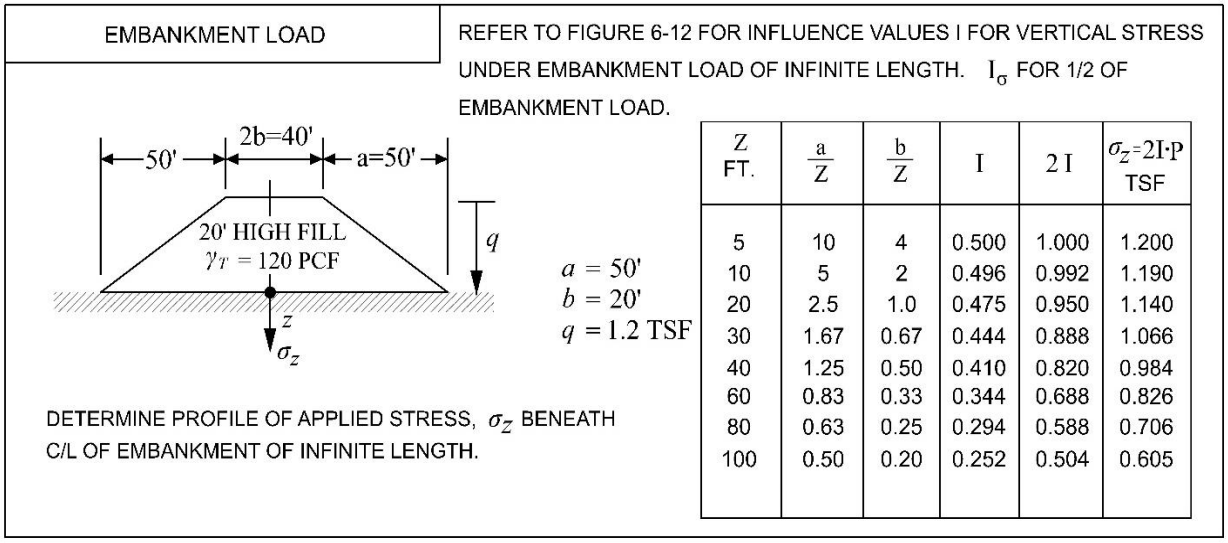
$$\Delta\sigma_{zfill} = 2\mathrm{I}_\sigma \cdot q \qquad (6\text{–}7)$$

where:

$\Delta\sigma_{zfill}$ = induced vertical stress (tsf), beneath centerline of embankment

I_σ = influence factor (dimensionless), beneath centerline of embankment, see Figure 6–12

q = unit load of embankment (tsf), = $\gamma_m \cdot H$, where γ_m = moist unit weight of embankment soil (tcf), and H = height of embankment at the centerline (ft)



Z FT.	a/Z	b/Z	I	2 I	σ_z=2I·P TSF
5	10	4	0.500	1.000	1.200
10	5	2	0.496	0.992	1.190
20	2.5	1.0	0.475	0.950	1.140
30	1.67	0.67	0.444	0.888	1.066
40	1.25	0.50	0.410	0.820	0.984
60	0.83	0.33	0.344	0.688	0.826
80	0.63	0.25	0.294	0.588	0.706
100	0.50	0.20	0.252	0.504	0.605

Figure 6–11. Influence values for vertical stress under embankment load of infinite length

b. Finite embankment. For finite embankments, the loading consists of two components: (1) the uniform areal load below the crest, and (2) the load under the sloping sections. The influence factor for the uniform areal load is determined using Figure 6–12, and is doubled since it is determined at the corner of the loaded area. The influence factor for the sloping portion is determined from Figure 6–13 and Figure 6–14.

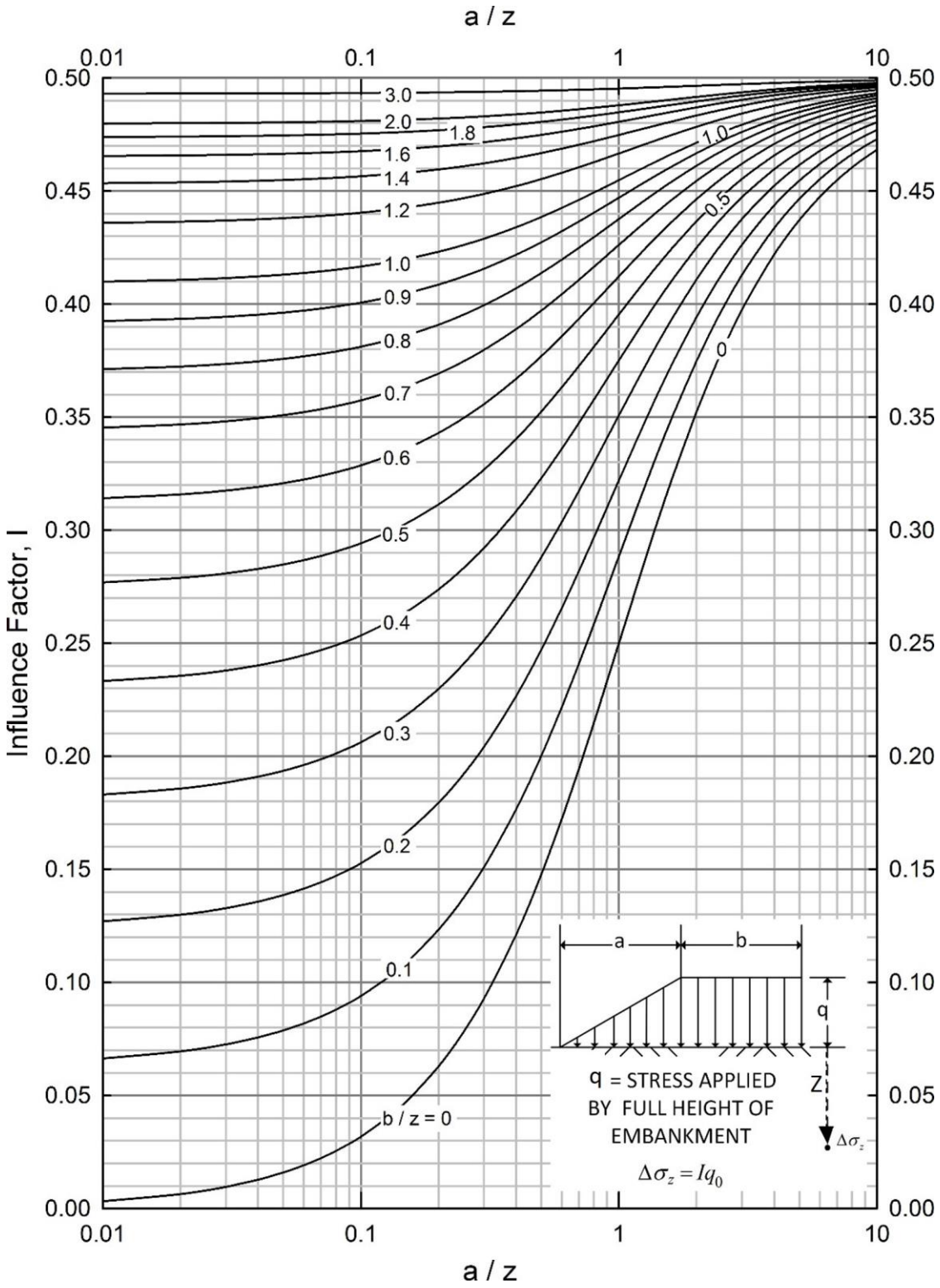


Figure 6–12. Influence chart for vertical stress embankment loading of infinite extent. Boussinesq case (Osterberg 1957)

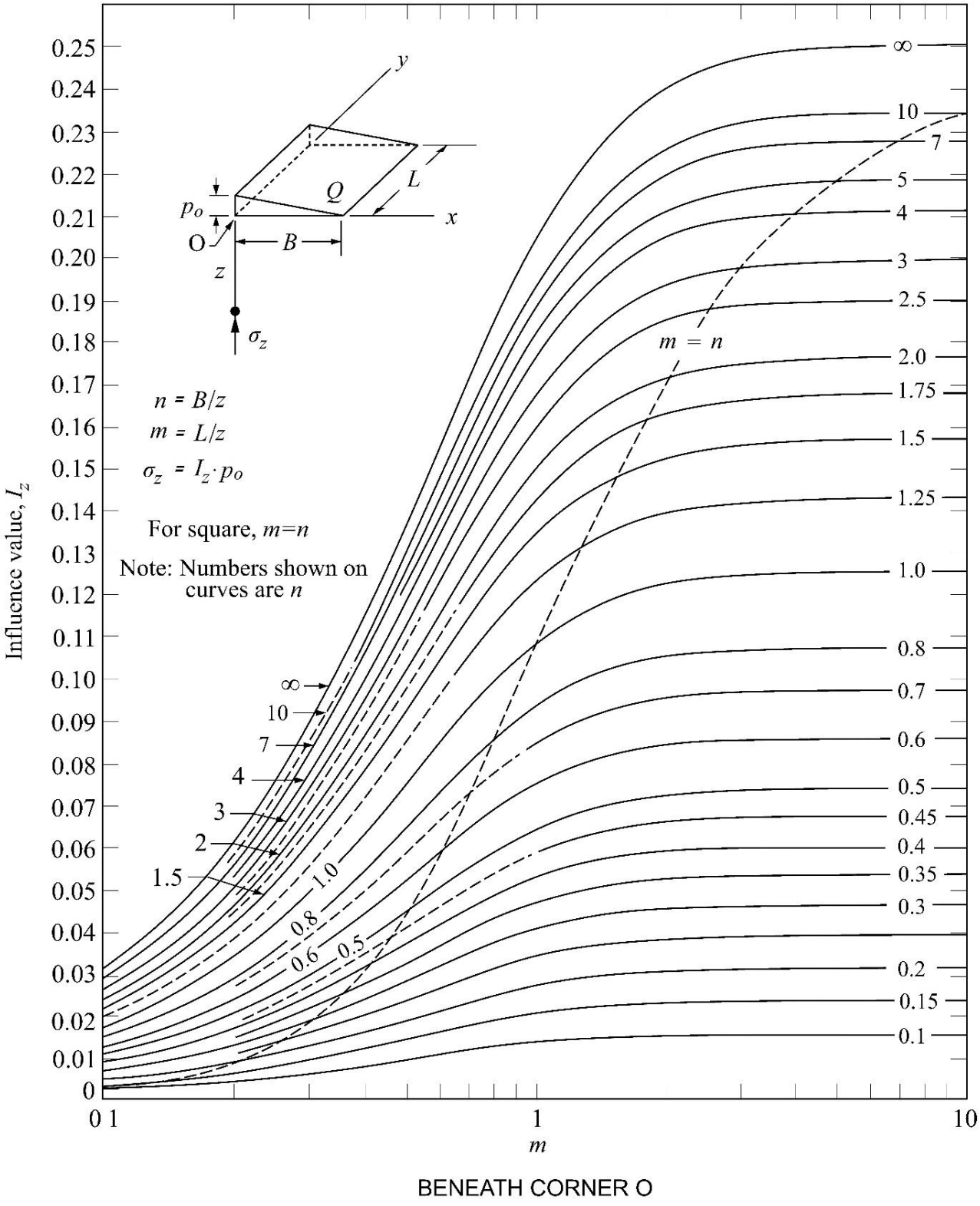


Figure 6–13. Influence chart beneath crest of slope (NAVFAC DM7.1 1982)

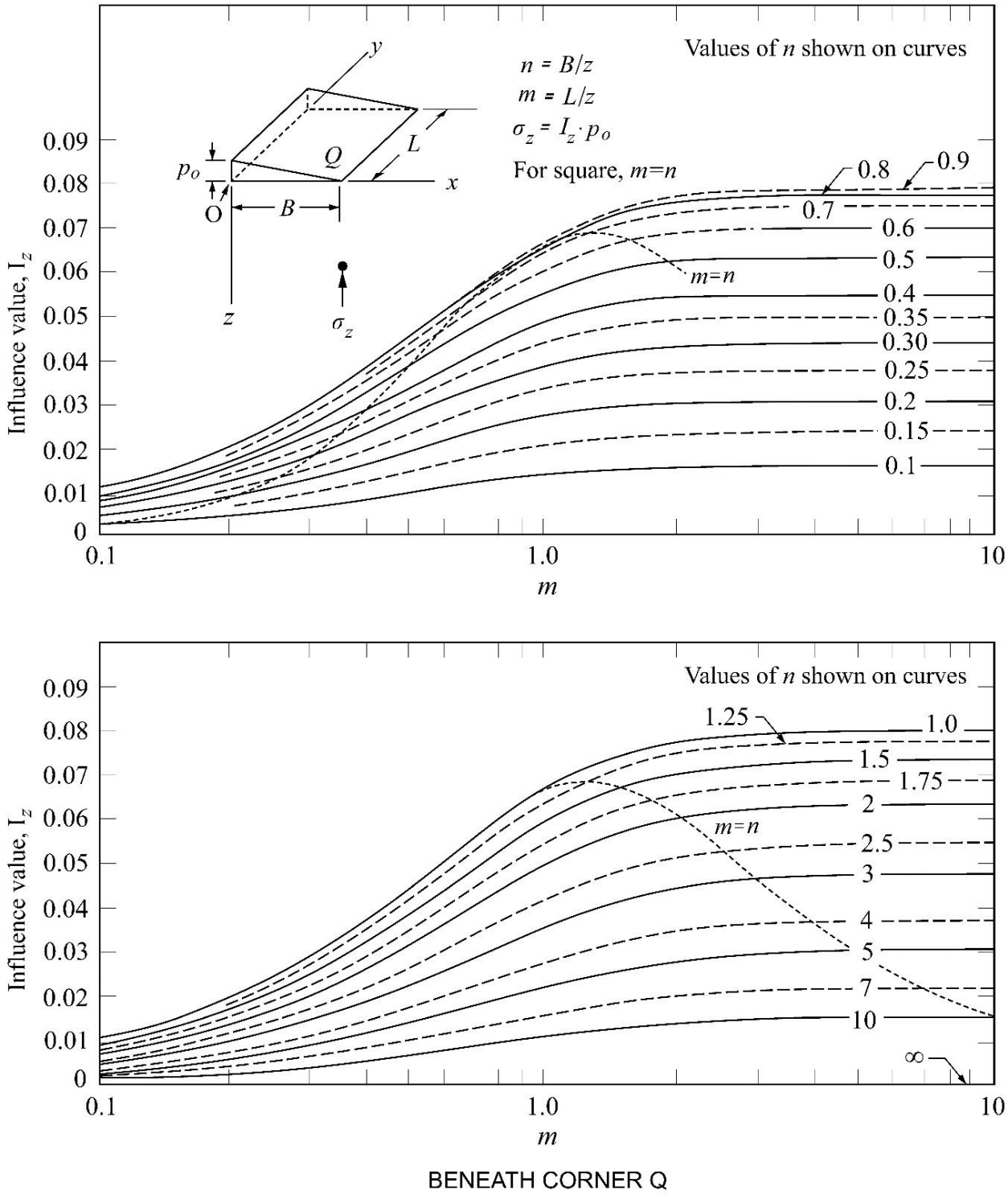


Figure 6–14. Influence chart beneath toe of slope (NAVFAC 7.1 1982)

6–9. Approximate methods for stress increase due to induced loads

Approximate methods and formulas can be used to calculate stresses due to induced loads. The approximate methods for stress distribution produce reasonable results in most applications and are useful for understanding the principles of stress distribution and for checking results obtained from elastic solutions or numerical analyses.

a. Boussinesq approximate solution. Approximate formulas based on the Boussinesq equations, equations 6–8 through 6–11, can be used to compute the influence factor (I_σ), beneath the center for the respective footing type. These equations produce the induced vertical stress ($\Delta\sigma_z$) beneath the footing per equation 6–6 within 5 percent of the Boussinesq solution (Poulos and Davis 1974).

$$\mathrm{I}_\sigma = 1 - \left(\frac{1}{1 + \left(\frac{B}{2z}\right)^2} \right)^{1.5} \qquad \text{(6–8)}$$

Circular Loaded Area

$$\mathrm{I}_\sigma = 1 - \left(\frac{1}{1 + \left(\frac{B}{2z}\right)^2} \right)^{1.76} \qquad \text{(6–9)}$$

Square Loaded Area

$$\mathrm{I}_\sigma = 1 - \left(\frac{1}{1 + \left(\frac{\mathrm{B}}{2z}\right)^{1.38+0.62\frac{B}{L}}} \right)^{2.60-0.84\frac{B}{L}} \qquad \text{(6–10)}$$

Rectangular Loaded Area

$$\mathrm{I}_\sigma = 1 - \left(\frac{1}{1 + \left(\frac{B}{2z}\right)^{1.38}} \right)^{2.60} \qquad \text{(6–11)}$$

Continuous (Strip) Loaded Area

where:

I_σ = influence factor beneath center of footing (dimensionless)
z = depth (ft), from base of foundation to point within the soil mass, typically at the center point of each layer evaluated for settlement
B = base width or diameter of footing (ft)
L = base length of rectangular footing (ft)

b. Approximate 2:1 method. With increasing depth below the base of a foundation, the area over which the load is applied at the surface increases at a 2:1 ratio and the magnitude of the loading stress decreases correspondingly (see Figure 6–15). For

example, the influence factor (I_σ) beneath the center of the loaded area for rectangular and circular footings is calculated per equation 6–12 and equation 6–13, respectively. The calculated influence factor can be used in equation 6–6 to calculate the induced vertical stresses ($\Delta\sigma_Z$) at the respective depth.

$$\mathrm{I}_\sigma = \frac{B \cdot L}{(B+z)(L+z)} \qquad \text{(6–12)}$$

Rectangular Footing

$$\mathrm{I}_\sigma = \frac{B^2}{(B+z)^2} \qquad \text{(6–13)}$$

Circular Footing

where:

I_σ = influence factor beneath center of a rectangular footing (dimensionless)
z = depth from base of foundation to point within the soil mass (ft), typically at the center point of each layer evaluated for settlement
B = base width of rectangular footing or diameter of circular footing (ft)
L = base length of rectangular footing (ft)

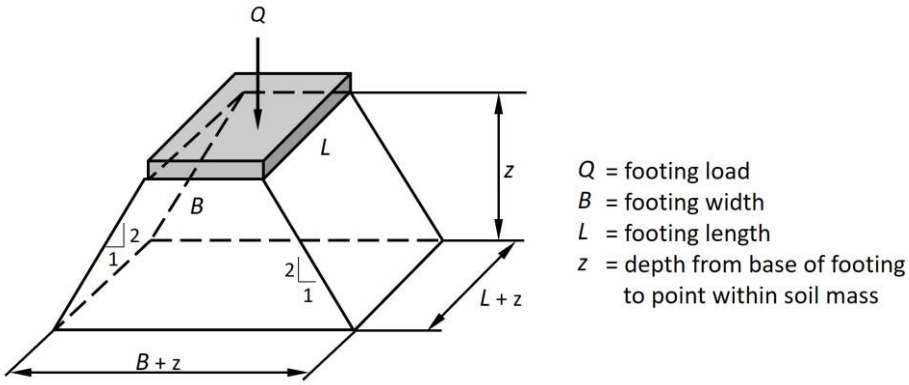


Figure 6–15. Approximate stress distribution by the 2:1 stress distribution method

c. Limitations of 2:1 method. The 2:1 method is not based on soil mechanics. As compared to Boussinesq theory, the 2:1 method tends to underestimate vertical stresses, especially for square footings at depths less than about 1.5B, where load influence is more significant. The error involved is not considerable; however, it is unconservative, and therefore, should be used only for preliminary estimates of induced stress. The 2:1 method should not be used to estimate stresses outside of loaded area (such as superposition).

Chapter 7
Settlement

7–1. Introduction

Stresses induced by foundation loads generate strain in the soil, resulting in vertical deformation of the soil, referred to as settlement. Foundation settlement can also result from changes in geostatic stress conditions, such as lowering of the groundwater table or the addition of fill. Guidance for estimating the settlement beneath shallow foundations due to changes in the soil's vertical effective stress and under constant loads due to creep are provided in this chapter in addition to foundation settlement influenced by excavation rebound. Static foundation loading conditions are assumed in this chapter. Settlement due to earthquake loading is addressed in Chapter 9.

7–2. Bearing pressure limitations

Settlement estimations are performed using the design loads described in paragraph 2–10, with adjustments to foundation size and bearing depth until a satisfactory gross or gross equivalent bearing pressure (q or q_{eq}), is obtained that produces tolerable settlement requirements. The methods presented are generally associated with vertical movements between 1 and 5 percent of the footing width. If estimated settlements exceed 5 percent of the footing width, the gross or gross equivalent bearing pressure should be reduced. The gross or gross equivalent bearing pressure (q or q_{eq}) used in settlement estimates must not exceed the ultimate bearing capacity (q_{ult}) divided by an adequate factor of safety. Gross and gross equivalent bearing pressures are defined in paragraph 5–2b.

7–3. Components of settlement

The total amount of settlement and how fast the settlement will occur is of interest in the design of foundations. The total settlement (δ_T) below a foundation, which is the response of effective stress applied to the soil, is the sum of the three components described by equation 7–1. Settlement estimates computed by the various methods in this chapter will be designated as settlement (δ), in lieu of distinguishing between immediate (δ_{i}) and primary consolidation (δ_{c}) settlement. Secondary settlement (δ_{s}) is considered separately.

$$\delta_T = \delta_{\mathrm{i}} + \delta_{\mathrm{c}} + \delta_{\mathrm{s}} \tag{7–1}$$

where:

δ_T = total settlement (inches)
δ_i = immediate or distortion settlement (inches)
δ_{c} = primary consolidation settlement (inches)
δ_{s} = secondary settlement or creep (inches)

a. Immediate or distortion settlement. For sand, immediate settlement (δ_{i}) is due to both volume change and distortion of the sand beneath the foundation load. For

saturated clay immediate settlement is due only to distortion; but for unsaturated clay, both distortion and volume change contribute to the immediate settlement. Some authors use the term “elastic settlement” and “immediate settlement” interchangeably. Immediate settlements generally occur within a period of about 7 days (Bowles 1996). Calculations of immediate settlement in cohesionless soils is complicated by a nonlinear stiffness that depends on the state of stress but may be estimated using elastic theory. Since the immediate settlement of clays is relatively small compared to primary consolidation settlement, it is often ignored.

b. Primary consolidation settlement. Primary consolidation settlement (δ_c) is the reduction in volume of soil due to an increase in effective stress from a reduction in the void ratio. Frequently, when engineers discuss consolidation settlement, they are referring to the expulsion of water of a fine-grain soil due to the application of a load, however the definition is broader and applies to coarse-grained soils as well as unsaturated, fine-grained soils that undergo void reduction due to an increase in effective stress. All soils can experience consolidation settlement when subjected to an increase in the effective stress. Since primary consolidation settlement results in a change in volume caused by an increase in effective stress in a drained condition, the drained soil properties should be used in computing the consolidation settlements in both sands and clays.

c. Secondary compression settlement. Secondary compression settlement (δ_s), also known as creep, is the reduction in volume of a soil mass caused by the application of a sustained load to the mass and due principally to the adjustment of the internal structure of the soil mass after most of the load has been transferred from the soil water to the soil solids (ASTM D653). Soils susceptible to significant secondary compression include highly compressible, high-plasticity soft clay and silt, and soil containing organic matter such as peat or Muskeg, which are typically too soft and unpredictable to support shallow foundations.

7–4. Accuracy of settlement estimates

Experience shows that predictions of settlement are reasonable and typically within 50 percent of actual settlements for many soil types. Time rates of settlement based on laboratory tests and empirical correlations may not be representative of the field because time rates are influenced by in situ fissures, existence of high-permeable sand or low-permeable bentonite seams, impervious boundaries, and non-uniform soil parameters as well as the time of construction. Accuracy of settlement estimates of secondary compression can be influenced by sample disturbance and variations within the soil profile. Sample disturbance tends to decrease the coefficient of secondary compression in the range of virgin compression.

7–5. Soil parameters for estimating settlement

Soil parameters for estimating settlement can be obtained from the results of (1) laboratory soil oedometer tests, or (2) correlations with in situ test results, to include the constrained modulus, M, from the flat plate DMT; equivalent soil modulus, (E_s), from the

CPT; the PMT modulus, (E_0), from the PMT; or relationships between compressibility and SPT N_{60} values.

a. Laboratory consolidation tests. Oedometer tests may be used to estimate the magnitude and rate of both primary and secondary settlement. The 1-D consolidation test (oedometer test) using incremental loading (ASTM D2435) is commonly used. Most of the existing empirical information on void ratio versus effective stress relationships of soft clays, including data on recompression (C_r), compression index, (C_c), and preconsolidation pressure (σ_p') originate from the incremental loading test. Continuous controlled-strain axial compression testing (ASTM D4186) may also be performed. Controlled-strain compression is typically referred to as constant rate of strain (CRS) testing.

b. In situ test correlations and strain rate effects. For important projects or where better estimates of foundation settlement are needed, correlations obtained from in situ "deformation" tests (such as DMT, PMT) that measure soil stiffness are preferred in lieu of penetration tests.

(1) *Penetration tests*. "Penetration" tests, such as the CPT and SPT strain the soil to failure (such as large shear strains greater than 10^{-1}), resulting in poor correlations with the constrained modulus (M) needed for estimating settlements at intermediate strain levels (Failmezger 2021). Where CPT testing is used, the constrained modulus should be correlated with the small-strain shear modulus (G_{max}) and corrected cone tip resistance (q_t) when calibrated with the constrained modulus computed from the DMT for the site-specific soils.

(2) *Deformation tests*. "Deformation" tests statically deform the soil at intermediate strain levels (such as shear strains between 10^{-5} to 10^{-3}), resulting in better correlations with the constrained deformation modulus and pressuremeter modulus as illustrated in Figure 7–1.

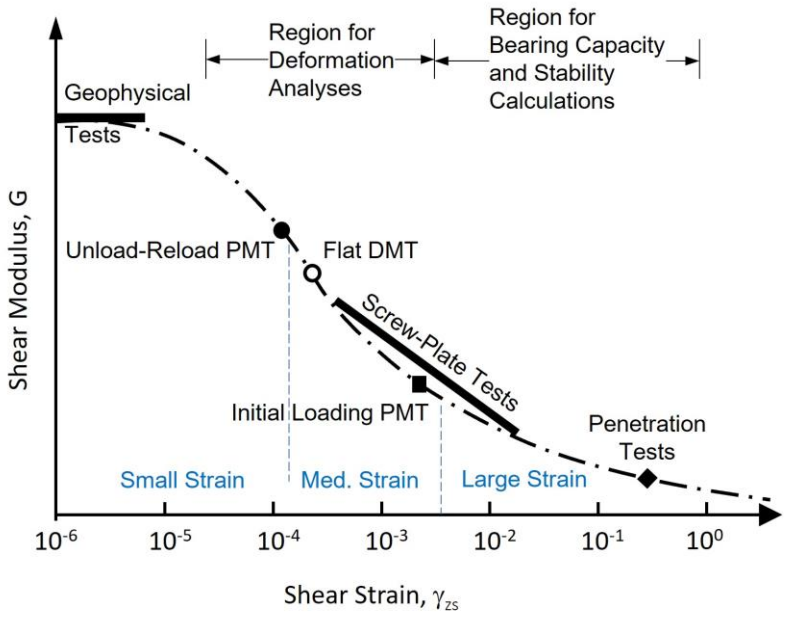


Figure 7–1. Variation of shear modulus with shear strain for various in situ tests as they pertain to deformation (settlement) and bearing capacity analysis (adapted from Mayne 2001)

7–6. Consolidation behavior of sands and clays

The fundamental difference between the consolidation of non-plastic silts and granular soils (such as sands and gravels) versus fine-grained cohesive soils (such as clays and plastic silts) is the rate at which the pore water dissipates from the soil under an increase in vertical effective stress. Consolidation (sometimes referred to as compression) in sands typically occurs during or soon after construction, whereas saturated clays typically compress for months, years, or decades, depending on soil permeability and the length of the flow path for dissipation of excess pore pressures.

7–7. Stress-strain models for one-dimensional consolidation settlement

Consolidation settlements are estimated based on 1-D strain conditions, also referred to as 1-D or confined compression, which can be simulated in the laboratory on relatively undisturbed or laboratory-compacted samples using an oedometer. Since the soil is constrained laterally during the consolidation test, the vertical stress is related to the vertical strain by the constrained modulus (M) as illustrated in Figure 7–2a. The soil-constrained modulus is usually considered a secant stiffness value, which means it represents the slope of the stress-strain curve between a small range of strain levels, where the stress-strain curve is represented by a straight line. However, it can also be approximated as a tangent stiffness value (or tangent line to the stress-strain curve) at a certain level of deformation.

a. Stress-strain plots of oedometer tests. Two common plots are the vertical effective stress (σ_z') versus strain (ε_z) with arithmetic scales for both axes as illustrated in Figure 7–2a and strain (ε_z), or void ratio (e) versus the logarithm of vertical effective

stress (σ_z'), also commonly referred to as the e-log-p curve, as illustrated in Figure 7–2b. Both the e-log-p (e-log-p method) and the linear σ_z' versus strain ε_z (incremental modulus method) curves are used for estimating settlements per paragraphs 7–9 and 7–10, respectively.

b. Relationship between stress-strain parameters for 1-D consolidation. As illustrated in Figure 7–2a, the 1-D constrained modulus (M) can be defined as the secant modulus of the stress-strain curve (Lambe and Whitman 1969) per equation 7–2. Values of C_c, C_r, and e_0 can be determined from the e-log-p curve as illustrated in Figure 7–2b. Since the constrained modulus is the inverse of the coefficient of volume compressibility, values of M, C_c, or C_r, and e_0 can be related by using equations 7–3a through 7–4. However, Figure 7–2a shows that m_v depends on vertical effective stress. For 1-D consolidation, other relations between stress-strain parameters are provided in Table 7–1.

$$M = \frac{\Delta\sigma_z'}{\Delta\varepsilon_z} = \frac{1}{m_v} \tag{7–2}$$

where:

M = constrained modulus in the range of loading (tsf); also denoted D'
$\Delta\sigma_z'$ = vertical effective stress increment (tsf)
$\Delta\varepsilon_z$ = unit deformation of the sample (in/in); also $\Delta e/(1+e_o)$

$\text{Normally Consolidated: } \sigma_z' \geq \sigma_p'$

$$M = \frac{(1+e_0)\Delta\sigma_z'}{\left[log\left(\frac{\sigma_f'}{\sigma_z'}\right)\right]C_c} \tag{7–3a}$$

$\text{Overconsolidated: } \sigma_f' < \sigma_p'$

$$M = \frac{(1+e_0)\Delta\sigma_z'}{\left[log\left(\frac{\sigma_f'}{\sigma_z'}\right)\right]C_R} \tag{7–3b}$$

$\text{Normally Consolidated to Overconsolidated: } \sigma_z' < \sigma_p' \; and \; \sigma_f' > \sigma_p'$

$$M = \frac{(1+e_0)\Delta\sigma_z'}{\left[log\left(\frac{\sigma_p'}{\sigma_z'}\right)\right]C_R + \left[log\left(\frac{\sigma_f'}{\sigma_p'}\right)\right]C_C} \tag{7–4}$$

where:

M = constrained modulus in the range of loading (tsf); also denoted D'
e_0 = initial void ratio (dimensionless)
$\Delta\sigma_z'$ = vertical effective stress increment (tsf)
σ_f' = final vertical effective stress (tsf)

σ_z' = vertical effective stress (tsf)
C_c = compression index (dimensionless)
C_r = recompression index (dimensionless)
σ_p' = preconsolidation stress (tsf)

Table 7–1
Relations between various stress-strain parameters for confined compression (adapted from Lambe and Whitman 1969)

	Constrained Modulus, *M*	**Coefficient of Volume Change, m_v**	**Coefficient of Compressibility, a_v**	**Compression Index, C_c**
Constrained modulus, M	$M = \frac{\Delta\sigma_z'}{\Delta\varepsilon_z}$	$M = \frac{1}{m_v}$	$M = \frac{1+e_o}{a_v}$	$M = \frac{(1+e_o)\sigma_{za}'}{0.434\ C_c}$
Coefficient of volume change, m_v	$m_v = \frac{1}{M}$	$m_v = \frac{\Delta\varepsilon_z}{\Delta\sigma_z'}$	$m_v = \frac{a_v}{1+e_o}$	$m_v = \frac{0.434C_c}{(1+e_o)\sigma_{za}'}$
Coefficient of compressibility, a_v	$a_v = \frac{1+e_o}{M}$	$a_v = (1+e_o)m_v$	$a_v = -\frac{\Delta e}{\Delta\sigma_z'}$	$a_v = \frac{0.434C_c}{\sigma_{za}'}$
Compression index, C_c	$C_c = \frac{(1+e_o)\sigma_{za}'}{0.434\ M}$	$C_c = \frac{(1+e_o)\sigma_{za}' \cdot m_v}{0.434}$	$C_c = \frac{a_v \cdot \sigma_{za}'}{0.434}$	$C_c = -\frac{\Delta e}{\Delta \log \sigma_z'}$

Notes:
e_0 denotes the initial void ratio.
σ_{za}' denotes the average of the initial and final stresses.

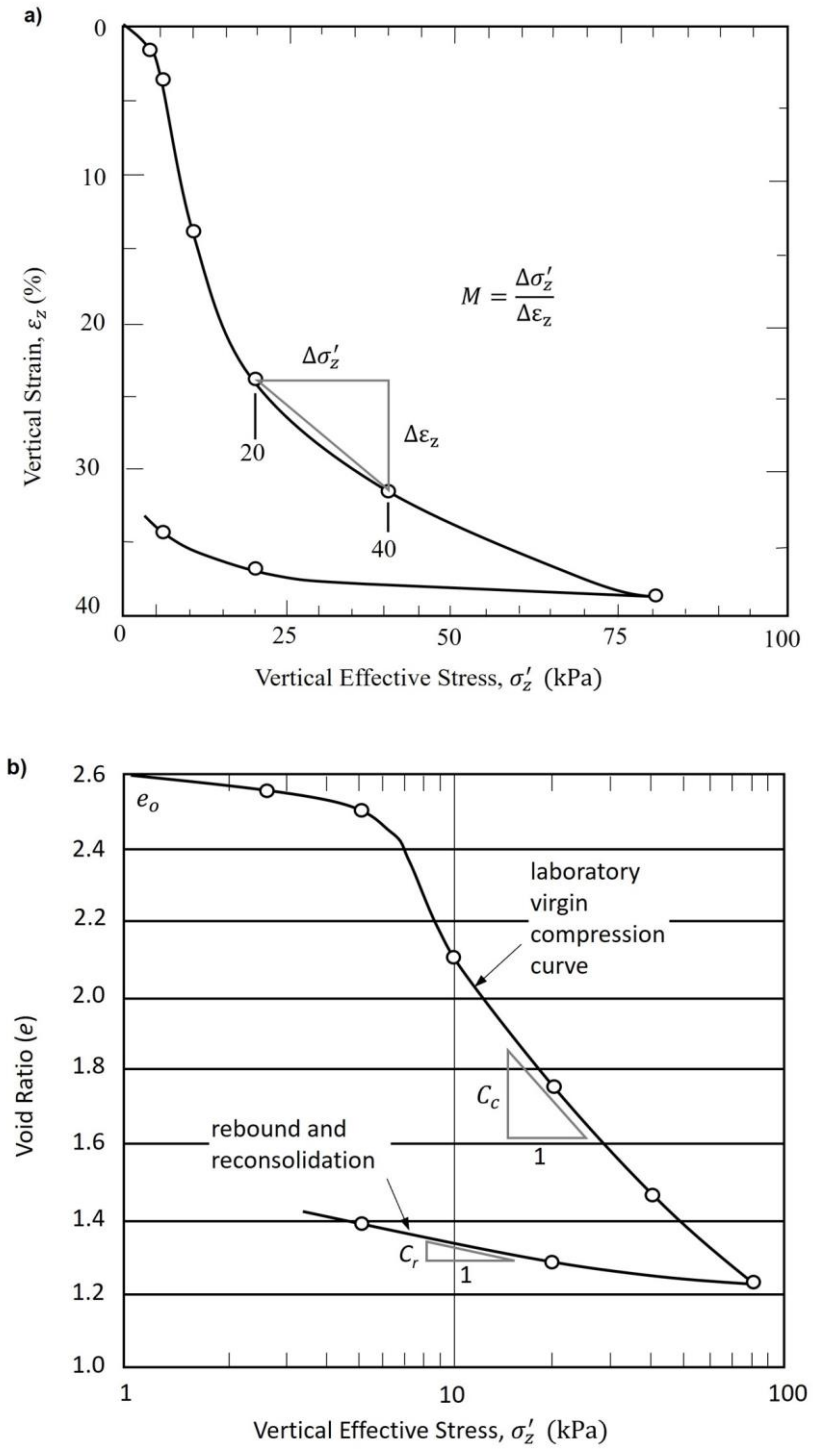


Figure 7–2. (a) Vertical strain (ε_z) versus vertical effective stress (σ_z') with arithmetic scales – soil stiffens as stress increases; (b) void ratio (e) versus the logarithm of vertical effective stress (σ_z') (both figures adapted from Holtz et al. 2011)

7–8. Settlement prediction methods

a. 1-D consolidation settlements. 1-D consolidation settlements can be estimated using either the constrained modulus (M) from the plot of vertical effective stress versus strain with arithmetic scales for both axes as illustrated in Figure 7–2a, or parameters obtained from the e-log-p curve as illustrated in Figure 7–2b. Both plots are developed based on the results of laboratory oedometer (consolidation) tests performed on relatively undisturbed cohesive samples or sometimes on compacted remolded materials.

b. Constrained modulus from in situ tests. Due to the difficulty in obtaining undisturbed samples and corresponding high degree of sample disturbance associated with granular soils (such as sands and gravels), oedometer tests are not usually performed on these materials. The constrained modulus (M) may be obtained from in situ DMT tests for all soil types to compute 1-D consolidation settlements. Settlements can also be estimated based on the results of in situ tests conducted in both fine-grained cohesive and cohesionless materials.

c. Fine-grained cohesive materials. The e-log-p method, Incremental Constrained Modulus method, or the PMT method should be used for estimating settlements in fine-grained materials, such as clays and plastic silts. Where undisturbed laboratory oedometer tests are performed adjacent to DMT tests within the same material, the constrained modulus (M), obtained from the oedometer test, and the DMT test can be compared (see paragraph 7–10b). Large differences in results could indicate poor sample quality or uncertainty between test results.

d. Cohesionless materials. The Incremental Constrained Modulus method, PMT method, Schmertmann's CPT method, or Burland and Burbidge SPT method may be used for estimating settlements in cohesionless materials such as sands, gravels, and non-plastic silts. Other SPT methods available in literature are discussed in Lutenegger and Degroot (1995) and Das and Sivakugan (2007) and may be used if experience has shown good results. Settlements estimated by methods using data from the CPT and SPT "penetration" tests tend to overpredict settlements. The accuracy and reliability of settlement estimates using SPT and CPT data are described in paragraph 7–14.

7–9. The e-log-p method

The e-log-p method uses the void ratio (e) versus vertical effective stress (σ_z') curve as illustrated in Figure 7–2b, from the results of oedometer tests to estimate settlements. The e-log-p method is appropriate for normally consolidated to lightly overconsolidated soft to medium-stiff sedimentary clays. For other natural soils, the Incremental Constrained Modulus method (paragraph 7–10) is more appropriate.

a. Laboratory e-log-p curve. To estimate 1-D primary consolidation settlement from the laboratory e-log-p curve, the preconsolidation stress (σ_p') may be estimated using the Casagrande (1936) graphical procedure or other suitable method. Using the Casagrande procedure, a "simplified" correction (Schmertmann 1955) to the virgin

compression curve can be made to reduce the effects of sample disturbance. The correction adjusts the laboratory curve to approximate the field virgin compression curve, referred to as the compression index (C_c). The recompression index (C_r), initial void ratio (e_0), and the initial vertical effective stress (σ'_{zo}) are also used to reconstruct the field virgin compression curve. Once the preconsolidation stress is estimated, the overconsolidation margin (equation 6–4) may be used to calculate values of the preconsolidation stress at other depths in the same stratum, provided it is of the same geologic origin.

(1) *Description of e-log-p field virgin compression versus laboratory consolidation curves*. The stress-strain history of a sedimentary clay during deposition, sampling, and reloading in the laboratory consolidation test is illustrated in Figure 7–3. It is from these observations that Casagrande developed his graphical procedure for defining the preconsolidation stress (σ'_p) and Schmertmann developed a graphical procedure to evaluate the slope of the field virgin compression Curve OAD, for determining the compression index (C_c). Casagrande's and Schmertmann's graphical procedures are described in paragraph 7–9a(2). Characteristics of the e-log-p field compression and laboratory consolidation curves are described as follows:

(a) Line OA represents the relationship between the void ratio (e) and logarithm vertical effective stress (σ'_z) of a particular element in the ground during deposition. The element has been consolidated to Point A.

(b) During sampling, the overburden is removed and the sample swells, or rebounds, as illustrated by the dashed Curve AB.

(c) Following removal of the sample from the Shelby tube and insertion into the oedometer ring, the sample is reloaded along Curve BCD, referred to as the laboratory consolidation test curve. The laboratory consolidation test Curve BCD eventually converges with the field virgin compression Curve OAD beyond Point D at a void ratio of $0.42e_0$.

(d) As disturbance of the sample increases, the laboratory consolidation test curve flattens, moving away from Point C in the direction of the arrow toward Point C′, resulting in the Curve BC′D making it difficult to estimate the preconsolidation stress (σ'_p).

(e) After a specified stress is reached, the specimen is rebounded along path DE, followed by reconsolidation of the specimen along Curve EF, eventually converging with the field virgin compression Curve OAD beyond Point D.

(2) *Determining settlement parameters from laboratory e-log-p curve*. Settlement parameters to be used in the estimation of 1-D primary consolidation can be obtained from the laboratory e-log-p curve. To correct for sample disturbance, the Casagrande 1936, and "Simplified" Schmertmann 1955, procedures discussed in Table 7–2 and Figure 7–4 can be used. However, the sample quality designation should be very good to excellent, per Table 4–2.

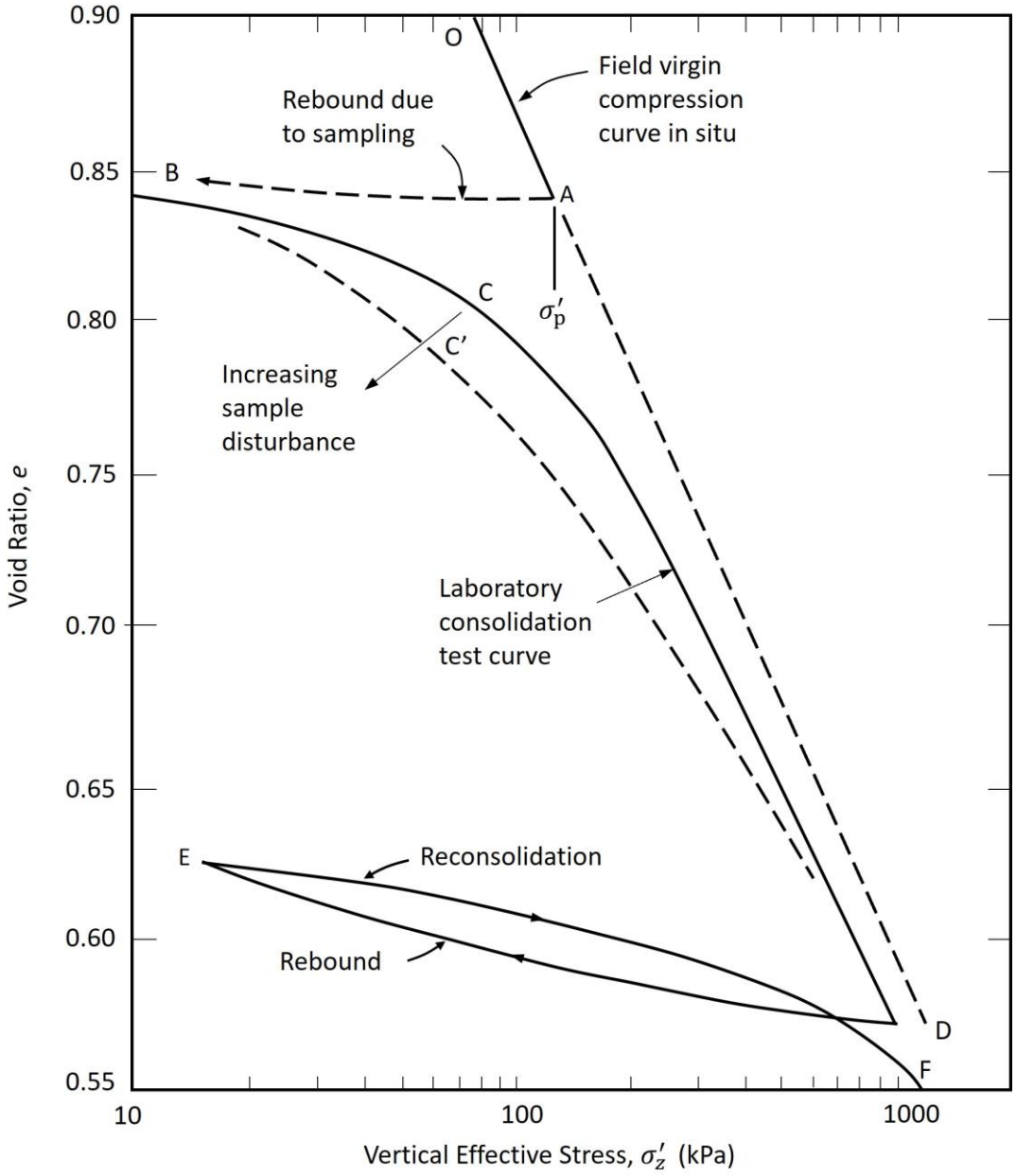


Figure 7–3. Laboratory e-log-p curve of sedimentary clay illustrating deposition, sampling (unloading), and reconsolidation in oedometer (adapted from Holtz et al. 2011)

Table 7–2
Procedure for determining settlement parameters from laboratory e-log-p curve (Casagrande 1936, and "Simplified" Schmertmann 1955, graphical procedures)

Step 1	Point B is plotted, which represents the effective stress and void ratio existing in the field at the sample location. The effective stress is usually the computed overburden pressure (σ'_{zo}). If the clay is not permitted to swell during storage, the field void ratio is the computed saturated initial void ratio (e_0) of the consolidation test sample.
Step 2	Because Point B represents the initial conditions for field compression, a perfect stress transfer from field to sampler to consolidometer would result in a test beginning with a constant e_0-line to Point B.
Step 3	The value of preconsolidation stress (σ'_p) is then determined by Casagrande's graphical procedure (Step 4). A constant stress line, x-x, is constructed at the chosen preconsolidation stress (σ'_p). Casagrande 1936, graphical construction procedure: (1) Plot Point C at the point of maximum radius of curvature (visually approximated) along the laboratory consolidation test curve. (2) Draw a horizontal line from Point C. (3) Draw a line tangent to the laboratory consolidation curve through Point C. (4) Draw the bisector between the horizontal and tangent lines. (5) Plot Point D at the intersection of the laboratory virgin compression line with the bisector. Point D indicates the maximum past pressure referred to as the preconsolidation stress (σ'_p).
Step 4	The geologic rebound is constructed parallel to the laboratory rebound, goes through Point B, and intersects line x-x at the Point F. For a normally consolidated soil where σ'_p equals σ'_{zo}, Point F will be identical to Point B without a C_r rebound curve.
Step 5	The initial laboratory virgin slope is extended as a straight line until it intersects the 42% e_0-line at Point G". The line G"-F is then constructed; this line represents the position and slope of the upper part of the estimated true virgin curve.
Step 6	A recompression curve is estimated for the geologic rebound line B-F. This curve probably merges with the estimated true initial virgin slope in such a manner that the graphical construction (Casagrande 1936) at the point of sharpest curvature will yield the preconsolidation point.
Step 7	Calculate the Recompression Index (C_r) per equation 7–5 which is the mean slope of the rebound-reconsolidation curve. Compute the mean slope between Points H and G of the rebound-reconsolidation curve (semilogarithmic plot).
Step 8	If the soil is overconsolidated ($\sigma'_{zo} < \sigma'_p$), calculate the compression index (C_c) using equation 7–6 (i.e. slope of the reconstructed field virgin compression curve.)
Step 9	Compute the overconsolidation margin (σ'_m) per equation 6–4.

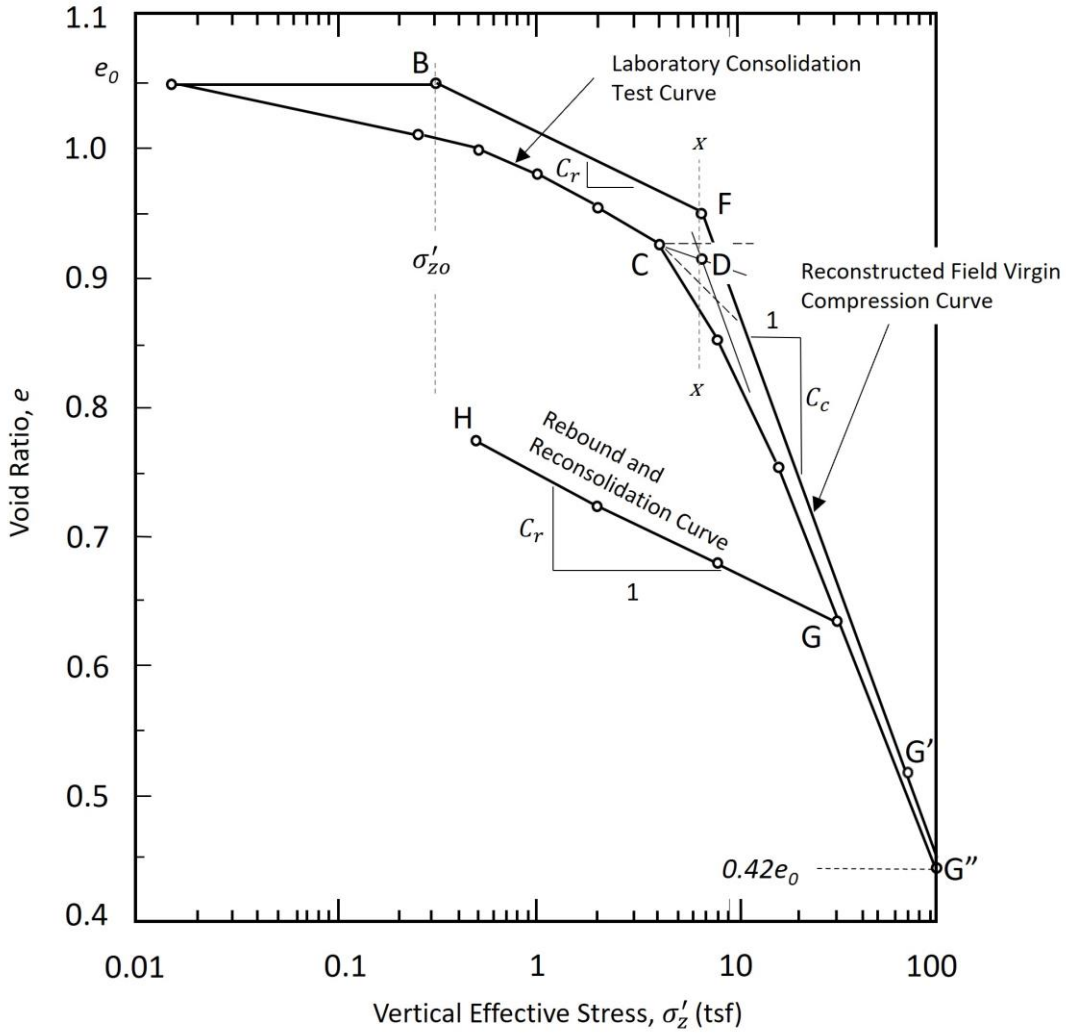


Figure 7–4. Example of reconstruction of the field virgin consolidation curve for overconsolidated soil using the "simplified" Schmertmann 1955, procedure

$$C_r = \frac{e_{\mathrm{H}} - e_{\mathrm{G}}}{\log_{10}\left(\sigma'_{z\mathrm{G}} \Big/ \sigma'_{z\mathrm{H}}\right)} \qquad (7\text{–}5)$$

Recompression Index, C_r

where:

C_r = recompression index (slope of the recompression curve)
e_H = void ratio at point of reload (Point H)
e_G = void ratio at intersection of unload-reload loop (Point G)
$\sigma'_{z\mathrm{G}}$ = vertical effective stress at Point G (tsf)
$\sigma'_{z\mathrm{H}}$ = vertical effective stress at Point H (tsf)

$$C_c = \frac{e_F - e_{G'}}{\log_{10}\left(\sigma'_{zG'} \Big/ \sigma'_{zF}\right)} \qquad (7\text{–}6)$$

Compression Index, C_c

where:

C_c = compression index (slope of the reconstructed field virgin compression curve)

e_F = void ratio at preconsolidation pressure on reconstructed curve (Point F)

$e_{G"}$ = void ratio at intersection of unload-reload loop (Point G)

$\sigma'_{G"}$ = vertical effective stress at Point G′; one-log cycle from Point F (tsf)

σ'_F = vertical effective stress at Point F (tsf)

b. Normally consolidated versus overconsolidated soil. At stresses less than the preconsolidation stress, soil will be relatively less compressible and its stress-strain, or e-log-p, behavior follows the recompression curve. At stresses greater than the preconsolidation stress, soil is more compressible and its stress-strain, or e-log-p, behavior follows the virgin curve.

(1) *Normally consolidated soil*. For normally consolidated soils, where the loading occurs along the reconstructed field virgin compressive curve as illustrated in Figure 7–5, the settlements are estimated per equation 7–7.

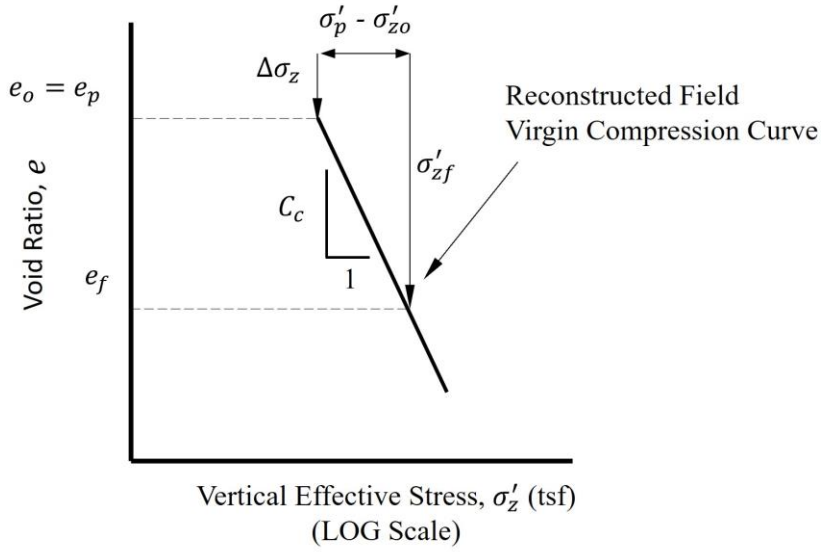


Figure 7–5. Typical e-log-p consolidation curve for normally consolidated soil

$$\delta_c = \sum \frac{C_c}{1+e_0} \mathrm{H} \cdot \log\left(\frac{\sigma'_{zf}}{\sigma'_{zo}}\right) \qquad \text{(7–7)}$$

Normally consolidated soil where $\sigma'_{zo} \approx \sigma'_p$

where:

δ_c = primary consolidation settlement (inches)
C_c = compression index (dimensionless)
e_0 = initial void ratio (dimensionless)
H = thickness of soil layer (ft)
σ'_{zo} = initial vertical effective stress at midpoint of soil layer, see paragraph 6–5a (tsf)
σ'_{zf} = final vertical effective stress at midpoint of soil layer = $\sigma'_{zo} + \Delta\sigma_z$ (tsf)
$\Delta\sigma_z$ = induced vertical stresses at the midpoint of soil layer, see paragraph 6–6 (tsf)

(2) *Overconsolidated soil*. For overconsolidated soils, where all the loading occurs along the rebound-recompression curve, settlements are estimated per equation 7–8. For overconsolidated soils, where part of the loading occurs along the rebound-recompression curve and part along the reconstructed field virgin compression curve, settlements are estimated per equation 7–9. Like normally consolidated soils, settlements can be significant when $\sigma'_{zf} > \sigma'_p$ as the slope of the virgin compressive curve (C_r), is greater than the slope of the rebound-reconsolidation curve (C_r). See Figure 7–6.

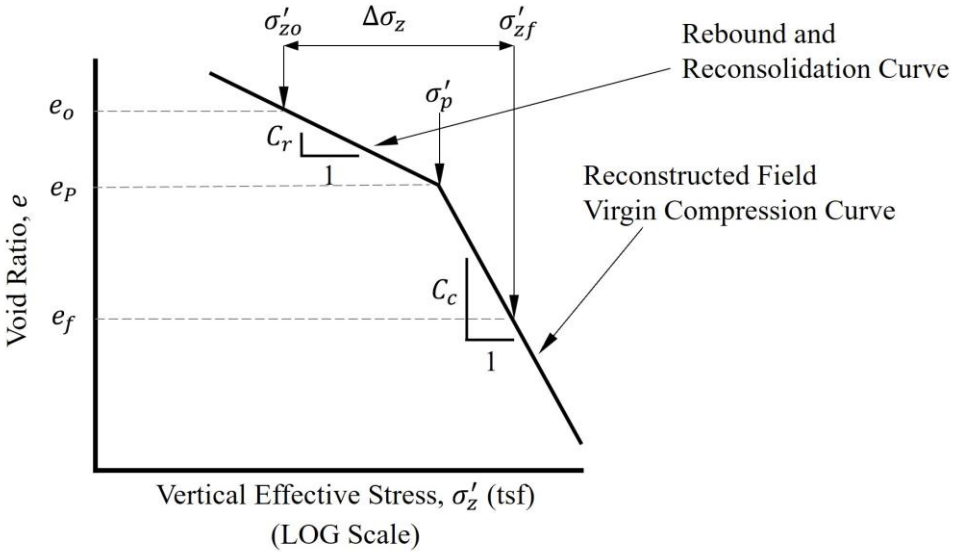


Figure 7–6. Typical *e*-log-p consolidation curve for overconsolidated soil

$$\delta_c = \sum \frac{C_r}{1+e_0} \mathrm{H} \cdot \log\left(\frac{\sigma'_{zf}}{\sigma'_{zo}}\right) \quad \text{(7–8)}$$

Overconsolidated Soil where $\sigma'_{zf} < \sigma'_p$

$$\delta_c = \sum \left[\frac{C_r}{1+e_0} \mathrm{H} \cdot \log\left(\frac{\sigma'_p}{\sigma'_{zo}}\right) + \frac{C_c}{1+e_0} \mathrm{H} \cdot \log\left(\frac{\sigma'_{zf}}{\sigma'_p}\right)\right] \quad \text{(7–9)}$$

Overconsolidated Soil where $\sigma'_{z0} < \sigma'_p < \sigma'_{zf}$

where:

δ_c = primary consolidation settlement (inches)
C_c = compression index (dimensionless)
e_0 = initial void ratio (dimensionless)
H = thickness of soil layer (ft)
σ'_{zo} = initial vertical effective stress at midpoint of soil layer (tsf), see paragraph 6–5a
σ'_{zf} = final vertical effective stress at midpoint of soil layer = $\sigma'_{zo} + \Delta\sigma_z$ (tsf)
$\Delta\sigma_z$ = induced vertical stress ($\Delta\sigma_z$) at the midpoint of each soil layer (tsf), see paragraph 6–6
C_r = recompression index (dimensionless)
σ'_p = preconsolidation stress at midpoint of soil layer (tsf)

c. Procedure for estimating 1-D primary consolidation settlement from laboratory e-log-p curves. The estimation of 1-D primary consolidation settlements from the laboratory e-log-p curves is shown in Table 7–3.

Table 7–3
Procedure for estimating 1-D primary consolidation settlements from laboratory e-log-p curve

Step 1	Determine e_o, σ'_m, σ'_p, C_c, C_r for each soil strata as illustrated in Figure 7–4 and Table 7–2. Empirical correlations for OCR, C_c, C_r,can be used when lab testing is not available.
Step 2	Divide the soil beneath the foundation into layers where the soil parameters are constant. The top layer should be at the base of the foundation and the bottom layer should be at a minimum depth of two times the footing width $2\mathrm{B}$ below the base of axisymmetric or rectangular footings, and six times the footing width $6\mathrm{B}$ below the base of strip footings, or until the increase in the induced vertical stress ($\Delta\sigma_z$) is negligible resulting in small acceptable increases in settlement, whichever is deeper. Since strains are nonlinear, multiple thinner layers will result in more accurate estimations of settlements. Layer thickness may be increased with depth. The number and thickness of layers should be adjusted to account for changes in soil properties.
Step 3	Compute the initial vertical effective stress (σ'_{zo}) at the midpoint of each layer reference paragraph 6–5a.
Step 4	From the overconsolidation margin (σ'_m) computed for each soil strata, compute the preconsolidation stress (σ'_p) at the midpoint of each layer per equation 6–4.
Step 5	Compute the gross bearing pressure (q) or gross equivalent bearing pressure (q_{eq}) for eccentrically loaded foundations per paragraph 5–2b.

Step 6	Compute the vertical effective stress at the bearing elevation of the footing (σ'_{zD}).
Step 7	Compute the influence factor (I_σ) for vertical stress per paragraph 6–7 and the corresponding induced vertical stress ($\Delta\sigma_z$) at the midpoint of each layer per equation 6–6. $\Delta\sigma_z = \mathrm{I}_\sigma(q - \sigma'_{zD})$ (6–6) where: σ'_{zD} = vertical effective stress at bearing elevation (tsf) from Step 6 q = use either the gross or gross equivalent bearing pressure (tsf), whichever is applicable; or the ultimate bearing capacity (q_{ult}) divided by factor of safety, whichever is less (tsf)
Step 8	Compute the final vertical effective stress (σ'_{zf}) at the midpoint of each layer per $\sigma'_{zf} = \sigma'_{zo} + \Delta\sigma_z$. where: σ'_{z0} = initial vertical effective stress at the midpoint of each layer (tsf) from Step 3 $\Delta\sigma_z$ = induced vertical stress at the midpoint of each layer (tsf) from Step 7
Step 9	Estimate the settlement of each layer using equations 7–7 through 7–9 as applicable. Sum the settlements of the soil layers within the zone of influence described in Step 2.

d. Corrections. The settlement calculation method presented in Table 7–3 applies to consolidation or volume change settlement under a perfectly flexible loaded area, for 1-D compression only. Corrections due to foundation rigidity are discussed in paragraph 6–6a(3). To account for 3-D and related overconsolidation effects, the Skempton and Bjerrum (1957) correction can also be applied as outlined in UFC 3-220-10.

7–10. Incremental Constrained Modulus method

The Incremental Constrained Modulus method utilizes the constrained (secant) modulus (M) obtained from the results of laboratory oedometer tests, as illustrated by the σ'_z versus ε_z curve in Figure 7–2a, or from in situ testing such as the CPT or DMT for estimating settlements. Because of the low- to medium-strain level of the soil during testing (Figure 7–1), the DMT is often the preferred deformation test method where speed, economy, and accuracy are desired in obtaining the constrained modulus.

a. Constrained modulus from dilatometer. The DMT constrained modulus (M or M_{DMT}) can be calculated per paragraph 3–9c(2). M_{DMT} is defined as the vertical, drained, confined (1-D) tangent modulus at the pre-insertion in situ vertical effective stress (σ'_{vo}) as shown in Figure 7–7 and is comparable with the secant constrained modulus (M) obtained from high-quality laboratory oedometer tests.

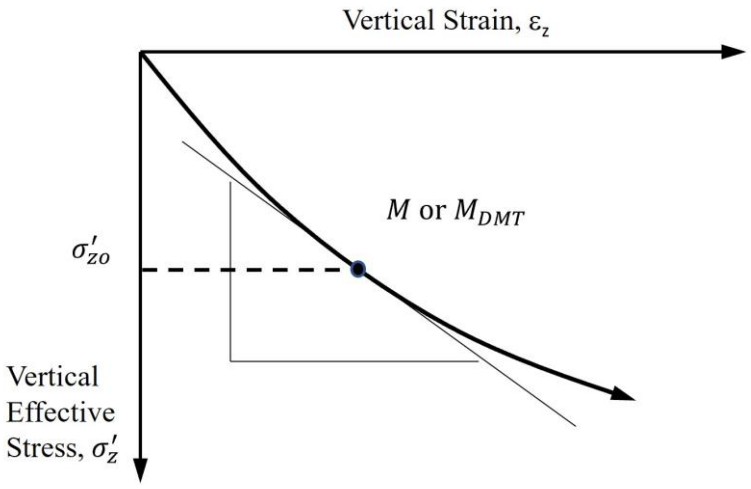


Figure 7–7. Definition of dilatometer test constrained (tangent) modulus, M_{DMT}

b. Comparison between M obtained from dilatometer and oedometer. The constrained modulus can be obtained from laboratory 1-D oedometer tests on very good or better samples or DMT data. Comparisons between M determined by DMT and high-quality oedometer testing is illustrated in Figure 7–8 (a and b). For routine practice, both methods are considered reasonably accurate and good agreement can be achieved between predicted versus measured settlements.

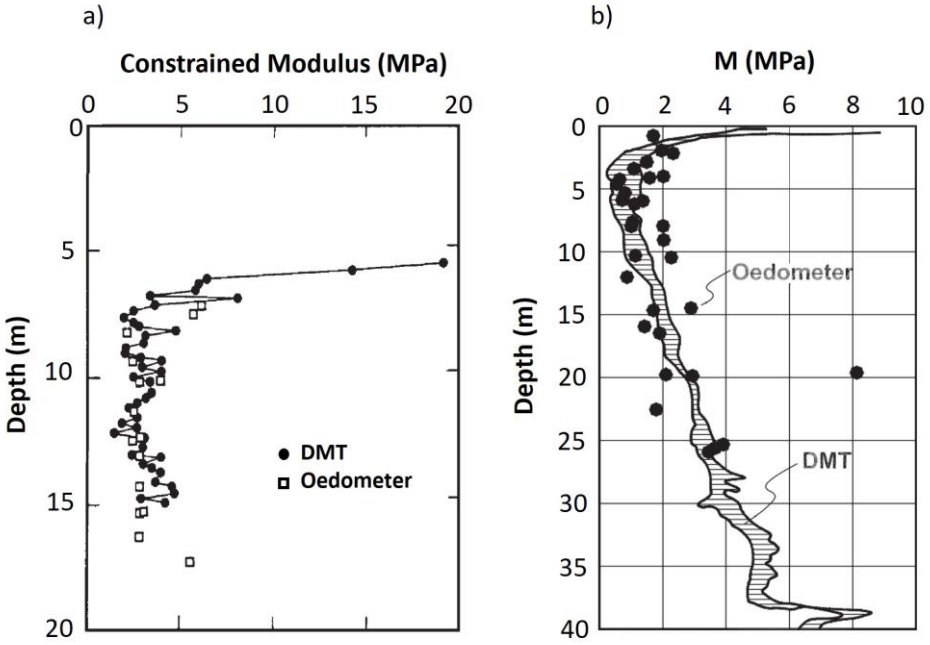


Figure 7–8. Comparison between M obtained from dilatometer test and high-quality oedometers from a) Tokyo Bay clay, Komatsugawa site, Japan (Iwasaki et al. 1991); and b) Onsoy clay, Norway (Lacasse 1986)

c. Estimating settlement by Incremental Constrained Modulus method using dilatometer.

(1) Induced vertical stress ($\Delta\sigma_z$) and settlement can be calculated as illustrated in Figure 7–9, using the techniques discussed in paragraphs 6–6 and 6–7 and using equation 7–10. The Incremental Constrained Modulus method using M_{DMT} is based on the settlement modulus concept developed by Janbu (1963, 1967). Settlement estimates using the M_{DMT} values can be calculated per Schmertmann's "Ordinary" method as described in Table 7–4.

(2) Schmertmann also developed a "Special" method which uses stress-history to modify the predicted constrained modulus. The "Ordinary" method predicts settlement based on the DMT modulus alone; whereas the "Special" method uses stress-history to modify the predicted constrained modulus. Settlements calculated from the "Ordinary" method are typically within 10 percent of the "Special" method for normal soils. For very soft or loose soils, the "Special" method typically results in reduced settlement, reference Schmertmann (1986).

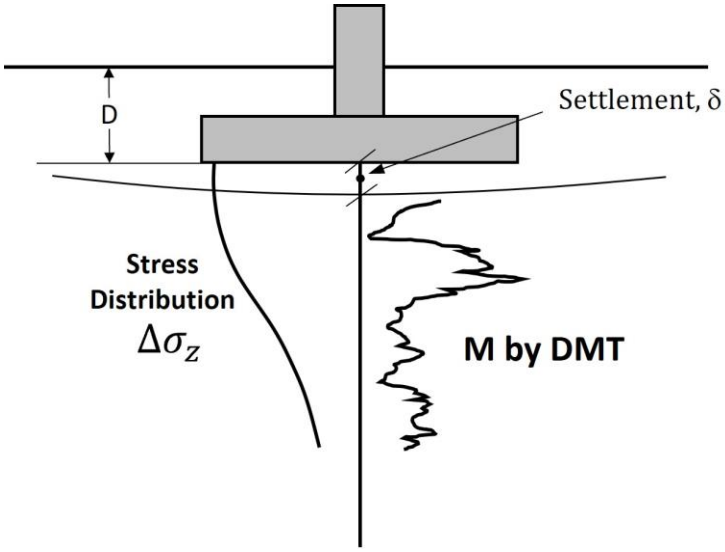


Figure 7–9. Illustration of constrained modulus (M_{DMT}) and change in induced vertical stress ($\Delta\sigma_z$) from dilatometer test beneath a footing (adapted from Marchetti 1997)

$$\delta = \sum \frac{I_\sigma (q - \sigma'_{zD})}{M} \mathrm{H} = \sum \frac{\Delta\sigma_z}{M} \mathrm{H} \qquad (7\text{–}10)$$

where:

δ = settlement (inches)

q = use either the gross or gross equivalent bearing pressure (tsf), whichever is applicable; or the ultimate bearing capacity (q_{ult}) divided by factor of safety, whichever is less (tsf)

σ'_{zD} = vertical effective stress at the bearing elevation of the footing (tsf)

I_σ = influence factor for vertical stress per paragraph 6–7 (dimensionless)

$\Delta\sigma_z$ = induced vertical stress at the midpoint of each layer per equation 6–6 (tsf)

M = constrained modulus for given soil layer (tsf)

H = thickness of soil layer (ft)

Table 7–4
Procedure for estimating settlements by Schmertmann's "Ordinary" dilatometer test method

Step 1	Obtain the constrained modulus (M) as a function of depth from DMT data.
Step 2	Divide the soil beneath the foundation into layers equal to depths in which the constrained modulus was obtained, typically 8-inch layers for DMT tests. The top layer should be at the base of the foundation and the bottom layer should be at a minimum depth of two times the footing widths $2\mathrm{B}$ below the base of axisymmetric or rectangular footings, and six times the footing width $6\mathrm{B}$ below the base of strip footings, or until the increase in the induced vertical stress ($\Delta\sigma_z$) is negligible, resulting in small acceptable increases in settlement, whichever is deeper.
Step 3	Assign a constrained modulus (M) for each layer, reference paragraph 3–9c(2).

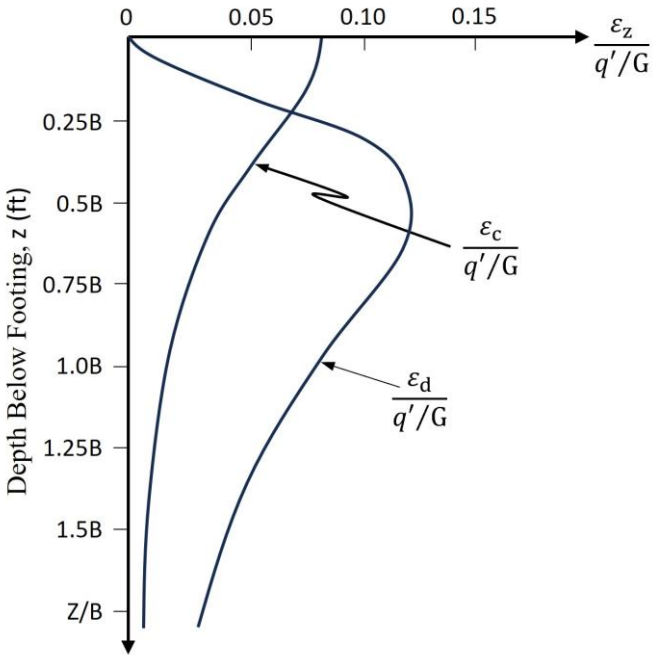
Step 4	Compute the gross bearing pressure (q), or gross equivalent bearing pressure (q_{eq}) for eccentrically loaded foundations (tsf) per paragraph 5–2b.
Step 5	Compute the vertical effective stress at the bearing elevation of the footing (σ'_{zD}).
Step 6	Compute the influence factor (I_σ) for vertical stress per paragraph 6–7 and the corresponding induced vertical stress ($\Delta\sigma_z$) at the midpoint of each layer per equation 6–6, $\Delta\sigma_z = I_\sigma(q - \sigma'_{zD})$. where: σ'_{zD} = vertical effective stress at bearing elevation (tsf) from Step 5 q = use either the gross or gross equivalent bearing pressure (tsf), whichever is applicable; or the ultimate bearing capacity (q_{ult}) divided by factor of safety, whichever is less (tsf)
Step 7	Estimate the settlement of each layer using equation 7–10 as applicable. Sum the settlements for each layer within the zone of influence described in Step 2.

d. Corrections. The Schmertmann (1986) "Ordinary" method applies to primary consolidation or volume change settlement of soil under a perfectly flexible loaded area, in 1-D compression only. Schmertmann (1986) does not recommend a correction for foundation rigidity, 3-D effects, creep, or aging for the DMT method, as these are often assumed to cancel each other. Marchetti (1997) also recommends not to account for 3-D effects, as the differences in settlement between 1-D and 3-D are small and the uncertainties associated with 3-D analysis reduce the precision of the settlement computation. Reference Schmertmann (1986) for a discussion of these corrections.

7–11. Ménard and Rousseau pressuremeter method

a. Shear and volumetric deformation. The theoretical background of the Ménard and Rousseau PMT method is composed of two settlement components: (1) settlement from the shear deformation (undrained) arising from the deviatoric strain tensor, and (2) settlement resulting from the volumetric deformation (consolidation) arising from the spherical strain tensor.

b. Vertical strain distribution. The vertical strain distribution, (ε_z), versus depth for both strain tensors are illustrated in Figure 7–10. The deviatoric strain, (ε_d), is significant to a depth of two footing widths, $2B$, below the footing bearing elevation, whereas the spherical strain, (ε_c), decreases rapidly with depth indicating that consolidation settlement is prevalent at shallow depths, on the order of one-half the footing width, $B/2$, below the footing bearing elevation. Therefore, for a wide footing over a thin compressible layer, the volumetric (consolidation) settlement will dominate, whereas for footings on deep relatively uniform deposits, the deviatoric settlement will dominate.



q' = net bearing pressure (ksf)
q = gross bearing pressure at base of footing (ksf)
σ'_{zD} = vertical effective stress at bearing elevation of footing at depth, *D*
G = shear modulus
ε_z = vertical strain
ε_d = deviatoric strain
ε_c = spherical strain

Figure 7–10. Strain distribution versus depth (adapted from Baguelin and Shields 1978)

c. Ménard and Rousseau method for estimating settlements using the pressuremeter.

(1) Equation 7–11 is the semi-empirical PMT settlement equation proposed by Ménard and Rousseau (1962), and is valid for footings bearing on deep, relatively uniform deposits. Equation 7–12 is used for estimating settlements when a soft/loose layer is encountered within a depth of two footing widths, $2B$, below the base of axisymmetric or rectangular footings, and six times the footing width, $6B$, below the base of strip footings, or until the increase in the induced vertical stress ($\Delta\sigma_z$), is negligible, resulting in small acceptable increases in settlement, whichever is deeper.

(2) The estimation of settlement using the Ménard and Rousseau procedure is described in Table 7–5. Steps 1 through 12 are for footings bearing on deep, relatively uniform deposits. Steps 1 through 17 are for estimating settlements when a soft/loose layer is encountered within a depth of $2B$ and $6B$ below the footing bearing elevation for axisymmetric and plane strain (strip) footings, respectively.

$$\delta = \text{deviatoric settlement} + \text{spherical (consolidation) settlement}$$

$$\delta = \left[\frac{2}{9\mathrm{E_d}}(q - \sigma'_{zD})\mathrm{B_0}\left(\lambda_\mathrm{d}\frac{B}{B_0}\right)^{\alpha_\mathrm{d}}\right] + \left[\frac{\propto_\mathrm{c}}{9\mathrm{E_c}}(q - \sigma'_{zD})\lambda_\mathrm{c}\cdot B\right] \quad (7\text{–}11)$$

General Equation
(Deep Relatively Uniform Deposits)

$$\delta = \text{deviatoric settlement} + \text{spherical (consolidation) settlement} + \text{soft/loose layer settlement}$$

$$\delta = \left[\left[\frac{2}{9\mathrm{E_d}}(q - \sigma'_{zD})B_0\left(\lambda_\mathrm{d}\frac{B}{B_0}\right)^{\alpha_\mathrm{d}}\right] + \left[\frac{\propto_\mathrm{c}}{9\mathrm{E_c}}(q - \sigma'_{zD})\lambda_\mathrm{c}\cdot B\right]\right] + \left[\sum \alpha_\mathrm{L}\left(\frac{1}{\mathrm{E_L}} - \frac{1}{\mathrm{E_d}}\right)\mathrm{I}_\sigma(q - \sigma'_{zD})\mathrm{H}\right] \quad (7\text{–}12)$$

Soft Layer Within $2B$

where:

δ = settlement (feet)
q = use either the gross or gross equivalent bearing pressure (tsf), whichever is applicable; or the ultimate bearing capacity (q_{ult}) divided by factor of safety, whichever is less (tsf)
σ'_{zD} = vertical effective stress at the bearing elevation of the footing (tsf)
I_σ = influence factor for vertical stress per paragraph 6–7 (dimensionless)
$\mathrm{E_d}$ = average pressuremeter modulus within the zone of the deviatoric tensor per equation 7–14 (tsf) to a depth of $8B$ below the footing bearing elevation (tsf)
$\mathrm{E_c}$ = average pressuremeter modulus within the zone of the spherical tensor to a depth of $B/2$below the footing bearing elevation (tsf)
$\mathrm{E_L}$ = average pressuremeter modulus of the soft/loose layer per the harmonic mean, equation 7–15 (tsf)
H = thickness of soft layer (feet)
B = base width of footing (feet)
B_0 = reference footing width = 2.0 ft
λ_d = shape factor for deviatoric term per Figure 7–11 (dimensionless)
λ_c = shape factor for spherical term per Figure 7–11 (dimensionless)
$\propto_\mathrm{d}$ = rheological factors for deviatoric term per (dimensionless)
$\propto_\mathrm{c}$ = rheological factors for spherical term per Table 7–6 (dimensionless)
$\propto_\mathrm{L}$ = rheological factors for soft/loose layer per Table 7–6 (dimensionless)

Table 7–5
Procedure for estimating settlements using the Ménard and Rousseau pressuremeter test method

Step 1	Calculate the pressuremeter modulus (E_o), reference paragraph 4–10d(4) for discussion of in situ test parameters. Designing beyond the yield pressure (P_y) is not recommended. If it needs to be done, then the modulus used should be the secant modulus related to the line joining the beginning of the straight line in the PMT curve to the point corresponding to the pressure applied in the geotechnical problem.
Step 2	Compute the ultimate bearing capacity (q_{ult}) per paragraph 5–5.
Step 3	Compute the gross bearing pressure (q) or gross equivalent bearing pressure (q_{eq}) for eccentrically loaded foundations (tsf) per paragraph 5–2b.
Step 4	Apply a factor of safety to Step 2 per Table 2–1 and choose the lesser of the values of Steps 2 and 3 for use in equation 7–11 or 7–12.
Step 5	Compute the vertical effective stress at the footing bearing elevation (σ'_{zD}).
Step 6	Divide the soil beneath the foundation into 16 elementary layers of $0.5B$ thickness to a depth of $8\mathrm{B}$ below the foundation and combine these sixteen elementary layers into five distinct layers, E_1, E_2, $E_{3/4/5}$, $E_{6/7/8}$, and $E_{9/16}$, as illustrated in Figure 7–12.
Step 7	Compute the average pressuremeter modulus ($\mathrm{E_k}$) of each of the five combined distinct layers by the harmonic mean technique, equation 7–13. $$\mathrm{E_k} = \frac{\mathrm{n}}{\frac{1}{\mathrm{E_1}} + \frac{1}{\mathrm{E_2}} + \cdots + \frac{1}{\mathrm{E_n}}} \quad (7\text{–}13)$$ where: $\mathrm{E_k}$ = average pressuremeter modulus of the distinct $\mathrm{k^{th}}$ layer per harmonic mean (tsf) n = number of E_0, values within a given layer $\mathrm{E_1, E_2, E_n}$ = PMT modus values, E_0, within distinct $\mathrm{k^{th}}$ layer (tsf) If a pressuremeter data is not available at deeper layers, estimate the modulus of these deeper layers based on other data available at the site. For example, if the modulus is increasing with depth but no modulus data is available for layers 9 through 16, estimate the modulus data from Layer 8 or Layers 9 through 16 at other test locations, if applicable.
Step 8	Compute the average pressuremeter modulus ($\mathrm{E_k}$) within the zone of influence of the spherical tensor ($\mathrm{E_c}$) to a depth of $0.5B$ (first layer) below the footing bearing elevation. Therefore, $\mathrm{E_c} = \mathrm{E_1}$ from Step 7.
Step 9	Compute the equivalent distortion modulus ($\mathrm{E_d}$) within the zone of influence of the deviatoric tensor to a depth of $8B$ (16 layers beneath the footing bearing elevation described in Step 6) per equation 7–14. $$\frac{1}{\mathrm{E_d}} = \frac{1}{4}\left(\frac{1}{\mathrm{E_1}} + \frac{1}{0.85\mathrm{E_2}} + \frac{1}{\mathrm{E_{3/4/5}}} + \frac{1}{2.5\mathrm{E_{6/7/8}}} + \frac{1}{2.5\mathrm{E_{9/16}}}\right) \quad (7\text{–}14)$$ where: $\mathrm{E_d}$ = equivalent distortion modulus ($\mathrm{E_d}$) within the zone of influence of the deviatoric tensor to a depth of $8B$. E_1, E_2, $E_{3/4/5}$, $E_{6/7/8}$, and $E_{9/16}$ are computed by the harmonic mean method from Step 7
Step 10	Obtain the rheological factors from Table 7–6. Rheological factors ($\propto_{\mathrm{d}}$ and $\propto_{\mathrm{c}}$) are a function of the soil type and state of consolidation from the ratio of $\mathrm{E_d}/\mathrm{p}_{ne}$ and $\mathrm{E_c}/\mathrm{p}_{ne}$, respectively. p_{ne} is the equivalent net limit pressure within the shear zone per equation 5–24 in tsf units.

Step 11	Obtain the deviatoric and spherical shape factors (λ_d and λ_c), from Figure 7–11. The shape factors are a function of the footing base length to width ratio (L/B). For a round footing λ_d and λ_c are equal to 1.
Step 12	Estimate the settlement per equation 7–11.
Step 13	If a soft/loose layer is encountered within a depth of $2B$ or $6B$ below the footing bearing elevation for an axisymmetric or strip footing, respectively, compute the average pressuremeter modulus of each of the soft/loose layers by the harmonic mean technique per equation 7–15. $$E_L = \frac{n}{\frac{1}{E_1} + \frac{1}{E_2} + \cdots \frac{1}{E_n}} \quad (7\text{–}15)$$ where: E_L = average PMT modulus of soft/loose layer per harmonic mean (tsf) E_1, E_2, E_n = PMT modulus values (E_0) within distinct soft/loose layer (tsf) n = number of E_0 values within a given layer
Step 14	Compute the influence factor (I_σ) for vertical stress per paragraph 6–7 and the corresponding induced vertical stress ($\Delta\sigma_z$) at the midpoint of the soft/loose layers per equation 6–6, $\Delta\sigma_z = I_\sigma(q - \sigma'_{zD})$. q is the value per Step 4.
Step 15	Compute the equivalent net limit pressure (p_{ne}) within the soft/loose layer per equation 5–24. $$s = p_{ne} = \sqrt[y]{P^*_{L1} \times P^*_{L2} \times \ldots P^*_{Ly}} \quad (5\text{–}24)$$ where: s = equivalent in situ testing parameter indicative of soil strength, in this case for the PMT P^*_L = net limit pressure = $P_L - P_o$ (ksf); see also equation 4–17 p_{ne} = equivalent net limit pressure in the shear zone (ksf) P^*_{L1} = 1st measured value of P^*_L within the bearing capacity shear zone (ksf) P^*_{L2} = 2nd measured value of P^*_L within the bearing capacity shear zone (ksf) P^*_{Ly} = y^{th} measured value of P^*_L within the bearing capacity shear zone (ksf)
Step 16	Obtain the rheological factor ($\propto_L$) from Table 7–6. $\propto_L$ is a function of the soil type and state of consolidation from the ratio of E_L/p_{ne}.
Step 17	Estimate the settlement per equation 7–12.

Table 7–6
Ménard's rheological factor (α) (Briaud 1992)

Soil Type	Peat		Clay		Silt		Sand		Sand and Gravel	
	E/P_L^*	α	E/P_L^*	α	E/P_L^*	α	E/P_L^*	α	E/P_L^*	α
Over-consolidated	–	–	>16	1	>14	2/3	>12	1/2	>10	1/3
Normally consolidated	For all values	1	9 to 16	2/3	8 to 14	1/2	7 to 12	1/3	6 to 10	1/4
Weathered and/or remolded	–	–	7 to 9	1/2	–	1/2	–	1/3	–	1/4
Rock	Extremely Fractured		Other				Slightly Fractured or Extremely Weathered			
	α = 1/3		α = 1/2				α = 2/3			

Notes:

$E = E_c$, E_d, and E_L as applicable (tsf), P_L^* = net limit pressure (tsf)

$\alpha = \alpha_C$, α_d, α_L rheological factors per $\frac{E_c}{p_{ne}}$, $\frac{E_d}{p_{ne}}$, and $\frac{E_L}{p_{ne}}$, respectively (dimensionless)

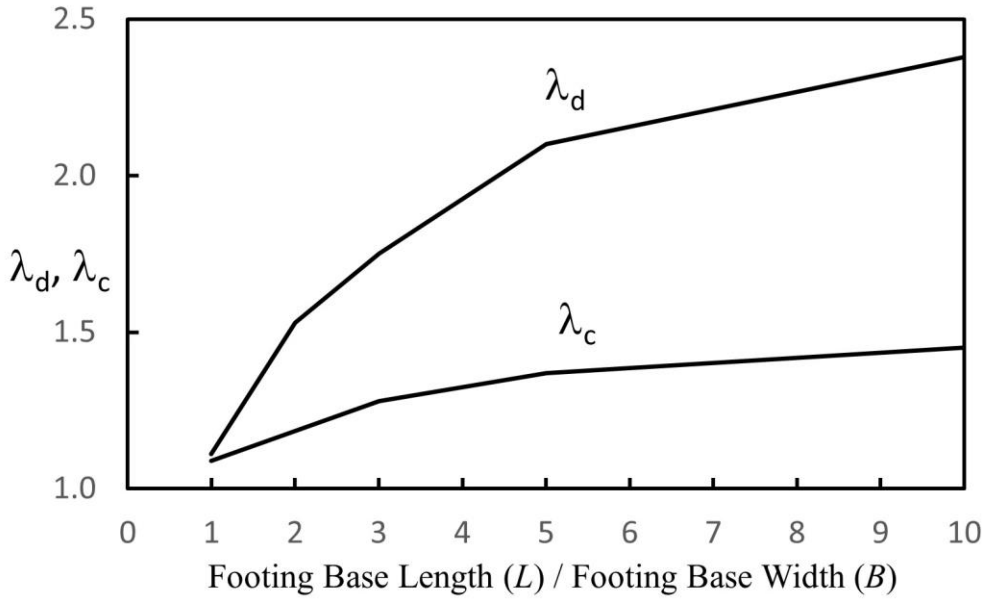


Figure 7–11. Ménard's shape factors (λ_d and λ_c) for deviatoric and spherical terms, respectively (Briaud 1992)

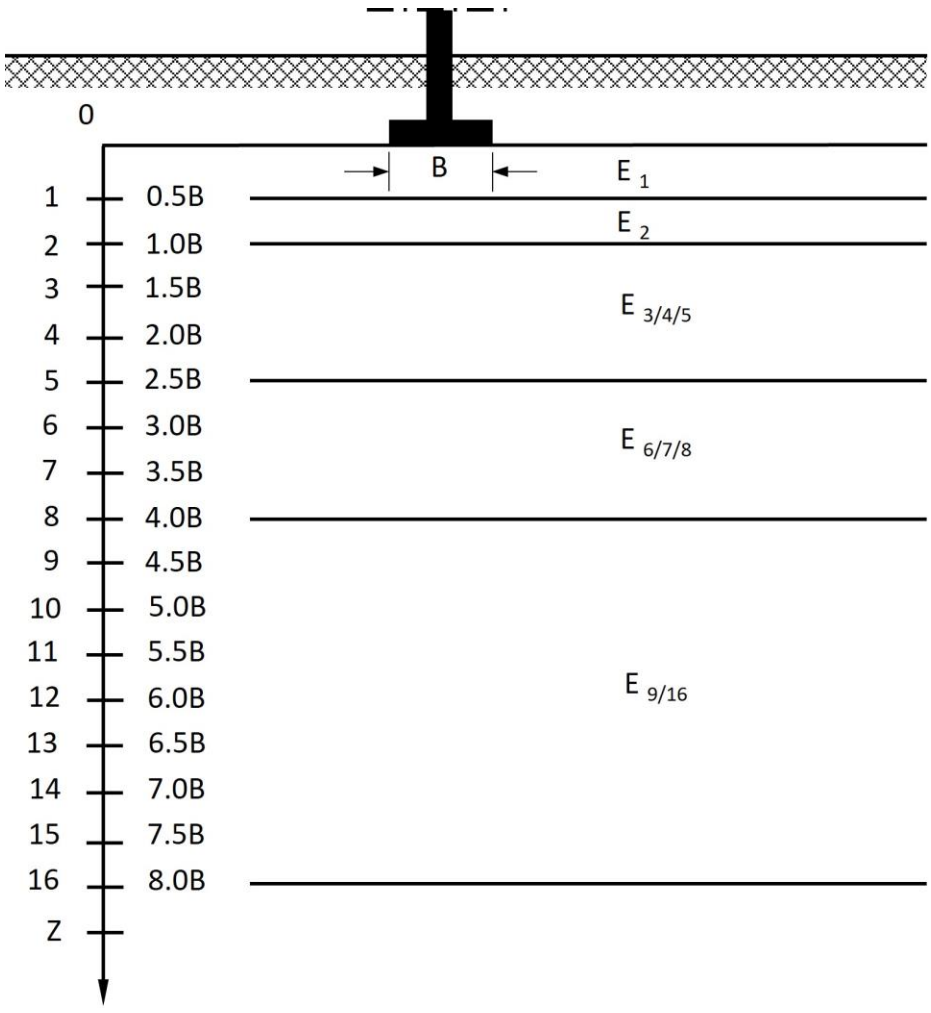


Figure 7–12. Decomposition of the soil into layers for settlement analysis (adapted from Briaud 1992)

7–12. Schmertmann's cone penetration test method for cohesionless soils

Schmertmann's method (Schmertmann 1970, and Schmertmann et al. 1978) was developed for estimating settlements in granular materials based on the results of the CPT. Schmertmann's method utilizes a strain distribution factor in lieu of a stress distribution factor. The analysis assumes that the distribution of vertical strain is compatible with a linear elastic half-space subjected to a uniform pressure and is calibrated with empirical data.

a. Equivalent Young's modulus from cone penetrometer test. An equivalent Young's modulus (E_s) is computed from the measured tip resistance (q_c), obtained from the CPT. The equivalent Young's modulus was calibrated with footing load tests, and therefore, includes 3-D stress effects from the footings. Since the stiffness of the soil is approximately 40 percent greater for plane strain compared to axisymmetric loadings, the equivalent Young's modulus (E_s) is computed by equation 7–16 for square and

axisymmetric footings and equation 7–17 for strip footings with a footing length-to-width ratio greater than 10. These equations were developed for young (< 100 years), normally consolidated, clean silica sands, and are generally conservative. For intermediate conditions (length to width ratios between 1 and 10), the equivalent Young's modulus (E_s) can be interpolated.

$$E_s = 2.5q_c \qquad (7\text{–}16)$$

Axisymmetric Footings

$$E_s = 3.5q_c \qquad (7\text{–}17)$$

Plane Strain Footings

where:

E_s = equivalent Young's modulus (tsf)
q_c = measured cone tip resistance from CPT (tsf)

b. Equivalent Young's modulus from standard penetration test N_{60}. The equivalent Young's modulus (E_s) can also be calculated from SPT N_{60} correlations as illustrated in Table 7–7. Because of the shortcomings inherent in the SPT, there is large scatter in correlating the SPT N_{60} data with the measured cone tip resistance (q_c). Therefore, CPT correlations should be used where available. The SPT correlations should not be corrected for depth or for the groundwater table.

Table 7–7
Correlations between measured cone tip resistance (q_c) and standard penetration test N_{60} (Schmertmann 1970)

Soil Type	Approximate E_s Value (ksf)	
	Axisymmetric ($L = B$)	Plane Strain ($L/B > 10$)
Silt, sandy silt, slightly cohesive silt-sand mixtures	$10\mathrm{N}_{60}$	$14\mathrm{N}_{60}$
Clean fine to medium sand, and slightly silty sand	$17.5\mathrm{N}_{60}$	$24.5\mathrm{N}_{60}$
Coarse sand and sand with little gravel	$25\mathrm{N}_{60}$	$35\mathrm{N}_{60}$
Sandy gravel	$30\mathrm{N}_{60}$	$42\mathrm{N}_{60}$

c. Strain influence factors. Schmertmann found that the maximum vertical strains do not occur directly at the bottom of the footing, as is the case for the induced vertical stress ($\Delta\sigma_z$), but instead occur at a depth of $B/2$ below the bearing elevation for square and circular "axisymmetric" footings, and at a depth of B below the bearing elevation for strip "plane strain" footings as illustrated in Figure 7–13, where B is the base width of the footing. The peak strain influence factor (I_{zp}), is computed per equation 7–18. The vertical strains extend to a depth of $2B$ and $4B$ for axisymmetric and plane strain footings, respectively. The vertical strain influence factor (I_{z}), at a given depth (z), is computed by equations 7–19 through 23. For other footing shapes, values of L/H are interpolated.

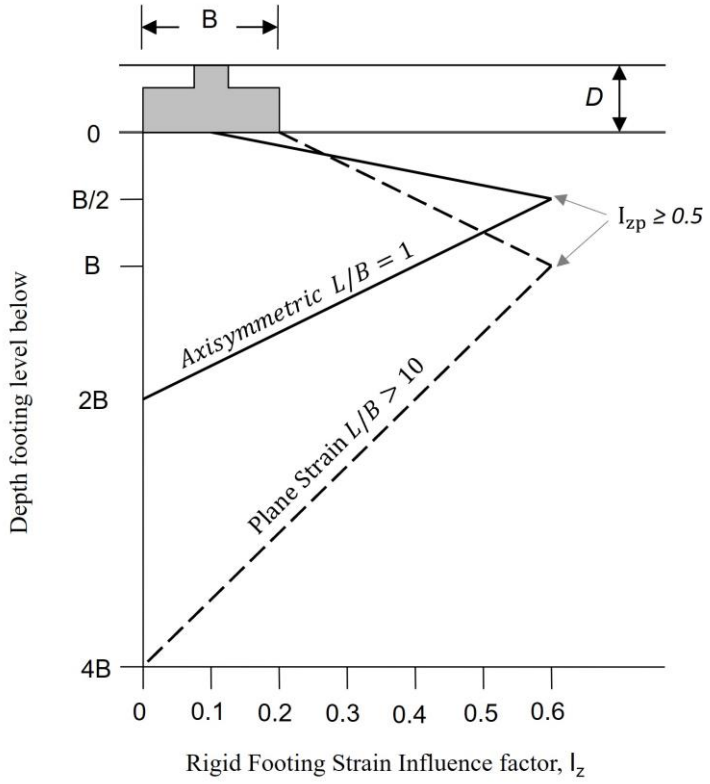


Figure 7–13. Rigid footing strain influence factor (Schmertmann et al. 1978)

$$I_{zp} = 0.5 + 0.1\sqrt{\frac{q - \sigma'_{zD}}{\sigma'_{zp}}} \tag{7–18}$$

where:

I_{zp} = peak strain influence factor (dimensionless)

q = use either the gross or gross equivalent bearing pressure (tsf), whichever is applicable; or the ultimate bearing capacity (q_{ult}) divided by factor of safety, whichever is less (tsf)

σ'_{zD} = vertical effective stress at the bearing elevation of the footing (tsf)

σ'_{zp} = vertical effective stress at the depth of peak strain influence factor (tsf), ($D + B/2$ for axisymmetric footings and $D + B$ for plane strain footings)

$$E_s = 2.5q_c \tag{7–16}$$

Axisymmetric Footings

$$I_{za} = 0.1 + \left(\frac{z}{B}\right)\left(2 \cdot I_{zp} - 0.2\right) \tag{7–19}$$

Axisymmetric Footings for $z = 0$ to $\frac{B}{2}$

$$I_{za} = 0.667 \cdot I_{zp}\left(2 - \frac{z}{B}\right) \quad \text{(7–20)}$$

Axisymmetric Footings for $z = \frac{B}{2}$ to $2B$

$$I_{zc} = 0.2 + \left(\frac{z}{B}\right)\left(I_{zp} - 0.2\right) \quad \text{(7–21)}$$

Plane Strain Footings for $z = 0$ to B

$$I_{zc} = 0.331 \cdot I_{zp}\left(4 - \frac{z}{B}\right) \quad \text{(7–22)}$$

Plane Strain Footings for $z = B$ to $4B$

$$I_{zr} = I_{za} + 0.111\left(I_{zc} - I_{za}\right)\left(\frac{L}{B} - 1\right) \quad \text{(7–23)}$$

Rectangular Footings

where:

z = depth from bottom of footing to midpoint of layer (ft)
I_{zp} = peak strain influence factor (dimensionless)
I_{za} = strain influence factor for axisymmetric footings (dimensionless)
I_{zc} = strain influence factor for plane strain footings (dimensionless)
I_{zr} = strain influence factor for rectangular footings (dimensionless)

d. Empirical embedment and creep correction factors. The empirical correction factor (C_1) is used to account for strain relief from embedment per equation 7–24. Traditionally, time rate of settlement in sand is not considered; however, it is observed in many published settlement records. The empirical correction factor (C_2) can be calculated from equation 7–25 for time, $t \geq 0.1$ years and is typically computed for $t = 50$ years resulting in $C_2 = 1.54$. For $t < 0.1$ years, $C_2 = 1.0$.

$$C_1 = 1 - 0.5\left(\frac{\sigma'_{zD}}{q - \sigma'_{zD}}\right) \geq 0.5 \quad \text{(7–24)}$$

$$C_2 = 1 + 0.2\log_{10}\left(\frac{t}{0.1}\right) \text{ for } t \geq 0.1 \text{ year} \quad \text{(7–25)}$$

where:

C_1 = embedment correction factor (dimensionless)
σ'_{zD} = vertical effective stress at the bearing elevation of the footing (tsf)
q = use either the gross or gross equivalent bearing pressure (tsf), whichever is applicable; or the ultimate bearing capacity (q_{ult}) divided by factor of safety, whichever is less (tsf)
C_2 = creep correction factor (dimensionless)
t = time after foundation loading (years)

e. Schmertmann procedure for estimating settlements using the cone penetration test. The estimation of settlements using the Schmertmann CPT method is calculated per equation 7–26 and as described in Table 7–8.

$$\delta = \mathrm{C}_1 \cdot \mathrm{C}_2(q - \sigma'_{zD}) \sum [(\mathrm{I}_z \cdot \mathrm{H})/E_s] \quad (7\text{–}26)$$

where:

δ = settlement (ft)

q = use either the gross or gross equivalent bearing pressure (tsf), whichever is applicable; or the ultimate bearing capacity (q_{ult}) divided by factor of safety, whichever is less (tsf).

σ'_{zD} = vertical effective stress at the bearing elevation of the footing (tsf)

I_z = strain influence value per equations 7–19 through 7–23 depending on footing type ($\mathrm{I}_{za}, \mathrm{I}_{zc}, \mathrm{I}_{zr}$) (dimensionless)

E_s = equivalent Young's modulus (tsf)

H = thickness of soil layer (ft)

C_1 = embedment correction factor per equation 7–24 (dimensionless)

C_2 = creep correction factor per equation 7–25 (dimensionless)

Table 7–8
Procedure for estimating settlement in cohesionless soils by Schmertmann's CPT method

Step 1	Obtain the measured tip resistance (q_c) obtained from the CPT test or SPT N_{60} values obtained from the SPT test.
Step 2	Calculate the equivalent Young's modulus (E_s) per equations 7–16 and 7–17 or Table 7–7. Divide the soil beneath the bearing elevation into layers depending on the variation in E_s versus depth.
Step 3	Compute the gross or gross equivalent bearing pressure (q or q_{eq}) at the base of the footing per paragraph 5–2b. The gross bearing pressure (q) must be less than or equal to the ultimate bearing capacity (q_{ult}) divided by the factor of safety.
Step 4	Compute the vertical effective stress at the footing bearing elevation (σ'_{zD}).
Step 5	Based on the footing type (axisymmetric or plane strain) and footing width, develop the strain influence factor diagram below the base of the footing per Figure 7–13.
Step 6	Compute the peak strain influence factor (I_{zp}) per equation 7–18.
Step 7	Compute embedment and creep correction factors per equations 7–24 and 7–25.
Step 8	Produce a table for each layer, use the strain influence factor diagram per Figure 7–13: a. the number of the layer b. depth of the layer (feet) c. depth below footing (midpoint of layer), z (feet) d. thickness of soil layer, H feet) e. q_c (tsf) or N_{60} values f. E_s (tsf) g. strain influence factor (I_z) depending on footing type ($\mathrm{I}_{za}, \mathrm{I}_{zc}, \mathrm{I}_{zr}$) h. $(\mathrm{I}_z \cdot H)/E_s$ (feet)
Step 9	Compute the vertical strain influence factor (I_z) at the midpoint of each layer per equations 7–19 through 7–23 depending on footing type.
Step 10	Compute $(\mathrm{I}_z \cdot H)/E_s$ at the midpoint of each layer and sum for the soil layers within the strain influence factor diagram per Figure 7–13.

Step 11	Compute settlement per equation 7–26.

7–13. Burland and Burbidge SPT method for sands and gravels

Burland and Burbidge (1985) presented a semi-empirical method using N_{60} values from SPT to compute settlements in granular soils. This method was based on the results of over 200 records of observed settlements of various footing sizes, tanks, and embankments on sands and gravels. The method accounts for several factors, to include shape and depth of foundation, depth of water table, grain size, and time. The following empirical equations are based on these observations.

a. Influence depth. The depth of influence (Z_I) is the thickness of sand and gravel beneath the foundation bearing elevation that is influenced by the loading from the foundation. Z_I is defined by equation 7–27 and is valid only when N_{60} increases or is constant with depth. Where N_{60} decreases with depth (Z_I) may be assumed as $2B$ or to the bottom of the loose layer, whichever is less.

$$Z_I = 1.43B^{0.7} \tag{7–27}$$

where:

Z_I = depth of influence (ft)
B = footing width (ft)

b. Soil compressibility index. The compressibility of soil can be represented by the soil compressibility index (I_c) as defined by equation 7–28.

$$I_c = \frac{0.35}{\bar{N}_{60}{}^{1.4}} \tag{7–28}$$

where:

I_c = soil compressibility index (tsf^{-1})
$\bar{N}_{60}$ = arithmetic mean of the measured N_{60} values within the depth of influence, Z_I

c. The arithmetic mean ($\bar{N}_{60}$). The arithmetic mean of the measured SPT N_{60} values measured within a depth of influence (Z_I) should not be corrected for effective overburden pressure. However, due to the possibility of dilation during driving of the split-spoon in dense or very dense, fine saturated sands, N_{60} > 15 blows per foot should be corrected per equation 4–7 prior to computing $\bar{N}_{60}$ for equation 7–28. In addition, N_{60} values may be increased by 25 percent (multiplied by 1.25) for gravel or sandy gravel prior to computing $\bar{N}_{60}$.

d. Settlement equations for axisymmetric footings (L/B =1). Estimate settlements for normally and overconsolidated granular soils per equations 7–29 through 7–31.

$$\Delta = C_1 \cdot C_2 \cdot C_3 \cdot (q + \bar{\sigma}'_{zo}) \cdot Z_I \cdot I_c \quad \text{(7–29)}$$

Normally Consolidated Granular Soil, $\bar{\sigma}'_{zo} = \sigma'_p$

$$\delta = C_1 \cdot C_2 \cdot C_3 (q + \bar{\sigma}'_{zo}) Z_I \frac{I_c}{3} \quad \text{(7–30)}$$

Overconsolidated Granular Soil, $(q + \bar{\sigma}'_{zo}) \leq \sigma'_p$

$$\delta = C_1 \cdot C_2 \cdot C_3 \left[(q + \bar{\sigma}'_{z0}) - \frac{2}{3}\sigma'_p \right] Z_I \cdot I_c \quad \text{(7–31)}$$

Overconsolidated Granular Soil, $(q + \bar{\sigma}'_{zo}) > \sigma'_p$

where:

δ = settlement (ft)

q = use either the gross or gross equivalent bearing pressure (tsf), whichever is applicable; or the ultimate bearing capacity (q_{ult}) divided by the factor of safety, whichever is less (tsf)

$\bar{\sigma}'_{zo}$ = initial average vertical effective stress within depth of influence (Z_I) beneath the footing, (tsf)

σ'_p = preconsolidation stress (tsf)

Z_I = depth of influence (ft) per equation 7–27

I_c = soil compressibility index (tsf^{-1}) per equation 7–28

C_1 = shape correction factor per equation 7–32

C_2 = layer thickness correction factor per equation 7–33

C_3 = time-dependent factor per equation 7–34

e. Shape correction factor. Settlements estimated by equations 7–29 through 7–31 are for axisymmetric footings (L/B = 1). For rectangular footings, the shape correction factor (C_1) is computed per equation 7–32. For axisymmetric footings, use C_1 = 1. For strip footings where L/B tends to infinity, use C_1 = 1.56.

$$C_1 = \left[\frac{1.25\left(\frac{L}{B}\right)}{\left(\frac{L}{B}\right) + 0.25} \right]^2 \geq 1 \quad \text{(7–32)}$$

f. Layer thickness correction factor. When the thickness of the granular layer (H_s), below the foundation is less than the zone of influence (Z_I), a layer thickness correction factor (C_2), is computed per equation 7–33.

$$C_2 = \frac{H_s}{Z_I}\left(2 - \frac{H_s}{Z_I}\right) \leq 1 \quad \text{(7–33)}$$

where:

H_s = thickness of the granular layer below base of foundation (ft)

Z_I = footing zone of influence (ft) per equation 7–27

g. Time-dependent factor. The time-dependent factor (C_3), is computed per equation 7–34. R_3 accounts for the time-dependent settlement within the first 3 years and R_t accounts for the time-dependent settlement that takes place after the first three years at a slower rate. For static loads, conservative values of R_3 equal to 0.3 and R_t equal to 0.2 may be assumed, resulting in C_3 = 1.5 for 30 years. For fluctuating loads such as bridges, conservative values of R_3 equal to 0.7 and R_t equal to 0.8 may be assumed, resulting in C_3 = 2.5 for 30 years.

$$C_3 = 1 + R_3 + R_t \cdot \log_{10} \frac{t}{3} \tag{7–34}$$

where:

C_3 = time-dependent factor
R_3 = time-dependent factor for first 3 years after foundation loading
R_t = time-dependent factor after first 3 years after foundation loading
t = time after foundation loading (years)

h. Burland and Burbidge procedure for estimating settlements using standard penetration test. The Burland and Burbidge procedure for estimating settlements using SPT N_{60} values is described in Table 7–9.

Table 7–9
Procedure for estimating settlements by the Burland and Burbidge standard penetration test method

Step 1	Obtain SPT N_{60} values from SPT tests. Correct N_{60} values greater than 15 bpf in very fine and silty sands located below the groundwater table per equation 4–7. For gravels or sandy gravels, multiply N_{60} values by 1.25.
Step 2	Compute the gross or gross equivalent bearing pressure (q or q_{eq}) at the base of the footing per paragraph 5–2b. The gross bearing pressure (q) must be less than or equal to the ultimate bearing capacity (q_{ult}) divided by the factor of safety.
Step 3	Compute depth of influence (Z_I) per equation 7–27. If N_{60} values decrease with depth, $Z_I = 2B$ or to the bottom of the loose layer, whichever occurs first.
Step 4	Compute the arithmetic mean of the measured N_{60} values measured within the depth of influence (Z_I).
Step 5	Compute the soil compressibility index (I_c) per equation 7–28.
Step 6	Compute shape (C_1), layer thickness (C_2), and time-dependent (C_3) factors per equations 7–32, 7–33 and 7–34.
Step 7	Compute the average initial vertical effective stress ($\bar{\sigma}'_{zo}$) within depth of influence (Z_I).
Step 8	Compute the preconsolidation stress (σ'_p) from available laboratory, in situ testing methods, or from N_{60} data (see paragraph 3–7e).
Step 9	Estimate settlements per equations 7–29, 7–30 or 7–31 as applicable.

7–14. Other empirical methods for cohesionless soils using standard and cone penetration tests

Many methods are available for estimating settlements in cohesionless soils. CSANDSET (Technical Report ITL-91-1) details 15 different methodologies involving SPT or CPT for estimating settlements in cohesionless soils. Tan and Duncan (1991) evaluated the accuracy and reliability of 12 SPT-based methods and the Schmertmann CPT method using more than 90 cases (Figure 7–14). The general results of this study highlight the fact that (1) most of the penetration methods used to calculate settlement of foundations on cohesionless soils tend to overpredict, and (2) the most accurate of these methods will make a “reliably” conservative estimate of settlement (greater than or equal to the actual value) about half the time. Where it is critical not to exceed the calculated settlement, methods with a higher reliability should be used.

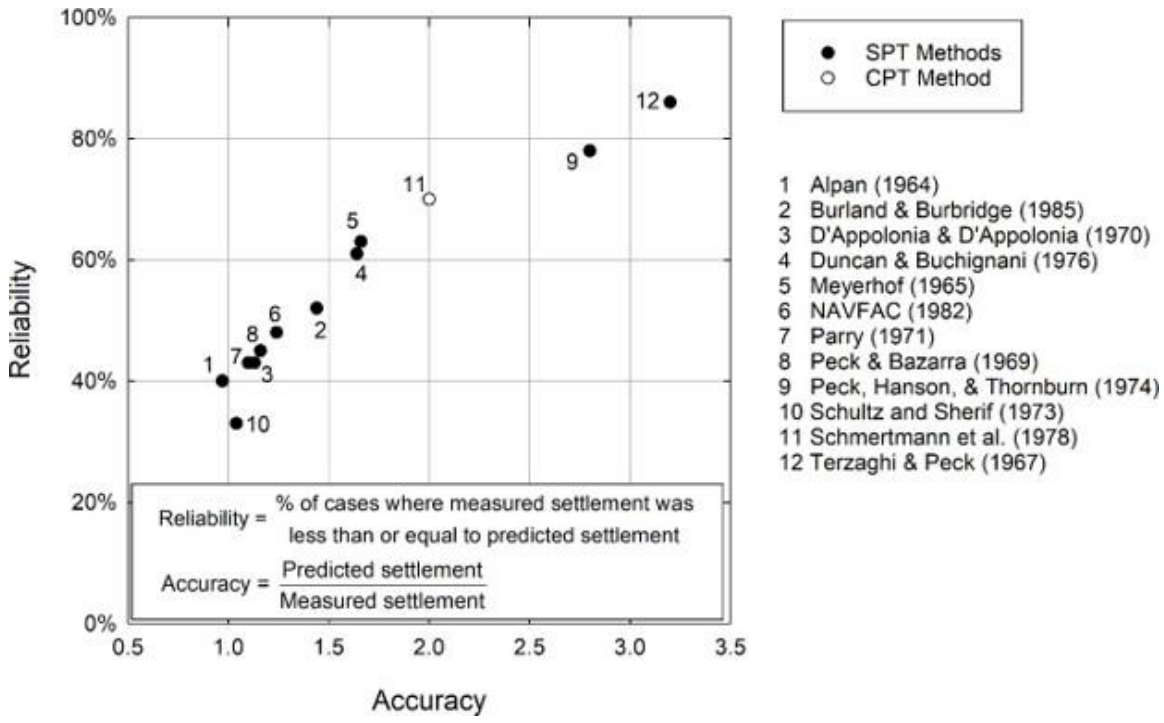


Figure 7–14. Comparison of settlement calculation methods for cohesionless soils using standard penetration test and cone penetration test (after Tan and Duncan 1991)

7–15. Settlements influenced by excavation rebound in cohesive soils

The removal of overburden above cohesive soils, such as from an excavation for a basement, reduces the loading on the soils beneath the excavation. This reduction in load and effective stress can lead to some elastic rebound of the cohesive soils below the excavation bottom prior to recompression from foundation loading. Total settlement should account for excavation rebound and recompression and can be evaluated using laboratory consolidation data as shown in Figure 7–15.

a. Floating foundations. When the stress from the loading of the new foundations is less than or equal to the removal of the overburden, the structure is termed to be "fully floating," and when the building loading is partially compensated by relief of the overburden, the building is termed to be "partially floating." The technique of floating the foundation reduces the net bearing pressure (contact stress) of the foundation, therefore, reducing settlements.

b. Soil rebound from excavation. When floating the foundation (whether fully or partially), the net bearing pressure of the foundation will occur along the recompression portion of the laboratory consolidation curve; similar to placing the structure on a preconsolidated soil.

c. Estimating settlements influenced by excavation rebound and subsequent recompression. A procedure for estimating the settlement of a foundation constructed on a normally consolidated cohesive soil following excavation rebound and subsequent recompression from foundation loading is provided in Table 7–10. Field evidence shows that the rate of heave is usually faster and of less magnitude than predicted. Based on a study of 43 building sites, the field heave typically amounted to about one-third of the computed heave. This is because only a very small fraction of settlement is due to elastic deformation of the soil grains. Where excavations are large and are open for a substantial time before foundation loadings are applied, the actual heave could be closer to the computed heave.

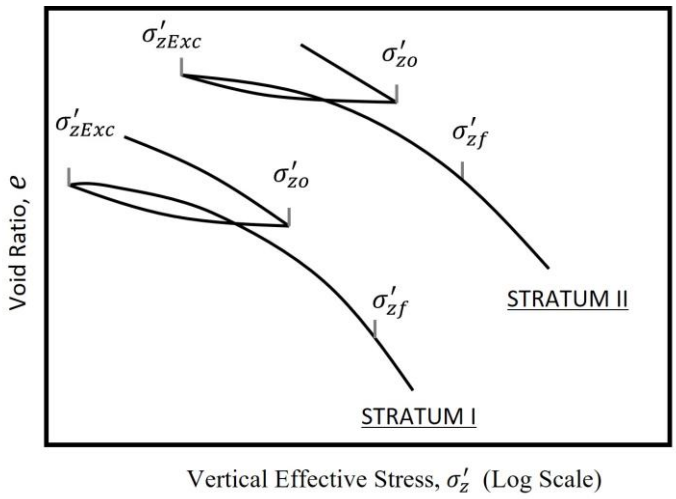


Figure 7–15. Example of e-log-p curve for several soil samples obtained at different depths prior to unloading and reloading; σ'_{zo} = initial vertical effective stress prior to excavation and dewatering, σ'_{zExc} = vertical effective stress following excavation and dewatering, σ'_{zf} = final vertical effective stress due to foundation load

Table 7–10
Procedure for estimating settlements due to excavation rebound and subsequent recompression

Step 1	Conduct laboratory oedometer tests for the compressible layers. Example oedometer test data is illustrated in Figure 7–15. A. The first unload increment should occur at the initial vertical effective stress (σ'_{zo}) where the sample was obtained in the ground prior to the excavation and lowering of the groundwater table (per Step 2). The sample is then unloaded to the vertical effective stress following the excavation and lowering of groundwater table (σ'_{zexc}) per Step 6. B. The soil is then reloaded beyond the final vertical effective stress (σ'_{zf}) per Step 9.
Step 2	Compute the initial vertical effective stress (σ'_{zo}) prior to the excavation and lowering of the groundwater table at the midpoint of each layer.
Step 3	If the water table is to be lowered (for example, to the base of the excavation for construction of foundations), compute vertical effective stress following the lowering of groundwater table (σ'_{zgw}) at the midpoint of each layer. Note that $\sigma'_{zgw} = \sigma'_{zo}$ when no groundwater is present to be lowered.
Step 4	Compute stress relief (q'_{exc}) due to the excavation per equation 7–35 following drawdown of the water table. $$q'_{exc} = \gamma_t \cdot \mathrm{D} \quad (7\text{–}35)$$ where: q'_{exc} = stress relief due to excavation (tsf) D = depth of excavation (ft) γ_t = total unit weight of the soil removed (tcf)
Step 5	Compute the vertical stress influence factor (I_σ) and corresponding unloaded vertical stress ($\Delta\sigma_{zexc}$) from the excavation at the midpoint of each layer per equation 7–36. $$\Delta\sigma_{zexc} = I_\sigma(q'_{exc}) \quad (7\text{–}36)$$ where: $\Delta\sigma_{zexc}$ = unloaded vertical stress (tsf) I_σ = influence factor for vertical stress per paragraph 6–7 (dimensionless) q'_{exc} = stress relief due to excavation per equation 7–35 (tsf)
Step 6	Compute the vertical effective stress following the excavation and lowering of groundwater table (σ'_{zexc}) at the midpoint of each layer per equation 7–37. $$\sigma'_{zexc} = \sigma'_{zgw} - \Delta\sigma_{zexc} \quad (7\text{–}37)$$ where: σ'_{zexc} = vertical effective stress (tsf), after excavation and dewatering σ'_{zgw} = vertical effective stress (tsf), after dewatering $\Delta\sigma_{zexc}$ = unloaded vertical stress due to excavation (tsf)

Step 7	Compute rebound (R) equation 7–38 for each stratum from the results of the oedometer tests. $$\mathrm{R} = \mathrm{H}\left(\frac{\Delta e}{1 + e_{\mathrm{zo}}}\right) \quad (7\text{–}38)$$ where: R = vertical effective stress (tsf), after excavation and dewatering H = vertical effective stress (tsf), after dewatering e_{zo} = void ratio (dimensionless) at initial vertical effective stress ($\sigma_{zo}^{'}$) prior to excavation and dewatering e_{zexc} = void ratio (dimensionless) at vertical effective stress following the excavation and lowering of groundwater ($\sigma_{zexc}^{'}$) Δe = $e_{\mathrm{zo}} - e_{\mathrm{zexc}}$
Step 8	Compute the gross bearing pressure (q) or gross equivalent bearing pressure (q_{eq}) for eccentrically loaded foundations (tsf) per paragraph 5–2b.
Step 9	Compute the vertical effective stress at the bearing elevation of the footing ($\sigma_{zD}^{'}$) at depth (D) (depth includes depth of excavation + depth of footing embedment).
Step 10	Compute the vertical stress influence factor (I_σ) per paragraph 6–7 and the corresponding induced vertical stress ($\Delta\sigma_Z$) beneath the footing at the midpoint of each layer per equation 6–6, $\Delta\sigma_z = \mathrm{I}_\sigma(q - \sigma_{zD}^{'})$. where: $\sigma_{zD}^{'}$ = vertical effective stress at bearing elevation (tsf) from Step 9 q = use either the gross or gross equivalent bearing pressure (tsf), whichever is applicable; or the ultimate bearing capacity (q_{ult}) divided by the factor of safety, whichever is less (tsf)
Step 11	Compute the final vertical effective stress ($\sigma_{zf}^{'}$) at the midpoint of each layer to a depth of the footing the zone of influence per equation 7–39. $$\sigma'_{zf} = \sigma'_{zo} + \Delta\sigma_z \quad (7\text{–}39)$$ where: $\sigma_{zf}^{'}$ = final vertical effective stress (tsf) $\sigma_{zo}^{'}$ = initial vertical effective stress (tsf) from Step 2 $\Delta\sigma_z$ = induced vertical stress beneath the footing (tsf) from Step 10

Step 12	Estimate the settlement due to the recompression of the soil per equation 7–40 for each stratum from the results of the oedometer tests. $$\delta_r = H\left(\frac{\Delta e}{1+e_{zexc}}\right) \quad (7\text{–}40)$$ where: δ_r = settlement due to recompression of soil (ft) H = thickness of soil layer (ft) e_{zExc} = void ratio (dimensionless) at vertical effective stress following the excavation and lowering of groundwater σ'_{zexc} e_{zf} = void ratio (dimensionless) at final vertical effective stress from the footing (σ'_{zf}) Δe = $e_{zExc} - e_{zf}$

7–16. Settlements beneath fill

Primary settlement beneath new fill is due to the induced vertical stress across the compressible layers from the footing ($\Delta\sigma_z$), and from the weight of the fill ($\Delta\sigma_{fill}$). Settlements below fill, with or without footings, can be estimated using either the results from laboratory oedometer tests or in situ testing.

a. Estimating settlements due to fill weight only. Since the base of fill is non-rigid and has length and/or width dimension greater than a conventional structure footing, settlements should be estimated at the center and the edges of the fill. The influence factor for vertical stress (I_σ), and corresponding induced vertical stress from the fill ($\Delta\sigma_{zfill}$), for the estimation of settlements at the edge and center of the fill should be computed per paragraph 6–8.

b. Estimating settlements due to fill plus footing loading. Where footings are to be constructed in conjunction with the placement of fill, the induced vertical stresses at the center of the compressible layer are the summation of the induced vertical stress from the weight of the fill ($\Delta\sigma_{zfill}$), plus the induced vertical stress from the footing ($\Delta\sigma_z$). The influence factor for vertical stress (I_σ), and corresponding induced vertical stresses from the fill and foundation are computed at the midpoint of each layer per paragraph 6–8.

7–17. Within-fill settlement

a. General. Settlements that occur within compacted fills due to self-weight plus any induced vertical stresses from footings are referred to as “within-fill” settlements. In service, the fill may be altered with time due to environmental changes (such as wetting), resulting in changes in compressibility behavior. During the compaction process, densification is achieved by air void reduction at constant water content. The amount of within-fill settlement will depend primarily on the magnitude of the induced vertical stresses from footings, lift thickness, compaction water content, the degree of compaction, and on the mode of compaction.

b. Estimating within-fill settlement. For compacted soils, within-fill settlements (resulting from the fill weight alone) will typically vary between 0.5 to 4.0 percent of the total fill height for confining pressures less than about 50 psi (approximately 100 ft). Within-fill settlement due to induced vertical stresses from footings can be estimated using the constrained modulus or e-log-p data obtained from oedometer tests conducted on laboratory compacted soil samples or from in situ testing (such as DMT) after the fill has been placed.

7–18. Time rate of consolidation settlement

Where fill placement or surcharge is required prior to the placement of foundations, time rate of consolidation settlement can be evaluated to calculate when foundations can be placed so that tolerable settlements are not exceeded. In addition, the time rate of settlement beneath a foundation without fill could also be of interest. Reference Holtz et al. (2011), for an in-depth discussion on the time rate of consolidation.

a. Average degree of primary consolidation settlement. In some cases, the average degree or percent consolidation of a soil layer is needed. The average degree of consolidation, $\mathrm{U\%}$, (also referred to as percent consolidation) for a compressible layer is directly related to the primary consolidation settlement at a given time after loading (δ_{ct}) as shown in equation 7–41. The time (t) required to achieve the average degree of primary consolidation ($\mathrm{U\%}$) under a given pressure is expressed by equation 7–42. The relationship between the average degree of consolidation $\mathrm{U\%}$ and the dimensionless time factor ($\mathrm{T_v}$) is provided in Figure 7–16.

$$\delta_{ct} = \frac{\mathrm{U\%}}{100}\delta_c \tag{7–41}$$

where:

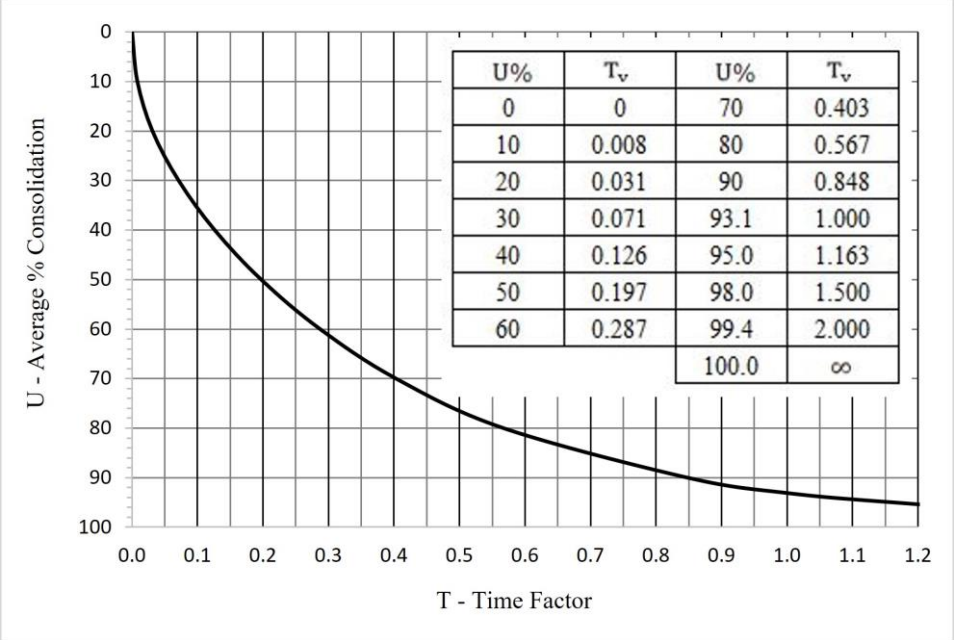
δ_{ct} = primary consolidation settlement at time, t (inches)
$\mathrm{U\%}$ = average degree (or percent) of primary consolidation at time, t (percentage)
δ_c = primary consolidation settlement per equations 7–7 through 7–9 (inches)

$$t = \frac{\mathrm{T_v} \cdot \mathrm{H_{dr}^2}}{C_v} \tag{7–42}$$

where:

t = time required for a given percentage of consolidation after loading (days)
$\mathrm{T_v}$ = dimensionless time factor for consolidation due to vertical drainage per Figure 7–16.
$\mathrm{H_{dr}}$ = longest distance to drainage boundary (ft); $\mathrm{H_{dr}}$ = one-half compressible layer thickness (H) for two-directional drainage (drainage from top and bottom surface); $\mathrm{H_{dr}}$ = compressible layer thickness (H) for one-directional drainage (drainage from one surface)

C_v = coefficient of consolidation (ft²/day) per paragraph 7–18c



U%	T_v	U%	T_v
0	0	70	0.403
10	0.008	80	0.567
20	0.031	90	0.848
30	0.071	93.1	1.000
40	0.126	95.0	1.163
50	0.197	98.0	1.500
60	0.287	99.4	2.000
		100.0	∞

Figure 7–16. Average degree of consolidation (U%) versus time factor (T_v) for linear distribution of excess pore water pressure with depth

b. Procedure for estimating time rate of settlement. The procedure for estimating the time required to achieve a specific percent of consolidation (or the percent consolidation for given time) is described in Table 7–11.

Table 7–11
Procedure for estimating time required for consolidation

Step 1	Estimate primary consolidation (δ_c or δ) of each compressible layer per paragraphs 7–9, 7–10, or 7–11.
Step 2	Determine coefficient of consolidation (C_v) from in situ test methods or from laboratory consolidation test data per Table 7–12 or Table 7–13. Select the appropriate value of C_v based on the final vertical effective stress (σ'_{zf}) at the midpoint of each compressible layer per Step 8 of Table 7–3. Preliminary estimates of coefficient of consolidation (C_v) may be made from empirical correlations.
Step 3	Compute the time (t) to achieve the specific percent of consolidation (U%) per equation 7–42 and Figure 7–16. Also, the percent consolidation U% for a specific time (t) can be computed by rearranging equation 7–42, solving for T_v and obtaining U% per Figure 7–16.
Step 4	To determine the time (t) and percent consolidation (U%) for a specific consolidation settlement at time (δ_{ct}), first compute U% per equation 7–41, then compute time (t) per Step 3.

c. Coefficient of consolidation. The coefficient of consolidation (C_v) can be estimated from either in situ tests or from laboratory oedometer time-deformation data. Two methods include: (1) logarithm-of-time method (log t) proposed by Casagrande and

Fadum (1940), or (2) the square-root-of-time method ($\sqrt{t}$) proposed by Taylor (1948) per Table 7–12 and Table 7–13, respectively. The $\sqrt{t}$ method typically results in higher values of C_v. Therefore, performing both methods will provide an upper and lower bound. The table procedures are performed for each applied stress increment conducted during the laboratory oedometer test, resulting in a separate C_v value for each stress increment. Approximate correlations of C_v with liquid limit are illustrated in Figure 3–12. The coefficient of consolidation (C_v) will generally increase with increasing permeability and decreasing compressibility.

Table 7–12
Procedure for estimating coefficient of consolidation (C_v) from Casagrande's logarithm (log t) graphical method (see Figure 7–17)

Step 1	Measure initial specimen height ($\mathrm{h_0}$). Measure dial reading (d) as a function of time (t). Determine final specimen height ($\mathrm{h_f}$) at end of stress increment. Plot the dial readings versus time ($\mathrm{h_f}$) for sample deformation for the given stress increment on semi-log paper.
Step 2	Plot Points P and Q on the upper portion of the consolidation curve which correspond to time t_1 and t_2, respectively. $t_2 = 4t_1$.
Step 3	The difference of the dial readings between P and Q is equal to x. Locate Point R, which is at a distance, x above Point P.
Step 4	Draw the horizontal line RS. The dial reading corresponding to this line is $\mathrm{d_0}$, which corresponds to 0% consolidation.
Step 5	Project the straight-line portions of the primary consolidation and the flatter portion toward the end of the consolidation curve to intersect at T. The dial reading corresponding to T is $\mathrm{d_{100}}$ (that is, 100 percent primary consolidation). The sample deformation beyond $\mathrm{t_{100}}$ is due to secondary compression.
Step 6	Determine the Point V on the consolidation curve, which corresponds to a dial reading of $(\mathrm{d_0}+\mathrm{d_{100}})/2 = \mathrm{d_{50}}$. The time corresponding to the Point V is $\mathrm{t_{50}}$ (50 percent consolidation).
Step 7	Compute the coefficient of consolidation (C_v), per equation 7–43. $$C_\mathrm{v} = \frac{T_{50}(\mathrm{H_{dr}})^2}{t_{50}} \quad (7\text{–}43)$$ where: C_v = coefficient of consolidation (in²/day) $\mathrm{T_{50}}$ = dimensionless time factor at 50 percent consolidation = 0.197 t_{50} = time at 50 percent consolidation from Step 6 (days) $\mathrm{H_{dr}}$ = longest distance to drainage boundary (inches). The standard oedometer test specimen has two-way drainage; therefore, $\mathrm{H_{dr}} = \bar{h} \,/\, 2$ (inches), see Step 1. $\bar{h}$ = avg. specimen height during stress increment = $0.5(h_0 + h_f)$ (inches), see Step 1

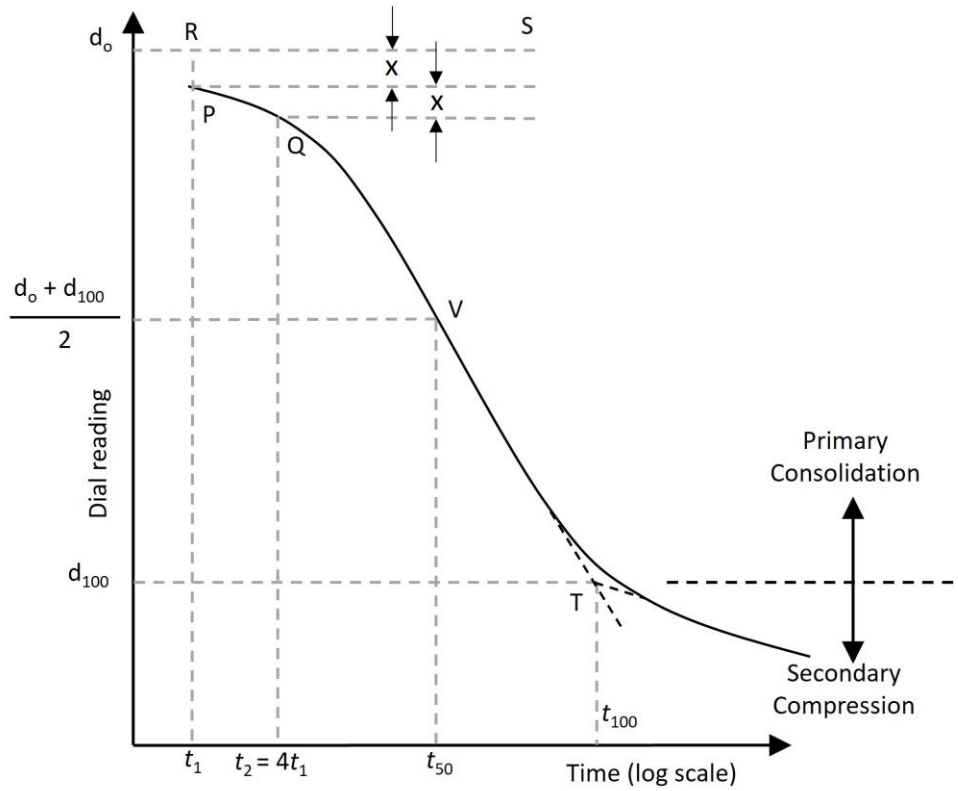


Figure 7–17. Determination of t_{50} by the Casagrande method for a given stress increment

Table 7–13
Procedure for estimating coefficient of consolidation (C_v) from Taylor's (1948) square-root-of-time method (see Figure 7–18)

Step 1	Measure initial specimen height (h_0). Measure dial reading (d) as a function of time (t). Determine final specimen height (h_f) at end of stress increment. Plot the dial readings versus corresponding square-root-of time ($\sqrt{t}$).
Step 2	Project a line backward from the consolidation curve back to Point P (zero time) to define d_0. Point P may be slightly lower than the initial dial reading d_0, observed in the laboratory due to immediate compression of the specimen.
Step 3	Draw the tangent PQ to the early portion of the plot.
Step 4	Draw a line PR such that OR = (1.15) (OQ).
Step 5	Obtain $\sqrt{t_{90}}$ (square-root-time for 90 percent consolidation) as the absisca of Point S.
Step 6	Compute the coefficient of consolidation (C_v) per equation 7–44. $$C_v = \frac{T_{90}(H_{dr})^2}{t_{90}} \quad (7–44)$$ where: C_v = coefficient of consolidation (in²/day) T_{90} = dimensionless time factor at 90 percent consolidation = 0.848 t_{90} = time at 90 percent consolidation from Step 1 (days) H_{dr} = longest distance to drainage boundary (inches). The standard oedometer test specimen has two-way drainage; therefore, $H_{dr} = \bar{h} / 2$ (inches), see Step 1. $\bar{h}$ = avg. specimen height during stress increment = $0.5(h_0 + h_f)$ (inches), see Step 1

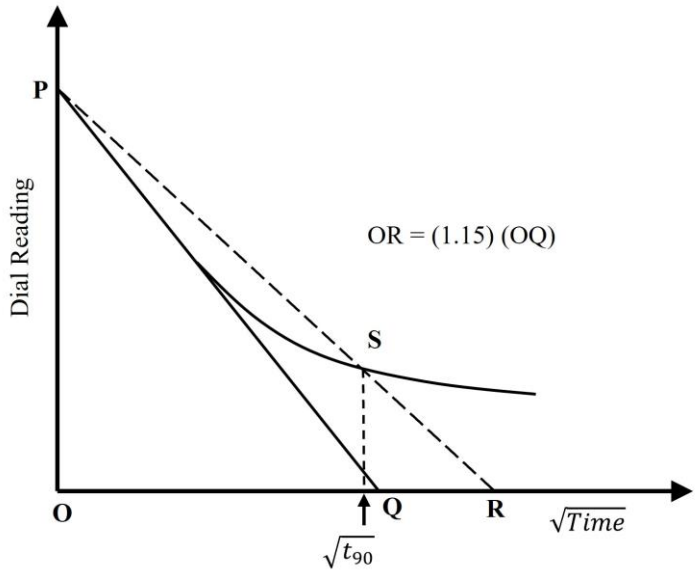


Figure 7–18. Determination of t_{90} by the Taylor Method for a given stress increment

d. Accuracy of time rate of settlement estimates. Time rates of settlement based on laboratory tests, in situ tests, and empirical correlations might not be representative of the field because time rates are influenced by in situ fissures, existence of high-permeable sand or low-permeable bentonite seams, impervious boundaries, and non-uniform soil parameters, as well as the rate of construction. In time rate computations, the drainage distance is squared, leading to a significant influence on time rate of settlement estimates that can often exceed significant changes in C_{v}.

e. Reducing settlement time. The most common methods to accelerate settlement and reduce the time to achieve the settlement are: (1) application of a surcharge greater than the loading to be subjected to the underlying soil by the footing or permanent fill, (2) installation of vertical wick drains into the foundation soils, or (3) or a combination of the two.

7–19. Estimation of secondary compression settlement

Research suggests that secondary compression can occur simultaneously with consolidation settlement. However, the approach generally adopted, and presented in this manual, assumes that it occurs after consolidation (after primary consolidation and at constant vertical effective stress with no subsequent change in pore water pressure).

a. Coefficient of secondary compression. The coefficient of secondary compression ($C_{\propto}$) is the slope of the void ratio-logarithm time plot for time exceeding that required for 100 percent of primary consolidation (t_{100}). t_{100} is arbitrarily established as the intersection of the tangent to the curve at the point of inflection with the tangent to the straight-line portion representing a secondary time effect as illustrated in Figure 7–19 for a given stress increment. The coefficient of secondary compression (C_{α}) can also be approximated for different soils based on correlations with the compression index (C_c) or natural moisture content, reference paragraph 3–7d.

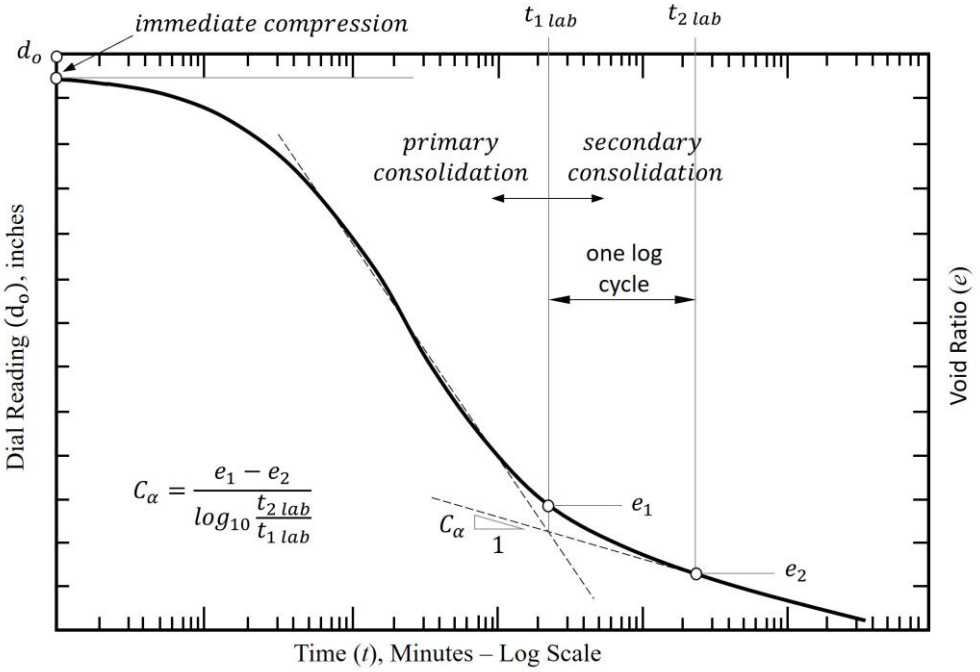


Figure 7–19. Oedometer void ratio—logarithm time plot for determination of C_α

b. Procedure for estimating secondary settlement. The procedure for estimating secondary settlement is described in Table 7–14.

Table 7–14
Procedure for estimating secondary compression settlement

Step 1	From the plot of the dial readings vs log-time plot completed in Step 1 of Table 7–12, plot void ratio (e) versus log-time t on semi-log paper as illustrated in Figure 7–19. For each dial reading (d) per elapsed time (t) the corresponding void ratio (e) is computed per equation 7–45. $e = \frac{(H_0 - H_s) - (d_0 - d_t)}{H_s}$ (7–45) where: e = void ratio at d_t H_o = original height of lab specimen (inches) H_s = height of solids = $H_0/(1 + e_0)$ (inches) d_0 = initial dial reading 10^{-4} inches) d_t = dial reading at time, t (10^{-4} inches)
Step 2	Project the straight-line portions of the primary consolidation and the flatter portion toward the end of the consolidation curve to intersect each other. The dial reading corresponding to the intersection is d_{100} (100 percent primary consolidation). The sample deformation beyond t_{100} is due to secondary compression.

Step 3	Calculate coefficient of secondary compression (C_α) slope of the void ratio-logarithm time plot for time exceeding that required for 100 percent of primary consolidation (t_{100}) per equation 7–46. $$C_\alpha = \frac{e_1 - e_2}{\log_{10}\frac{t_{2\,lab}}{t_{1\,lab}}} \quad (7\text{–}46)$$ where: C_α = coefficient of secondary compression $t_{1\,lab}$ = time after secondary compression begins, typically at t_{100} (minutes) $t_{2\,lab}$ = arbitrary time typically at one log cycle greater than t_{100} along the secondary compression straight line portion of the dial reading versus log time curve which occurs after primary consolidation is completed (minutes) e_1 = void ratio corresponding to $t_{1\,lab}$ (dimensionless) e_2 = void ratio corresponding to $t_{2\,lab}$ (dimensionless)
Step 4	Estimate secondary settlement per equation 7–47. $$\delta_s = \frac{C_\alpha}{1 + e_p} \mathrm{H} \cdot \log_{10}\frac{t_2}{t_1} \quad (7\text{–}47)$$ where: δ_s = secondary compression (feet) t_1 = time to complete primary consolidation in the consolidating layer in the field equation 7–42 (minutes) t_2 = service life of the structure or any other time of interest (minutes) C_α = coefficient of secondary compression per Step 3 e_p = void ratio at the time of complete primary consolidation in the test specimen of the compressible layer (dimensionless) H = thickness of compressible layer in field being evaluated (feet)

7–20. Numerical methods

Numerous commercially available software (SigmaW, Plaxis, OPTUM, etc.) are available to compute stress distribution, settlement, and time rate of settlement for variably loaded and irregular shapes, and for soils with variable stiffness properties. The analytical and chart-based solutions presented in Chapter 6 and the methods for estimating settlement in Chapter 7 are better suited for less complex subsurface profiles and loading conditions and are limited by the constraints and assumptions of each. Where more complex loading or subsurface conditions are encountered, the methods presented in Chapters 6 and 7 can be used as a first order approximation to the numerical methods considering more simpler loading and subsurface conditions. Reference UFC 3-220-10 for additional information regarding numerical solutions for stresses in soils and numerical analysis types.

Chapter 8
Mat Foundations

8–1. Introduction

This chapter presents considerations and methods for design of common mat foundations. Mats may be uniform or variable in thickness and are generally designed to carry the entire load of the superstructure to limit differential movements. Mat foundations have a complex soil-structure interaction in which the soil and mat deflections and stiffness are interdependent, and both influence the mat design. Soil reaction and the corresponding bending moments and shear forces in the mat and mat settlements (both total and differential) are often of primary importance in design.

8–2. Considerations for use

Mat foundations are commonly used for single-family homes, larger commercial structures (such as multi-story offices), parking garage structures, basements, pumping stations, and as equipment foundations. Typical reasons to choose mat foundations include more even distribution of structure settlement, need to span different foundation conditions, need to limit bearing pressures, need to resist hydrostatic uplift by balancing mat weight with hydrostatic pressures, need to resist or distribute expansive soil pressures and movements, and general constructability. In some situations, mat foundations are chosen to mitigate differential movement caused by seismically induced ground settlement or potential strength loss. The choice between a conventional spread footing design and mat design approach is generally made when footings become large and take up more than 30 percent of the plan area of the structure.

8–3. Mat foundation design methods

a. Design methods. Analysis methods should be carefully coordinated between the project geotechnical and structural engineers. Structural design approaches include closed-form analytical methods based on Conventional (Rigid) or ACI Approximately Flexible assumptions regarding the mat, or more rigorous soil-structure interaction (SSI) modeling using structural numerical finite difference, finite element, or finite grid models (ACI 336.2R).

(1) The SSI methods are characterized as pseudo-coupled or fully coupled as discussed in paragraphs 8–3c(1) and 8–3c(2). In both flexible and pseudo-coupled SSI methods, the foundation soil support is represented as vertical springs having a stiffness called the modulus of subgrade reaction (K_s).

(2) In fully coupled SSI modeling, fundamental engineering properties of the soil are used directly in the analysis. The method chosen by the designer should be based on the project details such that the design method assumptions are appropriate for the project. Analyses that consider flexural deflections in the mat and the resulting non-uniform soil bearing pressure usually produce more accurate estimates of the flexural stress in the mat and result in more reliable designs.

b. Rigid and ACI Approximately Flexible foundation analyses. In the Rigid and ACI Approximately Flexible mat design methods, soil support is modeled as continuous vertical springs that act independently in a vertical direction to support the mat bearing pressure, based on work by Winkler (1867). Typically, a single value of modulus of subgrade reaction is used and the 3-D stress distribution in the soil mass is not specifically considered, meaning the springs are considered uncoupled.

c. Soil-structure interaction modeling. The soil support in numerical soil-structure modeling (SSI) models can be modeled using pseudo-coupled or coupled approaches.

(1) *Pseudo-coupled*.

(a) It is common practice to perform SSI in a pseudo-coupled approach, in which soil support is modeled by the modulus of subgrade reaction and is approximated iteratively using the Discrete Area Method (Ulrich 1988). In pseudo-coupled mat analyses, separate geotechnical and structural analysis are performed in an iterative process until the structure and geotechnical models converge on a solution where the ground deflection magnitude and shape closely match the structure deformation deflection magnitude and shape. The analysis type is “pseudo-coupled,” since the geotechnical analyses considers the coupled relationships of 3-D stress distributions in the soil mass to develop simplified uncoupled vertical modulus of subgrade reaction spring constants for input into the structural design.

(b) Figure 8–1 illustrates the difference between coupled and uncoupled springs. In a pseudo-coupled analysis, the geotechnical portion of the analysis considers the coupled behavior of the ground (Figure 8–1b) and simplifies the behavior as uncoupled springs for use in the structural analysis (Figure 8–1a).

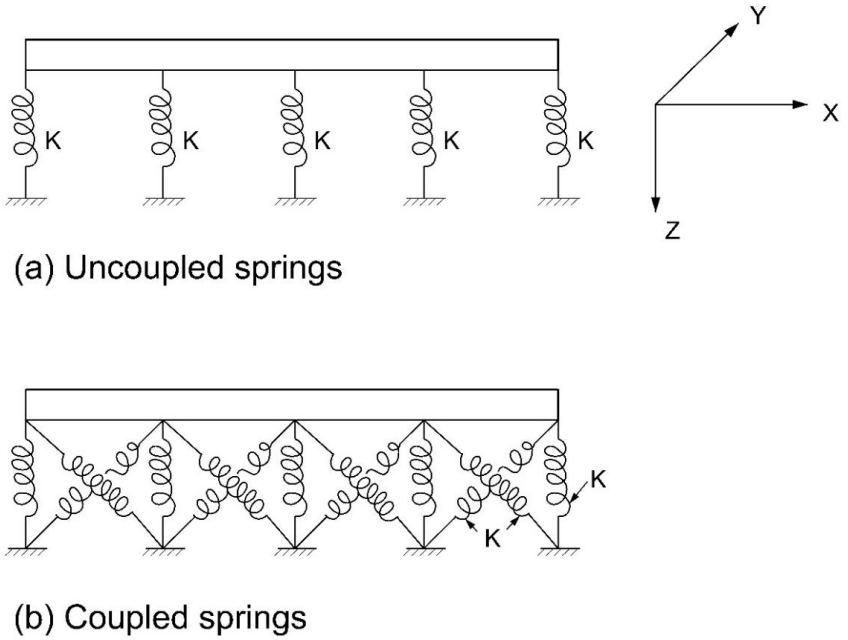


Figure 8–1. Coupled vs uncoupled soil springs as illustrated (ACI 336.2R)

(2) *Fully-coupled*. For fully-coupled soil-structure interaction numerical models, various constitutive soil and structural models may be used to more rigorously model the soil-structure system directly in a single model. This approach is uncommon due to the computational effort and the limited benefit to design, and because the modeler needs significant expertise in both the structural design, soil mechanics, and numerical modeling. For common mat design, pseudo-coupled analysis is adequate. Bowles (1996) has noted that the value of fully-coupled analyses to common mat design has not been demonstrated. A full discussion of the parameters required for fully-coupled SSI modeling is beyond the scope of this chapter.

d. Light structure mats. In lightly loaded structures (such as single-family homes), conventionally reinforced mats and post-tensioned mat foundations are sometimes selected as the foundation type and are used to resist and distribute expansive soil pressures and reduce differential movement to the superstructure caused by soil shrinkage or swelling pressures acting on the structure. These mats are generally uniform in thickness or ribbed, in which there is a grid of thickened beam sections. The Post-Tension Institute (PTI) has published prescriptive methods for mat design based on generalized climatic conditions and the soil shrinkage and swelling properties (Aalami and Bommer 1999). USACE Technical Report ITL-88-1 provides design procedures for conventionally reinforced mats as well as similar post-tension guidance as provided by PTI for post-tension mat design on expansive soils.

8–4. Modulus of subgrade reaction

Modulus of subgrade reaction (also denoted coefficient of soil subgrade reaction) is a model parameter specific to the structure being modeled and depends on bearing

pressure magnitude and distribution (which depends on the structural mat stiffness), mat geometry, and the soil properties. It is defined by equation 8–1 as the bearing pressure divided by the soil settlement at a given point (*x,y*) as illustrated in Figure 8–2. A common unit of expression is lb/in^2/in. The notation $k_{s(x,y)}$ denotes the modulus of subgrade reaction at a specific plan coordinate. The modulus of subgrade reaction is not a soil property. Some references name this term as the "coefficient of soil subgrade reaction." The term "modulus of subgrade reaction" used in this document has different meaning than that proposed by Vesic (1961), and the two cannot be used interchangeably.

$$k_{s(x,y)} = \frac{q_{(x,y)}}{\Delta_{(x,y)}} \tag{8–1}$$

where:

$k_{s(x,y)}$ = modulus of subgrade reaction at coordinate (*x, y*) (lb/in^2/in)
$q_{(x,y)}$ = bearing pressure at coordinate (*x,y*) (lb/in^2)
$\Delta_{(x,y)}$ = soil settlement at coordinate (*x,y*) (in)

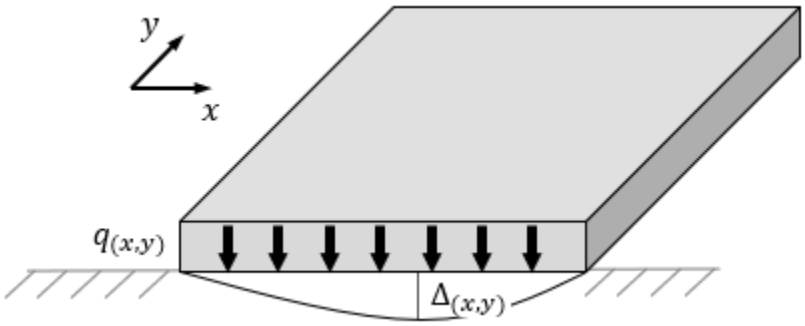


Figure 8–2. Mat bearing pressure (*q*) and ground settlement (Δ)

a. Application. For uncoupled and pseudo-coupled analyses, the soil supporting the mat is modeled as a vertically oriented, linear-elastic, Winkler-type spring, with the stiffness represented by the modulus of subgrade reaction.

b. Plate load test values.

(1) Terzaghi (1955) and other publications reported values from plate load tests. The reported values are often identified as k_1 or $k_{0.3}$, corresponding to 1 ft and 0.3-meter plate load tests and are primarily useful in pavement and lightly loaded slab-on-grade designs where wheel loading areas are similar in size to a plate load test. These values are different than k_s values used for structural mat designs, as k_s is a model parameter and specific to each structure and foundation condition.

(2) Several different researchers have proposed different methods to use the plate load test modulus to calculate a k_s suitable for building mat designs. Such approaches make many simplifications regarding soil uniformity and elastic behavior and mat

behavior that can introduce error. Sources of error include soil variability and material properties that vary with depth, variable in situ stress conditions and stress history, and potentially incorrect assumptions regarding the soil stress distribution. The simplified approaches to estimate a k_s for mat foundation design from textbook k_1 or $k_{0.3}$ should only be used for preliminary analyses.

8–5. Allowable bearing capacity

Allowable bearing capacity of the soil is calculated for the mat foundation following the bearing capacity analysis methods of Chapter 5 and safety factors presented in Table 8–1. General bearing capacity is checked considering the full mat geometry and total mat loads distributed over the mat.

a. The general bearing capacity is relatively insensitive to the precise shape of the bearing pressure, and it is uncommon that re-evaluation of bearing capacity is required after the final bearing pressure distribution is determined. General, local, and punching shear considerations should be examined in the bearing capacity analysis. Local bearing capacity considerations may not be significant for mat foundations in deep excavations where the excavation wall is at the mat edge; however, general, local, and punching shear considerations could be critical on projects where stratigraphy is layered with strata of major differences in strength, or on projects where the excavation wall does not extend to the mat edge and there are high lateral loads (Ulrich 1995).

b. Bearing capacity is rarely a controlling factor in mat foundations on sands and gravels but is more often critical for fine-grained soils. Bearing pressures under a mat foundation are not typically uniform. Sometimes local bearing pressures exceed allowable pressures calculated. In this situation, the bearing pressure under this local condition, typically at a wall or column load, should be checked considering the high-pressure area as its own footing, with the surrounding bearing pressure of the mat beyond higher pressure applied as a surcharge load in the bearing pressure equation. Often these locally higher bearing pressures have acceptable factors of safety.

8–6. Geotechnical aspects of mat design methods

Summaries of the Conventional (Rigid), ACI Approximately Flexible, and Discrete Area mat design methods are discussed in paragraphs 8–6a, 8–6b and 8–6c, respectively, with focus on the geotechnical aspects of the design. The Discrete Area Method may be used to estimate the modulus of subgrade reaction for mat designs. For simple loading conditions and generally small mats, the Conventional (Rigid) or ACI Approximately Flexible design methods are generally adequate and often more expedient.

a. Conventional (Rigid) Method.

(1) *General*. Rigid mats can be used as foundations for many kinds of projects, and are often used to support heavy machines, above-ground fuel tanks, and other similar structures. Rigid mats are designed such that the mat is so stiff that no differential settlement occurs between load concentrations relative to the mat, although the mat

might experience tilting if the load resultant is eccentric from the mat center, or the foundation support is uneven.

(2) *Application of Conventional (Rigid) Method*. The conventional method (Teng 1962; Bowles 1974) is the same as a conventional multi-column footing design extended in 2-D. The conventional method should be used only when (1) the variation in spacing of the columns and variation in the column loads is less than 20 percent, and (2) the column spacing is less than 1.75λ (ACI 336.2R); the stiffness evaluation factor (λ) is calculated per equation 8–2. The ACI Approximately Flexible Method or numerical methods should be used if the assumptions of the conventional method are not met.

$$\lambda = \sqrt[4]{\frac{k_s}{4EI}} \tag{8–2}$$

where:

λ = stiffness evaluation factor (in^{-1})
k_s = modulus of subgrade reaction (lb/in^2/in)
E = Young's modulus of mat material (lb/in^2)
I = moment of inertia of mat strip width (distance between column lines) (in^4)

(3) *Conventional (Rigid) Method assumptions*. Key assumptions made in the conventional method include:

(a) The mat is infinitely rigid, and therefore, mat-ground bearing pressures are not influenced by deformations in the mat,

(b) The bearing pressure below the mat follows a planar distribution where the centroid of the bearing pressure coincides with the line of action of the resultant force of the superstructure loads acting on the mat as shown in Figure 8–3, and

(c) The modulus of subgrade reaction can be simplified to a single constant value that independently deforms solely based on vertical pressure (such as uncoupled).

(4) *Limitations of the Conventional (Rigid) Method*. Due to the inherent flexibility of most mats, the conventional method could lead to erroneous estimates of mat bending moments. Some authors have recommended the conventional method be discontinued (Horvath 2002) due to general availability of better computational methods such as the Discrete Area Method.

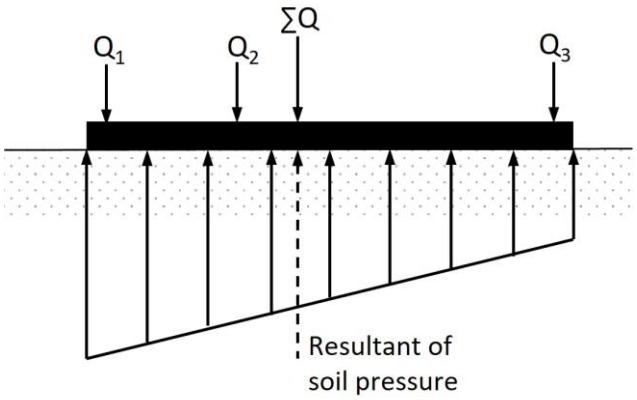


Figure 8–3. Conventional (Rigid) Method planar distribution of bearing pressure

(5) *Conventional (Rigid) Method steps*. The mat design should begin with calculation of allowable bearing capacity and estimation of a uniform modulus of subgrade reaction (k_s). Allowable bearing capacity (q_a) should be calculated according to paragraph 5–5, and the applied bearing pressure should be verified to be less than allowable. The settlement estimate should be made as would be made for a spread footing using service loads (see paragraph 2–10c). Different estimates of k_s for immediate and long-term settlement (consolidation) may be required, depending on the soil and loading conditions. The modulus of subgrade reaction (k_s) is used to verify relative mat rigidity per equation 8–2.

$$k_s = \frac{q}{\Delta} \tag{8–3}$$

where:

k_s = modulus of subgrade reaction (lb/in^2/in)
Δ = estimated settlement (in)
q = unfactored applied bearing pressure (psi)

(6) *Structural design steps*. Using the calculated modulus of subgrade reaction and allowable bearing pressure, the structural engineer follows a 7-step procedure to design the mat foundation. Geotechnical engineers should be familiar with the design steps to understand how the geotechnical parameters being provided are used in the structure design. The mat is designed into strips for evaluation, and the design procedure is summarized in Table 8–1.

Table 8–1
Procedure for Conventional (Rigid) Method

Step 1	Determine the line of action location for each of the column loads applied to the mat. For columns loading a mat, the loads are summed to calculate the total applied load as show in equation 8–4. Eccentricity from the x and y axes are calculated using equations 8–5 through 8–8. $$Q_t = \sum_{n=1}^{i} Q_n \quad (8–4)$$ $$x' = \frac{\sum_1^i Q_n \cdot x_n}{Q_t} \quad (8–5)$$ $$y' = \frac{\sum_1^i Q_n \cdot y_n}{Q_t} \quad (8–6)$$ $$e_x = x' - \frac{B}{2} \quad (8–7)$$ $$e_y = y' - \frac{B}{2} \quad (8–8)$$ where: Q_t = total applied load (lbs) i = number of columns Q_n = individual column load for column n (lbs) x' = location of resultant relative to the neutral location, x-axis (ft) y' = location of resultant relative to the neutral location, y-axis (ft) x_n = x-coordinate for column n (ft) y_n = y-coordinate for column n (ft) e_x = eccentricity from neutral axis about x-axis (ft) e_y = eccentricity from neutral axis about y-axis (ft) B = width of strip (ft)

Step 2	The mat bearing pressures (q) at points (x,y) beneath the mat are checked and compared to the allowable bearing pressures using equation 8–9. The bearing pressures must be less than the allowable bearing pressure. $$q = \frac{Q_t}{A} \pm \frac{M_x}{I_x} y \pm \frac{M_y}{I_y} x \quad (8–9)$$ $$I_x = \frac{B \cdot L^3}{12} \quad (8–10)$$ $$I_y = \frac{L \cdot B^3}{12} \quad (8–11)$$ $$M_x = Q_t \cdot e_y \quad (8–12)$$ $$M_y = Q_t \cdot e_x \quad (8–13)$$ where: q = applied bearing pressure (psf) Q_t = total load (lbs) A = area of mat (ft^2) M_x = moment of the column loads about x-axis (ft-lbs) M_y = moment of the columns loads about y-axis (ft-lbs) I_x = moment of inertia about x-axis (ft^4) I_y = moment of inertia about y-axis (ft^4) x = x-coordinate for column n (ft) y = y-coordinate for column n (ft) B = width of strip (ft) L = length of strip (ft) e_x = eccentricity from neutral axis about x-axis (ft) e_y = eccentricity from neutral axis about y-axis (ft)
Step 3	Proportion the mat thickness to resist column punching and shear failure.

Step 4	Divide the mat into strips in both the *x* and *y* directions. The width of a strip is notated as B_1 and the length as B as illustrated in Figure 8–4. The Q-values in Figure 8–3 represent column loads distributed throughout the mat. In practice, strip edges are generally set to bisect column lines, simplifying the analysis with loads centered on the strip.

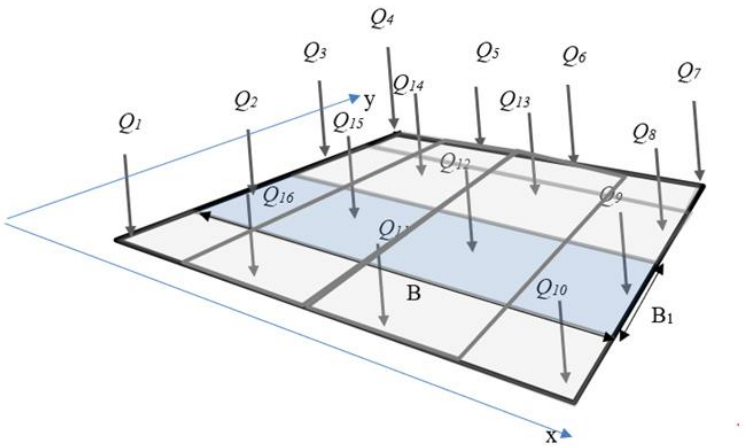


Figure 8–4. Rigid mat divided into strips

Step 5	For each strip, calculate the modified column loads by determination of the total loads on the strip (Q_s), equation 8–14, average bearing pressure on the strip (q_{av}), equation 8–15, average load on the strip (Q_{av}), equation 8–16, average modified bearing pressure (q_{avmod}), equations 8–17, and modification factor (F), equation 8–18.

$$Q_s = \sum_{1}^{i} Q_m \quad (8\text{–}14)$$

$$q_{av} = \frac{Q_s}{B \cdot B_1} \quad (8\text{–}15)$$

$$Q_{av} = \frac{Q_s + B \cdot B_1 \cdot q_{av}}{2} \quad (8\text{–}16)$$

$$q_{avmod} = \frac{Q_{av}}{B \cdot B_1} \quad (8\text{–}17)$$

$$F = \frac{Q_{av}}{Q_s} \quad (8\text{–}18)$$

where:

Q_s = total load on strip (lbs)
i = number of columns
Q_m = one of i column loads on the strip (lbs)
q_{av} = average bearing pressure (psf)
B = length of mat strip (ft)
B_1 = width of mat strip (ft)
Q_{av} = average column load (lbs)
q_{avmod} = average modified bearing pressure (psf)
F = modification factor (unitless)

Step 6 — Shear and moment loading diagrams for the strip are drawn with each column load multiplied by the factor (F), and the bearing pressure uniformly applied as (q_{avmod}). The highlighted strip from Figure 8–4 is selected for illustration on the mat strip loading diagram (Figure 8–5). Due to the simplifications required for the rigid mat design procedure, moment diagrams may not fully close to zero at each end.

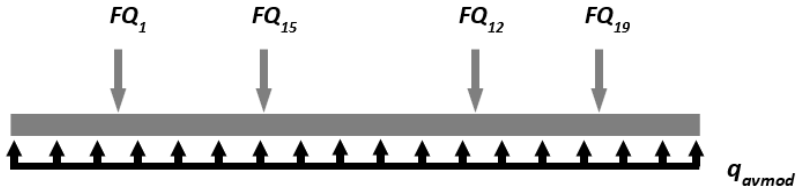


Figure 8–5. Mat strip loading diagram

Step 7 — Design the mat (such as dimensions, concrete properties, and reinforcing) for the shear and moments calculated.

b. ACI Approximate Flexible Method. For simple structures that have variable column loading, or for mats that are insufficiently stiff to be considered rigid, the ACI Approximately Flexible Method may be used to estimate developed moments and shears in the mat. This method is presented in ACI 336.2R, described in Bowles (1996), and adapted herein.

(1) *Limitations*. The assumptions made in this method can introduce error due to the assumption of a single value of modulus of subgrade reaction. For larger structures, computational finite grid, finite difference, or finite element methods should be used for full design of flexible mats.

(2) *ACI Approximately Flexible Method assumptions*. This method is based on theory of plates on an elastic foundation and makes the key assumptions that the foundation behaves as liner-elastic springs that are infinitely small and independent (such as uncoupled); the springs can resist both tension and compression; and the modulus of subgrade reaction can be simplified as a single uniform spring constant as shown in Figure 8–6. Like the conventional method procedure, the geotechnical engineer should calculate the allowable soil bearing capacity (q_a), estimate settlement (Δ), and approximate a soil modulus of subgrade reaction (k_s) in the same manner as described for the conventional method.

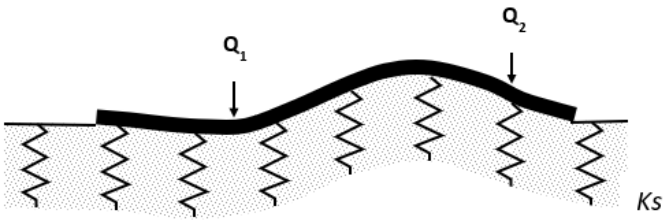


Figure 8–6. Principles of American Concrete Institute Approximate Flexible Method

(3) *Structural design steps*. The five design steps shown in Table 8–2 are typically performed by the structural designer in the ACI Approximately Flexible Method.

Table 8–2
Procedure for the American Concrete Institute Approximately Flexible Method

Step 1	Establish the thickness of the proposed mat required to resist punching and shearing at critical locations.
Step 2	Calculate the mat rigidity (D) per equation 8–19. $$D = \frac{E_C \cdot t^3}{12(1-\mu_C^2)} \quad (8–19)$$ where: D = flexural rigidity of the mat (lb-ft) E_C = Young's modulus of mat material (psf) t^3 = thickness of mat (ft) μ_c = Poisson's ratio of mat (for concrete, use 0.15)
Step 3	Compute the radius of effective stiffness (L) $$L = \sqrt[4]{\frac{D}{k_s}} \quad (8–20)$$ where: L = radius of effective stiffness (ft) D = flexural rigidity of the mat (lb-ft) k_s = modulus of subgrade reaction (psf/ft)

Step 4	Compute moments, shear, and deflection in the mat for each column load (P). The structural calculations are based on mat rigidity (D), radius of effective stiffness (L), and Z-factors obtained from chart solutions by Hetényi (1946) and shown on Figure 8–7. The equations 8–21 through 8–26 for moments, shear, and displacement are developed in polar coordinates. The equations for moments can be transformed to cartesian coordinates for design of x and y direction reinforcing design as shown on equations 8–27 and 8–28 where θ is the angle from x-axis. When load columns have overlapping radius of influence (L), principles of superposition can be applied to obtain the net effects. At mat edges where L extends past the edge, calculate the edge moment and shear, and apply edge loads with the opposite sign to balance the column load.

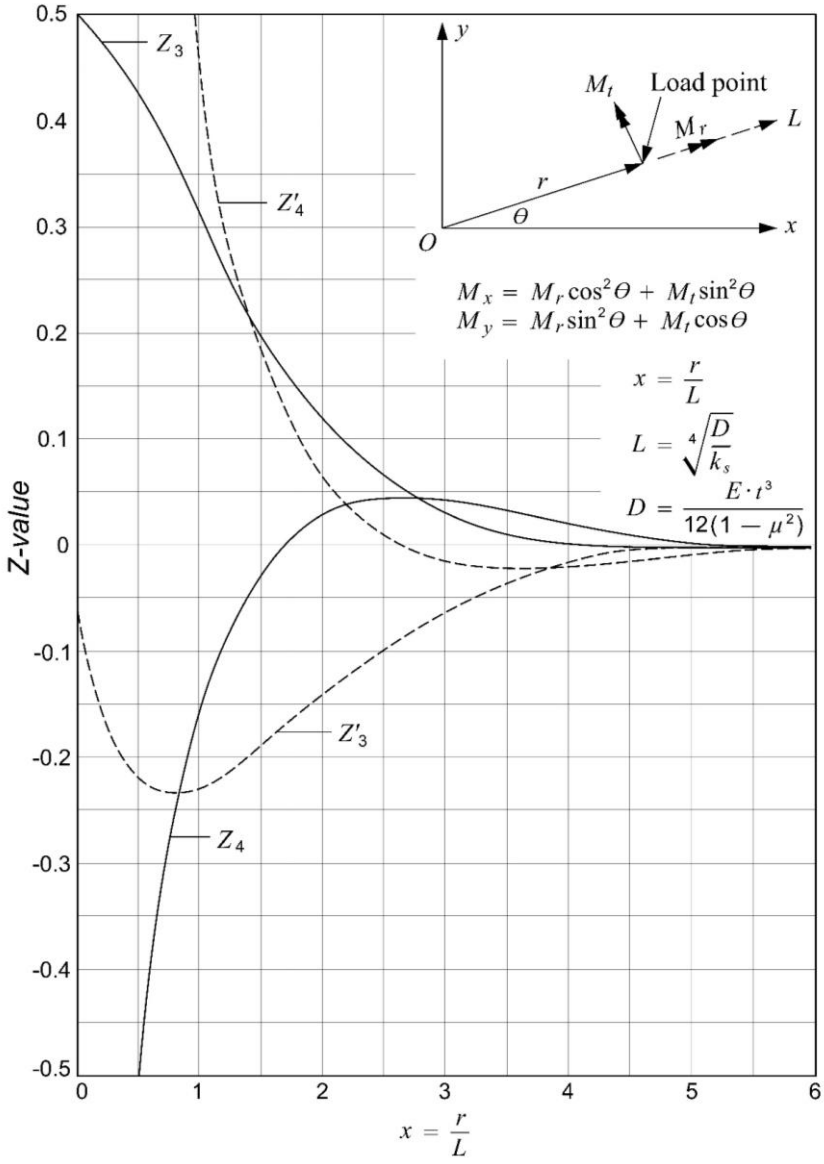


Figure 8–7. Z-factors for American Concrete Institute Approximately Flexible mat design procedure for use with equations 8–21 to 8–28 (Hetényi 1946)

	$$M_r = -\frac{P}{4}\left[Z_4 - \frac{1-\mu_c}{x}Z_3'\right] \quad (8\text{–}21)$$ $$M_t = \frac{P}{4}\left[\mu_c \cdot Z_4 + \frac{1-\mu_c}{x}Z_3'\right] \quad (8\text{–}22)$$ $$\Delta H = \frac{P \cdot L^2}{8D} \quad (8\text{–}23)$$ Vertical Displacement $$\Delta h = \frac{P \cdot L^2}{4D}Z_3 \quad (8\text{–}24)$$ At Distance r from Load $$V = -\frac{P}{4L}Z_4' \quad (8\text{–}25)$$ Shear $$x = {}^{r}/_{L} \quad (8\text{–}26)$$ $$M_x = M_r\, cos^2\,\theta + M_t sin^2\,\theta \quad (8\text{–}27)$$ $$M_y = M_r\, sin^2\,\theta + M_t\, cos^2\,\theta \quad (8\text{–}28)$$ where: M_r = radial moment at a distance, r from point load P (lb-ft) P = column load (lbf) Z_4 = Z_4-factor from Figure 8–7 (unitless), function of Z-value and x Z_4' = Z_4'-factor from Figure 8–7 (unitless), function of Z-value and x μ_c = Poisson's ratio of mat material (unitless) Z_3' = Z_3'-factor from Figure 8–7 (unitless), function of Z-value and x Z_3 = Z_3-factor from Figure 8–7 (unitless), function of Z-value and x x = distance ratio (r/L) shown in Figure 8–7 (dimensionless) M_t = tangential moment (lb-ft) ΔH = vertical displacement at column (ft) L = radius of effective stiffness (dimensionless), equation 8–20 D = flexural rigidity of the mat (lb-ft), equation 8–19 Δh = vertical displacement at distance r (ft) from load P V = shear per unit width (lbf/ft) at distance r (ft) from load P x = distance ratio as shown on Figure 8–7 r = radial distance from the column load (ft) M_x = M_r expressed in rectangular coordinate system θ = angle in radians from the x axis passing through the column load M_y = distance ratio as shown on Figure 8–7 (dimensionless)
Step 5	Proportion the mat.

c. Numerical design methods.

(1) *Non-Rigid Mat Foundations*. For non-rigid mat foundation designs, numerical finite grid, finite difference, or finite element computation routines should be used to determine the moments and shears in the mat and proportion the mat and design reinforcement. Numerical design methods require the input of the modulus of subgrade reaction, which is determined using the Discrete Area Method described in paragraph 8–6c(2). Details of the formulation of these numerical methods are beyond the scope of this chapter and are available in several commercially available software programs. Bowles (1996) or other publications may be referenced for the details regarding numerical formulation and details on the implementation of elements and boundary conditions. The ACI Approximately Flexible Method should be used to check reasonableness of numerical methods, and for preliminary mat thickness sizing.

(2) *Discrete Area Method*.

(a) The Discrete Area Method may be used to estimate the soil modulus of subgrade reaction for input into a numerical structural analysis of the mat foundation in a pseudo-coupled mat design approach. To use this method, the structural and geotechnical engineers perform separate parallel analyses. The structural engineer analyzes a mat supported by several independent springs; however, allows a different stiffness of each spring. The geotechnical engineer analyzes an elastic continuum with “discrete areas” of mat loading pressure to model stress distribution and ground deformation within the soil mass supporting the mat. The modulus of subgrade reaction is calculated by taking the mat bearing pressure divided by the settlement and is calculated at points throughout the mat area.

(b) The results are used to develop a contour plan of a variable modulus of subgrade reaction. Limitations of the analysis include assumptions about soil stress distribution, limitations of the structural numerical model, assumptions regarding soil properties, and limitations in the size of the discrete areas that can be practically modeled. The general steps in the Discrete Area Method iterative process between the structural engineer and the geotechnical engineer are described by Ulrich (1988) and adapted as shown in Table 8–3.

Table 8–3
Procedure for the Discrete Area Method

Step 1	The geotechnical engineer provides the allowable bearing pressure for the mat foundation based on rigid assumptions and the methods described in Chapter 8.5. Factors of safety should follow Table 2–1. If bearing pressures are near allowable limits, this step should be revisited with the final estimated bearing pressures. Often the bearing pressures are well below the allowable limits, and no additional analysis is necessary.
Step 2	The structural engineer proportions the mat thickness to satisfy general and local bearing capacity and punching requirements.

Step 3	The mat design continues using the numerical structural methods for determining internal mat stresses. For the first iteration of the numerical structural model, the geotechnical engineer provides a single modulus of subgrade reaction based on a calculated settlement for an idealized, uniform pressure on the soil. The bearing pressure used is simply the total loading from columns and walls divided by the total area of the mat.
Step 4	The structural engineer provides output of the mat/ground interaction analysis, including calculations of mat deflection and bearing pressures. The bearing pressure output is then idealized into discrete areas, modeled using the average pressure of the area. The discrete areas typically include corner areas, edge areas, center mat areas, and areas around large, concentrated loads (multiple columns or wall loads), but will generally be much larger than the structural analysis grids. The more areas modeled; the more computational difficulty is imposed on the geotechnical analysis. Limiting the number of discrete areas to 15 to 20 often results in reasonable convergence of the calculations. If the iterative mat design process does not result in reasonable mat deformation and ground settlement convergence, the discrete areas may need to be smaller, especially in areas of high rates of pressure change. Some geotechnical software programs are designed to allow direct importing of bearing pressure grided data from structure analysis software and allow very fine discrete areas, however, the computational effort for many discrete areas can be very high. Figure 8–8 illustrates an example discretization of the mat bearing pressure areas selected for geotechnical stress distribution and analysis, labeled A1 through A13, with white boundaries. The black gridlines are structural analysis elements.

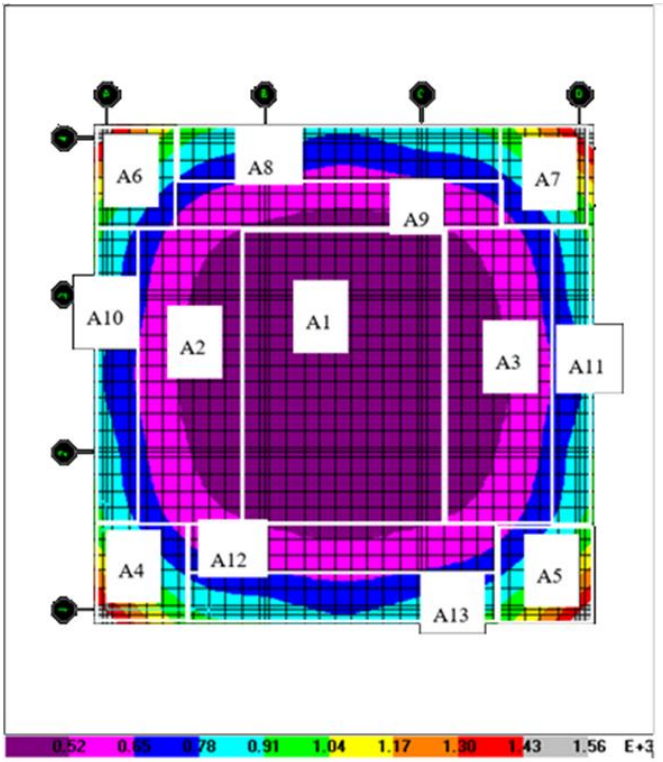


Figure 8–8. Example of area discretization of mat bearing pressure contours from structural analysis, plan view (Computers and Structures, Inc., SAP2000 computer software output, ksf); the mat is 110 ft square

Step 5	After the mat areas are divided into discrete areas and associated average bearing pressures are calculated, the soil stress distribution and settlement of the ground due to the applied contact pressures is calculated. Stresses and ground deformations are calculated in a 3-D grid across and below the mat, with each grid location stress and settlement calculation considering the applied pressures of each discrete area by principle of superposition and as described in equations 8–29 and 8–30. equation 8–30 is typically truncated at depths where settlement or stresses below are negligible to the design. Stress distribution and soil settlement models should be selected based on the site ground conditions and behavior. For example, for clays under sustained loads, the strain should be calculated using consolidation theory. For immediate settlement, strains may be calculated using elastic modulus soil parameters. Typically, computer programs are used to perform this work, although chart or analytical solutions may be used; however, such calculations are tedious. A profile plot of vertical stresses caused by the mat calculated at 2-ft soil grid locations is shown on Figure 8–9. $$\Delta\sigma_{xyz} = \sum_{n=1}^{i} \Delta\sigma_{xyz} f(A_n) \quad (8\text{–}29)$$ $$\rho_{xy} = \sum_{z=0}^{\infty} \varepsilon_{xyz} \cdot l \quad (8\text{–}30)$$ where: $\Delta\sigma_{xyz}$ = total increase in vertical stress at point (*x,y,z*) (ksf) i = number of loads $\Delta\sigma_{xyz}\mathrm{f(A_n)}$ = increase in vertical stress at point (*x,y,z*) due to discrete area load ($\mathrm{A_n}$) (ksf) ρ_{xy} = ground settlement at plan location (*x,y*) (ft) z = depth below mat (ft) ε_{xyz} = vertical strain at (*x,y,z*) due increase in vertical stress ($\Delta\sigma_{xyz}$) l = element length in *z*-direction (ft)

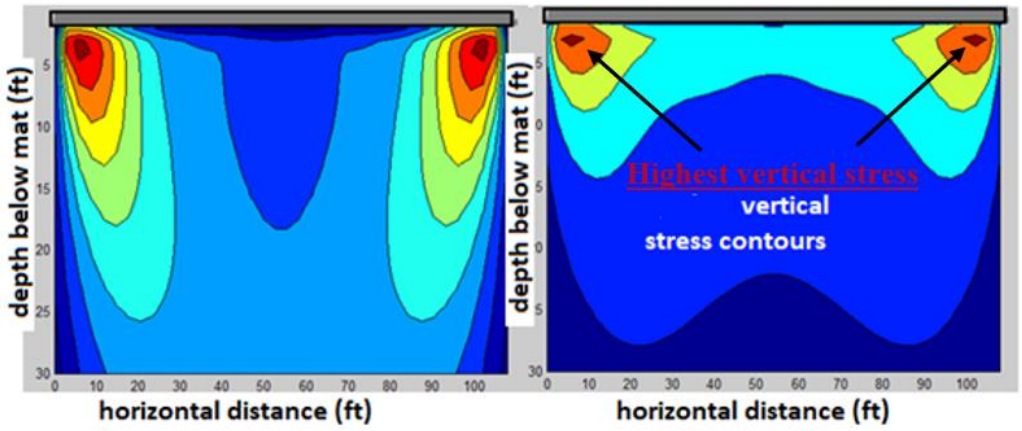


Figure 8–9. Example profile plot of increase in vertical soil stresses (psf, unlabeled) beneath the center (left) and edge (right) of a 110-ft-wide mat using Matlab v7.1 software; corresponds to mat bearing pressures shown in Figure 8–8

Step 6	The modulus of subgrade reaction is calculated at each plan grid point (*x,y*) by equation 8–30, and a contour plan of modulus of subgrade reaction is developed and input into the structural analysis. Figure 8–10 illustrates the calculation of modulus of subgrade reaction.

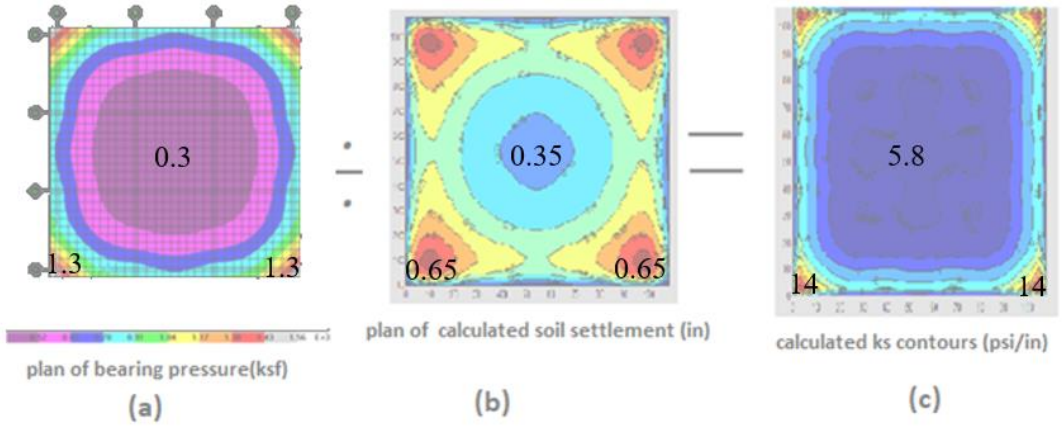


Figure 8–10. Determination of modulus of subgrade reaction contours: (a) is the mat bearing pressure calculated using SAP2000, (b) is a contour plan of settlement, and (c) is a contour plan of modulus of subgrade reaction, both plotted with Matlab v7.1

Step 7	The structural analysis is performed again with an updated modulus of subgrade reaction, and the estimated mat contact pressure and estimated settlement are plotted. If the mat deflections and bearing pressures are significantly different than was calculated at the end of Step 3, then Steps 4, 5, 6, and 7 are repeated until convergence is reached. Convergence is usually achieved in about 3 iterations with differences between the structural and geotechnical estimates of displacement on the order of 0.1 inch. For a uniformly loaded mat of uniform stiffness on an elastic foundation, the settlement along the edges and corners will be about 1/2 and 1/4 of the maximum settlement at the center of the mat, respectively. The corresponding k_s will vary accordingly, and at the edges and corners will be about 2 and 4 times the k_s calculated at the center of the mat, respectively. The flow chart in Figure 8–11 illustrates the iteratives steps for designing a mat foundation and how the structural and geotechnical engineers (SE and GE, respectively) work together to complete a mat design.

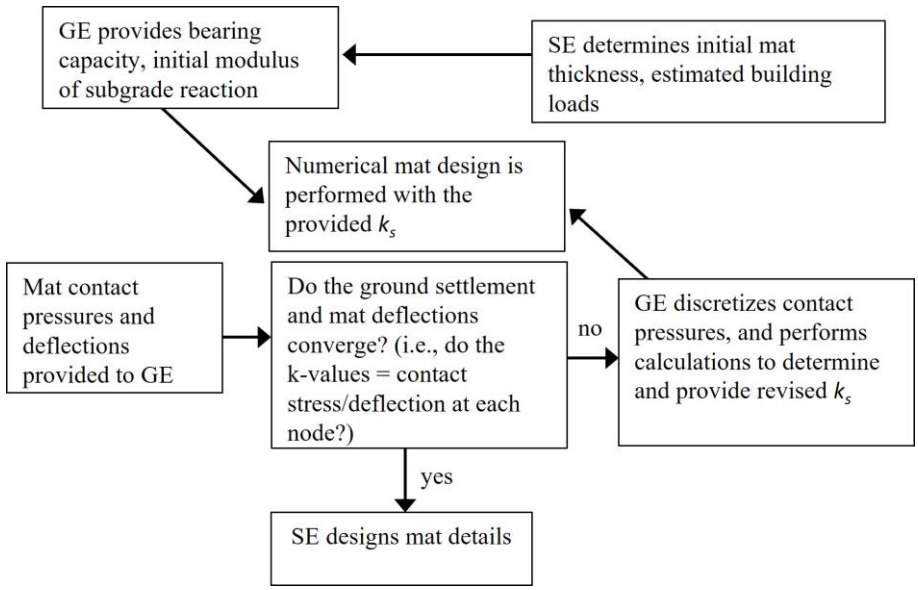


Figure 8–11. Flow chart of iterative steps in the Discrete Area Method for determination of modulus of subgrade reaction

8–7. Ground improvement considerations

Ground improvement methods such as aggregate piers, soil-cement mixing, grouting, or other methods are improvement techniques sometimes used to increase the soil stiffness properties. Ground improvement that incorporates isolated stiff elements (such as piles, aggregate piers) complicate the stress distribution and settlement distribution in the soil mass. Locally stiff areas will result in concentrated mat supports and highly variable modulus of subgrade reaction for structural analysis. Geotechnical software capable of modeling vertically oriented stiff elements is recommended to estimate the improved soil mass deformation response in determination of the ground settlement and modulus of subgrade reaction.

8–8. Performance monitoring

Mat designs should be supplemented by the observational method as described by Ulrich (1995) to collect data regarding mat settlement to help evaluate the structural performance and inform future designs. The mat performance should be monitored both during construction and post-construction to evaluate the performance. During construction, the building construction activities should be recorded to understand when new floors are added to the building, stockpiles are moved, or other loading changes occur. Geotechnical monitoring can include settlement plates installed at the soil/mat boundary, piezometers to monitor ground pore pressure dissipation, and floor level surveys to evaluate concrete mat defections and differential movement. For very tall buildings, monitoring of structural components should also be considered due to structural interactions of columns with mat.

8–9. Basement mat considerations

Mat foundations are commonly used where a basement for a large structure is planned. The removed load of the basement excavation is largely balanced by the building load. Unloading and rebound need to be considered in understanding the soil settlement under the new building loads as discussed in paragraph 7–15. If minimal rebound is anticipated or will be re-compressed before the mat cures, the mat average bearing pressure is assumed as the net pressure as defined in equation 8–31. In addition to distributing the building loads, the mat foundation also provides a water barrier for the basement.

$$Q_{net} = \frac{[W_s + W_m] - W_{exc}}{A} \tag{8–31}$$

where:

Q_{net} = net pressure (psf)
W_s = weight of structure (lbs)
W_m = weight of mat (lbs)
W_{exc} = weight of soil (lbs)
A = mat area (ft^2)

Chapter 9
Seismic Design

9–1. Introduction

Earthquake loading must be considered in shallow foundation design. Earthquake-induced loads could result in unsatisfactory foundation performance or failure. This chapter discusses the aspects of foundation performance due to seismic loads that should be considered during foundation design. Requirements for ultimate and serviceability limits states for seismic loading conditions are also provided.

9–2. Geologic and seismic hazards

Earthquake ground motions can affect shallow foundations in many ways. Load effects considered during foundation design should include: (1) cyclic softening, which can lead to bearing capacity failure, (2) inertial forces resulting in sliding or overturning failure, (3) soil liquefaction resulting in loss of shear strength and settlement, (4) ground mass instability (such as global stability, lateral spreading), and (5) fault rupture. As per American Society of Civil Engineers (ASCE) 7, an investigation report for a building structure must include the geologic and seismic hazards such as (a) slope instability, (b) liquefaction, (c) total and differential settlement, and (d) surface displacement caused by faulting or seismically induced lateral spreading or lateral flow. Seismic stability of sloping ground with structure is not covered in this manual.

9–3. Design standards

Design and loading conditions are determined according to the requirements of ER 1110-2-1806 and criteria provided in UFCs 3-301-01 and 3-301-02. Buildings are required to be designed according to ASCE 7, bridges are required to be designed according to the latest AASHTO code. Hydraulic structures (such as dams, floodwalls) are required to be designed according to ER 1110-2-1806 loading and associated performance criteria.

9–4. Design loads

Seismic loadings for shallow foundation design at a site are generally estimated based on site and regional seismic hazard, subsurface conditions, and the structure's natural frequency. Depending on the limit state or structural behavior evaluated, various parameters are used to characterize these loads.

a. Earthquake moment magnitude. Moment magnitude (M_w) is a measure of an earthquake's intensity and is used in liquefaction evaluations. Deaggregation methods can be used to review earthquake magnitudes that contribute to the earthquake hazard at a site. The modal M_w value has the highest contribution to the seismic hazard. The mean M_w value is calculated using the weighted average from all sources in the deaggregation. Using the modal or the mean values when there is more than one major contributing source is not always appropriate. In such cases, analysis should be

performed for multiple mean (M_w) values and their associated peak ground acceleration (PGA) to establish the critical scenario.

b. Response spectrum.

(1) The response spectrum is a plot of the peak accelerations experienced by a series of single-degree-of-freedom (SDOF) oscillators with varying fundamental periods of vibration (typically with an assumed structural damping of 5 percent) when excited by an input ground motion. Code-based procedures simplify the spectral shape such that it can be scaled with several parameters (such as S_S, S_1). Generally, response spectrum based on geotechnical studies are developed at the foundation elevation. It can be further used in seismic demand estimation and structural analyses. Spectral accelerations can be obtained from the applicable building code, U.S. Geological Survey (USGS), or from site-specific ground motion studies. Mapped accelerations should be adjusted (multiplied by appropriate site coefficients) depending on seismic site classification.

(2) PGA is the maximum acceleration that corresponds to the 0-second period (very low period such as 0.001 to 0.01 sec) on the response spectrum for a particular site. This parameter is used in several analyses (such as liquefaction evaluations). The USGS National Seismic Hazard Mapping Program (NSHMP) tools or site-specific seismic hazard analyses can be used to develop site PGA. Even though other seismic parameters such as peak ground velocity (PGV) (for near-source conditions), Arias Intensity (I_A), and duration (Dur) are important for seismic evaluations, PGA and spectral acceleration are the widely used parameters for seismic evaluations. However, for seismic evaluations of critical structures, other parameters should be evaluated during ground motion time series selection for seismic analyses.

c. Site classification. Sites should be classified as Site Class A through F based on the soil conditions in the upper 100 ft and determined according to the applicable design code (for example, ASCE 7). The site classification quantifies the soil's propensity to amplify surface ground motion propagating from underlying rock. The degree of site amplifications varies based on site soil conditions. Site classification is used to adjust mapped spectral accelerations (response spectrum) and for assigning the seismic design category. The site class should reflect the soil conditions that affect the ground motion input to the structure or a significant portion of the structure.

d. Site-specific ground motion studies. Code-based response spectrum data allow the use of a site-specific ground motion study. Site-specific studies can be performed by supplementing USGS data or independently performing site-specific seismic hazards analyses. For other DoD structures, the methods used for evaluation of seismic ground motion parameters is referenced in UFC. The decision to perform this study is typically based on (1) site classification (such as Site Class F requires site-specific study), (2) cost optimization (such as cost savings due to potential reduction or increase in seismic loading), (3) cases that require input not covered in general codes (such as power generation facility, storage tanks), and (4) development of site-specific parameters for critical structures.

9–5. Seismic design category

Building codes use the seismic design category (SDC) to progressively increase the requirements for seismic analysis and design as both the seismic hazard and consequences of failure increase. The SDC is a function of site class, occupancy, and the response spectrum. Coordination between the geotechnical engineer and structural engineer is necessary and should be implemented for assignment of the seismic design category. In some instances, minor design adjustments (such as lowering the building base, site-specific ground motion studies) could result in a more favorable design category, which could result in construction cost savings. Conversely, similar adjustments or a site-specific study could warrant a more conservative site design category assignment.

9–6. Performance requirements

The design should satisfy the performance requirements in the seismic design standards, see paragraph 9–3. Seismic performance requirements depend on type of structure and consequences of failure. As with static design, serviceability criteria are more subjective depending on building type and function; however, certain minimum requirements must be met and are generally guided by life-safety considerations (such as safe egress). Performance requirements are provided in paragraphs 9–10 and 9–11.

9–7. Liquefaction

Liquefaction occurs when the dynamic shear stresses from a seismic event are applied to saturated soils with lower relative densities, which causes contraction of the soil matrix. This contraction caused by the quickly applied loading causes the effective stresses supported by the soil matrix to be transferred to the pore water. Because a large volume (potentially laterally and vertically) of soil is subject to the dynamic shear stresses, dissipation of the excess pore pressures is not possible within a short time frame during or after the seismic event. The high excess pore pressures result in a substantial reduction in shear strength and stiffness of soils, termed liquefaction.

a. Liquefaction susceptibility. Liquefaction susceptibility is the ability of a soil to liquefy irrespective of the level of earthquake shaking. The assessment of soil susceptibility for liquefaction of coarse-grained soils or cyclic softening of fine-grained soils depends on the soil type, depth from surface, grain size distribution, Atterberg limits, water levels during earthquake, OCR, sensitivity, etc.

(1) *Fine-grained soils*. Non-plastic silts and clayey silts of low plasticity ($PI < 12$) at high water content-to-liquid limit ratios ($w_c/LL > 0.85$) can liquefy under significant cyclic loading. Clayey silts and silty clays of moderate plasticity ($12 < PI < 18$) at $w_c/LL > 0.8$ can undergo liquefaction when shaken intensely for a number of cycles of loading. Additionally, there could be cases where sensitive soils ($PI > 18$) undergo severe strength loss due to earthquake-induced straining (Bray and Sancio 2006). Per Boulanger and Idriss (2006), clay-like behavior can be expected for fine-grained soils that have $PI > 7$. Recent studies by Stuedlein et al. (2023) indicate that ($PI < 12$)

provides sufficient criterion for the evaluation of liquefaction susceptibility, which considers material characteristics alone. Clay-like soils are not susceptible to liquefaction; however, they could be susceptible to cyclic softening (see paragraph 9–8).

(2) *Geologic age and deposition*. Table 9–1 presents screening guidance for liquefaction susceptibility. This guidance can be used when deposition environment and age are known, with geologic mapping or through subsurface exploration. However, site-specific geotechnical data should be given higher importance for liquefaction assessment. The highest anticipated temporal phreatic surface should be considered.

Table 9–1
Liquefaction screening based on deposition and age (Youd and Perkins 1978)

Type of Deposit	Distribution of cohesionless sediments in deposit	Likelihood that cohesionless sediments, when saturated, would be susceptible to liquefication			
		< 500 years	Holocene	Pleistocene	Pre-Pleistocene
Continental					
River channel	Locally variable	Very high	High	Low	Very low
Floodplain	Locally variable	High	Moderate	Low	Very low
Alluvial fan and plains	Widespread	Moderate	Low	Low	Very low
Marine terraces and plains	Widespread	—	Low	Very low	Very low
Delta and fan delta	Widespread	High	Moderate	Low	Very low
Lacustrine and playa	Variable	High	Moderate	Low	Very low
Colluvium	Variable	High	Moderate	Low	Very low
Talus	Widespread	Low	Low	Very low	Very low
Dunes	Widespread	High	Moderate	Low	Very low
Loess	Variable	High	High	High	Unknown
Glacial till	Variable	Low	Low	Very low	Very low
Tuff	Rare	Low	Low	Very low	Very low
Tephra	Widespread	High	High	?	?
Residual soils	Rare	Low	Low	Very low	Very low
Sebkha	Locally variable	High	Moderate	Low	Very low
Coastal zone					
Delta	Widespread	Very high	High	Low	Very low
Estuarine	Locally variable	High	Moderate	Low	Very low
Beach—high wave energy	Widespread	Moderate	Low	Very low	Very low
Beach—low wave energy	Widespread	High	Moderate	Low	Very low
Lagoonal	Locally variable	High	Moderate	Low	Very low
Foreshore	Locally variable	High	Moderate	Low	Very low
Artificial fill					
Uncompacted fill	Variable	Very high	—	—	—
Compacted fill	Variable	Low	—	—	—

b. Liquefaction triggering analyses. Youd et al. (2001), Boulanger and Idriss (2014), and Cetin et al. (2018) can be used to evaluate liquefaction potential of sandy soils and low plasticity fine-grained soils using the SPT results. Youd et al. (2001), Moss et al. (2006), and Boulanger and Idriss (2015) can be used to evaluate liquefaction potential of sandy soils and low-plasticity, fine-grained soils using the CPT results. The Andrus and Stokoe (2000) and Kayen et al. (2013) methods can be used to evaluate liquefaction based on shear wave velocity results.

(1) *Subsurface exploration*. Field explorations should use techniques that can allow calculation of liquefaction resistance (such as shear wave velocity testing, SPTs, and

CPTs). EM 1110-1-1804 includes considerations for site investigations for seismic studies.

(2) *Gravelly soils*. In gravelly soils where behavior is dominated by sand or finer particles, the sand-based liquefaction triggering relationships may be used. Sand-dominant behavior typically occurs when the sand and finer particles content is at least 20 percent. For gravelly soils, the Becker Penetration Test (BPT) and large-diameter ring density tests can be used to evaluate the in situ density of gravelly soils and the results converted to equivalent SPT values for liquefaction assessment.
EM 1110-1-1804 includes guidance on investigation at gravelly sites. SPT and CPT could be unreliable or unfeasible for site investigations in gravelly sites for liquefaction assessment. SPT gravel corrections can be evaluated per paragraph 4–10a(2); however, results could be unreliable.

(3) *Liquefaction potential factor of safety*. Liquefaction potential is generally expressed in terms of factors of safety against liquefaction as the ratio of cyclic resistance ratio (CRR) to cyclic stress ratio (CSR). Where FS_{liq} < 1.0, liquefaction is likely, and where 1.0 < FS_{liq} < 1.4, liquefaction is likely to marginal. Liquefaction triggering analyses can be performed using different methods, understanding the limitations of each. The difference in results can indicate the level of uncertainty in the methods used.

(4) *Rigorous methods*. More rigorous methods such as 2-D site response analysis using stiffness and damping relationships can be performed to obtain site-specific CSR, which eliminates use of empirical stress reduction factor (r_D). FHWA-NHI-11-032 provides examples of some conditions that may warrant the use of more rigorous methods. These are:

(a) Sites where simplified liquefaction procedure shows liquefiable soils to depths greater than what is considered suitable for using the simplified procedures,

(b) Sites that have significant interlayering, particularly where interlayers comprise highly permeable soils or low-permeability layers, and

(c) Sites where the cost of ground improvement methods or deep foundation support to mitigate the impact of liquefaction is high.

9–8. Behavior of fine-grained soils

Ground failures and deformations in clays and plastic silts have also occurred with similar results to liquefied cohesionless soils. However, these occurrences are typically less common than liquefaction failures in cohesionless soils (sands and non-plastic silts). The term cyclic softening is used to describe strength loss and deformation in clays and plastic silts.

a. Susceptibility. Generally, normally consolidated to lightly overconsolidated clayey soils (OCR < 3 to 4) with medium to high sensitivity (sensitivity > 2) can be subject to cyclic strain softening due to an earthquake.

b. Cyclic softening soil behavior. Clay-like soils can generate excess pore pressures during cyclic loading and suffer a corresponding decrease in shear stiffness. These effects could lead to the accumulation of significant shear strains during cyclic loading or, in the case of sensitive clays, a significant drop in the available undrained strength. These effects should be included in seismic response and deformation analysis that includes potentially susceptible clay-like soils. Overconsolidation ratio, sensitivity ratio, peak or fully softened peak strength, and residual strength of fine-grained soils are required to characterize cyclic softening behaviors.

c. Cyclic softening resistance. Boulanger and Idriss (2007) provide a relationship to evaluate cyclic resistance of silts and clays that are likely to experience clay-like behavior and subject to potential cyclic softening. This relationship requires ratio of cyclic stress to undrained shear strength (s_u) for the number of equivalent uniform cycles representative of an M_w = 7.5. earthquake. Laboratory testing of undisturbed samples is required for use of this relationship and if the factor of safety of cyclic softening ($FS_{cs} \leq 1.0$), cyclic softening is likely.

9–9. Dynamic shear strength

During strong ground motions, liquefaction of sand-like soil and cyclic softening of clay-like soils could result in sudden loss of shear strength.

a. Soils not susceptible to shear strength loss.

(1) Dynamic strength of soils not susceptible to marginal liquefaction, liquefaction, or cyclic softening depends on rate effects, cyclic loading, and deformations. It also depends on duration and intensity of ground motions and response of soils. Various references suggest strength values from 80 percent to over 100 percent of the soil peak strength are expected during earthquake shaking. Some research has indicated that a 15 to 20 percent strength reduction based on static strength is appropriate. Other research indicates that this strength reduction can be ignored considering rapid earthquake loading. However, this assumption may not be appropriate for longer-duration earthquakes.

(2) In general, a peak to fully softened shear strength value should be used for evaluation, and selection of the parameter should be consistent with the amount of strain expected in the analysis. Static undrained strength reduced by 20 percent may also be used as the dynamic yield strength for non-liquefiable soils. This criterion is based on the Makdisi and Seed (1978) findings, which indicate that for clayey soils, dry or partially saturated cohesionless soils, or very dense saturated cohesionless soils, 80 percent of the static undrained strength represents an approximate threshold between large and small strains induced by cyclic loading.

b. Residual strength of soils susceptible to liquefaction. Residual strength (S_r) of potentially liquefiable soils can be estimated using empirical relationships. These empirical relationships have been developed using performance case histories of embankments that have experienced large deformations or flow slides. Several post-liquefaction residual strength relationships are available for use as shown in Table 9–2. The post-liquefaction residual strength should be estimated using different methods, understanding the limitations of each method.

Table 9–2
Post-liquefaction residual strength relationships

Description	Source
Residual shear strengths (S_r) for liquefied soils based on equivalent clean sand N-value $(N_1)_{60\text{cs}-\text{Sr}}$.	Seed and Harder (1990)
Residual shear strength to initial effective stress (σ'_{vo}) ratio (S_r/σ'_{vo}) based on stress normalized SPT N-value $(N_1)_{60}$.	Olson and Stark (2002)
Estimates S_r or S_r/σ'_{vo} based on $(N_1)_{60\text{cs}-\text{Sr}}$. Fines correction factors for Boulanger and Idriss (2015) are the same as Seed and Harder (1990) and different than fines correction factors for liquefaction triggering analysis. Relationship with void redistribution should be used.	Boulanger and Idriss (2015)
Estimates S_r as a function of $(N_1)_{60}$ and σ'_{vo}.	Kramer and Wang (2015)
Estimates S_r as a function of equivalent clean sand SPT N-value $(N_1)_{60\text{cs}-\text{Sr}}$ and σ'_{vo}.	Weber et al. (2022)

Note: Use the appropriate $(N_1)_{60}$ and $(N_1)_{60\text{cs}-\text{Sr}}$ procedures in residual strength estimation.

c. Residual excess pore pressures for marginally liquefiable soils. Residual excess pore pressure (r_u) is defined as the ratio of excess pore pressure to the total initial effective overburden stress. $r_u$can be generated when FS_{liq} > 1.0. As shown in Figure 9–1, when FS_{liq} < 1.4 and approaching FS_{liq} = 1.0 the expected r_u increases substantially compared to FS_{liq} > 1.4. Not accounting for this increase in pore pressures can greatly overestimate the shear strength. Figure 9–1 can be used to estimate excess pore pressures for marginally liquefiable soils, otherwise, residual undrained shear strengths should be used.

d. Fine-grained soils susceptible to cyclic softening. Shear strengths should be representative of large strains, which could be incrementally lowered from an initial reduced strength (80 percent of static strength) to residual strength of clay. If analyses are performed without ability to estimate strains, then the failure mode should be evaluated based on residual strength of clay. Laboratory cyclic triaxial shear or cyclic simple shear testing may also be used to evaluate the cyclic stress-strain behavior.

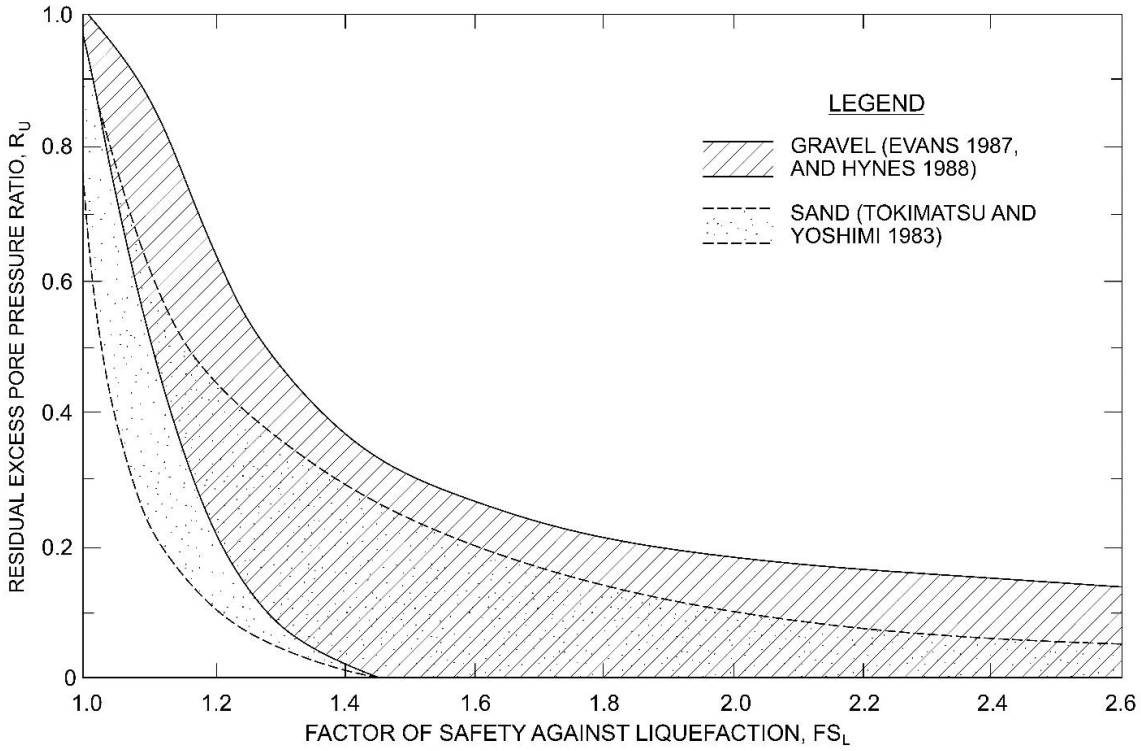


Figure 9–1. Relationship between residual excess pore pressure ratio and FS_{liq} for sand and gravel (Marcuson et al. 1990)

9–10. Ultimate limit state

Design for the ultimate limit state under seismic loading should represent a process that ensures the probability of collapse of a building and its associated risk to human life is at an acceptable level. Because the probability of a seismic event is lower than that assumed for working levels of stress, a lower factor of safety is often permitted and performance limits are generally less rigid.

a. Liquefaction and shear strength loss. With respect to liquefaction and SSL and to prevent foundation failure, shallow foundations should not be used where (1) soils within the bearing capacity shear zone are liquefiable, marginally liquefiable, or subject to significant SSL, and (2) significant lateral spreading (see paragraph 9–12) is expected.

b. Bearing capacity of soils not susceptible to shear strength loss. For soils within the bearing capacity shear zone that are not susceptible to SSL, the ultimate values for bearing capacity may be calculated using conventional static methods. The undrained shear strengths for cohesive soils and drained shear strengths for cohesionless soils should be used. Overstress factors may be applied where permitted by the building code. These values must not exceed one-third increase to the static allowable resistance. Overstress factors should not be applied to soils that are susceptible to SSL.

c. Bearing capacity of soils with shear strength loss. The general design principles in paragraph 9–10a should be adhered. Where additional evaluation may prove useful, the bearing capacity methods presented in this section can be used for foundations where the soils within the bearing capacity shear zone are susceptible to SSL during strong ground motions. Some of these procedures can be used for analysis of foundation systems with soils susceptible to SSL ($FS_{liq} \leq 1.4$ or $FS_{cs} \leq 1.0$) being within the bearing shear zone. The expected capacity of foundations can be calculated using the methods presented in the following sections.

(1) *Punching shear analysis*. Foundations supported by a non-liquefiable soil that is underlain by a potentially liquefiable layer should be evaluated for punching shear failure. For this analysis, the resistance to punching failure is provided by shear resistance of non-liquified soil acting on a vertical plane below the foundation edges as illustrated in Figure 9–2. The potentially liquefiable soil is assumed to provide no resistance to punching failure. The general forms of the equations are expressed in equations 9–1 and 9–2. The shear resistance will vary depending on material type and strength. The minimum factor of safety against a punching-type failure should be at least 3.0.

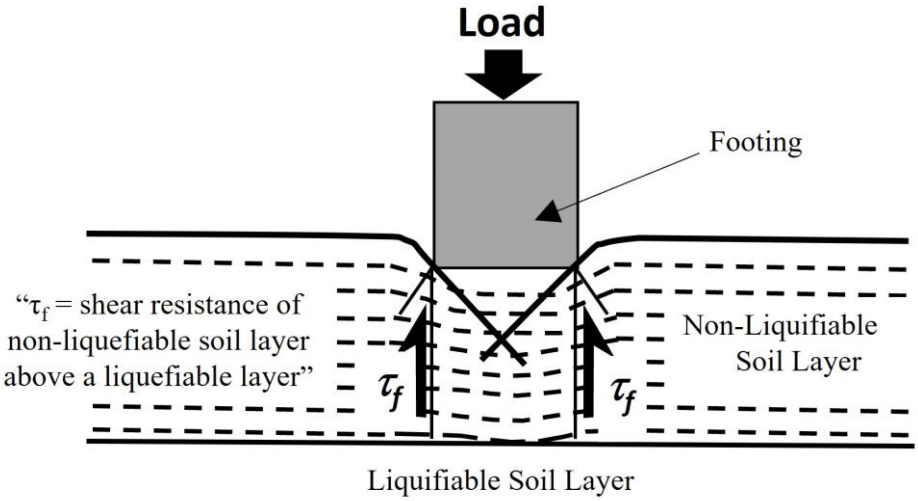


Figure 9–2. Punching shear resistance below a shallow footing (Day 2012)

$$FS = \frac{R}{P} = \frac{2T \cdot \tau_f}{P} \quad (9\text{–}1)$$

Continuous Footings

$$FS = \frac{R}{P} = \frac{2(B + L)\tau_f}{P} \quad (9\text{–}2)$$

Isolated Spread Footings

where:

FS = calculated factor of safety for punching shear failure
R = shear resistance of soil (lb/ft for continuous footings, lb for isolated spread footings)
P = applied foundation load (lb/ft for continuous footings, lb for isolated spread footings)
T = vertical distance from bottom of foundation to top of liquefiable layer (ft)
τ_f = shear resistance of non-liquefiable soil layer above liquefiable layer (psf)
B = base width of footing (ft)
L = base length of rectangular footing (ft)

(2) *Potentially liquefiable and non-liquefiable layers in soil stratum*. A total stress analysis is used to evaluate the bearing capacity of a cohesive soil underlain by a soil susceptible to liquefaction or cyclic SSL. The soil influencing bearing capacity is modeled as a layered profile and evaluated in a similar manner as described in paragraph 5–5a(2) (undrained loading of stiff clay over soft clay). The undrained shear strength of the upper, non-liquefiable cohesive soil is used to calculate the ultimate bearing capacity of the upper zone. The residual or reduced shear strength of the underlying soil that is susceptible to cyclic SSL is used to calculate the ultimate bearing capacity of the lower zone. If a cohesionless non-liquefiable soil is underlain by a soil susceptible to liquefaction or cyclic strength loss, drained strength parameters for the non-liquefiable soil should be used.

(3) *Sand-like soil with earthquake-induced excess pore water pressures*. Excess pore pressures generated in cohesionless soils that do not liquefy will effectively reduce the resistance provided in the soil effective weight term of the bearing capacity equation. Generally, this should be evaluated when a soil that is marginally liquefiable is in the shearing influence zone of a shallow foundation. A procedure presented by Day (2012) for determining ultimate bearing capacity that accounts for rise in excess pore pressures is shown in equations 9–3 and 9–4. This procedure utilizes the residual excess pore pressure ratio (r_u), reference paragraph 9–9c.

(a) Excess pore pressures influencing full shear zone. Equations 9–3 and 9–4 should be used when the water surface and the cohesionless soil layer with the excess pore pressures are within the bearing capacity shear zone. In this equation, the surcharge term has been removed as the shearing zone might not mobilize for a punching shear failure. Appropriate correction factors should be included as described in paragraph 5–5f of this manual.

$$q_u = \frac{1}{2}(1 - r_u)\gamma_b \cdot B' \cdot N_\gamma \qquad \text{(9–3)}$$
Strip Footings

$$q_u = 0.4(1 - r_u)\gamma_b \cdot B' \cdot N_\gamma \qquad \text{(9–4)}$$
Spread Footings

where:

q_u = ultimate bearing capacity (psf)
B' = effective base width of footing (feet)
r_u = residual excess pore pressure ratio (dimensionless), reference paragraph 9–9c
γ_b = buoyant or effective unit weight of soil within shear zone (pcf)
N_γ = bearing capacity factor for soil effective unit weight (dimensionless)

(b) Excess pore pressures partially influencing shear zone. For foundations partially influenced by excess pore pressures, the effective unit weight of the soil below the footing may be calculated using a weighted average of the soil susceptible to excess pore pressures and the soils not susceptible to generating excess pore pressures.

d. Sliding stability.

(1) Sliding stability should be evaluated using seismic loads applied in the direction that results in the most critical condition. The sliding shear zone is assumed to be similar to the static sliding shear zone described in paragraph 5–6a. Sliding should consider reduced resistance due to cyclic SSL.

(2) Technical Report ITL-92-11 provides procedures and examples for sliding stability with shallow founded retaining walls and for scenarios with and without excess pore pressure that can be applied to design of shallow foundations. Net lateral earth resistance (passive resistance minus active thrust) is typically calculated using the Mononobe-Okabe dynamic active and passive earth pressure coefficients as described in EM 1110-2-2502. Minimum factors of safety are provided in Table 2–1. Overstress factors can be applied if allowed by the building code. These values should generally not exceed one-third increase to the static allowable resistance. Overstress factors should not be applied to soils that are susceptible to SSL.

e. Overturning stability.

(1) For evaluation of overturning stability, seismic loads should be applied in the direction that results in the most critical condition. Similar to static conditions for overturning stability, eccentricity governs. The overturning stability requirements can vary from the static requirements that are presented in Table 2–1 for an extreme event. In the absence of specific criteria in building codes or design guidance, the following procedure should be used for seismic overturning stability of individual shallow footings during extreme events.

(2) Two criteria that must be adhered include: (1) the resultant of the footings should be located within the foundation footprint, and (2) the effective footing dimensions should be determined considering the eccentricity of the resultant in relation to the centroid of the foundation, and the effective foundation bearing pressure must not exceed the allowable bearing pressure.

f. Uplift. Uplift stability for a seismic load case is performed using the methodology in paragraph 5–7. For cases when a vertical seismic coefficient is used, the inertial loads on the structure should be adjusted based on the direction and magnitude of the inertial force. Uplift pressures acting on the base of foundations should be increased to include residual excess pore pressures (if present) that could be generated because of a seismic event (paragraph 9–9c). For load combinations that include seismic effects, the uplift factor of safety must be 1.5 or greater.

9–11. Seismically induced settlement

Settlement caused by ground shaking during earthquakes is often non-uniformly distributed and can cause differential deformations in structures, leading to major damage. Settlement can occur from compaction in moist or dry cohesionless soil and from dissipation of excess pore pressures induced in saturated soil by earthquake ground motions. For heavily loaded shallow footings that have stress influences in liquefiable layers, shear strain potential is increased. Ground deformations calculated for liquefiable and dry soils beneath foundations should be evaluated.

a. Applicability. The procedures presented in paragraphs 9–11d through 9–11f are specifically designed for free-field conditions and are most applicable for use with lightweight structures and mat foundations. The estimates do not capture shear-induced strains resulting from foundations bearing directly on liquefiable soils, shallow foundations with stress influences in a liquefiable zone, sites on sloping ground where flow or lateral spreading can occur, or where liquefaction-induced ground damage is predicted (paragraph 9–11f).

b. Differential settlement threshold values. Presumptive differential settlement thresholds for various structures referenced in ASCE 7 must be used when designing a shallow foundation system subject to SSL caused by liquefaction due to the maximum considered earthquake geometric mean peak ground acceleration (MCE_G) earthquake. These values assume the foundation and superstructure are designed to accommodate the differential settlements as detailed in ASCE 7. The threshold values provided for Risk Category II and III are intended to provide resistance to collapse. The limits for Risk Category IV are intended to result in less differential settlement than would cause doors to jam in the design earthquake. Adequate subsurface investigation and characterization data (such as at the foundation locations) should be available to properly analyze differential settlement potential.

c. Differential settlement due to liquefaction. Uniform liquefaction settlements could be estimated in rather uniform deposits. However, due to the dynamic loading conditions and natural soil variability, differential settlements should be expected. The preferred approach should be to estimate seismic settlement at the foundation locations based on site-specific data for differential settlement calculation. For uniform subsurface conditions between the foundations, the maximum differential settlement (due to liquefaction) between adjacent shallow foundations should be assumed to be on the order of 1/2 the uniform maximum liquefaction settlement assumed in the analysis. Where there is significant spatial variability and thickness of deposits, a minimum

differential settlement between adjacent supports of one-half to two-thirds of the maximum liquefaction settlement should be used unless more is determined by the analysis. If lateral spreading hazard is present and unmitigated, then maximum and differential settlements could be much greater than calculated (Martin and Lew 1999).

d. Settlement of unsaturated cohesionless soils (dry sand). Ground shaking due to earthquakes can induce settlement in dry or unsaturated cohesionless soil deposits. The procedure presented in Table 9–3 is recommended for estimating settlements of dry sand.

Table 9–3
Procedure for estimating settlement of dry sand (Tokimatsu and Seed 1984)

Step 1	Develop an idealized, representative soil profile for the analysis. This should consist of the corrected standard penetration resistance $(N_1)_{60}$ (blows per foot) and the unit weight of each soil layer. Use Youd et al. (2001) for $(N_1)_{60}$.
Step 2	Evaluate the total vertical stress (σ_v) and the mean normal total stress ($\sigma_m = \sigma'_m$) for unsaturated sand at several depths within the deposit at the time of exploration and for design. The design values should include stresses resulting from new structure or embankment loads. Outside of the structure footprint, the exploration and design values are generally the same unless grading is planned. For unsaturated, cohesionless soil $\sigma_m \approx 0.65\sigma_v$ assuming the at-rest earth pressure coefficient (K_0) is approximately 0.475.
Step 3	Evaluate the stress-reduction factor (r_d) using referenced methods in paragraph 9–7b for liquefaction resistance of cohesionless soils.
Step 4	Evaluate the normalized effective stress $\gamma_{eff}(G_{eff}/G_{max})$ using the methods developed by Tokimatsu and Seed (1984) in equation 9–5.
Step 5	Evaluate the effective shear strain (γ_{eff}) that occurs in response to the effective shear stress as a function of the mean principal effective stress (σ'_m) and $\gamma_{eff}(G_{eff}/G_{max})$ using Figure 9–3.
Step 6	Assuming $\gamma_{eff} \approx \gamma_c$, where γ_c = cyclic shear strain, evaluate the volumetric strain (ε_c) due to compaction as a function of γ_c and $(N_1)_{60}$ for an earthquake of magnitude 7.5 (15 cycles) using Figure 9–4.
Step 7	Correct the volumetric strain (ε_c) for earthquake magnitude other than M_w 7.5 using the correction factors reproduced in Table 9–4.
Step 8	Multiply the volumetric strain (ε_c) for each layer by 2.0 to correct for the multidirectional shaking effect, to get the representative volumetric strain for each layer.
Step 9	Calculate settlements of each layer by multiplying the layer thickness by representative volumetric strain evaluated in Step 8. Sum up the layer settlements to obtain the total settlement for the analyzed profile.

$$\gamma_{eff}\left(\frac{G_{eff}}{G_{max}}\right) = 0.65 r_d \left(\frac{\sigma_v}{G_{max}}\right)\left(\frac{a_{max}}{g}\right) \quad (9–5)$$

where:

γ_{eff} = effective vertical stress (psf)
G_{eff} = effective shear modulus at induced strain level (psf)
G_{max} = shear modulus at low strain level (psf)
r_D = stress-reduction factor (dimensionless)
σ_v = total vertical stress (psf)
a_{max} = peak ground acceleration (ft/s^2)
g = acceleration due to gravity (fps)

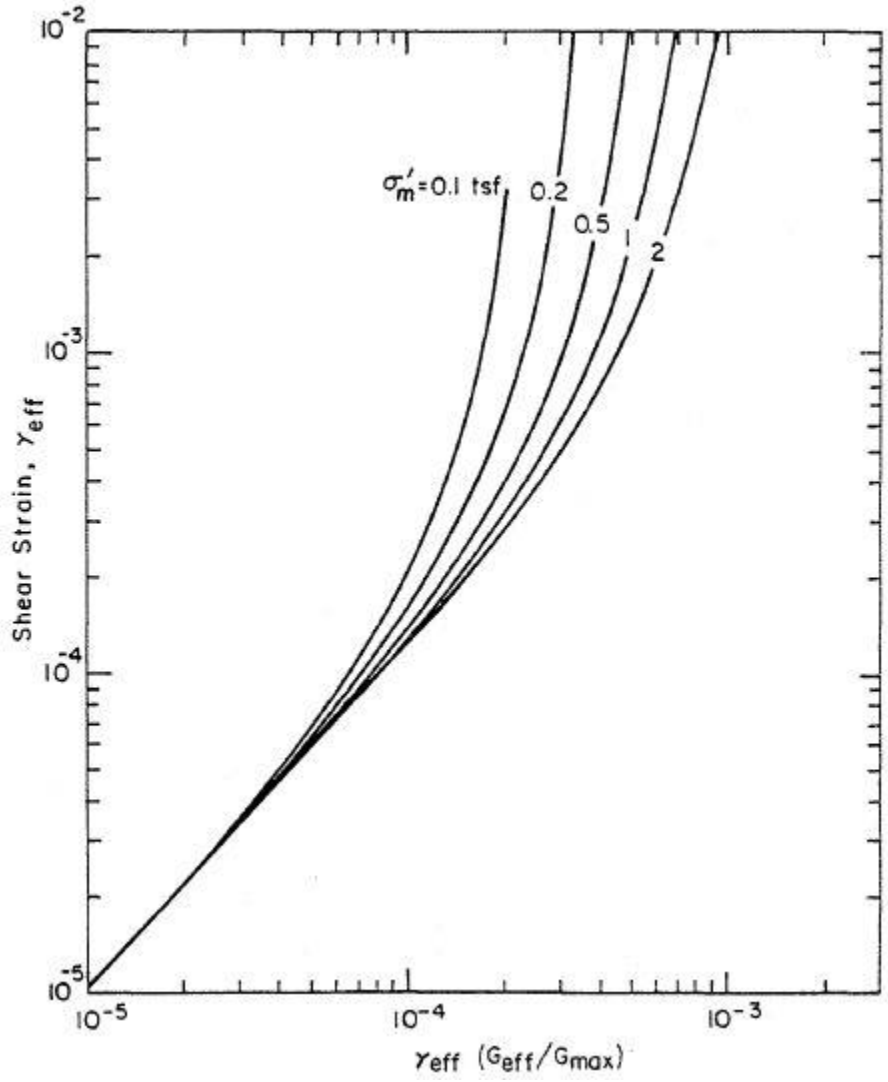


Figure 9–3. Plot for determination of earthquake-induced shear strain in sand deposits (Tokimatsu and Seed 1984)

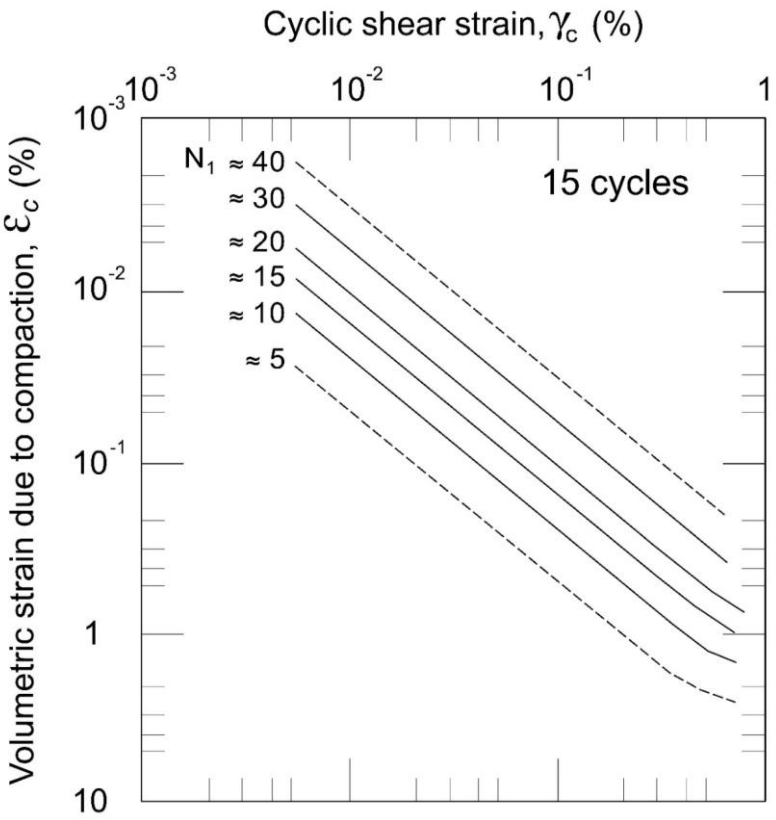


Figure 9–4. Relationship between volumetric strain, cyclic shear strain, and penetration resistance for unsaturated sands at N = 15 cycles (Tokimatsu and Seed 1984)

Table 9–4
Influence of earthquake magnitude on volumetric strain ratio for dry sands (after Tokimatsu and Seed 1984)

Earthquake Magnitude	Number of Representative Cycles (N) at $0.65\boldsymbol{\tau_{max}}$	Volumetric Strain Ratio $\boldsymbol{\varepsilon_{C,N}/\varepsilon_{C,N=15}}$
8.5	26	1.25
7.5	15	1.0
6.75	10	0.85
6.0	5	0.6
5.25	2 to 3	0.4

e. Settlement of liquefiable sands. Figure 9–5 provides the relationship between factor of safety against liquefaction and post liquefaction volumetric strain for different initial relative density values (D_R) from Ishihara and Yoshimine (1992). Normalized SPT blow counts, $(N_1)_{60}$ can be converted to D_R using relationships provided in paragraph 3–4. These volumetric settlement estimates or other relationships such as Cetin et al. (2009) are valid for saturated coarse-grained soils in relatively level-ground sites without potential for lateral spreading. The volumetric strain should be multiplied by the layer

thickness to estimate settlement. For lateral spreading sites, settlement estimates are likely to be larger than those for level-ground sites.

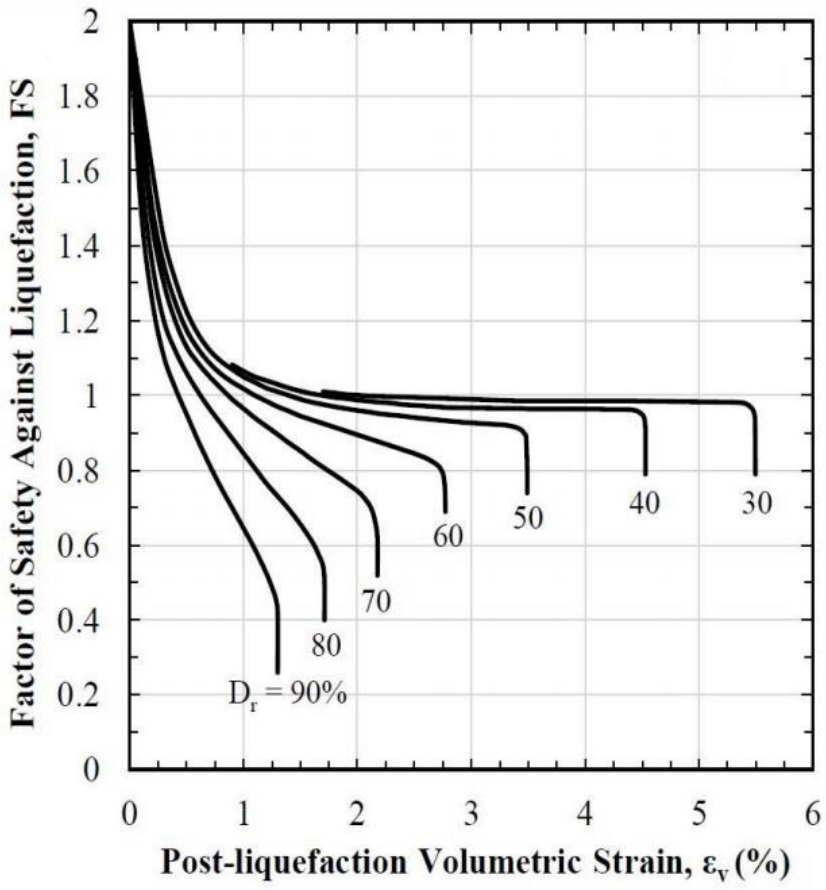


Figure 9–5. Post-liquefaction volumetric strains versus factor of safety against liquefaction (Ishihara and Yoshimine 1992)

f. Liquefaction-induced ground damage. Liquefaction can result in surface expressions in the form of sand boils and surface fissures, also referred to as ejecta. It is difficult to determine the amount of settlement due to the development of these features. Ishihara (1985) developed a series of charts presented in Figure 9–6 that can be used to determine the required thickness of a non-liquefiable soil surface layer (H_1) for a thickness of liquefiable soil layer (H_2) to mitigate such damages. The designer should verify that there is adequate non-liquefiable soil above any liquefiable zone. When designing shallow foundations, the required thickness of non-liquefiable soil is measured from the bottom of the foundation to the top of the non-liquefiable layer. If this cannot be provided, then additional fill can be placed, ground improvements performed, or deep foundation support provided. Seismic settlement in cases of shallow foundations over adequate non-liquefiable layer (thickness and stiffer or denser) overlying potentially liquefiable layers must be within allowable limits (per building code) to justify selection of a shallow foundation system.

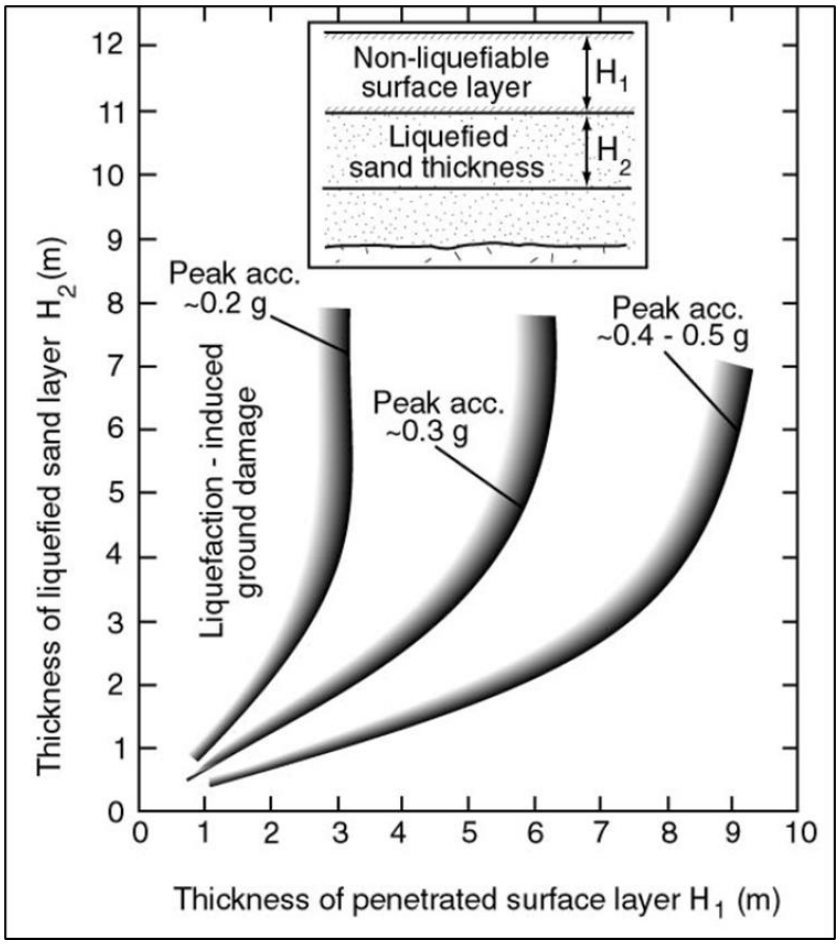


Figure 9–6. Relationship between H_1, H_2, and PGA and observed liquefaction-induced ground damage (from Obermeier et al. 2001; modified from Ishihara 1985)

g. Settlement due to rocking. Rocking can occur due to dynamic structural loads cyclically increasing the pressure to the foundation. The soil can deform in response to the additional foundation load. For cohesionless soils, rocking and volumetric compression can result in settlements. For cohesive soils, settlements due to rocking can occur when foundation soils are overloaded and strains due to mobilizing shear resistance occur. Settlement due to rocking can also occur when the existing vertical effective stress plus the dynamic loads subject the foundations soils to an effective stress beyond the preconsolidation pressures. The soils most susceptible to this are loose sands and gravels, sensitive cohesive soils, and soft, normally consolidated cohesive soils. To provide a sufficiently high factor of safety against bearing capacity failure, a factor of safety (for static loading conditions), $FS \geq 3.0$ is recommended.

h. Seismic settlement considering building loads. Structures on shallow foundations at liquefied sites typically settle more than the surrounding ground (such as free-field settlements). Bray and Macedo (2017) and Bullock et al. (2017) have developed procedures that may be used in a more detailed evaluation to estimate liquefaction-induced building settlement to account for three mechanisms for settlement.

These mechanisms are: (1) settlement resulting from loss of ground due to formation of sediment ejecta (D_e), (2) reconsolidation (volumetric) settlement (D_v), and (3) shear-induced settlement (D_s). For design, this process requires estimating the settlement due to ejecta from relevant case histories, evaluation of generated acceleration time histories, post-liquefaction bearing capacity analysis, and estimation of free-field volumetric settlement.

9–12. Faulting and lateral spreading

Seismic global stability is not covered in this manual; however, lateral spreading and faulting can occur on relatively flat to gently sloping sites. This section provides recommended procedures and serviceability criteria for these conditions.

a. Faulting. A fault is a fracture or zone of fractures between two blocks of rock or rock formations. Faults allow the blocks to move relative to each other. This movement may occur rapidly (in the form of an earthquake) or may occur slowly (in the form of creep). Faults are commonly considered to be active if they have moved one or more times in the last 11,000 to 35,000 years. The footprints for new structures should avoid active faults. ER 1110-2-1806 provides seismic loading criteria based on fault slip rates. It also has requirements for design in mapped fault areas and considerations for near-source effects on ground motions. Refer to EM 1110-1-1804 for fault identification in geotechnical exploration. California State guidelines based on Alquist-Priolo Earthquake Fault Zoning Act (1994) is an effective method to determine offset distances for design of structures in areas with known faults. It should be used in areas where active faults are mapped.

b. Lateral spreading. Lateral spreading occurs when the static plus seismically induced shear stresses exceed the resistance, often reduced by the earthquake, provided by the soil. In lateral spreading, the static factor of safety of the soil mass is greater than 1.0 but lateral movements accumulate during the earthquake. Displacement of a liquefied layer or displacement of an upper, non-liquefied layer above a liquefied layer can occur on gently sloping ground or on embankments. The potential for lateral movements due to lateral spreading is increased if there is a “free face,” such as a riverbank, in the laterally spreading mass. Usually, cracks are associated with movement of sliding masses in lateral spreading.

(1) Horizontal displacements from lateral spreading can be estimated using empirical and semiempirical methods. The multilinear regression (MLR) (Bartlett, Youd, and Hansen 2002) was developed for sites with sloping ground and for relatively level ground with a “free face” toward which lateral displacements could occur. The shear strain potential procedure by Zhang et al. (2004) developed methods to estimate lateral spreading that considers estimated shear strain from empirical methods, distance to free face, free-face height, and slope of the site. Bartlett, Youd, and Hansen (2002) suggested that a multiplier of 2 (applied to lateral spreading estimates) should be used for design. It is recommended to use appropriate relative density considering soil types when these simplified relationships are used. The resulting estimates of the simplified methods are applicable for up to 2 to 3 ft of lateral spreading.

(2) Horizontal displacements due to lateral spreading for buildings supported on conventional shallow-spread footings should be 3 inches or less. In some cases, the negative impacts from lateral spreading could reduce with increased distance from the free face. Building and foundations may be designed to accommodate greater lateral spreading displacements, up to the values shown in Table 9–5, using design techniques such as interconnecting individual foundations and integrating the foundations and slab. Specific design requirements are provided in ASCE 7.

Table 9–5
Upper limit on lateral spreading horizontal ground displacements with accommodating design features, from ASCE 7

General risk category description	Shallow-founded structures that represent low hazard to human life in the event of failure	Shallow-founded structures except for Risk Categories I, III, and IV	Shallow-founded structures that represent a substantial hazard to human life in the event of failure	Shallow-founded structures designated as essential structures
Risk Category	I	II	III	IV
Limit (inches) with accommodating design	18	18	12	4

Chapter 10
Problematic Ground Conditions

10–1. Introduction

a. Problematic ground conditions are those which do not fit conventional theories of soil and rock mechanics or that generally require avoidance or extensive mitigation for shallow foundation support. Such characteristics are often attributable to their composition or behavior, which can be inconsistent or unreliable. Common problematic ground conditions and their characteristics are discussed in this chapter.

b. Common problematic ground conditions are summarized in Table 10–1. Where encountered or suspected, geohazards (problematic ground) should be identified and documented during site exploration activities and the expected behavior addressed during foundation design. For problematic ground sites, soil behavior can be difficult to predict, and therefore, more qualitative assessments may be necessary. Often, the potential for unsatisfactory performance (and associated risk) cannot be eliminated.

Table 10–1
Some common problem ground conditions

Condition	Potential Engineering Challenges
Uncontrolled and undocumented fill	Could be inconsistent or unreliable. Could incorporate environmental contaminants. Extent can be difficult to evaluate.
Expansive soils	Often expand (heave) upon wetting and shrink upon drying. Can result in unacceptable foundation movements. Limiting extreme soil moisture changes could be difficult. Difficult to avoid in some geographic regions.
Collapsible soils	Could undergo large, sudden, irrecoverable collapse and significant loss of shear strength that is primarily wetting-induced. Difficult to avoid in some geographic regions.
Colluvium or colluvial soils	Can be highly variable with a loose soil matrix. Nearby global stability could be a major issue with proposed structure. Geotechnical investigations often hindered by variable-sized rock content.
Karst	Soil erosion into complex underground solution features is difficult to predict and nearly impractical to avoid in certain geographic regions. Site grading can aggravate natural solutioning processes. Could result in extreme soil movements and collapse of foundation-supporting soils.
Frozen ground	Complex engineering behavior aggravated by temperature fluctuations and heat transfer from structures which could result in extreme movements. Impractical to avoid in some geographic regions.
Scour and erosion	Hydraulic conditions can be complex and unreliable. Loss of bearing soils could result in excessive movement or structure collapse.

10–2. Uncontrolled and undocumented fill

Uncontrolled fills are materials placed by humans without engineering design and inspection. Undocumented fill could be either uncontrolled fill or otherwise acceptable fill; however, in some cases, a distinction cannot be easily made between the two because it was never documented, or documentation cannot be readily located.

a. Characteristics. Uncontrolled fills often contain soft or loose zones, trees, cobbles, boulders, cinders, or other construction or demolition debris. The consistency (or relative density) and moisture content of uncontrolled fill often varies significantly and within short distances. There is no practical amount of exploration or testing that can distinguish with certainty if an undocumented fill was placed in a controlled manner. In some cases, what may appear to be a controlled fill may include buried trees or other unsuitable material in unexplored areas. Often, uncontrolled fills are placed on unprepared and unsuitable subgrades containing large roots and organics, which aid in identification of uncontrolled fills.

b. Performance issues. Performance of uncontrolled fill is difficult to properly ascertain and quantify due to the possibility of large variations. Shallow foundations placed over uncontrolled fill could result in excessive settlement or collapse. According to Goodger and Leach (1991), the percentage of within-fill settlement (as a percentage of fill height alone) can exceed 3.5 percent for uncompacted sand, 12.0 percent for clay fill (pumped), and 30 percent for well-compacted mixed refuse. When placed over poor subgrades, additional long-term settlements could also occur in the underlying earth. If the uncontrolled fill incorporates hazardous waste or other environmental contaminants, disposal costs could be significant. Also, personnel exposure could result in sickness or death. Although it is possible to improve some uncontrolled fills, the possibility of unsatisfactory performance cannot be eliminated since certain unknown conditions will remain.

c. Mitigation techniques. Uncontrolled fills are typically mitigated by avoidance, extending footings to competent strata, or removal and replacement with suitable earth fill. In some instances, ground improvement techniques (such as dynamic compaction, aggregate piers) have been implemented in efforts to mitigate potential performance issues associated with uncontrolled fill; however, the degree of improvement can be difficult to quantify.

10–3. Expansive soils

Expansive soils are those which swell considerably on absorption of water and shrink upon drying. Expansive or swelling soils are found in many areas throughout the United States (see Figure 10–1) and the world and pose the greatest hazard in regions with pronounced wet and dry seasons.

a. Characteristics. Expansive soils often consist of plastic clays and clay shales that typically contain colloidal clay minerals (that expand upon wetting) such as the montmorillonites or smectite. They include marls, clayey siltstone, sandstone, and

saprolites. Some soils (such as dry, residual, clayey soil) may heave on wetting under low applied pressure but collapse at higher pressure. Other clayey soil may initially collapse on wetting, but heave over long periods of time as water slowly wets the less pervious clay particles. Desiccation can cause expansive soils to shrink. Additional engineering aspects such as site studies, identification, heave predictions, and foundation types are discussed in UFC 3-220-07.

b. Performance issues. Excessive movements (both heave and swell) can cause damages to structures that are not adequately designed and managed to accommodate soil volume changes. Of primary concern are vertical soil movements since they can cause the most damage to overlying structures. Damage can include impaired function, cracked walls, and jammed and misaligned doors and windows. Important factors that contribute to damages are: (1) the failure to recognize the presence of expansive soil, (2) failure to implement reasonable mitigation measures, and (3) failure to manage soil moisture in the vicinity of the structure in an appropriate manner.

c. Mitigation techniques. Foundations are commonly placed below zones of high volume change due to moisture fluctuations. Physicochemical stabilizers (such as lime, cement, fly ash), by-product stabilizers (such as ground blast furnace slag, kiln dust) can also be used to mitigate expansive soils. Other methods include moisture conditioning/compaction control, drainage/soil moisture control, pre-wetting, and removal and replacement with soil having low volume change characteristics. Refer to UFC 3-220-07 for mitigation techniques.

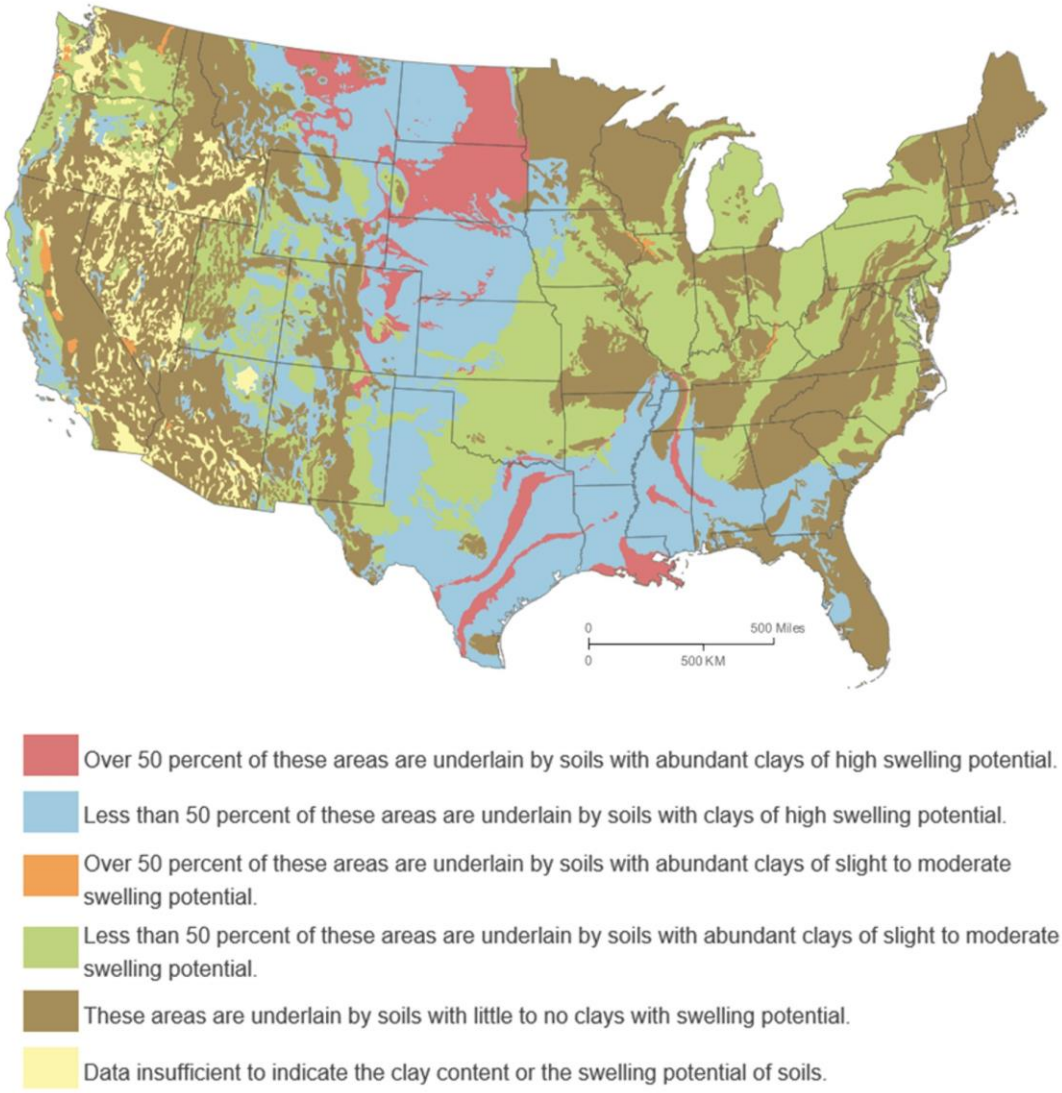


Figure 10–1. Swelling clays map of the conterminous United States (USGS, Map I-1940 1989)

10–4. Collapsible soils

Collapsible soils are unsaturated soils (typically silty sands with a loose structure) that undergo particle rearrangement upon wetting and additional loading (or both) and a significant decrease in volume. The primary triggering mechanism for volume reduction is an increase in moisture content, and collapse settlement cannot occur in soils below the water table. Numerous soil types can fall in the general category of collapsible soils as shown in Figure 10–2.

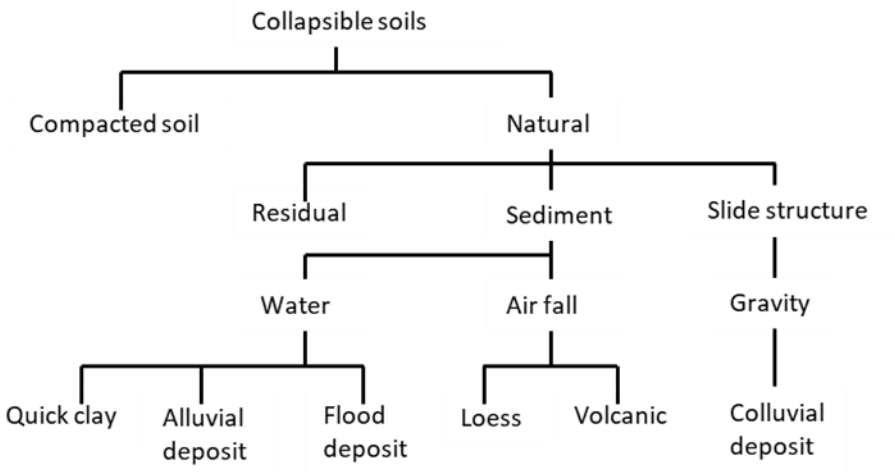


Figure 10–2. A family of collapsible soils (Rogers 1995)

a. Characteristics. Collapsible soils are generally characterized as having a relatively low dry-unit weight, high void ratio, and a low saturation percentage (usually between 13 and 39). However, the identification and prediction of soil collapse has proven difficult because no single criterion can be applied to all collapsible soils. Collapsible soils can be found in debris-flow deposits in young alluvial fans, aeolian deposits such as dunes and loess, lacustrine and colluvial deposits, residual soils and tuff, loose sand and gravels, dispersive clays, and fills compacted dry of optimum content (Owens and Rollins 1990; Mulvey 1992; and Santi 2005).

b. Performance issues. Collapsible soils are problematic for shallow foundation support due to the capacity to undergo large, sudden, and irrecoverable collapse and significant loss of shear strength that is primarily wetting induced. Collapsible soils can also be susceptible to liquefaction-induced settlements during earthquakes, and wetting under load has proven ineffective at reducing liquefaction potential. Reliance on laboratory classification and consolidation tests alone can fail to indicate soil that eventually could collapse in the field. In some geologic settings, avoidance of collapsible soils is not practical.

c. Mitigation techniques. Collapse potential can be assessed using ASTM D5333. If the collapsible soil layer is relatively thin and surficial, then removal and replacement with suitable soil or embedding foundations beneath is often preferred. Other alternatives include avoidance, pre-wetting (hydrocompaction) or controlled wetting to depths as great as 10 ft, avoidance of wetting, dynamic compaction, chemical stabilization or grouting, and differential-settlement-resistant foundations (such as continuous footing design with load-bearing beams).

10–5. Colluvium or colluvial materials

Colluvium is a general term that refers to loose, unconsolidated material that has been deposited by gravitational forces (especially those mobilized via mass wasting), usually collected at the base of gentle slopes or hillsides.

a. Characteristics. The characteristics of colluvial materials vary according to the characteristics of the parent material and the climate under which weathering and transport occur. Typically, colluvium is weakly stratified and consists of a heterogeneous mixture of soil and rock fragments ranging in size from clay particles to rocks that are several feet or more in diameter. In some cases, colluvium is not easily distinguished from alluvium (such as at the base of slopes and near alluvial deposits). Colluvium can accumulate slowly or result from large catastrophic earth movements (such as landslides) often attributed to intense rainfall, fire, or seismic activity. Subsurface drilling in colluvial soils is often difficult and test trenches can often provide better characterization than can be solely achieved with conventional drilling methods.

b. Performance issues. Placement of foundations on or near very loose and marginally stable colluvium could result in excessive movements. These movements could be due to settlement or instability of the colluvium itself, or due to adjacent slope movements (such as landslides). In some areas, landslides formed in colluvium are common and destructive. Commonly found along the lower portion of valley sides, they often need excavation for construction of structures, which could result in a more unstable configuration.

c. Mitigation techniques. Colluvial deposits and planned modifications (such as cut/fill) of these deposits should be carefully evaluated for slope stability, recognizing that the deposits often have highly variable characteristics. Mitigation techniques usually include avoidance or removal and replacement. Often, buttress slopes or earth retention near unstable slopes are also necessary to prevent further instability.

10–6. Organic soils and peat

Organic soils are formed by the accumulation of partially decomposed organic matter and are most common in water-saturated, humid environments such as low-lying areas or swamps. Organic materials contain large pore spaces in their cellular structures and can hold significant water (Terzaghi and Peck 1967). This higher water content typically results in lower shear strength and higher compressibility. For engineering classification, organic soils and peat are unique materials having more complex definition compared to other inorganic soils.

a. Characteristics. Organic soils and peat are generally characterized in Table 10–2a and Table 10–2b and may be qualitatively identified using the methods outlined in ASTM D2487/D2488. The amount of organic matter can be quantified by using a muffle furnace (ASTM D2974); however, some clay minerals contain bound water that could falsely indicate organic content (Arman and Munfakh 1970; Franklin et al. 1973). Engineering behavior is generally described in Table 10–3 (Edil and den Haan 1994).

Table 10–2a
Typical characteristics of organic soils and peat[1]

General Category	Organic Content (% by weight)	Distinguishing Characteristics for Visual Identification	Typical Range of Laboratory Test Values
Organic Matter			
Fibrous peat (such as woody, mats)	75 to 100% organics, either visible or inferred	Lightweight, spongy, and elastic at ω_n. Shrinks considerably on drying. Much water squeezes from sample. Threads cannot be rolled for plastic limit test.	ω_n = 500 to 1,200% γ_m = 60 to 70 pcf G_s = 1.2 to 1.8 $C_c/(1+e_0) > 0.4$
Fine-grained peat (amorphous)	75 to 100% organics, either visible or inferred	Lightweight, spongy but often not elastic at ω_n. Shrinks considerably on air drying. Much water squeezes from sample. Threads cannot be rolled for plastic limit test.	ω_n = 400 to 800% LL = 400 to 900% PI = 200 to 500 γ_m = 60 to 70 pcf G_s = 1.2 to 1.8 $C_c/(1+e_0) \geq 0.4$
Highly Organic Soils			
Silty peat	30 to 75% organics, either visible or inferred	Relatively lightweight, spongy. Thread usually weak and spongy near PL. Shrinks on air drying. Medium dry strength. Can usually squeeze water from sample readily. Slow dilatancy.	ω_n = 250 to 500% LL = 250 to 600% PI = 150 to 350 γ_m = 65 to 90 pcf G_s = 1.8 to 2.3 $C_c/(1+e_0)$ = 0.3 to 0.4
Sandy peat	30 to 75% organics, either visible or inferred	Sand fraction visible. Thread weak and friable near PL. Shrinks on air drying. Low dry strength. Can usually squeeze water from sample readily, high dilatancy, “gritty feel.”	ω_n = 100 to 400% LL = 150 to 300% PI = 50 to 150 Plots below A-Line γ_m = 70 to 100 pcf G_s = 1.8 to 2.4 $C_c/(1+e_0)$ = 0.2 to 0.3

Notes:

[1] Table is not intended for the purposes of soil classification.

[2] ASTM D2487 considers highly organic soils as peats. Soils are classified as highly organic based on the prevalence of organic matter, their dark color, and organic odor (not using a particular test or measurement).

Table 10–2b
Typical characteristics of organic soils and peat[1] (continued)

General Category	Organic Content (% by weight)	Distinguishing Characteristics for Visual Identification	Typical Range of Laboratory Test Values
Organic Soils			
Clayey organic Silt	5 to 30% organics, either visible or inferred (Note 2)	Often has strong hydrogen sulfide (H_2S) odor. Rolled thread may be tough at PL, depending on clay fraction. Medium dry strength. Slow dilatancy.	ω_n = 65 to 200% LL = 65 to 150% PI = 50 to 150 Usually plots near A-line γ_m = 70 to 100 pcf G_s = 2.3 to 2.6 $C_c/(1+e_0)$ = 0.20 to 0.35
Organic sand or silt	5 to 30% organics, either visible or inferred (Note 2)	Rolled threads weak and friable near PL, or in some cases threads cannot be rolled for PL test. Low dry strength. Medium to high dilatancy.	ω_n = 30 to 125% LL = 30 to 100% PI = Non-plastic to 40 Usually plots below A-Line γ_m = 70 to 110 pcf G_s = 2.3 to 2.6 $C_c/(1+e_0)$ = 0.2 to 0.35
Slightly Organic Soils			
Slightly organic sand, silt, or clay	Less than 5% organics, combined visible and inferred	Same as inorganic sand, silt, or clay.	Properties controlled by inorganic soil constituents.

Notes:
[1] Table is not intended for the purposes of soil classification.
[2] For the purposes of soil classification using ASTM D2487, a soil is termed organic if the liquid limit ratio (oven-dried/not dried) < 0.75. This results in a corresponding organic content that is approximately 16.5 percent (FHWA/IN/JTRP-2008/2).

Table 10–3
Engineering behavior of organic soils (Edil and den Haan 1994)

Organic Content (ASTM D2974)	Effects on Properties	Soil Category
< 5%	Little effect on behavior, considered inorganic soil.	Inorganic soil
≈ 6 to 20%	Effects properties but behavior is still like mineral soils; organic silts and clays.	Organic silts and clays
≈ 21 to 74%	Organic matter governs properties; traditional soil mechanics could be applicable; silty or clayey organic soils.	Silty or clayey organic soil
> 75%	Displays behavior distinct from traditional soil mechanics especially at low stresses.	Peat

b. Performance issues. Generally, soils containing large amounts of organic material (especially when greater than 5 percent) often exhibit poor engineering characteristics (such as high compressibility, low shear strength) and should not be used for shallow foundation support.

c. Mitigation techniques. If unsuitable organic soils are present at foundation bearing elevations, mitigation measures typically include removal and replacement with suitably compacted, engineered fill. In areas where removal and replacement are not a viable option (such as organic soils exist to great depths), preloading, surcharging, and staged construction could result in effective ground improvement. Due to the loss of surface hydration water in clayey soils at very high temperatures, ASTM D2974 can be used in conjunction with other qualitative identification methods (such as ASTM D2487/D2488) or laboratory testing to prevent mischaracterization and an over- or under-conservative design.

10–7. Karst terrain

Karst landforms constitute approximately 12 percent of the ice-free land surface and result from mildly acidic water acting on carbonate (such as limestone, dolostone) and sulfate (such as gypsum) rocks. The (carbonic) acid is formed as rain passes through the atmosphere and, over geologic time, dissolves the bedrock surface, enlarges fractures, and results in underground voids and drainage systems. Reference also EM 1110-1-2908.

a. Characteristics. Karst landscapes exhibit an irregular topography characterized by streamless valleys, sinkholes, and streams that disappear underground, all developed by the action of groundwater (Leet et al. 1978). Depressions, gullies, caves, underground drainage, and sinkholes are commonly found in karst terrain. Rock dissolution occurs primarily from the surface downward and along the surfaces of fissures (such as joints, bedding planes, and fractures) in the rock. The distribution of karst areas in the U.S. is shown in Figure 10–3.

b. Performance issues. Karst features (such as caves, sinkholes, and open fractures) cause severe damage for engineering structures worldwide. Major problems encountered by engineers in karst terrain include collapse of subsurface voids, subsidence of soil cover into rock voids or sinkholes, and excavation difficulties over pinnacled bedrock. In karst terrain, the potential for unsatisfactory performance cannot be eliminated. Karst can suddenly manifest at the surface when site development activities result in overburden removal, additional surface water infiltration, or more severe groundwater fluctuations.

c. Mitigation techniques. Several mitigation techniques have been used in karst terrain for shallow foundations. Subsurface grouting is a common technique; however, is difficult to verify and could disrupt groundwater flow. Other mitigation techniques have included structure relocation, using mat or grid foundations, grade raise by fill placement, inverted rock filters or plugs where shallow rock exists, and surface water management. In karst terrain, additional borings, rock coring, and geophysical surveys may be warranted to better characterize soil and bedrock conditions.

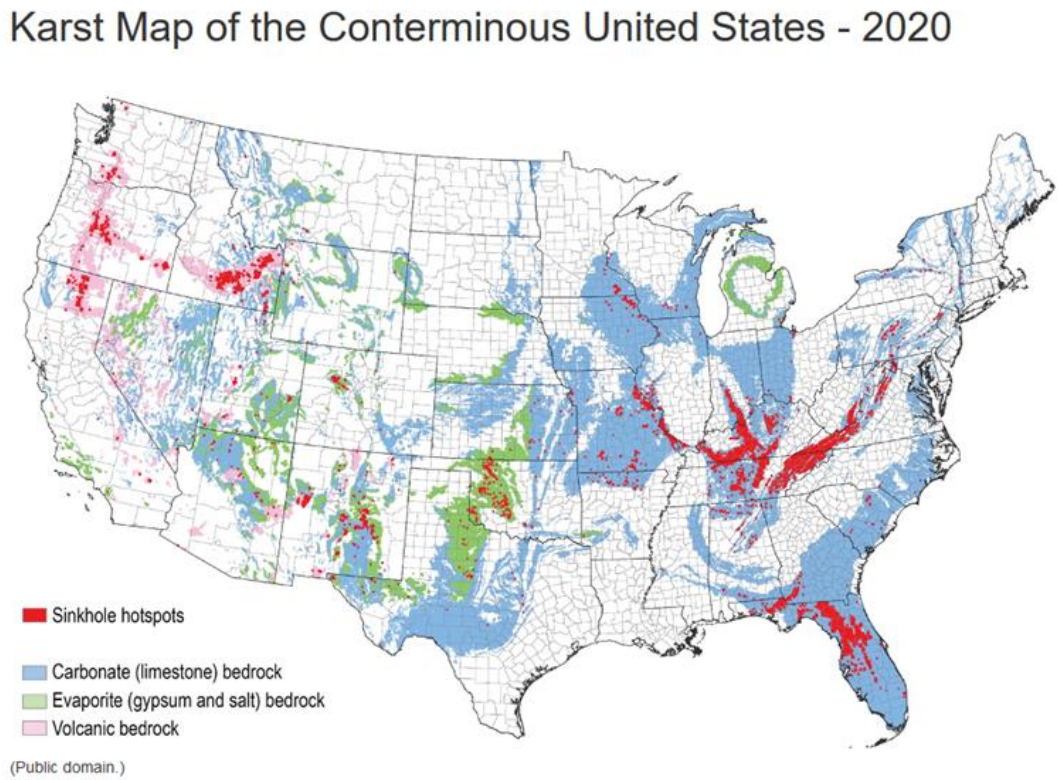


Figure 10–3. Distribution of karst areas in relation to carbonate and sulfate rocks in the U.S. (from USGS 2020)

10–8. Frozen ground

Frozen ground is defined as soil or rock having a temperature below 32 °F, regardless of water or ice content. How deeply and quickly the ground freezes depends on many other factors. In most cases, the ground temperature can be quite different from the air temperature. Permafrost, or perennially frozen ground, is defined as soil or rock having temperatures below 32 °F during at least two consecutive winters and the intervening summer (Brown and Kupsch 1974). Detailed requirements of engineering design in frozen ground areas are provided in UFC series, Arctic and Subarctic Construction (UFC 3-130-01 through UFC 3-130-07).

a. Characteristics. Frozen soil generally consists of four components: soil particles, ice, water, and voids, and can be described using the methods outlined in ASTM D4083 and ASTM D7099. The voids are filled with ice, unfrozen water, and air. Ice may be distributed evenly, or it may have accumulated in the form of irregular or stratified ice inclusions. The engineering properties of frozen soil is complex and can range from brittle to plastic, depending on the volume of ice, ground temperature, water content, and saline content. For the purposes of this manual, areas of significant frost penetration may be defined as those in which freezing temperatures occur in the ground to sufficient depth to be a significant factor in foundation design. Areas of significant frost penetration may be subdivided as follows:

(1) *Seasonal frost areas*. Significant ground freezing occurs in these areas during the winter season, but without development of permafrost. In northern Texas, significant seasonal frost typically occurs about once every 10 years. A little farther north, it occurs every year. Depth of seasonal freezing increases northward with decreasing mean annual and winter air temperatures until permafrost is encountered. With still further decrease of air temperatures, the depth of annual freezing and thawing becomes progressively thinner (especially when the ground is more permanently frozen).

(2) *Permafrost areas*. In these areas, perennially frozen ground is found below the annual frost zone. In North America, permafrost is found principally north of latitudes 55 to 65 degrees, although patches of permafrost are found much farther south on mountains where the temperature conditions are sufficiently low, including some mountains in the contiguous 48 States.

(3) *Permafrost distribution*. The geographic boundaries between zones of continuous permafrost, discontinuous permafrost, and seasonal frost without permafrost are poorly defined, but are represented approximately in Figure 10–4.

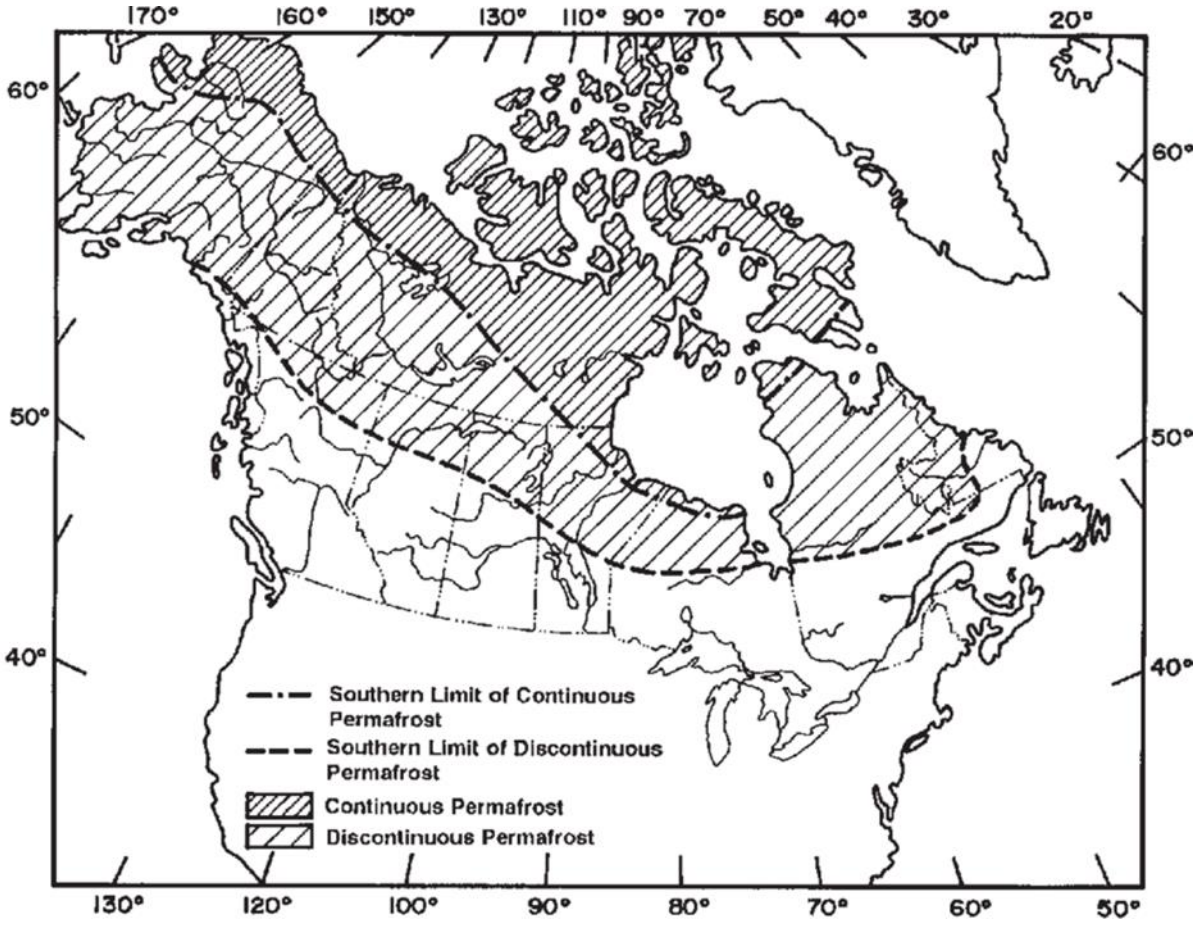


Figure 10–4. Permafrost distribution in North America

b. Performance issues.

(1) *Seasonal frost areas*. Design of foundations in areas of seasonal frost where the material below the frost line remains unfrozen presents certain problems. In the spring, thaw and settlement of frost-heaved material in the annual frost zone could occur differentially, and a very wet, condition with temporary but substantial loss of shear strength can occur.

(2) *Permafrost areas*. In permafrost areas, the same annual frost zone phenomena occur, but the presence of the underlying permafrost introduces additional problems, such as:

(a) Difficulty excavating and handling ground when it is frozen.

(b) Soft and wet ground conditions during thaw periods.

(c) Surface and subsurface drainage problems.

(d) Possible ice uplift and thrust action on foundations.

(e) Condensation on cold floors.

(f) In remote locations, limited availability of materials, support facilities, and labor (and reduced labor efficiency at low temperatures).

Note. Progressive freezing and frost heave of foundations can also develop under refrigerated warehouses and other facilities where sustained interior below-freezing temperatures are maintained.

(3) *Heat flow from buildings*.

(a) In permafrost areas, heat flow from buildings is a fundamental consideration. Change from natural conditions that results in warming of the ground beneath a structure can result in progressive thawing of the permafrost layer over time (typically years) that is known as degradation. If the permafrost contains ice in excess of the natural void or fissure space of the material when unfrozen, progressive downward thaw may result in extreme settlements of overlying soil and structures.

(b) This condition can be very serious because such subsidence is almost invariably differential, and hence, very damaging to a structure. Degradation can occur not only from building heat but also from solar heating, as under pavements, from surface water and groundwater flow, and from underground utility lines. Proper insulation can prevent degradation in some situations, but where a continuous source of heat is also available, thaw can eventually occur.

c. Mitigation techniques.

(1) *Seasonal frost areas*. For seasonal frost areas, foundation installation may be aided by ground thaw devices and designed using measures to protect against seasonal frost heave. Criteria for seasonal frost areas is provided in the UFC 3-130-04 and UFC 3-301-01. Design and construction standards can be found in ASCE 32.

(2) *Permafrost regions*. Detailed requirements of engineering design in frozen ground areas, to include mitigation techniques, is provided in UFC 3-130-04. When constructing shallow foundations in permafrost regions, passive or active techniques are used to protect structures from permafrost degradation. Passive methods maintain the frozen state of the soil and could include using insulation, ventilation, or refrigeration systems such as thermosyphons. Active methods include removing and pre-thawing frozen soils that may be suitable for foundation stabilization.

10–9. Scour and erosion

Scour and erosion are the localized loss of soil around a foundation due to flowing water. Shallow foundations most affected by scour and erosion are usually located in or near water bodies (such as rivers, beaches) or areas of flash flooding.

a. Characterization. Scour effects may be localized or result in the complete undermining of a building foundation. Scour effects typically result from flooding or storm surge type events.

b. Performance issues. Excessive removal of the material around a shallow foundation or undermining of a footing can cause excessive deformation or structure collapse.

c. Mitigation techniques. Scour and erosion countermeasures (such as structural features, river training, riprap, vegetation, geosynthetics) should be implemented for the design flood and the foundation constructed at sufficient depth beneath the maximum scour depth to provide sufficient bearing capacity and overall stability.

Chapter 11
Ground Improvement

11–1. Introduction

Problematic ground conditions often require modification to allow acceptable performance of shallow foundations in lieu of more expensive deep foundations; and in some instances, to facilitate excavation for foundation construction (such as ground thawing, dewatering). The application of an improvement technique depends on the ground conditions encountered and other project constraints. Some common ground improvement techniques are presented in this chapter. In some cases, the degree of improvement can be difficult or impossible to fully assess.

11–2. Design requirements

Ground improvement designs should be approved by the geotechnical engineer and include the parameters requiring improvement, methods considered for ground improvement, and the rationale for the method selected. Ground improvements can be designed to be either temporary or permanent. Design documents should indicate the proposed equipment, procedural details, quality control measures, method of evaluation, and quality acceptance criteria.

11–3. Improvement methods

a. General. Table 11–1 summarizes commonly used ground improvement methods, the type of treatment applied, soil suitability, engineering function, and some advantages and limitations of each method. Depending on the ground conditions encountered or other project constraints, ground improvement methods may need to be combined to achieve the project goals. In the following sections, more details are provided for methods applicable to shallow foundations, including applicability, implementation, design issues, and performance evaluation.

b. Improvement alternatives. Where problematic ground is encountered, design alternatives that should be considered include: (1) removal and replacement, (2) structure relocation, (3) foundation redesign to meet site constraints, and (4) ground modification (or ground improvement). In some situations, ground improvement methods should be excluded due to design constraints or combined to provide a more reasonable and cost-effective design. Where removal and replacement is selected, dewatering may be necessary. In addition, environmental testing and other controls may be required where uncontrolled fills or environmental contaminants are identified.

11–4. Evaluation and acceptance

a. Quality assurance. Field monitoring and quality assurance/quality control procedures should be implemented for successful installation of ground improvement projects.

b. Acceptance criteria. Ground improvement techniques have evolved mainly from contractor-led experience and have generally preceded development of engineering guidelines or theoretical treatment. For certain techniques, design methodologies are entirely proprietary and could require performance-based requirements for acceptance. Acceptance criteria should be clearly identified in the contract documents. For certain ground conditions, the degree of improvement can be difficult or impractical to fully quantify (such as karst, uncontrolled fill). Performance evaluation of ground improvement methods is not covered in this manual.

Table 11–1
Ground improvement categories and methods

Category	Function	Method
Densification	Increase soil density, bearing capacity, and reduce settlement. Increases resistance to liquefaction.	1) Vibro-compaction 2) Stone columns 3) Dynamic compaction 4) Dynamic replacement 5) Compaction grouting 6) Hydro-compaction
Consolidation	Surcharge or preloads applied to decrease settlement. Wick drains can accelerate drainage (settlement).	1) Preloading/surcharging 2) Drainage
Reinforcement	Increase in shear strength for foundation support, decreases soil compressibility.	1) Reinforced soil foundation 2) Cohesionless trenches
Chemical stabilization	Increases shear strength and overall bearing capacity. Reduction of shrink/swell potential in certain applications.	1) Cement/lime mixing 2) Cement grouting 3) Chemical grouting 4) Jet grouting 5) Deep mixing methods
Thermal stabilization	Strength increase.	1) Lime columns 2) Heating 3) Freezing 4) Active and passive measures
Load reduction	Reduction of load on foundation and corresponding settlement.	1) Lightweight fill
Expansive soil mitigation	Reduce water infiltration and swell potential of expansive soils.	1) Moisture barriers 2) Pre-wetting

11–5. Densification

a. Vibro-compaction. Vibro-compaction (also known as vibro-floatation) uses vibratory techniques to densify the soil, mainly by the inserting vibrating probes into the ground (Figure 11–1). The initial soil structure is broken down and the soil particles are moved to new packing arrangements using impact and vibration. In saturated cohesionless materials, micro-liquefaction is induced in the process. Densification of the in situ soil occurs rapidly. Ground improvement depths of 60 ft are routinely achieved by vibro-compaction. Depths of 100 ft or more have been attained in some cases.

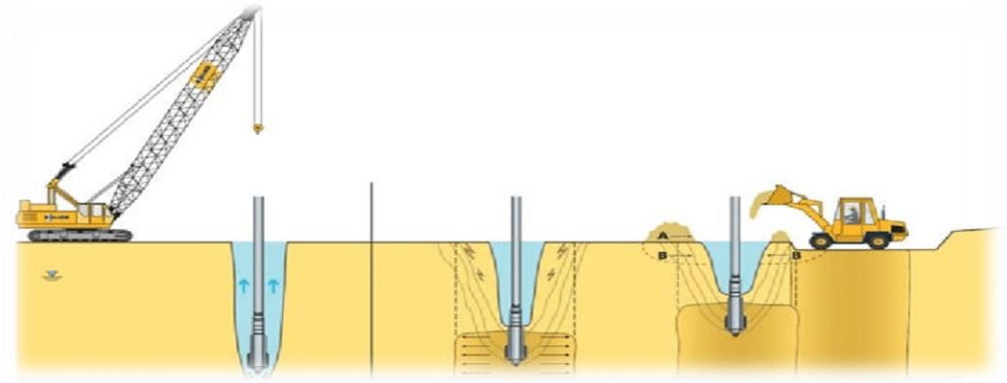


Figure 11–1. Vibro-compaction, penetration, compaction, backfilling (theengineeringcommunity.org)

(1) *Applicability*. Vibro-compaction is best suited for densification of cohesionless soils with non-plastic fines (passing No. 200 sieve) generally not exceeding 10 percent (Figure 11–2).

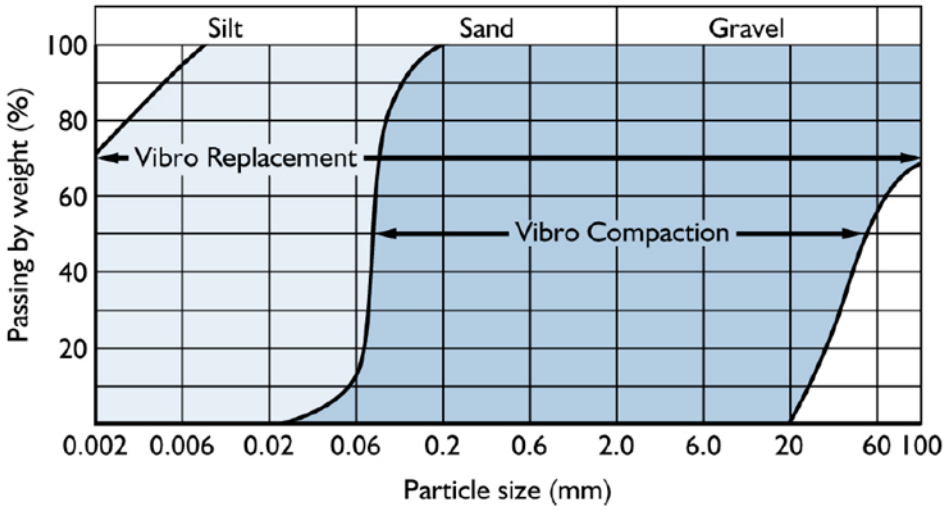


Figure 11–2. Suitable soils for vibro-compaction and vibro-replacement methods (McCabe et al. 2007)

(2) *Implementation*. Vibro-flot™, Terra probe™, vibro-rod, vibro-wing™, and vibrating I-beam are among the common equipment used in vibro compaction. Depending on the type of equipment employed, vibrating frequencies of greater than 30 Hz can be used; the vibration can be applied vertically, horizontally or in both directions. Compaction can be accomplished with or without the addition of cohesionless material (vibro-replacement), and with or without a water jet or compressed air which are applied at the tip of the probe to facilitate its penetration to greater depths and through hard zones. Upward-directed water jets along the sides of the probe have also been helpful.

(3) *Design issues*. Selection of probe pattern, spacing, and depth are required. These parameters depend on the type of soil, the foundation geometry, the required densification, and the equipment used. Triangular, rectangular, and square patterns can be used, and typical spacings are on the order of 3 to 7 ft. For preliminary design, Mitchell (1981) can be used.

b. Stone and concrete columns. Stone and concrete columns are constructed using the same equipment and procedure as vibro-compaction, and therefore, are discussed in sequence with vibro-compaction. There are similar products to stone columns, such as compacted aggregate piers, that can be considered for performance and cost-comparison purposes.

(1) *Applicability*. Both stone and concrete columns are applied to soft cohesive soils and loose sands for the purposes of (1) increasing bearing capacity, (2) reducing settlement and accelerating consolidation, (3) improving slope stability, and (4) mitigating liquefaction. About 10 to 35 percent of the soil mass can be replaced with stone (or sand) in the form of columns, essentially stiffening the soil. Typical column diameters are between 2.5 to 3.5 ft, and the column spacings are typically between 7 and 10 ft. The columns are typically placed in a triangular, square, or hexagonal grid pattern.

(2) *Implementation*. Stone columns are usually installed either using the dry bottom-feed or the wet, top-feed method. Figure 11–3 illustrates the more common wet, top-feed method of installation. Concreted stone columns are a bottom-feed procedure where a vibratory probe is lowered to the bottom of the hole, then concrete is installed under pressure to form a bulb at the bottom of the hole. The hole is then filled with concrete to the surface. The top-feed procedures involve pushing rock in from the surface. The bottom-feed procedure involves injecting material from the bottom by tremie-like methods.

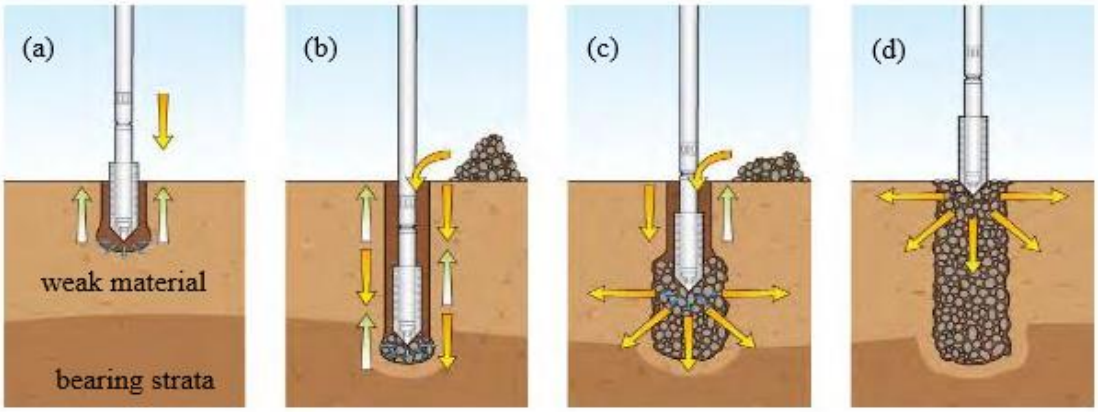


Figure 11–3. Wet, top-feed installation of stone columns, (a) equipment in position, feed pipe filled, valve open; (b) sinking the vibrator with compressed air (valve closed), and "pull down" assistance; (c) forming the stone column by oscillation; and (d) finishing the stone column

(3) *Design issues*. The composite mass of stone columns within the native soil has a lower compressibility and a higher shear strength than those of the native soil alone. The stone columns are stiffer than the soil they replace, however, their stiffness depends on the lateral support by the surrounding soil. If that support is inadequate, the column fails by bulging. The column could also fail as a pile because of insufficient skin friction and end bearing. A larger portion of the applied load can be transferred to the stone columns due to their higher stiffness. This improves the load-carrying capacity of the treated ground and reduces settlement.

c. Dynamic compaction. Dynamic compaction is used to increase the allowable bearing capacity and reduce liquefaction potential of the soil. It involves the repeated dropping of large weights by means of crawler cranes on a predetermined grid pattern. The repeated application of the high-energy impacts causes densification of the soil mass to depths ranging from 10 to 30 ft, while some contractors claim densification of 60 ft or more. Figure 11–4 illustrates the dynamic compaction process.

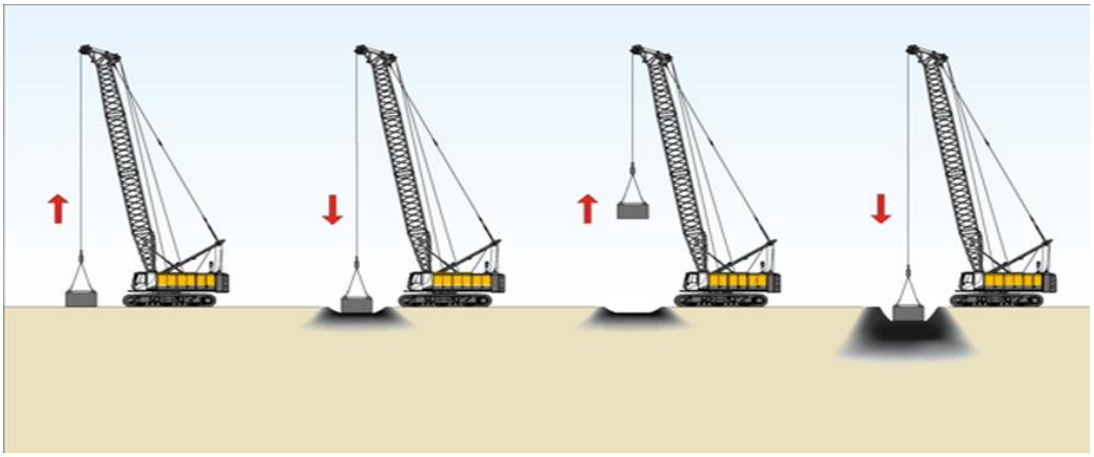


Figure 11–4. Dynamic compaction (Bauer.com)

(1) *Applicability*. Dynamic compaction can be used for densification of liquefiable, loose, saturated, cohesionless soils. The use of this method in fine cohesive soils can be uncertain, and therefore, the generation and dissipation of excess pore water pressures should be evaluated. Dynamic compaction should not be used for saturated, fine-grained soils with no additional methods for pore water pressure dissipation. Pore water dissipation in fine-grained soils can be facilitated by the forming drainage channels, by fissuring, or by vertical drains. Another common application can be stabilization of collapsible soils (Rollins and Kim 1994).

(2) *Implementation*. The weights used in dynamic compaction are usually square or circular in plan but can also be irregularly shaped and quite large in dimension. Drop weights typically consist of concrete blocks, steel plates, or thick steel shells filled with concrete or sand. They normally range from 6 to 35 tons, but heavier weights up to 200 tons have been used. The drop heights usually range from 40 to 80 ft, but heights up to 130 ft have been used. A typical treatment consists of an average of 2 to 3 blows per 10 ft^2 (Mitchell 1981). The drops are made at points spaced many feet apart in a grid pattern. For uniformity and high density in the near-surface zone, surface “ironing” is done by small weight drops over the treated area surface.

(3) *Design issues*.

(a) Tamper weight, grid pattern, drop heights, and depths of influence are calculated during design. A method for determining the depth of influence is provided by FHWA (see FHWA-SA-95-037) and depends on the soil stratification, degree of saturation, and the method of load application. For instance, a free drop is often more efficient than a crane drop. Soft cohesive layers or peat generally dampen the dynamic forces penetrating the soil, and thus the depth of influence is typically reduced.

(b) FHWA-SA-95-037 (Table 8) presents typical impact energy values per unit volume of treated soils. These values can be multiplied by the thickness of the treated soils to estimate the required applied energy at the surface. The estimated applied energy value can be used to determine the number of drops for a specified spacing or the minimum spacing for a particular number of drops. In addition to determining the energy required to achieve the required densification, the impact of the ground vibrations on adjacent structures should also be evaluated.

d. Dynamic replacement. Cohesionless inclusions using this improvement method are a form of soil reinforcement. However, because dynamic replacement can be an extension of dynamic compaction, it is discussed in the same sequence. In this method, tamping energy drives cohesionless material down into the craters developed in the compressible soils to form large-diameter soil reinforcement columns (with diameter around 6 to 12 ft). This method combines the advantages from both dynamic compaction and stone columns by creating large-sized inclusions with high internal shear resistance.

(1) *Applicability*. Dynamic replacement is appliable to cohesionless soils and fills. Dynamic replacement can be formed using a large variety of material. Material

recommended is a cohesionless fill material (generally up to 16-inch size), with low fines content (usually less than 15 percent), and free from vegetation or organic matter. The backfill material should be selected for each project depending on local availability. However, successful jobs have been completed with a large variety of material, including demolition debris (such as without wood, metal, and plaster) or crushed, cohesionless backfill material.

(2) *Implementation*. The dynamic replacement process typically usually uses weights with smaller sections as compared to dynamic compaction to improve the penetration and punching of the weight through the soft or loose layer. Through repeated tamping and backfilling phases, dynamic replacement columns are formed by dropping a 10- to 35-ton pounder from heights ranging from 33 to 100 ft. With this technique, replacement ratios of up to 20 to 25 percent can be achieved. Each dynamic replacement column can support loads of up to about 150 tons (Figure 11–5).

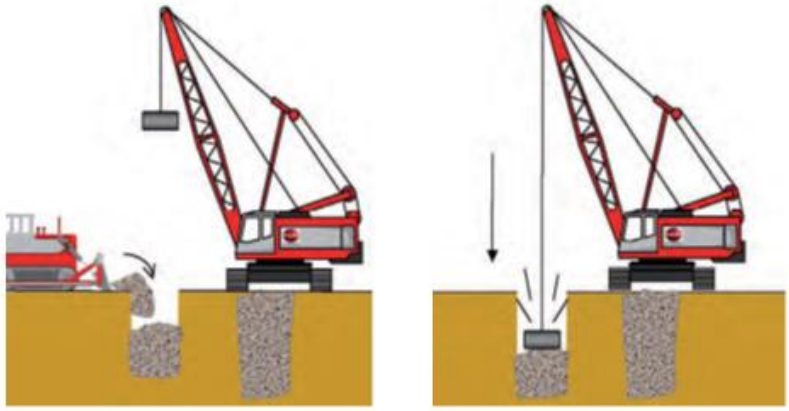


Figure 11–5. Dynamic replacement methodology (Ménard Group USA)

(3) *Design issues*. Dynamic replacement can be adapted to substantial loading conditions (up to 150 tons per column). It is possible to achieve replacement ratios of up to 25 percent. While vibro-stone columns have a limited range of application in organic soft soils (peat and organic clays), dynamic replacement columns can be used in peat or in soils with high organic content without the potential of bulging due to their relatively low slenderness (ratio of height over diameter). Dynamic replacement columns can increase the time rate of consolidation.

e. Compaction grouting. In this displacement grouting method, a stiff grout mixture is injected in the ground to densify loose soil formations. The grout generally does not enter the soil pores. Rather, it remains in a homogeneous mass that displaces and compresses the surrounding soil. Compaction grouting can be used to reduce differential settlements or to provide underpinning and ground strengthening. It can be considered more of a densification method than a chemical stabilization. The system can be used to fill voids.

(1) *Applicability*. Rubble fills, poorly placed fills, loosened or collapsible soils, soluble rocks, and liquefiable soils.

(2) *Implementation*. Compaction grouting equipment includes a mixing plant, force-feed piston pumps, hoses, and grout pipes. Compaction-grouted columns (piers) or bulbs are created under foundations to densify the ground and arrest potential ground and structural movements. Single or staggered bulbs are created through vertical or inclined grout pipes (see Figure 11–6). The grout pipes typically have a minimum inside diameter of 50 mm (4 inches) and are flush-jointed to avoid leakage. Soils which are dry of suitable moisture content can be pre-wetted to assist grout take. A recommended ground grout composition is silty sand (10 to 30 percent fines), cement (sometimes with fly ash), additives (such as fluidifiers, accelerators) and water. Coarse aggregates can be used, but they require larger pumps and grout lines.

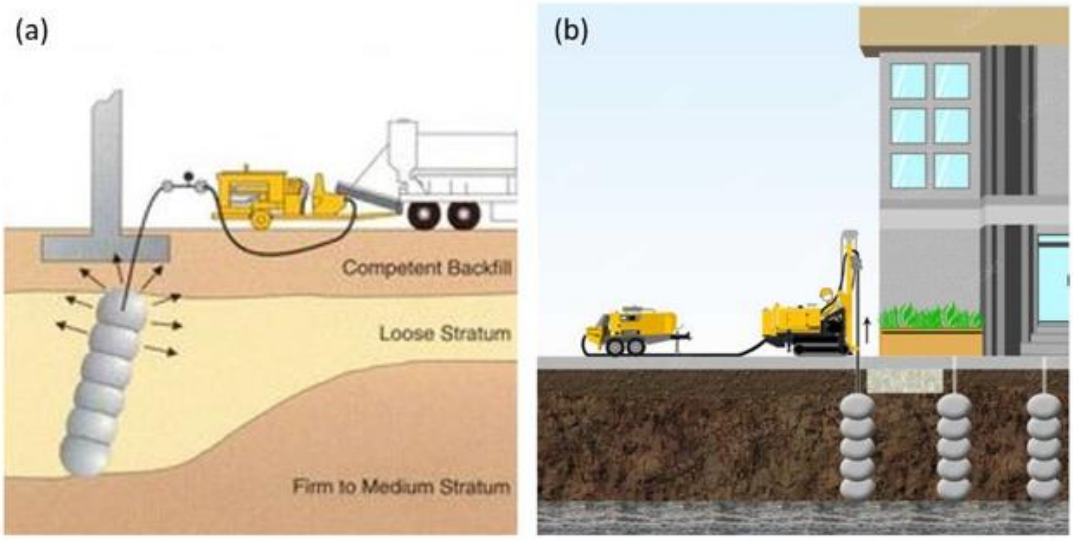


Figure 11–6. Compaction grouting (a) inclined installation; (b) vertical installation (Geo Grout, gggms.com)

(3) *Design issues*. The design of a compaction grouting involves locating the areas requiring compaction, defining the degree of densification needed, estimating the required grout volume, and selecting grouting layout. The compaction grouting design depends on experience, and a field test could be necessary.

f. Hydrocompaction. Collapsible soils settle when wetted or vibrated, therefore, the usual approach toward optimizing performance of structures on collapsible soil can be prewetting the construction site.

(1) *Applicability*. Hydrocompaction is applicable to collapsible soils.

(2) *Implementation*. Ponding or sprinkling water at the surface. Infiltration wells may be used to reach deeper soils.

(3) *Design issues*. When using hydrocompaction, infiltration wells should be used to wet soils to the required depth (Knodel 1992).

11–6. Consolidation

The engineering properties like shear strength and compressibility of a soft cohesive soil improve as the soil consolidates under a sustained static load. This improvement is accompanied by expulsion of water and an associated decrease in volume.

a. Preloading/surcharging and drainage. Preloading, or "surcharging," is used to protect against unacceptable consolidation settlement. Preloads should exceed the design loads so that the required settlement occurs prior to the construction of the structure or the more permanent fill. Preloading may also be used to shorten the duration needed to achieve primary as well as secondary consolidation. Drainage can be achieved by electro-osmosis. Electro-osmosis is similar to preloading in that effective stresses in the soil are increased and water is removed. The difference is that via electro-osmosis, the water is removed by application of electric potentials over the soil.

b. Prefabricated vertical drains. When the anticipated time for compression of the soil under the applied preload is significantly long, prefabricated vertical drains (PVDs) may be installed to accelerate drainage and consolidation settlement as well as increase shear strength. The initial application of vertical drains in the 1920s was to install sand drains. However, the sand drains can collapse during installation and can be costly to install.

(1) *Applicability*. Soils best suited for consolidation by preloading and electro-osmosis are saturated soft clays, compressible silts, organic silty clays, and peats. The vertical drains are of greatest effectiveness in inorganic clays and silts that exhibit little secondary consolidation.

(2) *Implementation*. Preloading may be accomplished by placing surcharge fills to elevations higher than design. Where necessary, the surcharge fill should be placed in stages to avoid stability failure, with each lift placed only after the soil has gained sufficient strength under the previous lift. For preloading or electro-osmosis, a drainage layer should be constructed to permit water drainage. Electro-osmosis does not involve large volumes of surcharge or stability issues; however, it is more complex in execution than preloading.

(3) *Design issues*. The design of a preloading program involves establishing (1) time-settlement curves under the design load, (2) time-settlement curves under the preload, and (3) evaluation of shear-strength increases under design load at each stage of surcharge. A critical issue with implementation of vertical drains is smear during installation, which can significantly increase the time to reach the required performance objective.

11–7. Reinforcement

a. Reinforced soil foundation. Geosynthetic-reinforced soil foundations (RSF) act in combination to increase the shear strength of the soil mass, to reduce its settlement under design loads, and to improve its resistance to liquefaction.

(1) *Applicability*. Reinforcement is applicable to clays, sands, silts, and gravels. Though fine-grained soils are more susceptible to long-term creep deformations, if the intent is to stabilize soils by increasing the confining stress, the reinforcement serves to stabilize the soils during placement and pore water pressure dissipation.

(2) *Implementation*. RSF systems consists of one or more layers of a geosynthetic reinforcement, typically rigid, biaxial geogrids and controlled fill placed below a shallow spread footing to create a composite material with improved performance characteristics (Chen et al. 2007). The presence of reinforcement geosynthetics under the foundation widens the stress bulb, reduces applied stresses to the foundation soils, and works as a stiffened platform or raft.

(3) *Design issues*. Research has shown that reinforcement produces the most benefit, or the highest bearing capacity ratio (BCR), when the reinforcement layers are within a depth, z equal to one-half of the footing width, B, $z = 0.5B$ (Adams and Collin 1997) (see Figure 11–7). BCR is the bearing capacity of a footing placed on an RSF divided by the bearing capacity of the same footing without reinforcement and can be between 1.5 and 2.0. For a three-layer system, z/B of 0.15, 0.375 and 0.5 found are recommended. The first layer at z/B of 0.15 can be optimal for settlement, s at a level of s/B of 0.5 percent. The RSF should extend 0.5 B past the edge of the footing (each side) with a minimum of 3 ft.

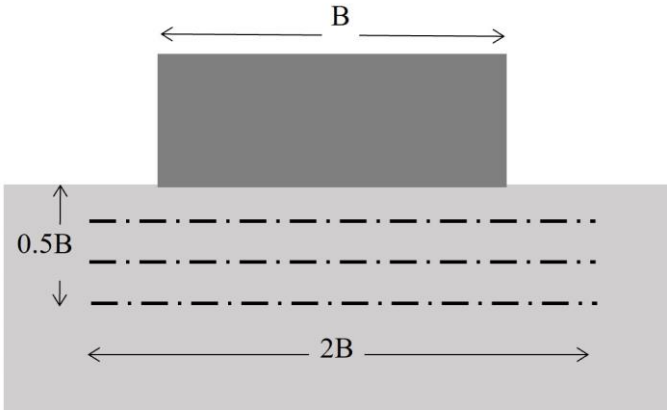


Figure 11–7. Reinforced soil foundation (after Adams and Collin 1997)

b. Granular trenches. To increase the allowable load-bearing capacity and reduce settlement, strip foundations can be constructed over granular trenches neatly excavated in soft clays, as shown in Figure 11–8.

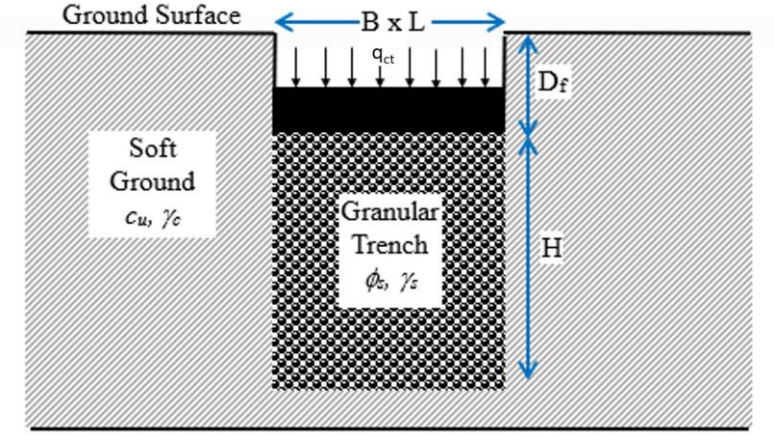


Figure 11–8. Strip footing in soft ground, stabilized with cohesionless trench (after Das 2017)

(1) *Applicability*. Granular trenches can be used to improve the load-bearing capacity of strip footings over soft soils.

(2) *Implementation*. Granular trenches in soft soil can be excavated with conventional equipment and do not require extensive excavation outside of the foundation footprint. Encapsulating the cohesionless trench with a geosynthetic layer can improve the performance of supported footings (Mathai and Unnikrishnan 2015).

(3) *Design issues*. For a cohesionless trench, the ultimate bearing capacity of a foundation can be computed using methods proposed by Das and Hanna (1988) and Das (1989), or commercially available software that models stress and deformation. Bearing capacity generally increases with depth of granular trench.

11–8. Chemical stabilization

a. Cement/lime mixing/cement grouting/chemical grouting. Cement, lime, fly ash, asphalt, silicate, and other materials are used to stabilize weak soils. They generally bind the soil particles together resulting in higher strength and lower compressibility.

(1) *Applicability*. Applicable soils vary by method. Cement can be used for sand and collapsing soils. Lime can be used for swelling clays. Cement grouting can be used for coarse soils (less than 15 percent fines), and chemical grouting can be used for fine sand and medium silt.

(2) *Implementation*. Chemical stabilization media (cement, lime, fly ash, etc.) is added to the soil. For clays, the soil dries, making the soil less susceptible to changes in water content (Afrin 2017). When at depth, the chemicals can be applied by injection or by deep mixing methods (DMM).

(3) *Design issues*. Organic matter, sulphates, sulfides, and carbon dioxide may adversely affect chemical stabilization. Sufficient moisture content must be present. For example, fully hydrated cement requires 20 percent of its own weight of water from the

surroundings, and Quicklime can take up to 32 percent of its own weight of water from the surroundings. The pozzolanic reaction is slower with lower temperatures. In cold regions, the soil should be stabilized during the warm season (Afrin 2017).

b. Jet grouting. This technique uses high-pressure fluids applied through a nozzle at the base of a drill pipe to erode the soil particles and mix them with cement grout as the drill bit is rotated and withdrawn, forming hard, impervious columns. Excess soil cuttings are carried to the surface in the form of a waste slurry. The grouted columns can be formed vertically, horizontally, or at an angle. A row of overlapping columns forms a wall (see Figure 11–9).

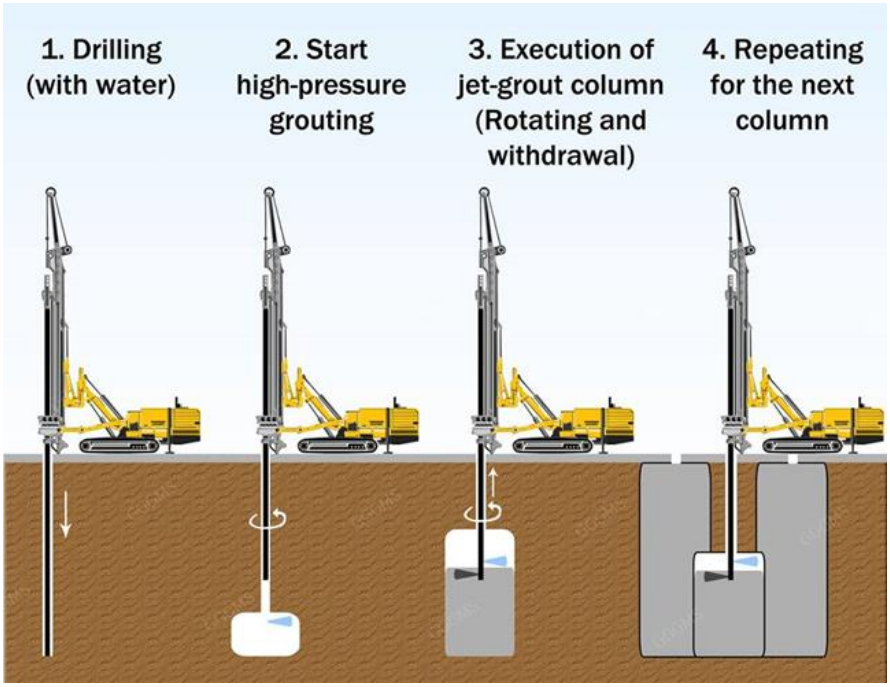


Figure 11–9. Jet grouting (Geo Grout, gggms.com)

(1) *Applicability*. Jet grouting works in most soils since the initial soil structure is broken down and replaced with a composite mass of soil and grout, sometimes called “soilcrete.”

(2) *Implementation*. Single-, double-, or triple-fluid jet grouting systems are generally used. In the single-fluid system, the fluid can be the grout. The jet simultaneously erodes and binds the soil particles. In the double-fluid system, a second fluid can be used in addition to the grout to enhance the cutting ability of the system. The second fluid can be either compressed air or water. In the triple- fluid system, both air and water are used. The column diameter increases in size from the single- to the triple-fluid system. Horizontal jet grouting uses only the first system. The jet-grouting plant includes a pump, a mixer, a compressed air source, and a water source. A minimum overlapping of 1 ft can be used for rows.

(3) *Design issues*. The design of a jet-grouted column involves the determination of the column diameter, strength, and permeability that would satisfy the project's requirements. Both the strength and the permeability of the grouted mass are influenced by the cement content in the grout mix. Plots relating strength to cement content are presented by Gallavresi (1992). These plots can be used in the design of the grout mix. The diameter of the constructed column is a function of operating parameters such as injection pressure, grout flow, and rod withdrawal and rotation rates. The slower the withdrawal and rotation rates, the larger the column diameter.

c. Deep mixing method. The DMM technique consists of mixing in-place soils with cement grout or other reagent slurries. Arranging the columns in various configurations permits strengthening of soft soils or for groundwater or liquefaction control.

(1) *Applicability*. DMM is applicable in soft to stiff silts and clays, sands, and gravels. In extremely dense soils, holes are predrilled using a powerful single-shaft auger to loosen up the ground along the DMM alignment. The obstructions in the soil are removed, if possible, and replaced with good quality fill prior to soil mixing. Although DMM and jet grouting have the same applications, there are two basic differences. Unlike jet grouting, which uses high pressure, high pressure is not used in DMM. In addition, the DMM columns are built to defined dimensions, while jet grouting dimensions are not as clearly defined.

(2) *Implementation*. Hollow-stem, multiple augers with mixing paddles penetrate the ground to the required depth. During penetration and withdrawal, cement grout can be pumped through the auger stem. The auger flights and mixing paddles mix the soil and the grout in place to form continuous, overlapping, soil-cement columns. If needed, steel reinforcement can be inserted in the column to provide bending resistance. A separate pump supplies the grout from the mixing plant to each of the injecting augers. A computer can be used to control grout mixing and flow.

(3) *Design issues*. The DMM design is influenced by the planned function of the system. When DMM is used for liquefaction control, the DMM can be performed in a block or lattice pattern to resist the stress from foundation or surcharge loadings when loose cohesionless soils liquefy during seismic ground shaking (Figure 11–10). For cut-off-wall construction, DMM can be designed to provide a required permeability. For excavation support, DMM usually has reinforcing members designed to resist bending moments and shear stress along the wall.

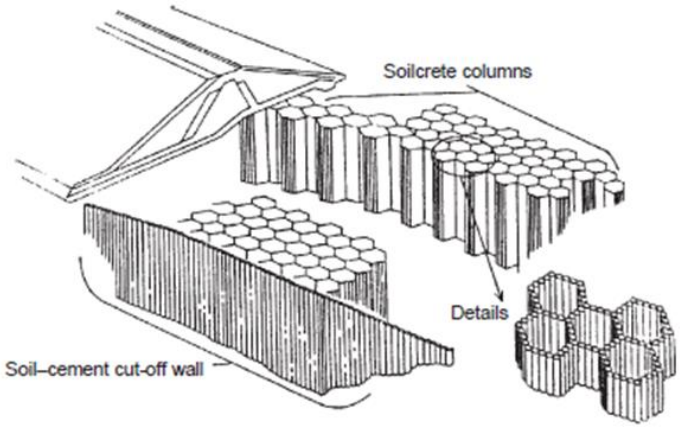


Figure 11–10. Deep mixing method for liquefaction control (Taki and Yang 1991)

11–9. Thermal stabilization

Heating of soils can reduce the water content and reduce the shrink-swell potential for expansive clays. Freezing of soils can also stabilize soils. In areas of permafrost, settlement can be significantly reduced by freezing existing soils that are subject to thawing. To protect soils from temperature variation in cold areas, the soil can be graded to reduce exposure of the soil to temperature variations, air convection can be used, or a heat drain can be installed (Doré et al. 2012).

a. Applicability. Thermal stabilization is applicable to expansive clays. Freezing is applicable to soils that are frozen for much of the year.

b. Implementation. There has been some advancement in ground freezing (Bayasan et al. 2008). Insulation is another option. For some applications, the soil can be frozen with pipes, refrigerants, and chillers.

c. Design issues. Heating can result in permanent changes to the soil properties, which turns expansive clays into non-plastic materials. Similarly, freezing can reduce settlement of existing foundations built on permafrost. In other applications, freezing will turn very soft materials into hard materials. Heating and freezing methods can be costly and energy intensive. For freezing methods, property changes remain only when the soil is frozen. Like seepage or consolidation analyses, a thermal analysis can be performed. Thermal conductivity, latent heat of fusion, heat of vaporization of soil water, and heat capacity of soil are needed for analysis.

11–10. Load reduction

Certain soil, rock, concrete, and man-made materials can be used to reduce settlement and can increase stability of foundation soils. The amount of settlement reduction is directly proportional to the reduction in load. Shear strength properties vary depending on the material selected.

a. Applicability. Lightweight fill is generally used above or as a replacement to soft and compressible natural soil.

b. Implementation. Lightweight fills are usually obtained from a controlled source (such as manufactured, processed) and can require proprietary placement methods.

c. Design issues. Granular lightweight fills typically have a shear strength like natural soils, while cemented fills and man-made products are generally characterized by compressive strength. Lightweight fill density can vary from about 1 to 65 pcf depending on the type selected.

11–11. Expansive soil mitigation

Removal of surface swell potential soils and replacement with non-expansive or low-expansive, low-permeable backfill can reduce heave at the ground surface and should be used in conjunction with surface drainage controls (such as directing drainage away from foundations). Lime modification can also be used to reduce volume changes and to increase the strength of foundation soils.

a. Applicability. Controlled backfill, lime modification, and drainage controls are used to reduce heave potential of expansive soils.

b. Implementation. Typically, 4 to 8 ft of surface expansive soils are removed and replaced with compacted soils that have a low degree of heave potential. Lime modification requires mixing lime with the natural soils and can typically be achieved over a depth of several feet.

c. Design issues. Controlled backfills (such as low-permeability soils with low heave potential) could be difficult to source. Where lime modification is used, excavation and replacement below foundations may be required where low-volume change material fill heights are insufficient to reduce heave potential to within tolerable levels.

Chapter 12
Earthwork Observation, Testing, and Documentation

12–1. Introduction

Accepting materials or procedures different from those specified or that deviate from the design intent can compromise foundation support and potentially lead to failure or other performance issues. Due to the possibility of unforeseen conditions or non-conformance with established requirements, comprehensive guidance for earthwork observation and testing is difficult to establish. Therefore, the designer should frequently reassess design concepts and assumptions in light of actual conditions observed in the field. This can be accomplished through earthwork observation and testing during construction.

12–2. Quality management

a. Quality assurance. Obtaining quality construction of shallow foundations and associated earthwork is a combined responsibility of the construction contractor and the government. The extent of quality assurance should be commensurate with the value and complexity of the project and the requirements of ER 1180-1-6.

b. Acceptance criteria. During design, the district geotechnical design element should confirm that the contract specifications include the appropriate earthwork and foundation observation, testing requirements, and associated acceptance criteria. Acceptance criteria may be prescriptive (exact details and instructions are provided) or performance based (desired results are stipulated).

12–3. Technical responsibilities

Technical responsibilities are provided in ER 1110-2-1150 and ER 1180-1-6 for USACE Project Delivery Team members (engineering and construction personnel). Management and administration of the construction contract is the responsibility of the construction team, to include the contracting officer or designated representative.

a. Geotechnical engineer responsibilities. The district geotechnical engineering element should review the project plans and specifications to ensure necessary earthwork and foundation testing requirements are stipulated. These basic services include:

(1) *Earthwork observation and testing*. During foundation and associated earthwork activities, the geotechnical engineer should remain engaged so that unforeseen conditions or unsatisfactory performance can be effectively managed (via the contracting officer) by the design team according to the design intent. The geotechnical engineer must approve the earthwork and foundation subgrades. For military and civil works projects, the designer of record (DOR) should prepare a short report titled “Engineering Considerations and Information for Field Personnel” or (ECIFP) document as outlined in ER 1110-3-12 and ER 1110-2-1150. This document covers the design intent and critical items to observe during foundation and associated earthwork activities.

(2) *Submittal review*. Designated submittals (such as specialty contractor designs, testing documents) received from the field should be reviewed and approved in a timely manner so that construction progress does not hinder removal, replacement, or modification of unacceptable or unforeseen conditions.

(3) *Construction site visits*. Construction site visits should be made during critical foundation and associated earthwork activities to observe and compare the field conditions with the design assumptions. For civil works projects, this should be performed according to ER 1110-2-112 and ER 1110-2-1150. Site visits may also be necessary to brief construction division personnel on issues affecting foundation construction that are not suitably conveyed via contract documents or the ECIFP.

b. Resident and project engineer. The designated Resident Engineer (RE) or Project Engineer (PE) are responsible for overall field supervision and quality assurance reporting. The RE/PE or their representative should perform follow-up reviews as needed to confirm/maintain continued acceptability. Field inspections and testing should follow the drawings and specifications and quality management procedures. The RE and PE are assisted by construction inspectors. The Area/Resident Engineer or other responsible individuals in the Area/Resident Office may also be delegated as the Contracting Officer's Representative (COR) for specific contracts (paragraph 12–4d).

c. Construction inspectors. Construction inspectors are responsible for implementing the field quality assurance procedures. Construction inspectors should understand the critical earthwork and foundation aspects of the project and be familiar with the requirements in the plans and specifications related to foundations and earthwork. Field inspectors should routinely coordinate with the DOR, RE, and PE. Prompt notification should be made of unfavorable or changed conditions that could negatively impact the design.

12–4. Earthwork observation and testing

Once construction begins and progresses, with the support of the construction inspector, the field conditions being encountered should be reviewed for compliance with those assumed for design. This can be done through field and laboratory verification testing and observations.

a. Foundation subgrade and bearing conditions. Foundation subgrades should be observed, and testing performed to confirm design conditions. Testing should be performed just before foundation placement to avoid degradation of otherwise acceptable soils by construction equipment or environmental factors (such as drainage). Suitable foundation tests may include probing the ground, using a dynamic cone penetrometer (DCP), performing density tests, or other similar methods. Common issues pertaining to shallow foundation support and engineering guidance include:

(1) *Groundwater*. Groundwater conditions can vary from those indicated in the original field exploration due to natural fluctuations. Saturated or wet conditions exposed at bearing elevations due to groundwater do not necessarily indicate unsatisfactory soils

are present. For constructability purposes, groundwater levels are typically lowered at least several feet below bearing elevations. This can typically be accomplished using ditches, sumps, or other dewatering methods. If groundwater seepage begins to exceed the capacity of the dewatering system, conditions should not be expected to improve unless the seepage is caused by recent hydrological events. Where seepage into foundation excavations becomes excessive, excavation operations should be halted until corrective measures are identified and approved by the geotechnical engineer. In some cases, footings can be placed in the wet if the subsurface conditions are acceptable and if the forms permit water to exit during placement (ACI 332).

(2) *Surface water*. Drainage of surface water into foundation excavations can result in ponded water and consequent delays in construction needed to dewater and de-muck excavation bottoms. Ditches or berms can be used in some cases to reduce water infiltration in foundation excavations due to rain, snow, and other surface water sources. Saturated or high moisture content soils could result in unsatisfactory performance, therefore, wet soils should be removed or modified in a manner to re-establish assumed design conditions.

(3) *Non-uniform bearing surfaces*. Non-uniform bearing surfaces and isolated areas of harder or firm material (such as cemented deposits, indurated clays, shallow rock) can be as detrimental as soft or loose soil conditions. Zones of harder materials could require removal and replacement with engineered fill to reduce potential differential settlements caused by non-uniform bearing conditions. The extent of removal and replacement, or other ground improvements, must be approved by the geotechnical engineer.

(4) *Non-level bearing surfaces*. Bearing surfaces should be relatively flat. Where foundation bases are inclined, the vertical resistance will decrease. Some exceptions include structures where lateral resistance governs (such as T-walls) and the increased lateral resistance is desired and considered during design. In general, bearing surfaces should have a slope no greater than 10H:1V (Horizontal to Vertical). Where necessary, footings can be stepped to change the elevation of the top surface of the footing or where the surface of the ground slopes more than 10H:1V. Stepped and sloping foundations to accommodate unanticipated ground conditions must be approved by the geotechnical and structural engineer.

(5) *Bearing elevation*. In some cases, foundation bearing elevations can be adjusted in the field to accommodate unsuitable bearing conditions (for example, it may be necessary to lower a foundation to a more competent bearing stratum). The normal method of extending foundations to greater depth is to use unreinforced concrete footing pedestals. Where used, the ratio of the unsupported height to the average least lateral dimension of plain concrete pedestals should not exceed 3. Use of unreinforced concrete pedestals to lower the bearing elevation must be approved by the geotechnical engineer.

b. Engineered fill supporting foundations. The specified fill and placement requirements should be achieved to perform as designed. Deviations from the earthwork specifications could result in unacceptable performance.

(1) *Fill suitability*. Prior to placement, proposed fill samples should be tested so that the desired engineering characteristics are established. Typically, soil gradation, plasticity, modified Proctor density and optimum moisture, and soil moisture content should be determined. These parameters aid in soil identification and provide an indication of anticipated moisture conditioning requirements needed during placement and compaction. In some cases, additional testing may be required (such as consolidation, organic content, unit weight) depending on the material source or the foundation soil strength and compressibility characteristics desired. Testing should be performed at suitable intervals to achieve the intended foundation support. Where on-site borrow is used, the testing frequency may need to be increased to account for its natural variability as compared to pit-mined or processed materials that typically exhibit more uniform characteristics.

(2) *Moisture content*. Proposed fill should be within the specified moisture range prior to compaction activities. Often it is impractical to adjust moisture content in the field by more than about 2 to 3 percent, and particularly where soils contain considerable silt or clay content. Where moisture conditioning (such as wetting, drying) of fill soils is necessary, discing, turning, and sprinkling can be used to achieve more uniform moisture. Indirect methods (such as nuclear gauge) for measuring soil moisture should be verified or adjusted based on direct heat or oven-dry measurements.

(3) *Fill placement*. Fill supporting foundations should be placed in relatively horizontal lifts that can support compaction equipment. In some instances, the location of fill placement (such as close to adjacent structures) can require reduction of lift thickness or use of alternative compaction equipment to achieve project requirements. After placement and compaction, fill material should be tested for density/relative compaction, and moisture content.

(a) Proofrolling. Where site space permits, proofrolling may be performed prior to and during fill placement as a visual aid in spotting pumping, soft, loose, or otherwise unsatisfactory conditions. Pumping soils normally contain excess moisture and require scarification, drying, and re-compacting until satisfactory performance is achieved, or the unsuitable soil replaced with suitable compacted material. Proofrolling is particularly useful in structure areas where highly variable fill material is used below foundations.

(b) Soil compaction. Compaction characteristics of the proposed fill should be established prior to fill placement. Generally, engineered fills are compacted to minimum 95 percent modified Proctor density for foundation support. The modified Proctor energy is comparable to that obtained with the heaviest rollers under favorable conditions (Sowers and Sowers 1979) and has a resulting energy that is about 4.5 times greater than can be achieved using the standard Proctor. The geotechnical engineer should approve modifications to compaction methodologies where the construction contracts use prescriptive specifications.

(c) Compaction of Free-Draining Soils. In some areas, compaction of relatively clean (less than 15 percent passing the No. 200 sieve) and free-draining sands may be necessary for foundation support. The U-shape of the compaction curves for relatively clean sands suggests that the maximum density could occur at either zero percent or 100 percent saturation. Because of the uncertainties associated with laboratory maximum and minimum tests required for determination of relative density, it is often considered desirable to develop compaction control methods based on Proctor values determined from oven-dry samples. For field compaction, clean sands are typically "flooded" (excess water is applied), and only moderate effort using conventional compaction equipment is required to achieve satisfactory compaction.

(d) Compaction of Variable fill. In some cases, it can be difficult to adequately identify fill by visual methods alone. In these cases, a "family," or series of compaction curves can be used for field identification in conjunction with the one-point compaction method. The one-point compaction test can identify the location of the appropriate compaction curve where moisture content of the sample is within a few (1 or 2) percentage points below optimum moisture content. If the point does not fall on a previously identified curve, additional points should be established until the maximum dry density and optimum moisture can be established.

c. Load and proof testing. Large-scale load testing is uncommon for verification of shallow foundation performance. Although small-scale load testing (plate load testing) can be performed somewhat efficiently, it is not recommended due to scale effects (limited depth of influence as compared to designed footings) and the potential for long-term consolidation that is not typically evaluated. Where ground improvement methods are used, the installation methods themselves can serve as proof tests. Verification of ground improvement methods is not covered in this manual.

12–5. Instrumentation and monitoring

Shallow foundations and associated earthwork instrumentation may include a combination of devices and methods such as pressure cells, telltales, survey monuments, and settlement plates. Instrumentation monitoring is unique to the needs of a particular project and must be approved by the geotechnical engineer. Application and design of geotechnical instrumentation is not covered in this manual. For additional instrumentation and monitoring guidance, reference EM 1110-2-1908.

12–6. Documentation and communication guidelines

Foundation and associated earthwork observation and testing documents should be maintained to provide a record describing field conditions as outlined in ER 1180-1-6. Test locations and elevations should be well documented. These documents are often required to evaluate claims, to implement design modifications (such as future expansion), and to assess foundation performance. Field reports should indicate if the work inspected was or was not completed in conformance to approved construction documents. Project geotechnical and materials completion report requirements are included in ER 1110-1-1901.

Appendix A
References

Section I

Required Publications

Unless otherwise indicated, USACE publications are available at https://armypubs.army.mil/ and https://www.publications.usace.army.mil. Unified Facilities Criteria (UFC) are available at https://www.wbdg.org/dod/ufc. American Society for Testing and Materials (ASTM) standards are available at https://www.astm.org.

Aalami and Bommer 1999
Aalami, B., and Bommer, A. 1999. *Design Fundamentals of Post-Tensioned Concrete Floors*. 1st ed. Phoenix, AZ: Post-Tensioning Institute. (Available at https://www.post-tensioning.org/publications)

Aas et. al 1988
Aas, G., Lacasse, S.M., Lunne, T.A., and Høeg, K. 1988. "Use of In situ Tests for Foundation Design on Clay." *International Journal of Rock Mechanics and Mining Sciences and Geomechanics Abstracts*. 2(1): A8 (Available at https://www.issmge.org)

AASHTO 2020
AASHTO. 2020. AASHTO LRFD Bridge Design Specifications, 9th ed. (Available at https://store.transportation.org/item/collectiondetail/202)

ACI 318
Building Code Requirements for Structural Concrete (Available at https://www.concrete.org/store/)

ACI 336.2R
Suggested Analysis and Design Procedures for Combined Footings and Mats, Reported by ACI Committee 336. (ACI 336.2R-88, Reapproved 2002) (Available at https://www.concrete.org/store/productdetail.aspx?ItemID=336288&Format=DOWNLOAD&Language=English&Units=US_AND_METRIC)

Adams and Collin 1997
Adams, M.T., and Collin, J.G. 1997. "Large Model Spread Footing Load Tests on Geosynthetic Reinforced Soil Foundations." *Journal of Geotechnical and Geoenvironmental Engineering*. 123(1): 66–72. (Available at https://doi.org/10.1061/(ASCE)1090-0241(1997)123:1(66))

Afrin 2017
Afrin, H. 2017. "A Review on Different Types Soil Stabilization Techniques." *International Journal of Transportation Engineering and Technology*. 3(2): 19–24. (Available at http://dx.doi.org/10.11648/j.ijtet.20170302.12)

Andrus and Stokoe 2000
Andrus, R.D., and Stokoe, K.H. 2000. “Liquefaction Resistance of Soils from Shear-Wave Velocity.” *Journal of Geotechnical and Geoenvironmental Engineering*. 126(11): 1015–1025. (Available at https://doi.org/10.1061/(ASCE)1090-0241(2000)126:11(1015))

ASCE 7-16
Minimum Design Loads for Buildings and Other Structures (Available at https://sp360.asce.org/PersonifyEbusiness/Merchandise/Product-Details/productId/233133882)

ASCE 32
Design and Construction of Frost-Protected Shallow Foundations (Available at https://ascelibrary.org/doi/book/10.1061/9780784405642)

ASTM D1556
Density and Unit Weight of Sol in Place by Sand-Cone Method

ASTM D1586
Standard Penetration Test (SPT) and Split-Barrel Sampling of Soils

ASTM D1587
Thin-Walled Tube Sampling of Soils for Geotechnical Purposes

ASTM D2435
One-Dimensional Consolidation Properties of Soils Using Incremental Loading

ASTM D2487
Classification of Soils for Engineering Purposes (Unified Soil Classification System)

ASTM D2488
Description and Identification of Soils (Visual-Manual Procedures)

ASTM D2573
Field Vane Shear Test in Saturated Fine-Grained Soils

ASTM D2974
Moisture, Ash, and Organic Matter of Peat and Other Organic Soils

ASTM D3441
Mechanical Cone Penetration Tests of Soil

ASTM D4083
Description of Frozen Soils (Visual-Manual Procedure)

ASTM D4186
One-Dimensional Consolidation Properties of Saturated Cohesive Soils, Controlled Strain Loading

ASTM D4220
Preserving and Transporting Soil Samples

ASTM D4253
Maximum Index Density and Unit Weight of Soils Using a Vibratory Table

ASTM D4254
Minimum Index Density and Unit Weight of Soils and Calculation of Relative Density

ASTM D4546
One-Dimensional Swell or Collapse of Cohesive Soils

ASTM D4633
Energy Measurement for Dynamic Penetrometers

ASTM D4719
Prebored Pressuremeter Testing of Soils

ASTM D4914
Density of Soil and Rock in Place by the Sand Replacement Method in a Test Pit

ASTM D5030
Density of Soil and Rock in Place by the Water Replacement Method in a Test Pit

ASTM D5333
Measurement of Collapse Potential of Soils

ASTM D5778
Electronic Friction Cone and Piezocone Penetration Testing of Soils

ASTM D6066
Determining the Normalized Penetration Resistance of Sands for Evaluation of Liquefaction Potential

ASTM D6635
Performing the Flat Plate Dilatometer

ASTM D7015
Obtaining Intact Block (Cubical and Cylindrical) Samples of Soils

ASTM D7099
Standard Terminology Relating to Frozen Soil and Rock

ASTM Special Technical Publication (STP) 654
Dynamic Geotechnical Testing, Use the SPT to Measure Dynamic Soil Properties

Baligh and Scott 1975
Baligh, M.M., and Scott, R.F. 1975. “Quasi-Static Deep Penetration in Clays.” *Journal of the Geotechnical Engineering Division*. 101(11): 1019–1133 (Available at https://doi.org/10.1061/AJGEB6.0000211)

Bartlett, Youd, and Hansen 2002
Bartlett, S.F., Youd, T.L., and Hansen, C.M. 2002. “Revised Multilinear Regression Equations for Prediction of Lateral Spread Displacement.” *Journal of Geotechnical and Geoenvironmental Engineering*. 128(12): 1007–1017. (Available at https://doi.org/10.1061/(ASCE)1090-0241(2002)128:12(1007))

Bayasan et al. 2008
Bayasan, R.M., Korotchenko, A.G., Volkov, N.G., Pustovoit, G.P., and Lobanov, A.D. 2008. “Use of two-phase heat pipes with the enlarged heat-exchange surface for thermal stabilization of permafrost soils at the bases of structures.” *Applied Thermal Engineering*. 28(4): 274–277. (Available at https://doi.org/10.1016/j.applthermaleng.2006.02.022)

Becker et al. 1987
Becker, D.E., et al. 1987. “Work As a Criterion for Determining In Situ and Yield Stresses in Clays.” *Canadian Geotechnical Journal*. 24:549–564. (Available at https://doi.org/10.1139/t87-070)

Boone 2010
Boone, S.J. 2010. “A Critical Reappraisal of ‘Preconsolidation Pressure’ Interpretations Using the Oedometer Test.” *Canadian Geotechnical Journal*. 47(3): 81–296. (Available at https://doi.org/10.1139/T09-093)

Boulanger and Idriss 2006
Boulanger, R., and Idriss, I.M. 2006. “Liquefaction Susceptibility Criteria for Silts and Clays.” *Journal of Geotechnical and Geoenvironmental Engineering*. 132(11): 1413–1426. (Available at https://doi.org/10.1061/(ASCE)1090-0241(2006)132:11(1413))

Boulanger and Idriss 2007
Boulanger, R., and Idriss, I.M. 2007. “Evaluation of Cyclic Softening in Silts and Clays.” *Journal of Geotechnical and Geotechnical Engineering*. 133(6): 641–652. (Available at https://doi.org/10.1061/(ASCE)1090-0241(2007)133:6(641))

Boulanger and Idriss 2015
Boulanger, R.W., and Idriss, I.M. 2015. “CPT- and SPT-Based Liquefaction Triggering Procedures.” *Journal of Geotechnical and Geoenvironmental Engineering*. 142(2). (Available at https://doi.org/10.1061/(ASCE)GT.1943-5606.0001388)

Bray and Sancio 2006
Bray, J.D., and Sancio, R.B. 2006. “Assessment of the Liquefaction Susceptibility of Fine-Grained Soils.” *Journal of Geotechnical and Geoenvironmental Engineering*. 132 (9): 1165–1177. (Available at https://doi.org/10.1061/(ASCE)1090-0241(2006)132:9(1165))

Briaud 1992
Briaud, Jean-Louis. 1992. *The Pressuremeter*. A.A. Balkema, Brookfield, VT. (Available at https://doi.org/10.1201/9780203736173)

Briaud 2013
Briaud, Jean-Louis. 2013. "Ménard Lecture – The Pressuremeter Test: Expanding its Use". *Proceedings of the 18th International Conference on Soil Mechanics and Geotechnical Engineering, Paris, 2013*. (Available at https://www.issmge.org/publications)

Bullock et al. 2017
Bullock, Z., et al. 2017. "Ground-Motion Prediction Equations for Arias Intensity Cumulative Absolute Velocity and Peak Incremental Ground Velocity for Rock Sites in Different Tectonic Environments." *Bulletin of the Seismological Society of America*. 107(2017): 2293–2309. (Available at https://doi.org/10.1785/0120160388)

Byrne et al. 1987
Byrne, P.M. et al. 1987. "Soil Parameters for Deformation Analysis of Sand Masses." *Canadian Geotechnical Journal*. 24(3): 366–376. (Available at https://doi.org/10.1139/t87-047)

Campanella et al. 1985
Campanella, R.G., Robertson, P.K., Gillespie D., and Greig, J. 1985. "Recent Developments In situ Testing in Soils." *Proceedings of XI ICSMFE, San Francisco, 1985*. (Available at https://www.issmge.org/publications)

Carter and Bentley 1991
Carter, M., and Bentley, S.P. 1991. *Soil Properties and Their Correlations*. London: Pentech Press. (Available at https://www.wiley.com/en-us/Soil+Properties+and+their+Correlations,+2nd+Edition-p-9781119130871)

Carter and Bentley 2016
Carter, M., and Bentley, S.P. 2016. *Soil Properties and Their Correlations*. 2nd ed. Newark: Wiley. https://public.ebookcentral.proquest.com/choice/publicfullrecord.aspx?p=4605109.

Cetin et al. 2009
Cetin, K.O., Bilge, H.T., Wu, J., et al. 2009. "Probabilistic Model for the Assessment of Cyclically Induced Reconsolidation (Volumetric) Settlements." *Journal of Geotechnical and Geoenvironmental Engineering*. 135(3): 387–398. (Available at https://doi.org/10.1061/(ASCE)1090-0241(2009)135:3(387))

Cetin et al. 2018
Cetin, K.O., Seed, R.B., Kayen, R.E., Moss, R.E.S., Bilge, H.T., Ilgac, M., and Chowdhury, K. 2018. "SPT-Based Probabilistic and Deterministic Assessment of Seismic Soil Liquefaction Triggering Hazard." *Soil Dynamics and Earthquake Eng*. 115: 698–709. (Available at https://doi.org/10.1016/j.soildyn.2018.09.012)

Chandler 1988
Chandler, R. 1988. “The In situ Measurement of the Undrained Shear Strength of Clays Using the Field Vane.” in *Vane Shear Strength Testing in Soils: Field and Laboratory Studies*, edited by A. Richards. West Conshohocken, PA: ASTM International. (Available at https://doi.org/10.1520/STP10319S)

Chapuis 2004
Chapuis, R.P. 2004. “Predicting the Saturated Hydraulic Conductivity of Sand and Gravel Using Effective Diameter and Void Ratio.” *Canadian Geotechnical Journal*. 41(5): 787–795. (Available at https://doi.org/10.1139/t04-022)

Chapuis and Aubertin 2003
Chapuis, R.P., and Aubertin, M. 2003. Predicting the Coefficient of Permeability of Soils Using the Kozeny-Carmen Equation. Montréal, QC: École Polytechnique Montréal. (Available at https://publications.polymtl.ca/2605/)

Charles and Skinner 2004
Charles, J.A., and Skinner, H.D. 2004. “Settlement and Tilt of Low-Rise Buildings.” *Proceedings of the Institution of Civil Engineers – Geotechnical Engineering 2004.* 157:2, 65–75. (Available at https://doi.org/10.1680/geng.2004.157.2.65)

Chen et al. 2007
Chen, Q., Abu-Farsakh, M.Y., Sharma, R., and Zhang, X. 2007. “Laboratory Investigation of Behavior of Foundations on Geosynthetic-Reinforced Clayey Soil.” *Transportation Research Record*. 2004(1): 28–38. (Available at https://doi.org/10.3141/2004-04)

Clarke 1994
Clarke, B.G. 1994. *Pressuremeters in Geotechnical Design*. 1st ed. CRC Press. (Available at https://doi.org/10.1201/9781482269345)

Coduto et al. 2016
Coduto, D.P., Kitch, W.A., and Yeung, M.R. 2016. *Foundation Design Principles and Practices.* 3rd ed. Boston, MA: Pearson. (Available at https://www.pearson.com)

Cubrinovski and Ishihara 1999
Cubrinovski, M., and Ishihara, K. 1999. “Empirical Correlation Between SPT N-Value and Relative Density for Sandy Soils.” *Soils and Foundations*. 39(5): 61–71**.** (Available at https://doi.org/10.3208/sandf.39.5_61)

D’Appolonia 1971
D’Appolonia, D.J. et al. 1971. “Initial Settlement of Structures on Clay.” *Journal of the Soil Mechanics and Foundations Division*. 97(10): 1359–1377. (Available at https://doi.org/10.1061/JSFEAQ.0001677)

D'Orazio and Duncan 1987
D'Orazio, T.B. and Duncan, J.M. 1987. "Differential Settlement in Steel Tanks." *Journal of Geotechnical Engineering*. 113(9). (Available at https://doi.org/10.1061/(ASCE)0733-9410(1987)113:9(967))

Darendeli 2001
Darendeli, M. 2001. "Development of a New Family of Normalized Modulus Reduction and Material Damping Curves." *PhD dissertation*. University of Texas at Austin, TX. (Available at https://apps.peer.berkeley.edu/lifelines/Workshop304/pdf/Stokoe_PlenaryPaper.pdf)

Das 2017
Das, B.M. 2017. *Shallow Foundations: Bearing Capacity and Settlement*. Boca Raton, FL: CRC Press. (Available at https://www.routledge.com/)

Das and Sivakugan 2007
Das, B.M., Sivakugan, N. 2007. "Settlement of Shallow Foundations on Granular Soil – An Overview." *International Journal of Geotechnical Engineering*. 1(1): 19–29. (Available at https://doi.org/10.3328/IJGE.2007.01.01.19-29)

Day 2012
Day, R.W. 2012. *Geotechnical Earthquake Engineering Handbook: With the 2012 International Building Code*. 2nd ed. New York, NY: McGraw Hill Education LLC. (Available at http://accessengineeringlibrary.com/)

Doré at al. 2012
Doré, G., Ficheur, A., Guimond, A., and Boucher, M. 2012. "Performance and Cost-Effectiveness of Thermal Stabilization Techniques." *Proceedings of the 15th International Specialty Conference on Cold Regions Engineering, Quebec City, Canada, 2012*. (Available at https://doi.org/10.1061/9780784412473.004)

Duncan 2000
Duncan, J.M. 2000. "Factors of Safety and Reliability in Geotechnical Engineering." *Journal of Geotechnical and Geoenvironmental Engineering*. 126(4): 307–316. (Available at https://doi.org/10.1061/(ASCE)1090-0241(2000)126:4(307))

Duncan and Chang 1970
Duncan, J.M., and Chang, Chin-Yung. 1970. "Nonlinear Analysis of Stress and Strain in Soils." *Journal of the Soil Mechanics and Foundations Division*. 96(5): 1629–1653. (Available at https://doi.org/10.1061/JSFEAQ.0001458)

Duncan and D'Orazio 1984
Duncan, J.M., and D'Orazio, T.B. 1984. "Stability of Steel Oil Storage Tanks." *Journal of Geotechnical Engineering*. 110(9): 1219–1238. (Available at https://doi.org/10.1061/(ASCE)0733-9410(1984)110:9(1219)).

EM 1110-1-1804
Geotechnical Investigations

EM 1110-1-2908
Rock Foundations

EM 1110-2-1901
Seepage Analysis and Control for Dams

EM 1110-2-1902
Slope Stability

EM 1110-2-1906
Laboratory Soils Testing

EM 1110-2-1908
Instrumentation of Embankment Dams and Levees

EM 1110-2-1913
Design and Construction of Levees

EM 1110-2-2100
Stability Analysis of Concrete Structures

EM 1110-2-2502
Floodwalls and other Hydraulic Retaining Walls

ER 10-1-55
Organization and Functions – Roles and Responsibilities, Risk Management Center

ER 1110-1-1807
Drilling in Earth Embankment Dams and Levees

ER 1110-1-1901
Engineering and Design, Construction Project Geotechnical and Materials Completion Report

ER 1110-2-112
Required Visits to the Construction Sites by Design Personnel

ER 1110-2-1150
Engineering and Design for Civil Works Projects

ER 1110-2-1806
Earthquake Design and Evaluation for Civil Works Projects

ER 1110-2-8159
Life Cycle Design and Performance

ER 1110-2-8161
Structural Design and Evaluation of Civil Works Buildings

ER 1110-345-100
Design Policy for Military Construction

ER 1180-1-6
Construction Quality Management

ER 1180-2-1156
Safety of Dams – Policy and Procedures

Failmezger 2021
Failmezger, R.A. 2021. “Using the Dilatometer to Make Accurate Settlement Predictions.” *Proceedings of the 6th International Conference on Geotechnical and Geophysical Site Characterization, Online Conference, 2021*. (Available at https://www.issmge.org/publications/author/roger-failmezger)

Fellenius 2009
Fellenius, B. 2009. *Basics of Foundation Design*. British Columbia, Canada: Lulu.com. (Available at https://www.fellenius.net)

FHWA Publication No. FHWA-NHI-11-032
LRFD Seismic Analysis and Design of Transportation Geotechnical Features and Structural Foundations (Available at https://www.fhwa.dot.gov/engineering/geotech/pubs/nhi11032/nhi11032.pdf)

FHWA Publication No. FHWA-SA-95-037
Geotechnical Engineering Circular No. 1. Dynamic Compaction (Available at https://www.fhwa.dot.gov/engineering/geotech/pubs/009754.pdf)

FHWA Report No. FHWA-IP-89-008
The Pressuremeter Test for Highway Applications (Available at https://library.ctr.utexas.edu/Presto/content/Detail.aspx?ctID=UHVibGljYXRpb25fMTE2MTA=&rID=NTM4MQ==&ssid=c2NyZWVuSURfMTQ2MDk=)

FHWA Report No. FHWA-PA-87-024+84-24
Schmertmann, J.H. 1988. Guidelines for Using the CPT, CPTu, and Marchetti DMT for Geotechnical Design: Volume III – DMT Test Methods and Data Reduction (Available at https://ntrl.ntis.gov/NTRL/dashboard/searchResults/titleDetail/PB88211669.xhtml)

FHWA Report No. FHWA-SA-02-054
Geotechnical Engineering Circular No. 6, Shallow Foundations. FHWA (2002). (Available at https://www.fhwa.dot.gov/engineering/geotech/pubs/010943.pdf)

FHWA Report No. FHWA-SA-91-044
The Flat Dilatometer Test (Available at https://www.marchetti-dmt.it/wp-content/uploads/bibliografia/briaud_1992_US_DOT_FHA_DMT_manual.pdf)

FHWA Report No. FHWA-TS-8-209
Guidelines for Cone Penetration Test (Performance and Design) (Available at https://rosap.ntl.bts.gov/view/dot/958)

FHWA/IN/FTRP-2011/12
Identification and Behavior of Collapsible Soils (Available at https://rosap.ntl.bts.gov/view/dot/23601/dot_23601_DS1.pdf)

FHWA/IN/JTRP-2008/2
Classification of Organic Soil. (Available at https://docs.lib.purdue.edu/context/jtrp/article/2676/viewcontent/jtrp1186.pdf)

Foster and Ahlvin 1954
Foster, C.R., and Ahlvin, R.G. 1954. "Stresses and Deflections Induced by a Uniform Circular Load." *Proceedings of the Thirty-Third Annual Meeting of the Highway Research Board, Washington, D.C., 1954*. (Available at https://onlinepubs.trb.org/Onlinepubs/hrbproceedings/33/33-037.pdf)

Franklin et al. 1973
Franklin, A.G., Orozco, L.F., and Semrau, R. 1973. "Compaction and Strength of Slightly Organic Soils." *Journal of the Soil Mechanics and Foundations Division*. 99(7): 541–557. (Available at https://doi.org/10.1061/JSFEAQ.0001900)

Gómez et al. 2008
Gómez, J.E., et al. 2008. "Sand-To-Concrete Interface Response to Complex Load Paths in a Large Displacement Shear Box." *Geotechnical Testing Journal*. 31(4): 358–369. (Available at https://doi.org/10.1520/GTJ100220)

Grant et al. 1974
Grant, et al. 1974. "Differential Settlement of Buildings." *Journal of the Geotechnical Engineering Division*. 100(9): 973–991. (Available at https://doi.org/10.1061/AJGEB6.0000101)

Hardin 1978
Hardin, B.O. 1978. "The Nature of Stress-Strain Behavior for Soils." *Proceedings of ASCE Geotechnical Engineering Division Specialty Conference, Earthquake Engineering and Soil Dynamics, Pasadena, CA, 1978*. (Available at https://cedb.asce.org/CEDBsearch/record.jsp?dockey=0026928)

Hatanaka and Uchida 1996
Hatanaka, M., and Uchida, A. 1996. "Empirical Correlation Between Penetration Resistance and Effective Friction of Sandy Soil." *Soils and Foundations*. 36(4): 1–9. (Available at https://doi.org/10.3208/sandf.36.4_1)

Holtz and Broms 1972
Holtz, R.D. and Broms, B. 1972. “Long-Term Loading Tests at Skå-Edeby, Sweden.” *Proceedings of the ASCE Specialty Conference on Performance of Earth and Earth-Supported Structures, West Lafayette, IN, 1972*. (Available at https://cedb.asce.org/CEDBsearch/record.jsp?dockey=0265089)

Holtz et al. 2011
Holtz, R.D., Kovacs, W.D., and Sheahan, T.C. 2011. *An Introduction to Geotechnical Engineering*. 2nd ed. Upper Saddle River NJ: Pearson. (Available at https://www.pearson.com/)

Instructional Report (K-82-7)
User’s Guide: Computer Program for Bearing Capacity Analysis of Shallow Foundations (CBEAR) (Available at https://apps.dtic.mil/sti/citations/tr/ADA118487)

Ishihara 1985
Ishihara, K. 1985. “Stability of Natural Deposits During Earthquakes.” *Proceedings of the 11th International Conference on Soil Mechanics and Foundation Engineering, San Francisco, CA, 1985*. (Available at https://www.issmge.org/publications/)

Ishihara and Yoshimine 1992
Ishihara, K., and Yoshimine, M. 1992. “Evaluation of Settlements in Sand Deposits Following Liquefaction During Earthquakes.” *Soils and Foundations*, JSSMFE. 32(1): 173–188. (Available at https://doi.org/10.3208/sandf1972.32.173)

Janbu 1969
Janbu, N. 1969. “The Resistance Concept Applied to Deformation of Soils.” *Proceedings of the 7th International Conference on Soil Mechanics and Foundation Engineering, Mexico City, Mexico, 1969*. (Available at https://www.issmge.org/publications)

Kayen et al. 2013
Kayen, R., et al. 2013. “Shear-Wave Velocity-Based Probabilistic and Deterministic Assessment of Seismic Soil Liquefaction Potential.” *Journal of Geotechnical and Geoenvironmental Engineering*. 139(3): 407–419. (Available at https://doi.org/10.1061/(ASCE)GT.1943-5606.0000743)

Knodel 1992
Knodel, P.C. and United States Bureau of Reclamation Denver Office Materials Engineering Branch. 1992. *Characteristics and Problems of Collapsible Soils*. Denver, CO: U.S. Dept. of the Interior Bureau of Reclamation Denver Office Research and Laboratory Services Division Materials Engineering Branch. (Available at https://www.usbr.gov/tsc/techreferences/rec/R9202.pdf)

Kokusho et al. 1995
Kokusho, T., and Tanaka, Y., et al. 1995. “Case Study of Rock Debris Avalanche Gravel Liquefied During the 1993 Hokkaido-Nansei-Oki Earthquake.” *Soils and Foundations*. 3(3): 83–94. (Available at https://doi.org/10.3208/sandf.35.83)

Kovacs and Salomone 1982
Kovacs, W.D., and Salomone, L.A. 1982. “SPT Hammer Energy Measurement.” *Journal of Geotechnical Engineering,* ASCE. 108(GT4): 599–620. (Available at https://doi.org/10.1061/(ASCE) 0733-9410(1984)110:4(555))

Kramer and Wang 2015
Kramer, S.L., and Wang, Chwen-Huan. 2015. “Empirical Model for Estimation of the Residual Strength of Liquefied Soil.” *Journal of Geotechnical Engineering*. 141(9): 555–561. (Available at https://doi.org/10.1061/(ASCE)GT.1943-5606.0001317)

Kulhawy and Chen 2007
Kulhawy, F., and Jie-Ru Chen. 2007. “Drilled Shaft Side Friction in Gravelly Soils [Discussion]”. *Journal of Geotechnical and Geoenvironmental Engineering*. 133(10): 1325–1335. (Available at https://doi.org/10.1061/(ASCE)1090-0241(2007)133:10(1325))

Kulhawy and Maynes 1990
Kulhawy, F.H., and Mayne, P.W. 1990. *Manual on Estimating Soil Properties for Foundation Design.* Report EL68000. Palo Alto, CA: EPRI. (Available at https://www.epri.com/research/products/EL-6800)

Ladd 1991
Ladd, C.C. 1991. “Stability Evaluation During Staged Construction.” *Journal of Geotechnical Engineering,* ASCE. 117(4): 540–615. (Available at https://doi.org/10.1061/(ASCE)0733-9410(1991)117:4(540))

Ladd and Foot 1974
Ladd, C.C., and Foot, R. 1974. “New Design Procedure for Stability of Soft Clays,” *Journal of the Geotechnical Engineering Division.* 100(7), 763–786. (Available at https://doi.org/10.1061/AJGEB6.0000066)

Ladd et al. 1977
Ladd, C.C. et al. 1977. “Stress-Deformation and Strength Characteristics.” *Proceedings of the 9th International Conference on Soil Mechanics and Foundation Engineering, Tokyo, 1977*. (Available at https://www.issmge.org/publications)

Lambe and Whitman 1969
Lambe, T.W., and Whitman, R.V. 1969. *Soil Mechanics*. New York: John Wiley & Sons. (Available at https://www.wiley.com)

Leonards 1976
Leonards, G.A. 1976. “Estimating Consolidation Settlements of Shallow Foundations on Overconsolidated Clay.” *Special Report 163, Transportation Research Board*. 13–16. (Available at http://onlinepubs.trb.org/Onlinepubs/sr/sr163/sr163-002.pdf)

Leonards and Girault 1961
Leonards, G.A., and Girault, P. 1961. “A study of the one-dimensional consolidation test.” *Proceedings of the 5th International Conference on Soil Mechanics and Foundation Engineering, Paris, 1961*. (Available at https://issmge.org/publications/)

Leshchinsky 2015
Leshchinsky, B. 2015. “Bearing Capacity of Footing Placed Adjacent to c′- φ' Slopes.” *Journal of Geotechnical and Geoenvironmental Engineering*. 141(6): 04015022. (Available at https://doi.org/10.1061/(ASCE)GT.1943-5606.0001306)

Leshchinsky and Xie 2017
Leshchinsky, B., and Xie, Y. 2017. “Bearing capacity for spread footings placed near c′ - φ' slopes.” *Journal of Geotechnical and Geoenvironmental Engineering*. 143 (1): 06016020. (Available at https://doi.org/10.1061/(ASCE)GT.1943-5606.0001578)

Liao and Whitman 1985
Liao, S.S.C., and Whitman, R.V. 1985. “Overburden Correction Factor for SPT in Sand.” *Journal of Geotechnical Engineering*, ASCE. 112(3): 373–377. (Available at https://doi.org/10.1061/(ASCE)0733-9410(1986)112:3(373))

Lu and Kaya 2013
Ning Lu, F., and Kaya, M. 2013. “Power Law for Elastic Moduli of Unsaturated Soil.” *Journal of Geotechnical and Geoenvironmental Engineering*. 140(1): 46–56. (Available at https://doi.org/10.1061/(ASCE)GT.1943-5606.0000990)

Lunne et al. 1997
Lunne, T., Robertson, P.K., and Powell, J.J.J. 1997. *Cone-Penetration Testing in Geotechnical Practice*. London: Blackie Academic and Professional. (Available at https://doi.org/10.1007/s11204-010-9072-x)

Lunne et al. 2006
Lunne, T., Berre, T., Andresen, K., Strandvik, S., and Sjursen, M. 2006. “Effects of Sample Disturbance and Consolidation Procedures on Measured Shear Strength of Soft Marine Norwegian Clays.” *Canadian Geotechnical Journal*. 43(7): 726–750. (Available at https://doi.org/10.1139/t06-040)

Makdisi and Seed 1978
Makdisi, F.I., and Seed, H.B. 1978. “Simplified Procedure for Estimating Dam and Embankment Earthquake-Induced Deformations.” *Journal of the Geotechnical Engineering Division*. ASCE. 104(7): 849–867. (Available at https://doi.org/10.1061/AJGEB6.0000668)

Marchetti 1980
Marchetti, S. 1980. “In Situ Tests by Flat Dilatometer.” *Journal of Geotechnical Engineering*. 106(3): 299–321. (Available at https://www.marchetti-dmt.it/)

Marchetti 1997
Marchetti, S. 1997. “The Flat Dilatometer: Design Applications.” *Proceedings of the Third International Geotechnical Engineering Conference, Cairo University, 1997*. (Available at https://www.marchetti-dmt.it/wp-content/uploads/bibliografia/marchetti_1997_SOA_cairo.pdf)

Marchetti and Crapps 1981
Marchetti, S., and Crapps, D.K. 1981. *Flat Dilatometer Manual*. Internal Report of G.P.E. Inc. (Available at https://www.marchetti-dmt.it/)

Marchetti and Totani 1989
Marchetti, S., and Totani, G. 1989. "C_h Evaluations from DMTA Dissipation Curves." *Proceedings from the 12th International Conference on Soil Mechanics and Foundation Engineering (ICSMFE), Rio de Janeiro, 1989*. (Available at https://www.marchetti-dmt.it/)

Marchetti et al. 2001
Marchetti, S., Monaco, P., Totani, G., and Calabrese, M. 2001. "The Flat Dilatometer Test (DMT) in Soil Investigations." *Proceedings of the 2nd International Conference on the Flat Dilatometer, Washington, DC. 2006*. (Available at https://www.marchetti-dmt.it/)

Marcuson et al. 1990
Marcuson III, W.F., Hynes, M.E., and Franklin, A.G. 1990. "Evaluation and Use of Residual Strength in Seismic Safety Analysis of Embankments." *Earthquake Spectra*. 6(3): 529–572. (Available at https://doi.org/10.1193/1.1585586)

Marsh and Thoeny 1999
Marsh, E.T. and Thoeny, S.A. 1999. "Damage and Distortion Criteria for Residential Slab-on-Grade Structures." *Journal of Performance of Constructed Facilities,* ASCE. 13(3): 121–127. (Available at https://doi.org/10.1061/(ASCE)0887-3828(1999)13:3(121))

Martin and Lew 1999
Martin, G.R., and Lew, M. 1999. *Recommended Procedures For Implementation of DMG Special Publication 117 Guidelines for Analyzing and Mitigating Liquefaction Hazard in California*. Los Angeles, USA: The Southern California Earthquake Center. (Available at http://scecinfo.usc.edu/resources/catalog/LiquefactionproceduresJun99.pdf)

Mayne 2005
Mayne, P.W. 2005. "Integrated Ground Behavior: In situ and Lab Tests." *Deformation Characteristics of Geomaterials*. Vol. 2 (Proc. Lyon): 155–177. (Available at https://doi.org/10.1201/9780203970812)

Mayne 2007a
Mayne, P.W. 2007. *Cone Penetration Testing, NCHRP Synthesis 368, A Synthesis of Highway Practices*. Washington, DC: Transportation Research Board. (Available at https://nap.nationalacademies.org/catalog/23143/cone-penetrating-testing)

Mayne 2007b
Mayne, P.W. 2007. "In-situ test calibrations for evaluating soil parameters." Characterization and Engineering Properties of Natural Soils. 3. 1601–1652. (Available at https://www.researchgate.net/publication/286462119_In-situ_test_calibrations_for_evaluating_soil_parameters.)

Mayne 2014
Mayne, P.W. and 2014 Congress on Geo-Characterization and Modeling for Sustainability Geo-Congress. 2014. “Generalized CPT Method for Evaluating Yield Stress in Soils.” *Geotechnical Special Publication.* N234 GSP (2014 01 01): 1336–1346. (Available at https://doi.org/10.1061/9780784413272.130)

McCabe et al. 2007
McCabe, B.A., McNeil, J.A., and Black, J.A. 2007. “Ground Improvement using the Vibro-Stone Column Technique.” *Proceedings of the joint meeting of Engineers Ireland West Region and the Geotechnical Society of Ireland, Galway, Ireland, 2007*. (Available at https://aran.library.nuigalway.ie/bitstream/handle/10379/6339/2007_%5BEI%5D_McCabe%2C_McNeill_and_Black_(IEI_West_Region_-_Geotech_Soc_lecture).pdf?sequence=1&isAllowed=y).

McGregor and Duncan 1998
McGregor, J.A., Duncan, J.M., and Virginia Polytechnic Institute and State University Center for Geotechnical Practice and Research. 1998. *Performance and Use of the Standard Penetration Test in Geotechnical Engineering Practice*. Blacksburg VA: Virginia Polytechnic Institute and State University the Charles E. Via Jr. Dept. of Civil Engineering Center for Geotechnical Practice and Research. (Available at https://cgpr.cee.vt.edu)

Menq 2003
Menq, F.Y. 2003. Dynamic properties of sandy and gravelly soils. *PhD Dissertation*. University of Texas 2003, Austin. (Available at https://repositories.lib.utexas.edu/bitstream/handle/2152/779/menqf032.pdf?sequence=2&isAllowed=y)

Mesri 1973
Mesri, G. 1973. “Coefficient of Secondary Compression.” *Journal of the Soil Mechanics and Foundation Division*, ASCE. 99(1): 123–137. (Available at https://doi.org/10.1061/JSFEAQ.0001840)

Mesri 1975
Mesri, G. 1975. “New Design Procedure for Stability of Soft Clays [Discussion].” *Journal of the Geotechnical Engineering Division*. 101(4): 409:412. (Available at https://doi.org/10.1061/AJGEB6.0005026)

Mesri and Godlewski 1977
Mesri, G., and Godlewski, P.M. 1977. “Time- and stress-compressibility interrelationship.” *Journal of the Geotechnical Engineering Division*, ASCE. 103(5): 417–430. (Available at https://doi.org/10.1061/AJGEB6.0000421)

Mesri et al. 1994
Mesri, G., Feng, T.W., Ali, S., and Hayat, T.M. 1994. “Permeability Characteristics of Soft Clays.” *Proceedings of the International Conference on Soil Mechanics and Foundation Engineering – International Society for Soil Mechanics and Foundation Engineering* 187–87. (Available at https://www.issmge.org/uploads/publications/1/32/1994_01_0042.pdf)

Meyerhof 1956
Meyerhof, G.G. 1956. “Penetration Tests and Bearing Capacity of Cohesionless Soils.” *Journal of the Soil Mechanics and Foundations Division*, ASCE. 82(1): 1–19. (Available at https://doi.org/10.1061/JSFEAQ.0000001)

Meyerhof 1957
Meyerhof, G.G. 1957. “Discussion on Research on Determining the Density of Sands by Spoon Penetration Testing.” *Proceedings of the 4th International Conference on Soil Mechanics and Foundation Engineering*. 1(110). (Available at https://www.issmge.org/publications/publication/research-on-determining-the-density-of-sands-by-spoon-penetration-testing)

Meyerhof 1963
Meyerhof, G.G. 1963. “Some recent research on the bearing capacity of foundations.” *Canadian Geotechnical Journal*. 1(1): 16–26. (Available at https://doi.org/10.1139/t63-003)

Meyerhof 1982
Meyerhof, G.G. 1982. “Limit State Design in Geotechnical Engineering.” *Structural Safety*. 1(1): 67–71. (Available at https://doi.org/10.1016/0167-4730(82)90015-7)

Mitchell 1981
Mitchell, J.K. 1981. “Soil Improvement: State-of-the-Art.” *Session 12. 10th International Conference on Soil Mechanics and Foundation Engineering, Stockholm Sweden, 1981*. (Available at https://www.issmge.org/uploads/publications/1/35/1981_04_0022.pdf)

MP GL-87-8
Andrus, R.D., and Youd, T.L. 1987. Subsurface Investigation of a Liquefaction-Induced Lateral Spread, Thousand Springs Valley, Idaho (Available at https://apps.dtic.mil/sti/citations/ADA182912)

MP GL-92-35
Gilbert, P.A. 1992. Effect of Sampling Disturbance on Laboratory-Measured Soil Properties (Available at https://apps.dtic.mil/sti/citations/ADA257268)

Mulvey 1992
Mulvey, W.E. 1992. *Engineering Geologic Problems Caused by Soil and Rock in Southwestern Utah*. Utah Geological Survey: Salt Lake City, UT. (Available at https://www.utahmapstore.com/products/uga-21)

Nishida 1956
Nishida, Y. 1956. “A Brief Note on Compression Index of Soil.” *Journal of Soil Mechanics and Foundation Engineering*, ASCE. 82(3): 1–14. (Available at https://doi.org/10.1061/JSFEAQ.0000015)

O’Neill and Ghazzaly 1977
O’Neill, M.W., and Ghazzaly, O.I. 1977. “Swell Potential Related to Building Performance.” *Journal of the Geotechnical Engineering Division*, ASCE. 103(12): 1363–1379. (Available at https://ascelibrary.org/doi/epdf/10.1061/AJGEB6.0000534)

Obermeier et al. 2001
Obermeier, S.F. et al. 2001. *Paleoliquefaction Studies in Continental Settings: Geologic and Geotechnical Factors in Interpretations and Back-Analysis*. U.S. Geological Survey Open-File Report 01-029. (Available at https://pubs.usgs.gov/of/2001/of01-029/)

Oikawa 1987
Oikawa, H. 1987. “Compression Curve of Soft Soils.” *Soils and Foundations*. 2(3): 99–104. (Available at https://doi.org/10.3208/sandf1972.27.3_99)

Olson and Stark 2002
Olson, S.M., and Stark, T.D. 2002. “Liquefied Strength Ratio from Liquefaction Flow Failure Case Histories.” *Canadian Geotechnical Journal*. 39(5): 629–647. (Available at https://doi.org/10.1139/t02-001).

Osterberg 1957
Osterberg, J.O. 1957. “Influence Values for Vertical Stresses in a Semi-infinite Mass Due to an Embankment Loading.” *Proceedings of the Fourth International Conference on Soil Mechanics and Foundation Engineering, London, England, 1957*. (Available at https://www.issmge.org/publications/)

Owens and Rollins 1990
Owens, R.L., and Rollins, K.M. 1990. *Collapsible Soil Hazard Map for the Southern Wasatch Front*. Utah: Utah Geological and Mineral Survey Miscellaneous Publication 90-1. (Available at https://www.utahmapstore.com/products/mp-90-1)

Peck et al. 1974
Peck, R.B., Hanson, W.H., and Thornburn, T.H. 1974. *Foundation Engineering*. 2nd ed. New York: John Wiley & Sons. (Available at https://www.wiley.com)

Polshin and Tokar 1957
Polshin, D.E. and Tokar, R.A. 1957. “Maximum Allowable Non-uniform Settlement of Structures.” *Proceedings Fourth International Conference on Soil Mechanics and Foundation Engineering, London, 1957*. (Available at https://www.issmge.org/publications)

Poulos and Davis 1974
Poulos, H.G., and Davis, E.H. 1974. *Elastic Solutions for Soil and Rock Mechanics*. New York: John Wiley & Sons. (Available at https://www.wiley.com)

Rix 1984
Rix, G.J. 1984. “Correlation of Elastic Moduli and Cone Penetration Resistance.” *Dissertation*. University of Texas at Austin. (Available at https://doi.org/10.3208/sandf1972.35.2_107)

Rix et al. 2018
Rix, G.J. and National Cooperative Highway Research Program. 2018. *Manual on Subsurface Investigations*. Washington DC: National Academies of Sciences Engineering Medicine: Transportation Research Board. (Available at https://nap.nationalacademies.org/catalog/25379/manual-on-subsurface-investigations)

Robertson 1990
Robertson, P.K. n.d. 1990. “Soil Classification Using the Cone Penetration Test.” *Canadian Geotechnical Journal*. 27(1):151–158. (Available at https://doi.org/10.1139/t90-014)

Robertson and Cabal 2022
Robertson, P.K., and Cabal, K.L. 2022. *Guide to Cone Penetration Testing for Geotechnical Engineering*. 7th ed. Signal Hill, CA: Gregg Drilling and Testing. (Available at https://www.cpt-robertson.com/PublicationsPDF/CPT-Guide-7th-Final-SMALL.pdf)

Rogers 1995
Rogers, C.D.F. 1995. “Types and Distribution of Collapsible Soils.” *In Genesis and Properties of Collapsible Soils*, edited by Derbyshire, E., Dijkstra, T., and Smalley, I.J. NATO ASI Series, Vol 468. Publisher location: Dordrecht: Springer. (Available at https://doi.org/10.1007/978-94-011-0097-7_1)

Rollins 1998
Rollins, K.M. et al. 1998. “Shear Modulus and Damping Relationships for Gravels.” *Journal of Geotechnical and Geoenvironmental Engineering*. 124(5): 393–405. (Available at https://doi.org/10.1061/(ASCE)1090-0241(1998)124:5(396))

Schmertmann 1955
Schmertmann, J.H. 1955. “The Undisturbed Consolidation Behavior of Clay.” *Transactions*, ASCE. 120(1): 1201–1233. (Available at https://doi.org/10.1061/TACEAT.0007231)

Schmertmann 1970
Schmertmann, J.H. 1970. “Static Cone to Compute Static Settlement Over Sand.” *Journal of the Soil Mechanics and Foundations Division.* 1011–43. (Available at https://doi.org/10.1061/JSFEAQ.0001418)

Schmertmann 1986
Schmertmann, J.H. 1986. “Dilatometer to Compute Foundation Settlement.” *Proceedings of the ASCE Specialty Conference, In situ ’86, VPI, Blacksburg, Virginia, 1986*. (Available at https://cedb.asce.org/CEDBsearch/record.jsp?dockey=0048890)

Schmertmann et al. 1978
Schmertmann, J.H., Hartman, J.P., and Brown, P.R. 1978. “Improved Strain Influence Factor Diagrams.” *Journal of the Geotechnical Engineering Division.* 1131–35. (Available at https://doi.org/10.1061/AJGEB6.0000683)

Schneider et al. 2008
Schneider, J.A., Randolph, M.F., Mayne, P.W., and Ramsey, N. 2008. “Analysis of Factors Influencing Soil Classification Using Normalized Piezocone Tip Resistance and Pore Pressure Parameters.” *Journal of Geotechnical and Geoenvironmental Engineering,* ASCE. 134(11):1569–1586. (Available at https://doi.org/10.1061/(ASCE)1090-0241(2008)134:11(1569))

Seed et al. 1984
Seed, H.B., National Science Foundation (U.S.), and University of California, Berkeley Earthquake Engineering Research Center. 1984. “Moduli and Damping Factors for Dynamic Analyses of Cohesionless Soils.” Berkeley, CA, Springfield, VA.: College of Engineering, University of California ; For sale by the National Technical Information Service, National Bureau of Standards, U.S. Dept. of Commerce. (Available at https://ascelibrary.org/doi/10.1061/%28ASCE%290733-9410%281986%29112%3A11%281016%29)

Seed et al. 1986
Seed, H.B. et al. 1986. “Moduli and Damping Factors for Dynamic Analyses of Cohesionless Soils.” *Journal of Geotechnical Engineering*. 112(11): 1016–1032. (Available at https://doi.org/10.1061/(ASCE)0733-9410(1986)112:11(1016))

Skempton 1944
Skempton, A.W. 1944. “Notes on the compressibility of clays.” *Quarterly Journal of the Geological Society of London*. 100: 119–135. (Available at https://doi.org/10.1144/GSL.JGS.1944.100.01-04.08)

Skempton 1986
Skempton, A.W. 1986. “Standard Penetration Test Procedures and the Effects in Sands of Overburden Pressure, Relative Density, Particle Size, Ageing and Overconsolidation.” *Géotechnique*. 36(3): 425–447. (Available at https://www.icevirtuallibrary.com/doi/abs/10.1680/geot.1986.36.3.425)

Skempton and Bjerrum 1957
Skempton, A.W. and Bjerrum, L. 1957. “A Contribution to the Settlement Analysis of Foundations on Clay.” *Géotechnique*. 168–78. (Available at https://doi.org/10.1680/geot.1957.7.4.168)

Skempton and MacDonald 1956
Skempton, A.W., and MacDonald, D.H. 1956. “The Allowable Settlement of Buildings.” *Proceedings of the Institution of Civil Engineers*. 5(6): 727–768. (Available at https://doi.org/10.1680/ipeds.1956.12202)

Stewart et al. 2014
Stewart, J.P., et al. 2014. *Guidelines for Performing Hazard-Consistent One-Dimensional Ground Response Analysis for Ground Motion Prediction*. PEER Rep. No. 2014/16. Pacific Earthquake Engineering Research Center. University of California, Berkeley, CA. (Available at https://peer.berkeley.edu/sites/default/files/webpeer-2014-16-jonathan_p._stewart_kioumars_afshari_and_youssef_m.a._hashash.pdf)

Technical Report ITL-88-1
Hartman, J.P., and James, B.H. 1988. Development of Design Formulas for Ribbed Mat Foundations on Expansive Soils. Computer-Aided Structural Engineering (Case) Project. Department of the Army. (Available at https://erdc-library.erdc.dren.mil/jspui/bitstream/11681/10861/1/TR-ITL-88-1.pdf)

Technical Report ITL-91-1
Knowles, V.R. 1991. Settlement of Shallow Footings on Sand: Report and User's Guide for Computer Program CSANDSET. Information Technology Laboratory. Department of the Army. (Available at https://apps.dtic.mil/sti/pdfs/ADA238950.pdf)

Technical Report ITL-92-11, NCEL TR-939
Ebeling, R.M., and Morrison, E.E. Jr. 1992. The Seismic Design of Waterfront Retaining Structures. Army Engineer Waterways Experiment Station Vicksburg MS Information Technology Lab. (Available at https://apps.dtic.mil/sti/citations/ADA262963)

Terzaghi 1943
Terzaghi, K. 1943. *Theoretical Soil Mechanics*. New York, NY: John Wiley & Sons. (Available at https://www.wiley.com)

Terzaghi 1955
Terzaghi, K. 1955. "Evaluation of Coefficients of Subgrade Reaction." *Géotechnique*. 5(4): 297–326. (Available at https://doi.org/10.1680/geot.1955.5.4.297)

Terzaghi and Peck 1948
Terzaghi, K., and Peck, R.B. 1948. *Soil Mechanics in Engineering Practice.* New York, NY: John Wiley & Sons. (Available at https://www.wiley.com)

Terzaghi and Peck 1967
Terzaghi, K., and Peck, R.B. 1967. *Soil Mechanics in Engineering Practice*, 2nd ed. New York: John Wiley & Sons. (Available at https://www.wiley.com)

Tokimatsu and Seed 1984
Tokimatsu, K., Seed, H.B., and University of California Berkeley Earthquake Engineering Research Center. 1984. *Simplified Procedures for the Evaluation of Settlements in Sands Due to Earthquake Shaking*. Berkeley Springfield, VA: University of California Earthquake Engineering Research Center. (Available at https://nehrpsearch.nist.gov/static/files/NSF/PB85197887.pdf)

UFC 1-200-01
DoD Building Code

UFC 3-130-01 through UFC 3-130-07
UFC Series, Arctic, and Subarctic Construction

UFC 3-220-07
Foundations in Expansive Soils

UFC 3-220-10
Soil Mechanics (DM 7.1)

UFC 3-301-01
Structural Engineering

UFC 3-301-02
Design of Risk Category V Structures

Ulrich 1995
Ulrich, E.J. 1995. “Subgrade Reaction in Mat Foundation Design.” *In Design and Performance of Mat Foundations, Special Publication 152*, edited by Edward J. Ulrich. Detroit, Michigan: ACI. (Available at https://www.concrete.org/store.aspx)

Uzielli et al. 2013
Uzielli, M., Mayne, P.W., Cassidy, M.J. 2013. “Probabilistic assessment of design strengths for sands from in situ testing data.” In: (series) *Advances in Soil Mechanics and Geotechnical Engineering. Vol. 1, Modern Geotechnical Design Codes of Practice*. Amsterdam: IOS-Millpress. (Available at https://ebooks.iospress.nl/publication/28737)

Vesic 1973
Vesic, A.S. 1973. “Analysis of ultimate loads of shallow foundations,” *Journal of the Soil Mechanics and Foundations Division*, ASCE. 99(1): 45–73. (Available at https://doi.org/10.1061/JSFEAQ.0001846)

Wair et al. 2012
Wair, B.R., DeJong, J.T., Shantz, T., and Pacific Earthquake Engineering Research Center. 2012. *Guidelines for Estimation of Shear Wave Velocity Profiles*. Berkeley: Pacific Earthquake Engineering Research Center. (Available at https://peer.berkeley.edu/publications/2012-08)

Wallace and Otto 1964
Wallace, G.B., and Otto, W.C. 1964. “Differential Settlement at Selfridge Air Force Base.” *Journal of the Soil Mechanics and Foundations Division*. 90(5): 197–220. (Available at https://doi.org/10.1061/JSFEAQ.0000652)

Weber et al. 2022
Weber et al. 2022. “Engineering Evaluation of Post-Liquefaction Residual Strength.” Geotechnical Research Report No. UCB/GT/22-01, Department of Civil and Environmental Engineering, University of California at Berkeley. (Available at https://geotechnical.berkeley.edu/sites/default/files/UCB-GT_22-01_Vol1.pdf)

Youd and Perkins 1978
Youd, T.L., and Perkins, D.M. 1978. “Mapping Liquefaction-Induced Ground Failure Potential.” *Journal of Geotechnical Engineering Division*, ASCE. 104(4): 433–446. (Available at https://doi.org/10.1061/AJGEB6.0000612)

Youd, et al. 2001
Youd, et al. 2001. “Liquefaction Resistance of Soils: Summary Report from the 1996 NCEER and 1998 NCEER Workshops.” *Journal of Geotechnical and Geoenvironmental Engineering*. 127(10): 817–833. (Available at https://doi.org/10.1061/(ASCE)1090-0241(2001)127:10(817))

Zhang et al. 2004
Zhang, G., Robertson, P.K., and Brachman, R.W.I. 2004. “Estimating Liquefaction-Induced Lateral Displacements Using the Standard Penetration Test or Cone Penetration Test.” *Journal of Geotechnical and Geoenvironmental Engineering*. 130(8):861–871. (Available at https://doi.org/10.1061/(ASCE)1090-0241(2004)130:8(861))

Section II

Related Publications

The following citations are provided as related references, but online versions of these references may not be available.

Arman and Munfakh 1970
Arman, A., and Munfakh, G. 1970. *Stabilization of Organic Soils with Lime*. Bulletin No. 103. Baton Rouge, LA: Louisiana State University, Division of Engineering Research.

Baguelin and Shields 1978
Baguelin, F.J., and Shields, D.H. 1978. *The Pressuremeter and Foundation Engineering*, 1st ed. Aedermannsdorf, Switzerland: Trans Tech Publications.

Bjerrum 1963
Bjerrum, L. 1963. “Allowable Settlements of Structures.” *Proceedings of the 3rd European Conference on Soil Mechanics and Foundation Engineering, Wiesbaden, Germany, 1963*.

Bjerrum 1972
Bjerrum, L. 1972. “Embankments on Soft Ground.” *Proceedings, Special Conference on Performance of Earth and Earth-Supported Structures, Lafayette, IN. 1972*.

Bjerrum 1973
Bjerrum, L. 1973. “Problems of Soil Mechanics and Construction on Soft Clays and Structurally Unstable Soils.” *Proceedings of the 8th International Conference on Soil Mechanics and Foundation Engineering, Session 4, Moscow, 1973*.

Boussinesq 1885
Boussinesq, J. 1885. *Application Des Potentiels À L'étude De L'équilibre Et Du Mouvement Des Solides Élastiques : Principalement Au Calcul Des Déformations Et Des Pressions Que Produisent Dans Ces Solides Des Efforts Quelconques Exercés Sur Une Petite Partie De Leur Surface Ou De Leur Intérieur : Mémoire Suivi De Notes Étendues Sur Divers Points De Physique Mathematique Et D'analyse*. Lille: L. Danel.

Bowles 1974
Bowles, J.E. 1974. *Analytical and Computer Methods in Foundation Engineering*. New York, NY: McGraw-Hill Book Co.

Bowles 1996
Bowles, J.E. 1996. *Foundation Analysis and Design*. 5th Ed. New York: McGraw-Hill Book Co.

Brown and Kupsch 1974
Brown, R.J.E. and Kupsch, W.O. 1974. *Permafrost Terminology, Technical Memorandum No. 11*. Ottawa, CA: National Research Council Canada, Associate Committee on Geotechnical Research.

Bray and Macedo 2017
Bray, J.D., and Macedo, J. 2017. "Simplified Procedure for Estimating Liquefaction-Induced Building Settlements." *Proceedings of the 19th International Conference on Soil Mechanics and Geotechnical Engineering, Seoul, 2017*.

Burland 1990
Burland, J.B. 1990. "On the Compressibility and Shear Strength of Natural Clays." *Géotechnique*. 40(3): 329–378.

Burland and Burbidge 1985
Burland, J.B., and Burbidge, M.C. 1985. "Settlement of Foundations on Sand and Gravel." *Proceedings of the Institution of Civil Engineers, Cambridge, 1974*.

Burland and Wroth 1975
Burland, J.B. and Wroth, C.P. 1975 Revision. "Settlement of Buildings and Associated Damage." *Proceedings of the British Geotechnical Society's Conference on Settlement of Structures, Session 5, Cambridge, 1974*.

Caquot and Kérisel 1948
Caquot, A.I., and Kérisel, J.L. 1948. *Tables for the calculation of passive pressure, active pressure and bearing capacity of foundations*. Paris: Gautier-Villars.

Casagrande 1936
Casagrande, A. 1936. "The Determination of the Preconsolidation Load and Its Practical Significance." *Proceedings of the 1st International Conference on Soil Mechanics, Vol. 3, Cambridge, MA, 1936*.

Casagrande and Fadum 1940
Casagrande, A., and Fadum, R.E. 1940. *Notes on Soil Testing for Engineering Purposes*. Cambridge, MA: Harvard University Graduate School of Engineering.

Clarke et al. 1979
Clarke, B.B., Carter, J.P., and Wroth, C.P. 1979. "In situ Determination of the Consolidation Characteristics of Saturated Clays." *Design Parameters in Geotechnical Engineering*. 2:207–211.

Das 1989
Das, B.M. 1989. "Bearing Capacity of Shallow Foundation on Granular Column in Weak Clay." *Proceedings of the 3rd International Symposium on Numerical Models in Geomechanics (NUMOG III), Niagara Falls, Canada, 1989*.

Das 2008
Das, B.M. 2008. *Advanced* Soil Mechanics. 2nd Ed. New York, NY: Taylor and Francis.

Das and Hanna 1988
Das, B.M., and Hanna, A.M. 1988. "Model Tests for Shallow Strip Foundation on Granular Soil." *Proceedings of a Session Sponsored by the Geotechnical Engineering Division, ASCE in Conjunction with the ASCE National Convention, Nashville, TN, 1988*.

Douglas and Olsen 1981
Douglas, B.J., and Olsen, R.S. 1981. "Soil Classification Using Electrical Cone Penetrometer." *Symposium on Cone Penetration Testing and Experiences, Geotechnical Engineering Division, ASCE, St. Louis, MO, 1981*.

Duncan 1980
Duncan, J.M. 1980. *Strength Stress-Strain and Bulk Modulus Parameters for Finite Element Analyses of Stresses and Movements in Soil Masses*. Berkeley Calif: College of Engineering Office of Research Services University of California.

Duncan and Buchignani 1976
Duncan, J.M., and Buchignani, A.L. 1976. *An Engineering Manual for Settlement Studies*. Berkeley: University of California Dept. of Civil Engineering.

Edil and den Haan 1994
Edil, T.B., and den Haan, E.J. 1994. "Settlement of Peats and Organic Soils." *Proceedings of a Session Sponsored by ASCE, Settlement '94, College Station, TX, 1994*.

Fadum 1948
Fadum, R.E. 1948. "Influence Values for Estimating Stresses in Elastic Foundations." *Proceedings of the 2nd International Conference on Soil Mechanics and Foundations, Rotterdam, Netherlands, 1948*.

Gallavresi 1992
Gallavresi, F. 1992. “Grouting Improvement of Foundation Soils.” *In Grouting, Soil Improvement and Geosynthetics.* Geotechnical Special Publication No. 30, edited by Borden, R.H., Holtz, R.D. and Juran, I. Reston, VA: American Society of Civil Engineers.

Goodger and Leach 1991
Goodger, H.K., and Leach, B.A. 1991. *Building on Derelict Land*. London, UK. Construction Industry Research and Information Association.

Gravesen 1960
Gravesen, S. 1960. “Elastic Semi-Infinite Medium Bounded by a Rigid Wall with a Circular Hole.” *Laboratoriet for Bygninsteknik, Danmarks Tekniske Højskole, Meddelelse* No. 10. Copenhagen.

Hansen and Inan 1970
Hansen, J.B., and Inan, S. 1970. *A Revised and Extended Formula for Bearing Capacity*. Copenhagen: Geoteknisk Institute.

Hetényi 1946
Hetényi, M. 1946. *Beams on Elastic Foundation; Theory with Applications in the Fields of Civil and Mechanical Engineering*. Ann Arbor: University of Michigan Press.

Iwasaki et al. 1991
Iwasaki, K. et al. 1991. “Applicability of the Marchetti Dilatometer Test to Soft Ground in Japan.” *Proceedings of the International Conference on Geotechnical Engineering for Coastal Development, Geo-Coast ’91. Yokohama. 1991*.

Janbu 1963
Janbu, N. 1963. “Soil Compressibility as Determined by Oedometer and Triaxial Tests.” *Proceedings of European Conference on Soil Mechanics and Foundation Engineering (ECSMFE), Wiesbaden. 1963*.

Janbu 1967
Janbu, N. 1967. “Settlement Calculations based on the Tangent Modulus Concept.” *Trondheim: Norwegian University of Science and Technology*.

Lacasse 1986
Lacasse, S. 1986. “In Situ Site Investigation Techniques and Interpretation for Offshore Practice.” *Norwegian Geotechnical Institute*. Report 40019-28.

Ladd and Edgers 1972
Ladd, C.C., and Edgers, L. 1972. “Consolidated-Undrained Direct-Simple Shear Tests on Saturated Clays.” Res. Report R72-82, No. 284. Cambridge, MA: Department of Civil Engineering, Mass. Inst. of Technology.

Ladd and Luscher 1965
Ladd, C.C., and Luscher, U. 1965. *Engineering Properties of the Soils Underlying the M.I.T. Campus. Res*. Report R65-68, No. 185. Cambridge, MA: Department of Civil Engineering, Mass. Inst. of Technology.

Leet et al. 1978
Leet, L.D., Judson, S., and Kauffman, M.E. 1978. *Physical Geology*. Englewood Cliffs, NJ: Prentice Hall.

Lutenegger and Degroot 1995
Lutenegger, A.J., and Degroot, D.J. 1995. *Settlement of Shallow Foundations on Granular Soils, Report of Research Conducted for Massachusetts Highway Department Transportation Research Project, Contract #6332, Task Order #4*. Amherst, MA: University of Mass. Transportation Center, College of Engineering.

Marchetti 1999
Marchetti, S. 1999. *On the Calibration of The DMT Membrane. L'Aquila University.* Unpublished report.

Mathai and Unnikrishnan 2015
Mathai, B., and Unnikrishnan, N. 2015. "Footings on Encapsulated Granular Trenches Under Repeated Loading." *Post Graduate Thesis.* College of Engineering Trivandrum.

Mayne 2001
Mayne, P. 2001. "Stress-Strain-Strength-Flow Parameters from Enhanced In Situ Tests." *Proceedings of the International Conference on In Situ Measurement of Soil Properties and Case Histories. Bali, IN. 2001*.

Ménard 1975
Techniques Louis Ménard and Ménard Inc. 1975. *The Ménard Pressuremeter: Interpretation and Application of Pressuremeter Test Results to Foundation Design*. Saulx-les-Chartreux (Essonne) France Monroeville Pa: Techniques Louis Ménard; Ménard.

Ménard and Rousseau 1962
Ménard, L., and Rousseau, J. 1962. "L'évaluation des tassements – Tendances nouvelles." [translation: The Assessment of Settlements – New Trends]. *Sols-Soils*. vol. I, no°1:13–29.

NAVFAC (formerly known as Bureau of Yards and Docks [BuDocks]), Contract No. NOY-88812 (1958)
Moran et al. 1959. Study of Deep Soil Stabilization by Vertical Sand Drains.

NAVFAC, DM 7.1
Soil Mechanics

Pacheco Silva 1970
Pacheco Silva, F. 1970. “A New Graphical Construction for Determination of the Preconsolidation Stress of a Soil Sample.” *Proceedings of the 4th Brazilian Conference on Soil Mechanics and Foundation Engineering, Rio de Janeiro, 1970*.

Perloff and Baron 1976
Perloff, W.H., and Baron, W. 1976. *Soil Mechanics: Principles and Applications*. New York: Ronald Press.

Rollins and Kim 1994
Rollins, K.M., and Kim, J.H. 1994. “U.S. Experience with Dynamic Compaction of Collapsible Soils.” *In Situ Deep Soil Improvements*. Geotechnical Special Publication No. 45: 26–43. ASCE, New York.

Salgado 2008
Salgado, R. 2008. *The Engineering of Foundations*. Boston: McGraw-Hill.

Santi 2005
Santi, P. 2005. “Recognition of Collapsible Soils Based on Geology, Climate, Laboratory Tests, and Structural Crack Patterns.” *Proceedings of the Annual Meeting of the Geological Society of America and Engineering Geology Division, Salt Lake City, UT, 2005*.

Seed and Harder 1990
Seed, R.B., and Harder Jr., L.F. 1990. “SPT-Based Analysis of Cyclic Pore Pressure Generation and Undrained Residual Strength.” In *Proceedings of the H.B. Seed Memorial Symposium*, edited by Duncan, J.M. Richmond, VA: Bi-Tech Publishing Ltd.

Sivakugan 1990
Sivakugan, N. 1990. “Development of Marshy Areas in Colombo, Sri Lanka.” *Proceedings of the Tenth Southeast Asian Geotechnical Conference, Taipei, 1990*.

Sowers and Sowers 1979
Sowers, G.B., and Sowers, G.F. 1979. *Introductory Soil Mechanics and Foundations: Geotechnical Engineering* Ed. 4 ed. New York London: Macmillan Publishing Company.

Taki and Yang 1991
Taki, O., and Yang, D.S. 1991. “Soil-Cement Mixed Wall Technique.” *Proceedings of the Geotechnical Engineering Congress, Boulder, CO, 1991*.

Tan and Duncan 1991
Tan, C.K., and Duncan, J.M. 1991. “Settlement of Footings on Sands – Accuracy and Reliability.” *Proceedings of the Geotechnical Engineering Congress, Boulder, CO, 1991*.

Taylor 1948
Taylor, D.W. 1948. *Fundamentals of Soil Mechanics*. New York, NY: John Wiley & Sons.

Teng 1962
Teng, W.C. 1962. *Foundation Design*. Englewood Cliffs N.J: Prentice-Hall.

Totani and Monaco 1998
Totani, G.C., and Monaco, P. 1998. “In Situ Determination of C_h by Flat Plate Dilatometer (DMT).” *Proceedings of the 1st International Conference on Site Characterization, Atlanta, GA, 1998*.

Tsytovich 1976
Tsytovich, N. 1976. *Soil Mechanics*. Moscow: Mir Publishers.

Ulrich 1988
Ulrich, E.J. 1988. “Geotechnical Considerations in Mat Foundation Design.” *In Mat Foundation Design*, “Soil Structure Interaction.” ASCE. Illinois Section, pp 1–33.

Vesic 1961
Vesic, A.B. 1961. “Beams on elastic subgrade and Winkler’s hypothesis.” Proceedings of the 5th International Conference on Soil Mechanics and Foundation Engineering. Paris. 845–850.

Westergaard 1939
Westergaard, H.M. 1939. *A Problem of Elasticity Suggested by a Problem in Soil Mechanics: Soft Material Reinforced by Numerous Strong Horizontal Sheets*. Cambridge, MA: Harvard University.

Winkler 1867
Winkler, E. 1867. *Die Lehre von Elastizitat und Festigkeit (on Elasticity and Fixity)*. Dominicus, Prague.

Section III

Prescribed Forms

This section contains no entries.

Appendix B
Design Examples: Geotechnical Ultimate Limit States for Static Analysis

Design examples provided in this appendix are described below. Settlement calculations for these same design examples are provided in Appendix C. No examples for uplift or global stability are presented in this appendix. Refer to EM 1110-2-2100 and EM 1110-2-1902, respectively.

Example B–1 Undrained vertical bearing capacity for isolated footing on clay.

Example B–2 Undrained vertical bearing capacity for continuous footing on clay.

Example B–3 Drained vertical bearing capacity for isolated footing on sand with water table below the shear zone. Dense over soft condition and shear within upper layer.

a. Solution for drained bearing capacity analysis, shear within surface layer.

b. Solution for undrained punch-through analysis.

Example B–4 Undrained vertical bearing capacity for isolated basement footing on clay, influence of construction dewatering.

a. Solution for the end-of-construction case (dewatered condition).

b. Solution for the long-term water level case (pre-construction water level re-established).

Example B–5 Undrained vertical bearing capacity for isolated footing on clay.

a. Solution for the analysis of bearing capacity based on undrained strength.

b. Solution for the bearing capacity analysis based on the pressuremeter.

c. Solution for punch-through analysis based on undrained strength.

d. Average safety factor from multiple methods.

Example B–6 Drained vertical bearing capacity for isolated footing on sand with water table in the shear zone, as compared to undrained punch-through failure into soft clay.

a. Solution for drained bearing capacity analysis within surface layer.

b. Solution for undrained punch-through analysis.

Example B–7 Drained vertical bearing capacity for isolated footing on sand, comparison of soil parameters and in situ tests.

a. Solution for drained bearing capacity analysis within surface layer.

b. Solution for the analysis based on in situ tests (PMT, CPT, SPT).

B–1. Example B–1. Undrained vertical bearing capacity of isolated footing on clay

a. Determine: Estimate the undrained vertical bearing capacity for an isolated footing on clay. Calculate the factor of safety for bearing capacity and compare to minimum requirements to see if footing is acceptable.

b. Given: A 7.0-ft by 7.0-ft isolated footing supporting a 40-kip column load bearing 2.0 ft below grade on the clay profile illustrated in Figure B–1. A summary of calculations is presented in Table B–1b. Correction factors for the USACE and AASHTO methods are presented In the calculations. Section 5–5f recommends performing analyses with both methods and average the results when assessing the safety factor.

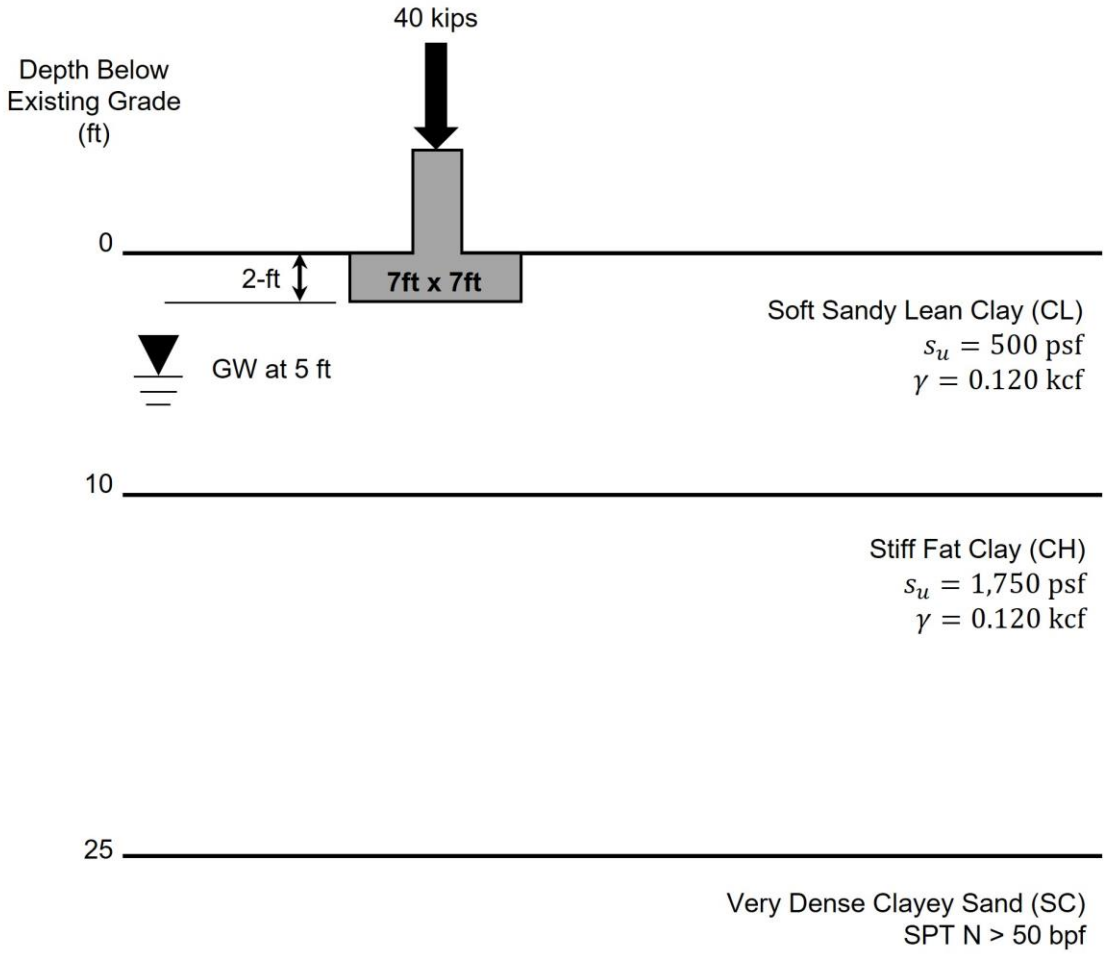


Figure B–1. Foundation and soil conditions for Example B–1

Table B–1a
Solution for undrained bearing capacity analysis (isolated footing)

Step 1	Collect geometric information for the problem and input to Table B–1b. There is no eccentricity ($e = 0$), so footing geometry is equal to effective footing geometry (see paragraph 5–4a). Footing width, $B = B' = 7$ ft (Row 1) Footing length, $L = L' = 7$ ft (Row 2) Footing effective area, $A' = B' \times L' = 7 \text{ ft} = 49 \text{ ft}^2$ (Row 3) Embedment depth of the footing, $D_f = 2$ ft (Row 4) Unit wt. between base of the footing and ground surface, $\gamma = 0.120$ kcf (Row 5)
Step 2	For undrained analysis use Eq. 5–22 for q_{ult} and correction factors in Table 5–4.
Step 3	For undrained analysis with $\phi = 0$, N_γ is zero. Cohesion c is taken as the undrained strength s_u. Eq. 5–22 simplifies to $q_{ult} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + s_q \cdot d_q \cdot i_q \cdot b_q \cdot g_q \cdot q_0 \cdot N_q$. Since $N_q = 1$ for $\phi = 0$, and all undrained correction factors are 1 (Table 5–4), Eq. 5–29 further simplifies to: $q_{ult} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + q_0$ The total stress at the base of the footing is: $q_o = \gamma \cdot D_f = 0.120 \text{ kcf} \cdot 2 \text{ ft} = 0.24 \text{ ksf}$ (Row 6) The shear zone is typically taken as 1/2 the footing width, or 3.5 ft. The bottom of the shear zone is estimated as $D_f + 0.5B = 5.5$ ft. If the soil in Layer 2 (10 ft to 25 ft) is weaker than soil in Layer 1 (0 ft to 10 ft), a deeper shear zone will need to be checked. The Layer 2 soil is stronger, so strength of Layer 1 soil is used for analysis. $c = s_u = 500 \text{ psf} = 0.5 \text{ ksf}$ (Row 7) The cohesion bearing capacity factor $N_c = 2 + \pi = 5.14$ (Row 8) The undrained shape factor for both the USACE and FHWA methods is (Table 5–4): $s_c = 1 + 0.2\left({B'}/{L'}\right) = 1 + 0.2\left({7 \text{ ft}}/{7 \text{ ft}}\right) = 1.2$ (Row 9) The undrained depth factor for the USACE methods is (Table 5–4): $d_c = 1 + 0.2\left({D_f}/{B'}\right) \leq 1.5 = 1 + 0.2\left({2 \text{ ft}}/{7 \text{ ft}}\right) = 1.06$ (Row 10) The FHWA method does not explicitly include a depth for undrained analysis, so $d_c = 1$. For this example, there is no load inclination, no base inclination, and no ground inclination. $i_c = 1$ (Row 11) $b_c = 1$ (Row 12) $g_c = 1$ (Row 13)
Step 4a	For the USACE method, bearing capacity pressure is calculated for Row 14. $q_{ult} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + q_0$. $q_{ult} = 1.2 \cdot 1.06 \cdot 1 \cdot 1 \cdot 1 \cdot 0.5 \text{ ksf} \cdot 5.14 + 0.24 \text{ ksf}$ $q_{ult} = 3.27 \text{ ksf} + 0.24 \text{ ksf} = 3.51 \text{ ksf}$

Step 4b	For the FHWA method, bearing capacity pressure is calculated for Row 14. $q_{ult} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + q_0$. $q_{ult} = 1.2 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 0.5 \text{ ksf} \cdot 5.14 + 0.24 \text{ ksf}$ $q_{ult} = 3.08 \text{ ksf} + 0.24 \text{ ksf} = 3.32 \text{ ksf}$
Step 5	The load on the soil is a combination of the column load and foundation weight and is expressed as a stress based on the effective area (Eq. 5–13). $q_{eq} = \frac{Q + W_f}{A'}$ Column load, $Q = 40 \text{ kips}$ (Row 15) Unit weight of concrete, $\gamma_{conc} = 0.150 \text{ kcf}$ Weight of foundation, $W_f = A \cdot D_f \cdot \gamma_{conc} = 49 \text{ ft}^2 \cdot 2 \text{ ft} \cdot 0.15 \text{ kcf} = 14.7 \text{ kips}$ (Row 16) Gross equivalent bearing pressure, (Eq. 5–13) (Row 17). $q_{eq} = \frac{Q + W_f}{A'} = \frac{40 \text{ kips} + 14.7 \text{ kips}}{49 \text{ ft}^2} = 1.12 \text{ ksf}$
Step 6a	The minimum factor of safety for ultimate bearing capacity is 3 (Chapter 2). For the USACE method, the factor of safety was calculated as (Eq. 5–23). $FS = \frac{q_{ult}}{q_{eq}} = \frac{3.51 \text{ ksf}}{1.12 \text{ ksf}} = 3.1 \geq 3 \text{ OK ü}$
Step 6b	For the FHWA method, the factor of safety was calculated as (Eq. 5–23). $FS = \frac{q_{ult}}{q_{eq}} = \frac{3.32 \text{ ksf}}{1.12 \text{ ksf}} = 3.0 \geq 3 \text{ OK ü}$

c. Discussion for Example B–1. Undrained analyses are performed in this example. For this case, there was a weaker layer above a stronger layer, so the bearing capacity in the weak layer controls resistance. The factors of safety using both the USACE and FHWA methods were close and met the minimum requirements from Chapter 2. The footing meets bearing capacity ultimate limit state for the undrained condition.

Table B–1b
Summary of calculations for Example B–1

	Undrained		
Row	Symbol	USACE	FHWA
1	B (ft)	7	7
2	L (ft)	7	7
3	A (ft^2)	49	49
4	D_f (ft)	2	2
5	γ (kcf)	0.120	0.120
6	q_o (ksf)	0.24	0.24
7	$c = s_u$ (ksf)	0.5	0.5
8	N_c	5.14	5.14
9	s_c	1.2	1.2
10	d_c	1.06	1
11	i_c	1	1
12	b_c	1	1
13	g_c	1	1
14	q_{ult} (ksf)	3.51	3.32
15	Q (kips)	40	40
16	W_f (kips)	14.7	14.7
17	q_{eq} (ksf)	1.12	1.12
18	FS	3.1	3
19	FS_{min}	3	3
20	Check	OK	OK

B–2. Example B–2. Undrained vertical bearing capacity for continuous footing on clay

a. Determine: Calculate the factor of safety for ultimate undrained bearing capacity for the 2-ft-wide continuous footing and compare to minimum requirements.

b. Given: In addition to constructing the 7.0-ft by 7.0-ft isolated footing in Example B–1, a 2-ft-wide continuous footing with a loading of 1.2 kips/foot is to be constructed adjacent to the isolated footing as illustrated in Figure B–2. Both footings are to bear 2 ft below existing grade.

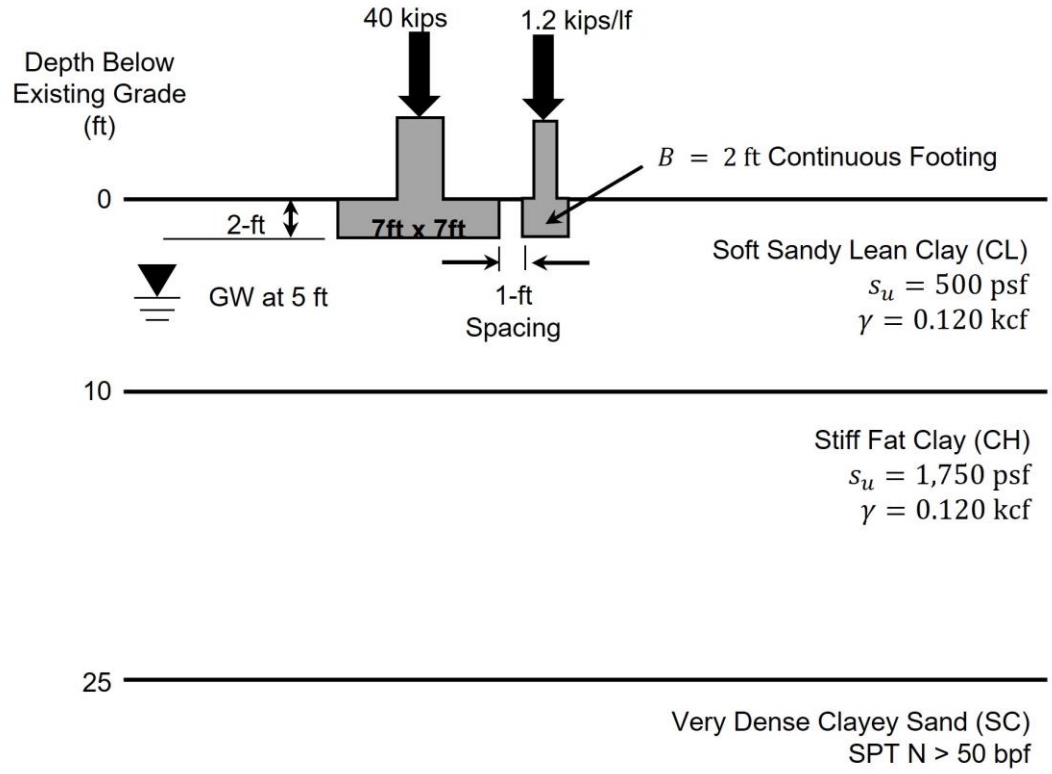


Figure B–2. Foundation and soil conditions for Example B–2

Table B–2a
Solution for undrained bearing capacity analysis (continuous footing)

Step 1	Collect geometric information for the problem and input to Table B–2b. There is no eccentricity ($e = 0$), so footing geometry is equal to effective footing geometry (see paragraph 5–4a). Footing width, $B = B' = 2$ ft (Row 1) For continuous footings, loads are considered per-unit length, and length is not specifically included in analysis; length L= $L = L' =$ not applicable (Row 2) Footing effective area, $A' = B' \cdot L' = 2$ ft $\cdot 1\ ft = 2\ ft^2/ft$ (Row 3) Embedment depth of the footing, $D_f = 2\ ft$ (Row 4) Unit wt. between base of the footing and ground surface, $\gamma = 0.120$ kcf (Row 5)
Step 2	Use Eq. 5–28 for undrained q_{ult} and correction factors in Table 5–4.

Step 3	For undrained analysis with $\phi = 0$, N_γ is zero. The value of cohesion c is taken as the undrained strength s_u. Eq. 5–22 simplifies to $q_{ult} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + s_q \cdot d_q \cdot i_q \cdot b_q \cdot g_q \cdot q_0 \cdot N_q$. Since $N_q = 1$ for $\phi = 0$, and all undrained correction factors are 1 (Table 5–4), Eq. 5–29 further simplifies to $q_{ult} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + q_0$. The total stress at the base of the footing is: $q_o = \gamma \cdot D_f = 0.120\ kcf \cdot 2\ ft = 0.24\ ksf$ (Row 6) The shear zone is typically taken as 1 footing width, or 2 ft. The bottom of the shear zone estimated as $D_f + B = 4\ ft$. The estimated shear zone is within Layer 1 (0 ft to 10 ft), so strength of Layer 1 soil is used for analysis. $c = s_u = 500\ psf = 0.5\ ksf$ (Row 7) The cohesion bearing capacity factor, $N_c = 2 + \pi = 5.14$ (Row 8) The undrained shape factor for a continuous footing for both the USACE and FHWA methods is 1 (Row 9). The undrained depth factor for the USACE method is (Table 5–4): $d_c = 1 + 0.2\left({}^{D_f}/_{B'}\right) \leq 1.5 = 1 + 0.2\left({}^{2\text{ ft}}/_{2\text{ ft}}\right) = 1.2$ (Row 10) The FHWA method does not explicitly include a depth for undrained analysis, so $d_c = 1$. For this example, there is no load inclination, no base inclination, and no ground inclination. $i_c = 1$ (Row 11) $b_c = 1$ (Row 12) $g_c = 1$ (Row 13)
Step 4a	For the USACE method, bearing capacity pressure is calculated for Row 14. $q_{ult} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + q_0$. $q_{ult} = 1 \cdot 1.2 \cdot 1 \cdot 1 \cdot 1 \cdot 0.5\ ksf \cdot 5.14 + 0.24\ ksf$ $q_{ult} = 3.08\ ksf + 0.24\ ksf = 3.32\ ksf$
Step 4b	For the FHWA method, bearing capacity pressure is calculated for Row 14. $q_{ult} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + q_0$. $q_{ult} = 1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 0.5\ ksf \cdot 5.14 + 0.24\ ksf$ $q_{ult} = 2.57\ ksf + 0.24\ ksf = 2.81\ ksf$

Step 5	The load on the soil is a combination of the column load and foundation weight and is expressed as a stress based on the effective area (Eq. 5–13). $q_{eq} = \frac{Q + W_f}{A'}$ Column load, $Q = 1.2$ kips/ft (Row 15) Unit weight of concrete, $\gamma_{conc} = 0.150\ kcf$ Weight of foundation, $W_f = A \cdot D_f \cdot \gamma_{conc} = 2\ ft^2/ft \cdot 2\ ft \cdot 0.15\ kcf = 0.6$ kips/ft (Row 16) Gross equivalent bearing pressure, (Eq. 5–13). $q_{eq} = \frac{Q+W_f}{A'} = \frac{1.2 \text{ kips/ft} + 0.6 \text{ kips/ft}}{2 \text{ ft}^2/\text{ft}} = 0.9$ ksf (Row 17)
Step 6a	The minimum factor of safety for ultimate bearing capacity is 3 (Chapter 2). For the USACE method, the factor of safety was calculated as (Eq. 5–23). $FS = \frac{q_{ult}}{q_{eq}} = \frac{3.32\ ksf}{0.9\ ksf} = 3.7\ \geq 3$ OK ü
Step 6b	For the FHWA method, the factor of safety was calculated as (Eq. 5–23). $FS = \frac{q_{ult}}{q_{eq}} = \frac{2.81\ ksf}{0.9\ ksf} = 3.1\ \geq 3$ OK ü

c. Discussion for Example B–2. An undrained analysis was performed on the footing. The factors of safety using both the USACE and FHWA methods were close and met the minimum requirements from Chapter 2. The footing meets undrained bearing capacity ultimate limit state requirements.

Table B–2b
Summary of calculations for Example B–2

		Undrained	
Row	Symbol	USACE	FHWA
1	$B\ (ft)$	2	2
2	$L\ (ft)$	NA	NA
3	$A\ (ft^2)$	2	2
4	$D_f\ (ft)$	2	2
5	$\gamma\ (kcf)$	0.12	0.12
6	$q_o\ (ksf)$	0.24	0.24
7	$c = s_u\ (ksf)$	0.5	0.5
8	N_c	5.14	5.14
9	s_c	1	1
10	d_c	1.2	1
11	i_c	1	1
12	b_c	1	1
13	g_c	1	1
14	$q_{ult}\ (ksf)$	3.32	2.81
15	Q (kips)	1.2	1.2
16	W_f (kips)	0.6	0.6
17	$q_{eq}\ (ksf)$	0.9	0.9
18	FS	3.7	3.1
19	FS_{min}	3	3
20	Check	OK	OK

B–3. Example B–3. Drained vertical bearing capacity for isolated footing on sand with water table below the shear zone

a. Determine: Calculate the factor of safety for drained ultimate bearing capacity for the 2-ft-wide continuous footing and compare to minimum requirements. Evaluate potential for punch through.

b. Given: Prior to constructing the footings, site grades are raised by placing 8 ft of areal fill throughout the entire site, including beneath the building (see Figure B–3). The fill material placed beneath the building and 10 ft beyond the building limits consists of approved structural fill material. A 7.0-ft by 7.0-ft isolated footing supporting a 90-kip column load will bear 2 ft below the new site grade.

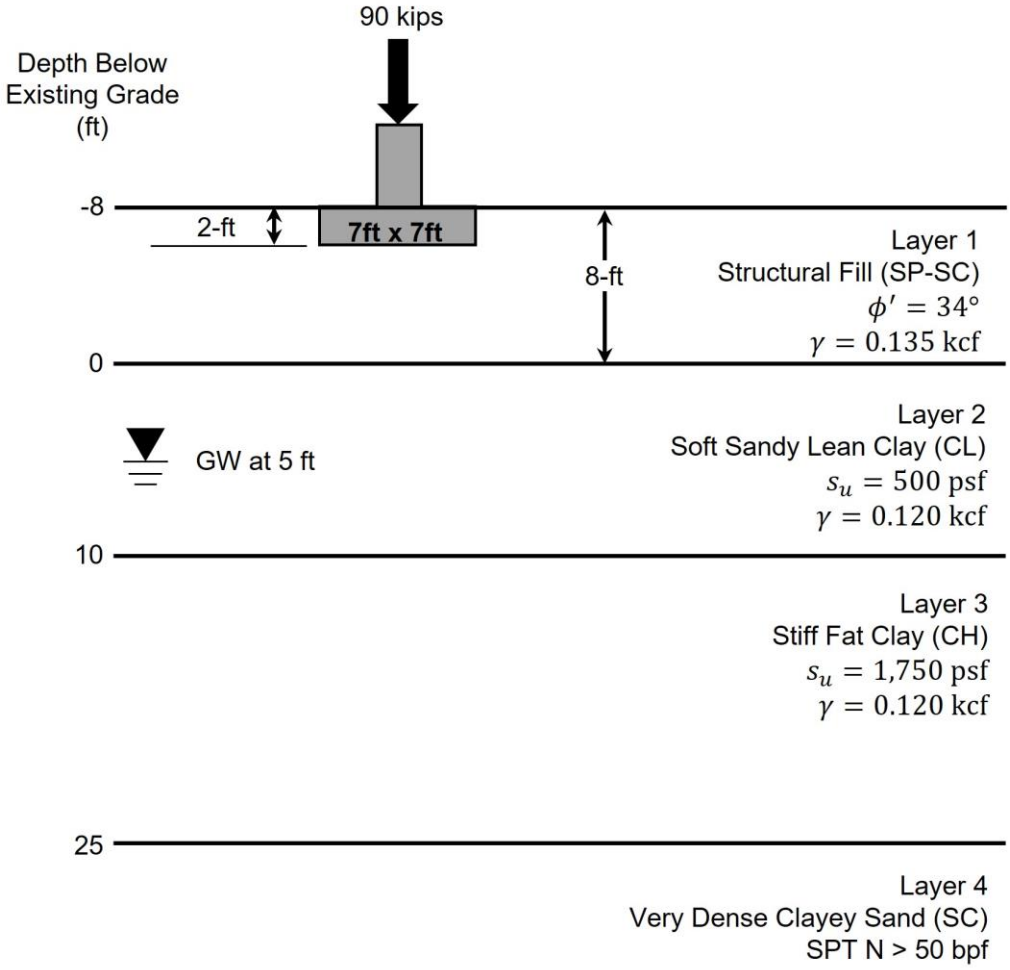


Figure B–3. Foundation and soil conditions for Example B–3

Table B–3a
Solution for drained bearing capacity analysis, shear within surface layer

Step 1	Collect geometric information for the problem and input to Table B–3c. There is no eccentricity ($e = 0$), so footing geometry is equal to effective footing geometry (see paragraph 5–4a). Footing width, $B = B' = 7\ ft$ (Row 1) Footing length, $L = L' = 7\ ft$ (Row 2) Footing effective area, $A' = B' \cdot L' = 7\ ft \cdot 7\ ft = 49\ ft^2$ (Row 3) Embedment depth of the footing, $D_f = 2\ ft$ (Row 4) Depth of water table, $D_w = 13\ ft$ (Row 5) Unit wt. between base of the footing and ground surface, $\gamma = 0.135\ kcf$ (Row 6)

Step 2	Identify that the problem involves free-draining soils (sands, silty sands, and non-plastic silts) in the layer immediately below the footing, and that the bearing capacity limit state must be assessed for drained conditions (Table 5–1). Use Eq. 5–22 for q_{ult} and correction factors in Table 5–2 and Table 5–3.
Step 3	For drained analysis with cohesion $c = 0$, Eq. 5–22 simplifies to: $q_{ult} = s_q \cdot d_q \cdot i_q \cdot b_q \cdot g_q \cdot q_0 \cdot N_q + s_\gamma \cdot d_\gamma \cdot i_\gamma \cdot b_\gamma \cdot g_\gamma \cdot \frac{1}{2} \cdot B' \cdot \gamma' \cdot N_\gamma$ The ratio of the effective unit weight in the shear zone to the total unit weight differs for the USACE and FHWA methods. For USACE, (Eq. 5–19) (Row 7): $\frac{\gamma'}{\gamma} \approx \left[0.45 + 0.55\left(\frac{\max(D_W - D_f, 0)}{B}\right)\right] \leq 1$ $\frac{\gamma'}{\gamma} \approx \left[0.45 + 0.55\left(\frac{\max(13\ ft - 2\ ft, 0)}{7\ ft}\right)\right] \leq 1 = 1$ For FHWA, (Eq. 5–20) (Row 7): $\frac{\gamma'}{\gamma} \approx C_{W_\gamma} = \left[0.5 + 0.5\left(\frac{D_W}{1.5B + D_f}\right)\right] \leq 1$ $\frac{\gamma'}{\gamma} \approx C_{W_\gamma} = \left[0.5 + 0.5\left(\frac{13\ ft}{1.5(7\ ft) + 2\ ft}\right)\right] \leq 1 = 1$ The effective stress at the base of the footing is: $q_o = \gamma_m \cdot D_f - u_o = 0.135\ kcf \cdot 2\ ft - 0\ ksf = 0.27\ ksf$ (Row 8) The shear zone is typically taken as 1 footing width, or 7 ft. The bottom of the shear zone estimated as $D_f + B = 9\ ft$. The estimated shear zone is within Layer 1 (0 ft to 10 ft), so strength of Layer 1 soil is used for analysis. Punch-through failure into the underlying weak layer will be checked in a separate analysis. $\phi' = 34°$ (Row 9) The flow number, or principal stress ratio is (Table 5–2) (Row 10): $N = \frac{1 + sin\phi'}{1 - sin\phi'} = \frac{1 + sin(34^o)}{1 - sin(34^o)} = \frac{1.56}{0.44} = 3.54$

Step 4	The dimensionless bearing capacity for surcharge is (Table 5–2) (Row 11): $N_q = N \cdot e^{\pi \cdot tan\phi'} = 3.54e^{\pi \cdot tan(34^o)} = 3.54e^{2.12} = 3.54 \cdot 8.32 = 29.46$ The drained shape factor for surcharge differs for the USACE and FHWA methods. For USACE, (Table 5–2) (Row 12): $s_q = 1 + 0.1N\left(\frac{B'}{L'}\right) = 1 + 0.1 \cdot 3.54\left(\frac{7\ ft}{7\ ft}\right) = 1.35$ For FHWA, (Table 5–2) (Row 12): $s_q = 1 + \left(\frac{B'}{L'}\right) tan\phi = 1 + \left(\frac{7\ ft}{7\ ft}\right) tan(34^o) = 1 + (1 \cdot 0.67) = 1.67$ The drained depth factor for surcharge differs for the USACE and FHWA methods. For USACE, (Table 5–2) (Row 13): $d_q = 1 + 0.1\sqrt{N}\left(\frac{D_f}{B'}\right) = 1 + 0.1\sqrt{3.54}\left(\frac{2\ ft}{7\ ft}\right) = 1 + 0.1 \cdot 1.88 \cdot 0.29 = 1.05$ For FHWA, (Table 5–2) (Row 13): $d_q = 1 + 2tan\phi' \cdot (1 - sin\phi')^2 \cdot atan\left(\frac{D_f}{B'}\right)$ $d_q = 1 + 2tan(34^o) \cdot \left(1 - sin(34^o)\right)^2 \cdot atan\left(\frac{2\ ft}{7\ ft}\right)$ $d_q = 1 + 2 \cdot 0.67 \cdot (1 - 0.56)^2 \cdot 0.28 = 1.07$ For this example, there is no load inclination, no base inclination, and no ground inclination. $i_c = 1$ (Row 14) $b_c = 1$ (Row 15) $g_c = 1$ (Row 16)

Step 5	The dimensionless bearing capacity for soil effective weight differs for the USACE and FHWA methods. For USACE, (Table 5–2) (Row 17): $N_\gamma = (N_q - 1) \cdot tan(1.4\phi') = (29.46 - 1) \cdot tan(1.4 \cdot 34^o)$ $N_\gamma = 28.46 \cdot tan(47.6^o) = 28.46 \cdot 1.095 = 31.17$ For FHWA, (Table 5–2) (Row 17): $N_\gamma = 2(N_q + 1) \cdot tan(\phi') = 2 \cdot (29.46 + 1) \cdot tan(34^o)$ $N_\gamma = 60.92 \cdot 0.675 = 41.09$ The drained shape factor for soil effective weight differs for the USACE and FHWA methods. For USACE, the shape factors for surcharge and soil effective weight are the same (Table 5–2) (Row 18): $s_\gamma = 1 + 0.1N\left(\frac{B'}{L'}\right) = 1 + 0.1 \cdot 3.54\left(\frac{7\ ft}{7\ ft}\right) = 1.35$ For FHWA, (Table 5–2) (Row 18): $s_\gamma = 1 - 0.4\left(\frac{B'}{L'}\right) = 1 - 0.4\left(\frac{7\ ft}{7\ ft}\right) = 1 - 0.4 = 0.6$ The drained depth factor for surcharge soil effective weight differs for the USACE and FHWA methods. For USACE, the shape factors for surcharge and soil effective weight are the same (Table 5–2) (Row 19): $d_\gamma = 1 + 0.1\sqrt{N}\left(\frac{D_f}{B'}\right) = 1 + 0.1\sqrt{3.54}\left(\frac{2\ ft}{7\ ft}\right) = 1 + 0.1 \cdot 1.88 \cdot 0.29 = 1.05$ For FHWA, (Table 5–2) (Row 19): $d_\gamma = 1$ For this example, there is no load inclination, no base inclination, and no ground inclination. $i_c = 1$ (Row 20) $b_c = 1$ (Row 21) $g_c = 1$ (Row 22)
Step 6a	For the USACE method, bearing capacity pressure is calculated for Row 23 as: $q_{ult} = s_q \cdot d_q \cdot i_q \cdot b_q \cdot g_q \cdot q_0 \cdot N_q + s_\gamma \cdot d_\gamma \cdot i_\gamma \cdot b_\gamma \cdot g_\gamma \cdot \frac{1}{2} \cdot B' \cdot \gamma' \cdot N_\gamma$ $q_{ult} = 1.35 \cdot 1.05 \cdot 1 \cdot 1 \cdot 1 \cdot 0.27\ ksf \cdot 29.46 + 1.35 \cdot 1.05 \cdot 1 \cdot 1 \cdot 1 \cdot \frac{1}{2} \cdot 7\ ft \cdot 0.135\ kcf \cdot 31.17$ $q_{ult} = 11.27\ ksf + 20.88\ ksf = 32.15\ ksf$
Step 6b	For the FHWA method, bearing capacity pressure is calculated for Row 23 as: $q_{ult} = s_q \cdot d_q \cdot i_q \cdot b_q \cdot g_q \cdot q_0 \cdot N_q + s_\gamma \cdot d_\gamma \cdot i_\gamma \cdot b_\gamma \cdot g_\gamma \cdot \frac{1}{2} \cdot B' \cdot \gamma' \cdot N_\gamma$ $q_{ult} = 1.67 \cdot 1.07 \cdot 1 \cdot 1 \cdot 1 \cdot 0.27\ ksf \cdot 29.46 + 0.6 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot \frac{1}{2} \cdot 7\ ft \cdot 0.135\ kcf \cdot 41.09$ $q_{ult} = 14.21\ ksf + 11.65\ ksf = 25.86\ ksf$

Step 7	The load on the soil is a combination of the column load and foundation weight and is expressed as a stress based on the effective area (Eq. 5–13). $q_{eq} = \frac{Q + W_f}{A'}$ Column load, Q = 90 kips (Row 24) Unit weight of concrete, $\gamma_{conc} = 0.150\ kcf$ Weight of foundation, $W_f = A \cdot D_f \cdot \gamma_{conc} = 49\ ft^2 \cdot 2\ ft \cdot 0.15\ kcf = 14.7$ kips (Row 25) Gross equivalent bearing pressure, (Eq. 5–13) (Row 26). $q_{eq} = \frac{Q + W_f}{A'} = \frac{90 \text{ kips} + 14.7 \text{ kips}}{49 \text{ ft}^2} = 2.14\ ksf$
Step 8a	The minimum factor of safety for ultimate bearing capacity is 3 (Chapter 2). For the USACE method, the factor of safety was calculated as (Eq. 5–23) (Row 27). $FS = \frac{q_{ult}}{q_{eq}} = \frac{32.15\ ksf}{2.14\ ksf} = 15\ \geq 3$ OK ü
Step 8b	For the FHWA method, the factor of safety was calculated as (Eq. 5–23) (Row 27). $FS = \frac{q_{ult}}{q_{eq}} = \frac{25.86\ ksf}{2.14\ ksf} = 12.1\ \geq 3$ OK ü

Table B–3b
Solution for undrained punch-through analysis

Step 1	Collect geometric information for the problem. There is no eccentricity ($e = 0$), so footing geometry is equal to effective footing geometry (see paragraph 5–4a). The geometry information is the same as that for the drained analysis of Layer 1. Footing width, $B = B' = 7\ ft$ (Row 1) Footing length, $L = L' = 7\ ft$ (Row 2) Footing effective area, $A' = B' \cdot L' = 7\ ft \cdot 7\ ft = 49\ ft^2$ (Row 3) Embedment depth of the footing, $D_f = 2\ ft$ (Row 4) Depth of water table, $D_w = 13\ ft$ (Row 5) Unit wt. between base of the footing and ground surface, $\gamma = 0.135\ kcf$ (Row 6)
Step 2	Identify that the problem of punch-through into the underlying soil involves low permeability soils (clayey soils and plastic silts), and that the bearing capacity limit state must be assessed for undrained conditions (Table 5–1). Use Eq. 5–22 for q_{ult} and correction factors in Table 5–4.

Step 3	For undrained analysis with $\phi = 0$, N_γ is zero. The value of cohesion, c, is taken as the undrained strength s_u. Eq. 5–22 simplifies to: $q_{ult} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + s_q \cdot d_q \cdot i_q \cdot b_q \cdot g_q \cdot q_0 \cdot N_q$ Since $N_q = 1$ for $\phi = 0$, and all undrained correction factors are 1 (Table 5–4), Eq. 5–29 further simplifies to: $q_{ult} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + q_0$. The total stress at the base of the footing is: $q_o = \gamma \cdot D_f = 0.135\ kcf \cdot 2\ ft = 0.27\ ksf$ (Row 8) The shear zone is typically taken as 1 footing width, or 7 ft, but may extend to approximately 2 footings widths when softer soils are at depth. The weaker soils in Layer 2 soil are used for the punch-through analysis. $c = s_u = 500\ psf = 0.5\ ksf$ (Row 9) The cohesion bearing capacity factor, $N_c = 2 + \pi = 5.14$ (Row 10) The undrained shape factor for both the USACE and FHWA methods is (Table 5–4): $s_c = 1 + 0.2\left(B'/L'\right) = 1 + 0.2\left(7\text{ ft}/7\text{ ft}\right) = 1.2$ (Row 11) The undrained depth factor for the USACE methods is (Table 5–4): $d_c = 1 + 0.2\left(D_f/B'\right) \leq 1.5 = 1 + 0.2\left(2\ ft/7\ ft\right) = 1.06$ (Row 12) The FHWA method does not explicitly include a depth for undrained analysis, so $d_c = 1$. For this example, there is no load inclination, no base inclination, and no ground inclination. $i_c = 1$ (Row 13) $b_c = 1$ (Row 14) $g_c = 1$ (Row 15)
Step 4a	For the USACE method, bearing capacity pressure for Layer 2 is calculated for Row 14. $q_{ult,2} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + q_0$. $q_{ult,2} = 1.2 \cdot 1.06 \cdot 1 \cdot 1 \cdot 1 \cdot 0.5\ ksf \cdot 5.14 + 0.27\ ksf$ $q_{ult,2} = 3.27\ ksf + 0.27\ ksf = 3.54\ ksf$
Step 4b	For the FHWA method, bearing capacity pressure for Layer 2 is calculated for Row 14. $q_{ult,2} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + q_0$. $q_{ult,2} = 1.2 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 0.5\ ksf \cdot 5.14 + 0.27\ ksf$ $q_{ult,2} = 3.08\ ksf + 0.27\ ksf = 3.35\ ksf$

Step 5a	The bearing capacity at the footing level for a punch-through failure is increased due to load spreading. The ratio of q_{ult} to the calculated q_{ult} for Layer 2 ($q_{ult,2}$) is determined from (Eq. 5–17). $\frac{q_{ult}}{q_{ult,2}} = \frac{(B' + H_s) \cdot (L' + H_s)}{B \cdot L} = \frac{(7\ ft + 6\ ft) \cdot (7\ ft + 6\ ft)}{7\ ft \cdot 7\ ft} = 3.45$ For the USACE method: $q_{ult} = \frac{q_{ult}}{q_{ult,2}} \cdot q_{ult,2} = 3.45 \cdot 3.54\ ksf = 12.21\ ksf$
Step 5b	For the FHWA method: $q_{ult} = \frac{q_{ult}}{q_{ult,2}} \cdot q_{ult,2} = 3.45 \cdot 3.35\ ksf = 11.56\ ksf$
Step 6	The load on the soil is a combination of the column load and foundation weight, therefore, is expressed as a stress based on the effective area (Eq. 5–13). $q_{eq} = \frac{Q + W_f}{A'}$ Column load, $Q = 90\ kips$ (Row 24) Unit weight of concrete, $\gamma_{conc} = 0.150\ kcf$ Weight of foundation, $W_f = A \cdot D_f \cdot \gamma_{conc} = 49\ ft^2 \cdot 2\ ft \cdot 0.150\ kcf = 14.7\ kips$ (Row 25) Gross equivalent bearing pressure, (Eq. 5–13) (Row 26). $q_{eq} = \frac{Q + W_f}{A'} = \frac{90\ kips + 14.7\ kips}{49\ ft^2} = 2.14\ ksf$
Step 7a	The minimum factor of safety for ultimate bearing capacity is 3 (Chapter 2). For the USACE method, the factor of safety was calculated as (Eq. 5–23) (Row 27). $FS = \frac{q_{ult}}{q_{eq}} = \frac{12.21\ ksf}{2.14\ ksf} = 5.7\ \geq 3$ OK ü
Step 7b	For the FHWA method, the factor of safety was calculated as (Eq. 5–23) (Row 27). $S = \frac{q_{ult}}{q_{eq}} = \frac{11.56\ ksf}{2.14\ ksf} = 5.4\ \geq 3$ OK ü

c. Discussion for Example B–3. For the layered sand over clay foundation in this example, the resistance to a punch-through failure into the clay foundation was less than that for a failure in the upper sand layer. The punch-through failure will control the bearing capacity design factor of safety. The factors of safety using both the USACE and FHWA methods met the minimum requirements from Table 2–1. The footing meets bearing capacity ultimate limit state requirements.

Table B–3c
Summary of calculations for Example B–3

	Layer 1: Drained bearing capacity				Layer 2: Undrained punch through		
Row	Symbol	USACE	FHWA		Symbol	USACE	FHWA
1	$B\ (ft)$	7	7		$B\ (ft)$	7	7
2	$L\ (ft)$	7	7		$L\ (ft)$	7	7
3	$A\ (ft^2)$	49	49		$A\ (ft^2)$	49	49
4	$D_f\ (ft)$	2	2		$D_f\ (ft)$	2	2
5	$D_w\ (ft)$	13	13		–	–	–
6	$\gamma\ (kcf)$	0.135	0.135		$\gamma\ (kcf)$	0.135	0.135
7	γ'/γ	1	1		–	–	–
8	$q_o\ (ksf)$	0.27	0.27		$q_o\ (ksf)$	0.27	0.27
9	ϕ' (deg)	34	34		$c = s_u\ (ksf)$	0.5	0.5
10	N	3.54	3.54		N_c	5.14	5.14
11	N_q	29.46	29.46		s_c	1.2	1.2
12	s_q	1.35	1.67		d_c	1.06	1
13	d_q	1.05	1.07		i_c	1	1
14	i_q	1	1		b_c	1	1
15	b_q	1	1		g_c	1	1
16	g_q	1	1		$q_{ult,2}\ (ksf)$	3.54	3.35
17	N_γ	31.17	41.09		H_s	6	6
18	s_γ	1.35	0.6		$q_{ult}/q_{ult,2}$	3.45	3.45
19	d_γ	1.05	1		–	–	–
20	i_γ	1	1		–	–	–
21	b_γ	1	1		–	–	–
22	g_γ	1	1		–	–	–
23	$q_{ult}\ (ksf)$	32.15	25.86		$q_{ult}\ (ksf)$	12.21	11.56
24	$Q\ (kips)$	90	90		$Q\ (kips)$	90	90
25	$W_f\ (kips)$	14.7	14.7		$W_f\ (kips)$	14.7	14.7
26	$q_{eq}\ (ksf)$	2.14	2.14		$q_{eq}\ (ksf)$	2.14	2.14
27	FS	15	12.1		FS	5.7	5.4
28	FS_{min}	3	3		FS_{min}	3	3
29	Check	OK	OK		Check	OK	OK

B–4. Example B–4. Undrained vertical bearing capacity for isolated basement footing on clay, influence of construction dewatering

a. Determine: Calculate the factor of safety for undrained ultimate bearing capacity for the isolated footing and compare to minimum requirements to see if footing is acceptable.

b. *Given*: The groundwater table is lowered from a depth of 6 to 12 ft below grade to support construction of a building. The water table will rise back to a depth of 6 ft for normal operation of the building (see Figure B–4). A 7.0-ft by 7.0-ft isolated footing supporting an 80-kip column load will bear 2 ft below the bottom of the excavation after construction and after the water has returned to a depth of 6 ft below the ground surface.

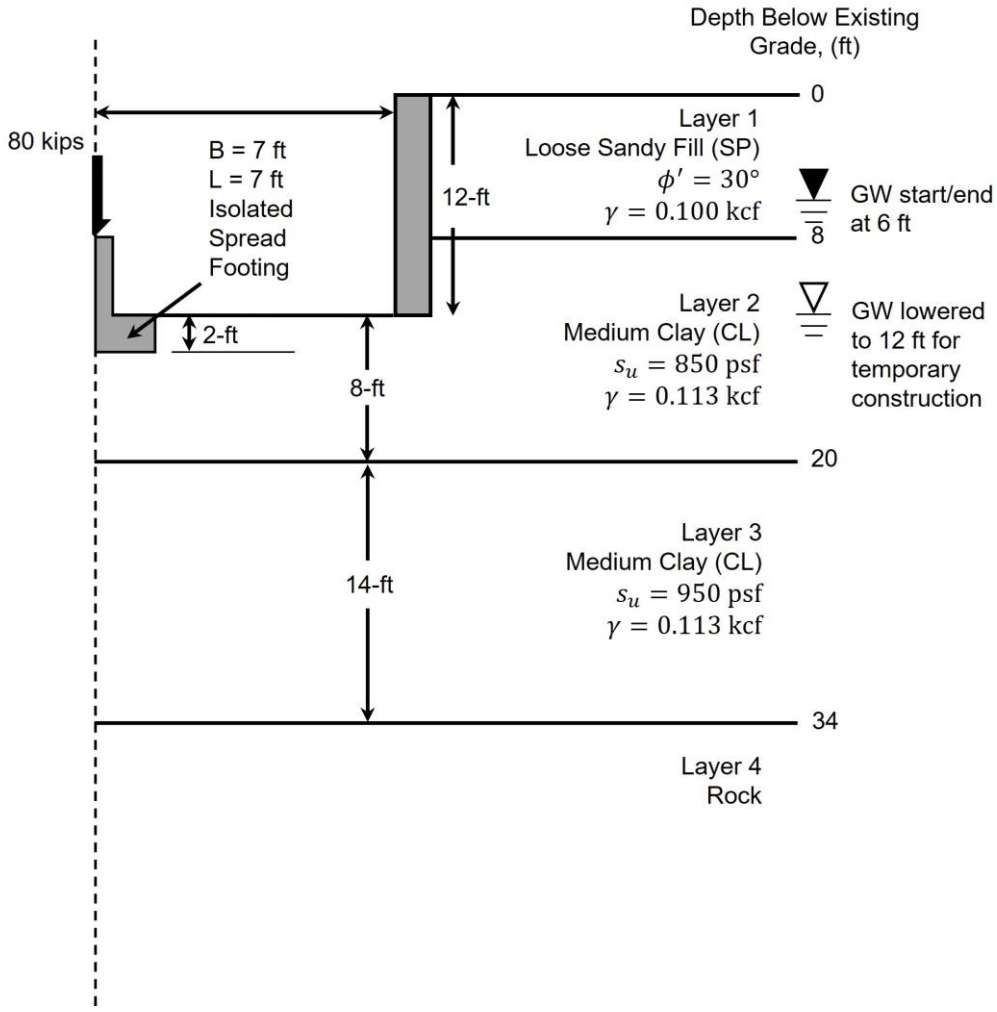


Figure B–4. Foundation and soil conditions for Example B–4

Table B–4a
Solution for the end-of-construction phase (dewatered case)

Step 1	Collect geometric information for the problem and input to Table B–4c. There is no eccentricity ($e = 0$), so footing geometry is equal to effective footing geometry (see paragraph 5–4a). Footing width, $B = B' = 7\ ft$ (Row 1) Footing length, $L = L' = 7\ ft$ (Row 2) Footing effective area, $A' = B' \cdot L' = 7\ ft \cdot 7\ ft = 49\ ft^2$ (Row 3) Embedment depth of the footing, $D_f = 2\ ft$ (Row 4) Unit wt. between base of the footing and ground surface, $\gamma = 0.133\ kcf$ (Row 5)
Step 2	Use Eq. 5–22 for undrained q_{ult} and correction factors in Table 5–4.
Step 3	For undrained analysis with $\phi = 0$, N_γ is zero. The value of cohesion, c, is taken as the undrained strength s_u. Eq. 5–22 simplifies to $q_{ult} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + s_q \cdot d_q \cdot i_q \cdot b_q \cdot g_q \cdot q_0 \cdot N_q$. Since $N_q = 1$ for $\phi = 0$, and all undrained correction factors are 1 (Table 5–4), Eq. 5–29 further simplifies to: $q_{ult} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + q_0$. The total stress at the base of the footing is: $q_o = \gamma \cdot D_f = 0.133\ kcf \cdot 2\ ft = 0.226\ ksf$ (Row 6) The shear zone is typically taken as 1 footing width, or 7 ft. The bottom of the shear zone estimated as $D_f = B = 9\ ft$. If the soil in Layer 2 (10 ft to 25 ft) is weaker than soil in Layer 1 (0 ft to 10 ft), a deeper shear zone will need to be checked. The Layer 2 soil is stronger, so strength of Layer 1 soil is used for analysis. $c = s_u = 850\ psf = 0.85\ ksf$ (Row 7) The cohesion bearing capacity factor $N_c = 2 + \pi = 5.14$ (Row 8) The undrained shape factor for both the USACE and FHWA methods is (Table 5–4): $s_c = 1 + 0.2\left({B'}/{L'}\right) = 1 + 0.2\left({7\text{ ft}}/{7\text{ ft}}\right) = 1.2$ (Row 9) The undrained depth factor for the USACE methods is (Table 5–4): $d_c = 1 + 0.2\left({D_f}/{B'}\right) \leq 1.5 = 1 + 0.2\left({2\text{ ft}}/{7\text{ ft}}\right) = 1.06$ (Row 10) The FHWA method does not explicitly include a depth for undrained analysis, so $d_c = 1$. For this example, there is no load inclination and no base inclination. $i_c = 1$ (Row 11) $b_c = 1$ (Row 12) The footing is in a basement, and an upward-sloping ground surface could be considered. However, since the footing is more than 2 widths from the edge of the excavation (see Table 5–4), the overlying soil will not influence the bearing capacity response and g_c can be considered as one. $g_c = 1$ (Row 13)

Step 4a	For the USACE method, bearing capacity pressure is calculated for Row 14. $q_{ult} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + q_0.$ $q_{ult} = 1.2 \cdot 1.06 \cdot 1 \cdot 1 \cdot 1 \cdot 0.85\ ksf \cdot 5.14 + 0.226\ ksf$ $q_{ult} = 5.56\ ksf + 0.226\ ksf = 5.78\ ksf$
Step 4b	For the FHWA method, bearing capacity pressure is calculated for Row 14. $q_{ult} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + q_0.$ $q_{ult} = 1.2 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 0.85\ ksf \cdot 5.14 + 0.226\ ksf$ $q_{ult} = 5.24\ ksf + 0.226\ ksf = 5.47\ ksf$
Step 5	The load on the soil is a combination of the column load and foundation weight and is expressed as a stress based on the effective area (Eq. 5–13). $q_{eq} = \frac{Q + W_f}{A'}$ Column load, $Q = 80\ kips$ (Row 15) Unit weight of concrete, $\gamma_{conc} = 0.150\ kcf$ Weight of foundation, $W_f = A \cdot D_f \cdot \gamma_{conc} = 49\ ft^2 \cdot 2\ ft \cdot 0.15\ kcf = 14.7\ kips$ (Row 16) Pore pressure at base of footing with water table lowered to 12 ft during construction: $u_D = 2\ ft \cdot 0.0624\ kcf = 0.125\ ksf$ Gross equivalent bearing pressure (Eq. 5–13) (Row 17). $q_{eq} = \frac{Q + W_f}{A'} - u_D = \frac{80\ kips + 14.7\ kips}{49\ ft^2} - 0.125\ ksf = 1.81\ ksf$
Step 6a	The minimum factor of safety for ultimate bearing capacity is 3 (Chapter 2). For the USACE method at the end of construction, the factor of safety was calculated as (Eq. 5–23). $FS = \frac{q_{ult}}{q_{eq}} = \frac{5.78\ ksf}{1.81\ ksf} = 3.2\ \geq 3$ OK ü
Step 6b	For the FHWA method, the factor of safety was calculated as (Eq. 5–23). $FS = \frac{q_{ult}}{q_{eq}} = \frac{5.47\ ksf}{1.81\ ksf} = 3.0\ \geq 3$ OK ü

Table B–4b
Solution for the long-term water level case

Step 1	Increasing the water table level does not change the soil resistance for an undrained analysis. All values in Rows 1 through 14 are the case as for the end of construction case. For the USACE method, bearing capacity pressure is calculated for Row 14. $q_{ult} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + q_0.$ $q_{ult} = 1.2 \cdot 1.06 \cdot 1 \cdot 1 \cdot 1 \cdot 0.85\ ksf \cdot 5.14 + 0.226\ ksf$ $q_{ult} = 5.56\ ksf + 0.226\ ksf = 5.78\ ksf$ For the FHWA method, bearing capacity pressure is calculated for Row 14. $q_{ult} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + q_0.$ $q_{ult} = 1.2 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 0.85\ ksf \cdot 5.14 + 0.226\ ksf$ $q_{ult} = 5.24\ ksf + 0.226\ ksf = 5.47\ ksf$
Step 2	The load on the soil is the same as for the end-of-construction case, and values in rows 15 and 16 are the same. Pore pressure at base of footing changes. The height of water above the base of the footing increases from 2 ft to 8 ft: $u_D = 8\ ft \cdot 0.0624\ kcf = 0.499\ ksf$ Gross equivalent bearing pressure (Eq. 5–13) (Row 17). $q_{eq} = \frac{Q + W_f}{A'} - u_D = \frac{80\ kips + 14.7\ k\text{–}ips}{49\ ft^2} - 0.499\ ksf = 1.43$ ksf
Step 3a	The minimum factor of safety for ultimate bearing capacity is 3 (Chapter 2). For the USACE method at the end of construction, the factor of safety was calculated as (Eq. 5–23). $FS = \frac{q_{ult}}{q_{eq}} = \frac{5.78\ ksf}{1.43\ ksf} = 4.0 \geq 3$ OK ü
Step 3b	For the FHWA method, the factor of safety was calculated as (Eq. 5–23). $FS = \frac{q_{ult}}{q_{eq}} = \frac{5.47\ ksf}{1.43\ ksf} = 3.8 \geq 3$ OK ü

c. Discussion for Example B–4. The undrained factors of safety using both the USACE and FHWA methods met the minimum requirements from Chapter 2 for both the end-of-construction and long-term loading case. As the water table rose, the buoyancy of the footing led to an increase in the bearing capacity safety factor. The footing meets undrained bearing capacity ultimate limit state requirements.

Table B–4c
Summary of calculations for Example B–4

	Undrained				Undrained, water at 6 ft		
Row	Symbol	USACE	FHWA		Symbol	USACE	FHWA
1	$B\ (ft)$	7	7		$B\ (ft)$	7	7
2	$L\ (ft)$	7	7		$L\ (ft)$	7	7
3	$A\ (ft^2)$	49	49		$A\ (ft^2)$	49	49
4	$D_f\ (ft)$	2	2		$D_f\ (ft)$	2	2
5	$\gamma\ (kcf)$	0.113	0.113		$\gamma\ (kcf)$	0.113	0.113
6	$q_o\ (ksf)$	0.226	0.226		$q_o\ (ksf)$	0.226	0.226
7	$c = s_u\ (ksf)$	0.85	0.85		$c = s_u\ (ksf)$	0.85	0.85
8	N_c	5.14	5.14		N_c	5.14	5.14
9	s_c	1.2	1.2		s_c	1.2	1.2
10	d_c	1.06	1		d_c	1.06	1
11	i_c	1	1		i_c	1	1
12	b_c	1	1		b_c	1	1
13	g_c	1	1		g_c	1	1
14	$q_{ult}\ (ksf)$	5.78	5.47		$q_{ult}\ (ksf)$	5.78	5.47
15	$Q\ (kips)$	80	80		$Q\ (kips)$	80	80
16	$W_f\ (kips)$	14.7	14.7		$W_f\ (kips)$	14.7	14.7
17	$u\ (ksf)$	0.125	0.125		$u\ (ksf)$	0499	0.499
18	$q_{eq}\ (ksf)$	1.81	1.81		$q_{eq}\ (ksf)$	1.43	1.43
19	FS	3.2	3		FS	4	3.8
20	FS_{min}	3	3		FS_{min}	3	3
21	Check	OK	OK		Check	OK	OK

B–5. Example B–5. Undrained vertical bearing capacity for isolated footing on clay

a. Determine: Calculate the factor of safety for undrained bearing capacity using both undrained strength and pressuremeter limit pressure and compare to minimum requirements to see if footing is acceptable. Evaluate potential for punch through.

b. Given: An 8-ft by 8-ft isolated footing supporting a 300-kip column load and bearing 2.5 ft below grade. Texas Rapid Assessment Method (TEXAM) PMT were conducted in clay soils as illustrated in Figure B–5. Groundwater was not encountered. The moist unit weight of the clay is 125 pcf.

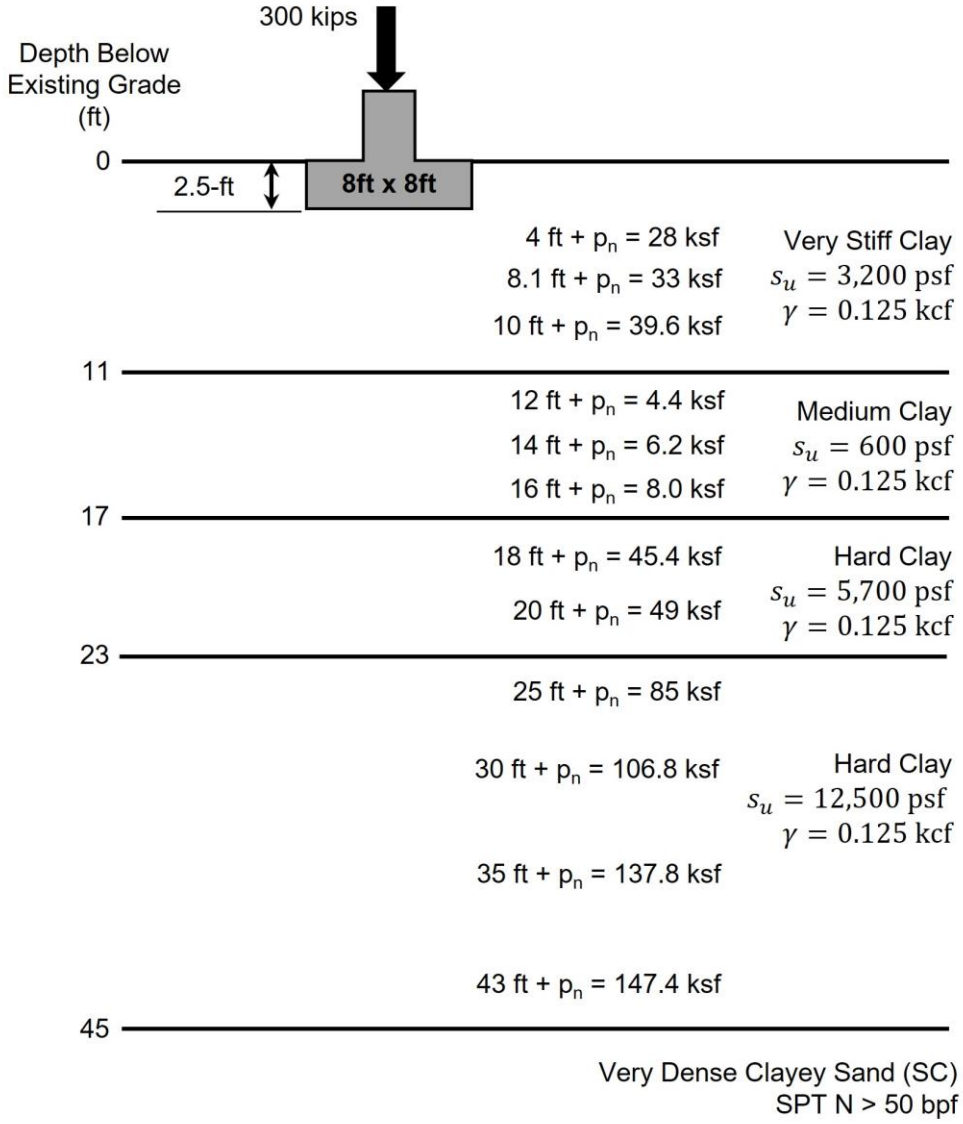


Figure B–5. Foundation and soil conditions for Example B–5

Table B–5a
Solution for the analysis of bearing capacity based on undrained strength

Step 1	Collect geometric information for the problem and input to Table B–5e. There is no eccentricity ($e = 0$), so footing geometry is equal to effective footing geometry (see paragraph 5–4a). Footing width, $B = B' = 8\ ft$ (Row 1) Footing length, $L = L' = 8\ ft$ (Row 2) Footing effective area, $A' = B' \cdot L' = 8\ ft \cdot 8\ ft = 64\ ft^2$ (Row 3) Embedment depth of the footing, $D_f = 2.5\ ft$ (Row 4) Unit wt. between base of the footing and ground surface, $\gamma = 0.125\ kcf$ (Row 5)
Step 2	Use Eq. 5–22 for undrained q_{ult} and correction factors in Table 5–4.
Step 3	For undrained analysis with $\phi = 0$, N_γ is zero. The value of cohesion, c, is taken as the undrained strength, s_u. Eq. 5–22 simplifies to $q_{ult} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + s_q \cdot d_q \cdot i_q \cdot b_q \cdot g_q \cdot q_0 \cdot N_q.$ Since $N_q = 1$ for $\phi = 0$, and all undrained correction factors are 1 (Table 5–4), Eq. 5–29 further simplifies to: $q_{ult} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + q_0$ The total stress at the base of the footing is: $q_o = \gamma \cdot D_f = 0.125\ kcf\ \cdot 2.5\ ft = 0.31\ ksf$ (Row 6) The shear zone is typically taken as 1 footing width, or 8 ft. The bottom of the shear zone estimated as $D_f + B = 10.5\ ft$, within the Layer 1 clay ($s_u = 3.2$ ksf). The undrained strength of 3.2 ksf will be used for these analyses. Since the soil in Layer 2 (11 ft to 17 ft) is weaker than soil in Layer 1 (0 ft to 10 ft), a deeper shear zone will also be checked in a separate punch-through analysis. $c = s_u = 3{,}200\ psf = 3.2\ ksf$ (Row 7) The cohesion bearing capacity factor, $N_c = 2 + \pi = 5.14$ (Row 8) The undrained shape factor for both the USACE and FHWA methods is (Table 5–4): $s_c = 1 + 0.2\left({B'}/{L'}\right) = 1 + 0.2\left({8\text{ ft}}/{8\text{ ft}}\right) = 1.2$ (Row 9) The undrained depth factor for the USACE methods is (Table 5–4): $d_c = 1 + 0.2\left({D_f}/{B'}\right) \leq 1.5 = 1 + 0.2\left({2.5\text{ ft}}/{8\text{ ft}}\right) = 1.06$ (Row 10) The FHWA method does not explicitly include a depth for undrained analysis, so $d_c = 1$. For this example, there is no load inclination, no base inclination, and no ground inclination. $i_c = 1$ (Row 11) $b_c = 1$ (Row 12) $g_c = 1$ (Row 13)

Step 4a	For the USACE method, bearing capacity pressure is calculated for Row 17. $q_{ult} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + q_0.$ $q_{ult} = 1.2 \cdot 1.06 \cdot 1 \cdot 1 \cdot 1 \cdot 3.2\ ksf \cdot 5.14 + 0.31\ ksf$ $q_{ult} = 20.92\ ksf + 0.31\ ksf = 21.23\ ksf$
Step 4b	For the FHWA method, bearing capacity pressure is calculated for Row 17. $q_{ult} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + q_0.$ $q_{ult} = 1.2 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 3.2\ ksf \cdot 5.14 + 0.31\ ksf$ $q_{ult} = 19.73\ ksf + 0.31\ ksf = 20.05\ ksf$
Step 5	The load on the soil is a combination of the column load and foundation weight; it is expressed as a stress based on the effective area (Eq. 5–13). $q_{eq} = \frac{Q + W_f}{A'}$ Column load, $Q = 300\ kips$ (Row 15) Unit weight of concrete, $\gamma_{conc} = 0.150\ kcf$ Weight of foundation, $W_f = A \cdot\ D_f \cdot \gamma_{conc} = 64\ ft^2 \cdot 2.5\ ft \cdot 0.15\ kcf = 24\ kips$ (Row 16) Pore pressure at base of footing, $u_D = 0\ ksf$ Gross equivalent bearing pressure, (Eq. 5–13) (Row 17). $q_{eq} = \frac{Q + W_f}{A'} - u_D = \frac{300\ kips + 24\ kips}{64\ ft^2} - 0\ ksf = 5.06\ ksf$
Step 6a	The minimum factor of safety for ultimate bearing capacity is 3 (Chapter 2). For the USACE method at the end of construction, the factor of safety was calculated as (Eq. 5–23). $FS = \frac{q_{ult}}{q_{eq}} = \frac{21.23\ ksf}{5.06\ ksf} = 4.2\ \geq 3$ OK ü
Step 6b	For the FHWA method, the factor of safety was calculated as (Eq. 5–29). $FS = \frac{q_{ult}}{q_{eq}} = \frac{20.05\ ksf}{5.06\ ksf} = 4.0\ \geq 3$ OK ü

Table B–5b
Solution for the bearing capacity analysis based on the pressuremeter

Step 1	All geometric information is the same as that for the analysis based on undrained strength. Rows 1 through 5 are the same.
Step 2	Identify that the problem involves low permeability soils (clayey soils and plastic silts) and that the bearing capacity limit state must be assessed for undrained conditions (Table 5–1). Use Eq. 5–26 for q_{ult} and correction factors in Table 5–4.
Step 3	All correction factors for the pressuremeter are the same as those for the USACE undrained method in Table 5–4. Rows 9 through 13 are the same as the USACE column for the undrained analysis.
Step 4	To evaluate soil strength (s), the pressuremeter method uses the geometric mean of the net limit pressure within $1.5B$ of the base of the footing (Eq. 5–24). The bottom of the zone is $D_f + 1.5B = 2.5\ ft + 1.5(8\ ft) = 14.5\ ft$. The zone for soil strength assessment includes the three measurements in the overlying strong layer and two measurements in the underlying weaker layer (Row 9). $s = p_{ne} = \sqrt[y]{p_{n1} \cdot p_{n2} \cdot \cdots p_{ny}} = \sqrt[5]{28\ ksf \cdot 33\ ksf \cdot 39.6\ ksf \cdot 4.4\ ksf \cdot 6.2\ ksf} = 15.8\ ksf$
Step 5	For undrained pressuremeter bearing capacity analysis, (Eq. 5–26) $q_{ult} = k \cdot s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s + q_0 = k \cdot s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot p_{ne} + q_0$ $q_o = \gamma \cdot D_f = 0.125\ kcf \cdot 2.5\ ft = 0.31$ (Row 6). The pressuremeter k-factor for continuous footings is 0.7 (Table 5–5) (Row 8) $q_{ult} = 0.7 \cdot 1.2 \cdot 1.06 \cdot 1 \cdot 1 \cdot 1 \cdot 15.8\ ksf + 0.31\ ksf$ $q_{ult} = 14.07\ ksf + 0.31\ ksf = 14.38\ ksf$
Step 6	The load on the soil is the same as the analysis based on undrained strength. Rows 18 through 21 are the same.
Step 7	The minimum factor of safety for ultimate bearing capacity is 3 (Chapter 2). For the pressuremeter method, the factor of safety was calculated as (Eq. 5–23). $FS = \frac{q_{ult}}{q_{eq}} = \frac{14.38\ ksf}{5.06\ ksf} = 2.8\ \leq 3$ NOT OK û

Table B–5c
Solution for punch-through analysis based on undrained strength

Step 1	All geometric information is the same as that for the analysis based on undrained strength. Rows 1 through 5 are the same.
Step 2	Identify that the problem involves low permeability soils (clayey soils and plastic silts) and that the bearing capacity limit state must be assessed for undrained conditions (Table 5–1). Use Eq. 5–22 for q_{ult} and correction factors in Table 5–4.
Step 3	All correction factors for the punch-through analysis are the same as those for the undrained method. Rows 8 through 13 are the same as the USACE column for the undrained analysis.
Step 4	The punch-through analysis assumes a failure in Layer 2, with an undrained strength of $600\ psf = 0.6\ ksf$ (Row 7).
Step 5a	For the USACE method, the punch-through bearing capacity for Layer 2 is calculated in Row 14. $q_{ult,2} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + q_0.$ $q_{ult,2} = 1.2 \cdot 1.06 \cdot 1 \cdot 1 \cdot 1 \cdot 0.6\ ksf \cdot 5.14 + 0.31\ ksf$ $q_{ult,2} = 3.92\ ksf + 0.31\ ksf = 4.23\ ksf$
Step 5b	For the FHWA method, the punch-through bearing capacity for Layer 2 is calculated in Row 14. $q_{ult,2} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + q_0.$ $q_{ult,2} = 1.2 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 0.6\ ksf \cdot 5.14 + 0.31\ ksf$ $q_{ult,2} = 3.70\ ksf + 0.31\ ksf = 4.01\ ksf$
Step 6	The bearing capacity at the footing level for a punch-through failure is increased due to load spreading. The ratio of q_{ult} to the calculated q_{ult} for Layer 2 ($q_{ult,2}$) is determined from (Eq. 5–17). $\frac{q_{ult}}{q_{ult,2}} = \frac{(B' + H_s) \cdot (L' + H_s)}{B \cdot L} = \frac{(8\ ft + 8.5\ ft) \cdot (8\ ft + 8.5\ ft)}{8\ ft \cdot 8\ ft} = 4.25$ For the USACE method: $q_{ult} = \frac{q_{ult}}{q_{ult,2}} \cdot q_{ult,2} = 4.25 \cdot 4.23\ ksf = 17.98\ ksf$ For the FHWA method: $q_{ult} = \frac{q_{ult}}{q_{ult,2}} \cdot q_{ult,2} = 4.25 \cdot 4.01\ ksf = 17.04\ ksf$
Step 7	The load on the soil is the same as the analysis based on undrained strength. Rows 18 through 21 are the same.
Step 8a	The minimum factor of safety for ultimate bearing capacity is 3 (Chapter 2). For the USACE method at the end of construction, the factor of safety was calculated as (Eq. 5–23). $FS = \frac{q_{ult}}{q_{eq}} = \frac{17.98\ ksf}{5.06\ ksf} = 3.6\ \geq 3$ OK ü

Step 8b	For the FHWA method, the factor of safety was calculated as (Eq. 5–23). $FS = \frac{q_{ult}}{q_{eq}} = \frac{17.04\ ksf}{5.06\ ksf} = 3.4\ \geq 3$ OK ü

Table B–5d
Average safety factor from multiple methods

For undrained strength-based analysis, the punch-through failure provided a lower factor of safety. The factors of safety for failure in the overlying strong layer are irrelevant, and not included in the final assessment. Undrained strength-based factors of safety are:

USACE FS = 3.6

FHWA FS = 3.4

The factor of safety based on pressuremeter analysis is more conservative since it does not explicitly consider the punch-through failure mechanism. PMT FS = 2.8

Section 5–5f recommends averaging the results of analyses performed when determining a design safety factor.

$FS = \frac{3.6 + 3.4 + 2.8}{3} = \frac{9.9}{3} = 3.3\ \geq 3$ OK ü

c. Discussion for Example B–5. The factors of safety using both the USACE and FHWA methods met the minimum requirements from Table 2–1. The punch-through failure mechanism gave a lower FS and was more critical. While the PMT method includes strength from the underlying weak layer in limit pressure averaging, it does not explicitly account for differences in the mechanism itself. The differences between the PMT and undrained strength mechanism indicate some additional uncertainty in results, however, the average factor of safety for the three methods was greater than 3. The footing meets undrained bearing capacity ultimate limit state requirements.

Table B–5e
Summary of calculations for Example B–5

	Undrained				Undrained in situ			Undrained punch through		
Row	Symbol	USACE	FHWA		Symbol	PMT		Symbol	USACE	FHWA
1	$B\ (ft)$	8	8		$B\ (ft)$	8		$B\ (ft)$	8	8
2	$L\ (ft)$	8	8		$L\ (ft)$	8		$L\ (ft)$	8	8
3	$A\ (ft^2)$	64	64		$A\ (ft^2)$	64		$A\ (ft^2)$	64	64
4	$D_f\ (ft)$	2.5	2.5		$D_f\ (ft)$	2.5		$D_f\ (ft)$	2.5	2.5
5	$\gamma\ (kcf)$	0.125	0.125		$\gamma\ (kcf)$	0.125		$\gamma\ (kcf)$	0.125	0.125
6	$q_o\ (ksf)$	0.31	0.31		$q_o\ (ksf)$	0.31		$q_o\ (ksf)$	0.31	0.31
7	$c = s_u\ (ksf)$	3.2	3.2		$s = p^*_{le}(ksf)$	15.8		$c = s_u(ksf)$	0.6	0.6
8	N_c	5.14	5.14		k	0.7		N_c	5.14	5.14
9	s_c	1.2	1.2		s_c	1.2		s_c	1.2	1.2
10	d_c	1.06	1		d_c	1.06		d_c	1.06	1
11	i_c	1	1		i_c	1		i_c	1	1
12	b_c	1	1		b_c	1		b_c	1	1
13	g_c	1	1		g_c	1		g_c	1	1
14	–	–	–		–	–		$q_{ult,2}\ (ksf)$	4.23	4.01
15	–	–	–		–	–		$H_s\ (ft)$	8.5	8.5
16	–	–	–		–	–		$q_{ult}/q_{ult,2}$	4.25	4.25
17	$q_{ult}\ (ksf)$	21.23	20.05		$q_{ult}\ (ksf)$	14.38		$q_{ult}(ksf)$	17.98	17.04
18	$Q\ (kips)$	300	300		$Q\ (kips)$	300		$Q\ (kips)$	300	300
19	$W_f\ (kips)$	24	24		$W_f\ (kips)$	24		$W_f(kips)$	24	24
20	$u\ (ksf)$	0	0		$u\ (ksf)$	0		$u\ (ksf)$	0	0
21	$q_{eq}\ (ksf)$	5.06	5.06		$q_{eq}\ (ksf)$	5.06		$q_{eq}(ksf)$	5.06	5.06
22	FS	4.2	4		FS	2.8		FS	3.6	3.4
23	FS_{min}	3	3		FS_{min}	3		FS_{min}	3	3
24	Check	OK	OK		Check	NOT OK		Check	OK	OK

B–6. Example B–6. Drained vertical bearing capacity for isolated footing on sand with water table in the shear zone, as compared to undrained punch-through failure into soft clay

a. Determine: Calculate the drained vertical bearing capacity for an isolated footing on sand with water table in the shear zone (as compared to undrained punch-through failure into soft clay). Calculate the factor of safety for bearing capacity and compare to minimum requirements to see if footing is acceptable.

b. *Given*: An 8-ft by 8-ft isolated footing supporting a 300-kip column load bears 2.5 ft below existing grade. Reference the mixed soil profile in Figure B–6. The moist unit weight for the clay and sand is 120 pcf.

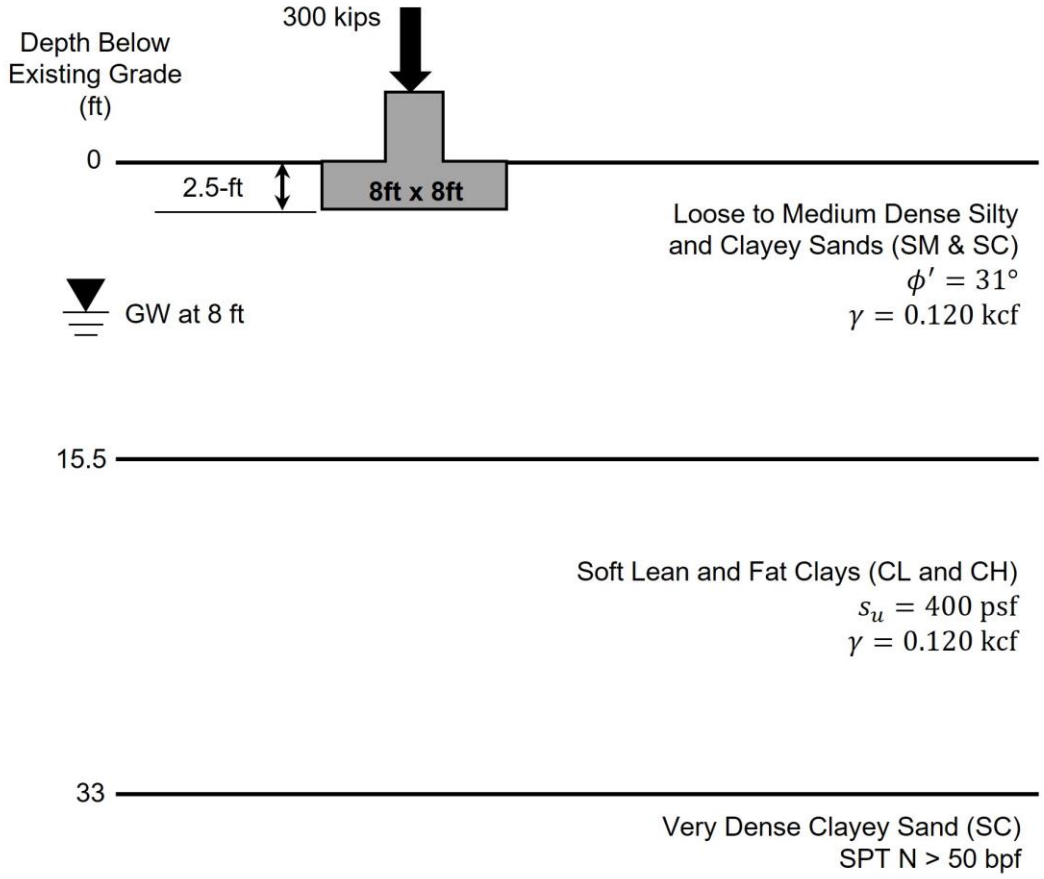


Figure B–6. Foundation and soil conditions for Example B–6

Table B–6a
Solution for drained bearing capacity analysis within surface layer

Step 1	Collect geometric information for the problem and input to Table B–6c. There is no eccentricity ($e = 0$), so footing geometry is equal to effective footing geometry (see paragraph 5–4a). Footing width, $B = B' = 8\ ft$ (Row 1) Footing length, $L = L' = 8\ ft$ (Row 2) Footing effective Area, $A' = B' \cdot L' = 8\ ft \cdot 8\ ft = 64\ ft^2$ (Row 3) Embedment depth of the footing, $D_f = 2.5\ ft$ (Row 4) Depth of water table, $D_w = 8\ ft$ (Row 5) Unit wt. between base of the footing and ground surface, $\gamma = 0.120\ kcf$ (Row 6)

Step 2	Identify that the problem involves free-draining soils (sands, silty sands, and non-plastic silts) in the layer immediately below the footing and that the bearing capacity limit state must be assessed for drained conditions (Table 5–1). Use Eq. 5–22 for q_{ult} and correction factors in Table 5–2 and Table 5–3.
Step 3	For drained analysis with cohesion $c = 0$, Eq. 5–22 simplifies to $q_{ult} = s_q \cdot d_q \cdot i_q \cdot b_q \cdot g_q \cdot q_0 \cdot N_q + s_\gamma \cdot d_\gamma \cdot i_\gamma \cdot b_\gamma \cdot g_\gamma \cdot \frac{1}{2} \cdot B' \cdot \gamma' \cdot N_\gamma$ The ratio of the effective unit weight in the shear zone to the total unit weight differs for the USACE and FHWA methods. For USACE (Eq. 5–19) (Row 7): $\frac{\gamma'}{\gamma} \approx \left[0.45 + 0.55\left(\frac{\max(D_W - D_f, 0)}{B}\right)\right] \leq 1$ $\frac{\gamma'}{\gamma} \approx \left[0.45 + 0.55\left(\frac{\max(8\ ft - 2.5\ ft, 0)}{8\ ft}\right)\right] \leq 1 = 0.45 + 0.55 \cdot 0.69 = 0.83$ For FHWA (Eq. 5–20) (Row 7): $\frac{\gamma'}{\gamma} \approx C_{W_\gamma} = \left[0.5 + 0.5\left(\frac{D_W}{1.5B + D_f}\right)\right] \leq 1$ $\frac{\gamma'}{\gamma} \approx C_{W_\gamma} = \left[0.5 + 0.5\left(\frac{8\ ft}{1.5(8\ ft) + 2.5\ ft}\right)\right] \leq 1 = 0.5 + 0.5 \cdot 0.55 = 0.78$ The effective stress at the base of the footing is: $q_o = \gamma \cdot D_f - u_D = 0.12\ kcf \cdot 2.5\ ft - 0\ ksf = 0.3\ ksf$ (Row 8) The shear zone is typically taken as 1 footing width, or 8 ft. The bottom of the shear zone estimated as $D_f + B = 10.5\ ft$. The estimated shear zone is within Layer 1 (0 ft to 15.5 ft), so strength of Layer 1 soil is used for analysis. Punch-through failure into the underlying weak layer will be checked in a separate analysis. $\phi' = 31°$ (Row 9) The flow number, or principal stress ratio, is (Table 5–2) (Row 10): $N = \frac{1 + sin\phi'}{1 - sin\phi'} = \frac{1 + sin(31^o)}{1 - sin(31^o)} = \frac{1.515}{0.485} = 3.12$

Step 4	The dimensionless bearing capacity for surcharge is (Table 5–2) (Row 11): $N_q = N \cdot e^{\pi \cdot tan\phi'} = 3.12e^{\pi \cdot tan(31^o)} = 3.12e^{1.89} = 3.12 \cdot 6.62 = 20.6$ The drained shape factor for surcharge differs for the USACE and FHWA methods. For USACE (Table 5–2) (Row 12): $s_q = 1 + 0.1N\left(\frac{B'}{L'}\right) = 1 + 0.1 \cdot 3.12\left(\frac{8\,ft}{8\,ft}\right) = 1.31$ For FHWA (Table 5–2) (Row 12): $s_q = 1 + \left[\left(\frac{B'}{L'}\right) tan\phi\right] = 1 + \left[\left(\frac{8\,ft}{8\,ft}\right) tan(31^o)\right] = 1 + (1 \cdot 0.6) = 1.6$ The drained depth factor for surcharge differs for the USACE and FHWA methods. For USACE, (Table 5–2) (Row 13): $d_q = 1 + 0.1\sqrt{N}\left(D_f/B'\right) = 1 + 0.1\sqrt{3.12}\left(\frac{2.5\,ft}{8\,ft}\right) = 1 + 0.1 \cdot 1.77 \cdot 0.31 = 1.06$ For FHWA, (Table 5–2) (Row 13): $d_q = 1 + 2 \cdot tan\phi' \cdot (1 - sin\phi')^2 \cdot atan\left(D_f/B'\right)$ $d_q = 1 + 2 \cdot tan(31^o) \cdot \left(1 - sin(31^o)\right)^2 \cdot atan\left({}^{2.5\text{ ft}}/_{8\text{ ft}}\right)$ $d_q = 1 + 2 \cdot 0.6 \cdot (1 - 0.515)^2 \cdot 0.30 = 1.09$ For this example, there is no load inclination, no base inclination, and no ground inclination. $i_c = 1$ (Row 14) $b_c = 1$ (Row 15) $g_c = 1$ (Row 16)
Step 5	The dimensionless bearing capacity for soil effective weight differs for the USACE and FHWA methods. For USACE, (Table 5–2) (Row 17): $N_\gamma = \left(N_q - 1\right) \cdot tan(1.4\phi') = (29.46 - 1) \cdot tan(1.4 \cdot 31^o)$ $N_\gamma = 19.6 \cdot tan(43.4^o) = 19.6 \cdot 0.945 = 18.5$ For FHWA, (Table 5–2) (Row 17): $N_\gamma = 2\left(N_q + 1\right) \cdot tan(\phi') = 2 \cdot (20.6 + 1) \cdot tan(31^o)$ $N_\gamma = 43.2 \cdot 0.6 = 26$ The drained shape factor for soil effective weight differs for the USACE and FHWA methods. For USACE, the shape factors for surcharge and soil effective weight are the same (Table 5–2) (Row 18): $s_\gamma = 1 + 0.1N\frac{B'}{L'} = 1 + 0.1 \cdot 3.12 \cdot \frac{8\,ft}{8\,ft} = 1.31$ For FHWA, (Table 5–2) (Row 18): $s_\gamma = 1 - 0.4\left(\frac{B'}{L'}\right) = 1 - 0.4\left(\frac{8\,ft}{8\,ft}\right) = 1 - 0.4 = 0.6$

	The drained depth factor for surcharge soil effective weight differs for the USACE and FHWA methods. For USACE, the shape factors for surcharge and soil effective weight are the same (Table 5–2) (Row 19): $d_\gamma = 1 + 0.1\sqrt{N}\left({}^{D_f}/_{B'}\right) = 1 + 0.1\sqrt{3.12}\left(\frac{2.5\ ft}{8\ ft}\right) = 1 + 0.1 \cdot 1.77 \cdot 0.31 = 1.06$ For FHWA, (Table 5–2) (Row 19): $d_\gamma = 1$ For this example, there is no load inclination, no base inclination, and no ground inclination. $i_c = 1$ (Row 20) $b_c = 1$ (Row 21) $g_c = 1$ (Row 22)
Step 6a	For the USACE method, bearing capacity pressure is calculated for Row 23 as: $q_{ult} = s_q \cdot d_q \cdot i_q \cdot b_q \cdot g_q \cdot q_0 \cdot N_q + s_\gamma \cdot d_\gamma \cdot i_\gamma \cdot b_\gamma \cdot g_\gamma \cdot \frac{1}{2} \cdot B' \cdot \gamma' \cdot N_\gamma$ $q_{ult} = 1.31 \cdot 1.06 \cdot 1 \cdot 1 \cdot 1 \cdot 0.3\ ksf \cdot 20.6 + 1.31 \cdot 1.06 \cdot 1 \cdot 1 \cdot 1 \cdot \frac{1}{2} \cdot 8\ ft \cdot 0.83 \cdot 0.12\ kcf \cdot 18.5$ $q_{ult} = 8.58\ ksf + 10.23\ ksf = 18.8\ ksf$
Step 6b	For the FHWA method, bearing capacity pressure is calculated for Row 23 as: $q_{ult} = s_q \cdot d_q \cdot i_q \cdot b_q \cdot g_q \cdot q_0 \cdot N_q + s_\gamma \cdot d_\gamma \cdot i_\gamma \cdot b_\gamma \cdot g_\gamma \cdot \frac{1}{2} \cdot B' \cdot \gamma' \cdot N_\gamma$ $q_{ult} = 1.6 \cdot 1.09 \cdot 1 \cdot 1 \cdot 1 \cdot 0.3\ ksf \cdot 20.6 + 0.6 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot \frac{1}{2} \cdot 8\ ft \cdot 0.78 \cdot 0.12\ kcf \cdot 26$ $q_{ult} = 10.78\ ksf + 5.85\ ksf = 16.6\ ksf$
Step 7	The load on the soil is a combination of the column load and foundation weight and is expressed as a stress based on the effective area (Eq. 5–13). $q_{eq} = \frac{Q + W_f}{A'}$ Column load, $Q = 300\ kips$ (Row 24) Unit weight of concrete, $\gamma_{conc} = 0.150\ kcf$ Weight of foundation, $W_f = A \cdot D_f \cdot \gamma_{conc} = 64\ ft^2 \cdot 2.5\ ft \cdot 0.15\ kcf = 24\ kips$ (Row 25) Gross equivalent bearing pressure, (Eq. 5–13) (Row 26). $q_{eq} = \frac{Q + W_f}{A'} - u_D = \frac{300\ kips + 24\ kips}{64\ ft^2} - 0\ ksf = 5.06\ ksf$
Step 8a	The minimum factor of safety for ultimate bearing capacity is 3 (Chapter 2). For the USACE method, the factor of safety was calculated as (Eq. 5–23) (Row 27). $FS = \frac{q_{ult}}{q_{eq}} = \frac{18.8\ ksf}{5.06\ ksf} = 3.7\ \geq 3$ OK ü
Step 8b	For the FHWA method, the factor of safety was calculated as (Eq. 5–23) (Row 27). $FS = \frac{q_{ult}}{q_{eq}} = \frac{16.6\ ksf}{5.06\ ksf} = 3.3\ \geq 3$ OK ü

Table B–6b
Solution for undrained punch-through analysis

Step 1	Collect geometric information for the problem and input to Table B–6c. There is no eccentricity ($e = 0$), so footing geometry is equal to effective footing geometry (see paragraph 5–4a). The geometry information is the same as that for the drained analysis of Layer 1. Footing width, $B = B' = 8\ ft$ (Row 1) Footing length, $L = L' = 8\ ft$ (Row 2) Footing effective area, $A' = B' \cdot L' = 8\ ft \cdot 8\ ft = 64\ ft^2$ (Row 3) Embedment depth of the footing, $D_f = 2.5\ ft$ (Row 4) Depth of water table, $D_w = 8\ ft$ (Row 5) Unit wt. between base of the footing and ground surface, $\gamma = 0.120\ kcf$ (Row 6)
Step 2	Identify that the problem of punch-through into the underlying soil involves low permeability soils (clayey soils and plastic silts) and that the bearing capacity limit state must be assessed for undrained conditions (Table 5–1). Use Eq 5–22 for q_{ult} and correction factors in Table 5–4.
Step 3	For undrained analysis with $\phi = 0$, N_γ is zero. The value of cohesion, c, is taken as the undrained strength s_u. Eq. 5–22 simplifies to $q_{ult} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + s_q \cdot d_q \cdot i_q \cdot b_q \cdot g_q \cdot q_0 \cdot N_q$. Since $N_q = 1$ for $\phi = 0$, and all undrained correction factors are 1 (Table 5–4), Eq. 5–29 further simplifies to: $q_{ult} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + q_0$. The total stress at the base of the footing is: $q_o = \gamma \cdot D_f = 0.12\ kcf \cdot 2.5\ ft = 0.3\ ksf$ (Row 8) The shear zone is typically taken as 1 footing width, or 7 ft, but may extend to approximately 2 footings widths when softer soils are at depth. The weaker soils in Layer 2 soil are used for the punch-through analysis. $c = s_u = 400\ psf = 0.4\ ksf$ (Row 9) The cohesion bearing capacity factor $N_c = 2\pi = 5.14$ (Row 10) The undrained shape factor for both the USACE and FHWA methods is (Table 5–4): $s_c = 1 + 0.2\left({B'}/{L'}\right) = 1 + 0.2\left({8\text{ ft}}/{8\text{ ft}}\right) = 1.2$ (Row 11) The undrained depth factor for the USACE methods is (Table 5–4): $d_c = 1 + 0.2\left({D_f}/{B'}\right) \le 1.5 = 1 + 0.2\left({2.5\text{ ft}}/{8\text{ ft}}\right) = 1.06$ (Row 12) The FHWA method does not explicitly include a depth for undrained analysis, so $d_c = 1$. For this example, there is no load inclination, no base inclination, and no ground inclination. $i_c = 1$ (Row 13) $b_c = 1$ (Row 14) $g_c = 1$ (Row 15)

Step 4a	For the USACE method, bearing capacity pressure for Layer 2 is calculated for Row 16. $q_{ult,2} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + q_0.$ $q_{ult,2} = 1.2 \cdot 1.06 \cdot 1 \cdot 1 \cdot 1 \cdot 0.4\ ksf \cdot 5.14 + 0.3\ ksf$ $q_{ult,2} = 2.62\ ksf + 0.3\ ksf = 2.92\ ksf$
Step 4b	For the FHWA method, bearing capacity pressure for Layer 2 is calculated for Row 16. $q_{ult,2} = s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s_u \cdot N_c + q_0.$ $q_{ult,2} = 1.2 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 0.4\ ksf \cdot 5.14 + 0.3\ ksf$ $q_{ult,2} = 2.47\ ksf + 0.3\ ksf = 2.77\ ksf$
Step 5	The bearing capacity at the footing level for a punch-through failure is increased due to load spreading. The ratio of q_{ult} to the calculated q_{ult} for Layer 2 ($q_{ult,2}$) is determined from (Eq. 5–17). (Row 18) $\frac{q_{ult}}{q_{ult,2}} = \frac{(B' + H_s) \cdot (L' + H_s)}{B \cdot L} = \frac{(8\ ft + 13\ ft) \cdot (8\ ft + 13\ ft)}{8\ ft \cdot 8\ ft} = 6.89$ For the USACE method, (Row 23): $q_{ult} = \frac{q_{ult}}{q_{ult,2}} \cdot q_{ult,2} = 6.89 \cdot 2.92\ ksf = 20.1\ ksf$ For the FHWA method, (Row 23): $q_{ult} = \frac{q_{ult}}{q_{ult,2}} \cdot q_{ult,2} = 6.89 \cdot 2.77\ ksf = 19.1\ ksf$
Step 6	The load on the soil is the same as the analysis based on undrained strength. Rows 24 through 27 are the same.
Step 7a	The minimum factor of safety for ultimate bearing capacity is 3 (Chapter 2). For the USACE method, the factor of safety was calculated as (Eq. 5–23) (Row 27). $FS = \frac{q_{ult}}{q_{eq}} = \frac{20.1\ ksf}{5.06\ ksf} = 4\ \geq 3$ OK ü
Step 7b	For the FHWA method, the factor of safety was calculated as (Eq. 5–23) (Row 27). $FS = \frac{q_{ult}}{q_{eq}} = \frac{19.1\ ksf}{5.06\ ksf} = 3.8\ \geq 3$ OK ü

c. Discussion for Example B–6. For the layered sand over clay foundation in this example, a punch-through failure into the clay foundation provided less resistance than that for a failure in the upper sand layer. The punch-through failure will control the design factor of safety. The factors of safety using both the USACE and FHWA methods met the minimum requirements from Chapter 2. The footing meets bearing capacity ultimate limit state requirements.

Table B–6c
Summary of calculations for Example B–6

	Drained with $c = 0$				Undrained punch through		
Row	Symbol	USACE	FHWA		Symbol	USACE	FHWA
1	$B\ (ft)$	8	8		$B\ (ft)$	8	8
2	$L\ (ft)$	8	8		$L\ (ft)$	8	8
3	$A\ (ft^2)$	64	64		$A\ (ft^2)$	64	64
4	$D_f\ (ft)$	2.5	2.5		$D_f\ (ft)$	2.5	2.5
5	$D_w(ft)$	8	8		–	–	–
6	$\gamma\ (kcf)$	0.12	0.12		$\gamma\ (kcf)$	0.12	0.12
7	γ'/γ	0.83	0.78		–	–	–
8	$q_o\ (ksf)$	0.3	0.3		$q_o\ (ksf)$	0.3	0.3
9	$\phi'\ (deg)$	31	31		$c = s_u\ (ksf)$	0.4	0.4
10	N	3.12	3.12		N_c	5.14	5.14
11	N_q	20.6	20.6		s_c	1.2	1.2
12	s_q	1.31	1.60		d_c	1.06	1
13	d_q	1.06	1.09		i_c	1	1
14	i_q	1	1		b_c	1	1
15	b_q	1	1		g_c	1	1
16	g_q	1	1		$q_{ult,2}\ (ksf)$	2.92	2.77
17	N_γ	18.5	26		$H_s(ft)$	13	13
18	s_γ	1.31	0.6		$q_{ult}/q_{ult,2}$	6.89	6.89
19	d_γ	1.06	1		–	–	–
20	i_γ	1	1		–	–	–
21	b_γ	1	1		–	–	–
22	g_γ	1	1		–	–	–
23	$q_{ult}\ (ksf)$	18.82	16.62		$q_{ult}\ (ksf)$	20.12	19.09
24	$Q\ (kips)$	300	300		$Q\ (kips)$	300	300
25	$W_f\ (kips)$	24	24		$W_f\ (kips)$	24	24
26	$u\ (ksf)$	0	0		$u\ (ksf)$	0	0
27	$q_{eq}\ (ksf)$	5.06	5.06		$q_{eq}\ (ksf)$	5.06	5.06
28	FS	3.7	3.3		FS	4.0	3.8
29	FS_{min}	3	3		FS_{min}	3	3
30	Check	OK	OK		Check	OK	OK

B–7. Example B–7. Drained vertical bearing capacity for isolated footing on sand, comparison of soil parameters, and in situ tests

a. Determine: Calculate the drained bearing capacity in sand. Compare soil parameters with in situ tests. Calculate the factor of safety for bearing capacity and compare to minimum requirements to see if footing is acceptable. Use methods based on friction angle, SPT, CPT, and PMT.

b. Given: An 8-ft by 8-ft isolated footing supporting a 400-kip column load bears 2.5 ft below existing grade as illustrated in Figure B–7. TEXAM PMT, CPT, and SPT were conducted on sandy soils as illustrated in Figure B–7. The moist unit weight of the sand is 125 pcf. Groundwater was not encountered.

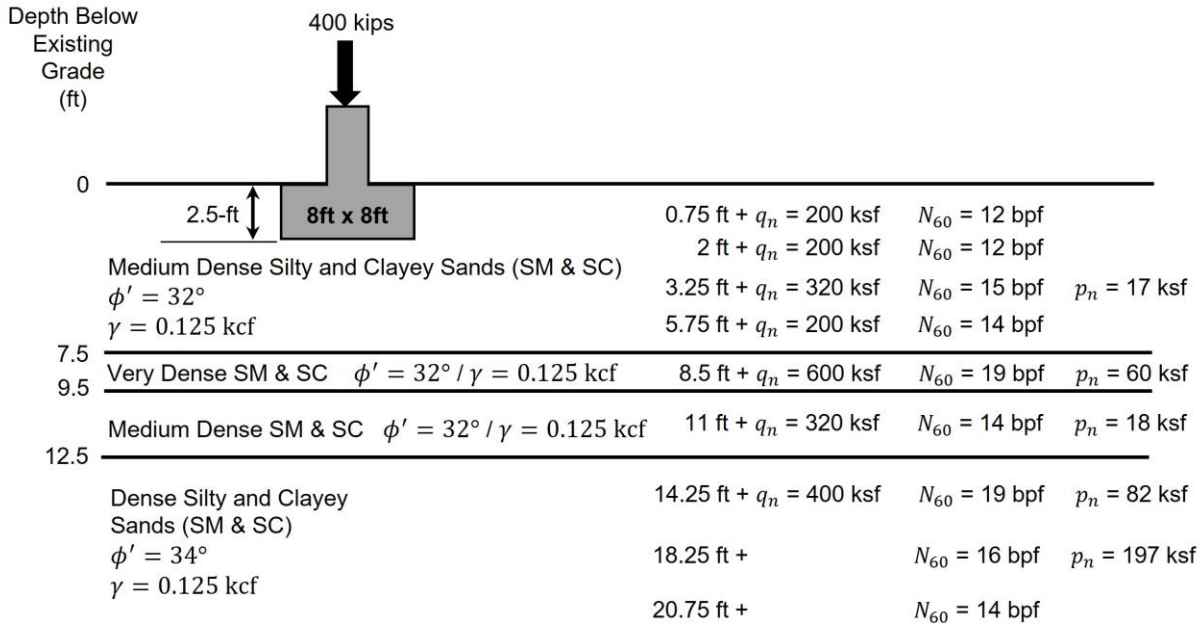


Figure B–7. Foundation and soil conditions for Example B–7

Table B–7a
Solution for drained bearing capacity analysis within surface layer

Step 1	Collect geometric information for the problem. There is no eccentricity ($e = 0$), so footing geometry is equal to effective footing geometry (see paragraph 5–4a). Footing width, $B = B' = 8\ ft$ (Row 1) Footing length, $L = L' = 8\ ft$ (Row 2) Footing effective area, $A' = B' \cdot L' = 8\ ft \cdot 8\ ft = 64\ ft^2$ (Row 3) Embedment depth of the footing, $D_f = 2.5\ ft$ (Row 4) Depth of water table, $D_w =$ not encountered (effective stress = total stress) Unit wt. between base of the footing and ground surface, $\gamma = 0.125\ kcf$ (Row 6)

Step 2	Identify that the problem involves free-draining soils (sands, silty sands, and non-plastic silts) in the layer immediately below the footing and that the bearing capacity limit state must be assessed for drained conditions (Table 5–1). Use Eq. 5–22 for q_{ult} and correction factors in Table 5–2 and Table 5–3.
Step 3	For drained analysis with $c = 0$, Eq. 5–22 simplifies to: $q_{ult} = s_q \cdot d_q \cdot i_q \cdot b_q \cdot g_q \cdot q_0 \cdot N_q + s_\gamma \cdot d_\gamma \cdot i_\gamma \cdot b_\gamma \cdot g_\gamma \cdot \frac{1}{2} \cdot B' \cdot \gamma' \cdot N_\gamma$ Water was not encountered within the upper 20 ft. When water is not expected for critical load cases, the total stress is equal to the effective stress and total unit weight is used for calculations ($\gamma'/\gamma = 1$) for both USACE and FHWA methods (Row 7). The effective stress at the base of the footing is: $q_o = \gamma \cdot D_f - u_o = 0.125\ kcf \cdot 2.5\ ft - 0\ ksf = 0.31\ ksf$ (Row 8) The shear zone is typically taken as 1 footing width, or 8 ft. The bottom of the shear zone estimated as $D_f + B = 10.5\ ft$. Since the soil are denser below 7.5 ft, it is assumed that the shear zone is isolated in the weaker layer with $\phi' = 32°$. $\phi' = 32°$ (Row 9) The flow number, or principal stress ratio, is (Table 5–2) (Row 10): $N = \frac{1 + sin\phi'}{1 - sin\phi'} = \frac{1 + sin(32^o)}{1 - sin(32^o)} = \frac{1.53}{0.47} = 3.26$

Step 4	The dimensionless bearing capacity for surcharge is (Table 5–2) (Row 11): $N_q = N \cdot e^{\pi \cdot tan\phi'} = 3.26e^{\pi \cdot tan(32^o)} = 3.26e^{1.96} = 3.26 \cdot 7.1 = 23.1$ The drained shape factor for surcharge differs for the USACE and FHWA methods. For USACE, (Table 5–2) (Row 12): $s_q = 1 + 0.1N\left(\frac{B'}{L'}\right) = 1 + 0.1 \cdot 3.26\left(\frac{8\,ft}{8\,ft}\right) = 1.33$ For FHWA, (Table 5–2) (Row 12): $s_q = 1 + \left(\frac{B'}{L'}\right) tan\phi = 1 + \left(\frac{8\,ft}{8\,ft}\right) tan(32^o) = 1 + (1 \cdot 0.62) = 1.62$ The drained depth factor for surcharge differs for the USACE and FHWA methods. For USACE, (Table 5–2) (Row 13): $d_q = 1 + 0.1\sqrt{N}\left(\frac{D_f}{B'}\right) = 1 + 0.1\sqrt{3.26}\left(\frac{2.5\,ft}{8\,ft}\right) = 1 + 0.1 \cdot 1.81 \cdot 0.31 = 1.06$ For FHWA, (Table 5–2) (Row 13): $d_q = 1 + 2 \cdot tan\phi' \cdot (1 - sin\phi')^2 \cdot atan\left(\frac{D_f}{B'}\right)$ $d_q = 1 + 2 \cdot tan(32^o) \cdot \left(1 - sin(32^o)\right)^2 \cdot atan\left(\frac{2.5\,ft}{8\,ft}\right)$ $d_q = 1 + 2 \cdot 0.62(1 - 0.53)^2 \cdot 0.30 = 1.08$ For this example, there is no load inclination, no base inclination, and no ground inclination. $i_c = 1$ (Row 14) $b_c = 1$ (Row 15) $g_c = 1$ (Row 16)

Step 5	The dimensionless bearing capacity for soil effective weight differs for the USACE and FHWA methods. For USACE, (Table 5–2) (Row 17): $N_\gamma = (N_q - 1) \cdot tan(1.4\phi') = (23.1 - 1) \cdot tan(1.4 \cdot 32^o)$ $N_\gamma = 22.1 \cdot tan(44.8^o) = 22.1 \cdot 0.99 = 21.9$ For FHWA, (Table 5–2) (Row 17): $N_\gamma = 2(N_q + 1) \cdot tan\phi' = 2 \cdot (23.1 + 1) \cdot tan(32^o)$ $N_\gamma = 48.2 \cdot 0.625 = 30.1$ The drained shape factor for soil effective weight differs for the USACE and FHWA methods. For USACE, the shape factors for surcharge and soil effective weight are the same (Table 5–2) (Row 18): $s_\gamma = 1 + 0.1N\left(\frac{B'}{L'}\right) = 1 + 0.1 \cdot 3.26\left(\frac{8\ ft}{8\ ft}\right) = 1.33$ For FHWA, (Table 5–2) (Row 18): $s_\gamma = 1 - 0.4\left(\frac{B'}{L'}\right) = 1 - 0.4\left(\frac{8\ ft}{8\ ft}\right) = 1 - 0.4 = 0.6$ The drained depth factor for surcharge soil effective weight differs for the USACE and FHWA methods. For USACE, the shape factors for surcharge and soil effective weight are the same (Table 5–2) (Row 19): $d_\gamma = 1 + 0.1\sqrt{N}\left(\frac{D_f}{B'}\right) = 1 + 0.1\sqrt{3.26}\left(\frac{2.5\ ft}{8\ ft}\right) = 1 + 0.1 \cdot 1.81 \cdot 0.31 = 1.06$ For FHWA, (Table 5–2) (Row 19): $d_\gamma = 1$ For this example, there is no load inclination, no base inclination, and no ground inclination. $i_c = 1$ (Row 20) $b_c = 1$ (Row 21) $g_c = 1$ (Row 22)
Step 6a	For the USACE method, bearing capacity pressure is calculated for Row 23 as: $q_{ult} = s_q \cdot d_q \cdot i_q \cdot b_q \cdot g_q \cdot q_0 \cdot N_q + s_\gamma \cdot d_\gamma \cdot i_\gamma \cdot b_\gamma \cdot g_\gamma \cdot \frac{1}{2} \cdot B' \cdot \gamma' \cdot N_\gamma$ $q_{ult} = 1.33 \cdot 1.06 \cdot 1 \cdot 1 \cdot 1 \cdot 0.31\ ksf \cdot 23.1 + 1.33 \cdot 1.06 \cdot 1 \cdot 1 \cdot 1 \cdot \frac{1}{2} \cdot 8\ ft \cdot 1 \cdot 0.125\ kcf \cdot 21.9$ $q_{ult} = 10.1\ ksf + 15.4\ ksf = 25.5\ ksf$
Step 6b	For the FHWA method, bearing capacity pressure is calculated for Row 23 as: $q_{ult} = s_q \cdot d_q \cdot i_q \cdot b_q \cdot g_q \cdot q_0 \cdot N_q + s_\gamma \cdot d_\gamma \cdot i_\gamma \cdot b_\gamma \cdot g_\gamma \cdot \frac{1}{2} \cdot B' \cdot \gamma' \cdot N_\gamma$ $q_{ult} = 1.62 \cdot 1.08 \cdot 1 \cdot 1 \cdot 1 \cdot 0.31\ ksf \cdot 23.1 + 0.6 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 0.5 \cdot 8\ ft \cdot 1 \cdot 0.125\ kcf \cdot 30.1$ $q_{ult} = 12.53\ ksf + 9.03\ ksf = 21.6\ ksf$

Step 7	The load on the soil is a combination of the column load and foundation weight and is expressed as a stress based on the effective area (Eq. 5–13). $q_{eq} = \frac{Q + W_f}{A'}$ Column Load, $Q = 400\ kips$ (Row 24) Unit weight of concrete, $\gamma_{conc} = 0.150\ kcf$ Weight of foundation, $W_f = A \cdot D_f \cdot \gamma_{conc} = 64\ ft^2 \cdot 2.5\ ft \cdot 0.15\ kcf = 24\ kips$ (Row 2) Gross equivalent bearing pressure, (Eq. 5–13) (Row 26). $q_{eq} = \frac{Q + W_f}{A'} - u_D = \frac{400\ kips + 24\ kips}{64\ ft^2} - 0\ ksf = 6.63\ ksf$
Step 8a	The minimum factor of safety for ultimate bearing capacity is 3 (Chapter 2). For the USACE method, the factor of safety was calculated as (Eq. 5–23) (Row 27). $FS = \frac{q_{ult}}{q_{eq}} = \frac{25.5\ ksf}{6.63\ ksf} = 3.8\ \geq 3$ OK ü
Step 8b	For the FHWA method, the factor of safety was calculated as (Eq. 5–23) (Row 27). $FS = \frac{q_{ult}}{q_{eq}} = \frac{21.6\ ksf}{6.63\ ksf} = 3.3\ \geq 3$ OK ü

Table B–7b
Solution for bearing capacity analysis based on in situ tests (PMT, CPT, SPT)

Step 1	All geometric information is the same as that for the analysis based on undrained strength. Rows 1 through 5 are the same.
Step 2	Identify that the problem involves free-draining soils (sands, silty sands, and non-plastic silts) in the layer immediately below the footing and that the bearing capacity limit state must be assessed for drained conditions (Table 5–1). Use Eq. 5–25 for q_{ult} and correction factors in Table 5–2 and Table 5–3.
Step 3	All correction factors for the in situ tests are the same as those for the USACE method in Table 5–2 and Table 5–3. However, the flow number, N = 3 (Row 10) is assumed. The drained shape factor for surcharge differs for the USACE and FHWA methods. For USACE, (Table 5–2) (Row 12): $s_q = 1 + 0.1N\left(\frac{B'}{L'}\right) = 1 + 0.1 \cdot 3\left(\frac{8\ ft}{8\ ft}\right) = 1.3$ The drained depth factor for surcharge differs for the USACE and FHWA methods. For USACE, (Table 5–2) (Row 13): $d_q = 1 + 0.1\sqrt{N}\left(\frac{D_f}{B'}\right) = 1 + 0.1\sqrt{3}\left(\frac{2.5\ ft}{8\ ft}\right) = 1 + 0.1 \cdot 1.73 \cdot 0.31 = 1.05$ For this example, there is no load inclination, no base inclination, and no ground inclination. $i_{bg} = 1$ (Row 14) $b_q = 1$ (Row 15) $g_q = 1$ (Row 16)
Step 4	The in situ test method uses the geometric mean of the net strength parameter (s) within the shear zone below the footing (Eq. 5–24). The shear zone is estimated to be within the weaker Layer 1 soils. For the pressuremeter, the shear zone has one p_n measurement of 17 ksf (Row 9). $s = p_{ne} = \sqrt[y]{p_{n1} \cdot p_{n2} \cdot \cdots p_{ny}} = \sqrt[1]{17\ ksf} = 17\ ksf$ For the CPT, the shear zone has four q_t^* measurements of 17 ksf (Row 9). $s = q_{ne} = \sqrt[y]{q_{n1} \cdot q_{n2} \cdot \cdots q_{ny}} = \sqrt[4]{200\ ksf \cdot 200\ ksf \cdot 320\ ksf \cdot 200\ ksf}$ $s = 225\ ksf$ For the SPT, the shear zone has four N_{60} measurements. $N_{60,e} = \sqrt[y]{N_{60,1} \cdot N_{60,2} \cdot \cdots N_{60,n}} = \sqrt[4]{12 \cdot 12 \cdot 15 \cdot 14} = 13.2$ bpf The $N_{60,e}$-value needs to be multiplied by a reference stress of 2 ksf (Row 9): $s = N_{60,e} \cdot p_{ref} = 13.2\ \text{bpf} \cdot 2\ ksf = 26.4\ ksf$

Step 5a	For drained pressuremeter bearing capacity analysis (Eq. 5–25) $q_{ult} = k \cdot s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s + q_0 = k \cdot s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot p_{ne} + q_0$ $q_o = \gamma \cdot D_f = 0.125\ kcf \cdot 2.5\ ft = 0.31$ (Row 8) The pressuremeter k-factor for a continuous footing (drained condition) is 0.9 (Table 5–5) (Row 11) $q_{ult} = 0.9 \cdot 1.3 \cdot 1.05 \cdot 1 \cdot 1 \cdot 1 \cdot 17\ ksf + 0.31\ ksf$ $q_{ult} = 20.9\ ksf + 0.31\ ksf = 21.2\ ksf$
Step 5b	For drained CPT bearing capacity analysis, (Eq. 5–25) $q_{ult} = k \cdot s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s + q_0 = k \cdot s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot q_{ne} + q_0$ $q_o = \gamma \cdot D_f = 0.125\ kcf \cdot 2.5\ ft = 0.31$ (Row 8) The CPT k-factor for continuous footing (drained analysis) is 0.1 (Table 5–5) (Row 11) $q_{ult} = 0.1 \cdot 1.3 \cdot 1.05 \cdot 1 \cdot 1 \cdot 1 \cdot 225\ ksf + 0.31\ ksf$ $q_{ult} = 30.7\ ksf + 0.31\ ksf = 31\ ksf$
Step 5c	For drained SPT bearing capacity analysis, (Eq. 5–25) $q_{ult} = k \cdot s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot s + q_0 = k \cdot s_c \cdot d_c \cdot i_c \cdot b_c \cdot g_c \cdot N_{60,e} \cdot p_{ref} + q_0$ $q_o = \gamma \cdot D_f = 0.125\ kcf \cdot 2.5\ ft = 0.31$ (Row 8) The SPT k-factor for a continuous footing (drained analysis) is 0.6 (Table 5–5) (Row 11) $q_{ult} = 0.6 \cdot 1.3 \cdot 1.05 \cdot 1 \cdot 1 \cdot 1 \cdot 26.4\ ksf + 0.31\ ksf$ $q_{ult} = 21.6\ ksf + 0.31\ ksf = 21.9\ ksf$
Step 6	The load on the soil is the same as the analysis based on undrained strength. Rows 18 through 21 are the same.
Step 7	The minimum factor of safety for ultimate bearing capacity is 3 (Chapter 2). For the in situ test methods, the factor of safety was calculated as (Eq. 5–23). Pressuremeter: $FS = {q_{ult}}/{q_{eq}} = {21.2\ \text{ksf}}/{6.63\ \text{ksf}} = 3.2\ \geq 3$ OK ü CPT: $FS = {q_{ult}}/{q_{eq}} = {31\ \text{ksf}}/{6.63\ \text{ksf}} = 4.7\ \geq 3$ OK ü SPT: $FS = {q_{ult}}/{q_{eq}} = {21.9\ \text{ksf}}/{6.63\ \text{ksf}} = 3.3\ \geq 3$ OK ü

c. Discussion for Example B–7. For the foundation on sandy soil in this example, the factors of safety using the USACE, FHWA, PMT, CPT, and SPT methods all met the minimum requirements from Table 2–1. The footing meets bearing capacity ultimate limit state requirements.

Table B–7c
Summary of calculations for Example B–7

	Drained with $c = 0$				Drained in situ			
Row	Symbol	USACE	FHWA		Symbol	PMT	CPT	SPT
1	$B\ (ft)$	8	8		$B\ (ft)$	8	8	8
2	$L\ (ft)$	8	8		$L\ (ft)$	8	8	8
3	$A\ (ft^2)$	64	64		$A\ (ft^2)$	64	64	64
4	$D_f\ (ft)$	2.5	2.5		$D_f\ (ft)$	2.5	2.5	2.5
5	D_w	20	20		–	–	–	–
6	$\gamma\ (kcf)$	0.125	0.125		$\gamma\ (kcf)$	0.125	0.125	0.125
7	γ'/γ	1	1		–	–	–	–
8	$q_o\ (ksf)$	0.31	0.31		$q_o\ (ksf)$	0.31	0.31	0.31
9	$\phi'\ (deg)$	32	32		$s\ (ksf)$	17	225	26.4
10	N	3.25	3.25		N	3	3	3
11	N_q	23.1	23.1		k	0.9	0.1	0.6
12	s_q	1.33	1.62		s_q	1.3	1.3	1.3
13	d_q	1.06	1.08		d_q	1.05	1.05	1.05
14	i_q	1	1		$i_{q\gamma}$	1	1	1
15	b_q	1	1		b_q	1	1	1
16	g_q	1	1		g_q	1	1	1
17	N_γ	21.9	30.1		–	–	–	–
18	s_γ	1.33	0.60		–	–	–	–
19	d_γ	1.06	1		–	–	–	–
20	i_γ	1	1		–	–	–	–
21	b_γ	1	1		–	–	–	–
22	g_γ	1	1		–	–	–	–
23	$q_{ult}\ (ksf)$	25.5	21.6		$q_{ult}\ (ksf)$	21.2	31	21.9
24	$Q\ (kips)$	400	400		$Q\ (kips)$	400	400	400
25	$W_f\ (kips)$	24	24		$W_f\ (kips)$	24	24	24
26	$u\ (ksf)$	0	0		$u\ (ksf)$	0	0	0
27	q_{eq} (ksf)	6.63	6.63		$q_{eq}\ (ksf)$	6.63	6.63	6.63
28	FS	3.8	3.3		FS	3.2	4.7	3.3
29	FS_{min}	3	3		FS_{min}	3	3	3
30	Check	OK	OK		Check	OK	OK	OK

Appendix C
Design Examples: Settlement for Static Analysis

Design examples provided in this appendix are described below. The evaluation of ultimate limit states for these same design examples are provided in Appendix B. No examples of secondary settlement (creep) are presented in this appendix.

Example C–1	Settlement of isolated footing on clay (*e*-log-p method).
Example C–2	Settlement due to superimposed foundation loads on clay (*e*-log-p method).
Example C–3a	Settlement of soft clay due to areal fill load only (*e*-log-p method).
Example C–3b	Settlement of soft clay layer due to areal fill and isolated footing (*e*-log-p method).
Example C–3c	Time required to obtain 1 inch or less of settlement due to areal fill and foundation placed on soft clay (*e*-log-p method).
Example C–4	Recompression settlement of isolated footing on clay (*e*-log-p method).
Example C–5	Settlement of isolated footing on clay (PMT method).
Example C–6a	Settlement of isolated footing on mixed soils (DMT Incremental Constrained Modulus method).
Example C–6b	Comparison of dilatometer constrained modulus M_{DMT} with laboratory oedometer constrained modulus M.
Example C–7a	Settlement of isolated footing on sand (DMT Incremental Constrained Modulus method).
Example C–7b	Settlement of isolated footing on sand (Ménard and Rousseau PMT method).
Example C–7c	Settlement of isolated footing on sand (Schmertmann's method for cohesionless soils).
Example C–7d	Settlement of isolated footing on sand (Burland and Burbidge SPT method for sands and gravels).

C–1. Example C–1. Settlement of isolated footing on clay (e-log-p method)

a. Determine: Estimate settlement of an isolated footing on clay using the *e*-log-p method.

b. Given: A 7.0-ft by 7.0-ft isolated spread footing supporting a 20-ton column load bearing 2.0 ft below grade on the clay profile illustrated in Figure C–1.

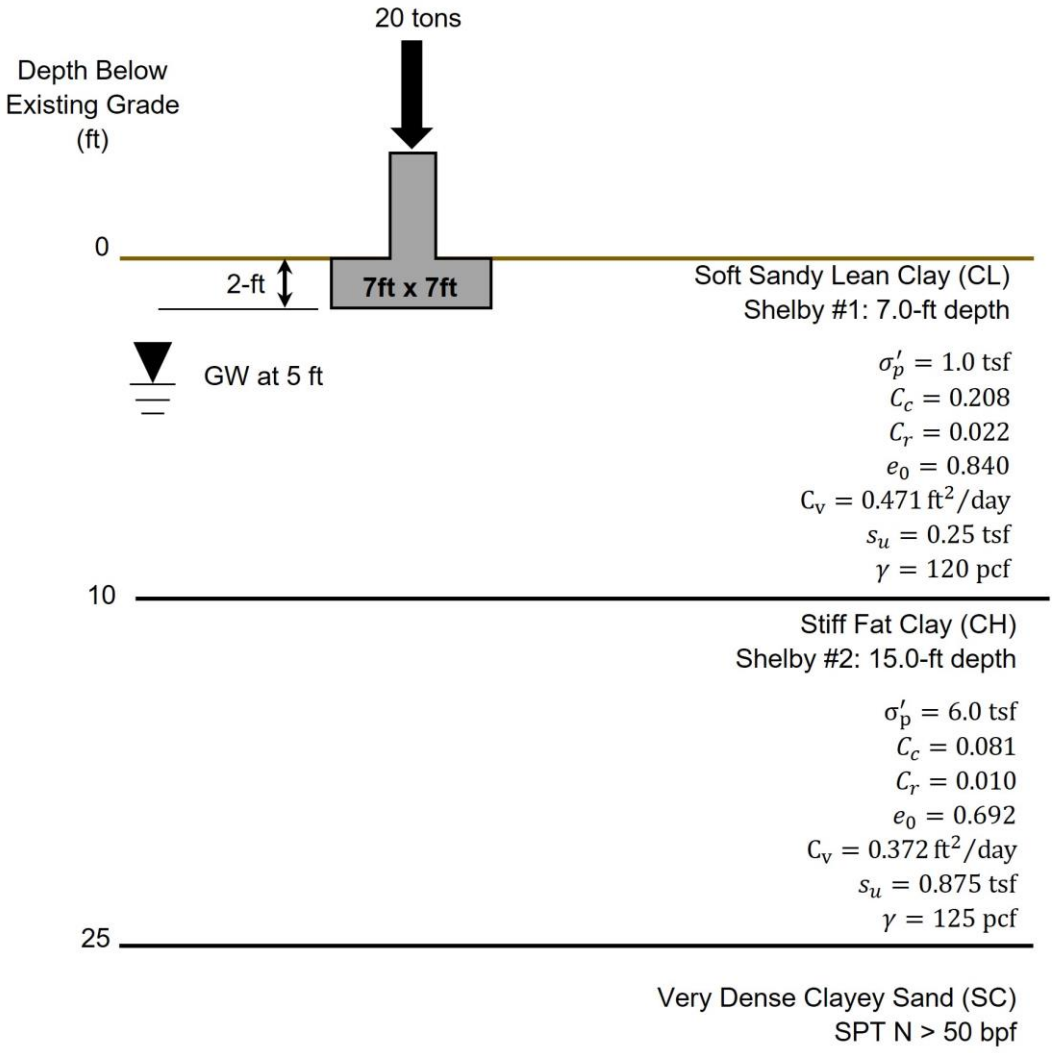


Figure C–1. Foundation and soil conditions for Example C–1

Table C–1
Solution for settlement of isolated footing on clay (*e*-log-p method)

Step 1	e_0, σ_p', C_c, and C_r are given for each soil strata as illustrated in Figure C–1. Compute σ_{z0}' and σ_m' for each Shelby tube sample. paragraph 6–5a: $\sigma_{z0}' = \sum \gamma \cdot \mathrm{H} - u$ Eq. 6–4: $\sigma_m' = \sigma_p' - \sigma_{z0}'$ Shelby #1: 7.0-ft depth (Strata 1): $\sigma_{z0}' = \frac{(7.0\ \text{ft} \cdot 120\ \text{pcf}) - (2.0\ \text{ft} \cdot 62.4\ \text{pcf})}{2{,}000} = 0.36\ \text{tsf}$ $\sigma_m' = 1.0\ \text{tsf} - 0.36\ \text{tsf} = 0.64\ \text{tsf}$ Shelby #2: 15.0-ft depth (Strata 2): $\sigma_{z0}' = \frac{(10.0\ \text{ft} \cdot 120\ \text{pcf}) + (5.0 \cdot 125\ \text{pcf}) - (10.0\ \text{ft} \cdot 62.4\ \text{pcf})}{2{,}000} = 0.60\ \text{tsf}$ $\sigma_m' = 6.0\ \text{tsf} - 0.60\ \text{tsf} = 5.40\ \text{tsf}$
Step 2	Divide the soil beneath the foundation into layers where the soil parameters are constant. The top layer should be at the base of the foundation and the bottom layer should be at a minimum depth of two times the footing width $2B$ below the base of the isolated spread footing, or until the increase in the induced vertical stress ($\Delta\sigma_z$) is negligible, resulting in small acceptable increases in settlement, whichever is deeper. Therefore, depth of bottom layer below grade = $D + (2B) = 2.0\ \text{ft} + (2 \cdot 7.0\ \text{ft}) = 16\ \text{ft}$.
Step 3	Compute initial vertical effective stress (σ_{z0}') at the midpoint of each layer per paragraph 6–5a. $\sigma_{z0}' = \sum \gamma \cdot \mathrm{H} - u$
Step 4	From the overionization margin (σ_m') computed for each soil strata, compute the preconsolidation stress (σ_p') at the midpoint of each layer per Eq. 6–4. $\sigma_p' = \sigma_m' + \sigma_{z0}'$
Step 5	Compute the gross bearing pressure q (or q_{eq}),at the base of the footing per paragraph 5–2b. (The ultimate bearing capacity with a factor of safety of 3 exceeds the gross bearing pressure as illustrated in Appendix B, Example B–1. $q = \frac{Q + W_f}{A} - u_d = \frac{20\ \text{T} + \left[\frac{150\ \text{pcf}}{2{,}000} \cdot (7.0\ \text{ft})^2 \cdot 2.0\ \text{ft}\right]}{(7.0\ \text{ft})^2} - 0 = 0.56\ \text{tsf}$
Step 6	Compute vertical effective stress at the bearing elevation of the footing (σ_{zD}') $\sigma_{zD}' = \frac{(2.0\ \text{ft} \cdot 120\ \text{pcf})}{2{,}000} = 0.12\ \text{tsf}$

<table>
<tr><td>Step 7</td><td>Compute the influence factor (I_σ) and the corresponding induced vertical stress ($\Delta\sigma_z$) at the midpoint of each layer per paragraph 6–7.
Eq. 6–6: $\Delta\sigma_z = I_\sigma(q - \sigma'_{zD})$
Eq. 6-9: $I_\sigma = 1 - \left(\frac{1}{1+\left(\frac{B}{2z}\right)^2}\right)^{1.76}$
<table><tr><th>Depth Below Existing Ground (ft)</th><th>Depth Below Footing, z (ft)</th><th>I_σ</th><th>$\Delta\sigma_z = I_\sigma(q - \sigma'_{zD})$
$\Delta\sigma_z = I_\sigma(0.558 - 0.12)$ (tsf)</th></tr><tr><td>2.5</td><td>0.5</td><td>1.0</td><td>0.44</td></tr><tr><td>4.0</td><td>2.0</td><td>0.92</td><td>0.40</td></tr><tr><td>6.0</td><td>4.0</td><td>0.63</td><td>0.28</td></tr><tr><td>8.5</td><td>6.5</td><td>0.36</td><td>0.16</td></tr><tr><td>11.5</td><td>9.5</td><td>0.20</td><td>0.09</td></tr><tr><td>14.5</td><td>12.5</td><td>0.12</td><td>0.05</td></tr></table>Note: Eq. 6–9 is used in this example to determine I_σ. Other methods could also be used to determine I_σ as explained in paragraphs 6–7, 6–8, and 6–9.</td></tr>
<tr><td>Step 8</td><td>Compute final applied vertical effective stress (σ'_{zf}) at the midpoint of each layer.
$\sigma'_{zf} = \sigma'_{z0} + \Delta\sigma_z$</td></tr>
</table>

Step 9

The soil layers are overconsolidated $\sigma'_{zf} < \sigma'_p$. Therefore, estimate settlement for each layer per Eq. 7–8. Sum the settlements for each layer within the zone of influence of $2B$ beneath the footing bearing elevation per Step 2.

Eq. 7–8: Overconsolidated soil where $\sigma'_{zf} < \sigma'_p$

$$\delta_c = \sum \frac{C_r}{1+e_0} \mathrm{H} \cdot \log\left(\frac{\sigma'_{zf}}{\sigma'_{z0}}\right)$$

Layer No.	Midpoint of Soil Layer					
	Layer Thickness (ft)	Depth Below Existing Ground (ft)	Depth Below Footing, z (ft)	$\frac{C_c}{1+e_o}$	$\frac{C_r}{1+e_o}$	σ'_{zo} (tsf) Step 3
1	1	2.5	0.5	0.113	0.012	0.15
2	2	4.0	2.0	0.113	0.012	0.24
3	2	6.0	4.0	0.113	0.012	0.33
4	3	8.5	6.5	0.113	0.012	0.40
5	3	11.5	9.5	0.048	0.006	0.49
6	3	14.5	12.5	0.048	0.006	0.59

Layer No.	Midpoint of Soil Layer					δ_c (ft) Step 9
	σ'_m (tsf)	σ'_p Step 4	$\Delta\sigma_z$ Step 7	σ'_{zf} Step 8	Case Step 9	
1	0.64	0.79	0.44	0.59	Eq. 7–8	0.0071
2	0.64	0.88	0.40	0.64	Eq. 7–8	0.0102
3	0.64	0.97	0.28	0.61	Eq. 7–8	0.0064
4	0.64	1.04	0.16	0.56	Eq. 7–8	0.0053
5	5.40	5.89	0.09	0.58	Eq. 7–8	0.0013
6	5.40	5.99	0.05	0.64	Eq. 7–8	0.0006

$\sum \delta_c$ = 0.0309 ft (0.37 inches)

C–2. Example C–2. Settlement due to superimposed foundation loads on clay (e-log-p method)

a. Determine: Estimate the primary consolidation settlement at points B and E beneath the center of both footings using the principle of superposition as described in paragraph 6–6e(2) and Figure 6–8.

b. *Given*: Reference the soil profile illustrated in Figure C–2a. In addition to constructing the 7.0-ft x 7.0-ft isolated spread footing in Example 1, a 2.0-ft-wide strip footing with a loading of 0.6 tons per linear foot is to be constructed adjacent to the isolated spread footing as illustrated in Figures C–2b. Both footings are to bear 2.0 ft below grade.

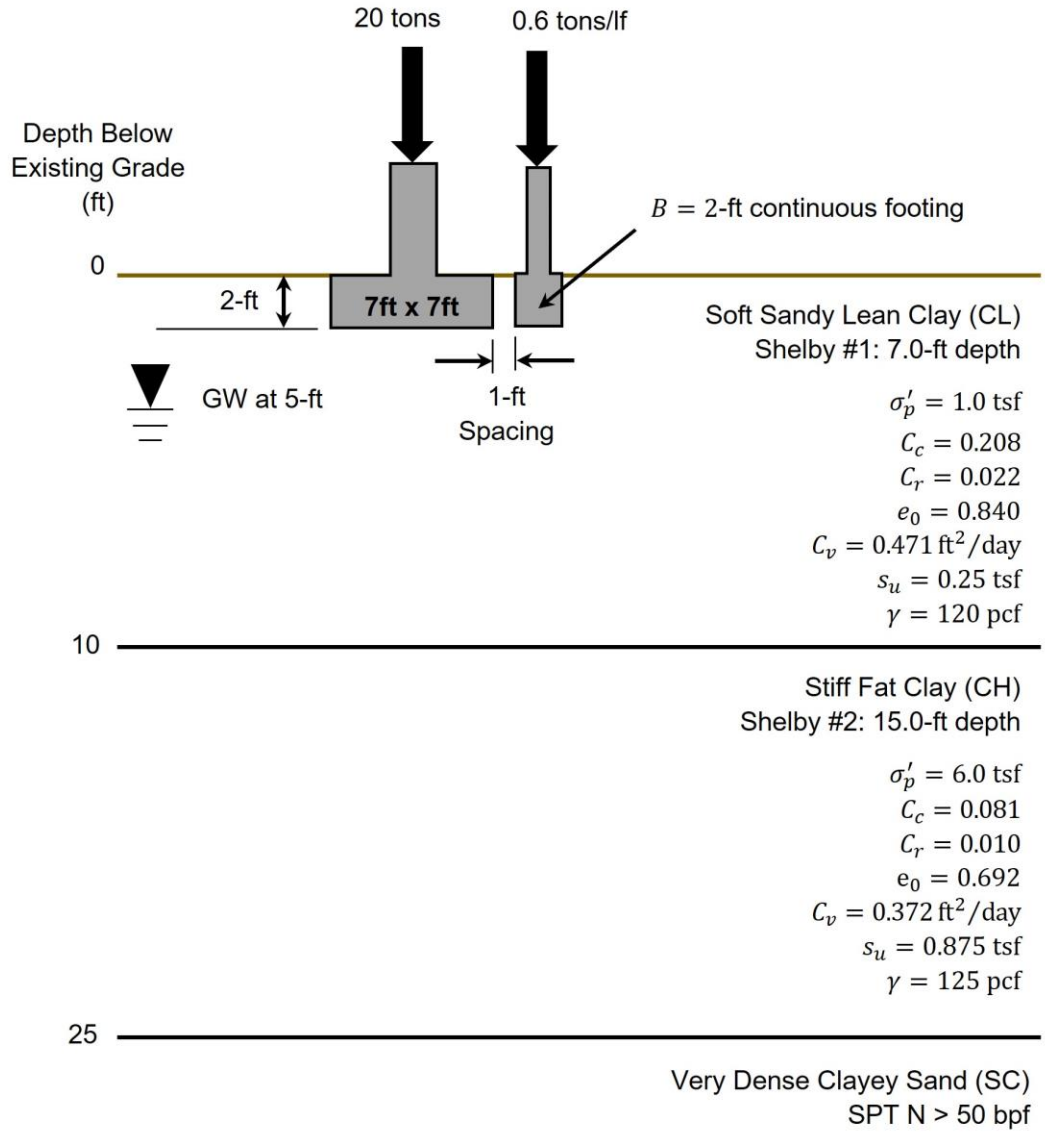


Figure C–2a. Foundation and soil conditions for Example C–2

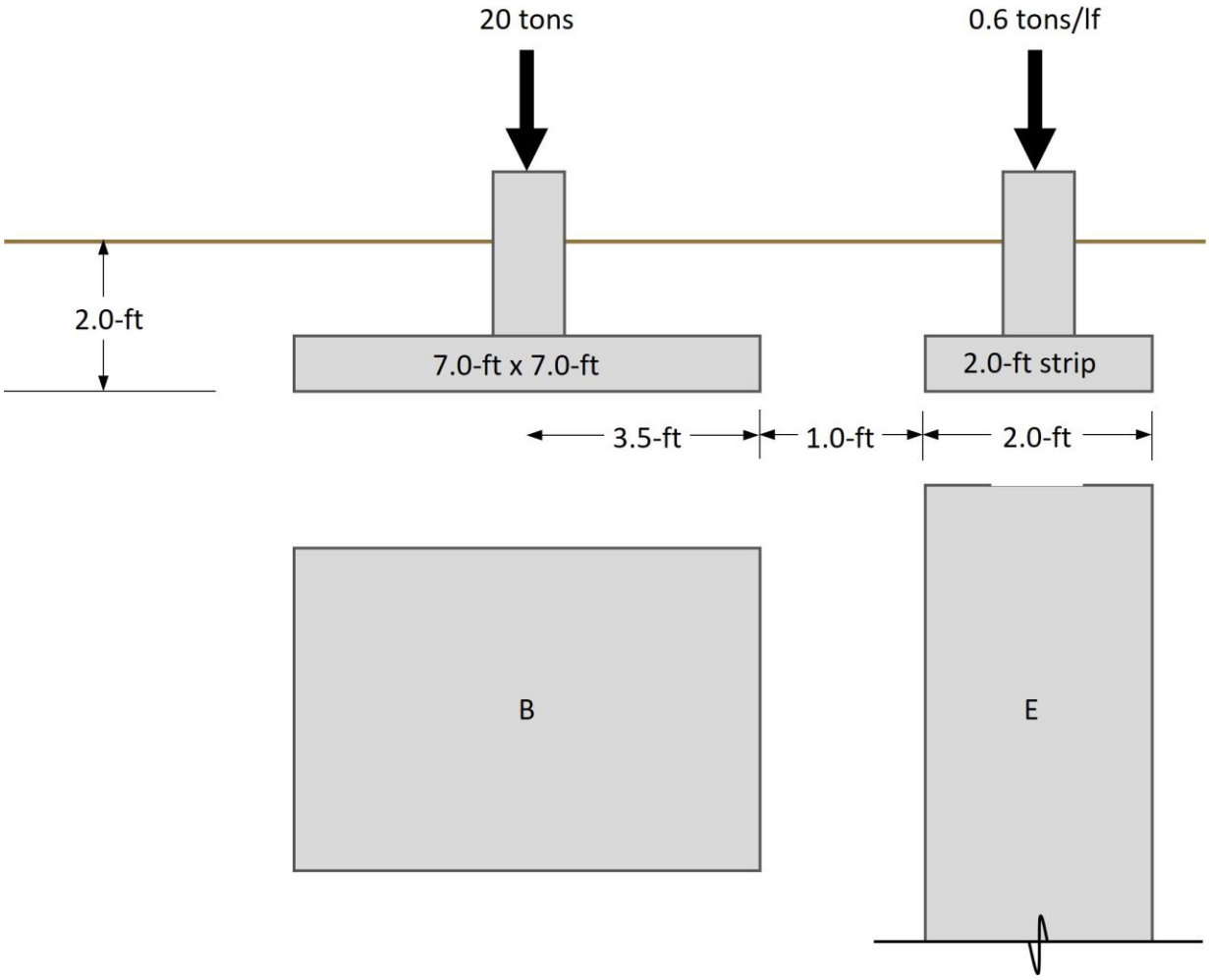


Figure C–2b. Foundation and soil conditions for Example C–2

Table C–2
Solution for settlement due to superimposed foundation loads on clay (*e*-log-p method)

Step 1	e_0, σ'_p, C_c, and C_r are provided for each soil strata as illustrated in Figure C–1. Compute σ'_{z0} and σ'_m for each Shelby tube sample. paragraph 6–5a: $\sigma'_{z0} = \sum \gamma \cdot \mathrm{H} - u$, and Eq. 6–4: $\sigma'_m = \sigma'_p - \sigma'_{z0}$ Shelby #1: 7.0-ft depth (Strata 1): $\sigma'_{z0} = \frac{(7.0\ \mathrm{ft} \cdot 120\ \mathrm{pcf}) - (2.0\ \mathrm{ft} \cdot 62.4\ \mathrm{pcf})}{2{,}000} = 0.36\ \mathrm{tsf}$ $\sigma'_m = 1.0\ \mathrm{tsf} - 0.36\ \mathrm{tsf} = 0.64\ \mathrm{tsf}$ Shelby #2: 15.0-ft depth (Strata 2): $\sigma'_{z0} = \frac{(10.0\ \mathrm{ft} \cdot 120\ \mathrm{pcf}) + (5.0\ \mathrm{ft} \cdot 125\ \mathrm{pcf}) - (10.0\ \mathrm{ft} \cdot 62.4\ \mathrm{pcf})}{2{,}000} = 0.60\ \mathrm{tsf}$ $\sigma'_m = 6.0\ \mathrm{tsf} - 0.60\ \mathrm{tsf} = 5.40\ \mathrm{tsf}$
Step 2	Divide the soil beneath the foundation into layers where the soil parameters are constant. The top layer should be at the base of the foundation and the bottom layer should be at a minimum depth of two times the footing width $2B$ below the base of the isolated spread footing and six times the footing width $6B$ below the base of strip footings, or until the increase in the induced vertical stress ($\Delta\sigma_z$) is negligible, resulting in small acceptable increases in settlement, whichever is deeper. Therefore, depth of bottom layer below grade: 1. Isolated spread footing: $D + (2B) = 2.0\ \mathrm{ft} + (2 \cdot 7.0\ \mathrm{ft}) = 16\ \mathrm{ft}$ 2. Strip footing: $D + (6B) = 2.0\ \mathrm{ft} + (6 \cdot 2.0\ \mathrm{ft}) = 14.0\ \mathrm{ft}$
Step 3	Compute initial vertical effective stress (σ'_{z0}) at the midpoint of each layer per paragraph 6–5a. $\sigma'_{z0} = \sum \gamma \cdot \mathrm{H} - u$
Step 4	From the overionization margin (σ'_m) computed for each soil strata, compute the preconsolidation stress (σ'_p) at the midpoint of each layer per Eq. 6–4. $\sigma'_p = \sigma'_m + \sigma'_{z0}v$
Step 5	Compute the gross bearing pressure q (or q_{eq}) at the base of both footings per paragraph 5–2b. The ultimate bearing capacity with a factor of safety of 3 exceeds the gross bearing pressure as illustrated in Appendix B, Examples B–1 and B–2. 1. Isolated spread footing: $q = \frac{Q + W_f}{A} - u_d = \frac{20\ \mathrm{T} + \left[\frac{150\ \mathrm{pcf}}{2{,}000} \cdot (7.0\ \mathrm{ft})^2 \cdot 2.0\ \mathrm{ft}\right]}{(7.0\ \mathrm{ft})^2} - 0 = 0.56\ \mathrm{tsf}$ 2. Strip footing: $q = \frac{Q + W_f}{B} - u_d = \frac{0.6\ \mathrm{T} + \left[\frac{150\ \mathrm{pcf} \cdot 2.0\ \mathrm{ft} \cdot 2.0\ \mathrm{ft}}{2{,}000}\right]}{(2.0\ \mathrm{ft})^2} - 0 = 0.45\ \mathrm{tsf}$

Step 6	Compute the vertical effective stress at the bearing elevation of the footing (σ'_{zD}) $\sigma'_{zD} = \frac{(2.0 \text{ ft} \cdot 120 \text{ pcf})}{2{,}000} = 0.12 \text{ tsf}$
Step 7a	Compute the influence factor (I_σ) and the corresponding induced vertical stress ($\Delta\sigma_z$) beneath the center of the isolated spread footing at point B from the loading of the isolated spread footing plus the loading from the adjacent strip footing using the principle of superposition per Figure C–2c, paragraph 6–6e(2), and Figure 6–8.

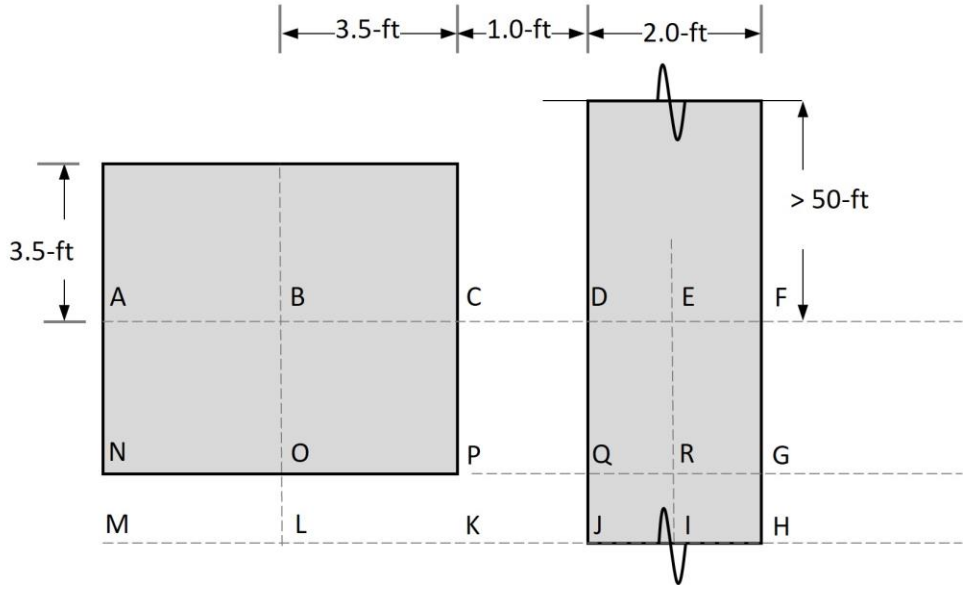


Figure C–2c. Induced stress below point B by method of superposition

1. The induced vertical stress ($\Delta\sigma_z$) from the isolated spread footing below point B from the isolated spread footing is calculated from the rectangle area (ABON) and multiplied by 4.

Induced Stress from Spread Footing per Figure 6–8				
Depth Below Footing, z (ft)	$B = L = 3.5$ ft		I_σ	$\Delta\sigma_z = 4I_\sigma(q - \sigma'_{ZD})$ $= 4I_\sigma(0.558 - 0.12)$ tsf
	$B/_Z = m$	$L/_Z = n$		
0.5	7.0	7.0	0.249	0.44
2.0	1.75	1.75	0.226	0.40
4.0	0.88	0.88	0.158	0.28
6.5	0.54	0.54	0.093	0.16
9.5	0.37	0.37	0.053	0.09
12.5	0.28	0.28	0.033	0.06

2. The induced vertical stress ($\Delta\sigma_z$) from the adjacent strip footing below point B is calculated from the difference of the rectangle areas $[(\mathrm{BFHL}) - (\mathrm{BDJL})]$ and multiplied by 2.

Induced Stress from Adjacent Strip Footing per Figure 6–8						
Depth Below Footing, z (ft)	$B = 6.5$ ft; $L \gg B \gg z$		$B = 4.5$ ft; $L \gg B \gg z$		$I_{\sigma net} = I_{\sigma 1} - I_{\sigma 2}$	$\Delta\sigma_z = 2I_\sigma(q - \sigma'_{zD})$ $= 2I_{\sigma net}(0.45 - 0.12)$ tsf
	$B/z = m$	$I_{\sigma 1}$	$B/z = m$	$I_{\sigma 2}$		
0.5	13.0	0.250	9.0	0.250	0	0
2.0	3.25	0.247	2.25	0.243	0.004	0.003
4.0	1.63	0.233	1.13	0.213	0.020	0.013
6.5	1.00	0.205	0.69	0.171	0.034	0.022
9.5	0.68	0.170	0.47	0.132	0.038	0.025
12.5	0.52	0.141	0.36	0.106	0.035	0.023

3. The total induced vertical stress ($\Delta\sigma_z$) below point B from the isolated spread and adjacent strip footings.

Depth Below Footing, z (ft)	Spread Footing $\Delta\sigma_z = 4I_\sigma(q - \sigma'_{zD})$ $= 4I_\sigma(0.558 - 0.12)$ tsf	Adjacent Strip Footing $\Delta\sigma_z = 2I_\sigma(q - \sigma'_{zD})$ $= 2I_{\sigma net}(0.45 - 0.12)$ tsf	Spread + Adjacent Strip Footing $\sum\Delta\sigma_z$ tsf
0.5	0.44	0	0.44
2.0	0.40	0.003	0.40
4.0	0.28	0.013	0.29
6.5	0.16	0.022	0.18
9.5	0.09	0.025	0.12
12.5	0.06	0.023	0.18

Step 7b

Compute the influence factor (I_σ) and the corresponding induced vertical stress ($\Delta\sigma_z$) beneath the center of the strip footing at point E from the loading of the strip footing plus the loading from the adjacent isolated spread footing using the principle of superposition per Figure C–2c, paragraph 6–6e(2), and Figure 6–8.

1. The induced vertical stress ($\Delta\sigma_z$) from the strip footing below point E from the strip footing is calculated from the rectangle area (EFHI) and multiplied by 4.

Induced Stress from Strip Footing per Figure 6–8			
Depth Below Footing, z (ft)	$B = 1.0$ ft		$\Delta\sigma_z = 4I_\sigma(q - \sigma'_{ZD})$ $= 4I_\sigma(0.45 - 0.12)$ tsf
	$B/_Z = m$	I_σ	
0.5	2.00	0.240	0.32
2.0	0.50	0.137	0.18
4.0	0.25	0.076	0.10
6.5	0.15	0.048	0.06
9.5	0.11	0.033	0.04
12.5	0.08	0.025	0.03

2. The induced vertical stress ($\Delta\sigma_z$) from the adjacent isolated spread footing below point E is calculated from the difference of the rectangle areas $[(\mathrm{AERN}) - (\mathrm{CERP})]$ and multiplied by 2.

Induced Stress from Adjacent Spread Footing per Figure 6–8							
Layer No.	Depth Below Footing, z (ft)	$B = 9.0$ ft; $L = 3.5$ ft			$B = 2.0$ ft; $L = 3.5$ ft		
		$B/_Z = m$	$L/_Z = n$	$I_{\sigma 1}$	$B/_Z = m$	$L/_Z = n$	$I_{\sigma 1}$
1	0.5	18.0	7.0	0.250	4.0	7.0	0.248
2	2.0	4.5	1.75	0.236	1.0	1.75	0.198
3	4.0	2.25	0.88	0.190	0.50	0.88	0.115
4	6.5	1.39	0.54	0.137	0.31	0.54	0.060
5	9.5	0.95	0.37	0.093	0.21	0.37	0.032
6	0.72	0.72	0.28	0.065	0.16	0.28	0.020

Layer No.	Depth Below Footing, z (ft)	$I_{\sigma net} = I_{\sigma 1} - I_{\sigma 2}$	$\Delta\sigma_z = 2I_\sigma(q - \sigma'_{zD})$ $= 2I_{\sigma net}(0.558 - 0.12)$tsf
1	0.5	0.002	0.002
2	2.0	0.038	0.03
3	4.0	0.075	0.07
4	6.5	0.077	0.07
	0.72	0.045	0.04

3. The total induced vertical stress ($\Delta\sigma_z$) below point E from the strip and adjacent isolated spread footings.

Depth Below Footing, z (ft)	Adjacent Strip Footing $\Delta\sigma_z = 4I_\sigma(q - \sigma'_{zD})$ $= 4I_{\sigma net}(0.45 - 0.12)$tsf	Spread Footing $\Delta\sigma_z = 2I_\sigma(q - \sigma'_{ZD})$ $= 2I_\sigma(0.558 - 0.12)$ tsf	Strip + Adjacent Spread Footing $\sum\Delta\sigma_z$ tsf
0.5	0.32	0.002	0.32
2.0	0.18	0.03	0.21
4.0	0.10	0.07	0.17
6.5	0.06	0.07	0.13
9.5	0.04	0.05	0.09
12.5	0.03	0.04	0.07

Step 8a

Compute the final vertical effective stress (σ'_{zf}) at the midpoint of each layer beneath the isolated spread footing.

$\sigma'_{zf} = \sigma'_{z0} + \Delta\sigma_z$

where:

σ'_{zo} from Step 3

$\Delta\sigma_z$ from Step 7A

Step 8b

Compute the final vertical effective stress (σ'_{zf}) at the midpoint of each layer beneath the strip footing.

$\sigma'_{zf} = \sigma'_{z0} + \Delta\sigma_z$

where:

σ'_{z0} from Step 3

$\Delta\sigma_z$ from Step 7B

Step 9

Estimate the settlement for each layer beneath the isolated spread and strip footings. Soils are overconsolidated $\sigma'_{zf} < \sigma'_p$. Therefore, estimate settlement for each layer per Eq. 7–8 and sum the settlements of all the layers within the zone of influence beneath the footings per Step 2.

Eq. 7–8: Overconsolidated soil, where $\sigma'_{zf} < \sigma'_p$

$$\delta_c = \sum \frac{C_r}{1+e_0} \mathrm{H} \cdot \log\left(\frac{\sigma'}{\sigma'_{z0}}\right)$$

1. Settlement beneath isolated spread footing:

Layer No.	Midpoint of Soil Layer					
	Layer Thickness (ft)	Depth Below Existing Ground (ft)	Depth Below Footing, z (ft)	$\frac{C_c}{1+e_o}$	$\frac{C_r}{1+e_o}$	σ'_{zo} (tsf) Step 3
1	1	2.5	0.5	0.113	0.012	0.15
2	2	4.0	2.0	0.113	0.012	0.24
3	2	6.0	4.0	0.113	0.012	0.33
4	3	8.5	6.5	0.113	0.012	0.40
5	3	11.5	9.5	0.048	0.006	0.49
6	3	14.5	12.5	0.048	0.006	0.59

Layer No.	Midpoint of Soil Layer					
	σ'_m (tsf)	σ'_p Step 4	$\Delta\sigma_z$ Step 7A	σ'_{zf} Step 8A	Case Step 9	δ_c (ft) Step 9
1	0.64	0.79	0.44	0.59	Eq. 7–8	0.0071
2	0.64	0.88	0.40	0.64	Eq. 7–8	0.0102
3	0.64	0.97	0.29	0.62	Eq. 7–8	0.0066
4	0.64	1.04	0.18	0.58	Eq. 7–8	0.0058
5	5.40	5.89	0.12	0.61	Eq. 7–8	0.0017
6	5.40	5.99	0.08	0.67	Eq. 7–8	0.001
$\sum \delta_c$ = 0.0324 ft (0.39 inches)						

2. Settlement beneath strip footing:

Layer No.	Midpoint of Soil Layer					
	Layer Thickness (ft)	Depth Below Existing Ground (ft)	Depth Below Footing, z (ft)	$\frac{C_c}{1+e_o}$	$\frac{C_r}{1+e_o}$	σ'_{zo} (tsf) Step 3
1	1	2.5	0.5	0.113	0.012	0.15
2	2	4.0	2.0	0.113	0.012	0.24
3	2	6.0	4.0	0.113	0.012	0.33
4	3	8.5	6.5	0.113	0.012	0.40
5	3	11.5	9.5	0.048	0.006	0.49
6	3	14.5	12.5	0.048	0.006	0.59

Layer No.	Midpoint of Soil Layer					
	σ'_m (tsf)	σ'_p Step 4	$\Delta\sigma_z$ Step 7B	σ'_{zf} Step 8B	Case Step 9	δ_c (ft) Step 9
1	0.64	0.79	0.32	0.47	Eq. 7–8	0.0060
2	0.64	0.88	0.21	0.45	Eq. 7–8	0.0066
3	0.64	0.97	0.17	0.50	Eq. 7–8	0.0043
4	0.64	1.04	0.13	0.53	Eq. 7–8	0.0044
5	5.40	5.89	0.09	0.58	Eq. 7–8	0.0013
6	5.40	5.99	0.07	0.66	Eq. 7–8	0.0009

$\sum \delta_c$ = 0.0235 ft (0.28 inches)

C–3. Example C–3. Settlement due to isolated footing and areal fill on soft clay and surcharge duration

a. Determine:

(1) Settlement of clay layer due to areal fill loading only per Table 7–3 and paragraph 7–16.

(2) Settlement of the soft clay layer due to areal fill plus footing loading per Table 7–3 and paragraph 7–16b. Assume within-fill settlements are negligible due to material type and compaction control.

(3) Time required after placement of 8 ft of fill before the footing can be constructed within the fill to obtain settlements (footing plus fill) less than 1.0 inch per Table 7–11.

b. Given: Prior to constructing the footings, site grades are raised 8.0 ft by placing areal fill throughout the entire site, including beneath the building (see Figure C–3). The fill material placed beneath the building and 10 ft beyond the building limits consists of well-compacted, engineered fill (assume structural fill will not settle due to the induced loadings from the footings). Estimate the following:

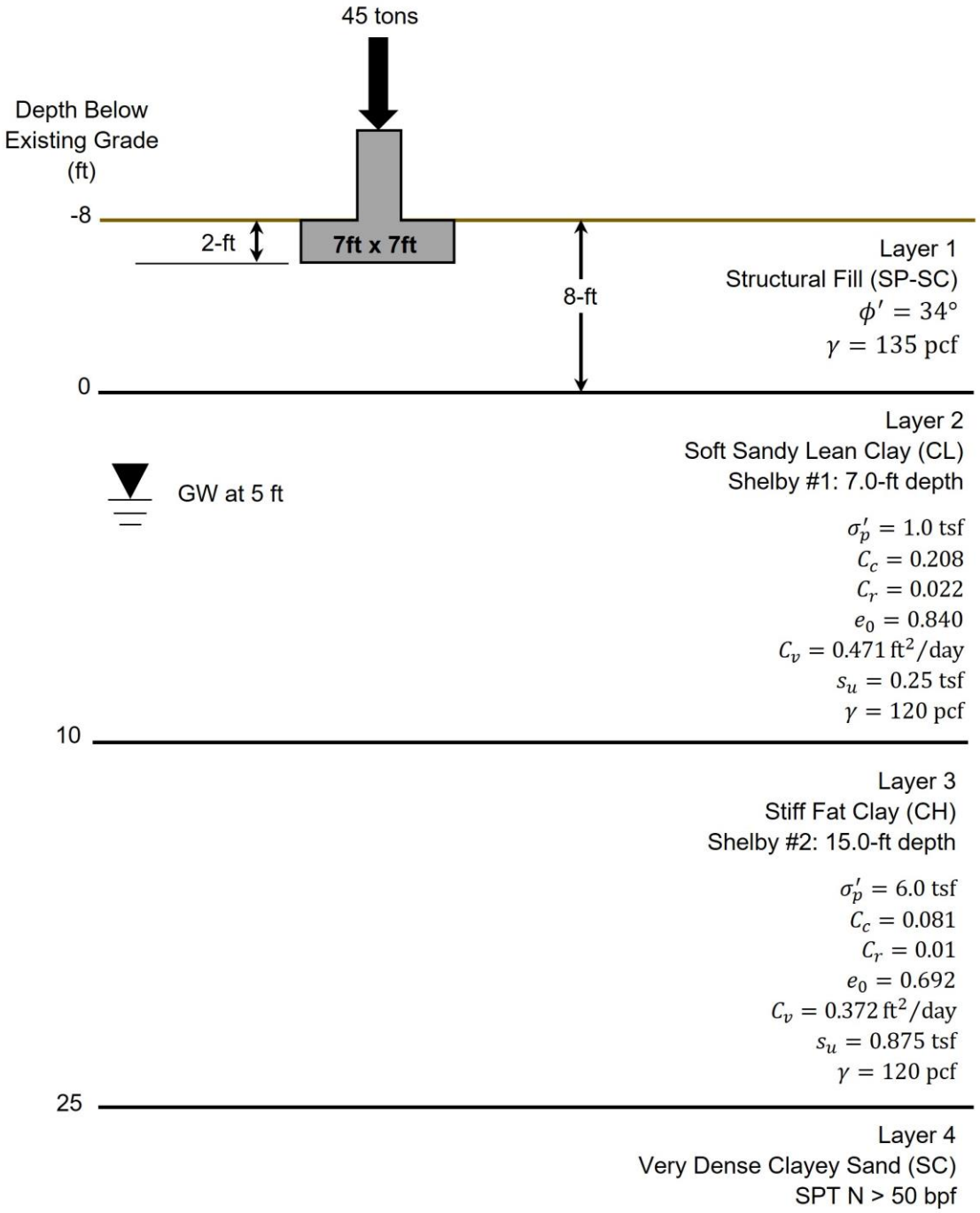


Figure C–3. Foundation and soil conditions for Example C–3

Table C–3a
Solution for settlement of soft clay layer due to areal fill loading only

Step 1	e_0, σ_p', C_c, and C_r are provided for each soil strata as illustrated in Figure C–3. Compute σ_{z0}' and σ_m' for each Shelby tube sample. paragraph 6–5a: $\sigma_{z0}' = \sum \gamma \cdot \mathrm{H} - u$ Eq. 6–4: $\sigma_m' = \sigma_p' - \sigma_{z0}'$ Shelby #1: 7.0-ft depth (Strata 1): $\sigma_{z0}' = \frac{(7.0\ \mathrm{ft} \cdot 120\ \mathrm{pcf}) - (2.0\ \mathrm{ft} \cdot 62.4\ \mathrm{pcf})}{2{,}000} = 0.36\ \mathrm{tsf}$ $\sigma_m' = 1.0\ \mathrm{tsf} - 0.36\ \mathrm{tsf} = 0.64\ \mathrm{tsf}$ Shelby #2: 15.0-ft depth (Strata 2): $\sigma_{zo}' = \frac{(10.0\ \mathrm{ft} \cdot 120\ \mathrm{pcf}) + (5.0\ \mathrm{ft} \cdot 125\ \mathrm{pcf}) - (10.0\ \mathrm{ft} \cdot 62.4\ \mathrm{pcf})}{2{,}000} = 0.60\ \mathrm{tsf}$ $\sigma_m' = 6.0\ \mathrm{tsf} - 0.60\ \mathrm{tsf} = 5.4\ \mathrm{tsf}$
Step 2	Divide the soil beneath the foundation into layers where the soil parameters are constant. The top layer should be at the base of the areal fill and the bottom layer should extend to an incompressible layer, or until the increase in the induced vertical stress $\Delta\sigma_{zfill}$ is negligible, resulting in small acceptable increases in settlement, whichever is deeper. Settlements will be estimated to a depth of 25 ft below existing grade, where a very dense clayey sand is encountered.
Step 3	Compute initial vertical effective stress (σ_{z0}') at the midpoint of each layer per paragraph 6–5a. $\sigma_{z0}' = \sum \gamma \cdot \mathrm{H} - u$
Step 4	From the overionization margin (σ_m') computed for each soil strata, compute the preconsolidation stress (σ_p') at the midpoint of each layer per Eq. 6–4. $\sigma_p' = \sigma_m' + \sigma_{z0}'$
Step 5	Compute the areal fill load (P) at the center line of the areal fill. $\mathrm{P} = \gamma_{m\,fill} \cdot \mathrm{H_{fill}} = \frac{135\ \mathrm{pcf} \cdot 8.0\ \mathrm{ft}}{2{,}000} = 0.54\ \mathrm{tsf}$
Step 6	Compute stress distribution ($\Delta\sigma_{zfill}$) beneath the centerline of the areal fill per Figure 6–12 and Eq. 6–7. Settlements due to fill are negligible below a depth of 25 ft (Layer 4) since the areal fill width and length are much greater than the values of depth (z) used in the analysis $b \gg 25$ ft, $I_\sigma \approx 0.5$ for all depths. Eq. 6–7: $\Delta\sigma_{zfill} = 2I_\sigma \cdot \mathrm{P} = 2 \cdot 0.5 \cdot 0.54\ \mathrm{tcf} = 0.54\ \mathrm{tsf}$ for depths < 25 ft per Step 2.
Step 7	Compute the final applied vertical effective stress (σ_{zf}') at the midpoint of each layer. $\sigma_{zf}' = \sigma_{z0}' + \Delta\sigma_{zfill} = \sigma_{z0}' + 0.54\ \mathrm{tcf}$

Step 8

Estimate the settlement for each layer beneath the areal fill. Soils are overconsolidated $\sigma'_{z0} < \sigma'_p$ and $\sigma'_{zfill} < \sigma'_p$ in all soil layers. Therefore, estimate settlement for each layer per Eq. 7–8 to a depth of 25 ft below existing grade, where a very dense clayey sand is encountered per Step 2.

Eq. 7–8: Overconsolidated soil, where $\sigma'_{zf} < \sigma'_p$.

$$\delta_c = \sum \frac{C_r}{1+e_0} \mathrm{H} \cdot \log\left(\frac{\sigma'_{zf}}{\sigma'_{z0}}\right)$$

Layer No.	At Midpoint of Soil Layer					
	Layer Thickness (ft)	Depth Below Existing Ground (ft)	$\frac{C_c}{1+e_o}$	$\frac{C_r}{1+e_o}$	σ'_{zo} (tsf) Step 3	σ'_m (tsf)
1	2	1.0	0.113	0.012	0.06	0.64
2	1	2.5	0.113	0.012	0.15	0.64
3	2	3.5	0.113	0.012	0.21	0.64
4	2	6.0	0.113	0.012	0.33	0.64
5	3	8.5	0.113	0.012	0.40	0.64
6	3	11.5	0.048	0.006	0.49	5.40
7	3	14.5	0.048	0.006	0.59	5.40
8	4	18.0	0.048	0.006	0.69	5.40
9	5	22.5	0.048	0.006	0.84	5.40

Layer No.	At Midpoint of Soil Layer				
	σ'_p (tsf) Step 4	I_σ Step 6	$\Delta\sigma_{zfill}$ Step 6	σ'_{zf} Step 7	δ_c (ft) Step 8
1	0.70	0.5	0.54	0.60	0.024
2	0.79	0.5	0.54	0.69	0.008
3	0.85	0.5	0.54	0.75	0.013
4	0.97	0.5	0.54	0.87	0.010
5	1.04	0.5	0.54	0.94	0.013
6	5.89	0.5	0.54	1.03	0.006
7	5.99	0.5	0.54	1.13	0.005
8	6.09	0.5	0.54	1.23	0.006
9	6.24	0.5	0.54	1.38	0.006

$\sum \delta_c$ = 0.092 ft (1.10 inches)

Table C–3b
Solution for settlement of the soft clay layer due to areal fill plus footing loading

Step 1	Same as Example C–3a.
Step 2	Divide the soil beneath the foundation into layers where the soil parameters are constant. 1. Footing: The top layer should be at the base of the foundation and the bottom layer should be at a minimum depth of two times the footing width ($2B$) below the base of the isolated spread footing, or until the increase in the induced vertical stress $\Delta\sigma_z$ is negligible, resulting in small acceptable increases in settlement, whichever is deeper. Therefore, settlements will be estimated to a minimum depth of $2B$ = 14 ft below the bearing elevation. Since the structural fill is assumed to not settle due to the induced loadings from the footings, settlements from the footing will be significant only to a depth of approximately 8 ft (14 ft to 6 ft) below existing grade, where 6 ft is the depth of structural fill beneath footing. 2. Fill: The top layer should be at the base of the areal fill and the bottom layer should extend to an incompressible layer, or until the increase in the induced vertical stress $\Delta\sigma_{zfill}$ is negligible, resulting in small acceptable increases in settlement, whichever is deeper. Therefore, settlements will be estimated to a depth of 25 ft below existing grade, where a very dense clayey sand is encountered.
Step 3 & 4	Same as Example C–3a above.
Step 5a	Compute the gross bearing pressure q (or q_{eq}) at the base of the footing per paragraph 5–2b. (The ultimate bearing capacity with a factor of safety of 3 exceeds the gross bearing pressure as illustrated in Appendix B, Example B–3. $q = \frac{Q + W_f}{A} - u_D = \frac{45\text{ T} + \left[\frac{150\text{ pcf}}{2{,}000} \cdot (7.0\text{ ft})^2 \cdot 2.0\text{ ft}\right]}{(7.0\text{ ft})^2} - 0 = 1.07\text{ tsf}$
Step 5b	Compute the areal fill load P at the center line of the areal fill. $P = \gamma_{m,fill} \cdot H_{fill} = \frac{135\text{ pcf} \cdot 8.0\text{ ft}}{2{,}000} = 0.54\text{ tsf}$
Step 6	Compute the vertical effective stress at the bearing elevation of the footing (σ'_{zD}). $\sigma'_{zD} = \frac{(2.0\text{ ft} \cdot 135\text{ pcf})}{2{,}000} = 0.14\text{ tsf}$

<table>
<tr><td>Step 7a</td><td>Compute the influence factor I_σ and the corresponding induced vertical stress ($\Delta\sigma_z$) at the midpoint of each layer per paragraph 6–7 or 6–8.
Eq. 6–6: $\Delta\sigma_z = I_\sigma(q - \sigma'_{zD})$
Eq. 6–9: $I_\sigma = 1 - \left(\frac{1}{1+\left({}^{B}/_{2z}\right)^2}\right)^{1.76}$
<table><tr><th>Depth Below Existing Ground (ft)</th><th>Depth Below Footing, z (ft)</th><th>I_σ</th><th>$\Delta\sigma_z = I_q(q - \sigma'_{zD})$
$\Delta\sigma_z = I_q(1.07 - 0.12)$ tsf</th></tr><tr><td>1.0</td><td>7.0</td><td>0.32</td><td>0.31</td></tr><tr><td>2.5</td><td>8.5</td><td>0.24</td><td>0.23</td></tr><tr><td>3.5</td><td>9.5</td><td>0.20</td><td>0.19</td></tr><tr><td>6.0</td><td>12.0</td><td>0.13</td><td>0.13</td></tr><tr><td>8.5</td><td>14.5</td><td>0.09</td><td>0.09</td></tr><tr><td>11.5</td><td>17.5</td><td>0.07</td><td>0.06</td></tr><tr><td>14.5</td><td>20.5</td><td>0.05</td><td>0.05</td></tr><tr><td>18.0</td><td>24.0</td><td>0.04</td><td>0.03</td></tr><tr><td>22.5</td><td>28.5</td><td>0.03</td><td>0.02</td></tr></table>Note: Eq. 6–9 is used in this example to determine I_σ. Other methods could also be used to determine I_σ as explained in paragraphs 6–7, 6–8, and 6–9.</td></tr>
<tr><td>Step 7b</td><td>Compute stress distribution ($\Delta\sigma_{zfill}$) beneath the centerline of the areal fill per Figure 6–12 and Eq. 6–7.
Settlements due to fill are negligible below a depth of 25 ft (Layer 4). Since the areal fill width (B) and length (L) are much greater than the maximum value of depth used in the analysis ($z = 25$ ft), $I_\sigma \approx 0.5$ for all depths.
Eq. 6–7: $\Delta\sigma_{zfill} = 2I_\sigma \cdot \mathrm{P} = 2 \cdot 0.5 \cdot 0.54$ tcf $= 0.54$ tsf to a depth of 25.0 ft per Step 2.</td></tr>
<tr><td>Step 8</td><td>Compute the final applied vertical effective stress (σ'_{zf}) at the midpoint of each layer.
$\sigma'_{zf} = \sigma'_{z0} + \Delta\sigma_z + \Delta\sigma_{zfill}$</td></tr>
</table>

Step 9

The soil layers are overconsolidated $\sigma'_{z0} < \sigma'_p$. Therefore, estimate settlement for each layer per Eq. 7–8 and 7–9 as applicable. Sum the settlements for each layer within the zone of influence of $2B$ beneath the footing bearing elevation and to the incompressible layer at 25 ft for the fill per Step 2.

Eq. 7–8: Overconsolidated soil, where $\sigma'_{zf} < \sigma'_p$

$$\delta_c = \sum \frac{C_r}{1+e_0} \mathrm{H} \cdot \log\left(\frac{\sigma'_{zf}}{\sigma'_{z0}}\right)$$

Eq. 7–9: Overconsolidated soil, where $\sigma'_{z0} < \sigma'_p < \sigma'_{zf}$

$$\delta_c = \sum \left[\frac{C_r}{1+e_0} \mathrm{H} \cdot \log\left(\frac{\sigma'_p}{\sigma'_{z0}}\right) + \frac{C_c}{1+e_0} \mathrm{H} \cdot \log\left(\frac{\sigma'_{zf}}{\sigma'_p}\right)\right]$$

Layer No.	Midpoint of Soil Layer					
	Layer Thickness (ft)	Depth Below Existing Ground (ft)	Depth Below Footing, z (ft)	$\frac{C_c}{1+e_o}$	$\frac{C_r}{1+e_o}$	σ'_{zo} (tsf) Step 3
1	2	1.0	7.0	0.113	0.012	0.06
2	1	2.5	8.5	0.113	0.012	0.15
3	2	3.5	9.5	0.113	0.012	0.21
4	2	6.0	12.0	0.113	0.012	0.33
5	3	8.5	14.5	0.113	0.012	0.40
6	3	11.5	17.5	0.048	0.006	0.49
7	3	14.5	20.5	0.048	0.006	0.59
8	4	18.0	24.0	0.048	0.006	0.69
9	5	22.5	28.5	0.048	0.006	0.84

Layer No.	Midpoint of Soil Layer						δ_c (ft) Step 9
	σ'_m (tsf)	σ'_p (tsf) Step 4	$\Delta\sigma_z$ (tsf) Step 7A	$\Delta\sigma'_{zfill}$ (tsf) Step 7B	σ'_{zf} (tsf) Step 8	Case Step 9	
1	0.64	0.70	0.31	0.54	0.91	Eq. 7–9	0.051
2	0.64	0.79	0.23	0.54	0.92	Eq. 7–9	0.016
3	0.64	0.85	0.19	0.54	0.94	Eq. 7–9	0.025
4	0.64	0.97	0.13	0.54	1.00	Eq. 7–9	0.014
5	0.64	1.04	0.09	0.54	1.03	Eq. 7–8	0.015
6	5.40	5.89	0.06	0.54	1.09	Eq. 7–8	0.006
7	5.40	5.99	0.05	0.54	1.18	Eq. 7–8	0.005
8	5.40	6.09	0.03	0.54	1.26	Eq. 7–8	0.006
9	5.40	6.24	0.02	0.54	1.40	Eq. 7–8	0.007

$\sum \delta_c$ = 0.145 ft (1.74 inches)

Table C–3c
Solution for time required to obtain 1 inch or less of settlement due to areal fill and foundation placed on soft clay (see Table 7–11)

Step 1	Estimate total primary consolidation (δ_c) of each compressible layer per Table 7–3.
Step 2	Coefficient of consolidation C_v provided in Figure C–3. $\bar{c_v} = \frac{\left(0.471\frac{\text{ft}^2}{\text{day}} \cdot 10\text{ft}\right) + \left(0.372\frac{\text{ft}^2}{\text{day}} \cdot 15\text{ft}\right)}{25\text{ft}} = 0.412\frac{\text{ft}^2}{\text{day}}$
Step 3	Compute percent consolidation (U%) of areal fill to obtain total consolidation settlements less than 1.0 inch. Total consolidation settlement from areal fill only per Example C–3a = 1.10 inch. Total consolidation settlement from areal fill plus footing per Example C–3b = 1.74 inch. To obtain total consolidation settlement from the fill plus footing loading $\leq 1.0\text{in}$, the soil beneath the fill is calculated to settle 0.74 inch prior to construction of the footing. Rearranging Eq. 7–41, the percent consolidation of the underlying soil from the induced stresses from the areal fill: $\text{U\%} = \frac{\delta_{ct}}{\delta_c} \times 100$ $\text{Therefore, U\%} = \frac{0.74}{1.10} \times 100 = 67\% \approx 70\%\ \text{consolidation due to fill}$ Per Figure 7–16, $T_v = 0.403$ for an average degree of consolidation of 70%.
Step 4	Compute time required to achieve 70% consolidation from areal fill Per Eq. 7–42. $t = \frac{T_v \cdot H_{dr}^2}{C_v} = \frac{0.403\left(\frac{25\,ft}{2}\right)^2}{0.412\,ft^2/day} = 152.8\,days$

C–4. Example C–4. Recompression settlement of isolated footing on clay (*e*-log-p method)

a. Determine: Estimate excavation rebound and subsequent recompression (primary consolidation) settlement of the building foundation at the center of a 34-ft by 34-ft-wide by 12-ft-deep excavation per Table 7–10.

b. Given: Reference Figure C–4a. A 12-ft-deep excavation plus the lowering of the groundwater table from 6 ft to 12 ft below grade is to be conducted in a normally consolidated clay prior to construction of the footing (the permanent groundwater level will return to 6 ft below grade following construction).

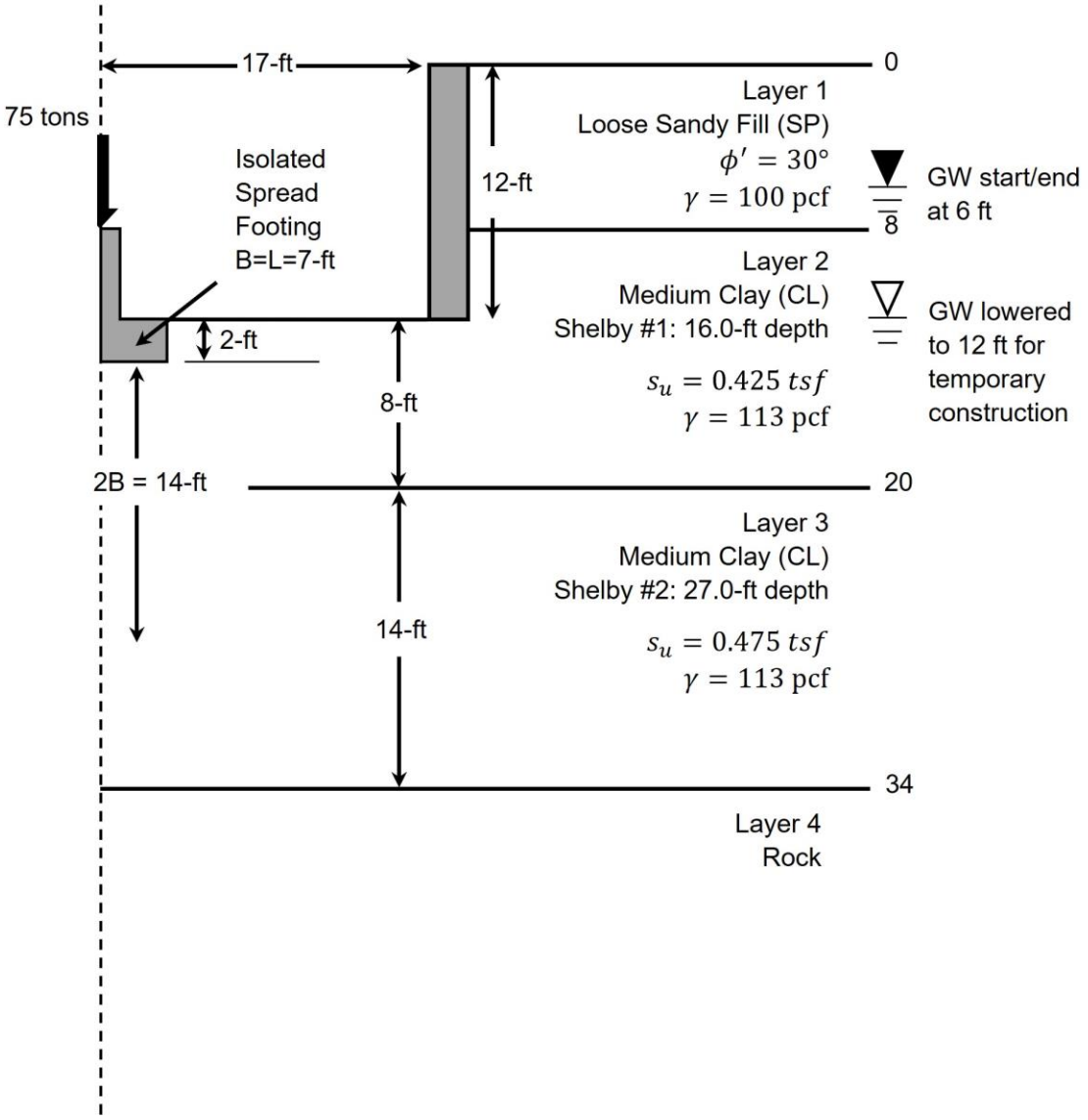
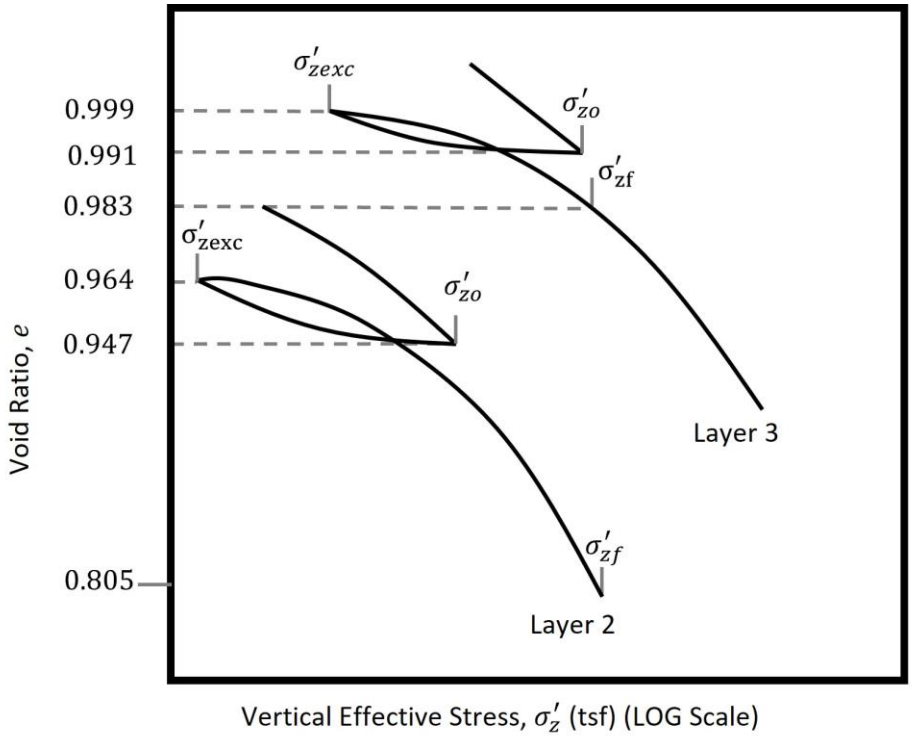


Figure C–4a. Foundation and soil conditions for Example C–4

Table C–4
Solution for recompression settlement of isolated footing on clay (*e*-log-p method)

Step 1	Conduct laboratory oedometer tests for the compressible layers. Reference Figure C–4b. 1. The first unload increment should occur at the initial vertical effective stress (σ'_{z0}) where the sample was obtained in the ground prior to the excavation and lowering of the groundwater table (per Step 2). The sample is then unloaded to the vertical effective stress following the excavation and lowering of groundwater table (σ'_{zexc}) per Step 6. 2. The soil is then reloaded beyond the final vertical effective stress (σ'_{zf}) per Step 11.



σ'_{zo} = initial vertical effective stress prior to excavation and dewatering
σ'_{zexc} = vertical effective stress, after excavation groundwater lowering (tsf)
σ'_{zf} = final vertical effective stress due to footing load

Figure C–4b. Plot of vertical effective stress and void ratio for layers 2 and 3

Step 2	Compute the initial vertical effective stress (σ'_{z0}) prior to the excavation and lowering of the groundwater table at the midpoint of each layer per paragraph 6–5a. Section 6–5a: $\sigma'_{z0} = \sum \gamma \cdot \mathrm{H} - u$ Layer 2: $\sigma'_{zo} = \frac{(8\ \mathrm{ft} \cdot 100\ \mathrm{pcf}) + (8.0\ \mathrm{ft} \cdot 113\ \mathrm{pcf}) - (10\ \mathrm{ft} \cdot 62.4\ \mathrm{pcf})}{2{,}000} = 0.54\ \mathrm{tsf}$ Layer 3: $\sigma'_{zo} = \frac{(8\ \mathrm{ft} \cdot 100\ \mathrm{pcf}) + (19.0\ \mathrm{ft} \cdot 113\ \mathrm{pcf}) - (21\ \mathrm{ft} \cdot 62.4\ \mathrm{pcf})}{2{,}000} = 0.82\ \mathrm{tsf}$

<table>
<tr><td>Step 3</td><td>Compute the effective vertical overburden stress (σ_z') after the groundwater table is lowered from 6 to 12 ft below grade prior to the excavation at points A and B per paragraph 6–5a.
Layer 2:
$\sigma'_{zgw} = \frac{(8\text{ ft} \cdot 100\text{ pcf}) + (8.0\text{ ft} \cdot 113\text{ pcf}) - (4\text{ ft} \cdot 62.4\text{ pcf})}{2{,}000} = 7.3\text{ tsf}$
Layer 3:
$\sigma'_{zgw} = \frac{(8\text{ ft} \cdot 100\text{ pcf}) + (19.0\text{ ft} \cdot 113\text{ pcf}) - (15\text{ ft} \cdot 62.4\text{ pcf})}{2{,}000} = 1.01\text{ tsf}$</td></tr>
<tr><td>Step 4</td><td>Compute stress relief (q'_{exc}) due to the excavation per Eq. 7–35 following drawdown of the water table.
Eq. 7–35: $q'_{exc} = \gamma \cdot \mathrm{D}$
$q'_{exc} = \frac{(8\text{ ft} \cdot 100\text{ pcf}) + (4.0\text{ ft} \cdot 113\text{ pcf})}{2{,}000} = 0.63\text{ tsf}$</td></tr>
<tr><td>Step 5</td><td>Compute the influence factor for vertical stress (I_σ) per Figure 6–8 and the corresponding unloaded vertical stress ($\Delta\sigma_{zexc}$) from the excavation at the midpoint of each layer per Eq. 7–36.
Eq. 7–36: $\Delta\sigma_{zexc} = I_\sigma(q'_{exc})$
<table><tr><th>Layer No.</th><th>Depth Below Excavation, z (ft)</th><th>$m = n$</th><th>I_σ</th><th>$\Delta\sigma_{zexc} = 4I_\sigma \cdot 0.63$ tsf</th></tr><tr><td>2</td><td>4</td><td>4.25</td><td>0.248</td><td>0.63</td></tr><tr><td>3</td><td>15</td><td>1.13</td><td>0.188</td><td>0.47</td></tr><tr><td colspan="5">Per Figure 6–8:
$B = L = \frac{34\text{ ft}}{2} = 17\text{ ft};\ m = \frac{B}{z} = n = \frac{L}{z}$</td></tr></table>Note: Figure 6–8 is used in this example to determine I_σ. Other methods could also be used to determine I_σ as explained in paragraphs 6–7, 6–8, and 6–9.</td></tr>
<tr><td>Step 6</td><td>Compute the vertical effective stress following the excavation and lowering of groundwater table (σ'_{zexc}) at the midpoint of each layer per Eq. 7–37.
Eq. 7–37: $\sigma'_{zexc} = \sigma'_{zgw} - \Delta\sigma_{zexc}$
Layer 2: $\sigma'_{zexc} = 0.73\text{ tsf} - 0.63\text{ tsf} = 0.10\text{ tsf}$
Layer 3: $\sigma'_{zexc} = 1.05\text{ tsf} - 0.47\text{ tsf} = 0.58\text{ tsf}$</td></tr>
</table>

Step 7

Compute rebound (R) per Eq. 7–38 for each layer from the results of the oedometer tests per Figure C–4b.

$R = H\left(\frac{\Delta e}{1 + e_{zo}}\right)$ per Eq. 7-38

Layer No.	Layer Thickness, H (ft)	σ'_{z0} (tsf) Step 2	σ'_{zexc} (tsf) Step 6	e_{z0} Figure C–4b	e_{zexc} Figure C–4b	$\Delta e = e_{zexc} - e_{z0}$	R (ft)
2	8.0	0.54	0.10	0.947	0.964	0.017	0.07
3	14.0	0.82	0.58	0.991	0.999	0.008	0.06

$\sum \delta_c$ = 0.13 ft (1.56 inches)

Step 8

Compute the gross bearing pressure q (or q_{eq}) at the base of the footing per paragraph 5–2b. The ultimate bearing capacity with a factor of safety of 3 exceeds the gross bearing pressure as illustrated in Appendix B, Example B–4.

$$q = \frac{Q + W_f}{A} - u_D$$

$$q = \frac{40\ \text{T} + \left[\frac{150\ \text{pcf}}{2{,}000} \cdot (7.0\ \text{ft})^2 \cdot 2.0\ \text{ft}\right]}{(7.0\ \text{ft})^2} - \frac{(14.0\ \text{ft} - 6.0\ \text{ft})62.4\ \text{pcf}}{2{,}000} = 0.72\ \text{tsf}$$

Step 9

Compute the vertical effective stress at the bearing elevation of the footing, σ'_{zD}.

$$\sigma'_{zD} = \frac{(8.0\ \text{ft} \cdot 100\ \text{pcf}) + 6.0\ \text{ft} \cdot 113\ \text{pcf}) - (8.0\ \text{ft} \cdot 62.4\ \text{pcf})}{2{,}000} = 0.49\ \text{tsf}$$

Step 10

Compute the influence factor for vertical stress (I_σ) per Eq. 6–9 and corresponding induced vertical stress ($\Delta\sigma_z$) beneath the footing at the midpoint of each layer per Eq. 6–6.

Eq. 6-6: $\Delta\sigma_z = \text{I}_\sigma(q - \sigma'_{zD})$

Eq. 6–9: $\text{I}_\sigma = 1 - \left(\frac{1}{1+\left(\frac{B}{2z}\right)^2}\right)^{1.76}$

Layer No.	Depth Below Footing, z (ft)	Layer Thickness, (ft)	I_σ	$\Delta\sigma_z = I_\sigma(q - \sigma'_{zD})$ $\Delta\sigma_z = I_\sigma(1.43 - 0.49)$ tsf
2	2.0	6	0.92	0.87
3	13.0	8	0.12	0.11

Note: Eq. 6–9 is used in this example to determine I_σ. Other methods could also be used to determine I_σ as explained in paragraphs 6–7, 6–8, and 6–9.

Step 11	Compute the final applied vertical effective stress (σ'_{zf}) at midpoint of each layer to a depth of the zone of influence $2B$ (14 ft) below the footing per Eq. 7–39. Eq. 7–39: $\sigma'_{zf} = \sigma'_{z0} + \Delta\sigma_z$ Layer 2: $\sigma'_{zf} = 0.54 \text{ tsf} + 0.87 \text{ tsf} = 1.41 \text{ tsf}$ 6 ft into Layer 3 (depth of $2B$ below footing): $\sigma'_{zf} = \frac{(8 \text{ ft} \cdot 100 \text{ pcf}) + (15 \text{ ft} \cdot 113 \text{ pcf}) - (17 \text{ ft} \cdot 62.4 \text{ pcf})}{2{,}000} + 0.11 \text{ tsf} = 0.83 \text{ tsf}$
Step 12	Estimate the settlement due to the recompression of the soil per Eq. 7–40 for each layer from the results of the oedometer tests. Eq. 7–40: $\delta_r = H\left(\frac{\Delta e}{1+e_{zexc}}\right)$

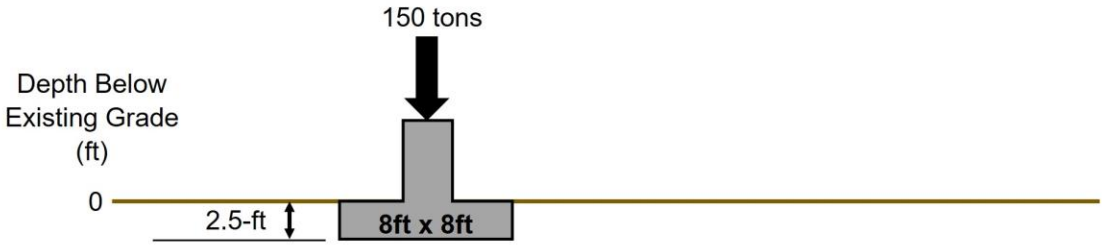
Layer No.	Layer Thickness, H (ft)	σ'_{zexc} (tsf) Step 6	σ'_{zf} (tsf) Step 11	e_{zexc} Figure C–4b	e_{zf} Figure C–4b	$\Delta e = e_{zexc} - e_f$	δ_r (ft)
2	6.0	0.10	1.41	0.964	0.805	0.159	0.49
3	8.0	0.58	0.83	0.999	0.983	0.016	0.06

$\sum \delta_c$ = 0.55 ft (6.6 inches)

C–5. Example C–5. Settlement of isolated footing on clay (PMT method)

a. Determine: Estimate the total settlement beneath a footing using the Ménard and Rousseau PMT method per Table 7–5.

b. Given: An 8.0-ft by 8.0-ft isolated spread footing supporting a 150-ton column load bearing 2.5 ft below grade on the clay profile illustrated in Figure C–5. TEXAM PMT conducted in clay soils as illustrated in Figure C–5. Groundwater was not encountered. Assume secondary settlement is negligible.



	Depth Below Footing, z (ft)	P_L^*	E_o(tsf)
Very Stiff to Hard Fat Clay $\gamma = 120$ pcf	4.0	14.0	225
	8.1	16.5	292
	10.0	19.8	394
Soft to Medium Stiff Clay $\gamma = 135$ pcf	12.0	2.2	25
	14.0	3.1	33
	16.0	4.0	52
Very Stiff to Hard Fat Clay $\gamma = 120$ pcf	18.0	22.7	349
	20.0	24.5	287
	25.0	40.5	678
	30.0	53.4	1,071
	35.0	68.9	1,220
	43.0	73.7	1,724

45.0

Very Dense Clayey Sand (SC)
SPT N > 50 bpf

Figure C–5. Settlement of isolated footing on clay (pressuremeter test method)

Table C–5
Solution for settlement of isolated footing on clay (pressuremeter test method) per Table 7–5

Step 1	PMT modulus E_0 per depth from the corrected pressuremeter test curve for each PMT test is illustrated in Figure C–5.
Step 2 & 3	Compute the gross bearing pressure q (or q_{eq}) at the base of the footing per paragraph 5–2b. The ultimate bearing capacity with a factor of safety of 3 exceeds the gross bearing pressure as illustrated in Appendix B, Example B–5. $q = \frac{Q + W_f}{A} - u_D = \frac{150\ \mathrm{T} + \left[\frac{150\ \mathrm{pcf}}{2{,}000} \cdot (8.0\ \mathrm{ft})^2 \cdot 2.5\ \mathrm{ft}\right]}{(8.0\ \mathrm{ft})^2} - 0 = 2.53\ \mathrm{tsf}$
Step 4	Choose the lesser of the values from Steps 2 and 3 for estimating settlements per Eq. 7–11. Therefore, $q = 2.53$ tsf.
Step 5	Compute the vertical effective stress at the bearing elevation of the footing (σ'_{zD}). $\sigma'_{zD} = \frac{(2.5\ \mathrm{ft} \cdot 125\ \mathrm{pcf})}{2{,}000} = 0.156\ \mathrm{tsf}$
Step 6	Divide the soil beneath the foundation into sixteen elementary layers of $B/2$ thickness to a depth of $8B$ (64 ft) below the foundation and combine these sixteen elementary layers into five distinct layers, E1, E2, E3/4/5, E6/7/8, and E9/16 as illustrated in Figure 7–12.
Step 7	Compute the average pressuremeter modulus of each of the five combined distinct layers (E_1, E_2, $\mathrm{E}_{3/4/5}$, $\mathrm{E}_{6/7/8}$, and $\mathrm{E}_{9/16}$) per Step 6 by the harmonic mean technique per Eq. Figure 7–13. Eq. 7–13: $\mathrm{E_k} = \frac{\mathrm{n}}{\frac{1}{\mathrm{E_1}} + \frac{1}{\mathrm{E_2}} + \cdots + \frac{1}{\mathrm{E_n}}}$ Layer E_1: $B/2$ below footing (4 ft below footing): $\mathrm{E}_1 = 225$ tsf Layer E_2: $B/2$ to B below footing (4 to 8 ft below footing): $\frac{2}{\mathrm{E}_2} = \frac{1}{225\mathrm{tsf}} + \frac{1}{292\mathrm{tsf}}$; therefore, $\mathrm{E}_2 = 254$ tsf Layer $\mathrm{E}_{3/4/5}$: B to 2.5B below footing (8 to 20 ft below footing): $\frac{7}{\mathrm{E}_{3/4/5}} = \frac{1}{292\ \mathrm{tsf}} + \frac{1}{394\ \mathrm{tsf}} + \frac{1}{25\ \mathrm{tsf}} + \frac{1}{33\ \mathrm{tsf}} + \frac{1}{52\ \mathrm{tsf}} + \frac{1}{349\ \mathrm{tsf}} + \frac{1}{287\ \mathrm{tsf}}$ therefore, $\mathrm{E}_{3/4/5} = 69$ tsf Layer $\mathrm{E}_{6/7/8}$: 2.5B to 4B below footing (20 to 32 ft below footing): $\frac{3}{\mathrm{E}_{6/7/8}} = \frac{1}{287\ \mathrm{tsf}} + \frac{1}{678\ \mathrm{tsf}} + \frac{1}{1{,}071\ \mathrm{tsf}}$; therefore, $\mathrm{E}_{6/7/8} = 509$ tsf Layer $\mathrm{E}_{9/16}$: 4B to 8B below footing (32 to 64 ft below footing): $\frac{2}{\mathrm{E}_{9/16}} = \frac{1}{1{,}220\ \mathrm{tsf}} + \frac{1}{1{,}724\ \mathrm{tsf}}$; therefore, $\mathrm{E}_{9/16} = 1{,}429$ tsf
Step 8	Compute the average pressuremeter modulus within the zone of influence of the spherical tensor ($\mathrm{E_c}$) to a depth of $0.5B$ (first layer) below the footing bearing elevation. Therefore, $\mathrm{E_c} = \mathrm{E}_1 = 225$ tsf from Step 7.

Step 9	Compute the equivalent distortion modulus (E_d) within the zone of influence of the deviatoric tensor to a depth of $8B$ (16 layers beneath the footing bearing elevation per Step 6) per Eq. 7–14: Eq. 7-14: $\frac{1}{E_d} = \frac{1}{4}\left(\frac{1}{E_1} + \frac{1}{0.85E_2} + \frac{1}{E_{3/4/5}} + \frac{1}{2.5E_{6/7/8}} + \frac{1}{2.5E_{9/16}}\right)$ where: E_1, E_2, $E_{3/4/5}$, $E_{6/7/8}$, and $E_{9/16}$ are computed by harmonic mean method, Step 7. $\frac{1}{E_d} = \frac{1}{4}\left(\frac{1}{E_1} + \frac{1}{0.85E_2} + \frac{1}{E_{3/4/5}} + \frac{1}{2.5E_{6/7/8}} + \frac{1}{2.5E_{9/16}}\right)$ $\frac{1}{E_d} = \frac{1}{4}\left(\frac{1}{225\text{ tsf}} + \frac{1}{0.85 \cdot 254\text{ tsf}} + \frac{1}{69\text{ tsf}} + \frac{1}{2.5 \cdot 509\text{ tsf}} + \frac{1}{2.5 \cdot 1429\text{ tsf}}\right)$ Therefore, $E_d = 162$tsf.
Step 10	Obtain the rheological factor $\propto_d$ and $\propto_c$ from Table 7–6. α is determined based on the soil type and state of consolidation from the ratio of E_d/p^*_{Le} and E_c/p^*_{Le}, respectively. $\frac{E_d}{p^*_{Le}} = \frac{162\text{ tsf}}{10.0\text{ tsf}} = 16.2$ From Table 7–6, $\propto_d = 1$ $\frac{E_c}{p^*_{Le}} = \frac{225\text{ tsf}}{10.0\text{ tsf}} = 22.5$ From Table 7–6, $\propto_c = 1$
Step 11	Obtain the deviatoric and spherical shape factors λ_d and λ_c from Figure 7–11. The shape factors are a function of the base length to width ratio L/B of the footing. $\frac{L}{B} = 1$ From Figure 7– 11, λ_d and $\lambda_c = 1.1$
Step 12	Estimate the total settlement per Eq. 7–11. Eq. 7-11: $\delta = \left[\frac{2}{9E_d}(q - \sigma'_{zD})B_0\left(\lambda_d\frac{B}{B_0}\right)^{\alpha_d}\right] + \left[\frac{\propto_c}{9E_c}(q - \sigma'_{zD})\lambda_c \cdot B\right]$ $\delta = \left[\frac{2}{9 \cdot 162\text{ tsf}} \cdot (2.53\text{ tsf} - 0.156\text{ tsf}) \cdot 2.0\text{ ft}\left(1.1 \cdot \frac{8.0\text{ ft}}{2.0\text{ ft}}\right)^1\right] + \left[\frac{1}{9 \cdot 225\text{ tsf}} \cdot (2.53\text{ tsf} - 0.156\text{ tsf}) \cdot 1.1 \cdot 8.0\text{ ft}\right]$ $\delta = 0.029\text{ ft} + 0.010\text{ ft} = 0.039\text{ ft} = 0.47\text{ in}$
Step 13	Compute the average pressuremeter modulus of each of the soft/loose layers by the harmonic mean technique per Eq. 7–15. Eq. 7-15: $E_L = \frac{n}{\frac{1}{E_1} + \frac{1}{E_2} + \cdots + \frac{1}{E_n}}$ $E_L = \frac{3}{\frac{1}{25} + \frac{1}{33} + \frac{1}{52}} = 33.5\text{ tsf}$

Step 14	Compute the influence factor for vertical stress (I_σ) per Eq. 6–9 and corresponding induced vertical stress ($\Delta\sigma_z$) beneath the footing at the midpoint of each layer per Eq. 6–6. Eq. 6–6: $\Delta\sigma_z = I_\sigma(q - \sigma'_{zD})$ Eq. 6-9: $\mathrm{I}_\sigma = 1 - \left(\frac{1}{1+\left(\frac{\mathrm{B}}{2z}\right)^2}\right)^{1.76}$

Depth Below Footing, z (ft)	Layer Thickness, H (ft)	I_σ	$\Delta\sigma_z = I_\sigma(q - \sigma'_{zD})$ $\Delta\sigma_z = I_\sigma(2.53 - 0.156)$ tsf
14.0	4	0.129	0.306

Note: Eq. 6–9 is used in this example to determine I_σ. Other methods could also be used to determine I_σ , as explained in paragraphs 6–7, 6–8, and 6–9.

Step 15	Compute the equivalent net limit pressure (p_{ne}) within the soft/loose layer per Eq. 5–24. $p_{ne} = \sqrt[\mathrm{n}]{\mathrm{p}^*_{\mathrm{L1}} \cdot \mathrm{p}^*_{\mathrm{L2}} \cdot \cdots \cdot \mathrm{p}^*_{\mathrm{Ly}}}$ $p_{ne} = \sqrt[3]{2.2 \text{ tsf} \cdot 3.1 \text{ tsf} \cdot 4.0 \text{ tsf}} = 3.01 \text{ tsf}$
Step 16	Obtain the rheological factor $\propto_{\mathrm{L}}$ from Table 7–6. $\frac{\mathrm{E_L}}{p_{ne}} = \frac{33.5 \text{ tsf}}{3.01 \text{ tsf}} = 11.1$ From Table 7–6, $\alpha_L = 2/3$
Step 17	Estimate the total settlement per Eq. 7–12. $\delta = \left[\left[\frac{2}{9\mathrm{E_d}}(q - \sigma'_{zD})\mathrm{B}_0 \cdot \left(\lambda_\mathrm{d}\frac{B}{\mathrm{B}_0}\right)^{\alpha_\mathrm{d}}\right] + \left[\frac{\propto_\mathrm{c}}{9\mathrm{E_c}}(q - \sigma'_{zD})\lambda_\mathrm{c} \cdot B\right]\right] + \left[\sum \alpha_\mathrm{L}\left(\frac{1}{\mathrm{E_L}} - \frac{1}{\mathrm{E_d}}\right)\mathrm{I}_\sigma \cdot (q - \sigma'_{zD})\mathrm{H}\right]$ Therefore, $\delta = 0.039 \text{ ft (from Step 12)} + \alpha_L\left(\frac{1}{\mathrm{E_L}} - \frac{1}{\mathrm{E_d}}\right) \cdot I_\sigma \cdot (q - \sigma'_{zD}) \cdot \mathrm{H}$ $\delta = 0.039 \text{ ft} + \frac{2}{3}\left(\frac{1}{33.5} - \frac{1}{162}\right) 0.306 \text{ tsf} \cdot 4.0 \text{ ft} = 0.058 \text{ ft} = 0.70 \text{ in}$

C–6. Example C–6. Settlement of isolated foundation on mixed soils (DMT method)

a. Determine:

(1) Estimate total settlement beneath a footing using the flat plate DMT Incremental Constrained Modulus method per Table 7–4.

(2) Compute the constrained modulus from the results of laboratory oedometer performed on the undisturbed sample obtained at 21.0 ft below grade. Compare oedometer modulus M with the dilatometer modulus M_{DMT} obtained at the same test depth.

b. *Given*: An 8.0-ft by 8.0-ft isolated spread footing supporting a 150-ton column load bearing 2.5 ft below grade on a mixed soil profile, reference Figure C–6a. A DMT and SPT test boring were performed side by side. A laboratory oedometer (consolidation) test performed on the undisturbed clay sample obtained at a depth of 21 ft below grade is illustrated in the *e*-log-p curve in Figure C–6b. Assume secondary settlement is negligible.

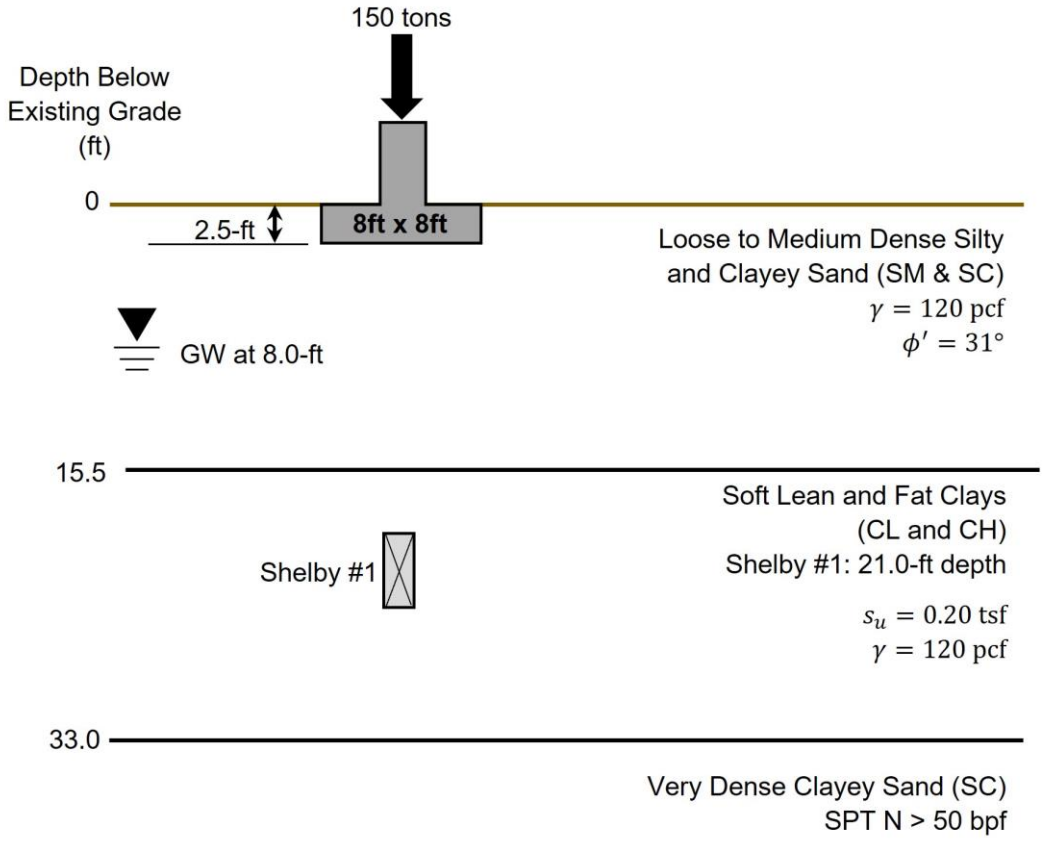
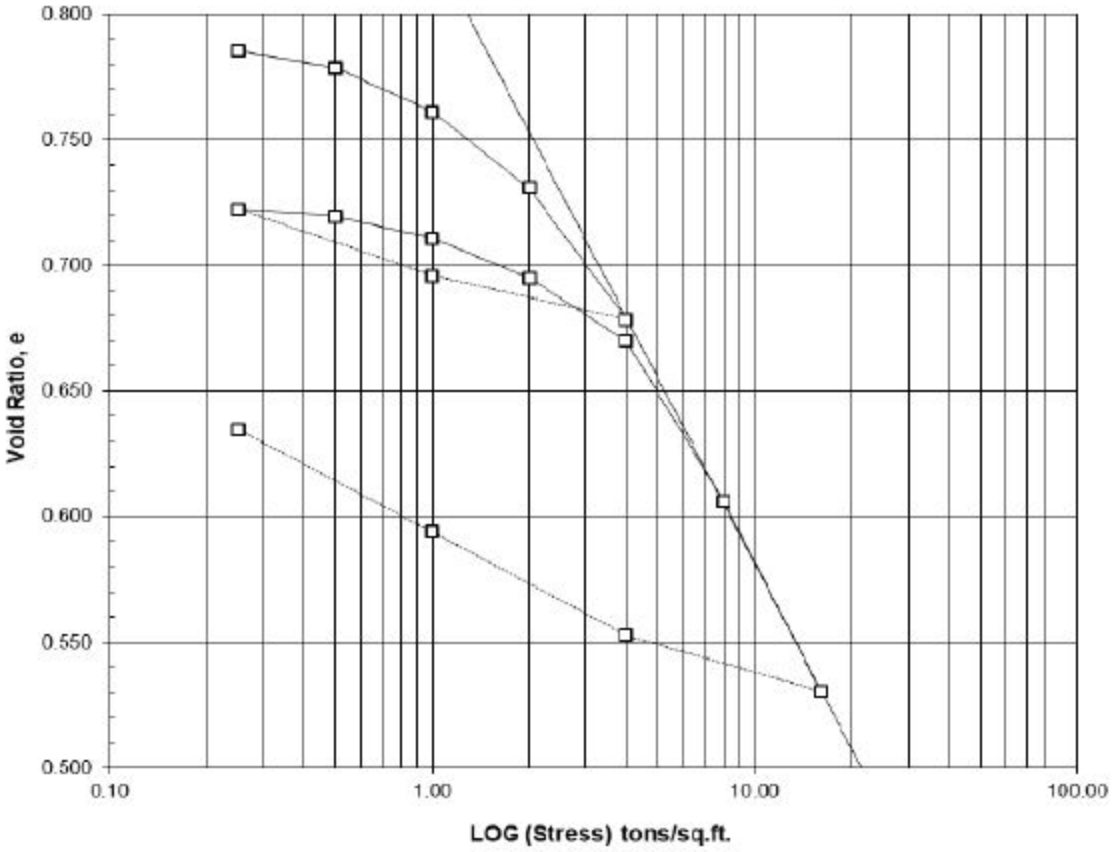


Figure C–6a. Foundation and soil conditions for Example C–6

Table C–6a
Dilatometer test data for Example C–6

Depth Below Grade (ft)	Depth Below Footing, z (ft)	DMT Constrained Modulus, M_{DMT} (tsf)	Depth Below Grade (ft)	Depth Below Footing, z (ft)	DMT Constrained Modulus, M (tsf)
3.2	0.7	769	18.6	16.1	39
4.1	1.6	770	19.2	16.7	61
4.8	2.3	920	19.9	17.4	48
5.5	3.0	158	20.5	18.0	45
6.1	3.6	717	21.2	18.7	70
6.8	4.3	917	21.9	19.4	88
7.4	4.9	898	22.5	20.0	115
8.1	5.6	985	23.2	20.7	167
8.7	6.2	2264	23.8	21.3	138
9.4	6.9	2132	24.5	22.0	78
10.0	7.5	1559	25.1	22.6	190
10.7	8.2	1457	25.8	23.3	135
11.4	8.9	1115	264	23.9	199
12.0	9.5	789	27.1	24.6	179
12.7	10.2	858	27.8	25.3	40
13.3	10.8	858	28.4	25.9	35
14.0	11.5	459	29.1	26.6	122
14.6	12.1	412	29.7	27.2	55
15.3	12.8	1059	30.4	27.9	39
15.9	13.4	517	31.0	28.5	38
16.6	14.1	72	31.7	29.2	31
17.3	14.8	58	32.3	29.8	33
17.9	15.4	29	33.0	30.5	32



Stress, σ_v' (tsf)	Void Ratio, e	Stress, σ_v' (tsf)	Void Ratio, e
0.25	0.786	1.0	0.711
0.5	0.779	2.0	0.695
1.0	0.761	4.0	0.670
2.0	0.731	8.0	0.606
4.0	0.678	16.0	0.530
Rebound 1.0	0.696	Rebound 4.0	0.553
0.25	0.722	1.0	0.594
Reload 0.5	0.720	0.25	0.635

Figure C–6b. Laboratory consolidation test data for Shelby Sample #1

Table C–6b
Solution for settlement of isolated footing on mixed soils using the dilatometer test Incremental Constrained Modulus method per Table 7–4

Step 1	The constrained modulus M_{DMT} as a function of depth from DMT data is illustrated in Figure C–6a.
Step 2	Divide the soil beneath the foundation into layers equal to depths in which the constrained modulus was determined, typically 8-inch layers for DMT tests. The top layer should be at the base of the foundation and the bottom layer should be at a minimum depth of two times the footing width $2B$ (16.0 ft) below the base of the isolated spread footing, or until the increase in the induced vertical stress $\Delta\sigma_z$ is negligible, resulting in small acceptable increases in settlement, whichever is deeper.
Step 3	Determine constrained modulus M for each layer.
Step 4	Compute the gross bearing pressure q (or q_{eq}) at the base of the footing per paragraph 5–2b. The ultimate bearing capacity with a factor of safety of 3 exceeds the gross bearing pressure as illustrated in Appendix B, Example B–6, Steps 4 and 6). $q = \frac{Q + W_f}{A} - u_D = \frac{150\ \text{T} + \left[\frac{150\ \text{pcf}}{2{,}000} \cdot (8.0\ \text{ft})^2 \cdot 2.5\ \text{ft}\right]}{(8.0\ \text{ft})^2} - 0 = 2.53\ \text{tsf}$
Step 5	Compute the vertical effective stress at the bearing elevation of the footing (σ'_{zD}). $\sigma'_{zD} = \frac{(2.5\text{ft} \cdot 120\text{pcf})}{2{,}000} = 0.15\ \text{tsf}$
Step 6	Compute the influence factor for vertical stress (I_σ) per Eq. 6–9 and corresponding induced vertical stress ($\Delta\sigma_z$) beneath the footing at the midpoint of each layer per Eq. 6–6. Eq. 6–6: $\Delta\sigma_z = I_\sigma(q - \sigma'_{zD})$ Eq. 6-9: $\mathrm{I}_\sigma = 1 - \left(\frac{1}{1+\left(\frac{\mathrm{B}}{2\mathrm{z}}\right)^2}\right)^{1.76}$ *Note.* Eq. 6–9 is used in this example to determine I_σ. Other methods could also be used to determine I_σ as explained in paragraphs 6–7, 6–8, and 6–9.

Step 7

Estimate the settlement of each layer using Eq. 7–10 as applicable. Sum the settlements for each layer within the zone of influence beneath the footing per Step 2.

Eq. 7-10: $\delta = \sum \frac{I_\sigma (q - \sigma'_{zD})}{M} \cdot H = \sum \frac{\Delta \sigma_z}{M} \cdot H$

Depth Below Footing, z (ft)	Layer Thickness, H (ft)	M_{DMT} (tsf) Step 3	I_σ Step 6	$\Delta \sigma_z$ Step 6	$\delta = \frac{\Delta \sigma_z}{M} H \times \left(\frac{12 \text{ in}}{1 \text{ ft}}\right)$ Step 7
0.7	0.7	769	0.998	2.34	0.026
1.6	0.9	770	0.969	2.27	0.032
2.3	0.7	920	0.914	2.14	0.020
3.0	0.7	158	0.834	1.95	0.104
3.6	0.6	717	0.757	1.77	0.018
4.3	0.7	917	0.666	1.56	0.014
4.9	0.6	898	0.593	1.39	0.011
5.6	0.7	985	0.516	1.21	0.010
6.2	0.6	2264	0.458	1.07	0.003
6.9	0.7	2132	0.399	0.93	0.004
7.5	0.6	1559	0.356	0.83	0.004
8.2	0.7	1457	0.313	0.73	0.004
8.9	0.7	1115	0.277	0.65	0.005
9.5	0.6	789	0.250	0.58	0.005
10.2	0.7	858	0.223	0.52	0.005
10.8	0.6	858	0.202	0.47	0.004
11.5	0.7	459	0.182	0.43	0.008
12.1	0.6	412	0.167	0.39	0.007
12.8	0.7	1059	0.151	0.35	0.003
13.4	0.6	517	0.139	0.33	0.005
14.1	0.7	72	0.127	0.30	0.035
14.8	0.7	58	0.117	0.27	0.040
15.4	0.6	29	0.109	0.25	0.063
16.1	0.7	39	0.100	0.23	0.050
16.7	0.6	61	0.094	0.22	0.026

Depth Below Footing, z (ft)	Layer Thickness, H (ft)	M_{DMT} (tsf) Step 3	I_σ Step 6	$\Delta\sigma_z$ Step 6	$\delta = \frac{\Delta\sigma_z}{M} H \times \left(\frac{12\text{ in}}{1\text{ ft}}\right)$ Step 7
17.4	0.7	48	0.087	0.20	0.035
18.0	0.6	45	0.081	0.19	0.030
18.7	0.7	70	0.076	0.18	0.021
19.4	0.7	88	0.071	0.17	0.016
20.0	0.6	115	0.067	0.16	0.010
20.7	0.7	167	0.062	0.15	0.007
21.3	0.6	138	0.059	0.14	0.007
22.0	0.7	78	0.056	0.13	0.014
22.6	0.6	190	0.053	0.12	0.005
23.3	0.7	135	0.050	0.12	0.007
23.9	0.6	199	0.047	0.11	0.004
24.6	0.7	179	0.045	0.11	0.005
25.3	0.7	40	0.043	0.10	0.021
25.9	0.6	35	0.041	0.10	0.020
26.6	0.7	122	0.039	0.09	0.006
27.2	0.6	55	0.037	0.09	0.011
27.9	0.7	39	0.035	0.08	0.018
28.5	0.6	38	0.034	0.08	0.015
29.2	0.7	31	0.032	0.08	0.020
29.8	0.6	33	0.031	0.07	0.016
30.5	0.7	32	0.030	0.07	0.018

$\sum \delta_c$ = 0.081 inches

Table C–6c
Solution for comparison of dilatometer constrained modulus M_{DMT} with laboratory oedometer constrained modulus M per paragraph 7–10b

Step 1	From the laboratory 1-D oedometer tests provided in Figure C–6b, plot the vertical effective stress (σ_z') versus the cumulative vertical strain (ε_z) with arithmetic scales for both axes. See Figure C–6b.
Step 2	Compute the initial vertical effective stress (σ_{zo}') where the oedometer sample was obtained from the ground per paragraph 6–5a. $\sigma_{zo}' = \frac{(21.0 \text{ ft} \cdot 125 \text{ pcf}) - (13.0 \text{ ft} \cdot 62.4 \text{ pcf})}{2{,}000} = 0.91 \text{ tsf}$
Step 3	Locate the initial vertical effective stress (σ_{zo}') along the plot in Step 1 and draw a line tangent to the curve.
Step 4	Compute the constrained modulus $M = \Delta\sigma_z'/\Delta\varepsilon_z$ defined as the slope of the tangent along the curve at initial vertical effective stress (σ_{zo}') from Step 3. $M = \frac{\Delta\sigma_v'}{\Delta\varepsilon_v} = \frac{1.3 \text{ tsf} - 0.7 \text{ tsf}}{0.064 - 0.06} = 150 \text{ tsf}$
Step 5	Compare M_{lab} from the results of the oedometer test in Step 4 with M_{DMT} from the adjacent DMT at the same depth. $M_{lab} = 150 \text{ tsf} \cong M_{DMT} = 138 \text{ tsf} @ 21.0 \text{ ft}$

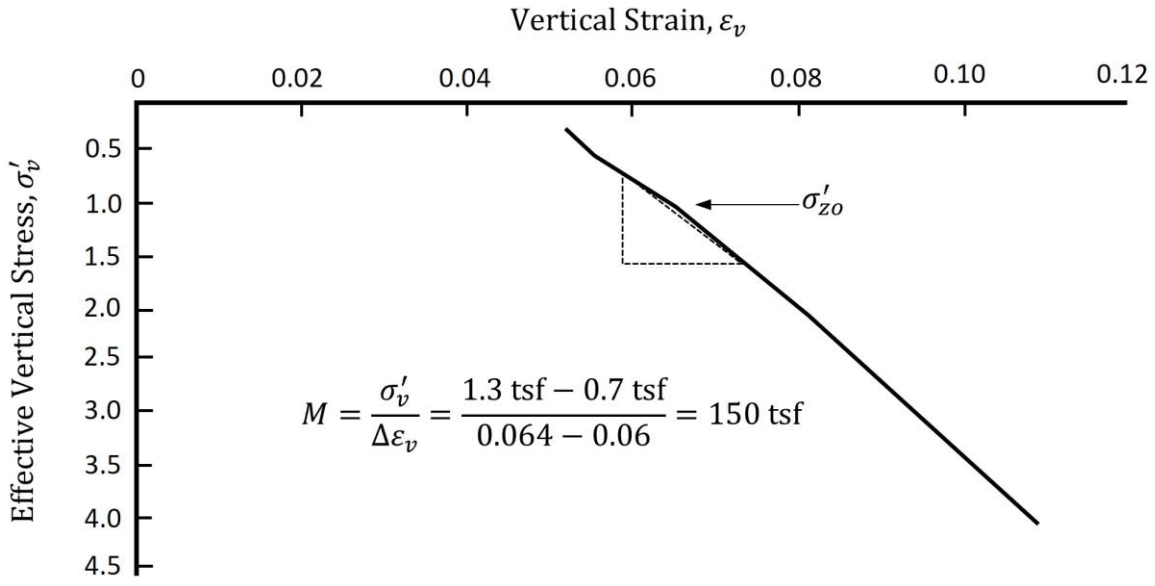


Figure C–6c. Arithmetic plot of effective vertical stress versus strain from Laboratory 1-D Oedometer Test

C–7. Example C–7. Settlement of isolated footing on sand (DMT method)

a. Determine: Estimate the settlement beneath a footing using the following methods:

(1) DMT Incremental Constrained Modulus method per Table 7–4.

(2) Ménard and Rousseau PMT method per Table 7–5.

(3) Schmertmann's CPT method for cohesionless soils per Table 7–8.

(4) Burland and Burbidge SPT method for sands and gravels per Table 7–9.

b. Given: An 8.0-ft x 8.0-ft square footing bearing supporting a 200-ton column load bearing 2.5 ft below grade on the sand profile illustrated in Figure C–7a. Time (t) after foundation loading = 50 years. Drained ultimate bearing capacity computed per paragraph 5–5 and shown in Appendix B, Example B–7. Flat plate DMT, TEXAM PMT, CPT, and SPT were performed on sandy soils as illustrated in Figure C–7a. Groundwater was not encountered.

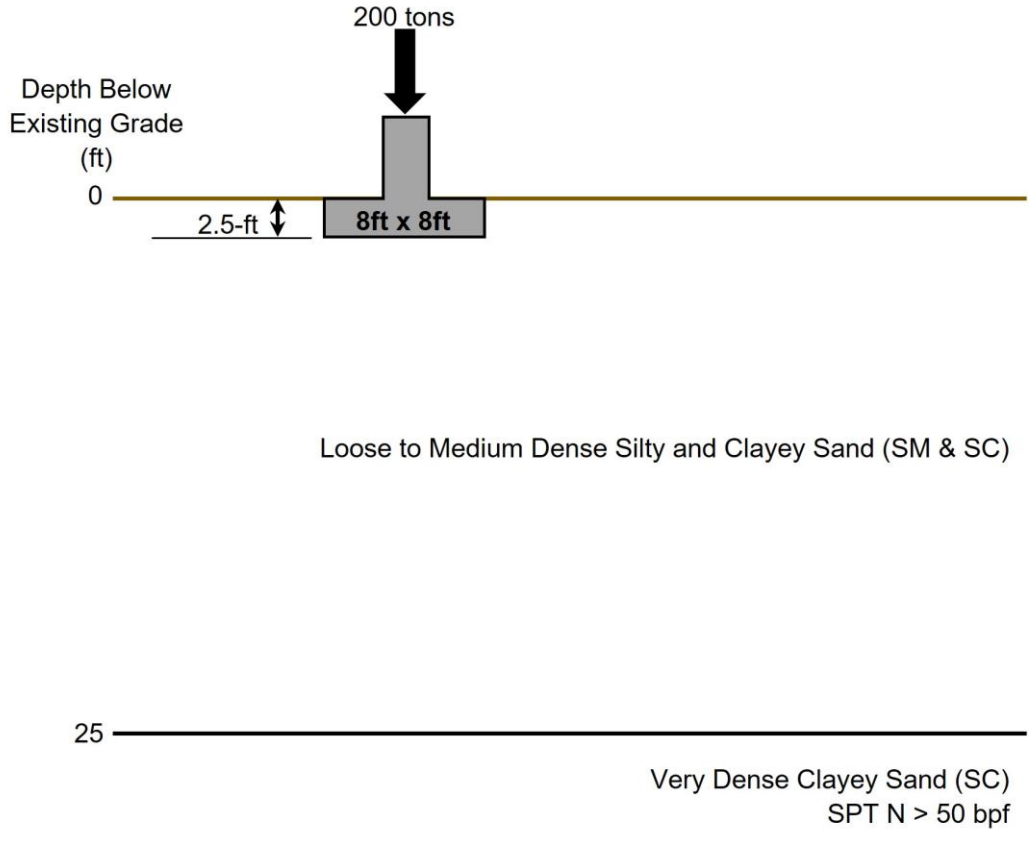


Figure C–7a. Foundation and soil conditions for Example C–7

Table C–7a
In situ test data for Example C–7

Depth Below Footing (ft)	DMT M (tsf)	PMT P_L^* (tsf)	PMT E_0 (tsf)	CPT q_c (tsf)	SPT N_{60} (bpf)
0.7	769	–	–	–	–
0.75	–	–	–	100	12
1.3	770	–	–	–	–
2.0	920	–	–	100	–
2.6	158	–	–	–	–
3.25		–	–	160	15
3.3	717	–	–	–	–
3.9	917	–	–	–	–
4.0	–	8.5	78	–	–
4.6	898	–	–	–	–
5.2	985	–	–	–	–
5.75	–	–	–	100	14
5.9	2264	–	–	–	–
6.6	2132	–	–	–	–
7.2	2266	–	–	–	–
7.9	1487	–	–	–	–
8.1	–	30.2	238	–	–
8.25	–	–	–	–	19
8.5	1740	–	–	300	–
9.2	1760	–	–	–	–
9.8	2138	–	–	–	–
10.5	2220	–	–	–	–
10.75	–	–	–	–	20
11.0	–	–	–	160	–
11.2	2212	–	–	–	–
11.8	2284	–	–	–	–
12.5	2134	–	–	–	–
13.1	2076	–	–	–	–
13.25	–	–	–	–	14
13.6	–	40.8	320	–	–
13.8	2583	–	–	–	–
14.25	–	–	–	200	–
14.4	2357	–	–	–	–
15.1	2343	–	–	–	–
15.7	1842	–	–	–	–
15.75	–	–	–	–	19
16.4	1570	–	–	–	–
17.1	1357	–	–	–	–
17.7	1264	–	–	–	–
18.25	–	–	–	–	16
18.4	1218	98.4	826	–	–

Depth Below Footing (ft)	DMT M (tsf)	PMT P_L^* (tsf)	PMT E_0 (tsf)	CPT q_c (tsf)	SPT N_{60} (bpf)
19.0	1155	–	–	–	–
19.7	887	–	–	–	–
20.3	636	–	–	–	–
20.75	–	–	–	–	14
21.0	822	–	–	–	–
21.7	543	–	–	–	–
22.3	630	–	–	–	–
23.0	914	–	–	–	–
23.25	–	–	–	–	15
23.3	–	142.8	1419	–	–
23.6	1507	–	–	–	–
24.3	1350	–	–	–	–
24.9	1307	–	–	–	–
25.6	1293	–	–	–	–
26.2	895	–	–	–	–
26.9	802	–	–	–	–
27.6	1398	–	–	–	–
28.0	–	123.9	–	–	–
28.2	1267	–	–	–	–
28.9	388	–	–	–	–
29.5	501	–	–	–	–
30.2	904	–	–	–	–
30.8	984	–	–	–	–
31.5	1460	–	–	–	–
33.2	–	155.0	2508	–	–
38.1	–	123.6	1059	–	–
43.2	–	120.4	1128	–	–

Table C–7b
Solution for dilatometer test Incremental Constrained Modulus method per Table 7–4

Step 1	Constrained modulus (M) as a function of depth from DMT data is illustrated in Figure C–7a.
Step 2	Divide the soil beneath the foundation into layers equal to depths in which the constrained modulus was determined, typically 8-inch layers for DMT tests. The top layer should be at the base of the foundation and the bottom layer should be at a minimum depth of two times the footing width $2B$ (16.0 ft) below the base of the isolated spread footing, or until the increase in the induced vertical stress ($\Delta\sigma_z$) is negligible, resulting in small acceptable increases in settlement, whichever is deeper.
Step 3	Determine constrained modulus (M) for each layer.
Step 4	Compute the gross bearing pressure q (or q_{eq}) at the base of the footing per paragraph 5–2b. The ultimate bearing capacity with a factor of safety of 3 exceeds the gross bearing pressure as illustrated in Appendix B, Example B–7, Table B–7a. $q = \frac{Q + W_f}{A} - u_D = \frac{200\ \text{T} + \left[\frac{150\ \text{pcf}}{2{,}000} \cdot (8.0\ \text{ft})^2 \cdot 2.5\ \text{ft}\right]}{(8.0\ \text{ft})^2} - 0 = 3.31\ \text{tsf}$
Step 5	Compute the vertical effective stress at the bearing elevation of the footing (σ'_{zD}). $\sigma'_{zD} = \frac{(2.5\ \text{ft} \times 125\ \text{pcf})}{2{,}000} = 0.156\ \text{tsf}$
Step 6	Compute the influence factor for vertical stress (I_σ) per Eq. 6–9 and corresponding induced vertical stress ($\Delta\sigma_z$) beneath the footing at the midpoint of each layer per Eq. 6–6. Eq. 6–6: $\Delta\sigma_z = I_\sigma(q - \sigma'_{zD})$ Eq. 6-9: $I_\sigma = 1 - \left(\frac{1}{1+\left(\frac{B}{2z}\right)^2}\right)^{1.76}$ *Note.* Eq. 6–9 is used in this example to determine I_σ. Other methods could also be used to determine I_σ as explained in paragraphs 6–7, 6–8, and 6–9.

Step 7

Estimate the settlement of each layer using Eq. 7–10 as applicable. Sum the settlements for each layer within the zone of influence beneath the footing per Step 2.

Eq. 7-10: $\delta = \sum \frac{I_\sigma (q - \sigma'_{zD})}{M} \cdot H = \sum \frac{\Delta\sigma_z}{M} \cdot H$

Depth Below Footing, z (ft)	Layer Thickness, H (ft)	M (tsf) Step 3	I_σ Step 6	$\Delta\sigma_z$ (tsf) Step 6	$\delta = \frac{\Delta\sigma_z}{M} H \times \left(\frac{12\text{ in}}{1\text{ ft}}\right)$ (inches) Step 7
0.7	0.7	769	0.998	3.13	0.034
1.3	0.6	770	0.984	3.09	0.029
2.0	0.7	920	0.941	2.96	0.027
2.6	0.6	158	0.882	2.77	0.126
3.3	0.7	717	0.796	2.5	0.029
3.9	0.6	917	0.718	2.25	0.018
4.6	0.7	898	0.629	1.97	0.018
5.2	0.6	985	0.559	1.75	0.013
5.9	0.7	2264	0.486	1.53	0.006
6.6	0.7	2132	0.423	1.33	0.005
7.2	0.6	2266	0.377	1.18	0.004
7.9	0.7	1487	0.331	1.04	0.006
8.5	0.6	1740	0.297	0.93	0.004
9.2	0.7	1760	0.263	0.82	0.004
9.8	0.6	2138	0.238	0.75	0.003
10.5	0.7	2220	0.212	0.67	0.003
11.2	0.7	2212	0.190	0.60	0.002
11.8	0.6	2284	0.174	0.55	0.002
12.5	0.7	2134	0.158	0.50	0.002
13.1	0.6	2076	0.145	0.46	0.002
13.8	0.7	2583	0.132	0.42	0.001
14.4	0.6	2357	0.123	0.39	0.001
15.1	0.7	2343	0.113	0.35	0.001
15.7	0.6	1842	0.105	0.33	0.001
16.4	0.7	1570	0.097	0.30	0.002

$\sum \delta_c = 0.034$ inches

Table C–7c
Solution for settlement of isolated footing on sand using the Ménard and Rousseau pressuremeter test per Table 7–5

Step 1	PMT modulus (E_0) per depth from the corrected pressuremeter test curve for each PMT test is illustrated in Figure C–7a.
Step 2 & 3	Compute the gross bearing pressure q (or q_{eq}) at the base of the footing per paragraph 5–2b. The ultimate bearing capacity with a factor of safety of 3 exceeds the gross bearing pressure as illustrated in Appendix B, Example B–7, Tables B–7a and B–7b. $q = \frac{Q + W_f}{A} - u_D = \frac{200\ \mathrm{T} + \left[\frac{150\ \mathrm{pcf}}{2{,}000} \cdot (8.0\ \mathrm{ft})^2 \cdot 2.5\ \mathrm{ft}\right]}{(8.0\ \mathrm{ft})^2} - 0 = 3.31\ \mathrm{tsf}$
Step 4	Choose the lesser of the values from Steps 2 and 3 for estimating settlements per Eq. 7–11 (Deep Relatively Uniform Deposits). Therefore, $q = 3.31$ tsf.
Step 5	Compute the vertical effective stress at the bearing elevation of the footing (σ'_{zD}). $\sigma'_{zD} = \frac{(2.5\mathrm{ft} \cdot 125\mathrm{pcf})}{2{,}000} = 0.156\ \mathrm{tsf}$
Step 6	Divide the soil beneath the foundation into sixteen elementary layers of $B/2$ thickness to a depth of $8B$ (64.0 ft) below the foundation and combine these sixteen elementary layers into five distinct layers, E_1, E_2, $\mathrm{E}_{3/4/5}$, $\mathrm{E}_{6/7/8}$, and $\mathrm{E}_{9/16}$ as illustrated in Figure 7–12.
Step 7	Compute the average pressuremeter modulus of each of the five combined distinct layers (E_1, E_2, $\mathrm{E}_{3/4/5}$, $\mathrm{E}_{6/7/8}$, and $\mathrm{E}_{9/16}$) per step 6 by the harmonic mean technique per Eq. 7–13. Eq. 7-13: $\mathrm{E_k} = \frac{\mathrm{n}}{\frac{1}{\mathrm{E}_1} + \frac{1}{\mathrm{E}_2} + \cdots + \frac{1}{\mathrm{E_n}}}$ Layer E_1: $B/2$ below footing (4 ft below footing): $\mathrm{E}_1 = 78\ \mathrm{tsf}$ Layer E_2: $B/2$ to B below footing (4 to 8 ft below footing): $\frac{2}{\mathrm{E}_2} = \frac{1}{78\ \mathrm{tsf}} + \frac{1}{238\ \mathrm{tsf}}; \mathrm{therefore}, \mathrm{E}_2 = 118\ \mathrm{tsf}$ Layer $\mathrm{E}_{3/4/5}$: B to $2.5B$ below footing (8 to 20 ft below footing): $\frac{3}{\mathrm{E}_{3/4/5}} = \frac{1}{238\ \mathrm{tsf}} + \frac{1}{320\ \mathrm{tsf}} + \frac{1}{826\ \mathrm{tsf}}; \mathrm{therefore}, \mathrm{E}_{3/4/5} = 351\ \mathrm{tsf}$ Layer $\mathrm{E}_{6/7/8}$: $2.5B\ to\ 4B$ below footing (20 to 32 ft below footing): $\frac{2}{\mathrm{E}_{6/7/8}} = \frac{1}{1{,}419\ \mathrm{tsf}} + \frac{1}{1{,}065\ \mathrm{tsf}}; \mathrm{therefore}, \mathrm{E}_{6/7/8} = 1{,}217\ \mathrm{tsf}$ Layer $\mathrm{E}_{9/16}$: $4B\ to\ 8B$ below footing (32 to 64 ft below footing): $\frac{3}{\mathrm{E}_{9/16}} = \frac{1}{2{,}508\ \mathrm{tsf}} + \frac{1}{1{,}059\ \mathrm{tsf}} + \frac{1}{1{,}128\ \mathrm{tsf}}; \mathrm{therefore}, \mathrm{E}_{9/16} = 1{,}346\ \mathrm{tsf}$
Step 8	Compute the average pressuremeter modulus within the zone of influence of the spherical tensor $\mathrm{E_c}$ to a depth of $0.5B$ (first layer) below the footing bearing elevation. Therefore, $\mathrm{E_c} = \mathrm{E}_1 = 78\ \mathrm{tsf}$ from Step 7.

Step 9	Compute the equivalent distortion modulus $\mathrm{E_d}$ within the zone of influence of the deviatoric tensor to a depth of 8B (16 layers beneath the footing bearing elevation per Step 6) per Eq. 7–14: Eq. 7-14: $\frac{1}{\mathrm{E_d}} = \frac{1}{4}\left(\frac{1}{\mathrm{E_1}} + \frac{1}{0.85\mathrm{E_2}} + \frac{1}{\mathrm{E_{3/4/5}}} + \frac{1}{2.5\mathrm{E_{6/7/8}}} + \frac{1}{2.5\mathrm{E_{9/16}}}\right)$ where: $\mathrm{E_1}$, $\mathrm{E_2}$, $\mathrm{E_{3/4/5}}$, $\mathrm{E_{6/7/8}}$, and $\mathrm{E_{9/16}}$ are computed by the harmonic mean method, Step 7. $\frac{1}{\mathrm{E_d}} = \frac{1}{4}\left(\frac{1}{\mathrm{E_1}} + \frac{1}{0.85\mathrm{E_2}} + \frac{1}{\mathrm{E_{3/4/5}}} + \frac{1}{2.5\mathrm{E_{6/7/8}}} + \frac{1}{2.5\mathrm{E_{9/16}}}\right)$ $\frac{1}{\mathrm{E_d}} = \frac{1}{4}\left(\frac{1}{78\ \mathrm{tsf}} + \frac{1}{0.85 \cdot 118\ \mathrm{tsf}} + \frac{1}{351\ \mathrm{tsf}} + \frac{1}{2.5 \cdot 1{,}217\ \mathrm{tsf}} + \frac{1}{2.5 \cdot 1{,}346\ \mathrm{tsf}}\right)$ $\mathrm{E_d} = 152\ \mathrm{tsf}$
Step 10	Obtain the rheological factor $\propto_{\mathrm{d}}$ and $\propto_{\mathrm{c}}$ from Table 7–6. α is determined based on the soil type and state of consolidation from the ratio of $\mathrm{E_d}/p_{ne}$ and $\mathrm{E_c}/p_{ne}$, respectively. p_{ne} is calculated using Eq. 5–24. $\frac{\mathrm{E_d}}{p_{ne}} = \frac{152\ \mathrm{tsf}}{8.5\ \mathrm{tsf}} = 9.5$ from Table 7–6, $\alpha_d = 0.5$ $\frac{\mathrm{E_c}}{p_{ne}} = \frac{78\ \mathrm{tsf}}{8.5\ \mathrm{tsf}} = 4.9$ from Table 7–6, $\alpha_c = 0.33$
Step 11	Obtain the deviatoric and spherical shape factors λ_{d} and λ_{c} from Figure 7–11. The shape factors are a function of the base length-to-width ratio L/B of the footing. $\frac{\mathrm{L}}{\mathrm{B}} = 1;$ from Figure 7– 11, λ_{d} and $\lambda_{\mathrm{c}} = 1.1$
Step 12	Estimate the total settlement per Eq. 7–11. $\delta = \left[\frac{2}{9\mathrm{E_d}}(q - \sigma'_{zD})B_0 \cdot \left(\lambda_{\mathrm{d}}\frac{B}{B_0}\right)^{\alpha_{\mathrm{d}}}\right] + \left[\frac{\propto_{\mathrm{c}}}{9\mathrm{E_c}}(q - \sigma'_{zD})\lambda_{\mathrm{c}} \cdot B\right]$ $\delta = \left[\frac{2}{9 \cdot 152\ \mathrm{tsf}} \cdot (3.31\ \mathrm{tsf} - 0.156\ \mathrm{tsf}) \cdot 2.0\mathrm{ft} \cdot \left(1.1 \cdot \frac{8.0\ \mathrm{ft}}{2.0\ \mathrm{ft}}\right)^{0.5}\right]$ $+ \left[\frac{0.33}{9 \cdot 78\ \mathrm{tsf}} \cdot (3.31\mathrm{tsf} - 0.156\ \mathrm{tsf}) \cdot 1.1 \cdot 8.0\mathrm{ft}\right]$ $\delta = 0.015\ \mathrm{ft} + 0.013\ \mathrm{ft} = 0.028\ \mathrm{ft} = 0.38\ \mathrm{in}$

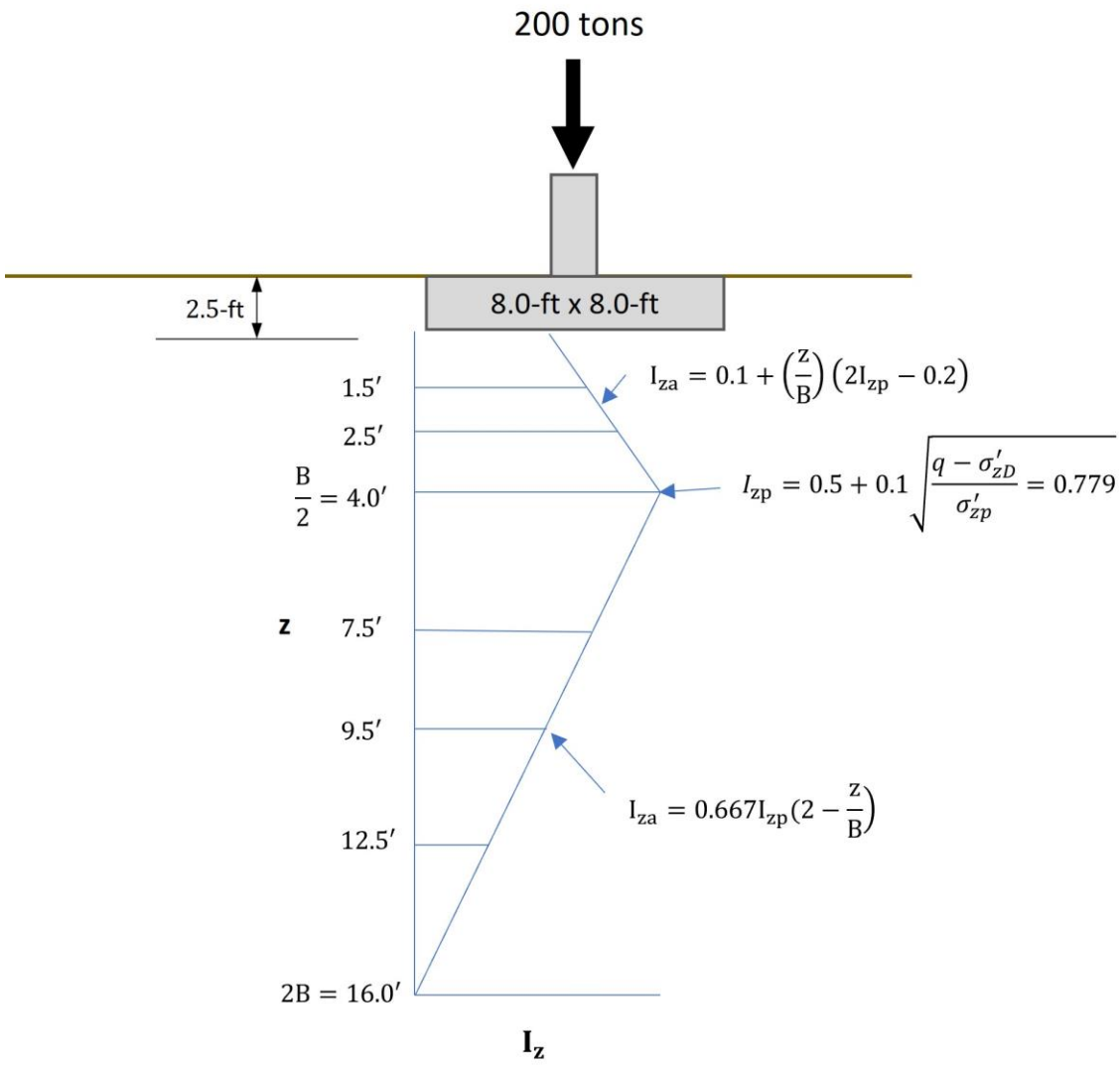


Figure C–7b. Strain influence diagram using the Schmertmann et al. (1978) method, per Table 7–8

Table C–7d
Solution for settlement of isolated footing on sand using Schmertmann's cone penetration test method

Step 1	Measured tip resistance q_c as a function of depth from CPT data is illustrated in Figure C–7a.
Step 2	Determine the equivalent Young's modulus $\mathrm{E_s}$ per Eq. 7–16. Divide the soil beneath the bearing elevation into layers depending on the variation in $\mathrm{E_s}$ versus depth.
Step 3	Compute the gross bearing pressure q (or q_{eq}) at the base of the footing per paragraph 5–2b. The ultimate bearing capacity with a factor of safety of 3 exceeds the gross bearing pressure as illustrated in Appendix B, Example B–7, Tables B–7a and B–7b. $q = \frac{Q + W_f}{A} - u_D = \frac{200\ \mathrm{T} + \left[\frac{150\ \mathrm{pcf}}{2{,}000} \cdot (8.0\ \mathrm{ft})^2 \cdot 2.5\ \mathrm{ft}\right]}{(8.0\ \mathrm{ft})^2} - 0 = 3.31\ \mathrm{tsf}$
Step 4	Compute the vertical effective stress at the bearing elevation of the footing (σ'_{zD}). $\sigma'_{zD} = \frac{(2.5\mathrm{ft} \cdot 125\ \mathrm{pcf})}{2{,}000} = 0.156\ \mathrm{tsf}$
Step 5	Develop strain influence factor diagram below the base of the footing per Figure 7–13. Reference Figure C–7b.
Step 6	Compute the peak strain influence factor ($\mathrm{I_{zp}}$) per Eq. 7–18: $\mathrm{I_{zp}} = 0.5 + 0.1\sqrt{\frac{q - \sigma'_{zD}}{\sigma'_{zp}}}$ $\mathrm{I_{zp}} = 0.5 + 0.1\sqrt{\frac{3.31\ \mathrm{tsf} - 0.156\ \mathrm{tsf}}{6.5' \times \frac{125\ \mathrm{pcf}}{2{,}000}}} = 0.779$
Step 7	Compute embedment ($\mathrm{C_1}$) and creep correction ($\mathrm{C_2}$) factors per Eq. 7–24 and 7–25. Eq. 7-24: $\mathrm{C_1} = 1 - 0.5\left(\frac{\sigma'_{zD}}{q - \sigma'_{zD}}\right) \geq 0.5$ $\mathrm{C_1} = 1 - 0.5\left(\frac{0.156\ \mathrm{tsf}}{3.31\ \mathrm{tsf} - 0.156\ \mathrm{tsf}}\right) = 0.975 \geq 0.5$ Eq. 7-25: $\mathrm{C_2} = 1 + 0.2\log_{10}\left(\frac{t}{0.1}\right)$ $\mathrm{C_2} = 1 + 0.2\log_{10}\left(\frac{50}{0.1}\right) = 1.539$

Step 8	Produce a table for each layer as follows within the strain influence factor diagram per Figure C–7b. Reference table in Step 10. *a.* The number of the layer *b.* Depth of the layer (feet) *c.* Depth to the midpoint of each layer, z (feet) *d.* The layer thickness in feet *e.* q_c (tsf) or N60 (bpf) values *f.* $\mathrm{E_s}$ (tsf) *g.* Strain influence factor ($\mathrm{I_a}$) for axisymmetric footing *h.* $\frac{\mathrm{I_z \cdot H}}{\mathrm{E_S}}$ feet
Step 9	Compute the vertical strain influence factor ($\mathrm{I_{za}}$) at the midpoint of each layer per Eq. 7–19 and 7–20 for the axisymmetric footing. Reference Table in Step 10. Eq. 7–19: Axisymmetric footings for $\mathrm{z} = 0 \text{ to } 0.5B$ $\mathrm{I_{za}} = 0.1 + \left(\frac{\mathrm{z}}{\mathrm{B}}\right)\left(2 \cdot \mathrm{I_{zp}} - 0.2\right)$ Eq. 7–20: Axisymmetric footings for $\mathrm{z} = 0.5B \text{ to } 2B$ $\mathrm{I_{za}} = 0.667 \cdot \mathrm{I_{zp}}(2 - \frac{\mathrm{z}}{\mathrm{B}})$
Step 10	Compute $\frac{\mathrm{I_z \cdot H}}{\mathrm{E_s}}$ at the midpoint of each layer and sum for the layers within the strain influence factor diagram per Figure C–7b.

Layer No.	Depth of Layer Below Footing (ft)	Depth at Layer Midpoint, z (ft)	Layer Thickness, H (ft)	Tip Resistance, q_c(tsf)	E_s (tsf)	I_z	$\frac{I_z \cdot H}{E_s}$ (ft³/tons)
1	0 to 1.5	0.75	1.5	100	250	0.227	0.0014
2	1.5 to 2.5	2.0	1.0	100	250	0.439	0.0018
3	2.5 to 4.0	3.25	1.5	160	400	0.651	0.0024
4	4.0 to 7.5	5.75	3.5	100	250	0.665	0.0093
5	7.5 to 9.5	8.5	2.0	300	750	0.489	0.0013
6	9.5 to 12.5	11.0	3.0	160	400	0.324	0.0024
7	12.5 to 16.0	14.25	3.5	200	500	0.114	0.008

$\sum \frac{I_z \cdot H}{E_s} = 0.0266$

Step 11	Compute total settlement per Eq. 7–26. $\delta = C_1 \cdot C_2 (q - \sigma'_{zD}) \sum \frac{I_z \cdot H}{E_s}$ $\delta = 0.975 \cdot 1.539 \cdot (3.31 \text{ tsf} - 0.156 \text{ tsf}) \cdot 0.0266 \frac{\text{ft}^3}{\text{tons}} = 0.125 \text{ ft} = 1.5 \text{ in}$

Table C–7e
Solution for settlement of isolated footing on sand using the Burland and Burbidge standard penetration test method for sands and gravels per Table 7–9

Step 1	N_{60} data as a function of depth from SPT results is illustrated in Figure C–7a.
Step 2	Compute the gross bearing pressure q (or q_{eq}) at the base of the footing per paragraph 5–2b. The ultimate bearing capacity with a factor of safety of 3 exceeds the gross bearing pressure as illustrated in Appendix B, Example B–7, Tables B–7a and B–7b. $q = \frac{Q + W_f}{A} - u_D = \frac{200 \text{ T} + \left[\frac{150 \text{ pcf}}{2{,}000} \cdot (8.0 \text{ ft})^2 \cdot 2.5 \text{ ft}\right]}{(8.0 \text{ ft})^2} - 0 = 3.31 \text{ tsf}$
Step 3	Compute depth of influence (Z_I) per Eq. 7–27. $Z_I = 1.43 B^{0.7}$ $Z_I = 1.43 \cdot 8.0^{0.7} = 6.13 \text{ ft}$
Step 4	Compute the arithmetic mean of the measured N_{60} values measured within the depth of influence $Z_I = 6.13$ ft. $\bar{N}_{60} = \frac{12 + 15 + 14}{3} = 14 \text{ bpf}$
Step 5	Compute the soil compressibility index I_c per Eq. 7–28. $I_c = \frac{0.35}{\bar{N}_{60}^{1.4}}$ $I_c = \frac{0.35}{14^{1.4}} = 0.0087 \text{ tsf}^{-1}$
Step 6	Compute shape (C_1) layer thickness (C_2) and time-dependent (C_3) factors per Eq. 7–32, 7–33, and 7–34. Eq. 7-32: $C_1 = \left[\frac{1.25\left(\frac{L}{B}\right)}{\left(\frac{L}{B}\right)+0.25}\right]^2 \geq 1$ Eq. 7-33: $C_2 = \frac{H_s}{Z_I}\left(2 - \frac{H_s}{Z_I}\right) \leq 1$ $C_2 = \frac{6.13}{6.13}\left(2 - \frac{6.13}{6.13}\right) \leq 1 = 1.0$ Eq. 7-34: $C_3 = 1 + R_3 + R_t \cdot \log_{10} \frac{t}{3}$ $C_3 = 1 + 0.3 + 0.2 \log_{10} \frac{50}{3} = 1.54$

Step 7	Compute the average initial vertical effective stress (σ'_{zo}) within depth of influence (Z_I). Depth: 2.5 ft below grade: $\sigma'_{z0} = \frac{2.5 \text{ ft} \cdot 125 \text{ pcf}}{2{,}000} = 0.156 \text{ tsf}$ Depth: 8.63 ft below grade (2.5 ft + 6.13 ft): $\sigma'_{zo} = \frac{8.63 \text{ ft} \cdot 125 \text{ pcf}}{2{,}000} = 0.539 \text{ tsf}$ $\bar{\sigma}'_{zo} = \frac{0.156 \text{ tsf} + 0.539 \text{ tsf}}{2} = 0.35 \text{ tsf}$
Step 8	Compute the preconsolidation stress (σ'_p) from Eq. 3–23. Eq. 3–23: $\sigma'_p = 0.468 P_a \cdot N_{60}^m$ $\sigma'_p = 0.47 \cdot 1.0 \cdot 14^{0.8} = 3.88 \text{ tsf}$
Step 9	$\bar{\sigma}'_{zo} = 0.35$ tsf, $\sigma'_p = 3.88$ tsf, and $(q + \bar{\sigma}'_{z0}) = 3.31 \text{ tsf} + 0.35 \text{ tsf} = 3.66 \text{ tsf}$ Since $\bar{\sigma}'_{z0} \neq \sigma'_p$ and $(q + \bar{\sigma}'_{z0}) \leq \sigma'_p$, estimate settlements per Eq. 7–30. Eq. 7–30: $\delta = C_1 \cdot C_2 \cdot C_3 (q + \bar{\sigma}'_{z0}) Z_I \frac{I_c}{3}$ $\delta = 1.0 \cdot 1.0 \cdot 1.54 \cdot (3.31 \text{ tsf} + 0.35 \text{ tsf}) \cdot 6.13 \text{ ft} \cdot \frac{0.0087 \text{ tsf}^{-1}}{3} = 0.10 \text{ ft} = 1.2 \text{ in}$

Appendix D
Common Unit Conversion Factors

Table D–1
Unit of Angle Measure

To Convert	To	Multiply By
Radian	degree arc	57.29578

Table D–2
Units of Distance

To Convert	To	Multiply By
ft	m	0.3048
in	mm	25.4
in	cm	2.54
m	ft	3.281
mile	km	1.609
mm	in	0.03937
yard	m	0.9144

Table D–3
Units of Force

To Convert	To	Multiply By
kg	N	9.807
kg	ton (metric)	1/1000
kip	kN	4.448
kip	lb	1000
kN	k	0.2248
lb	kg	0.4536
lb	N	4.448
lb	kip	1/1000
lb	ton (short)	1/2000
lb	ton (long)	1/2240
N	kg	0.1020
N	lb	0.2248
ton (long)	lb	2240
ton (metric)	kg	1000
ton (short)	lb	2000

To Convert	To	Multiply By
ton (short)	kg	907.1847
ton (short)	ton (metric)	0.907185

Table D–4
Units of Volume

To Convert	To	Multiply By
ft^3	in^3	1728
in^3	cm^3	16.3871
m^3	ft^3	35.3147

Table D–5
Units of Stress and Pressure

To Convert	To	Multiply By
atm	lb/ft^2	2116.2
atm	kPa	101.325
bar	kPa	100
kg/cm^2	kPa	98.07
kg/cm^2	lb/ft^2	2048.2
kPa	atm	9.869×10^{-3}
kPa	bar	0.01
kPa	kg/cm^2	1.0197×10^{-2}
kPa	lb/ft^2	20.885
kPa	lb/in^2	0.1450
kPa	ton (metric)/m^2	0.1020
lb/ft^2	atm	4.725×10^{-4}
lb/ft^2	kPa	0.04788
lb/ft^2	lb/in^2	1/144
lb/in^2	kPa	6.895
lb/in^2	lb/ft^2	144
lb/in^2	MPa	6.895×10^{-3}
ton (metric)/m^2	kPa	9.807
MPa	lb/in^2	145.04

Table D–6
Units of Unit Weight

To Convert	To	Multiply By
kN/m^3	lb/ft^3	6.366
lb/ft^3	kN/m^3	0.1571
lb/ft^3	kg/m^3	16.0185
lb/ft^3	lb/in^3	1/1728
lb/in^3	lb/ft^3	1728
ton (short)/yd^2	kg/m^3	1186.553

Glossary of Terms

Term	Definition
AASHTO	Association of State Highway and Transportation Officials
ACI	American Concrete Institute
ASCE	American Society of Civil Engineers
ASD	Allowable Stress Design
ASTM	American Society for Testing and Materials
BCR	Bearing Capacity Ratio
BPT	Becker Penetration Test
CBEAR	Computer Program for Bearing Capacity Analysis of Shallow Foundations
CIU	Isotropically Consolidated Undrained Triaxial Test
COR	Contracting Officer's Representative
CPT	Cone Penetration Test
CPTU	Piezocone Penetration Test
CRR	Cyclic Resistance Ratio
CRS	Constant Rate of Strain
CSR	Cyclic Stress Ratio
CSSM	Critical State Soil Mechanics
DCP	Dynamic Cone Penetrometer
DM	Design Manual
DMM	Deep Mixed Material
DMT	Dilatometer Test
DoD	Department of Defense
DOR	Designer Of Record
DPT	Dynamic Penetration Testing
DR	Relative Density
DSO	Dam Safety Officer
DSS	Direct Simple Shear
ECIFP	Engineering Considerations and Information for Field Personnel
ECSMFE	European Conference on Soil Mechanics and Foundation Engineering
EM	Engineer Manual
ER	Engineer Regulation
FC	Facility Criteria
FHWA	Federal Highway Administration
FS	Factor of Safety
FVST	Field Vane Shear Testing
GL	Geotechnical Laboratory
H2S	Hydrogen Sulfide

Term	Definition
I_A	Arias Intensity
IBC	International Building Code®
ICR	Inside Clearance Ratio
ICSMFE	International Conference on Soil Mechanics and Foundation Engineering
IEBC	International Existing Building Code®
ISSMGE	International Society for Soil Mechanics and Geotechnical Engineering
ITL	Information Technology Laboratory
L/H	Length to Height Ratio
LL	Lessons Learned
LRFD	Load and Resistance Factor Design
LSO	Levee Safety Officer
MCE_G	Maximum Considered Earthquake Geometric Mean Peak Ground Acceleration
MLR	Multilinear Regression
MP	Miscellaneous Paper
NA	Not Applicable
NATO	North Atlantic Treaty Organization
NAVFAC	Naval Facilities Engineering Command
NC	Normally Consolidated
NCEER	National Center for Construction Education and Research
NSHMP	National Seismic Hazard Mapping Program
OC	Overconsolidated
OCR	Overconsolidation Ratio
OSHA	Occupational Health and Safety Administration
PE	Project Engineer
PGA	Peak Ground Acceleration
PGV	Peak Ground Velocity
PI	Plasticity Index
PMT	Pressuremeter Test
PTI	Post-Tension Institute
PVD	Prefabricated Vertical Drain
QC	Quality Control
RE	Resident Engineer
RMC	Risk Management Center
RSF	Reinforced Soil Foundations
SAP	Structural Analysis Program
SBT	Soil Behavior Type
SCPTu	Seismic Cone Penetration Testing

Term	Definition
SDC	Seismic Design Category
SDMT	Seismic Dilatometer
SDOF	Single-Degree-of-Freedom
SHANSEP	Stress History and Normalized Soil Engineering Properties
SHT	Strain Hold Test
SLS	Serviceability Limit State
SPT	Standard Penetration Test
SQD	Sample Quality Designation
SSI	Soil-Structure Interaction
SSL	Shear Strength Loss
STP	Special Technical Publication
TEXAM	Texas Rapid Assessment Method
U.S.	United States
UFC	Unified Facilities Criteria
ULS	Ultimate Limit State
USACE	U.S. Army Corps of Engineers
USGS	U.S. Geological Survey
VST	Vane Shear Test
WSD	Working Stress Design